Plasma-Materials Interactions

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A list of titles in this series appears at the end of this volume.

Plasma Diagnostics

Volume 1

Discharge Parameters and Chemistry

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Preface

The study of plasma–material interactions has evolved into an important and dynamic field of research. An understanding of the basic physical and chemical processes underlying these interactions is vital to the development of microelectronics, surface modification, fusion, space, and other key technologies of our age. Plasma processing is a critical technology for leading-edge microelectronics. For example, ultra large scale integrated circuits (ULSI) cannot be manufactured without plasma-assisted etching and plasma chemical vapor deposition. Similarly, the various plasma–surface phenomena—physical sputtering, chemical etching, particle trapping in solid walls etc.—must be understood and controlled to achieve self-sustained fusion reactions in future commercial power plants. Plasma interactions with surfaces of spaceships can produce harmful degradation, such as the undesirable etching of thermal blankets observed in the cargo bay of the Space Shuttle. These effects could jeopardize long-term missions in space. All of these problems are now being investigated by scientists and engineers around the world.

Unfortunately, scientific and technical information in these diverse fields is often published in journals aimed at a narrow specialized audience. One of the chief goals of this series on "Plasma–Materials Interactions", which we are now initiating, is to provide an interdisciplinary forum. We hope to disseminate knowledge of basic and applied physicochemical processes and plasma-processing art to the global community. The series is structured to make this information readily accessible to scientists, engineers, students and technical personnel in universities, industry, and national laboratories. We consider plasma–materials interactions to be one of the pivotal fields of research that will contribute to the technological revolution now under way. Therefore, we hope that this series will encourage the pursuit of new ideas and expand the horizons of science and technology in allied interdisciplinary fields.

Diagnostics and characterization techniques are prerequisites for understanding plasmas and solid surfaces exposed to plasmas. Unfortunately the

necessary know-how is scattered throughout the literature, often in a form that is difficult to use. Consequently, we begin this series with an authoritative and up-to-date treatment of plasma and surface diagnostics written for an interdisciplinary audience. The authors are renowned specialists who explain how to set up, make, and interpret measurements and how to assess the validity of diagnostic data and detect complications. Finally, they present the theoretical background necessary to understand each technique with references to recent literature. Because the material is fairly comprehensive, the book is divided into two volumes.

Volume 1 contains seven chapters on the important diagnostic techniques for plasmas and details their use in particular applications. This part includes (1) optical diagnostics for low-pressure plasmas and plasma processing, (2) plasma diagnostics for electrical discharge light sources, (3) Langmuir probes, (4) mass spectroscopy of plasmas, (5) microwave diagnostics, (6) paramagnetic resonance diagnostics, and (7) diagnostics in thermal plasma processing.

Volume 2 covers diagnostics of surfaces exposed to plasmas and includes chapters on (1) quartz crystal microbalances for studies of plasma–surface interactions, (2) elemental analysis of treated surfaces by electron and ion spectroscopies, (3) spectroscopic ellipsometry in plasma processing, (4) ion beam analysis of plasma-exposed surfaces (Rutherford backscattering, elastic recoil detection, particle-induced X-ray emission and nuclear reaction analysis), (5) the interpretation of plasma probe data in fusion experiments, and (6) non-destructive photoacoustic and photothermal techniques for the analysis of plasma-exposed surfaces.

We hope that these, and subsequent books in this series, will be valuable to experts and newcomers alike. "Plasma–Materials Interactions" volumes on plasma etching technology and on plasma deposition and etching of polymers are now in press. We would welcome your suggestions for future volumes.

Orlando Auciello
Daniel L. Flamm

January, 1989

1 Optical Diagnostic Techniques for Low Pressure Plasmas and Plasma Processing

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I. Introduction 1
II. Plasma-Induced Optical Emission 3
III. Absorption Spectroscopy 15
A. Ultraviolet-Visible Absorption 15
B. Infrared Absorption 17
IV. Laser Techniques 19
A. Laser-Induced Fluorescence (LIF) 19
B. Raman Scattering 28
C. Coherent Anti-Stokes Raman Spectroscopy (CARS) 29
D. Tunable Infrared Laser Absorption 30
E. Optogalvanic Effects 34
V. Optical Techniques for Plasma-Surface Interactions 35
A. Laser Interferometry 36
B. Ellipsometry 37
C. Second Harmonic Generation (SHG) 39
D. Laser Raman Scattering From Surfaces 40
E. Surface Absorption and Photoacoustic Spectroscopy 41
VI. Summary 42
VII. Acknowledgments 43
References 43

I. Introduction

Advances in microelectronics technology over the last two decades have exceeded even the most optimistic expectations. The rapid progress in the manufacturing of complex devices such as dynamic random access memory integrated circuits has been made possible, in part, by advances in processing and process control. Low pressure glow discharges are used in many of

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these steps (Flamm, Donnelly, and Ibbotson, 1984). For example, radio-frequency plasmas in SiH_4/N_2O and SiH_4/NH_3 are used to deposit SiO_2 and silicon nitride thin films for applications including gate-oxides, device isolation and encapsulation, diffusion masks, and etch masks. Oxygen atoms formed in O_2 discharges are employed to clean wafer surfaces and remove photoresists. However, plasma processing has had the largest impact in silicon integrated circuit technology by making it possible to etch submicron-sized features with vertical side walls in silicon, metals, and insulators. The precise control of both etching and deposition processes often requires some in situ diagnostic probe to measure the rates or to detect the endpoint when a film has been etched through to expose a sensitive, underlying material. Several optical techniques have been successfully applied to this task.

Empirical optimization of plasma parameters (gas composition, pressure, power, etc.) has produced many successful etching and deposition "recipes." However, with the advent of increasingly sophisticated diagnostic techniques and computer modeling, it is now possible to obtain very detailed information on the chemistry and physics of these processing plasmas and their interactions with surfaces. Of all the diagnostics, optical techniques have yielded the most detailed information. With the availability of multichannel detectors, narrow bandwidth, tunable lasers, Fourier transform IR, and other state-of-the-art equipment, many studies have already revealed details of ion dynamics, radical formation and loss processes, and electron impact excitation mechanisms.

More fundamental studies (Donnelly, Ibbotson, and Flamm, 1984) designed to elucidate mechanisms can be placed in two categories. Since the plasma is a very complex chemical environment, one *ex situ* technique is to isolate part of the etching or deposition process from the discharge. Discharge flow tube (Donnelly and Flamm, 1980; Flamm, et al., 1981; Mucha et al., 1981; Ryan and Plumb, 1984; Danner and Hess, 1986) and molecular beam (Coburn, 1985) experiments have determined many of the reactions of both neutral and charged species with other gas-phase species as well as with materials exposed to the discharge. A second, in situ approach employs discharge conditions comparable to those in the plasma processing environment but with equipment designed to be amenable to various diagnostic probes. While these studies are often carried out under less well controlled conditions, they are nonetheless extremely valuable because they provide information more closely related to actual processing conditions.

Optical probes are particularly well suited for in situ plasma diagnostic studies because they are nonintrusive, species-selective, and can yield both space- and time-resolved information. This paper will review relevant

optical diagnostic techniques and cite examples of applications to process monitoring and fundamental studies. While many of the techniques described can be applied to other types of plasmas (e.g., thermal plasmas and microwave discharges), most of the examples are of low pressure, radio frequency discharges of the type used in microelectronics materials processing.

II. Plasma-Induced Optical Emission

Optical emission eminating from gas discharges has long been used as a qualitative diagnostic of plasma physics and chemistry. The segmenting of emission into bright and dark regions provides information on the motions of ions and electrons in response to the spatially varying electric fields, while the identification of the spectrally resolved features establish the presence of radicals and ions formed by reactions in the discharge.

The apparatus required for this technique is one of the least complicated. It requires a monochromator to disperse the plasma emission, a set of optics to image light from the discharge onto the detector, and a photodetector to measure the dispersed fluorescence. A 0.25 m focal length monochromator can provide ~ 0.05 nm resolution, which is sufficient for most studies. When resolution of atomic fine structure, molecular rotational levels, or detailed line shape measurements are necessary, a long focal length monochromator (e.g., 1 m focal length, ~ 0.01 nm resolution) or Fabre-Perot interferometer (~ 0.0005 nm resolution) can be used. The Fabre-Perot interferometer must be used in conjunction with a monochromator, since the scanning range of an interferometer is typically less than 1 nm.

Monochromatic light emerging from the exit slit of the monochromator can be detected with a photomultiplier tube used in an analog (current measuring) or digital (photon counting) mode. Alternatively, the exit slit can be replaced with an optical multichannel analyzer (OMA) placed in the image plane of the exit slit. The advantage of an OMA is that a portion of the emission spectrum (defined by the monochromator dispersion) can be recorded simultaneously without scanning the monochromator. The most sensitive OMAs have a quantum efficiency nearly as good as photomultiplier tubes and so can provide much faster data collection. Most species of interest emit in the spectral region between 200 and 900 nm, where GaAs and S-20 detectors have adequate sensitivity. Both shorter and longer wavelength detectors and optics are available. Below ~ 190 nm, vacuum enclosures are required to prevent absorption by O_2. Hence, studies in the region are less common. Longer wavelength (> 900 nm) emissions from

electronic transitions are relatively weak and less common and hence not as useful for most applications. IR emission from vibrationally excited levels can provide useful information. However, signals are many orders of magnitude weaker than visible-UV emission, while blackbody background emission requires all surfaces "seen" by the IR-detector to be cooled, preferably to liquid nitrogen temperature. Consequently, very few studies have been published in the IR region.

Light from the discharge can be imaged onto the monochromator entrance slit with UV-grade fused silica lenses and UV-coated aluminum mirrors for best average response between 200 and 900 nm. Care should be taken in determining what region of the discharge is imaged onto the slit, since this carries added information and is important in comparing studies under various operating conditions and, in particular, in different reactor geometries.

Spectrally resolved optical emission has been reported for a variety of discharges, under conditions employed in both etching and deposition. For example, emission from atomic fluorine is observed from CF_4/O_2 discharges (Harshberger et al., 1977; Flamm, 1978; Donnelly et al., 1984). Intensity increases with O_2 content (up to $\sim$ 50% O_2) and decreases with the area of silicon exposed to the plasma. From these observations one concludes that oxygen consumes CF_x species to liberate F-atoms, which in turn react with silicon to form volatile products.

Plasma-induced emission can arise from electron impact excitation,

$$A + e \rightarrow A^* + e, \tag{1}$$

electron impact dissociation,

$$AB + e \rightarrow A^* + B + e, \tag{2}$$

or an ion impact process,

$$A^+ + e(+M) \rightarrow A^*(+M), \tag{3}$$

where A and B are atoms or molecules, * indicates the excited, emitting species, and $e(+M)$ may be a neutral species, a negative ion, an electron plus a third body, or a surface. Each of these processes has been found to occur in low pressure glow discharges. For example, reaction (1) is responsible for emission from excited F-atoms in CF_4/O_2 discharges (Gottscho and Donnelly, 1984), and also from Cl in Cl_2 discharges under most conditions (Gottscho and Donnelly, 1984), while process (2) has been observed by time-resolved emission in the momentary cathode sheath of a radio-frequency Cl_2 discharge (Gottscho and Donnelly, 1984; Donnelly et al., 1985; Flamm and Donnelly, 1986). Excitation of H-atoms in a H_2-DC discharge has been ascribed in part to reaction (3) (Capelli et al., 1985; Benesch and Li, 1985).

Molecular species can also be excited by chemiluminescent recombination reactions,

$$A + BC \rightarrow AB^* + C. \tag{4}$$

For example, such processes occur in Cl_2 discharges due to Cl-atom recombination and in F-atom containing discharges when silicon is present (Donnelly and Flamm, 1980; Flamm et al., 1981), from the reaction

$$SiF_2 + F\ (\text{or } F_2) \rightarrow SiF_3^*(+F). \tag{5}$$

Emission from SiF_3^* has been used as a probe of etching kinetics (Mucha et al., 1981; Donnelly and Flamm, 1980; Flamm et al., 1981), and also as a spatially localized endpoint detector (Donnelly and Flamm, 1983). Recently, the same emission has been found from the etching of Si by XeF_2 and in the vacuum UV photodissociation of SiF_3H (Suto et al., 1987). The latter experiment provides strong corroborating evidence that the emitting species is SiF_3^*, as previously reported (Donnelly and Flamm, 1980; Flamm et al., 1981).

Recombination reactions involving uncharged species can easily be identified by time resolving emission with respect to the applied field in rf discharges (or by pulsing DC-discharges), provided that the fluorescence lifetime of the excited species is short compared to the modulation frequency. Lifetimes of (ground state) neutral species are limited by diffusion ($\sim 10^{-4}$–10^{-2} seconds for a typical distance of $\sim$ 10 cm at 3–300 mtorr) and/or reactions ($> 10^{-3}$–10^{-5} seconds at these pressures). Hence reactive species concentrations are not significantly modulated at rf periods below 10^{-5} seconds, and chemiluminescence due to recombination reactions such as (4) are not modulated. Figure 1 shows emission spectra from a SiF_4 plasma (Donnelly and Flamm, 1980; Flamm et al., 1981) (actually recorded downstream from the discharge zone). The 13.5 MHz rf was switched off and on at a frequency of 1 kHz with a 50% duty cycle. The banded emission is modulated at 1 kHz because it is excited in the plasma by electron impact processes, while the continuum emission is not modulated because it is due to the recombination reaction (5).

Metastable species can also excite emission in plasmas. Krogh et al. (1986) have reported enhancement of H-Balmer-α emission caused by the addition of Ar to H_2 discharges at 50 and 100 mtorr total pressure. Since Ar has a lower ionization potential and a larger collision cross section, the electron energy distribution should be lower, hence electron impact excitation of H-Balmer-α emission should *decrease*. The observed *enhancement* is ascribed to metastable energy transfer between Ar metastables (11.723 and 11.548 eV) and hot H-atoms to form H at 12.09 eV. More recently, Scheller et al. (1988) have ascribed part of the emission from BCl ($A^1\Pi$) in

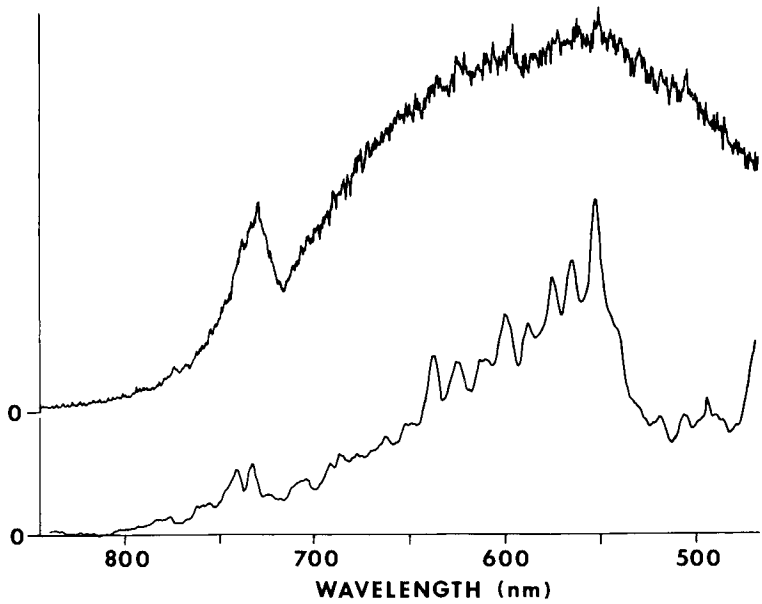


FIGURE 1. Emission spectrum recorded ~ 5 cm downstream form a 1 torr SiF_4 discharge, capacitively coupled by external electrodes on a 2 cm diameter alumina tube (Donnelly and Flamm, 1980; Flamm et al., 1981). The 13 MHz rf was switched off and on at 1 kHz. Top: DC component of the emission spectrum. Bottom: emission in phase with the 1 kHz chopping frequency. (The feature near 7300 Å is an experimental artifact).

a $< 1\%$ BCl_3/Ar discharge to participation by Ar metastables. Specifically, Ar metastables cause Penning ionization of BCl_3. Subsequent recombination of BCl_3^+ with electrons leads to formation of $BCl(A^1\Pi)$. This conclusion is based in part on the fact that the time-resolved BCl emission is phase shifted with respect to the applied voltage waveform and possesses a DC component. Since the lifetime of BCl $(A^1\Pi)$ is short compared to the rf period, and electron energy relaxation is fast, emission arising strictly from electron impact would not possess a DC component.

Unlike neutral species, electron and ion concentrations and energies change dramatically during an rf cycle, leading to a high degree of modulation in emissions excited by ion and electron impact. Ion motion becomes limited at ~ 3 MHz for typical ions (Cl_2^+, CF_3^+, etc.), while electron energy can be modulated up to ~ 100 MHz (Flamm and Donnelly, 1986). Consequently, time-resolved emission spectroscopy can distinguish between the various processes (1)–(4) that give rise to emission.

In addition, lineshape measurements can distinguish process (1) from all others. This is indicated schematically in Figure 2. Since the electron mass

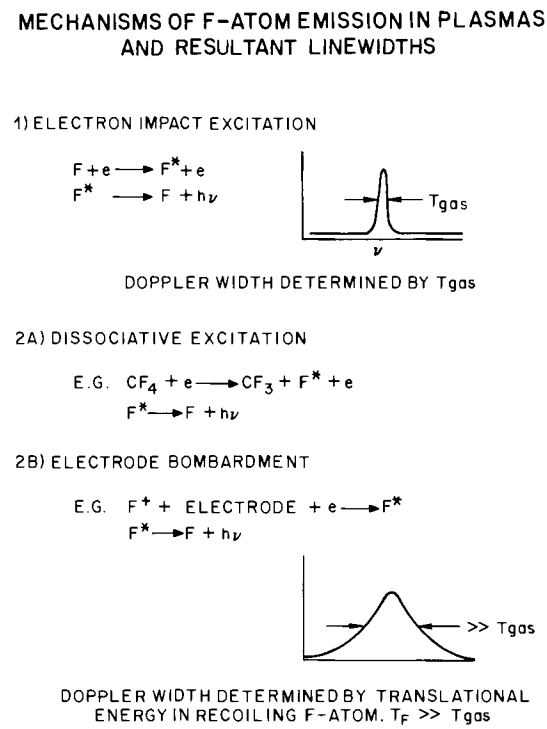


FIGURE 2. Mechanisms of atomic emission in plasmas and resulting linewidths—hypothetical example: F-atom emission in CF_4/O_2 plasmas.

is small compared to atomic or molecular masses, the velocity distribution of the emitting species formed in reaction (1) is the same as the ground state, which is in turn equilibrated with the gas temperature. On the other hand, excited species formed by other impact processes will be created with significant translational energy. Hence a Doppler width corresponding to the measured gas temperature is evidence for reaction (1), while a broadened emission corresponds to some other pathway (e.g., reactions (2)–(4)).

Emission from a CF_4/O_2 rf-discharge is shown in Figure 3. The narrow lineshape is indicative of reaction (1). (The splitting of the fluorine line is due to hyperfine structure.) On the other hand, time-resolved Cl-atom emission (Figure 4) from the cathodic sheath consists of a narrow peak (due to process (1), superimposed on a broad feature (most likely due to reaction (2)). The area ratios for these two features indicate that the process responsible for the broadened emission comprises ~ 70% of the total emission under these conditions.

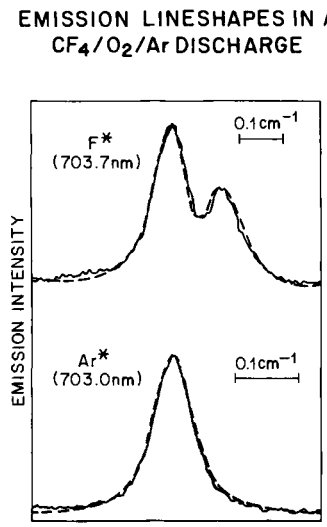


FIGURE 3. Time-averaged emission lineshapes in a $CF_4/O_2/Ar$ discharge 1 mm from the electrode (Gottscho and Donnelly, 1984). Frequency = 50 kHz, pressure = 0.3 torr.

Lineshapes can also be used to determine excited state translational energies. For example, Cappelli et al. (1985) have detected H-Balmer-α emission in a direction parallel to the electric field vector by looking through a small hole in one of the electrodes of a small parallel-plate reactor. Emission from a 50 kHz, 0.5 torr H_2 discharge is shown in Figure 5 for the point in the rf cycle when the near electrode is at its peak negative applied voltage. The central, intense peak is assigned to electron-impact excitation of H (reaction 1), and its width is limited by the instrument resolution. The shoulders between 0 and $\pm$ 0.1 nm are ascribed to formation of excited H by electron impact dissociation of H_2 (i.e., reaction 2). Finally, the Doppler-shifted emission extending out to ± 0.7 nm is thought to be due to reaction (3):

$$H_3^+ + H \rightarrow H_3^+ + H(n = 3) \tag{6}$$

$$H^+ + H \rightarrow H^+ + H(n = 3) \tag{7}$$

$$H^+ \text{ (or } H_3^+) + \text{surface} \rightarrow H(n = 3). \tag{8}$$

The most blue-shifted emission corresponds to H atoms with 540 eV or 85% of the applied voltage amplitude.

If emission arises from reaction (1), then the intensity is proportional to

$$I_{A^*} \propto n_{A^*} \propto n_A \int_{E^0}^{\infty} \sigma_A(E)\epsilon(E)\, dE, \tag{9}$$

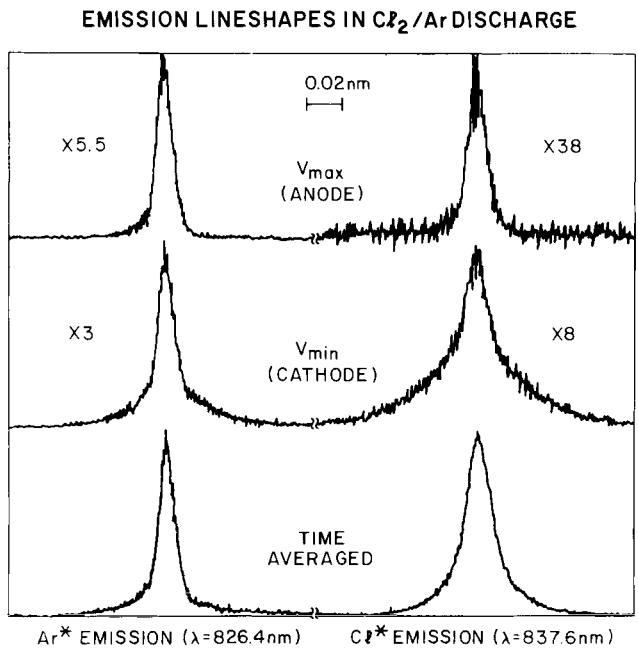


FIGURE 4. Emission lineshapes in a Cl_2/Ar discharge (Gottscho and Donnelly, 1984). The top and middle traces are sampled 1 mm from one electrode for brief periods (2 μsec) near the applied voltage maximum and minimum. The bottom trace is a time-averaged emission near the electrode.

where n_{A^*} is the number density of the excited species A^*, n_A is the ground state number density of A, $\sigma_A(E)$ is the energy-dependent electron impact excitation cross section, $\epsilon(E)$ is the electron energy distribution function, and E^0 is the threshold energy. Electron energy distributions in rf discharges are complex and are not easily measured or calculated. Consequently, many researchers have employed an actinometry technique (Donnelly et al., 1984; Gottscho and Donnelly, 1984; Donnelly et al., 1985; Flamm and Donnelly, 1986; Coburn and Chen, 1980; d'Agostino et al., 1981, 1982, 1984; Ibbotson et al., 1983), where small quantities of an inert gas X (e.g., Ar) are added to the discharge and emission is detected from excited states formed in direct electron impact excitation (reaction (1)). The intensity is given by equation (9) with the appropriate constants. The inert gas is chosen such that the threshold energy and cross-section energy dependence are similar to that of the species of unknown concentration. However, as pointed out by d'Agostino et al. (1984), this constraint is not necessary if it can be shown that the emission intensity of actinometer X_1,

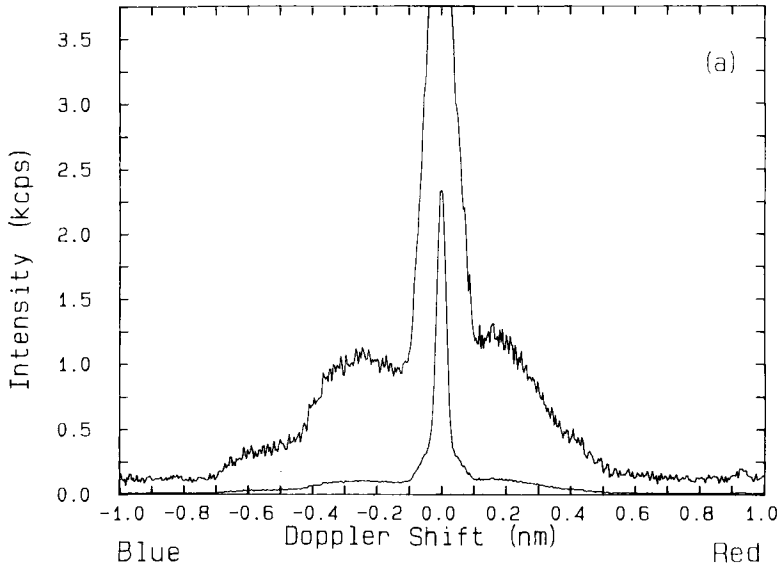


FIGURE 5. H-Balmer-α emission lineshapes from a 0.5 torr, 50 kHz H_2 plasma, measured through a small hole in the electrode (Capelli et al., 1985). The near electrode was the mementary cathode during the point in the cycle over which emission was recorded.

with threshold E_1^0 below E^0, tracks emission from actinometer X_2 with threshold E_2^0, which is above E^0.

The concentration then becomes

$$n_A = cn_x\left(\frac{I_A}{I_x}\right), \tag{10}$$

where c is a proportionality constant. This technique has been verified for CF_4/O_2 and NF_3/Ar discharges by chemical titrations (Donnelly et al., 1984) and (for CF_4/O_2) by lineshape measurements (Gottscho and Donnelly, 1984). A plot of F-atom emission (normalized to a suitably chosen Ar line) vs. n_F determined downstream by the chemiluminescent titration reaction with Cl_2, is shown in Figure 6. The normalized emission is directly proportional to n_F over a wide range of conditions, regardless of gas composition. Emission from $Br(5p^2P°)$ normalized to Ar* has also been found to track n_{Br} (Figure 7), determined by optical absorption of Br_2 (Ibbotson et al., 1983). The frequency was varied from 0.1 to 13 MHz, causing a factor of ~ 2 increase in n_{Br}. Changing frequency over this range results in dramatic changes in the time-averaged electron energy distribution and in the processes that sustain the discharge, yet the normal-

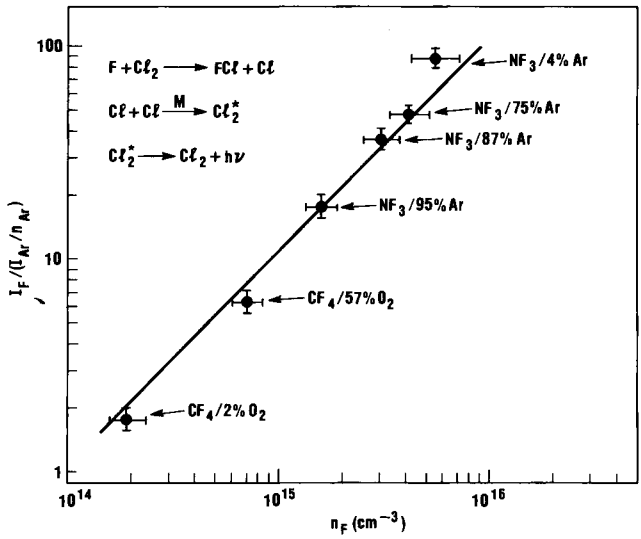


FIGURE 6. F-atom emission, normalized to Ar emission and number density, vs. fluorine atom number density, n_F, determined by chemical titration in 0.30 torr, 250 kHz CF_4/O_2, and NF_3/Ar discharges (Donnelly et al., 1984). The titration reactions are given in the figure.

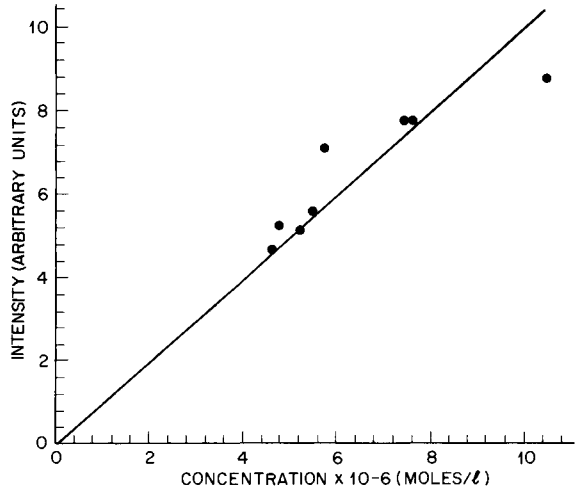


FIGURE 7. Br($5p^2P°$) emission scaled to Ar emission vs. n_{Br} determined by Br_2 absorption (see below and Ibbotson et al., 1983), for a 0.3 torr Br_2 discharge. Frequency was varied from 0.1 to 15 MHz.

ized emission continues to track n_{Br}. In general, actinometry works best for uncharged atoms with excited, emitting states roughly 10 eV above the ground state. It is generally not reliable for charged species, since excited ions are often formed by electron impact ionization-excitation of the neutral species, instead of electron impact excitation of the ground state ion. Low-lying (< 5 eV) excited states should also be avoided, since population of these states by electron impact dissociative excitation (i.e.,

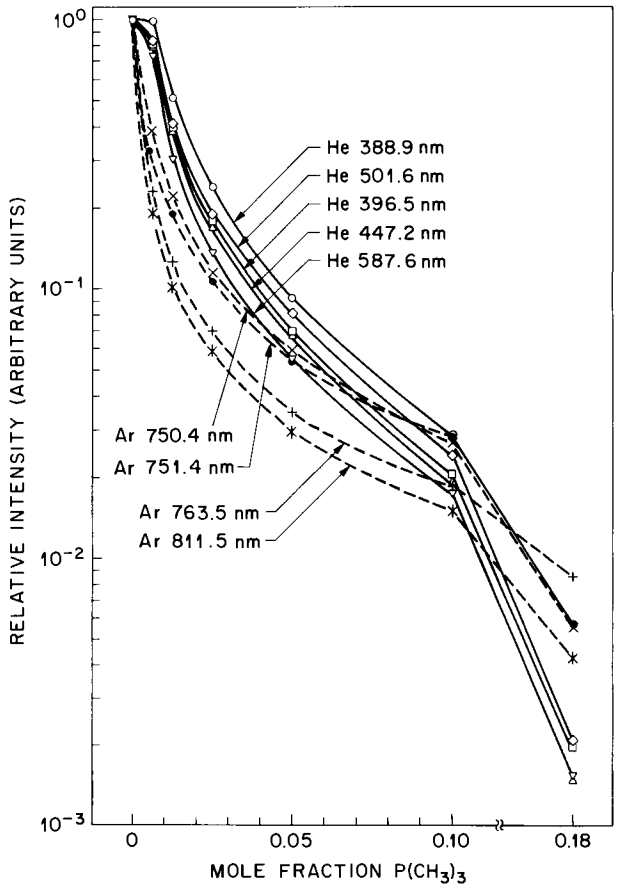


FIGURE 8. Atomic emission from a He/0.10% Ar discharge as a function of $P(CH_3)_3$ mole fraction (Brown et al., 1987). All emissions are normalized to unity at 0% added $P(CH_3)_3$. Total pressure = 0.80 torr, total flow rate = 370 sccm. Above 0.10 mole fraction $P(CH_3)_3$, the total flow rate was reduced to 98 sccm. Frequency = 13 MHz, power = 2.0 W, electrode diameter = 3.8 cm, electrode spacing = 1.0 cm.

reaction (2)) becomes statistically more likely. Because of this, application of actinometry to most molecular species is questionable without verification that emission is arising from electron impact excitation of the ground state of the molecule and not by a dissociation or recombination process. As with the fluorine atom emission studies cited above, one reliable method to determine the excitation mechanism would be to measure the width of a single rotational line and show that it corresponds to the Doppler width for the gas temperature.

Optical emission can also provide information on electron energy distributions in rf discharges. Emission from small quantities of Ar and He added to the discharge can be used to sense relative number densities of electrons in various energy windows. For example, adding a species with a low ionization potential and large inelastic collision cross section to a discharge in a high ionization-potential rare gas will quench high-energy electrons. This is observed when trimethyl phosphine, $P(CH_3)_3$ (IP = 8.5 eV) is added to a He (IP = 24.6 eV) discharge (Brown et al., 1987). Optical emission from Ar (E_{th} = 13 eV) and He (E_{th} = 23 eV) coming

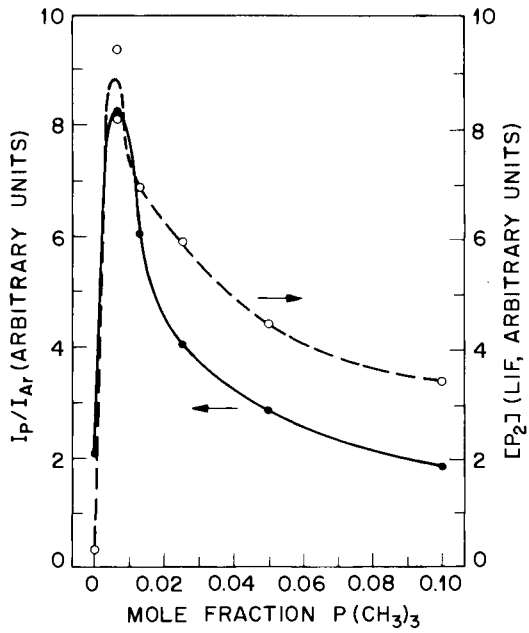


FIGURE 9. Atomic P-emission at 253.5 and 255.4 nm divided by Ar-emission at 811.5 nm (Brown et al., 1987), and P_2 concentrations measured by ArF excimer laser induced fluorescence vs. $P(CH_3)_3$. Conditions identical to Figure 8.

from the center of a parallel-plate discharge is quenched rapidly with added $P(CH_3)_3$ (Figure 8). Since optical emission is excited by electrons with energies above E_{th} (and possibly by He-metastables, which, in turn, are formed by high energy electrons), the quenching of optical emission reflects the suppression of high energy electrons by $P(CH_3)_3$. This cooling of the electron energy distribution also suppresses the decomposition of $P(CH_3)_3$, to form P (estimated by actinometry) and P_2 (determined by laser-induced fluorescence), hence they peak at very low $P(CH_3)_3$ concentrations (Figure 9).

Time-resolved optical emission provides further insights into the discharge dynamics (Flamm and Donnelly, 1986). Figure 10 shows time-

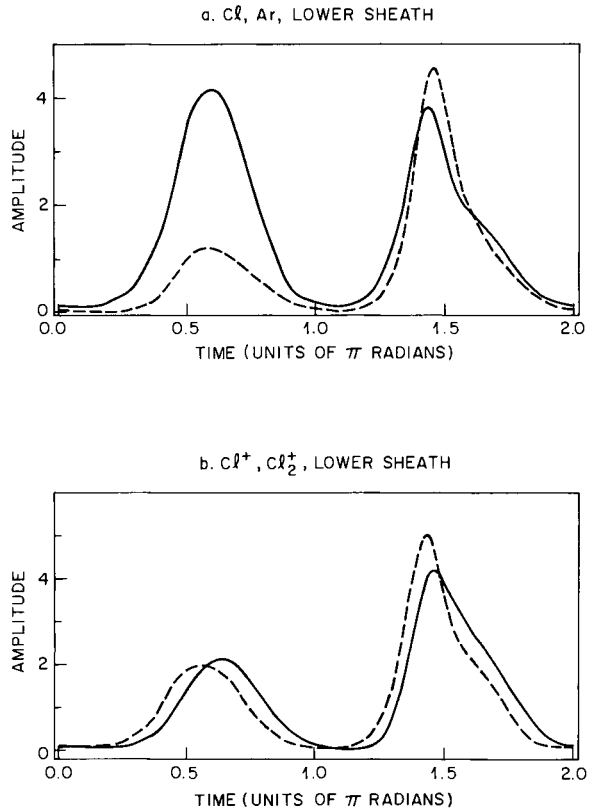


FIGURE 10. Excitation rates for (a): Cl (solid line) and Ar (dashed line) atoms, and (b): Cl^+ (solid line) and Cl_2^+ (dashed line) in the sheath 0.54 mm above the grounded electrode (Flamm and Donnelly, 1986). Frequency = 220 kHz, pressure = 0.30 torr.

resolved emission from the grounded electrode sheath region of a Cl_2/Ar discharge. Intensity peaks twice per cycle near the maximum positive and negative rf voltage applied to the opposite electrode. The 0.3 μsec lag time between the peak positive applied voltage (or peak negative voltage at the momentary "cathodic" grounded electrode) corresponds to the time required to accelerate positive ions across the sheath toward the grounded electrode. These ions can excite emission directly or can eject secondary electrons from the electrode, which are rapidly accelerated across the sheath, where they cause electron impact induced emission. During the opposite part of the cycle, electron current flows toward the grounded electrode and induces emission. The peak in the flux of energetic electrons that give rise to this emission slightly leads the maximum applied voltage.

In addition to the more fundamental information obtained from plasma excited optical emission highlighted above, spectrally resolved emission has been widely used to measure relative etch rates and control many plasma processes. The most common application has been the determination of etching endpoints by sensing changes in emission from etch products or reactants. For example, endpoints for etching of thin films of aluminum and photoresist have been determined by detecting emission from AlCl (Curtis and Brunner, 1978) and CO (Degenkolb et al., 1976) etch products, respectively, while Si etching endpoints have been sensed by observing an increase in the intensity of emission from F-atom etchants (Harshbarger et al., 1977).

III. Absorption Spectroscopy

A. Ultraviolet-Visible Absorption

Low pressure rf discharges are quite amenable to absorption spectroscopy, yet relatively few plasma diagnostic studies have been carried out using this technique. The principle advantages to absorption spectroscopy are that it is relatively simple to implement in most cases and that it yields absolute concentrations, provided that the absorption cross section is known. The sensitivity is determined by Beer's law

$$\frac{I}{I_0} = e^{-\sigma \ell n}, \tag{11}$$

where I_0 is the incident light intensity and I is the light transmitted through a path length ℓ of a species with an absorption cross section σ and number density n.

The apparatus used for optical emission measurements can easily be adapted for absorption measurements. Light from a lamp or laser is directed through the discharge and onto the entrance slit of the monochromator. The optimum light source depends on the species one wishes to detect. For diatomic and polyatomic species, a continuum light source is best. Between ~ 160 and ~ 350 nm, a deuterium lamp is used, while in the visible region a tungsten lamp is preferred. The output of lamp can be chopped and phase-sensitive detection used to suppress the large background from the plasma.

In the ultraviolet and visible wavelength region, typical electronic excitation cross sections for polyatomic species are $\sim 10^{-16}$ cm^2. A 1% absorption is easily detected, which for a path length of 10 cm, corresponds to $\sim 5 \times 10^{15}$ cm^{-3} or ~ 0.1 torr. This limit can be extended one to two orders of magnitude by employing extremely stable light sources, signal averaging, or long path lengths (including multipass techniques). Consequently, parent compounds, and reactant and product species present at moderate to high concentrations, can be detected in the discharge. The main disadvantage is spectral congestion—visible and ultraviolet absorption is broad and often structureless. Multicomponent absorption requires deconvolution, which reduces accuracy and reliability.

Atoms typically have larger cross sections, however, most species of interest (halogen atoms, O and N) have transitions in the vacuum ultraviolet region (1100–1600 Å), requiring vacuum enclosures for the source, monochromator, and detector. The best source for atomic absorption is a resonance lamp containing a small quantity of the atom one wishes to detect. The partial pressure of the atom in the lamp must be small to avoid line reversal (a bleaching out of the center of the emission line due to absorption in the lamp). Vacuum ultraviolet atomic absorption has been employed in gas-phase kinetic and photochemistry studies. Such techniques could also serve as diagnostic probes in many low-pressure radio frequency discharges.

Figure 11 demonstrates the kind of information one can obtain with absorption spectroscopy (Ibbotson et al., 1983). Light from a tungsten filament lamp was chopped and passed through a bromine discharge. The transmitted beam was dispersed with a scanning monochromator and phase-sensitive detection was used to discriminate against light from the discharge. Entire visible absorption spectra of Br_2 were recorded as a function of frequency. Integrated absorption coefficients were then measured and used with Beer's law to determine n_{Br_2} (and hence n_{Br}). It is important that one measure the *integrated* absorption over the entire band to remove the effects of temperature broadening and shifting (Ibbotson et al., 1983). One sees from Figure 11 that Br-atom production increases by

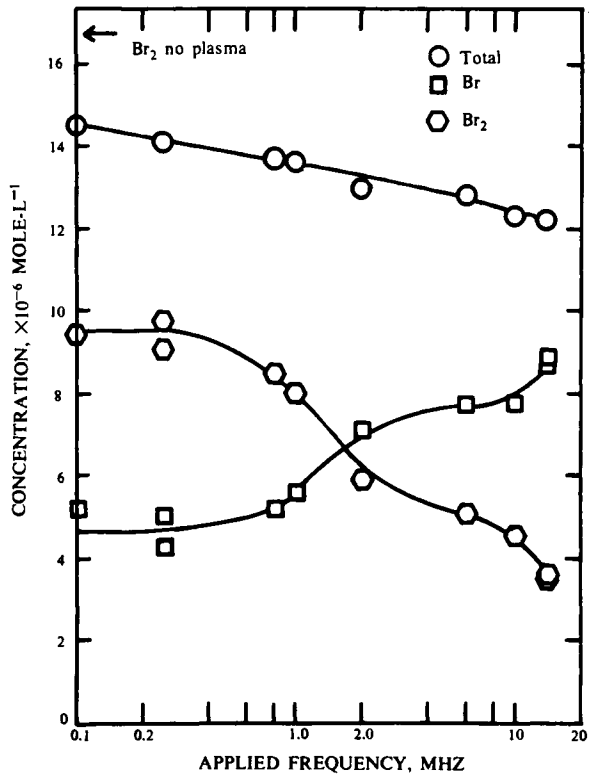


FIGURE 11. Br_2 and Br concentration in a 0.30 torr bromine discharge, determined by visible absorption by Br_2 (Ibbotson et al., 1983) as a function of frequency.

a factor of about two, with frequency between 0.1 and 15 MHz and goes through an inflection point at ~ 1 MHz.

B. Infrared Absorption

Since IR absorption coefficients are typically three orders magnitude weaker than UV-visible values, conventional IR absorption spectroscopy is best suited for strong absorbers present at high concentrations. For example, Hamasaki et al. (1982) have used IR absorption to measure the dissociation efficiency of SiH_4 in a DC discharge as a function of power and flow rate. They find that SiH_4 is 100% dissociated at 10 W for 30 sccm SiH_4 flow rate and at 30 W for 50 sccm SiH_4 flow rate.

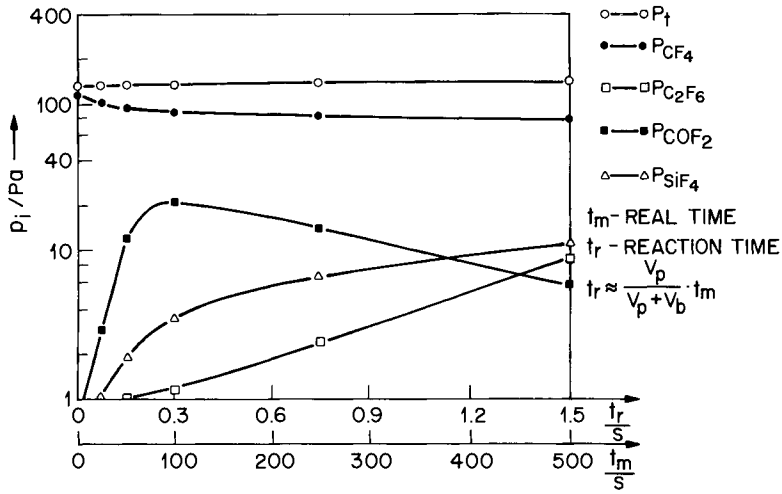


FIGURE 12. Species partial pressures P_i in Pascals determined by IR absorption for CF_4/O_2 plasma etching of Si vs. reaction time in seconds (see text). Applied RF frequency = 50 kHz, 800 V pp, 15 mA pp. (from Poll et al., 1976, 1982a, 1982b).

Poll and coworkers (1976, 1982a, 1982b) have reported detailed IR absorption of fluorocarbon/oxygen plasmas used to etch silicon. They have recorded spectra in two configurations. In one case, the IR absorption cell was located just downstream of a parallel-plate discharge operating in a flowing mode. Alternately, the absorption cell was connected to a side arm on a valved-off plasma reactor. Data taken in this latter configuration for a CF_4/O_2 discharge etching silicon is shown in Figure 12. The reaction time (t_r) is defined as the real time (t_m) multiplied by the ratio of the volume between the two electrodes divided by the total chamber volume. The total pressure remained relatively constant. However, the partial pressure of CF_4 decreased with time and eventually reached a steady state, while C_2F_6 and SiF_4 etch product increased with time. COF_2 created by reactions of O with CF_3 (Ryan and Plumb, 1984)

$$O + CF_3 \rightarrow COF_2 + F \tag{12}$$

initially increases, but then falls off with increasing time, possibly due to competition of reaction (12) with reactions between O and unsaturated species such as C_2F_4.

More sophisticated techniques, such as second derivative tunable diode laser absorption or Fourier transform-infrared (FT-IR), can extend detection limits by several orders of magnitude, making it possible to probe

atoms and free radicals present at lower levels. IR-laser absorption techniques are discussed in more detail below. The advantages to FT-IR are rapid acquisition of the full IR spectrum and improved signal-to-noise, which extends the sensitivity limit. Cleland and Hess (1987) have reported FT-IR studies of N_2O discharges and of a Cl_2 plasma etching aluminum. The plasma reactor was positioned inside the sample chamber of the spectrometer. Percent dissociations of N_2O were measured in a 13 MHz discharge. At 30 W, N_2O was found to be 66% dissociated. They also detected AlCl, $AlCl_3$, and Al_2Cl_6 while etching Al in a 3 MHz, 40 W Cl_2 discharge. Formation of AlCl and $AlCl_3$ was ascribed to fragmentation of Al_2Cl_6, the primary etch product.

IV. Laser Techniques

A. Laser-Induced Fluorescence (LIF)

This technique employs a laser (usually a tunable dye laser) to excite a resonance between a lower state, (in most instances a single rotational level or group of rotational levels of the ground vibronic state) and a level in an excited electronic state. Continuously tunable dye lasers are pumped by N_2 lasers, Nd-YAG lasers, excimer lasers, or flash lamps for pulsed output, or by Ar^+ or Kr^+ lasers for CW radiation. Dye lasers pumped by either an excimer laser or a Nd-YAG laser are highest in peak power and are now the most common tunable pulsed sources. Excimer laser pumped dye lasers provide output from 330–900 nm, while frequency doubled and tripled Nd-YAG lasers can be used to pump dye lasers between ~ 380 and 900 nm. Frequency doubling in crystals such as KDP can extend the UV output to ~ 200 nm. The laser beam is usually in a horizontal configuration, parallel to the electrode surface (Figure 13). In this manner, the discharge can be probed in the direction perpendicular to the electrode surface by translating the beam and detection optics. Fluorescence is detected at right angles with a monochromator or a filter in front of a photomultiplier tube or by an optical multichannel analyzer (with the monochromator).

The advantages of laser-induced fluorescence are that it is extremely sensitive (number densities below 10^8 cm^{-3} are often detectable), selective (usually no detectable interference from other species), and can provide temporal and three-dimensional spatial resolution. The disadvantage is that it is only applicable to those species with bound excited electronic states that can be reached by optically allowed transitions from the ground state. Uncharged species such as CCl (Gottscho et al., 1983; Cleland and Hess,

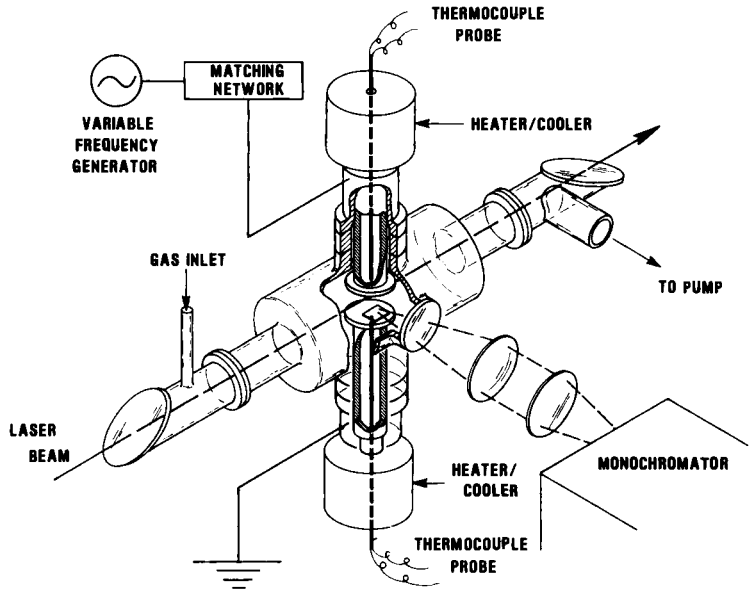


FIGURE 13. Laser-induced fluorescence (or laser Raman scattering) apparatus for probing of a parallel-plate plasma reactor.

1987), BCl (Moore et al., 1984; Gottscho and Mandich, 1985), CF (Ninomiya et al., 1986), CF_2 (Ninomiya et al., 1986; Hargis and Kusher, 1982), and SiH (Matsumi et al., 1986) have been detected in rf and microwave discharges, and their relative concentrations measured as a function of position and external parameters.

The application of LIF as a quantitative tool is possible, provided the detected fluorescence intensity is corrected for laser intensity variations and for quenching. For the single photon process in the absence of saturation, fluorescence intensity varies linearly with laser intensity. If the fluorescence collection optics are focused on the center of the beam path inside the reactor and Beer's law absorption is assumed and if the detection electronics are set to capture the entire fluorescence pulse (or a CW laser is used), the corrected fluorescence intensity is given by

$$F = F_t \sqrt{\frac{\left(I_0^{\text{in}}\right)^2}{I_t^{\text{out}} \cdot I_t^{\text{in}}}}, \tag{13}$$

where F_t, I_t^{in}, and I_t^{out} are the fluorescence intensity, input, and output laser intensities at a time t after the beginning of a set of measurements, and I_0^{in} is the reference beam intensity at the beginning of the measure-

ments ($t = 0$). For multiple photon excitation via a real or virtual intermediate state, the fluorescence intensity varies superlinearly with laser intensity and the normalization factor should be determined empirically by measuring fluorescence intensity as a function of laser power.

The excited state produced by absorption of laser light may relax via fluorescence or nonradiative collision processes (quenching). For a simple two-state quenching model, the fluorescence intensity is given by

$$F_{\nu_f} = C_0 \frac{k_r}{k_r + \sum_M k_{q_M} P_M} [AB] \tag{14}$$

where F_{ν_f} is the fluorescence intensity at frequency ν_f, $[AB]$ is the concentration of the ground state, k_r is the radiative rate constant for the excited state (i.e., reciprocal of the radiative lifetime), k_{q_M} is the quenching rate constant of AB by species M, and P_M is the partial pressure of species M. C_0 is a proportionality constant containing geometric, spectroscopic, and kinetic factors related to excitation and detection efficiency. A full treatment of the factors required to directly relate fluorescence intensity to *absolute* number densities has been given elsewhere (Donnelly et al., 1982). For pressures encountered in many semiconductor processing plasmas, quenching is important, and the constants k_r and k_{q_M} in equation (14) must be measured. This can be done using pulsed excitation-time resolved fluorescence.

One must be careful in relating fluorescence intensities to quantitative, relative concentrations using equation (14), since k_{q_M} is a function of not only gas composition, but also of gas temperature (a $T^{-1/2}$ dependence, if the quenching cross section has no temperature dependence, as is often the case) and, in some cases, position in the discharge and phase of the rf cycle. The latter two effects can be very pronounced for ions, since the large electric fields present in the sheaths close to the electrodes cause the laser-excited ion to suffer many more collisions in its radiative lifetime. For example, the quenching rate constant of a 250 eV ion would be 100 times that of a thermalized, room temperature ion.

Another way to minimize errors in relating signals to relative concentrations caused by quenching (at least for uncharged species) when using pulsed laser excitation is to choose a detector gate width that is coincident with the laser pulse and short compared with the fluorescence lifetime. For example, for $k_{q_M} M \gg k_r$, $k_{q_M} = 5 \times 10^{-10}$ cm^3 sec^{-1} and $M = 3 \times 10^{16}$ cm^{-3} (1 torr, room temperature), the gate width should be much less than 50 nsec.

Relative number densities measured by laser induced fluorescence can also be in error due to changes in internal energy. In the laser-induced

fluorescence method, the laser is often exciting a single rotational level of one vibrational level of a particular excited electronic state. However, one is often interested in determining species concentrations as a function of variables (e.g., power, pressure, frequency, or position and phase in the rf cycle), irrespective of rotational or vibrational level. Often, researchers simply record fluorescence intensity while changing plasma parameters, with the laser frequency fixed. However, unless rotational and vibrational temperatures are also determined, then the fluorescence intensity cannot be related to even relative number densities. The population of level (v, J) of a diatomic molecule with number density N is given by (neglecting nuclear spin, anharmonicities, and centrifugal distortions)

$$P_{v,J} = N\frac{(e^{-(v+1/2)\omega/kT_v})(2J+1)e^{-BJ(J+1)/kT_r}}{Q_{\text{rot}}Q_{\text{vib}}}, \tag{15}$$

where Q_{rot} and Q_{vib} are the rotational and vibrational partition functions. So, strictly speaking, one must evaluate the rotational and vibrational temperatures for each set of conditions to determine overall relative number densities. In practice, if one chooses a rotational level near but above the Maxwell-Boltzman maximum, then the population of that level is relatively insensitive to temperature, and so the error in neglecting changes in temperature is relatively small, provided the temperature excursions are not extreme.

Positive ions play a critical role in achieving anisotropic etching with the high aspect ratios necessary for replication of submicron features. Ion bombardment is also believed to be beneficial in promoting surface mobility in adsorbates, which is needed for growth of high quality films and conformal coverage of a patterned substrate. Because of its high sensitivity, laser-induced fluorescence is particularly well suited to study the motion of ions, which are present at levels below 10^{11} cm^{-3}. Such studies have been carried out on Cl_2^+ and N_2^+, yielding detailed information on the motion of ions (Donnelly et al., 1982; Gottscho et al., 1984). Data from some early studies of Cl_2^+ in a chlorine discharge (Donnelly et al., 1982) are shown in Figure 14. Time-averaged Cl_2 number density is plotted vs. applied frequency for several power densities, along with the rf voltage. The laser probed a spot in the center of the discharge, equidistant from two 3.8 cm diameter electrodes, spaced 1 cm apart. At power densities below ~ 0.5 W/cm^2, the number density dips near 1 MHz, due to the time varying electric field, which accelerates ions toward the electrode surface, where they are lost. Above ~ 5 MHz the ions can no longer respond to the rapidly time varying fields. At the high power densities, ion densities rise, the sheaths contract, and electric fields do not penetrate to the same extent to the center of the discharge, hence ion densities are nearly frequency independent.

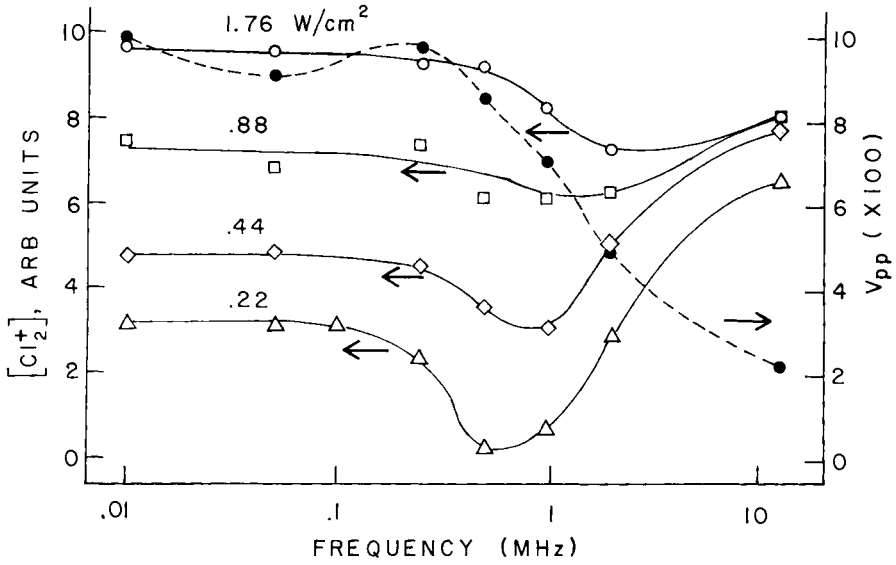


FIGURE 14. $C\ell_2^+$ concentration in the center of a chlorine discharge and peak-to-peak voltage vs. applied frequency. Electrode separation = 1.0 cm, pressure = 0.30 torr. Eight arbitrary units corresponds to $\sim 1 \times 10^{10} C\ell_2/cm^3$ (Donnelly et al., 1982).

Neutral radicals are present at levels several orders of magnitude higher than ions and are hence of primary importance in chemical aspects of etching and deposition. CF_2 radicals play an important role in fluorocarbon plasmas such as CHF_3 used to etch SiO_2 selectively over Si or for plasma deposition of teflon-like films. Hargis and Kushner (1982) have reported detection of CF_2 in CF_4 and C_2F_6 radio frequency discharges by exciting the $\tilde{A}^1B_1$ $(0,6,0) \leftarrow \tilde{X}^1A_1$ $(0,0,0)$ transition with a 248 nm KrF laser. They find that CF_2 increases linearly with power and that C_2F_6 yields roughly four times as much CF_2 as CF_4 at any given discharge power. This effect is attributed first to the larger electron impact dissociation cross section for C_2F_6, second, to the fact that there are two CF_2 groups per C_2F_6 vs. one for CF_4, and third, to the assumption that the higher fluorine atom densities in a CF_4 discharge will suppress CF_2 by the reaction

$$CF_2 + F + M \rightarrow CF_3 + M. \tag{16}$$

More recently, Ninomiya et al. (1986) have extended the CF_2 LIF technique to microwave discharges in CF_4 and C_3F_8. Not surprisingly, they find that C_3F_8 discharges yield much more CF and CF_2 than do CF_4 microwave discharges. They also observe a maximum in CF and CF_2 concentration as a function of power and pressure. They attribute the falloff in concentration

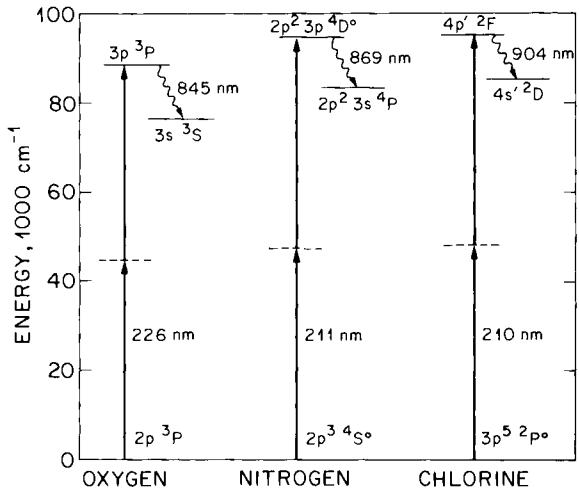


FIGURE 15. Energy level diagrams for two-photon laser induced fluorescence detection of O, N (Bischel et al., 1981) and Cl atoms (Heaven et al., 1982).

at high pressure to electron energy relaxation and recombination reactions. The falloff at high power is somewhat more puzzling. Perhaps the plasma volume expanded somewhat or the gas temperature started to increase dramatically at these high powers.

Laser-induced fluorescence can also be extended to the detection of some species with high-lying electronic states by employing two-photon excitation. This technique has been demonstrated for atomic and diatomic species, including O and N (Bischel et al., 1981) and has been used to detect O atoms in an O_2 plasma (Di Mauro et al., 1984) and Cl atoms downstream from a Cl_2 discharge (Heaven et al., 1982). The energy levels involved in these two-photon processes are shown in Figure 15.

Etching products have also been detected by laser-induced fluorescence. One of the more controversial questions in silicon etching involves the formation of SiF_2 as a primary reaction product. The formation of SiF_2 was first proposed from chemiluminescence studies (see Section II), and has recently been confirmed by LIF studies of Matsumi et al. (1986). An upstream microwave discharge was used to dissociate CF_4 (with and without added O_2) and generate reactive species, the most important being F-atoms. Silicon was etched, forming SiF_2, which was detected above the surface by exciting vibronic bands in the 222–225 nm region. The SiF_2 LIF intensity was found to track the etch rate directly.

Laser-induced fluorescence can also be used to measure velocity distributions of atoms and small molecules in plasmas. A narrow bandwidth laser is tuned through an atomic resonance line or a single rotational level molecular transition. The Doppler-shifted transition frequency (ν_0') can be related to the component of the velocity component along the axis (Wright et al., 1981, 1982)

$$\nu_0' = \nu_0\left(1 + \frac{v\cos\theta}{c}\right), \tag{17}$$

where v is the velocity of the species traveling at a polar angle θ with respect to the surface normal and c is the speed of light. The perpendicular velocity component $v_\perp$ measured in the laser-induced fluorescence experiment is related to the transition frequency by

$$\nu_0' = \nu_0\left(1 + \frac{v_\perp}{c}\right). \tag{18}$$

The transformation of measured laser induced fluorescence lineshapes into velocity distributions has been treated in detail for atoms by Wright et al. (1981, 1982) and Husinsky (1985). The measured velocity distribution, $N(v_\perp)$ is related to the actual velocity distribution, $n_1^0(v_\perp')$ by

$$N(v_\perp)\,dv_\perp \alpha \left\{\int_0^x n_1^0(v_\perp')Y(v_\perp')L(v_\perp - v_\perp')\,dv_\perp'\right\} dv_\perp\,, \tag{19}$$

where $L(v_\perp - v_\perp')$ is a Lorentzian lineshape function given by

$$L(v_\perp - v_\perp') = \frac{2\Gamma(v_\perp')/\pi}{(v_\perp - v_\perp')^2 + \Gamma(v_\perp')^2}, \tag{20}$$

where Γ is the half width at half maximum for the Lorentzian function. $Y(v_\perp')$ is the fluorescence yield (photons/atom/pulse for pulsed laser excitation or photons/atom for CW laser excitation). Both Γ and $Y(v_\perp')$ depend on the number of levels involved in the overall excitation process, the degeneracies and lifetimes of the levels involved, the laser power density, the transit time of the species in the beam, as well as the beam spatial and temporal profile and spectral bandwidth. These functions have been evaluated for a three-level atomic system under both pulsed and CW conditions (Wright et al., 1981, 1982). They have applied Doppler-shifted laser spectroscopy to determine velocity distributions of atoms sputtered

from target surfaces subjected to ion bombardment. The same technique can, of course, be used to study ion stimulated desorption processes occurring in plasma etching or to measure ion (and neutral species) translational energies.

Recently, laser excitation has been used to measure time and space-resolved electric fields by taking advantage of the Stark effect. A narrow bandwidth laser is used to populate a single level in an excited state. Changes in the detected fluorescence intensity due to Stark mixing (Moore et al, 1984; Gottscho and Mandich, 1985) or optogalvanic effects induced

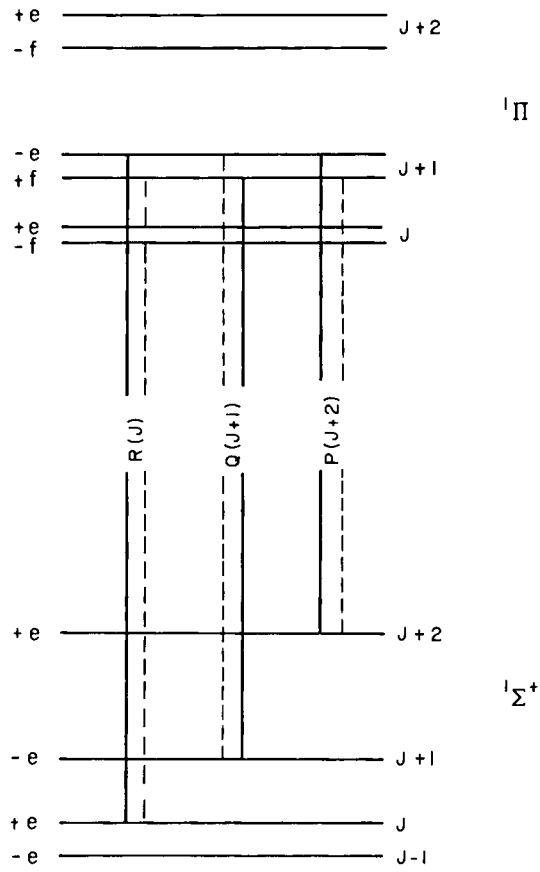


FIGURE 16. Energy level diagram for a $^1\Pi \leftarrow {}^1\Sigma^+$ transition of a heteronuclear diatomic molecule (e.g., BCl). The dashed lines refer to parity forbidden transitions in the absence of an electric field (see Moore et al., 1984; Gottscho and Mandich, 1985).

by Stark shifting (Lawler et al., 1984; Doughty et al., 1984; Doughty and Lawler, 1984; Ganguly and Garscadden, 1985), can be directly related to the electric field through calibrations or calculations. Gottscho and coworkers (Moore et al, 1984; Gottscho and Mandich, 1985) have mapped out electric fields in BCl_3 rf discharges by detecting parity mixing in BCl, while Lawler and coworkers (Lawler et al., 1984; Doughty et al., 1984; Doughty and Lawler, 1984) and Ganguly and Garsaddon (1985) have looked at He and Ne DC discharges in the cathode fall region by optogalvanic detection of Stark shifting.

The parity mixing mechanism (Moore et al., 1984; Gottscho and Mandich, 1985) for BCl is shown in Figure 16. Since the transition is $^1\Pi \leftarrow {}^1\Sigma^+$, the rotational levels have alternating + and − parity in the lower $^1\Sigma^+$ state, and both parities in the upper $^1\Pi$ state. The selection rule for parity in electric dipole allowed transitions is $+ \leftrightarrow -$, hence if one excites the $J' + 1$

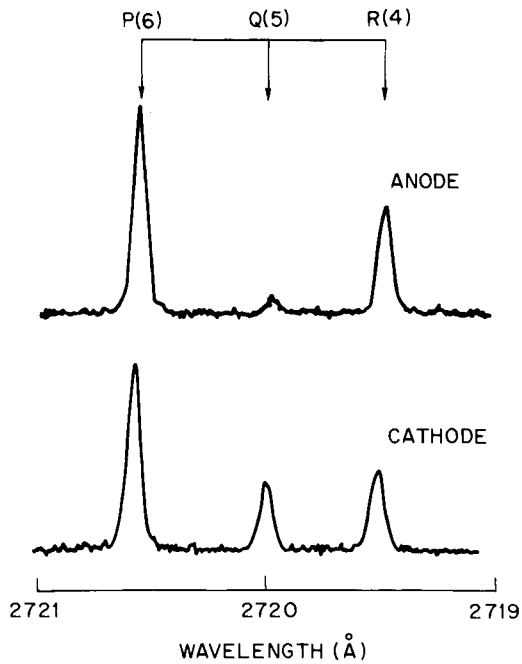


FIGURE 17. Spectrally resolved laser induced fluorescence obtained at a fixed laser frequency, showing the effects of electric field on the P, Q, and R-branch intensities (see Moore et al., 1984; Gottscho and Mandich, 1985). The laser is positioned 1 mm above the powered electrode in a 0.3 torr BCl_3 discharge at 13.56 MHz. Top: laser fired at voltage maximum. Bottom: laser fired at voltage minimum.

level, transitions will occur to the J'' (R-branch) and $J'' + 2$ levels (P-branch). The Q-branch is forbidden by the parity selection rule. However, in the presence of an electric field, the + and − levels of $J' + 1$ mix by the Stark interaction and the Q-branch gains intensity at the expense of the P and R branches. This effect is shown in Figure 17. When the laser is positioned near the electrode and is fixed at the peak anodic part of the cycle, the Q-branch is nearly absent due to the relatively weak electric field. However, if the laser is instead pulsed on at the peak cathodic part of the cycle, then the Q-branch is quite intense, due to a relatively intense electric field. The ratio of the forbidden-to-allowed line intensities for a particular J' level can be directly related to the electric field from first principles or by integrating the field across the gap between the electrodes and simply measuring the voltage drop across the electrodes. Using this technique, Gottscho and coworkers (Moore et al, 1984; Gottscho and Mandich, 1985) have measured sheath fields in BCl_3 discharges as a function of a wide variety of parameters.

B. Raman Scattering

Light of frequency ν_e is weakly scattered at frequency $\nu_e - \omega$, where ω is a frequency corresponding to an energy difference between rotational, vibrational, or electronic levels. Raman scattering cross sections for visible light are extremely small ($\lesssim 10^{-22}$ cm^2). Consequently, conventional Raman spectroscopy carried out with visible Ar^+ CW lasers is only applicable to species present at high concentrations. However, Raman cross sections increase as ν^4, hence UV lasers offer an attractive advantage in sensitivity. The configuration for laser-Raman probing is very similar to that used for laser-induced fluorescence (Figure 13). However, since the Raman signals are weak, and close in wavelength to the exciting light, a double monochromator is often required. Alternatively, a single monochromator with a long-wavelength-pass cutoff filter (Hargis, 1981) can sometimes provide sufficient scattered light rejection and provides additional advantages of large collection efficiency and simplicity.

Hargis and coworkers (1981) have used pulsed UV lasers (KrF excimer and third harmonic of Nd-YAG) to observe Raman scattering from a number of species, including CF_4, C_2F_6, and SF_6, with detection limits down to several mtorr. The pulsed-UV source provides larger signals (due to the ν_e^4 dependence of the scattering cross section) and better signal-to-noise, since gated detection electronics can be used to discriminate against background light from the discharge and other sources. They have recently extended these studies to measure N_2 and SF_6 decomposition in rf dis-

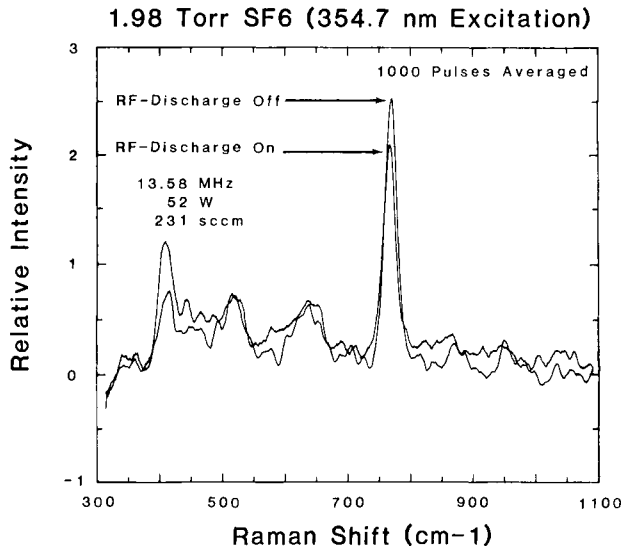


FIGURE 18. Raman spectrum of a 2 torr SF_6 discharge. The spectrum was obtained with the third harmonic of a Nd-YAG laser at 355 nm (Greenberg and Hargis, 1987). A background spectrum of 2 torr SF_6 is also shown.

charges (Greenberg and Hargis, 1987). A Raman spectrum from a 2 torr SF_6 discharge is shown in Figure 18. Contrary to theoretical predictions, they find only 10%–15% dissociation of SF_6. A disadvantage to the pulse UV Raman technique is that quite often the laser creates excited species that fluoresce and interfere with the Raman signal, either by direct excitation of species into electronically excited states or indirectly by photodissociation and/or multiple photon absorption to produce excited, fluorescing species.

C. Coherent Anti-Stokes Raman Spectroscopy (CARS)

CARS is a three wave mixing process in which light at frequency ω_1 induces a transition from an initial level (usually the ground vibrational level of the ground electronic state) to a virtual state which is stimulated to emit when a second laser is tuned in resonance with a transition at ω_2 ($\omega_2 < \omega_1$) back down to a real state at energy ΔE above the initial level. The laser at frequency ω_1 excites the molecule from this real state to a second virtual

state from which coherent anti-stokes emission is detected at frequency $\omega_1 + \Delta E$. Several excellent reviews of the CARS technique have been written (see for example Tolles, et al., 1977, Nibler and Knighten, 1977, and Druet and Taran, 1979). The technique requires two pulsed lasers. Usually the doubled output of a Nd-YAG laser at 532 nm is used to provide frequency ω_1 and pump a dye laser to provide frequency ω_2. The two beams are directed co-linearly and focussed into the reactor. The signal appears as a beam, co-linear with the two laser beams. The signal beam is separated from the two laser beams by prisms and a monochromator.

CARS has advantages over conventional Raman scattering in that there is very little background to contend with; the signal is in a beam and therefore can be detected far away from the isotropically emitting plasma. Depending on circumstances, it can therefore be more sensitive than Raman spectroscopy, although not as sensitive as laser induced fluorescence. It can be applied to virtually all species (i.e. all species with Raman active modes). The main disadvantage of CARS is its experimental complexity, requiring two high power lasers to be spatially and temporally overlapped. Nonetheless CARS has been applied to several plasma diagnostic studies. McDonald (1980) was perhaps the first to report a CARS spectrum of a species in a plasma. He measured rotational and vibrational populations of D_2 in a deuterium discharge lamp. More recently, CARS has been used to measure percent dissociation of SiH_4 in a DC discharge as a function of position in the discharge and pressure (Hata et al., 1983).

D. Tunable Infrared Laser Absorption

Tunable infrared (IR) lasers are finding increasing applications as sensitive probes of species important in plasma processing. The most commonly employed continuously tunable IR laser source is the diode laser. Diode laser spectroscopy has been used to detect a number of stable molecules, free radicals, and ions. Difference-frequency lasers, color-center lasers, and Raman shifted dye lasers have also been employed and will be covered briefly below.

Tunable IR laser spectroscopy of ions has received considerable attention in recent years. The goal of these studied was to assign IR absorption spectra of various molecular ions and not to use the technique as a probe of plasma chemistry and physics. For example, Foster and McKellar (1984) have recorded IR absorption spectra of HN_2^+, DN_2^+, and DCO^+ in a pulsed DC discharge by using a multipass, tunable diode laser technique. The total path length in their experiment was 11 m. By pulsing the discharge off and on at $\sim$ 9 kHz and using phase sensitive detection, they

were able to obtain considerable improvements in sensitivity over DC conditions.

Narrow bandwidth, tunable IR between 2250 and 4400 cm^{-1} can also be generated by a difference-frequency technique (Pine, 1974 and 1976), where radiation from an Ar^+ laser is mixed with the output of a tunable dye laser in a $LiNbO_3$ crystal. An output power of $\sim$ 10 μW in the 2250–4400 cm^{-1} region has been obtained at ($\nu_A - \nu_D$), the difference in frequency between the Ar^+ laser (ν_A) and the dye laser (ν_D) (Watson et al., 1984). This source has been used to detect IR absorption by H_3^+ in a DC H_2 discharge. In this experiment, the IR frequency was dithered by modulating the cavity length of the Ar^+ laser, and phase sensitive detection was employed. The effective pathlength of their multipass system was $\sim$ 32 m. They realized higher sensitivities by modulating the discharge, instead of the IR frequency, and still higher sensitivities by employing a velocity modulation technique. Velocity modulated IR spectroscopy was originally developed by Sakally and coworkers (Gudeman et al., 1983). This technique employs an AC discharge and takes advantage of the modulated Doppler shifts resulting from the time varying drift velocity of ions in the AC field. Phase sensitive detection in synchronous with the applied voltage frequency is employed. This technique seems to be more sensitive than other IR-laser absorption methods and has the additional advantage of providing ion energy and direction from a more detailed analysis of the peak shape. While these studies were not intended as plasma diagnostic investigations, such techniques could be employed for this purpose in cases where long pathlengths ($\gtrsim$ 1 m) or multipass configurations could be employed.

IR laser absorption techniques have been employed to detect F and Cl atoms at levels comparable to those found in plasmas ($\leq 10^{14}$ cm^{-3}) (Stanton and Colb, 1980; Laguna and Beattle, 1982; Dagenais et al., 1979; Davies and Russel, 1979). Recently, Wormhoudt et al. (1986) have extended this work to measure atomic chlorine number densities in CF_3Cl and Cl_2 discharges. Cl has a weak absorption at 882 cm^{-1} due to the $^2P_{1/2} \leftarrow {}^2P_{3/2}$ transition. In their experimental configuration, the laser beam was passed through a 0.25 m monochromator to select a single mode and then was reflected back and forth 20 times through the plasma formed between two 12.5 cm diameter parallel plate electrodes. The total path length was therefore $\sim$ 250 cm. The exiting beam was detected with a HgCdTe detector. An absorption spectrum in the region of the $^2P_{1/2} \leftarrow {}^2P_{3/2}$ transition is shown in Figure 19. The strong feature apparent in both the raw absorption scan and in the second derivative is due to the $2 \leftarrow 1$ transition for both isotopes of Cl.

A translational temperature can be calculated from the width of the absorption line, while the absolute Cl-atom number density can be derived

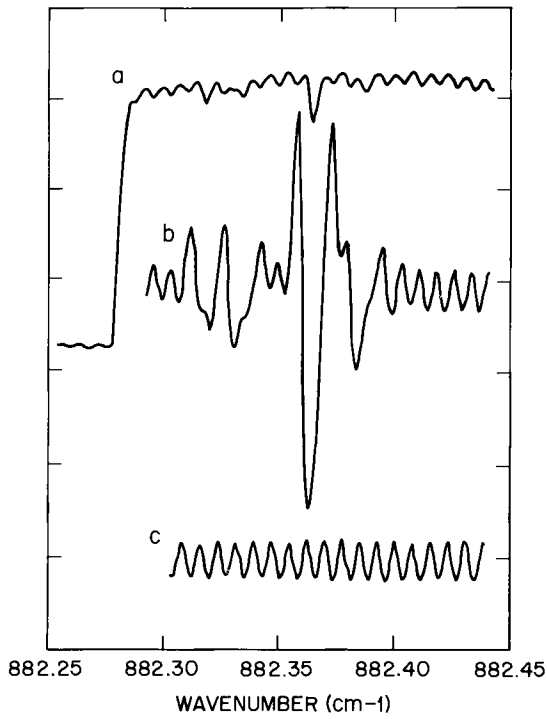


FIGURE 19. Tunable IR diode laser absorption spectrum of the $^2P_{1/2} \leftarrow {}^2P_{3/2}$ transition of Cl in a Cl_2 discharge (Wormhoudt et al., 1986). Pressure = 0.50 torr, frequency = 13.5 MHz.

by independently measuring absolute IR absorption cross sections and then applying Beer's law. Absolute Cl-atom number densities were determined by this method as a function of discharge power, frequency, and total pressure for a 95% Cl/5% Ar plasma. The dependence on applied frequency is shown in Figure 20. They find that Cl concentration increases with frequency, in contrast to what is found by actinometry (Donnelly et al., 1985) (see Figure 21). While the reason for this discrepancy is still unknown, one likely cause considered by Wormhoudt et al. (1986) is that the $^2P_{1/2}$-to-$^2P_{3/2}$ population ratio is not in equilibrium with the gas temperature (i.e., the ratio is much higher). Since the IR absorption technique detects only $^2P_{3/2}$, they would "miss" a large fraction of the atomic Cl. However, they argue that the $^2P_{1/2} \rightarrow {}^2P_{3/2}$ quenching rate is much faster than the Cl_2 dissociation rate, hence the population of the two states should be in equilibrium. Consequently, this explanation cannot easily explain the discrepancy with actinometry.

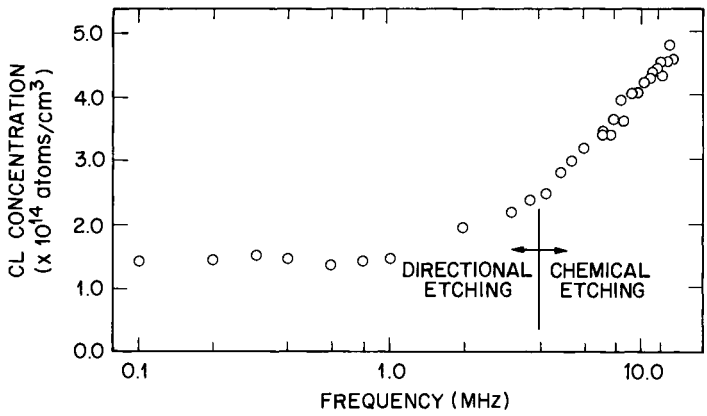


FIGURE 20. Cl-atom density vs. applied frequency in a Cl_2 discharge, determined by tunable diode laser absorption (Wormhoudt et al., 1986). Other conditions identical to Figure 19.

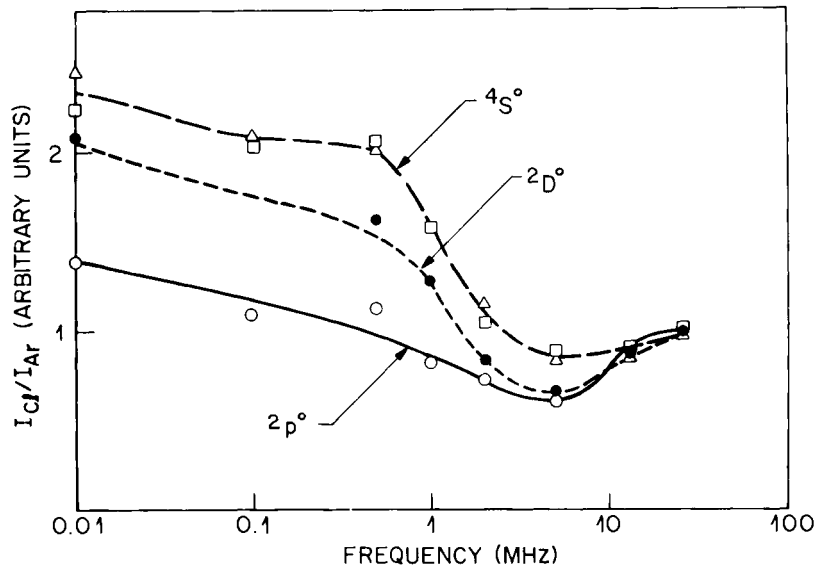


FIGURE 21. Cl-to-Ar emission ratio vs. applied frequency in a Cl_2 discharge (Donnelly et al., 1985). Pressure = 0.30 torr.

One disadvantage to tunable diode laser spectroscopy applied to plasma diagnostics is that it is very cumbersome to tune over a wide spectral range, hence the technique is not very amenable to multiple-species detection. Also, since these are CW lasers, time-dependent studies such as those carried out with pulsed laser induced fluorescence cannot be performed in the interesting time regime below 100 nsec. Raman-shifted pulsed dye lasers offer a promising alternative to diode lasers for more easily tunable sources, which can also be used to provide time-resolved information. Tunable IR laser radiation has been generated by Raman scattering in H_2 (Rabinowitz et al., 1986), and metal vapors (Harris et al., 1984; Hanna et al., 1979; Bethume et al., 1983; Harris and Levinos, 1987). The metal vapor technique has advantages of higher powers and a wider tuning range (2.8–16 μm).

E. Optogalvanic Effects

Despite the fact that laser probing techniques are essentially nonintrusive, pulsed lasers can in fact be used to cause transient perturbations that give rise to a change in the discharge current and voltage. This effect can be used to obtain information on ion and neutral species concentration and discharge dynamics. Both increases and decreases in discharge current have been observed. When the laser is tuned to a transition from a metastable state to higher state, which, in turn, relaxes rapidly to the ground state (Figure 22), a decrease in current is observed due to the depletion of the

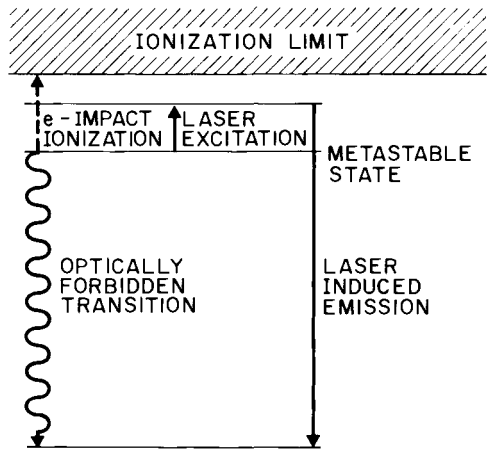


FIGURE 22. Optogalvanic effect in which more easily ionizable metastable species are depleted, causing a decrease in discharge current.

more easily ionized metastable species (Kenty, 1950; Meissner and Miller, 1953; Doughty and Lawler, 1983).

Walkup et al. (1983) have observed a transient increase in current in a N_2 DC-discharge when a pulsed laser is tuned in resonance with the N_2 ($B^2\Sigma_u^+ \leftarrow X^2\Sigma_g^+$) transition. They attribute this to a lower charge exchange collision cross section for the excited state relative to resonant charge exchange with the ground state due to a breaking of the energy resonance condition. Therefore, the excited ions have increased mobility in the cathode sheath and lead to a transient increase in current.

Optogalvanic effects can also result from negative ion detachment. For example, Gottscho and Gaebe (1986) found a transient current pulse in a 50 kHz rf discharge in Cl_2 when a pulsed UV laser was passed near the momentary anodic electrode. They attributed this effect to dissociative attachment followed by photo-detachment

$$Cl_2 + e \rightarrow Cl^- + Cl \tag{21}$$

$$Cl^- + h\nu \rightarrow Cl + e. \tag{22}$$

From the sign of the signal, they showed that the plasma potential was actually negative with respect to the anode.

V. Optical Techniques for Plasma-Surface Interactions

Optical techniques have been successfully applied to many aspects of gas-phase chemistry occurring in plasmas. However, plasma-surface interactions are equally important in developing a fuller understanding of plasma processes. To date, essentially all studies of "plasma"-surface interactions have been carried out with electron spectroscopy techniques, either after plasma exposure (i.e., the sample is transferred to a UHV analysis chamber) or in real time by using molecular beams of ions, electrons, and/or neutral species to simulate the plasma environment. While such studies have provided valuable information, the delay time between plasma exposure and analysis makes it difficult to draw more quantitative conclusions on what species reside on the surface during exposure to the plasma. Similarly, the low flux conditions of molecular beam studies make extrapolations to plasma conditions questionable.

Traditional electron spectroscopy techniques such as Auger electron spectroscopy, X-ray photoelectron spectroscopy, and electron energy loss spectroscopy are not possible at pressures above $\sim 10^{-5}$ torr due to electron collisions in the gas. Recently, several optical techniques have

emerged and show promise for applications in high pressure environments. A few studies have already been carried out in plasmas, and no doubt many more will be reported in the future.

A. Laser Interferometry

The simplest of surface optical diagnostics is laser interferometry. While this technique provides no chemical information, it does give an extremely accurate measure of material deposition or etching rates. The technique is simple to implement and interpret and is routinely applied in commercial processing. It is required that the film be at least partially transparent and that it be deposited on a smooth substrate. The film and substrate must have different enough optical constants so that the interface reflects a fraction of the incident light. For a laser beam striking the surface of a transparent film at angle of incidence θ_i with respect to normal incidence,

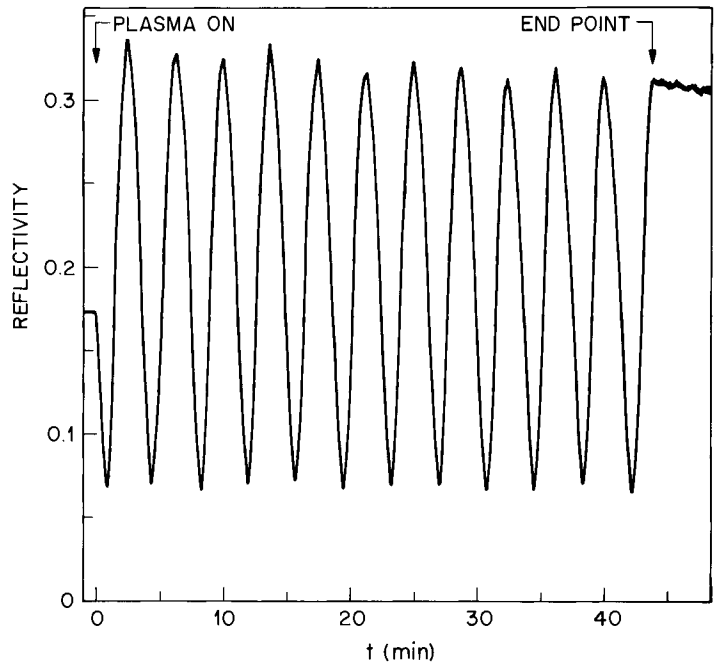


FIGURE 23. Interferogram of SiO_2 being etched off a Si substrate in a 0.30 torr, 250 kHz CF_4/O_2 discharge.

the reflected beam amplitude will sinusoidally oscillate, due to constructive and destructive interference, with a period (Δd) given by

$$\Delta d = \frac{\lambda}{2\sqrt{n^2 - \sin^2 \theta_i}}. \tag{23}$$

Maximum depth resolution is obtained at normal incidence. A sample interferogram is shown in Figure 23 for etching of SiO_2 on Si by a low frequency CF_4/O_2 plasma. A He-Ne laser was reflected off the sample at normal incidence. From the positions of the maxima and minima, an etch rate of 575 Å/minute obtained for this set of conditions. At the inflection points, a 1% change in reflectivity can easily be measured and corresponds to the removal of only 5 Å. This limit could be extended to even smaller thickness changes by using a shorter wavelength laser and splitting the beam to provide a reference.

B. Ellipsometry

For measuring small thickness changes or thin layers (< 30 Å), ellipsometry is a very useful technique, which is quite easy to adopt to a plasma processing environment. Ellipsometry measures the complex reflectance ratio (Maissel and Glang, 1970)

$$\rho = (\tan \psi) e^{i\Delta}, \tag{24}$$

where $\tan \psi = \rho_p/\rho_s$, $\Delta = \Delta_p - \Delta_s$, ρ_p and ρ_s are the amplitudes of parallel and perpendicular components of radiation, respectively, and Δ_p and Δ_s are the total phase changes on reflection of the parallel and perpendicular components, respectively.

The measurement requires determining the polarization angle of the reflected light. The interval required to measure this angle determines the time resolution of the technique. Mechanically rotating analyzers are most common and measure the angle within $\leq$ 10 msec, while piezobirefringent elements can achieve a $\sim$ 20 μsec measurement time (Muller, 1976; Drevillon et al., 1982). Recently, this limit has been extended to $\sim$ 1 nsec (Jellison and Lowndes, 1985). Consequently, ellipsometry has more than enough time resolution to be used as a real-time monitor of plasma etching or deposition. This subject has been reviewed by Aspnes and Chang in a previous publication (1984) and in a chapter in this book and so will only be covered briefly here.

Real-time ellipsometry has been used to study thin film deposition of Si on Si_3N_4 in a thermal CVD process (Hottier and Theeten, 1980). Δ, ψ

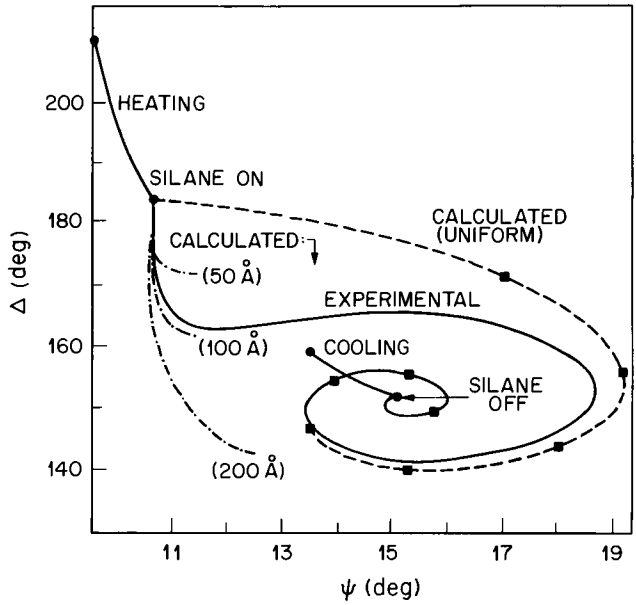


FIGURE 24. In situ ellipsometric data for pyrolytic deposition of amorphous Si on Si_3N_4 (Hottier and Theeten, 1980).

trajectories were measured as a function of time (i.e, film thickness), as shown in Figure 24. The sample was first heated to the deposition temperature. The silane gas was then turned on and growth of amorphous Si began. After growing a film, the gas was turned off and the sample was allowed to cool to room temperature. The experimental curve can be compared with a family of theoretical curves assuming uniform film growth or nucleation of hemispherical islands of various sizes. It is obvious from these curves that the initial stages of film growth occurs by nucleation of islands with an average size of ~ 90 Å.

Similar measurements have been carried out more recently in plasma enhanced CVD of amorphous Si and Ge, deposited from SiH_4 and GeH_4/H_2 DC discharges (Antoine and Drevillon, 1987, Drevillon, et al., 1987). In this case, nuclei of 30–70 Å average spacing coalescence to form a continuous film. The distance between nuclei is inversely proportional to ion bombardment energy, due to ion stimulated mobility of adsorbates. Obviously, the information on nucleation mechanisms makes ellipsometry a very attractive diagnostic technique for basic studies of plasma processing chemistry and also for routine process monitoring.

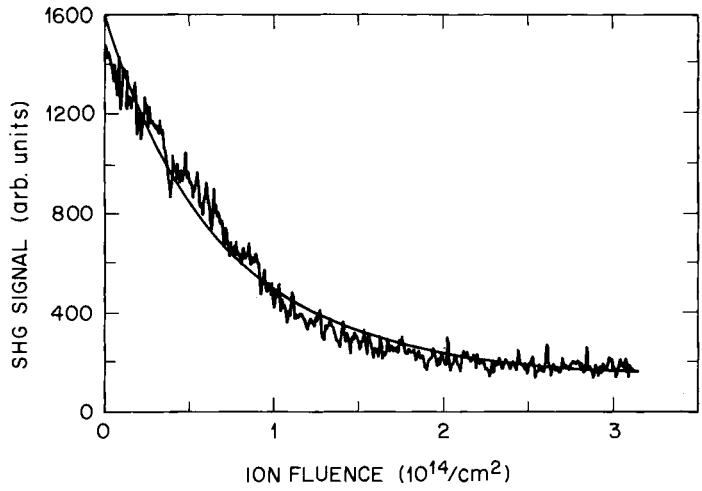


FIGURE 25. Decay of Si (111) 7 × 7 surface order due to 5 keV Ar^+ ion bombardment, determined by second harmonic generation (Heinz et al., 1985).

C. Second Harmonic Generation (SHG)

Laser light of frequency ν incident on an ordered surface of a single crystal will generate a small amount of light at frequency 2ν. The second harmonic signal is easily detected by spectral filtering. SHG is allowed for the surface but is forbidden for the bulk due to inversion symmetry in the bulk (Shen, 1985). Consequently, the technique is very surface sensitive. By recording SHG signal as a function of angle between the electric field vector of the pump beam and principle axes of the crystal, the symmetry of the surface structure can be determined. For example, a (111) Si surface exhibits a twofold symmetry SHG pattern corresponding to the (2 × 1) structure and then transforms to a threefold pattern upon heating, indicative of the (7 × 7) reconstructed surface (Heinz et al., 1985).

SHG signal intensities have been used to follow the decay of surface order following exposure of Si (111) to O_2 and also to Ar^+ ion bombardment (Heinz et al., 1985). The latter experiment is shown in Figure 25. The solid line is an exponential decay fit, which is used to obtain a cross section of 105 $Å^2$ for disordering of the surface. Such measurements could conceivably also be made on a surface during exposure to a plasma. For example, it has been speculated that fluorine atoms quickly disorder a silicon surface during etching due to their ability to be pulled into the bulk by the Mott-Cabrera mechanism (Winters and Haarer, 1987). Chlorine

atoms are expected to be less likely to penetrate into the bulk due to their larger size and lower electronegativity. This mechanism has been invoked to explain the much slower reactivity of Cl vs. F with Si and the smooth surfaces obtained etching Si by ion-bombardment assisted etching in Cl-atom generating discharges. These effects may be evident in SHG from silicon exposed to a Cl-containing discharge. Also, there are questions concerning the mechanisms for ion bombardment enhancement of etching, specifically whether a disordered surface created by ion bombardment is responsible for enhanced rates in the direction parallel to the sheath electric field. SHG measurements may be able to discern whether such a highly disordered surface is the steady-state surface under etching conditions.

D. Laser Raman Scattering From Surfaces

Surface Raman spectroscopy provides vibrational frequencies and chemical composition and concentrations of many molecular species adsorbed on surfaces. Laser-induced Raman scattering from adsorbates can be coarsely divided into two categories. In one case, the surface has little effect on the scattering cross section, which is comparable to that for the gas-phase molecule. Adsorbates on a smooth surface behave in this fashion. However, if the surface of a metal possesses roughness on a scale comparable to the wavelength of the exciting light, then the Raman scattering cross sections are enhanced a factor of up to 10^7 (Otto, 1982).

Although the unenhanced signals are very weak, several researchers have demonstrated monolayer sensitivities for several well studied systems (Stencel et al., 1978; Stencel and Bradley, 1978). Apparently, no one has made similar measurements on a sample immersed in a plasma. Large background emission would make such measurements difficult. Pulsed lasers would offer a large advantage in this regard. UV-pulsed lasers would be more surface sensitive than visible lasers on technologically important materials such as Si and GaAs because of the shorter penetration depths in the UV. Nonresonant Raman cross sections increase as ν^4, and additional increases can be realized due to resonant enhancement from nearby electronic states. Hence, pulsed UV laser Raman scattering would be the best approach for detecting adsorbates in a plasma environment.

Surface enhanced Raman scattering from roughened metal surfaces would certainly be detectable from samples in a plasma. However, the quantification of signal intensities is much more difficult than in the unenhanced case. Under circumstances where unenhanced Raman scattering can be detected, increased intensities due to roughening could be modeled to provide information on surface microstructures that sometimes develop in plasma etch-

ing and also could possibly provide insights into mechanisms of island growth in plasma enhanced CVD, much like what one can obtain from ellipsometry. Murphy and Brueck (1983) have demonstrated such effects for silicon exposed to a CF_4/O_2 plasma, where randomly distributed ellipsoidal Si features with $\sim 0.1\ \mu m$ radii were generated. The enhancement factor for this surface was ~ 40 over a smooth silicon surface.

E. Surface Absorption and Photoacoustic Spectroscopy

IR absorption by adsorbates is potentially one of the more informative techniques for studying plasma-surface interactions. Since the pressure range of interest is typically 10^{-3}–10 torr, electron energy loss spectroscopy is not applicable. Most IR spectroscopy of adsorbates has been studied by measuring changes in intensity of light reflected off a surface (see, for example, King et al., 1986). In one configuration, light is simply reflected off the surface at glancing angle to increase the absorption pathlength. This method is relatively easy to implement but suffers from a lack of sensitivity. To date, most studies have been of strong IR absorbing adsorbates (e.g., CO) on well characterized surfaces under UHV conditions (Chesters, 1986). Absorbances of $\lesssim 1\%$ are typical for strong absorbers. Single beam IR absorption of AlO has been reported for oxidation cleaning of aluminum exposed to O_2 after UHV cleaning (Bermudez et al., 1986).

Large enhancements in sensitivity may be realized when adsorbates are on an IR-transparent substrate such as Ge. In this case, IR light can be coupled into the substrate by prisms, gratings, or sloped edges on the substrate (Chesters, 1986). The IR light is then totally internally reflected within the substrate and can make ~ 100 reflections off the surface-plasma interface. Absorption by adsorbates at specific frequencies will attenuate the beam on each reflection. Therefore, large increases in sensitivity over single reflection techniques can be realized. Claassen and Dieleman (1987) have recently reported multireflection IR absorption of SiO on a Si surface exposed to O-atoms from an oxygen discharge. The thin Si substrate was pressed into optical contact with a Ge prism. Light coupled into the Ge passed back and forth through the Si sample and therefore underwent multiple reflections off the Si surface-gas interface.

Absorption techniques are limited in sensitivity, since one is trying to measure a very small difference between two large numbers. In photoacoustic spectroscopy, the absorbed light is measured directly by detecting sound waves created by the small amount of heating resulting from absorption of light. Recently, Higashi et al. (1985) and Highashi and Rothberg (1985)

have developed IR-photoacoustic spectroscopy for detection of submonolayer coverages of trimethyl aluminum (TMA) adsorbed on an IR-transparent substrate (sapphire). A transducer was bonded to the sapphire substrate, which is transparent to IR light in the region of interest. The IR source was a pulsed, tunable dye laser Raman shifted in H_2 to provide tunable IR in the 2800–3800 cm^{-1} region. The asymmetric stretch of CH_3 groups on TMA were detected at $\sim$ 2960 cm^{-1}. Upon exposure to 193 nm ArF excimer laser irradiation, the signal due to CH_3 was reduced in intensity due to photolysis of adsorbed TMA, liberating CH_3 into the gas phase and leaving behind Al on the surface. Similar studies could also be carried out in a plasma environment to study plasma generated species adsorbed on IR transparent substrates.

VI. Summary

No single optical method can be regarded as a universal plasma diagnostic. However, the variety of complementary techniques are now providing detailed information on a variety of processing plasmas. Of all the gas-phase techniques, optical emission is easiest to implement and can provide qualitative data on species concentrations. This real-time information can be used to improve control of etching and deposition processes. Optical emission combined with time resolution, actinometry, and high-resolution line width measurements yields a wealth of often quantitative information on relative number densities, as well as discharge kinetics and dynamics.

Optical absorption is a largely underutilized technique, which introduces a somewhat more complex degree of instrumentation but can provide absolute concentration measurements and is capable of both spatial and temporal resolution. In particular, Fourier transform IR and tunable diode laser IR techniques are finding increasing applications as important plasma diagnostics.

Laser-induced fluorescence is relatively complex, and many species cannot be detected due to a lack of excited, bound states at energies accessible to tunable dye lasers. However, it is by far the most sensitive technique and so is uniquely suited to detection of trace species, including ions. In addition, lasers can be used to measure electric fields and induce optogalvanic effects.

Finally, Raman and CARS spectroscopy with pulsed, UV lasers, while lacking the sensitivity of laser-induced fluorescence, promises to be a more universal probe of absolute concentrations of gas-phase species present at higher concentrations and so is complementary to laser induced fluorescence.

Optical diagnostic techniques are also beginning to emerge as important in situ probes of plasma-surface interactions. In particular, studies are underway, utilizing ellipsometry and multipass IR-absorption to study nucleation and oxidation in plasma environments. These techniques, as well as UV-Raman scattering and second harmonic generation, will no doubt provide important insights into plasma processing mechanisms in the years to come.

VII. Acknowledgments

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References

Antoine, A. M. and Drevillon, B. (1987). *Mater. Res. Soc. Symp.* **75**, 333.
Aspnes, D. E. and Chang, R. P. (1984). *Mat. Res. Soc. Symp. Proc.* **29**, 217.
Benesch, W. and Li, E. (1984). *Optics Letters* **9**, 338.
Bermudez, V. M., Hylden, J. L., and Bulter, J. E. (1986). *J. Electron. Spectros. and Related Phenomena*, **38**, 143.
Bethune, D. S., Schell-Sorokin, A. J., Loy, M. M., Lankard, J. R., and Sorokin, P. P. (1983). *Adv. in Laser Spectroscopy*, Vol. 2 (B. A. Garetz and J. R. Lombardi, eds., Wiley, New York), pp. 1–43.
Bischel, W. K., Perry, B. E., and Crosley, D. R. (1981). *Chem. Phys. Lett.* **82**, 85.
Brown, L., Donnelly, V. M., and McCrary, V. R. (1987). *Plasma Chem. and Plasma Processing* **7**, 395.
Cappelli, A. L., Gottscho, R. A., and Miller, T. A. (1985). *Plasma Chemistry and Plasma Processing*, **5**, 317.
Chesters, M. A. (1986). *J. Electron Spectros, and Related Phenomena* **38**, 123.
Claassen, W. C. and Dieleman, J. (1987). *Topical Meeting on Microphysics of Surfaces, Beams, and Adsorbates*, Sante Fe, NM, Feb. 16–18, TuB4-1.
Cleland, T. A. and Hess, D. W. (1987). Conference on Emerging Technologies in Materials, Minneapolis, MN, Aug. 18–20; (1987) *Plasma Chem. Plasma Proc.* **7**, 379.
Coburn, J. W. (1982). *Plasma Chemistry and Plasma Processing* **2**, 1; Winters, H. F. (1985). *J. Vac. Sci. Technol.* **B3**, 9.
Coburn, J. W. and Chen, M. (1980). *J. Appl. Phys.* **51**, 3134.
Curtis, B. J. and Brunner, H. J. (1978). *J. Electrochem. Soc.* **127**, 234.
Dagenais, M., Johns, J. W., and McKeller, A. R. (1976). *Can. J. Phys.* **54**, 1438.
d'Agostino, R., Cramarossa, F., DeBenedictus, S., and Ferraro, G. (1981). *J. Appl. Phys.* **52**, 1259.
d'Agostino, R., Cramarossa, F., and DeBenedictus, S. (1982). *Plasma Chem. Plasma Proc.* **2**, 213.
d'Agostino, R., Cramarossa, F., De Benedictus, S., and Ferraro, G. (1984). *Plasma Chem. Plasma Proc.* **4**, 163.
Danner, D. A. and Hess, D. W. (1986). *J. Appl. Phys.* **59**, 940.

Davies, P. B. and Russell, D. K. (1979). *Chem. Phys. Lett.* **67**, 440.
Degenkolb, E. O., Mogab, C. J., Goldrick, M. R, and Griffiths, J. R. (1976). *Appl. Spectros.* **30**, 520.
DiMauro, L. F., Gottscho, R. A., and Miller, T. A. (1984). *J. Appl. Phys.* **56**, 2007.
Donnelly, V. M. and Flamm, D. L. (1980). *J. Appl. Phys.* **51**, 5273.
Donnelly, V. M. and Flamm, D. L. (1983). U.S. Patent 4,377,436, Mar. 22, 1983.
Donnelly, V. M., Flamm, D. L., and Collins, G. (1982). *J. Vac. Sci. Technol.* **21**, 817.
Donnelly, V. M., Flamm, D. L., and Bruce, R. H. (1985). *J. Appl. Phys.* **58**, 2135.
Donnelly, V. M., Flamm, D. L., Dautremont-Smith, W. C., and Werder, D. J. (1984). *J. Appl. Phys.* **55**, 242.
Donnelly, V. M., Ibbotson, D. E., and Flamm, D. L. (1984). In *Ion Beam Modification of Surfaces: Fundamentals and Applications*, O. Auciello and R. Kelly, ed., Elsevier, Amsterdam, Chap. 8.
Doughty, D. K. and Lawler, J. E. (1983). *Phys. Rev. A* **28**, 773.
Doughty, D. K., Salih, S., and Lawler, J. E. (1984). *Appl. Phys. Lett.* **103A**, 41.
Doughty, D. K. and Lawler, J. E. (1984). *Appl. Phys. Lett.* **45**, 611.
Drevillon, B., Godet, C., and Antoine, A. M. (1987). *Mater. Res. Soc. Symp.* **75**, 341.
Drevillon, B., Perrin, J., Marbot, R., Violet, A., and Dalby, J. L. (1982). *Rev. Sci. Instrum.* **53** 969.
Druet, S. and Taran, J. P. (1979). In *Chemical and Biological Applications of Lasers*, C. B. Moore, ed. Academic Press, New York.
Flamm, J. (1978). *Appl. Phys.* **49**, 3796.
Flamm, D. L. and Donnelly, V. M. (1986). *J. Appl. Phys.* **59**, 1052.
Flamm, D. L., Donnelly, V., and Ibbotson, D. E. (1984). In *VLSI Electronics Microstructure Science, Vol. 8, Plasma Processing for VLSI*, N. G. Einspruch and D. M. Brown, ed., Academic Press, Orlando, FL, Chap. 8.
Flamm, D. L., Donnelly, V. M., and Mucha, J. A. (1981). *J. Appl. Phys.* **52**, 3633.
Foster, S. C. and McKellar, A. R. (1984). *J. Chem. Phys.* **81**, 3424.
Ganguly, B. N. and Garscadden, A. (1985). *Appl. Phys. Lett.* **46**, 540.
Gottscho, R. A. and Donnelly, V. M. (1984). *J. Appl. Phys.* **56**, 245.
Gottscho, R. A. and Gaebe, C. E. (1986). *J. Vac. Sci. Technol.* **A4**, 1795.
Gottscho, R. A. and Mandich, M. L. (1985). *J. Vac. Sci. Technol.* **A3**, 617.
Gottscho, R. A., Davis, G. P., and Burton, R. H. (1983). *J. Vac. Sci. Technol.* **A1**, 622; *Plasma Chem. Plasma Proc.* **3**, 193.
Gottscho, R. A., Burton, R. H., Flamm, D. L., Donnelly, V. M., and Davis, G. P. (1984). *J. Appl. Phys.* **55**, 2707.
Greenberg, K. E. and Hargis, P. J. (1987). CLEO.
Gudeman, C. S., Begemann, M. H., Pfaff, J., and Saykally, R. J. (1983). *Phys. Rev. Lett.* **50**, 727.
Hamasaki, T., Hirose, M., and Osaka, Y. (1982). *Proc. 7th ICVM*, Tokyo, Japan, p. 432.
Hanna, D. C., Yuratich, M. A., and Cotter, D. (1979). *Nonlinear Optics of Free Atoms and Molecules*, Springer Series in Optical Sciences, Vol. 17, Springer-Verlag, New York.
Hargis, P. J., Jr. (1981). *Appl. Opt.* **20**, 149.
Hargis, P. J., Jr. and Kushner, M. J. (1982). *Appl. Phys. Lett.* **40**, 779.
Harris, A. L., Brown, J. K., Berg, M., and Harris, C. B. (1984). *Opt. Lett.* **9**, 47.
Harris, A. L. and Levinos, N. J. (1987). *Appl. Opt.* **26**, 3996.
Harshbarger, W. R., Porter, R. A., Miller, T. A., and Norton, P. (1977). *Applied Spectroscopy* **31**, 201.
Hata N., Matsuda, A., Tanaka, K., Kajiyama, K., Moro, N., and Sajiki, K. (1983). *Jpn. J. Appl. Phys.* **22**, L1.

Heaven, M., Miller, T. A., Freeman, R. R., White, J., and Bokar, J. (1982). *Chem. Phys. Lett.* **86**, 458.
Heinz, T. F., Loy, M. M., and Thompson, W. A. (1985). *Phys. Rev. Lett.* **54**, 63; *J. Vac. Sci. Technol.* **B3**, 1467.
Higashi, G. S. and Rothberg, L. J. (1985). *J. Vac. Sci. Technol.* **B3**, 1460.
Higashi, G. S., Rothberg, L. J., and Fleming, C. G. (1985). *Chem. Phys. Lett.* **115**, 167.
Hottier, F. and Theeten, J. B. (1980). *J. Cryst. Growth* **48**, 644.
Husinsky, W. (1985). *J. Vac. Sci. Technol.* **B3**, 1546.
Ibbotson, D. E., Flamm, D. L., and Donnelly, V. M. (1983). *J. Appl. Phys.* **54**, 5974.
Jellison, G. E., Jr. and Lowndes, D. H. (1985). *Appl. Phys. Lett.* **47**, 718.
Kenty, C. (1950). *Phys. Rev.* **80**, 95.
King, D. A., Richardson, N. V., and Holloway, S., ed. (1985). *Vibrations at Surfaces*, Elsevier, Amsterdam.
Krogh, O., Widser, T., and Chapman, B. (1986). *J. Vac. Sci. Technol.* **A4**, 1976.
Laguna, G. A. and Beattle, W. H. (1982). *Chem. Phys. Lett.* **88**, 439.
Lawler, J. E., Doughty, D. K., and Salih, S. (1984). Abs. no. K6, 37th Gaseous Electronics Conference, Univ. of Colorado (Oct. 9–12, 1984), published by the American Physical Society.
Matsumi, Y., Hayashi, T., Hoshikawa, H., and Komiya, S. (1986). *J. Vac. Sci. Technol.* **A4**, 1786.
Matsumi, Y., Toyoda, S., Hayashi, T., Miyamma, M., Yoshikawa, H., and Komiya, S. (1986). *J. Appl. Phys.* **60**, 4102.
Maissel, L. I. and Glang, R. ed., (1970). *Handbook of Thin Film Technology*, McGraw Hill, New York, p. 11–22.
McDonald, J. R. (1980). In *Laser Probes for Combustion Chemistry*, D. R. Crosley, ed. American Chemical Society Symposium Series 134, pp. 19–58.
Meissner, K. W. and Miller, W. F. (1953). *Phys. Rev.* **92**, 896.
Moore, C. A., Davis, G. P., and Gottscho, R. A. (1984). *Phys. Rev. Lett.* **52**, 538.
Mucha, J. A., Donnelly, V. M., Flamm, D. L., and Webb, L. M. (1981). *J. Phys. Chem.* **85**, 3529.
Muller, R. H. (1976). *Surf. Sci.* **56**, 19.
Murphy, D. V. and Brueck, S. R. (1983). *Opt. Lett.* **8**, 494.
Nibler, J. W. and Knighten, G. V. (1977). In *Topics in Current Physics*, A. Weber, Springer Verlag, Stuttgart, Germany, Chap. 7.
Ninomiya, K., Suzuki, K., Nishimatsu, S., and Okada, O. (1986). *J. Vac. Sci. Technol.* **A4**, 1971.
Otto, A. (1982). *Surf. Sci.* **117**, 330.
Pine, A. S. (1974). *J. Opt. Soc. Am.* **64**, 1683; (1976) **66**, 97.
Poll, H. U., Arzt, M., and Wickleder, K. H. (1976). *Eur. Polym. J.* **12**, 505.
Poll, H. U., Hinze, D., and Schlemm, H. (1982a). *Appl. Spectros.* **36**, 445.
Poll, H. U. and Schlemm, H. (1982b). *Beiträge, aus der Plasma Physik* **22**, 195.
Rabinowitz, P., Perry, B. N., and Levinos, N. (1986). *IEEE J. Quantum Electronics*, **QE-22**, 797.
Ryan, K. R. and Plumb, I. C. (1984). *Plasma Chemistry and Plasma Processing* **4**, 141.
Scheller, G. R., Gottscho, R. A., and Intrator, T. (1988). *J. Appl. Phys.*, in press.
Shen, Y. R. (1985). *J. Vac. Sci. Technol.* **B3**, 1464.
Stanton, A. C. and Kolb, C. F. (1980). *J. Chem. Phys.* **72**, 6637.
Stencel, J. M. and Bradley, E. B. (1979). *J. Raman Spectros.* **8**, 203; *Spectros. Lett.* (1978) **11**, 563.
Stencel, J. M., Noland, D. M., Bradley, E. B., and Frenzel, C. A. (1978). *Rev. Sci. Instrum.* **49**, 1163.

Suto, M., Mitchell, M. J., Lee, L. C., and Chuang, T. J. (1987). *Topical Meeting on Microphysics of Surfaces, Beams, and Adsorbates, Sante Fe NM*, Feb. 16–18, 1987, paper TuB3-1.
Tolles, W. M., Nibler, J. W., McDonald, J. R., and Harvey, A. B. (1977). *Appl. Spectrosc.*, ed. **31**, 253.
Walkup, R., Dreyfus, R. W., and Arouris, P. H. (1983). *Phys. Rev. Lett.* **50**, 1846.
Watson, J. K., Foster, S. C., McKellar, A. R., Bernath, P., Amano, T., Pan, F. S., Crofton, M. W., Altman, R. S., and Oka, T. (1984). *Can. J. Phys.* **62**, 1875.
Winters, H. F. and Haarer, D. (1987). *Phys. Rev.*, in press.
Wormhoudt, J., Stanton, A. C., Richards, A. D., and Sawin, H. H. (1986). *Mat. Res. Soc. Symp. Proc.* **68**, 115.
Wright, R. B., Pellin, M. J., and Gruen, D. M., *Surf. Sci.* **110**, 151.
Wright, R. B., Pellin, M. J., and Gruen, D. M. (1982). *J. Vac. Sci. Technol.* **20**, 510.

2 Plasma Diagnostics in Electric Discharge Light Sources

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I. Introduction 47
A. Types of Electric Discharge Lamps 47
II. Diagnostic Techniques Employed 57
A. Langmuir Probe Diagnostics in Low-Pressure Discharges 57
B. Microwave Diagnostics 72
C. Spectroscopic Diagnostics for Low-Pressure Lamps 73
D. Spectroscopic Diagnostics of HID Lamps 88
E. Laser Diagnostics in High-Pressure Discharge Lamps 104
F. Continuum Measurements 107
G. Stark Shift 107
III. Summary and Conclusions 107
IV. Acknowledgements 108
References 108

I. Introduction

A. Types of Electric Discharge Lamps

Light sources in which electric power is converted to visible light by means of electric discharge in gases are extremely important in virtually all areas of lighting except residential. Their commercial importance stems from the fact that they are typically three or more times as efficient in this conversion as the common household incandescent bulb. Since the cost of electric energy to operate a light source over its lifetime is typically 80% of the total cost of light (i.e., many times the cost of the bulb itself), in the commercial environment, where life-cycle costs are the dominant consideration, discharge light sources find wide application, despite their relatively high first cost in comparison to incandescent.

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There is a large variety of different types of discharge lamps, depending on the application requirements. From the standpoint of plasma physics, and plasma diagnostics, these may be divided into two general classes: low-pressure, nonequilibrium; and high-pressure, approximately in local thermal equilibrium (LTE). In this section will be discussed the principal features and properties of the several different families of discharge lamps, together with the diagnostic information which is most often sought as a guide to product understanding, improvement, or new product development.

Unlike the situation in many plasma physics problems, the application of plasma diagnostics to discharge lamps frequently involves discharges for which relatively sophisticated mathematical models are available. Thus, comparison of measured to calculated quantities can be made on a relatively detailed basis, serving to validate both model and diagnostic technique. For these applications, the present state of agreement between model calculations and diagnostic results is quite satisfying.

In subsequent sections will be discussed the diagnostic techniques which have been employed in discharge lamps. Only representative examples will be discussed in any detail, for reasons of limitations of space; other work will be incorporated by reference in the bibliography. A review of some of the topics covered here is also given by van den Hoek (1983).

1. Low-pressure, Nonequilibrium

The low-pressure, nonequilibrium family of discharge lamps includes the fluorescent lamp and the low pressure sodium lamp. It is characterized by a discharge in a mixture of a noble-gas buffer (at a few hundred pascals [few torr] pressure) plus a resonance-radiation generating species (at a few pascals [few millitorr] pressure). These discharges are ambipolar-diffusion controlled, collision-dominated discharges which are far from local thermal equilibrium. The electron energy distribution is approximately Maxwellian, but the electron temperature is substantially greater than the excited state temperature and many times the heavy-particle kinetic temperature. The effective electron temperature is sufficiently low that the buffer gas is not excited or ionized during steady-state operation; only the low-pressure resonance radiation emitting species participates in the radiative and ionization balances.

Discharge modelling requires simultaneous solution of a Boltzmann equation (or approximation thereto), excited state balance equations, ionization balance equations, and total energy balance equations. Commonly, rate constants must be calculated for some two dozen processes (including a

number of lumped-state approximations) to solve these equations. See Waymouth and Bitter (1956); Cayless (1962); Vriens (1973 and 1977); Polman and Drop (1972); Morgan and Vriens (1980); Lagushenko and Maya (1984); and Dakin (1986). Of these, only Cayless and Dakin have attempted to calculate the radial dependence of discharge parameters, for which experimental data has only recently become available.

a. Fluorescent Lamps. Fluorescent lamps employ a discharge in a mixture of a rare gas (typically argon, neon, krypton, or mixtures thereof) at a few hundred pascals, plus mercury vapor at a few pascals, in elongated tubes with an aspect ratio greater than 10 : 1. The power density is low, about a few tens of milliwatts per centimeter cubed. The principal output radiations are the 254 nm and 185 nm resonance lines of mercury; up to 60% of the positive column power is emitted in the former and up to 10% in the latter. The other principal energy loss mechanism is elastic scattering of electrons by the buffer gas, amounting to 25%. The ultraviolet light is converted to visible light, with near-unity quantum efficiency, by a phosphor coating on the walls of the tube; there is, however, a large energy loss in the process because the energy of a visible photon is only about 2.5 ev,while that of the UV resonance photon is nearly 5 ev. Thus, the overall energy efficiency of the device is about 25%, nearly five times that of the household incandescent lamp. Light emission of a variety of colors, or blends thereof, is possible by selection of the appropriate phosphor.

Efficiency is dependent on mercury vapor pressure, tube diameter, the nature and pressure of the rare gas, and current density. The several dependencies are reasonably well understood as a result of more than a quarter century of diagnostic experiments combined with ever improving theoretical modelling. A summary of the state of the art as of 1971 is given by Waymouth, 1971; for more recent developments, see Lagushenko and Maya, 1984. Associative ionization processes have in the past few years been demonstrated to make a significant contribution to the ionization, placing a premium on diagnostic measurements of metastable and resonance level densities in such plasmas. Moreover, since all aspects of the discharge behavior depend on electron collision rates, especially inelastic, the electron energy distribution is a key parameter for understanding, not to mention controlling, the discharge. Probe measurements of electron energy distributions have figured heavily in the R & D effort devoted to this product family.

b. Low-pressure Sodium Lamps. Low-pressure sodium lamps make use of a discharge in a mixture of a rare gas, usually neon, at a pressure of a few hundred pascals, plus sodium vapor at a few pascals pressure. The principal

radiant emission is in the sodium resonance lines 589.0–589.6 nm (D-lines), at an energy efficiency of up to 40%. Because no energy losses due to photon conversion by a phosphor are required and because the emitted wavelength is very near the eye sensitivity maximum, the luminous efficacy of these lamps is very high, as much as 200 + lumens per watt (13 times that of the household incandescent bulb). The emitted spectrum being almost monochromatic, visual sensation of the illuminated area is essentially devoid of color, i.e., like a black-and-white photograph. Such lamps are primarily utilized for street and highway lighting, where color discrimination is unimportant.

The discharge is ambipolar diffusion controlled, collision dominated, and far from LTE. It is similar in many respects to that of the fluorescent lamp, and very similar approaches to discharge modelling have been employed to study it. A major difference is that the required vapor pressure of sodium is obtained only at tube-wall temperatures about 250°C; this temperature is achieved by thermal insulation of the discharge tube, employing an infrared reflecting evacuated outer jacket to minimize the input power required for reaching of proper wall temperature. Power loadings are, nevertheless, triple the values used in fluorescent lamps.

As a result, low-pressure sodium lamps typically operate at several times the current densities of fluorescent lamps. The radial pumping of ions to the wall by ambipolar diffusion fields, together with their return diffusion as neutral atoms with much lower diffusion coefficients, results in significant depletion of neutral sodium atom density in the center of the positive column relative to the walls. Important diagnostic information developed in recent years has been the neutral sodium density profile as a function of radial position. Since lamps are normally operated on alternating current, the depletion phenomena are time-dependent as well.

Modelling calculations have been somewhat less successful in predicting device behavior and performance characteristics than in the case of fluorescent lamps, in part because of the time-dependent complexity; the work of van Tongeren and Heuvelmans (1974) appears to be closest to the mark.

2. *High-pressure, Approximate Local Thermal Equilibrium*

This discharge lamp family is characterized by discharges in gases at pressures of one to several atmospheres; electron densities are sufficiently high that the principal excited states of the radiating atoms are in Boltzmann equilibrium with the local electron temperature. Electron/

heavy-particle collision frequencies are high enough that heavy particle temperatures are locally a few degrees to a few tens of degrees less than the electron temperature. Accordingly, one may speak of *the* temperature, which is a function of radial position, being a maximum at the axis of the discharge and decreasing continuously toward the wall. Although the tube wall is much cooler than the core of the discharge, it is still quite hot, 1000°K–1500°K depending on lamp type. Accordingly, the discharge vessel must be made of highly refractory light transmissive materials. To date, only fused silica glass (commonly, and incorrectly, referred to as "quartz") and translucent polycrystalline alumina (PCA) have been found to be satisfactory.

The suitability of an LTE model for the discharge then makes knowledge of the radial temperature profile in the arc the *sine qua non* for understanding the properties of the lamp. Extensive modelling efforts are devoted to calculating the temperature profile; necessarily, diagnostics for measuring the temperature profile have also been extremely important. Equally important, however, is diagnostic information about concentrations of neutral and ionic species as a function of radial position. Spectroscopic techniques, emission as well as absorption, have proven to be invaluable for both diagnostics. To date, there have not been any direct diagnostics for electron densities. For most emission lines, other broadening mechanisms predominate over Stark broadening; for most lamps, there are a number of other mechanisms which give rise to continuous spectra, as well, making brehmstrahlung measurements of uncertain value.

The two principal power dissipations from high-pressure arc sources are radiation and thermal conduction to the wall through the relatively large gradient in heavy-particle temperature. The latter depends only linearly on core temperature, while radiation depends exponentially. The higher the input power per unit volume for a given gas mixture, the higher the discharge temperature and the greater the ratio of radiative losses to thermal conduction losses. As a consequence, efficiency for such lamps, in general, increases with increasing specific power input. The practical factor limiting specific power input (commonly referred to as *wall loading*) is the maximum temperature supportable by the wall of the discharge vessel, both from purely thermomechanical considerations as well as from limitations due to chemical reactivity.

Because efficiency increases with increasing power per unit volume, high pressure lamps are designed to be as small as possible for a given power input (consistent with the limitations above). Consequently, they find applications where large amounts of light are needed in small physical volumes: outdoor lighting, street lighting, industrial lighting, commercial

lighting for high ceiling mounting heights, sports lighting, and the like. The high specific light output makes these sources extremely bright, and they are commonly referred to in the industry by the generic title *HID lamps* (acronym for *high intensity discharge*).

a. High-pressure Mercury Lamps. The oldest form of HID lamp in current use is the high-pressure mercury lamp, consisting of a discharge at a power density about 20 watts per cubic centimeter in mercury vapor at 3–10 atmospheres pressure in a fused silica discharge vessel (*arc tube*) contained within a glass outer envelope. Figure 1 shows a photograph of a typical mercury vapor lamp. The discharge is well described by an LTE model, with an axis temperature about 6000°K and arc tube wall temperature about 1000°K. The principal radiant emission consists of the line spectrum of mercury; the most important of these lines are characterized by an optical depth of unity, i.e., neither optically thick nor optically thin. The ultraviolet resonance lines are, of course, very thick and strongly self absorbed, thereby constituting a negligible fraction of the total radiant power.

Seventy percent of the input power is dissipated from the arc as radiation power, the balance being dissipated by thermal conduction from the plasma column and by electrode losses. Because of the unfavorable distribution of mercury lines between the near-UV and the visible, however, only 23% of the total radiation is in the visible part of the spectrum, providing an energy efficiency of 16%, still three times that of a household incandescent lamp. The spectrum is deficient in red, leading to very poor color rendering properties. However, present-day mercury lamps usually employ a phosphor coating on the inside surface of the outer bulb, which converts some of the otherwise useless ultraviolet light to visible red light. As a result, the total efficacy of the lamp is increased by 10%, and the color rendering property of the illumination is vastly improved. Mercury lamps have been in commercial service for 50 years and have progressed far down the learning curve of cost reduction; they are the least costly of the HID family. They have, moreover, prodigious operating lifetimes, 24,000 hours (six years in typical street lighting service). They still find wide application in many industrial and commercial situations, but are gradually being displaced by higher efficacy sources, such as high-pressure sodium and metal halide lamps.

The physics of the discharge is well understood as a result of many years of diagnostic and modelling efforts; it is of interest here primarily because many of the spectroscopic techniques developed for mercury lamps have found wide application in the other members of the HID family. The state of the art is well summarized by Elenbaas (1951, 1972).

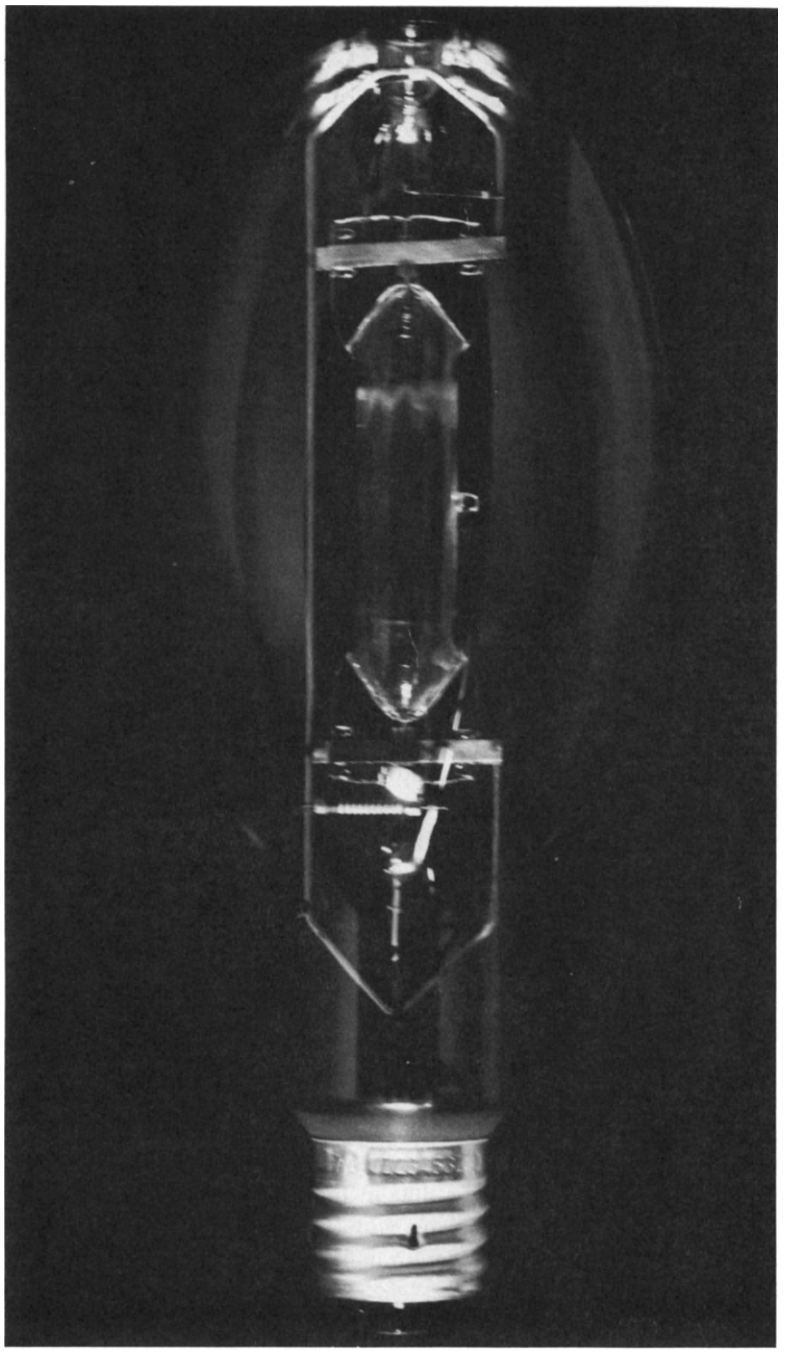


FIGURE 1. Photo of high-pressure mercury lamp (courtesy of GTE Electrical Products).

b. High-pressure Sodium Lamps. High-pressure sodium (HPS) lamps incorporate a discharge at a power density about 100 watts/cm^3 in a mixture of mercury and sodium vapors, at 0.5–1 atmosphere and 0.1–0.2 atmosphere pressure, respectively, in an arc tube of polycrystalline alumina (PCA), which in turn is contained in an evacuated outer glass envelope. A photograph showing an HPS lamp is given in Figure 2. The arc can be described reasonably well by an LTE model, with axis temperatures 3500°K–4200°K and wall temperatures about 1500°K.

At these temperatures, the mercury component of the vapor is substantially not excited nor ionized and serves primarily as a buffer gas. The relatively high mole fraction of mercury in the vapor reduces the thermal conductance of the gas mixture, thereby reducing the thermal conduction loss to the arc tube wall; it also increases the electron collision frequency, reducing the electrical conductivity of the plasma. This permits input of power at relatively low current and high voltage to the device, minimizing the fraction of power dissipated as electrode losses and ballast losses (both of which increase with current).

The radiant power output is dominated by the resonance-broadened sodium resonance lines (D-lines), which are strongly self reversed by the radial temperature gradient. Although the optical depth at line center is about 10^5, there is a large range of wavelengths over which the optical depth is near unity because of the very wide wings to the line. Twenty-five percent of the input power to the lamp can be radiated in the broadened sodium resonance line, over the range from 550 nm to 700 nm. Much of this power being in the yellow portion of the spectrum, where the eye sensitivity is high, the luminous efficacy of the total radiation is high: up to 140 lumens per watt, nine to ten times that of incandescent.

Not only does the broadened D-line span a much wider range of wavelengths, but other emission lines in the sodium spectrum also contribute in the blue and green. Therefore, the HPS lamp offers much improved color rendering in comparison to the low-pressure sodium lamp. There are, in addition, radiations from Na-Xe and Na-Hg excimers in the visible and near infrared, and continua of uncertain (but probably molecular) origin radiated in the visible and infrared, at least to the limit of transmission of the tube wall.

The principal diagnostic techniques which have been employed are spectroscopic, both emission and absorption. Because very little of the spectrum is optically thin, most diagnostic methods have of necessity dealt with optically-thick radiations or with optical depths near unity. Such techniques have been used to determine the plasma temperature profile, as well as species concentrations.

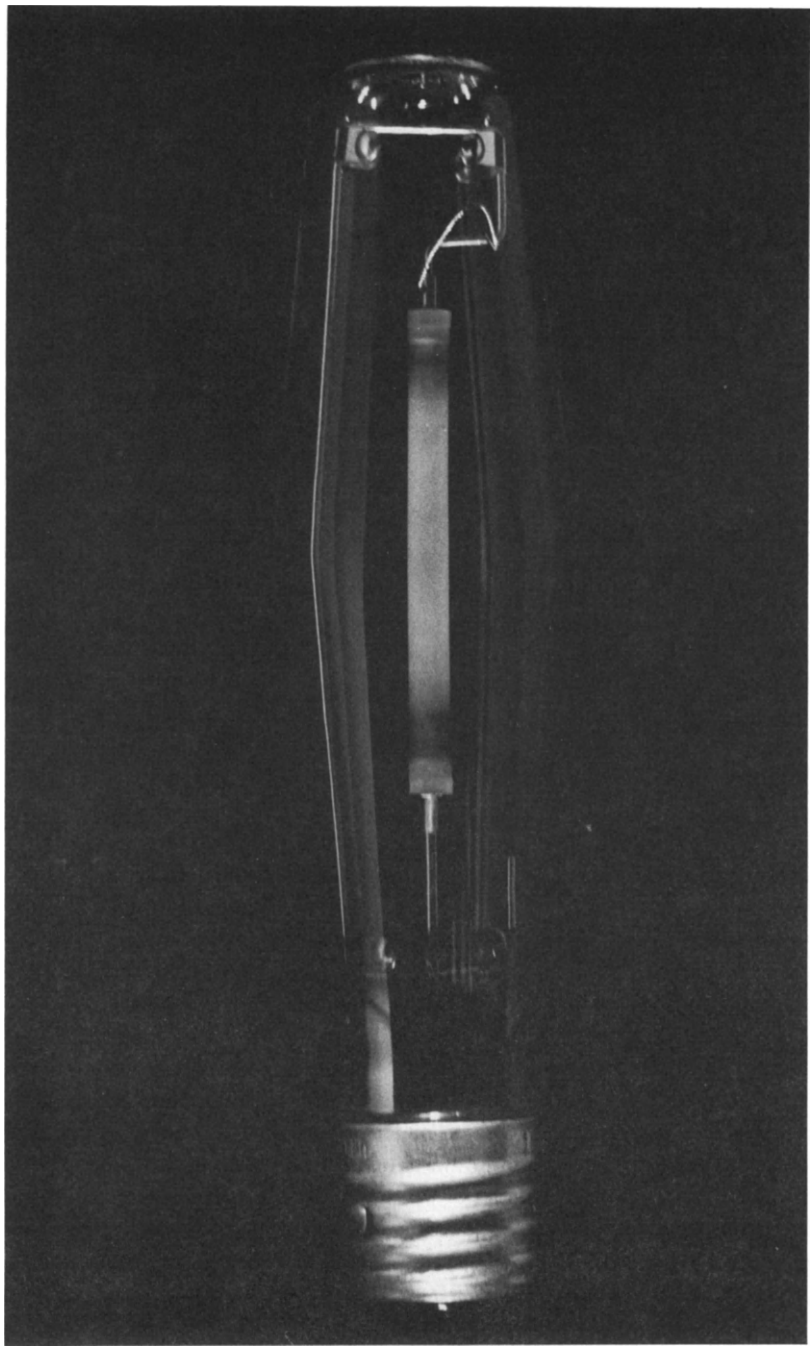


FIGURE 2. Photo of high-pressure sodium lamp (courtesy of GTE Electrical Products).

The present understanding of the physics of the discharge is relatively complete, as a result of several decades of experimental and theoretical efforts. An excellent review has recently been published by de Groot and van Vliet (1986).

c. Metal Halide Lamps. Metal halide lamps employ a discharge at a power density about 25–30 watts/cm^3, 3–30 atmospheres pressure, in a mixture of mercury vapor and the vapors of one or more metal halides in a quartz arc tube inside a glass outer jacket. The physical appearance of a metal halide lamp is very similar to that of a mercury lamp of the same wattage.

In operation, the metal halides are vaporized from the heated arc tube walls; the molecules diffuse or are convected into the central plasma, where they dissociate, freeing up the metal atoms to be excited and ionized. Because of the relatively low ionization and excitation potentials of most metals in comparison to mercury, radiations from these metals can dominate the output spectrum, even though they are present in concentrations thousands of times smaller than mercury.

Again, the discharge can be well described by LTE models, providing a formal basis for calculation of radiant emissions. Arc axis temperatures are 5500°K–5750°K, with tube wall temperatures 1100°K–1200°K. However, to the usual energy balance and radiation transport equations must be added the coupled thermochemical equilibrium equations describing the dissociation of molecular and free-radical species to yield metal atoms and ions and halogens. The nature and pressure of the added metals exerts a dominant role on the radial temperature profile, as does the halogen stoichiometry. These may serve to "constrict" as well as "fatten" the radial temperature profile in a manner which is understood qualitatively, but which has not yet been modelled in a completely satisfactory way.

An enormous variety of potential metal additives are possible, encompassing nearly every metal of the periodic table, singly and in combination. This is both the advantage of the metal halide system and its curse: advantage in that it affords enormous flexibility to the lamp designer to tailor the output spectrum for the application; curse in that there is a semi-infinite set of possibilities from which to choose, with little or no theoretical guidance. In commercial applications for general lighting there are at least five mixtures in use in various different countries; however, in the United States, the only formula in widespread use incorporates a blend of scandium and sodium iodides.

Scandium provides a multiline spectrum analogous to the rare earths and would result in a highly constricted, non-wall-stabilized arc if used by itself. The addition of sodium adds a strong component of the sodium D-lines, reducing the apparent source color temperature into the 3000°K–4500°K

range most desirable for general lighting. It also cancels the constricting effect of scandium and results in a "fat," wall-stabilized radial arc temperature profile. Such lamps combine high efficacy (up to 125 lumens per watt, eight times that of the household incandescent lamp) with excellent color balance and color rendition. They are widely used for indoor commercial lighting and outdoor lighting in color-critical applications for which HPS lamps are unsuitable. Sports lighting is another important application.

Qualitative descriptions of metal halide lamp phenomena have been given by Waymouth (1971); a review article by Keeffe (1980) provides further information.

Because of the lack of truly satisfactory *ab initio* models of discharge behavior, progress in understanding such discharges has relied heavily on diagnostic experiments. Spectroscopic diagnostics, both emission and absorption, have figured heavily in this study. Diagnostic information required includes radial temperature profile and concentrations of atomic, ionic, free radical, and molecular species. Radiation transport ranges from optically thick to optically thin, and diagnostic techniques encompassing the entire range must be employed.

II. Diagnostic Techniques Employed

A. Langmuir Probe Diagnostics in Low-Pressure Discharges

Langmuir probe diagnostics have been extensively employed in low-pressure rare gas mercury discharges, such as those used in fluorescent lamps, and have also been utilized in low-pressure sodium-neon discharges: Easley (1951); Waymouth (1959); Verweij (1961); Waymouth (1964); Little and Waymouth (1966); Barnes (1966); Kalyazim, Milenim, and Timofev (1961); Cornelissen and Merks-Eppingbroek (1986); Godyak and Lagushenko (1986).

Probes have been used in fluorescent lamp discharges to measure electron density, plasma potential, and electron energy distributions. In low-pressure sodium lamps, probes have been used to provide measurements of electron temperature and independent estimates of electron density for comparison with spectroscopic diagnostic measurements.

1. Elementary Concepts

A Langmuir probe is an electrically conducting electrode (usually a wire or disk) of "negligibly small area" inserted into a plasma, the potential of

which may be varied with respect to the plasma. When the probe is strongly negative with respect to the plasma, it repels all plasma electrons, and collects only positive ions. At some less negative potential, it still collects positive ions, but also collects all electrons having kinetic energy of motion normal to the probe surface greater than the retarding potential energy. The probe electron current component is determined by subtracting the positive ion current from the total probe current. The potential at which the positive ion current is equal in magnitude to electron current (but of opposite sign!) results in zero net current and is called the *floating potential.* At zero potential with respect to the plasma, the probe collects the plasma random currents of both ions and electrons. At positive potential with respect to the plasma, it collects only electrons. Since random electron current is many-fold larger than the random ion current, the probe current at plasma potential is negative and essentially equal to random electron current in the plasma. This can be interpreted to yield the electron density in the plasma, provided the electron energy distribution is known. The current voltage characteristic for a probe negative with respect to plasma is a measure of the density of electrons having energy greater than negative probe potential and is thus a measure of the electron energy distribution; for the special case of a Maxwellian distribution of electron energies, the logarithm of the probe electron current plotted against probe potential is a straight line, with slope inversely proportional to electron temperature. For any isotropic electron energy distributions, the second derivative with respect to probe voltage of the probe electron current times the square root of the electron energy is proportional to the electron energy distribution.

The general theory of probe measurements, as well as the effects of deviations from ideality are covered completely in the book by Swift and Schwar (1969), and will not be reproduced here. See also Chen (1965) and Chung, Talbot, and Touryan (1975) for detailed descriptions of probe theory.

2. *Problems in Application of Probe Techniques to Discharge Lamps*

Four principal problems arise in the application of Langmuir probes to rare-gas mercury, or rare-gas sodium type discharges. First is the fact that if data is taken in a DC mode, with probe potential slowly swept from negative to positive, probe temperature varies with probe potential, being lowest when the probe is negative with respect to the plasma and collecting only small currents, and highest when the probe is at, or positive with

respect to, plasma potential. The power dissipation due to the relatively large electron current strongly heats the probe, which is deliberately kept very small to minimize the perturbation of the plasma. There is a very strong tendency for mercury (or sodium) to condense on the probe when it is cold and evaporate when the probe is hot. This results in a change in work function of the probe and hence of the contact potential difference between probe and plasma. Since such a change may be a continuous function of probe potential, it appears as a distortion of the probe electron current volt-ampere characteristic and therefore of the apparent electron energy distribution. Other contaminants which affect the probe work function may also be present in the gas atmosphere of such lamps, evolved from the cathodes or desorbed from the phosphor surface. The sorption and desorption of these contaminants as probe temperature changes may also result in work function changes, distorting the probe V-I characteristic.

Second, discharge lamps operate on alternating current, and the time dependence of many of the parameters is required; many of the investigations have been forced to develop time-resolved methods of probe measurements. Third, elementary probe analysis is simplest and most straightforward when the probe is small in comparison to the mean free paths of plasma particles, but large in comparison to the Debye length. Because buffer gas pressures are typically several torr, mean free paths are on the order of one millimeter or less, and probes (and probe supports) are not always conveniently made small in comparison. Thus, perturbation of plasma densities by the probe and its support is not easy to avoid. Finally, noise due to oscillations or striations can introduce problems, especially in time-resolved measurements.

Easley (1951) and Kenty, Easley, and Barnes (1951) dealt with the first problem by "flashing" the probe prior to obtaining each data point, operating it positive with respect to plasma potential to heat it to red heat, subsequently switching it to a measuring circuit and obtaining the measurement for a single potential before it cooled down. Any mercury or contaminant which might have been condensed on the probe was driven off by heating, and therefore the work function for each of the measurements was the same. With this technique, Kenty, Easley, and Barnes (1951) concluded that electron energy distribution in the positive column at the discharge conditions appropriate at the time to 40 watt fluorescent lamps (3.5 torr argon pressure, 0.006 torr Hg pressure, 425 ma discharge current in a 1.5 inch diameter tube) was Maxwellian to about 8 volts, with a radially averaged electron density of 2×10^{11} per cm cubed. Figure 3 shows Easley's data, illustrating the changes in shape of the probe I-V characteristic depending on the state of the probe surface and the straight line nature of the clean probe characteristic.

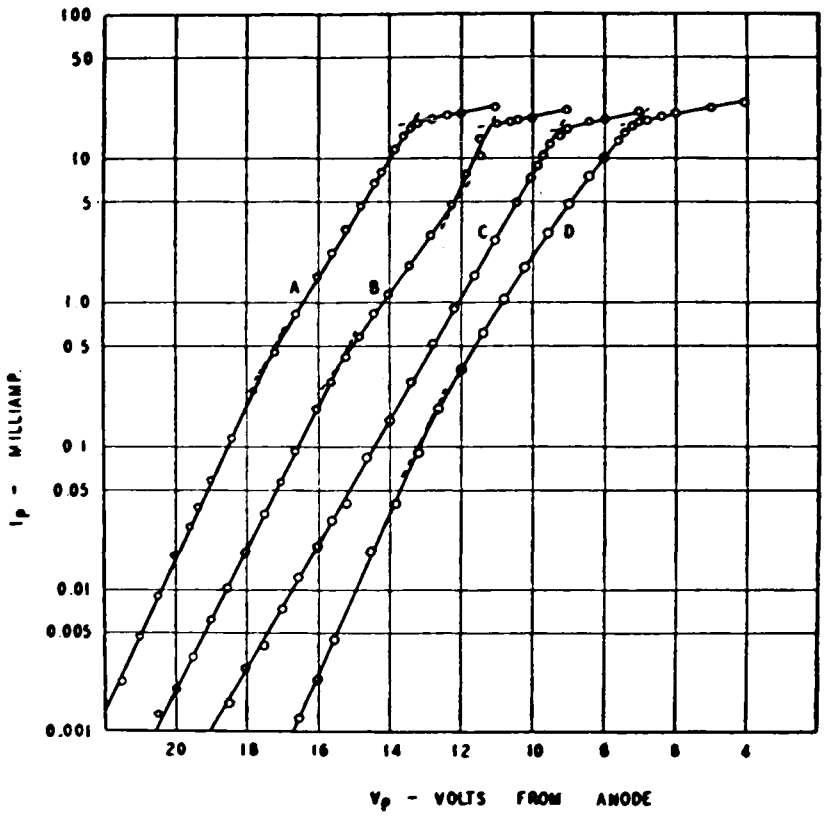


FIGURE 3. Probe V-I characteristics (Easley, 1951; reprinted with permission of the Journal of Applied Physics). A, no probe cleaning, gradual voltage change from positive toward negative; B, same as A, except voltage change from negative toward positive; C, probe heated by electron bombardment before each reading; D, repeat of A, taken after C.

An additional source of changes in probe work function was identified by Waymouth (1959), who used probe measurements to study plasma potential, electron density, and electron energy distribution in the negative glow region near the oxide-coated cathode electrode. Contaminants evaporated from the cathode deposited on the probe, creating semi-insulating layers across which a sufficient potential drop existed, proportional to net probe current, to introduce substantial distortion of the apparent electron current volt-ampere characteristic.

In addition, Waymouth wished to resolve the time-dependent behavior in the negative glow of a discharge lamp operating on alternating current. For this purpose, he developed a pulsed-probe technique, in which the probe

was maintained at a strongly negative potential with respect to the plasma, except for a brief period each cycle, in which it was rapidly ramped in potential from negative to positive, with the instantaneous current displayed on an oscilloscope as a function of instantaneous potential. The excursion through the significant range of probe potentials required about 10 microseconds. During the period when the probe was negative, it was bombarded by high energy ions, and the resulting sputtering kept the probe surface free of contaminating deposits; the rapid excursion through the potential region of interest did not allow time for any probe heating so that the surface was always covered by condensed mercury.

Adriaansz (1973) has also employed a rapid sweep of probe voltage in short pulses synchronized with pulsed lamp current in a neon-sodium discharge to study the time dependence of electron temperature during the current pulse.

With regard to perturbation of plasma densities by probe and probe supports, Waymouth (1964) analyzed the depression in plasma density near the probe by a diffusion model similar to that of Davidoff and Zmanovskaya (1936), except for specifically including the effect of ion temperature. Waymouth pointed out that a major perturbation frequently encountered with cylindrical probes was that a substantial portion of the collecting area lay within the volume of plasma perturbed by diffusion to the probe shield. Waymouth reports relatively simple formulas for correcting probe-measured densities for the effects of the perturbation. Static depression of plasma density due to an insulating shield for the probe wire support was considered, as well as perturbation dependent on probe potential.

Little and Waymouth (1966) report experimental measurements with a very small probe of plasma density depression in the vicinity of a large probe in a rare-gas mercury-vapor plasma and find reasonable agreement with the calculations of Waymouth, except for discrepancies resulting from asymmetric effects due to the presence of unidirectional current flow past the probe in a DC discharge. Barnes (1962) has also observed "wake" effects downstream of a probe in pure rare-gas discharges.

With respect to the problem of noise and striations causing interference with probe measurements, especially those involving pulsed-probe or double-differentiation techniques, Godyak (1985) has reported a unique noise suppression circuit based on earlier results by Boyd and Twiddy (1959) and Gulidov, Kagan, Kolokolov, and Milenin (1969). This circuit samples the plasma noise potential with a reference probe and feeds back an equal and opposite potential to the discharge operating circuit to insure that the plasma at the site of the test probe is at ground potential with respect to noise voltages. The noise suppression feature is extended to much higher frequencies than prior work and is combined with a fast pulse measurement

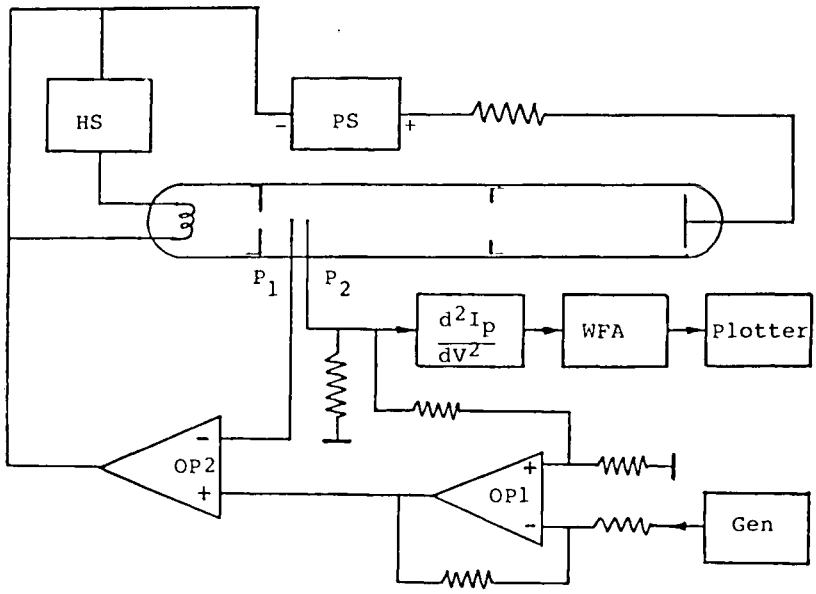


FIGURE 4. Noise suppression circuit (Godyak, 1985). Fluctuations in plasma potential due to noise, oscillations, etc. are sampled by floating probe P1 and fed back through operational amplifier OP2 to entire discharge circuit, insuring that plasma at location of probe P2 is maintained with high accuracy at ground potential with respect to noise voltages. Noise suppression of 40 db or greater is achieved for all frequencies less than 100 kHz.

(< 0.5 msec) to minimize probe work function changes. Figure 4 shows a schematic diagram of Godyak's circuit.

3. *Principal Results*

The most extensive series of probe measurements in the mercury/rare-gas discharge have been those of Verweij (1960). He measured electron temperature, axial electron density, and positive column field strength for an argon pressure range 0–20 torr, mercury pressure range 0.0006–0.06 torr, current density range 0.01–0.08 amp/cm^2. A diagram of the extremely minute probe employed by Verweij is shown in Figure 5. The probe was designed to avoid significant plasma perturbation by the probe up to the highest argon pressures employed. The Waymouth formulas applied to this probe design show essentially no depletion due to the shield at any argon pressure. Maximum depletion due to electron and ion current flow to the

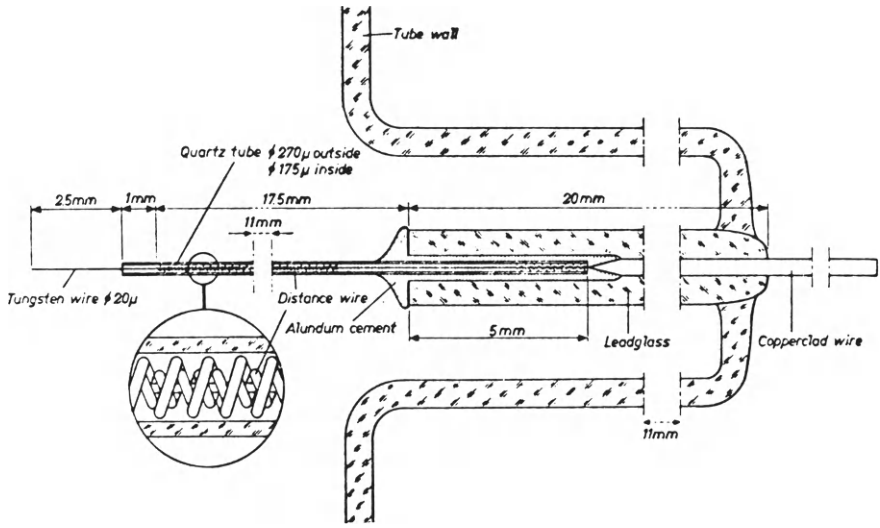


FIGURE 5. Probe construction (reprinted from Verweij, 1960, with the permission of the author and of Philips Journal of Research).

probe itself at plasma potential is about a factor 2.5, at argon pressure of 20 torr.

Verweij used the "flashing" technique of Easley (1951) to insure clean probes. The principal results obtained by Verweij are shown in Figures 6–9. Because of the very small probes he employed, his measurements of probe electron current characteristic were limited to 4–5 volts negative with respect to plasma potential; over this range, significant deviations from a Maxwellian distribution were not found (nor would they be expected).

As already noted, Easley observed Maxwellian distributions to eight or more volts of electron energy. The validity of the data beyond 5–6 volts is in some doubt, because it depends on the subtraction of an extrapolated ion current from total probe current to determine the net electron current. For wire probes, ion current has significant dependence on negative probe voltage, but Easley does not discuss the details of the function chosen for ion current vs. negative probe voltage. Barnes (1966) obtained distributions depleted below the Maxwellian for electron energies greater than about five volts, for very similar conditions, except 3.0 torr argon pressure vs. 3.5 torr employed by Easley.

The two-electron-group model calculations of the electron energy distribution by Vriens (1973), Ligthart and Keijser (1980), and Morgan and Vriens (1980), as well as the Boltzmann equation calculations of Lagushenko and Maya (1985), indicate that the electron energy distribution is

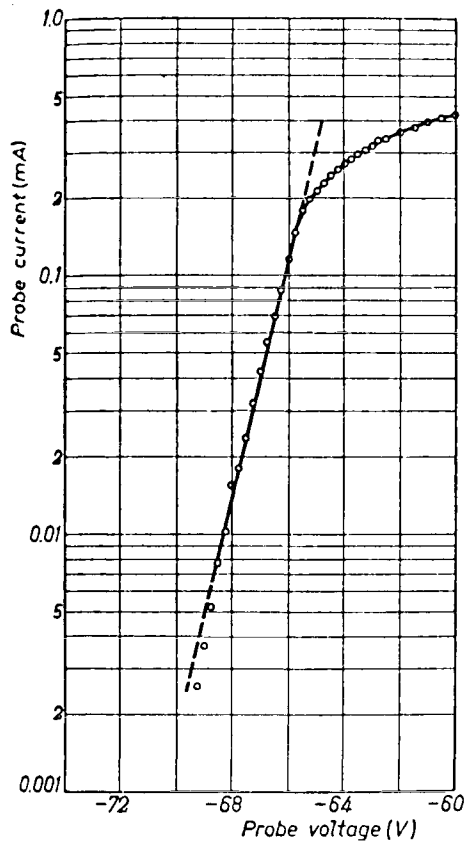


FIGURE 6. Probe V-I characteristics (reprinted from Verweij, 1960, with the permission of the author and of Philips Journal of Research).

in general depleted below the Maxwellian above the first excitation potential of mercury as a result of the large energy losses by inelastic collisions of electrons. Only at high electron densities are Maxwellian distributions expected.

Godyak and Lagushenko (1986) have reported extensive probe measurements in the positive column of a neon-mercury discharge, using the method of Godyak (1985) already described. A pulsed-sweep, double-differentiation circuit automatically obtains and plots the second derivative of probe current as a function of probe voltage; the noise suppression circuit already cited is used to eliminate the effect of oscillations and striations. A factor of 10^3 range of the electron distribution function is accessible.

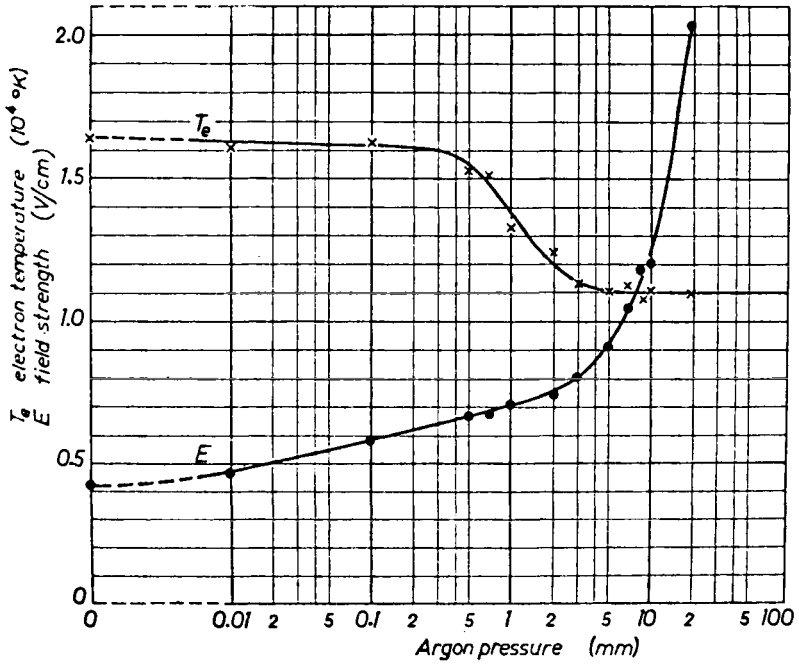


FIGURE 7. Electron temperature and axial electric field vs. argon filling pressure; Tube ID 37 mm, discharge current 400 ma, condensed mercury temperature 42°C (reprinted from Verweij, 1960, with the permission of the author and of Philips Journal of Research).

Figures 10 and 11 illustrate Godyak's principal experimental results for a neon-mercury discharge. These are plots of the logarithm of second derivative of probe current (proportional to the logarithm of electron energy distribution) vs. probe voltage. Maxwellian distributions on this plot would be straight lines. As can be seen, all of these distributions are deficient in high-energy electrons compared to the Maxwellian, but the plots all become more linear as neon pressure increases and as discharge current increases. That is, as electron density increases, energy distributions become more Maxwellian. This trend is entirely in accord with intuition and with Boltzmann equation calculations. Figure 12 shows the comparison between Godyak's measurements and Lagushenko's calculation for a specific case.

Although, as seen above, depleted Maxwellian distributions are common in the positive column, electron energy distributions with an excess of high-energy electrons may be obtained in certain circumstances. Waymouth (1959) observed strongly "super-Maxwellian" electron energy distributions in the negative glow, near the cathode, with an excess of high-energy

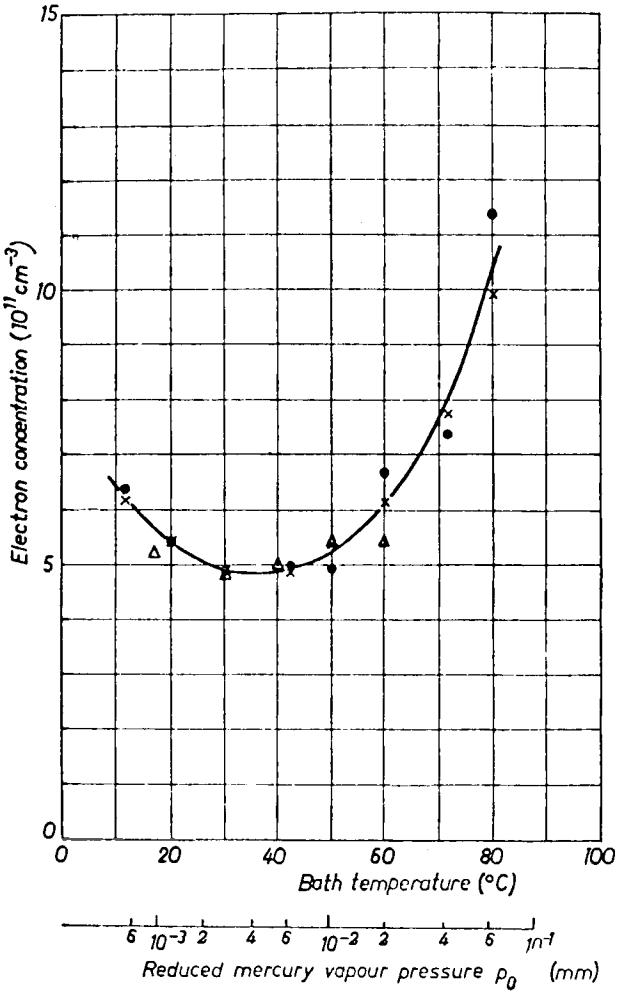


FIGURE 8. Electron concentration at the axis vs. condensed mercury temperature, 3 torr argon pressure, 400 ma discharge current, tube ID 37 mm. (reprinted from Verweij, 1960, with the permission of the author and of Philips Journal of Research).

electrons: "primary" electrons emitted from the cathode and injected into the negative glow with the full energy of the cathode sheath potential, superposed on a larger population of secondary electrons with an approximate Maxwellian distribution having a temperature about one electron volt. Far from the cathode, near the beginning of the positive column, "sub-Maxwellian" distributions with a deficiency of high-energy electrons were found. Waymouth's data, with probe currents scaled from photographs of

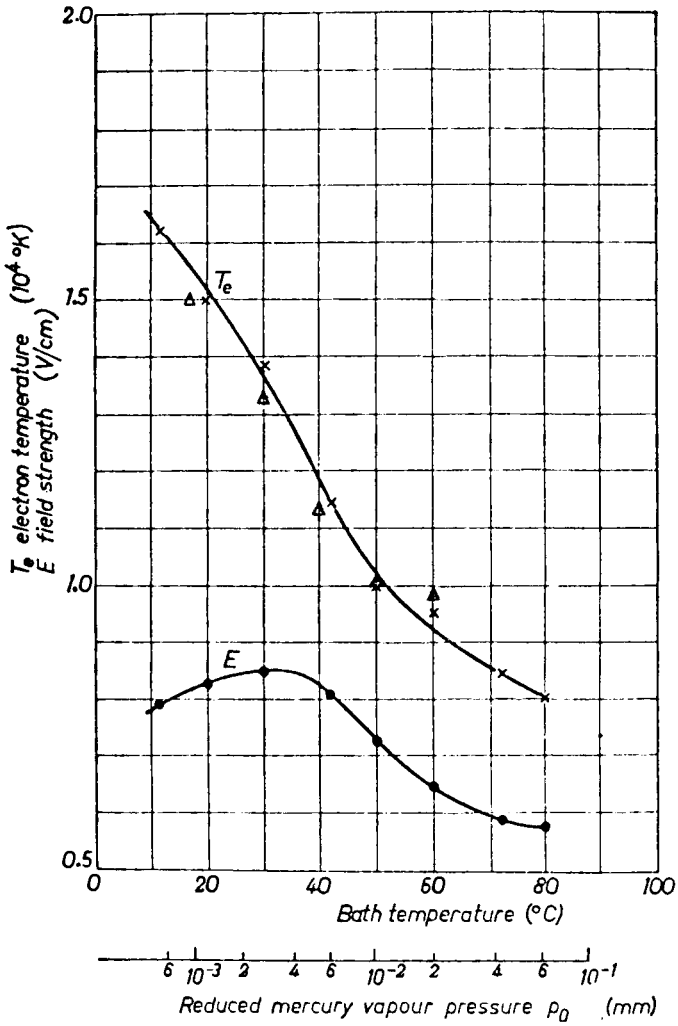


FIGURE 9. Electron temperature and axial electric field vs. condensed mercury temperature, 3 torr argon pressure, 400 ma discharge current, tube ID 37 mm (reprinted from Verweij, 1960, with the permission of the author and of Philips Journal of Research).

oscilloscope traces, were not sufficiently accurate to permit quantitative measurement of the distribution by double differentiation. Figure 13 shows plots of the logarithm of probe current vs. probe voltage (from Waymouth (1971)) for various distances from the cathode, together with electron density and plasma potential as functions of position in the negative glow. Figure 14 shows the result of a double differentiation, illustrating qualita-

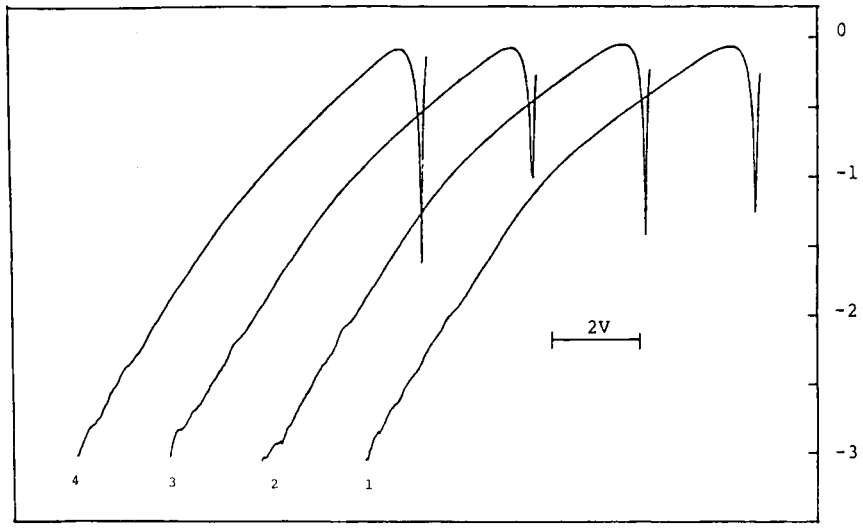


FIGURE 10. Electron energy distributions, 1.9 torr neon, 40°C condensed mercury temperature, tube ID 36 mm; 1) 0.1 amp discharge current, 2) 0.2 amp, 3) 0.4 amp, 4) 0.8 amp; curves offset on voltage axis for clarity. Plasma potential is identified as that potential for which d^2I/dV^2 crosses zero (Godyak and Lagushenko, 1986).

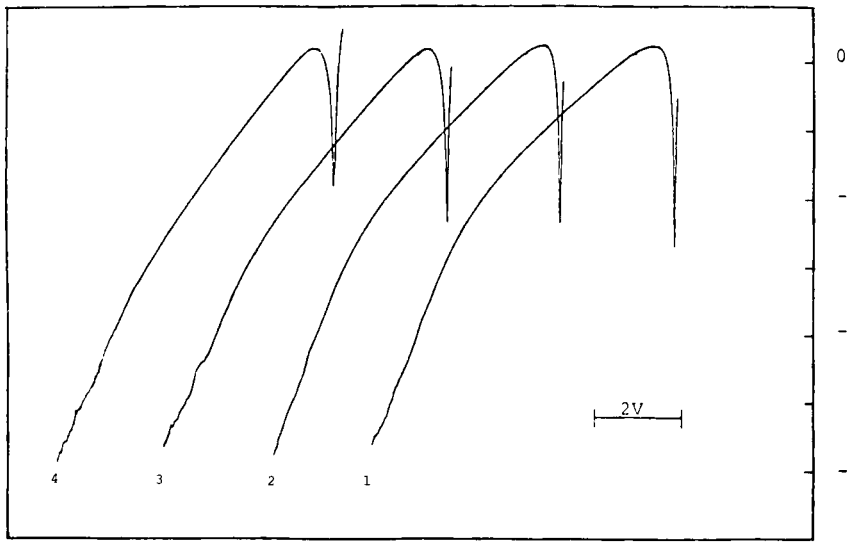


FIGURE 11. Electron energy distributions. Same as Figure 11, except condensed mercury temperature is 60°C (Godyak and Lagushenko, 1986).

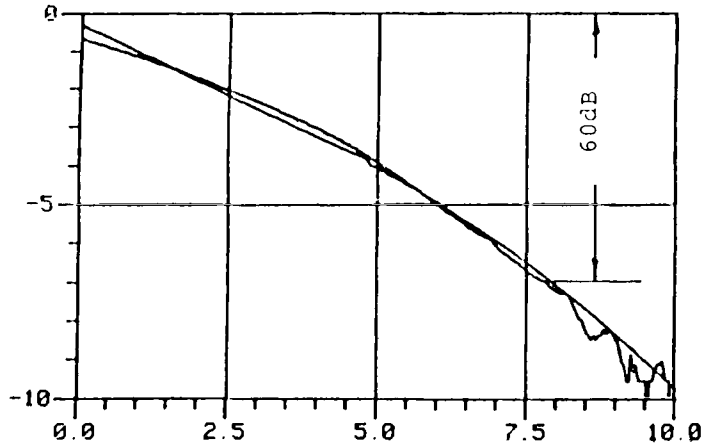


FIGURE 12. Measured and calculated electron energy distributions (Godyak & Lagushenko, 1986).

tively the double-humped nature of the electron energy distribution, demonstrating the existence of a high-energy group of primary electrons.

Godyak, May, Lagushenko, and Pai (1986) have studied a novel mercury/rare-gas discharge with a "segmented positive column," in which a series of apertured partitions at regular intervals divide the otherwise uniform column region of the discharge into a series of segments. The smaller cross-section area at the aperture requires a higher electric field to maintain the continuity of current. Depending on the choice of ratio of aperture area to tube cross-section area, the potential gradient in this region may be made great enough that the potential difference across the "energy-exchange distance" in the plasma becomes comparable to the excitation potential of mercury.

Under these conditions, the entire electron energy distribution may be translated upward in energy by several volts, with minimal inelastic-collision depletion of the high-energy group in passage through this high-field region. Thus, the electron energy distribution becomes substantially enriched in high-energy electrons relative to the Maxwellian. By suitable choice of rare gas and mercury pressures, the energy exchange distance may be made sufficiently long that the "super-Maxwellian" distribution persists for a significant distance downstream of the partition. Figure 15 illustrates plasma potential as a function of axial position through the partition region, while Figure 16 shows the electron energy distributions which result for various axial positions. The entire light-generating properties of the

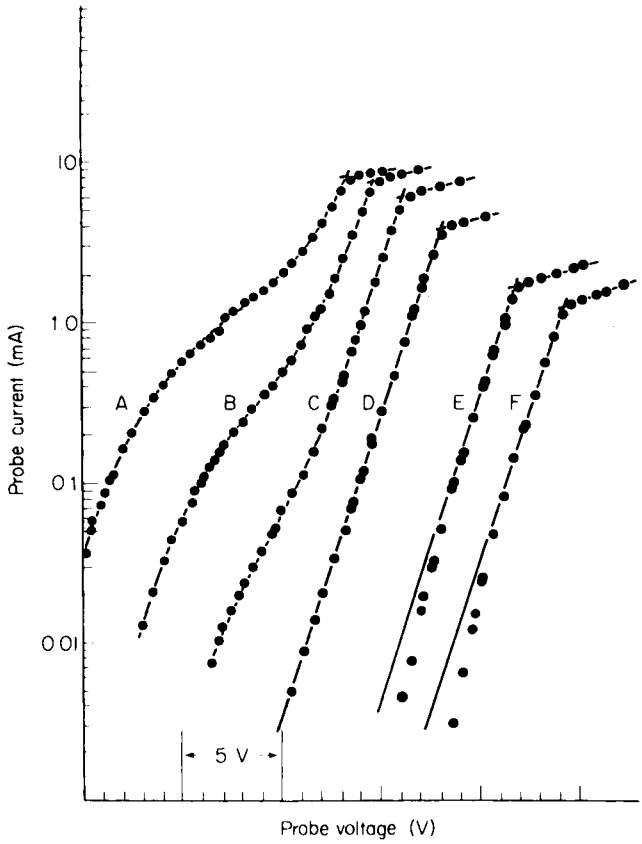


FIGURE 13a. Probe V-I characteristics, negative glow; 1.0 torr, neon pressure, condensed mercury temperature 40°C; A, 0.25 cm from cathode; B, 0.75; C, 1.5; D, 2.0; E, 3.0; F, 4.25 (reprinted from Waymouth, 1971, with the permission of the author and copyright holder).

discharge being determined by the electron energy distribution, this finding affords the lamp designer an extremely important new dimension of control, which is now being exploited.

Cornelissen (1986) and Cornelissen and Merks-Eppingbroek (1986) report the results of probe measurements in low-pressure sodium-neon discharges such as are employed in low-pressure sodium lamps. A probe design similar to Verweij (1960) was used, except that the glass shield was 1 mm in diameter and the wire probe extended only 1 mm past the shield. A pulsed-probe technique was employed, with the probe maintained posi-

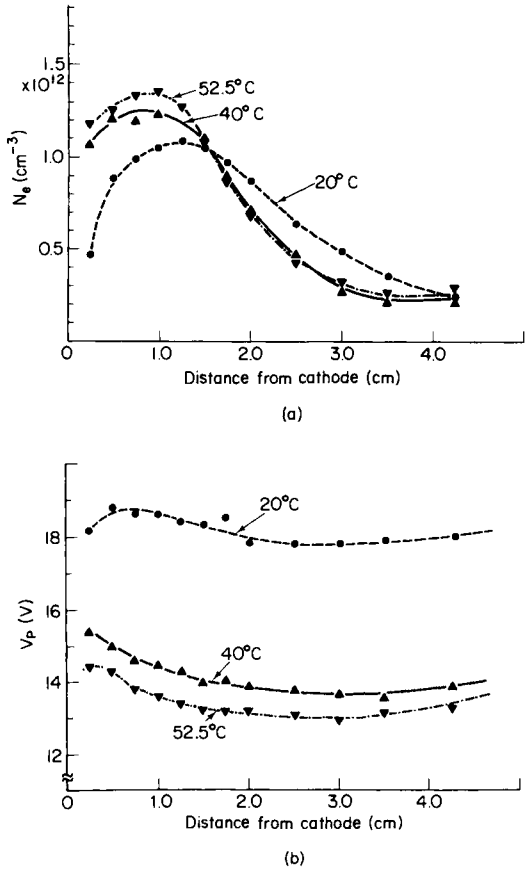


FIGURE 13b. Electron density and plasma potential vs. position in negative glow (reprinted from Waymouth, 1971, with the permission of the author and copyright holder).

tive with respect to plasma potential between pulses, and ramped negative over a time span of approximately 1.3 milliseconds; instantaneous probe current and voltage are digitized during the sweep for determination of the I-V characteristic. Thus, the probe is maintained hot at all times, and sodium is prevented from condensing on the probe surface and on the probe support.

Maxwellian, or nearly-Maxwellian, electron energy distributions were found for all conditions, in agreement with two-electron-group model calculations. However, electron densities obtained from the probe data were

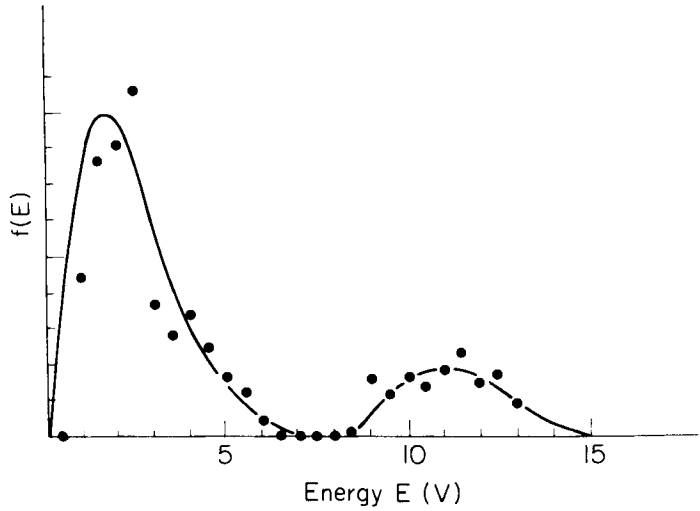


FIGURE 14. Electron energy distribution calculated by graphical double differentiation from curve A, Figure 13a, illustrating presence of high-energy group superposed on approximately Maxwellian group of lower energy (secondary) electrons. Cathode fall was about 15 volts (reprinted from Waymouth, 1971, with the permission of the author and copyright holder).

approximately 2.5 times less than those obtained from analysis of neutral sodium atom depletion (obtained from laser absorption; see Section II.C.1.b). Cornelissen et al. conclude that perturbation of the plasma of the probe shield is the principal cause of the electron density discrepancy. The Waymouth formula (1964) for depression of plasma density by the shield applied to the conditions of these authors predicts that apparent electron density is 37% of actual.

B. Microwave Diagnostics

Polman, van der Werf, and Drop (1972) have measured the average electron density as a function of phase angle in a strongly modulated DC low-pressure mercury-neon discharge with a 4 mm microwave interferometer for several different modulation frequencies. They also determined the electron temperature as a function of phase angle, using a 1.5 Ghz microwave radiometer to measure the noise temperature of the plasma.

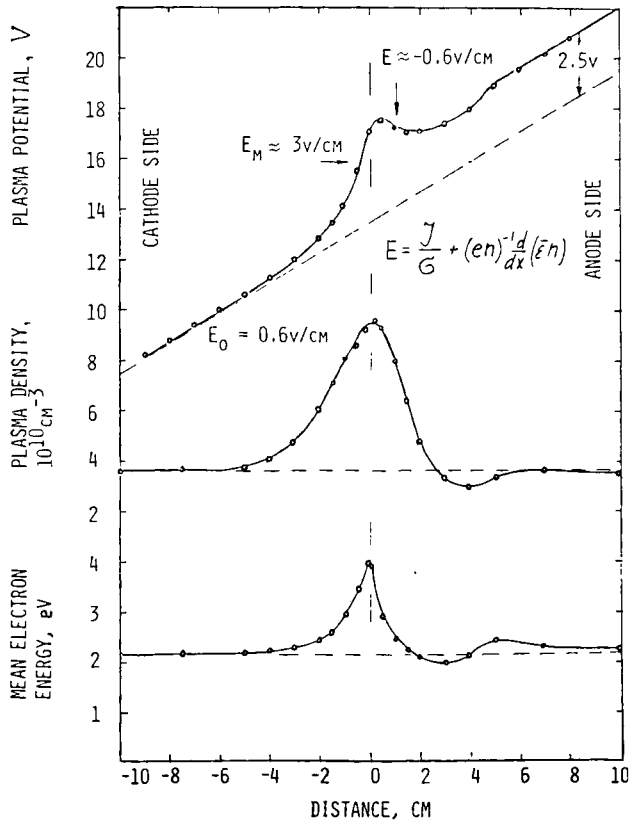


FIGURE 15. Plasma potential vs. position in partitioned discharge (Godyak, Lagushenko, Maya, and Pai, 1986).

Polman and Drop (1972) used the noise temperature technique to measure the electron temperature decay in the afterglow of an interrupted mercury/rare-gas discharge.

C. Spectroscopic Diagnostics for Low-Pressure Lamps

Emission, absorption, laser-induced fluorescence, and optogalvanic spectroscopy have all been utilized for diagnostics of low-pressure discharge lamp plasmas. Doppler-free LIF spectroscopy has recently been used to

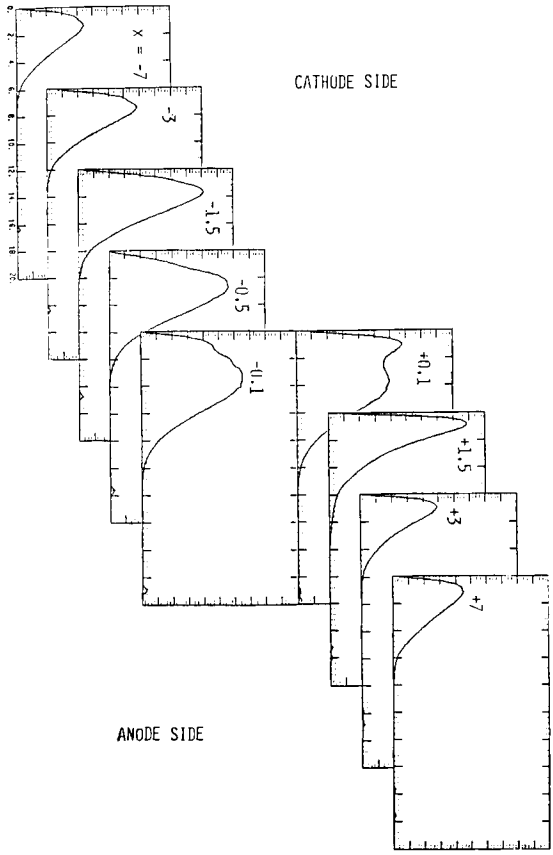


FIGURE 16. Electron energy distribution vs. position in partitioned discharge; 1.0 torr Ne, 40°C condensed mercury temperature, 0.2 amp discharge current. Distances in centimeters from orifice. (Godyak, Lagushenko, Maya, and Pai, 1986).

determine the Stark broadening and shift of the sodium 4D levels to measure electron density in the low-pressure sodium/rare-gas discharge. These are discussed in detail below.

1. Optical Absorption

a. Determination of Excited State Densities. Early work in optical absorption measurements in mercury-rare gas plasmas used absorption of the 405 nm, 436 nm, and 546 nm lines of mercury to determine the concentration of the 3P-0, 3P-1, 3P-2 excited states of mercury (Kenty, 1950; Koedam

and Kruithof, 1962). Since these investigations were carried out in the pre-laser era, the line widths of the probing light source were not small in comparison to the absorption line width. This, of course, greatly multiplies the difficulty of analysis of the results, because different fractions of the probing beam encounter different values of optical depth. The measured absorption is therefore an integration over the wavelength range of the incident line, and the results must be deconvoluted to obtain the desired number densities of absorbers. Kenty utilized a source discharge operating under "identical" conditions to the probed discharge. Under such circumstances, the incident line and the absorption line have the same shape and width, and the absorption integrated over the entire line can be obtained analytically. Kenty used a formula originally developed by Ladenburg and Reiche (see Mitchell and Zemansky (1932) pp 118 ff), assuming Doppler-broadened lines of Gaussian profile. The actual profile is now known to be a convolution of a Gaussian profile due to Doppler broadening and a Lorentzian profile due to collisional broadening. Such a profile is known as a Voigt profile.

A further problem with mercury is the complex isotope-shift/hyperfine structure of the absorption and emission lines, which results in each of these lines having five or more components separated by amounts comparable to their widths. Koedam and Kruithof (1962) treated the problem by numerical integration over incident and absorption line profiles. Neither author explicitly dealt with the fact that the concentrations of all three of the excited states vary with radial position and that their radial dependencies differ, the metastable levels having flatter radial profiles than the resonance level.

Nevertheless, although the absolute magnitudes of the spatially-averaged concentrations may be somewhat suspect, the qualitative dependencies obtained by both authors are in reasonable agreement. In particular, the concentrations of the metastables are found to exhibit a pronounced "saturation" effect as a function of current, illustrating the importance of collisions with slow electrons which quench excited states. Such so-called *collisions of the second kind* compete with radiative transitions in dissipating excitation energy and tend to bring excited atom density into LTE with the electron gas as electron density increases.

Post (1985 and 1986) has used laser absorption measurements of the mercury 1P-1 excited state density as a function of time during the afterglow of an interrupted Hg-A discharge to determine the imprisonment time for 184.9 nm mercury resonance radiation.

Van de Weijer and Cremers (1982) have recently reported the application of the "hook" method (Rozhdestvenskii, 1912) to the mercury/rare-gas plasma, measuring the dispersion of the index of refraction in the vicinity of

an absorption line. The source beam in this case is a broad-band dye laser, with band width much wider than the absorption line. The method depends on an interference pattern of fringes, which are displaced in a characteristic "hook" shape in the vicinity of the absorption line. The wavelength difference between the maxima and minima of the fringe patterns is proportional to the number density of the absorbers. The advantage of this technique is that these extrema occur at wavelength deviations from the line center that are large in comparison to the isotope and hfs splittings and are in the far wings of the line; there is a very simple relationship between hook separation and number density of absorbing states.

Figure 17 shows results obtained by van de Weijer and Cremers for spatially-averaged concentrations of the triplet P states, for three different discharge currents, as a function of condensed mercury temperature.

Uvarov and Fabrikant (1965) have also employed the hook method to measure concentrations of the 3P-1, 3P-2 excited states of mercury as a function of radial position in a low-pressure mercury, as well as in a low-pressure mercury-argon, discharge. No experimental details are given. The axial densities obtained by Uvarov and Fabrikant for 3P-1 appear to be only about one half the average values found by van de Weijer and Cremers under comparable conditions.

Moskowitz (1987) has very recently reported a unique modulated linear absorption (MLA) method of determining spatially-resolved concentrations of triplet excited atoms of mercury in a mercury/rare-gas discharge. A "pump" beam and "probe" beam, obtained from the same dye laser (with line width small in comparison to the absorption line) transect the plasma at right angles, intersecting in one local volume. The pump beam is intense enough to saturate the absorption, dropping the concentration of absorbing atoms essentially to zero at all points along its path. The probe beam is weak enough to remain in the linear absorption domain at all points along its path. The pump beam is modulated on-off, and the presence of amplitude modulation in the probe beam is synchronously detected.

The only source of pump-beam modulation frequency signals in the probe beam is from the alternate depletion of absorbing-atom concentration in the common intersection volume when the pump beam is on, and the restoration by plasma collision processes in the same volume when the pump beam is off. Thus, the strength of the pump-beam modulation frequency signal in the detected probe beam is proportional to the unperturbed concentration of absorbing atoms in the common intersection volume.

In application, the pump beam transects the plasma along a diameter, and the probe beam intersection is traversed across the discharge to determine the radial dependence of concentration of absorbing species. The

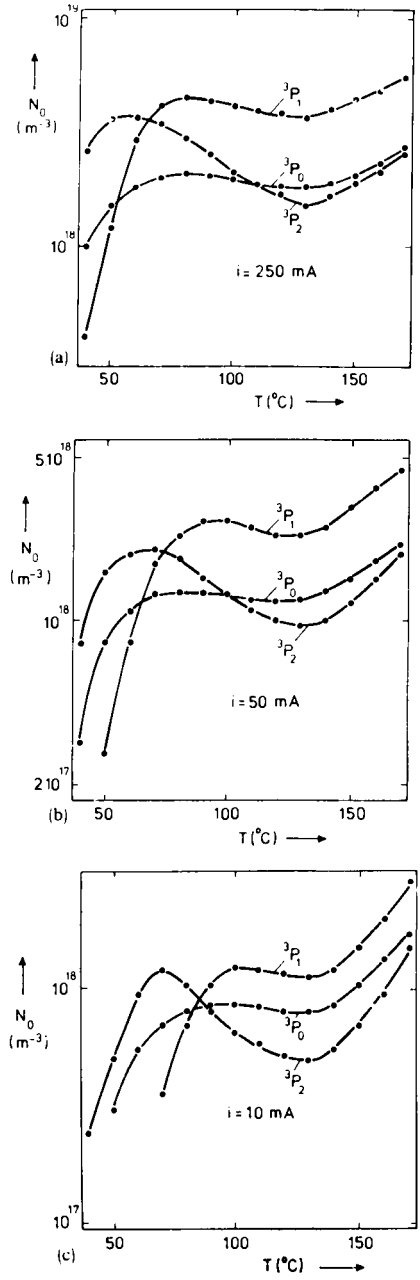


FIGURE 17. Hg triplet-P concentrations vs. Hg pressure A, 250 ma discharge current; B, 50 ma; C, 10 ma (reprinted from van de Weijer and Cremers, 1982, with the permission of the author and of the Journal of Applied Physics).

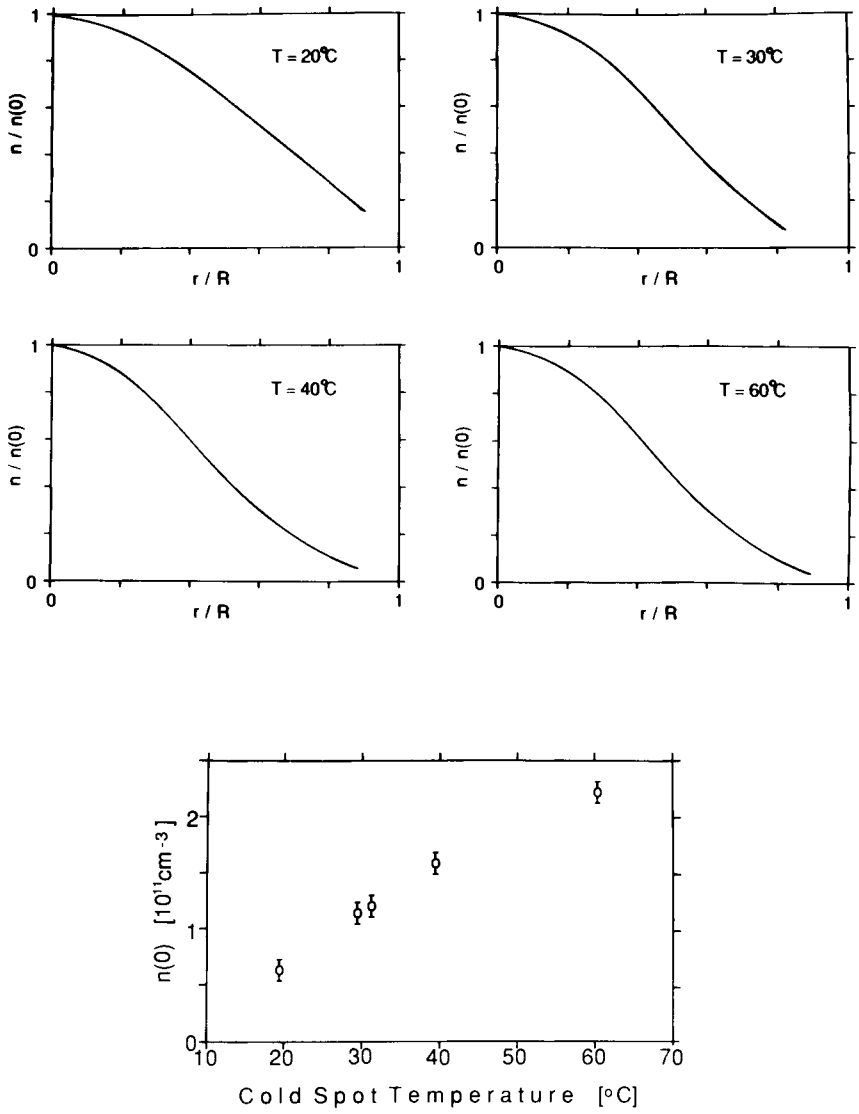


FIGURE 18. Hg 3P-1 excited state concentration vs. radial position (reprinted from Moskowitz, 1987, with the permission of the author and of Applied Physics Letters).

technique, therefore, yields spatially resolved concentrations of absorbing species without the necessity for Abel inversion of chord-integrated data. Figure 18 shows representative early data for the radial dependence of the concentration of Hg 3P-1 atoms; a discharge employing only the Hg-198 isotope was employed for this case to avoid the complexity of the multi-component line of natural mercury. Extension to natural mercury and to the determination of absolute concentrations was in process at the time of this writing.

Bigio (1988) has even more recently described a "saturated absorption" laser technique for measuring the radial dependence of the concentrations of 3P-0, 1, 2 excited mercury atoms in an argon-mercury discharge for several different mercury vapor pressures. The total power absorbed from a pulsed laser beam of sufficient intensity to excite every absorbing atom along its path is proportional to the total number of absorbing atoms along that path at $t = 0$ before the pulse, provided the pulse is short enough in comparison to plasma time constants that the absorbing population cannot be regenerated. Bigio used an axial beam, adjustable to different radial positions, to determine the relative number density of excited mercury atoms as a function of radius. The absolute scale was determined by the hook method for a limited number of cases.

Bigio's results agree well with Moskowitz's (1986) for 3P-1 excited atoms at 6 millitorr mercury pressure, 400 ma discharge current. There is presently an unexplained difference at 1.2 millitorr mercury pressure. Bigio's radial profile is much flatter than Moskowitz's.

b. Determination of Ground-State Densities. Cornelissen (1986) and Cornelissen and Merks-Eppingbroek (1986) have measured the radial density dependence of ground-state neutral sodium atoms in a low-pressure sodium-neon discharge, employing laser absorption. The laser was detuned into the far wings of the line to reduce the optical depth and to operate at a wavelength where the line profile was Lorentzian, and determined by a single mechanism, collisional broadening by neon. Chord-integrated absorption measurements were made as a function of distance from the arc center, and the results were fitted to a three-parameter function. This was necessary because they were unable to obtain sufficiently accurate Abel inversion routines for dealing with a radial profile having a pronounced minimum in the center and maxima at the walls.

The neutral sodium atom concentration in the center of such a discharge is depleted as a result of ionization and ambipolar diffusion to the walls, followed by neutralization at the walls and ordinary diffusion of neutrals to the center. Cornelissen and Mercks-Eppingbroek point out that in this circumstance the following relationship holds between local electron density

and local density of neutral atoms:

$$n_e(r) - n_e(R_W) = \frac{D_0}{D_a}\{n_0(R_W) - n_0(r)\},$$

where $n_e(r, R_W)$ is electron density at r, R_W, respectively, and $n_0(r, R_W)$ is neutral atom density at r, R_W. D_0 is the free diffusion coefficient of neutral sodium atoms, and D_a is the ambipolar diffusion coefficient. At the partial pressures of sodium and neon employed, the dominant collision process is with neon atoms, and these diffusion coefficients can be well calculated, provided the electron temperature is known. As a result, from the measured radial density distribution of neutral sodium concentration, the radial distribution of electron density can be uniquely, if indirectly, determined. The electron temperature in this work was determined from Langmuir probe measurements (see Section II.A.3).

Figure 19 shows results on neutral sodium concentration as a function of radial position for two different discharge currents, illustrating the very substantial depletion in the discharge, especially at the higher current. Figure 20 shows the values of axial electron density deduced from the

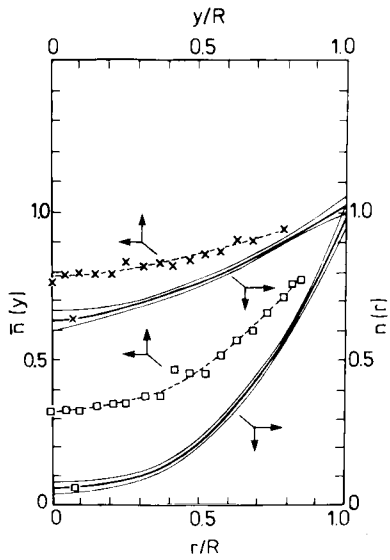


FIGURE 19. Concentration of neutral sodium vs. radial position in low-pressure sodium-neon discharge; Points, measured chord-integrated average densities (×, 200 ma, □ 500 ma). Lines, calculated radial profiles of Na ground-state densities (reprinted from Cornelissen and Merks-Eppingbroek, 1986, with the permission of the author and the Journal of Applied Physics).

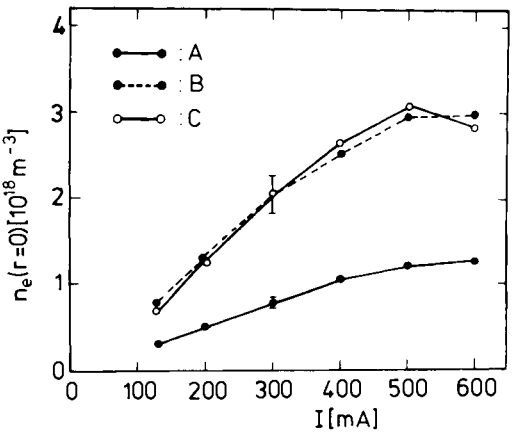


FIGURE 20. Axial electron density vs. current; A, values from probe measurements; B, values determined from laser-absorption measurements; C, values calculated from measured electrical conductivity (reprinted from Cornelissen and Merks-Eppingbroek, 1986, with the permission of the author and of the Journal of Applied Physics).

neutral atom concentration as a function of current, compared with the values measured by Langmuir probe.

2. *Laser-induced Fluorescence*

Van de Weijer and Cremers (1985a) have used the decay of laser-induced fluorescence in a low-pressure mercury/rare-gas discharge to measure the imprisonment time of 253.7 nm mercury resonance radiation as a function of the various parameters of the discharge. Absorption of a dye-laser pulse of 405 nm radiation by mercury atoms in the 3P-0 metastable state pumps these to the 7-3S state; prompt radiation from this state at 435.8 nm creates an overpopulation of 3P-1 resonance levels, which decay to the ground state by the emission of 253.7 nm resonance radiation. Measuring the decay time of the laser-induced fluorescence of 253.7 nm radiation determines the imprisonment time. Figure 21 shows the imprisonment time as a function of the product of line-center absorption coefficient times tube radius. Van de Weijer and Cremers calculated the resonance radiation flux as a function of mercury density from the concentrations of 3P-1 excited atoms and the measured decay times. Good agreement with measured fluxes was found for a pure mercury discharge using the excited atom densities determined by Uvarov and Fabrikant (1965). Calculated fluxes exceeded measured fluxes for Hg/A discharges by about 20% when the excited atom densities

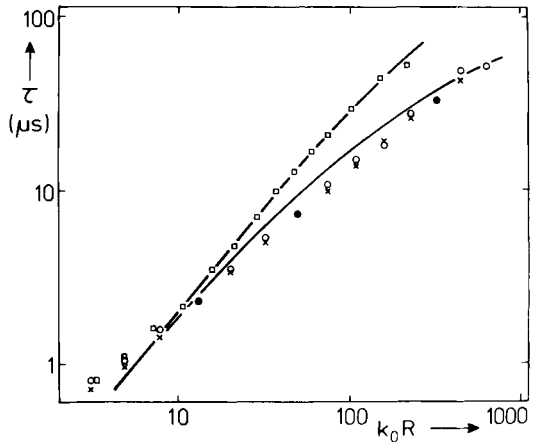


FIGURE 21. Imprisonment time of 254 nm resonance radiation versus line-center optical depth; squares, mercury only; open circles, mercury-argon; ×'s, mercury krypton (reprinted from van de Weijer and Cremers, 1985a, with the permission of the author and of Optics Communications).

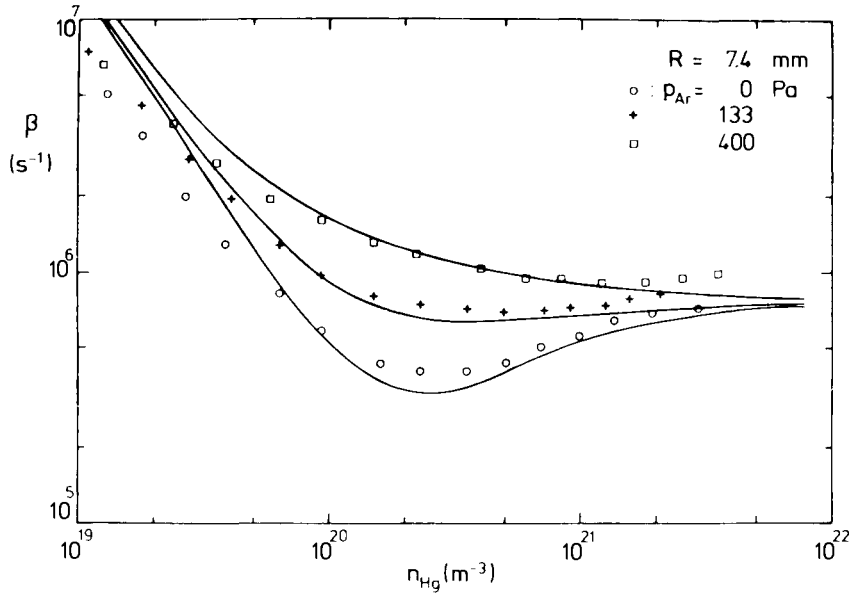


FIGURE 22. Decay rate (inverse of imprisonment time) of 185 nm resonance radiation; solid lines are theoretical calculations (reprinted from Post, 1986, with the permission of the author and of Physical Review).

determined by Van de Weijer and Cremers (1982) were used (see Section II.C.1.a).

Post (1985 and 1986) has also measured the imprisonment time of 184.9 nm mercury resonance radiation from low-pressure mercury-rare-gas discharges by a laser-induced fluorescence technique. Mercury atoms in the 3P-2 metastable state are pumped by a pulse of laser radiation at 365.5 nm to the 3D-2 state, from which they decay to the 1P-1 resonance state by the prompt emission of 577.0 nm radiation. The time dependence of the subsequent decay of 184.9 nm resonance radiation is determined. Good agreement with values determined by laser absorption in the afterglow (Section II.C.1.a) was obtained. Figure 22 gives the decay rate for mercury-

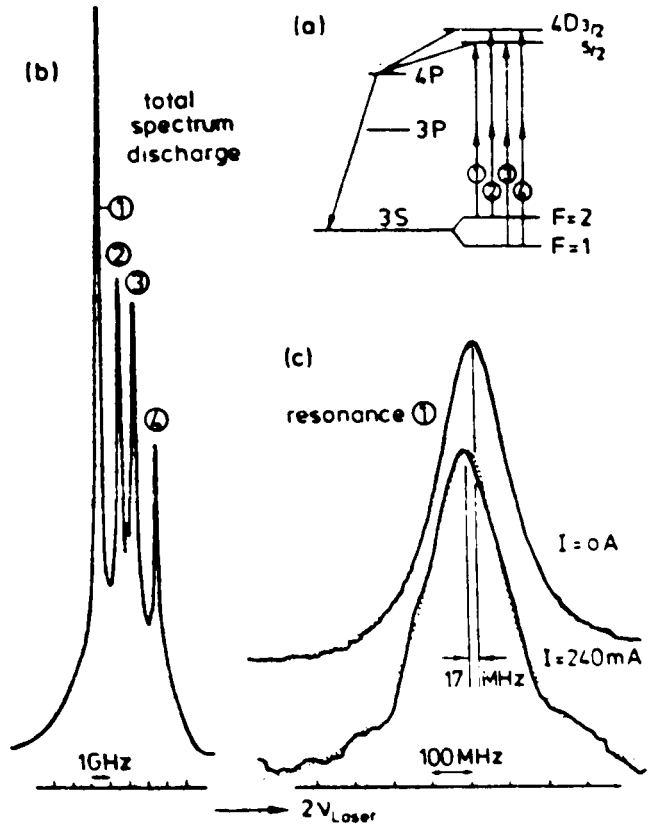


FIGURE 23. Observed line shapes: two-photon Doppler-free spectroscopy in low-pressure sodium-neon discharge; A, energy-level diagram; B, overall line shape; C, resonance (1) at zero current and 240 ma, illustrating 17 Mhz shift (reprinted from Cornelissen and Burgmans, 1982, with the permission of the author and of Optics Communications).

argon discharges as a function of mercury density in a tube with 35.4 mm inside diameter. The experimental results are in good agreement with theoretical calculations, assuming only partial spectral redistribution of absorbed photons before re-emission; the common assumption of complete spectral redistribution yields decay rates that are too large by a factor as much as eight. In contrast, complete spectral redistribution is an excellent approximation for the 253.7 nm resonance transition of mercury.

3. *Two-photon Doppler-Free Spectroscopy*

Cornelissen and Burgmans (1982) have employed two-photon Doppler-free spectroscopy to measure the Stark shift of a resonance transition in sodium vapor to determine the electron density in a low-pressure sodium-neon discharge. In this technique, the 4D excited states of sodium are excited from the ground state by the simultaneous absorption of two oppositely directed photons of 578.73 nm. The Doppler shift from the atom velocity adds to the frequency of one and subtracts from the frequency of the other, thereby cancelling. The absorption line width is therefore free of Doppler broadening and is dominated by collisional and Stark broadening and shift. The occurrence of an absorption event is detected by the subsequent emission of a cascade photon at 330 nm. The plot of intensity of 330 nm radiation versus the wavelength of the exciting photons then contains the desired linewidth and shift information. An example is shown in Figure 23. Figure 24 gives results for axial electron density as a function of discharge current. They are about 25% higher than those cited by Cornelissen and

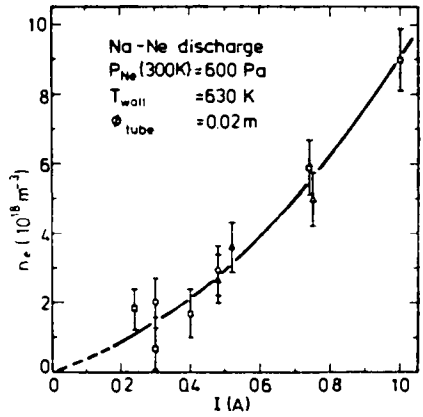


FIGURE 24. Axial electron density (reprinted from Cornelissen and Burgmans, 1982, with the permission of the author and of Optics Communications).

Merks-Eppingbroek (1986), but cannot be directly compared therewith, since the latter authors did not state the sodium vapor pressure at which their results were obtained.

4. *Optogalvanic Spectroscopy*

Van de Weijer and Cremers (1985b) measured the optogalvanic effect in low-pressure mercury/rare-gas discharges. Pumping with 546 nm radiation transfers atoms from the 3P-2 metastable to 7-3S, from whence they decay to the three 3P states; thus the net effect is depleting the population of the upper metastable, which figures heavily in the two-stage ionization process. The OGE effect for this wavelength is therefore positive. Pumping with 405 nm or 436 nm radiation transfers atoms from the lower metastable or the resonance level to the upper metastable 3P-2, effectively increasing the ionization rate; consequently, the OGE effect is negative. Van de Weijer and Cremers also observed that the time constant for the decay of the OGE effect is strongly influenced by the magnitude of the ballast resistor in the DC lamp operating circuit; larger resistance results in faster decay.

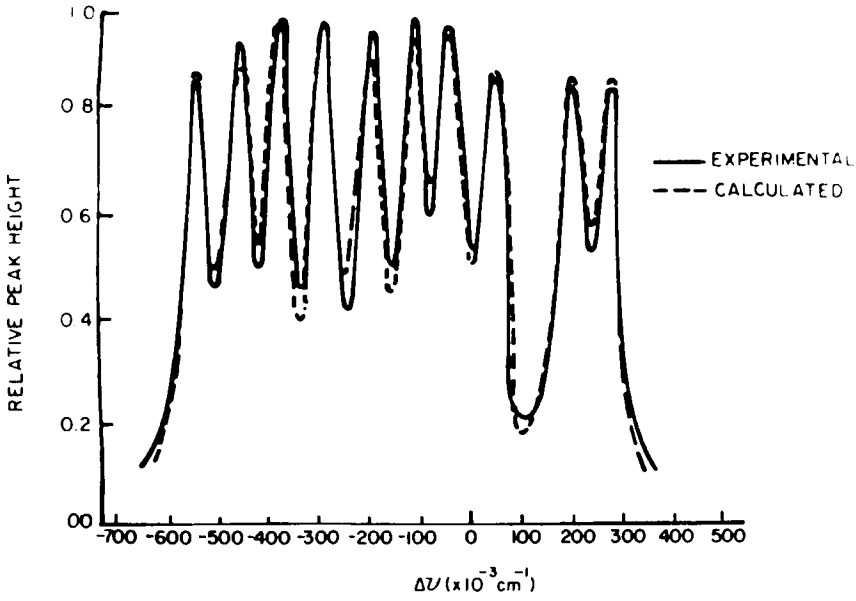


FIGURE 25. Observed and calculated spectral profile of 254 nm resonance line emitted from low-pressure Hg-A discharge; center wavelength of Hg-198 component is frequency zero (reprinted from Lagushenko, Grossman, and Maya, 1985, with the permission of the author and of Chemical Physics Letters).

5. *High-resolution Fabry-Perot Spectroscopy*

Because of the multi-isotopic composition of mercury, the 253.7 nm resonance line consists of a number of components separated from each other by amounts comparable to the line widths, themselves a composite of Doppler and collisional broadening. Collisional redistribution between absorption-emission processes and resonant energy transfer between isotopes

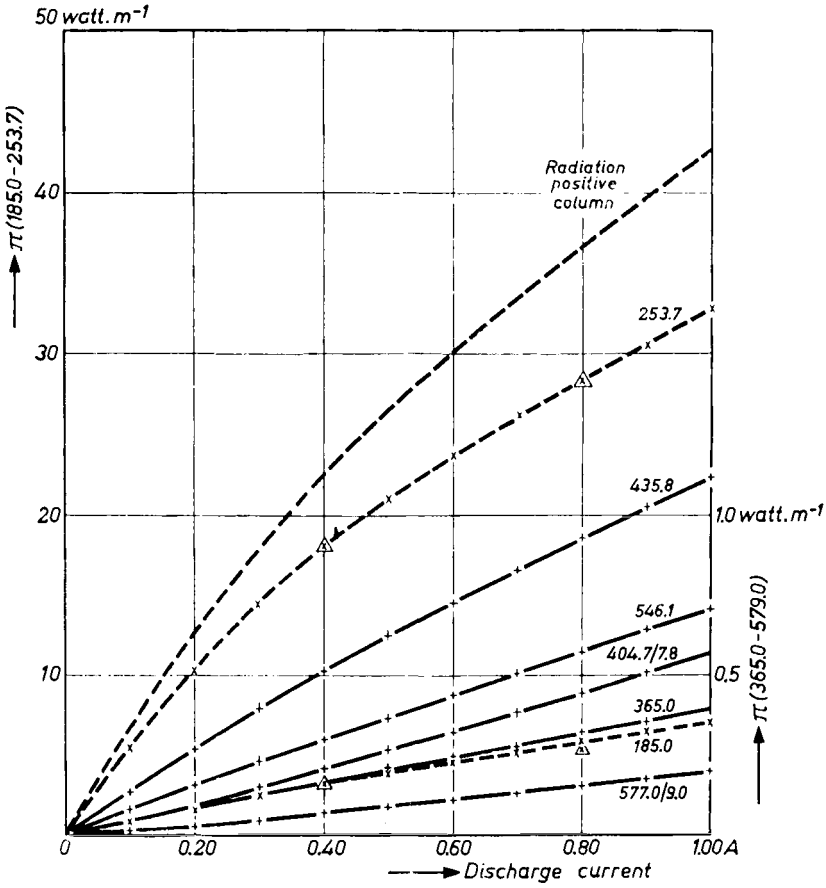


FIGURE 26a. Absolute spectral power radiated in several important spectral lines of mercury from low-pressure-mercury/rare-gas discharge; condensed mercury temperature 42°C, argon pressure 3 torr, 37 mm tube ID. Triangles are values measured by Barnes (reprinted from Koedam, Kruithof, and Riemens, 1963, with the permission of the author and of Physica).

strongly affect the imprisonment time. *Ab initio* calculation of the dependence of imprisonment time on isotopic composition requires knowledge of the cross sections for these processes. The same parameters also affect the composite spectral line shape of the radiation emergent from the discharge tube wall. Thus a comparison of the calculated spectral line shapes with experimentally-measured values affords an independent estimate of the cross sections.

Lagushenko, Grossman, and Maya (1985) have measured the emission line shape with a high-resolution Fabry-Perot interferometer and compared

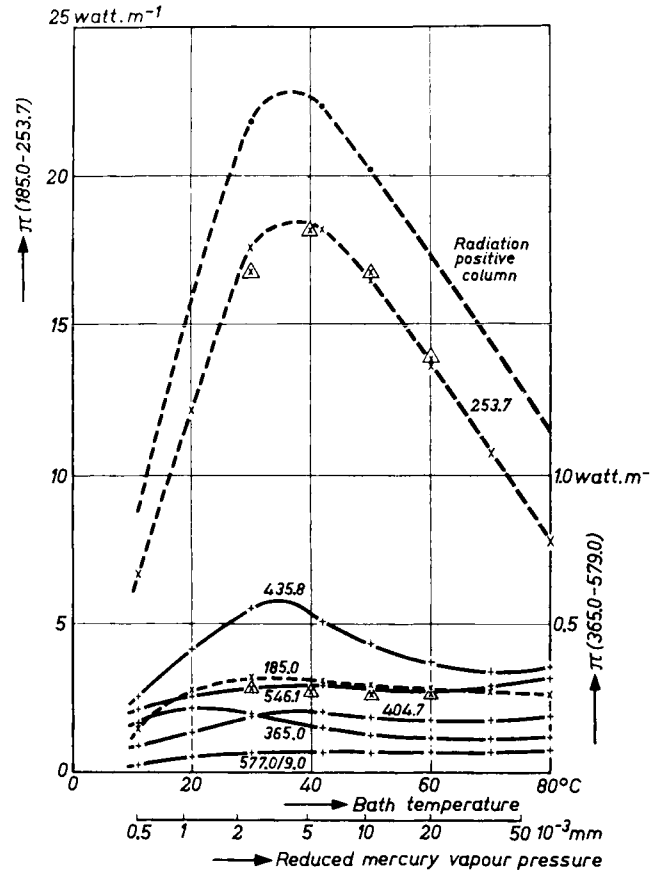


FIGURE 26b. Same as Figure 26a, except discharge current fixed at 400 ma, condensed mercury temperature variable.

it with calculated values, as shown in Figure 25. The minima between major peaks are primarily determined by the overlaps, i.e., the widths of the individual components; thus they are most sensitive to the value of the collisional broadening cross section. The envelope of the amplitudes of the major peaks is primarily sensitive to interisotopic transfer, i.e., to the cross section for resonant energy transfer. This is found to be about 10^{-13} cm^2, 1000 square Å.

6. *Absolute Measurements of Spectral Power*

Absolute measurements of spectral power as a function of discharge conditions are usually required for comparison with discharge model calculations in order to validate the discharge model. Koedam, Kruithof, and Riemens (1963) have measured the absolute spectral power radiated in the principal mercury lines per meter of arc length for a low-pressure mercury/rare-gas discharge for a 38 mm ID discharge tube as a function of mercury vapor pressure at 0.4 ampere discharge current and as a function of discharge current at 0.6 millitorr Hg pressure.

This measurement requires extremely painstaking calibration of absolute spectrometer sensitivity as a function of wavelength for the determination of absolute spectral radiance. Furthermore, numerical integration over a carefully measured angular distribution of the spectral radiance is required for each emission line for the determination of total spectral power. None of the spectral lines exhibit a Lambertian dependence, and all the lines have different angular distributions. Figure 26 shows typical results obtained by Koedam et al.

D. Spectroscopic Diagnostics of HID Lamps

Because of the high pressures, high temperatures, and short mean free paths, probes have been unsuitable diagnostic tools in plasmas in HID lamps, except for measuring axial field strengths in uniform columns from the differences in floating potentials of two probes. The principal diagnostic tools for these lamps have employed radiation, either in emission or absorption. The diagnostic situations span the entire range between cases where the line integral of optical absorption coefficient over the optical path length is very much less than unity (optically thin) to those where it is very much larger than unity (optically thick). The preferred diagnostic is usually optically thin spectroscopy, because it gives information which is more easily analyzed to deduce plasma parameters. However, in many cases,

suitable transitions are not available; therefore, a number of techniques have been developed for analysis of spectroscopic information in the optically thick case as well. Not surprisingly, these draw heavily on the mathematics of radiation transport originally developed in the astrophysics arena. Many of the techniques are closely related to those in other high-pressure arc diagnostic situations; see the article in this volume by Pierre Fauchais. See also the already cited review article by van den Hoek (1983).

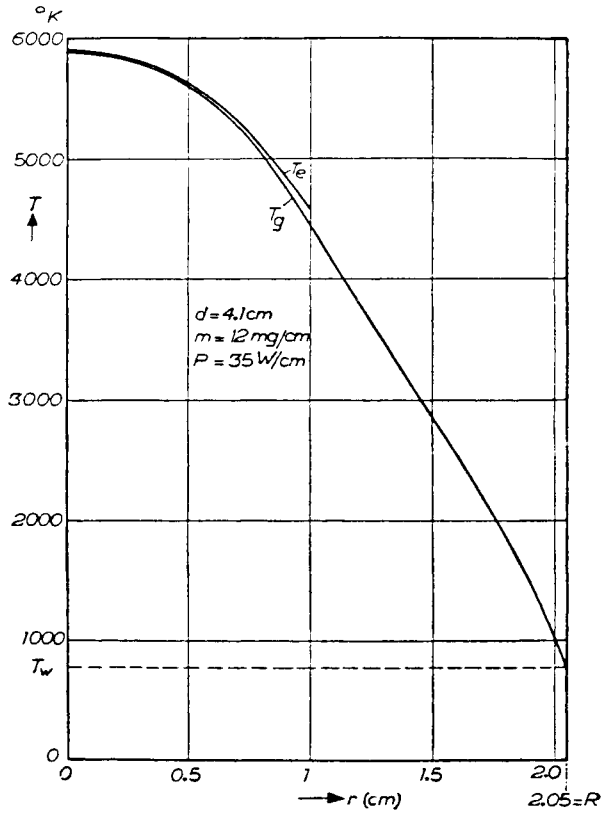
1. Optically-Thin Emission Spectroscopy

This very powerful technique involves a measurement of the local emission coefficient for the radiation in question and analysis of that local emission coefficient to determine the concentration of upper states for the transition. For atomic line emission, of course, the local emission coefficient is equal to the concentration of upper state atoms times the transition probability times the photon energy. For a discharge in LTE, the upper state concentration is simply related to the ground-state concentration by a Boltzmann factor. Thus, if the ground-state concentration is known (from knowledge of partial pressure of the radiating species, for example), the local temperature can be deduced from the result.

A complication to the experimental determination of local emission coefficient is that the spectroscopic apparatus integrates the emission along a line-of-sight through the plasma. This integration may be deconvoluted for axially-symmetric plasmas by means of the Abel procedure (see Lochte-Holtgreven, 1969). Measurements are made of the integrated intensities along a series of chords through the plasma, at equally-spaced increments of perpendicular distance of chord from the arc axis. Clearly, the outermost chord contains information about excited atom densities only in the outermost fringe of the plasma; each successive chord moving inward toward the axis contains the information about excited state densities in plasma sampled by previous chords plus the information about the new increment of volume at the new closest distance of approach to the arc center. Mathematical extraction of this information requires integration of derivatives of experimental data and suffers from considerable propagation of errors. It usually requires smoothing of experimental data to yield useful results.

A second experimental complication to the extraction of useful information from the optically-thin spectroscopic technique is that absolute measurements of spectral radiance and accurate knowledge of the transition probability are required.

a. Determination of Radial Temperature Profile. One of the earliest examples of the application of the optically-thin technique is the measurement of radial temperature profile of a high-pressure mercury arc reported by Elenbaas (1951) using the optically-thin 577 nm line of mercury. In such discharges, the total quantity of mercury in the vapor state is known. Thus, the local emission coefficient yields the local temperature. Elenbaas measured the relative radial dependence of excited state concentration, determining the absolute value at the axis so that calculated values of the total radiation power per unit length of the emission line equaled measured total radiation power. Figure 27 shows the results obtained by Elenbaas. A further limitation of the technique is that it can yield information about



Electron temperature and gas temperature of the discharge

FIGURE 27. Radial temperature profile in high-pressure mercury lamp (reprinted from Elenbaas, 1951, with the permission of the author and of North-Holland Publishing Co.). T_e measured from optically thin 557 nm line; T_G calculated from energy balance.

local temperature only in the luminous region of the plasma, out to relative radii of 0.5–0.6. Typically, the balance of the temperature profile to the wall is deduced from the solution of the energy balance equation from that point to the known wall temperature.

The technique has also been widely used for determination of radial temperature profile in metal halide lamps, which employ fully-vaporized mercury vapor as a dominant species. See Keeffe, Morris, and Walter (1978); Zollweg and Liebermann (1978); Karabourniotis and Damelincourt (1982); or Dakin (1986) for a few examples.

Another technique which may be used for radial temperature determination, in some cases derives from a measurement of the local emission coefficients for two optically-thin lines having upper states of different excitation energy. The ratio of the upper state concentrations being given by a Boltzmann factor, the experimental ratio can be used to determine the local temperature directly. Since only ratios are involved, absolute measurements of spectral radiances are not required. See, for example, Dakin (1986). The technique suffers from the problem that it may be difficult to find two transitions which are both optically thin and of sufficient strength over the entire temperature range for which information is needed. Moreover, a relatively large (> 2–$3\ kT/e$) difference in upper state excitation potentials is required for best accuracy.

b. Determination of Species Concentrations. When the radial temperature profile is known, for example by the application of the techniques above, the measurement of local emission coefficient for a spectral line can be worked backwards to give the concentration of particular atomic species in the plasma. This technique has been used by Rothwell and Keeffe (1980), based on earlier work by Waymouth, Keeffe, et al. (1975), to measure the concentrations of scandium, sodium, and thorium neutral and ionic species in scandium-sodium metal halide lamps. These investigators were primarily interested in a rapid diagnostic to be used to analyze rates of metal consumption of large numbers of commercial lamps in life testing. Therefore, they used a single spatially-integrated spectral radiance measurement, together with a two-temperature arc model for interpretation of data, determining concentrations averaged over the central zone of the arc.

It is, however, much more powerful than that and has been used for determining radial and axial dependences of concentrations of a large variety of species in metal halide lamps of all types. Stormberg (1980) has used this technique to determine radial concentration profiles of sodium and thallium in a metal-halide arc for comparison with theoretical models; Zollweg and Liebermann (1978) have made use of it to determine radial and axial profiles of scandium and sodium concentrations in a scandium-

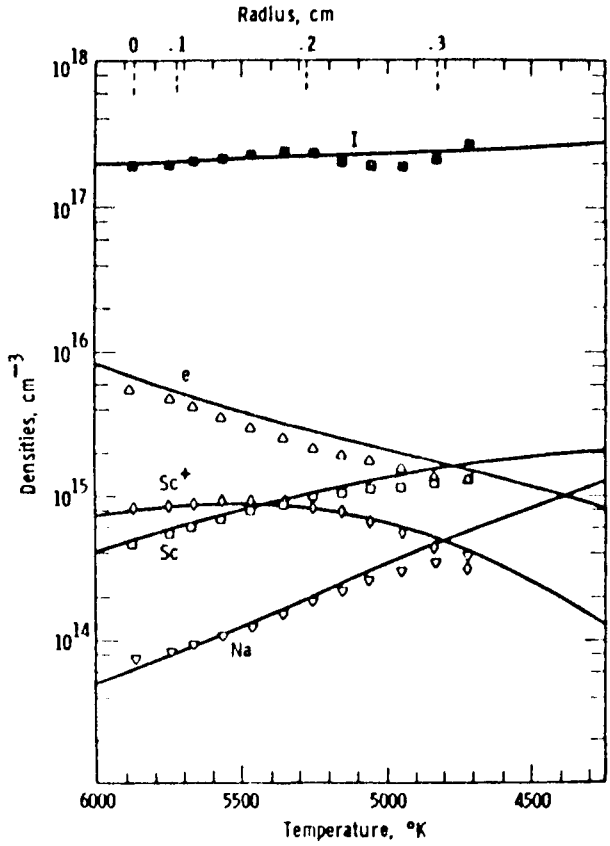


FIGURE 28. Concentrations of Sc, Sc + , Na vs. radial position in scandium-sodium metal halide lamp; electron concentration determined from Sc + /Sc ratio (reprinted from Zollweg and Liebermann, 1978, with the permission of the author and of the Electrochemical Society, Inc.).

sodium metal halide arc, to cite only two of the many examples. Figure 28 shows a sample of results from Zollweg and Liebermann.

c. Determination of Electron Density. Electron concentrations in high-pressure discharge light sources can be determined from the ratio of the intensities of selected optically-thin ionic and atomic spectral lines. Rothwell and Keeffe (1980) use the ratio of scandium neutral to scandium ionic line intensities and the ratio of thorium neutral to thorium ionic line intensities, together with the assumption of a Saha equilibrium between ion/neutral concentrations and the electron density to determine the elec-

tron densities in a scandium-sodium metal halide discharge. Good agreement is found for the scandium-determined and thorium-determined values, a few times 10^{15} per cubic centimeter averaged across the arc column. Zollweg and Liebermann (1978) used the Sc + /Sc ratio for the same purpose; see Figure 28.

The lack of any other satisfactory methods of measuring electron concentration in high-pressure discharge light sources precludes validating this measurement by comparison with others.

2. *Non-Optically-Thin Emission Spectroscopy*

In HID lamps, radiation transport in the domain of optical thickness about unity is extremely important, not only for diagnostics but also for the performance of the device. The majority of the useful radiation from the lamp is emitted from the region of the plasma having an optical depth of unity. It is the objective of the lamp designer to adjust dimensions and partial pressures of species as to provide an optical depth of about unity over the greatest possible fraction of the volume and the desired spectral range. Consequently, diagnostics applicable to optical depths of unity are extremely important in research and development of high-pressure arc light sources.

a. Radiation Transport in Non-optically-thin Plasmas. The fundamental equation describing radiation transport in an absorbing medium is the Kirchoff-type equation,

$$\frac{dI(\nu)}{dx} = S(\nu) - \alpha(\nu)I(\nu),$$

in which $I(\nu)$ is the spectral radiance in watts per unit area per unit frequency interval per unit solid angle, and $\alpha(\nu)$ is absorption coefficient. $S(\nu)$ is the so-called *source function*, which represents the rate of generation of radiation frequency (ν) in watts per unit volume per unit frequency interval per unit solid angle. For the special case of plasma in local thermal equilibrium (LTE), the source function can be represented as the product of the absorption coefficient times the spectral radiance of a blackbody at frequency (ν) and local temperature T. For this case, which is usually encountered in HID lamps,

$$\frac{dI(\nu)}{dx} = \alpha(\nu)\{I_{BB}(\nu, T) - I(\nu, x)\}.$$

A spectral radiance at frequency (ν), measured from a line-of-sight through the plasma, can therefore be represented as the integral along that

line-of-sight of the radiation transport equation and provides information about both the absorption coefficient and the temperature profile along that line of sight.

Examination of this equation shows that where local spectral radiance is less than blackbody spectral radiance, local spectral radiance will increase with increasing distance along the line of sight, with a rate that increases with absorption coefficient. If local spectral radiance exceeds blackbody spectral radiance, the derivative is negative, and its magnitude is proportional again to the absorption coefficient. Considering the special case of temperature independent of position, one can see that for path lengths times absorption coefficient (optical depths) greater than three or four, the local spectral radiance becomes equal to the blackbody spectral radiance. Consequently, in the case of varying temperature along the line-of-sight, at large optical depths, the emergent spectral radiance is approximately equal to the blackbody spectral radiance at the exit temperature; in discharge lamps, this is frequently the wall temperature.

For the optically thin case, the absorption term in the radiation transport equation is negligible, but the source function is also small. Thus, although there is little or no absorption of radiation along the line-of-sight, the spectral radiance never reaches a significant fraction of the blackbody radiance. At optical depths around unity, however, the spectral radiance of a ray traversing a plasma reaches a significant fraction of the blackbody intensity at the maximum temperature along the line-of-sight, but still has a transmittance of about $1/e$ (40%) in escaping. Thus, the emergent radiation contains significant information about the maximum temperature encountered along its traverse through the plasma.

Non-optically-thin spectroscopic diagnostics used in high-pressure discharge lamps make use of these facts either for determination of maximum temperature along a line-of-sight through a plasma, for the determination of the concentration of a given species, or both.

b. Bartels' and Related Methods for Axis Temperature Determination. Bartels (1950a, 1950b) developed the first such diagnostic technique, still in widespread use today, employing a self-reversed, nonresonant line for the determination of the maximum temperature along a line-of-sight through a plasma. A self-reversed spectral line in an LTE plasma (as illustrated in Figure 29) is characterized by an optical depth very much greater than unity at line center, such that the spectral radiance at line center is approximately equal to the blackbody radiance at all points along the ray transit of the plasma. Its observed value exiting from the plasma is near to the blackbody radiance at the temperature at which the plasma becomes optically thin. Because the absorption coefficient decreases with the wave-

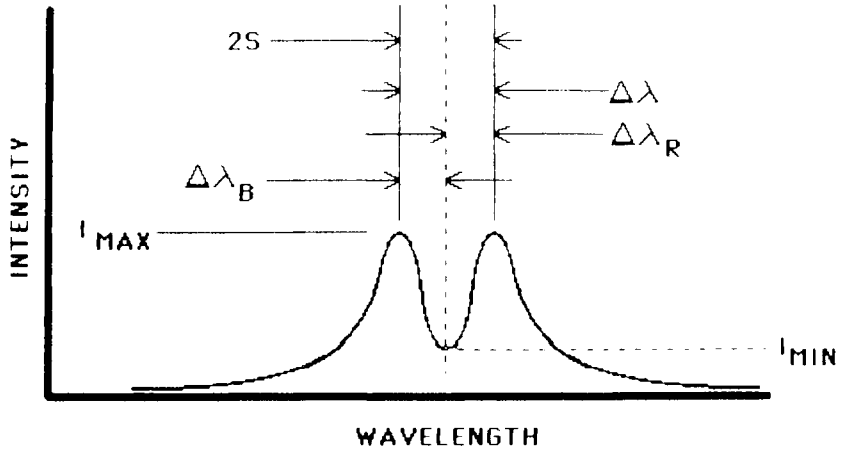


FIGURE 29. Self-reversed spectral line (schematic) together with nomenclature used by various authors.

length difference from line center, at a rate dependent on the line-broadening processes, the optical depth likewise decreases; in the vicinity of the reversal maxima, optical depth is about unity, and the spectral radiance approaches a significant fraction of the maximum value along the line-of-sight. For greater difference in wavelength from line center, absorption coefficient and optical depth decrease still further, until in the far wings, the plasma is optically thin. Consequently, a single self-reversed line spans the entire range of radiation transport conditions from optically thick to optically thin.

The requirement for nonresonant lines in Bartels' method means that the lower state of the transition pair (which contributes the absorption) is occupied only in the high-temperature portion of the line-of-sight. Consequently, the method yields information about the highest temperature region along the line-of-sight. The technique employs the experimental measurement of the maximum absolute spectral radiance in the wings of the self-reversed line; Bartels relates this measured spectral radiance to the blackbody spectral radiance at the maximum temperature as follows:

$$I(\max) = MY(\max) I_{BB}(\lambda_0, T_m).$$

M and Y are functions of excitation and ionization potentials of the radiating atoms; their product is typically 50–70%. A complete discussion of these parameters is given by deGroot and Jack (1973a), together with a discussion of the sensitivity to radial temperature profiles and limiting

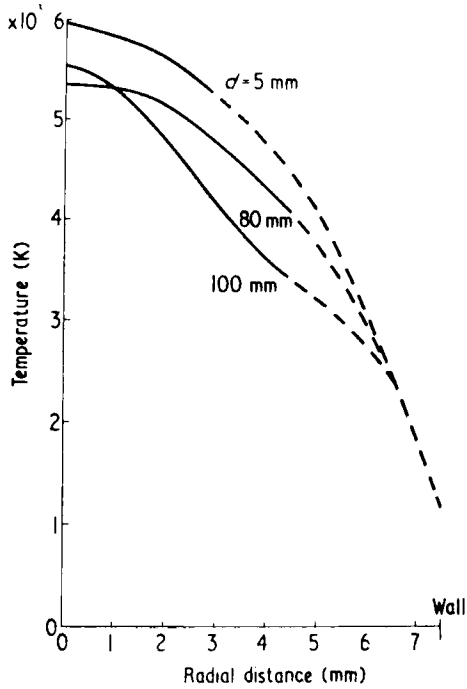


FIGURE 30. Radial temperature profile for tin-halide mercury arc, determined by Bartels' method, for various axial distances from the upper electrode of a vertically-burning discharge (reprinted from de Groot and Jack, 1973b, with the permission of the author and of the Journal of Physics D, Applied Physics).

approximations to their values that are applicable to most cases of interest in discharge lamp diagnostics.

Because of the conditions described above, the resultant maximum spectral radiance is remarkably insensitive to temperature profile as long as that profile is monotonic; thus, this is an extremely useful diagnostic for maximum temperature. Using measurements at different chordal distances from arc center permits tracing a radial temperature profile, although it is sometimes difficult to find a single nonresonant line which is self-reversed along lines-of-sight more than a limited range of chordal distances from arc center. The earliest application of this technique of discharge lamps known to this author was that of deGroot and Jack (1973b), who measured radial temperature in mercury/tin-iodide arcs. Their results are shown in Figure 30. Another early application was the use by deGroot (1974) to measure radial temperature profiles in a high-pressure sodium arc to reduced radius

of 0.7. The method has been widely used by many other investigators since that time.

Wesselink, deMooy, and van Gemert (1983) have described a modified Bartels' method employing two self-reversed lines. The ratio of the line-reversal maxima of the two lines is given by the ratio of the Bartels constants times the ratio of the blackbody radiances at the two wavelengths at the axis temperature. Thus, the experimental measurement of this ratio provides a determination of the maximum temperature along the line-of-sight. A distinct advantage of this method is that absolute calibration is not required, only relative spectral calibration. A secondary advantage is that the method is even more insensitive to radial temperature profile, since substantially the same deviations occur in the Bartel's constants for both lines and cancel out in taking the ratio. The disadvantage is that it is frequently difficult to find two lines which satisfy the required conditions. However, in many metal halide lamps, the 404 nm, 436 nm, and 546 nm visible lines of mercury are self-reversed over a reasonable range of plasma conditions and serve as diagnostic tracers. See Dakin and Gilliard (1986) for a recent example.

A number of authors have compared optically thin and Bartels' methods of temperature measurements, with good results. See, for example, Seehawer and Zauter (1973), deGroot (1974), and Dakin and Gilliard (1986). Seehawer and Zauter (1973) have also given a detailed discussion of the effect of refraction by the tube wall upon the path of a ray through a plasma and its container.

A diagnostic method related to Bartel's method, but yielding more information, is that developed by Karabourniotis and Damelincourt (1982), who derive information about the plasma from the maximum spectral radiance, the wavelength separation between maxima, and the ratio of maximum to line-center minimum of a self-reversed nonresonance spectral line. Whereas the spectral radiance at line reversal maxima contains information about maximum temperature, the spectral radiance at the reversal minimum contains information about the temperature at which the plasma becomes optically thin, and the wavelength separation between reversal maxima contains information about the concentration of the radiating species.

Letting $R = I_{max}/I_{min}$ represent the ratio of maximum to minimum intensities, and $2s$ be the wavelength separation between maxima, experimental data determining these values are obtained for chordal scans at various distances from arc axis. The slope of a $\log(R)$ vs. $\log(2s)$ plot for these data can be used to determine the "inhomogeneity parameter," n, (as defined by Cowan and Dieke, 1948), which for a given emission line

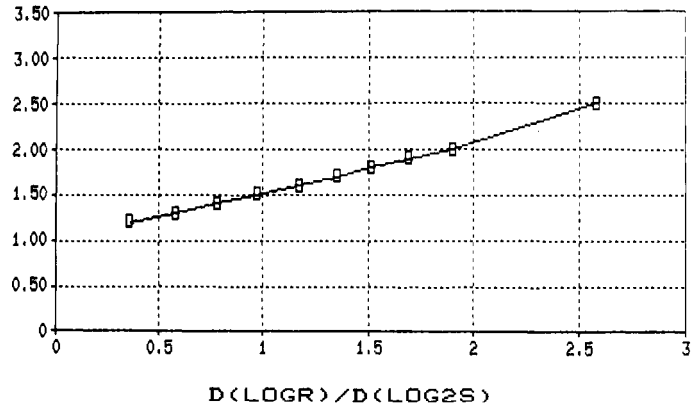


FIGURE 31. Plot of $d \log(R)/d \log(s)$ versus inhomogeneity parameter n, from Karabourniotis and Damelincourt (1982), Table I.

represents a one-parameter descriptor of the radial temperature profile. The larger the value of this parameter, the more constricted is the temperature profile. The magnitude of the ratio R for a diametral scan can then be used to determine the line-center optical depth once the inhomogeneity parameter is known. Figure 31 is a plot of data in Karabourniotis and Damelincourt (1982), Table I, while Figure 32 is a reproduction of Figure 2 from that reference. Figure 33 shows experimental data on ratio $\log(R)$ vs. $\log(2s)$ for the 546 nm line of mercury in a high-pressure mercury arc, the slope of which determines n from Figure 33.

Finally, the absolute spectral radiance at the line wing maximum can be used to determine the maximum plasma temperature along the line-of-sight, as in the Bartel's method. Knowledge of the line-center optical depth, transition probability, and line-broadening parameters for the line in question permits calculation of the partial pressure of the radiating species. Thus, from relatively simple experimental measurements can be determined the axis temperature, a radial temperature profile, and the partial pressure of the radiating species.

The technique was first used by Karabourniotis, Karras, Drakakis, and Damelincourt (1982) to determine axis temperatures in a mercury arc and by Karabourniotis and Damelincourt (1982) to determine mercury pressure in the same discharge. It has since been used by Karabourniotis and a series of coworkers for diagnostics in a variety of metal halide arcs. Somewhat surprisingly, despite its relative simplicity, no other workers have adopted it. See the bibliography under Karabourniotis for references.

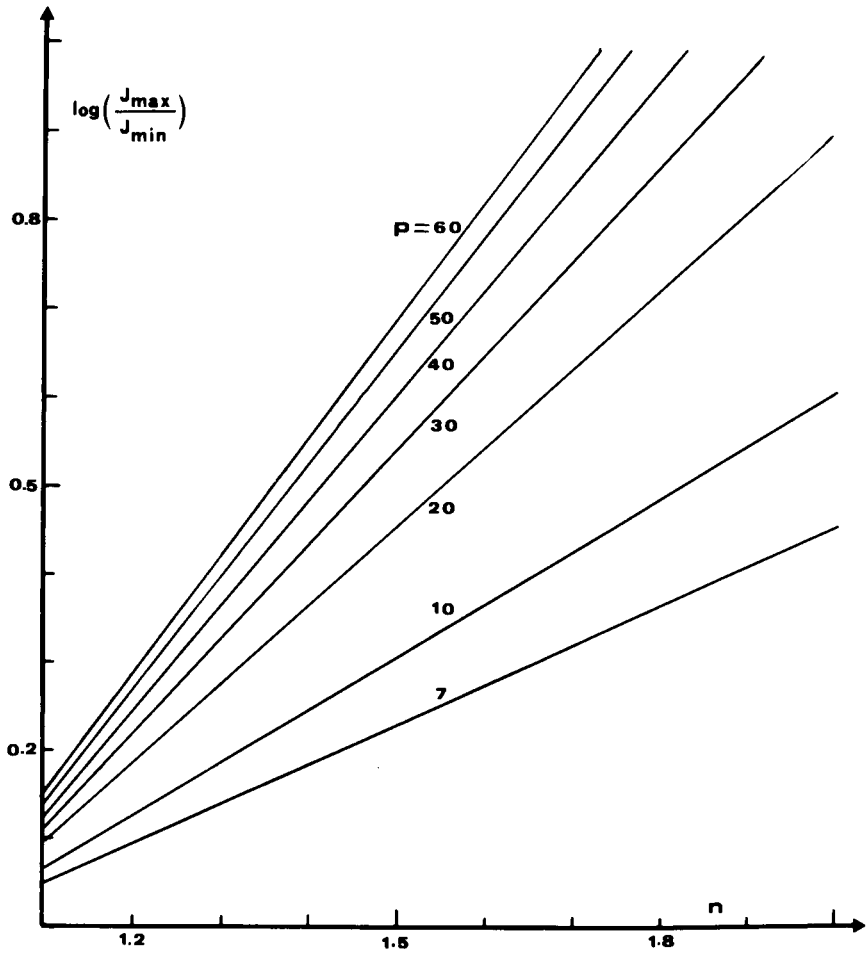


FIGURE 32. Log(R) for diametral scan versus inhomogeneity parameter n, with line-center optical depth p as parameter, reprinted from Karabourniotis and Damelincourt (1982), with the permission of the author and of the Journal of Applied Physics.

c. Diagnostics Employing Resonance Lines. Because of the technical importance of high-pressure sodium (HPS) lamps, a great deal of attention has been given to optical diagnostics employing the self-reversed resonance lines of sodium (D-lines). Ozaki (1971) first deduced empirically a relationship between the separation between reversal maxima and partial pressures of sodium. This relationship results from the fact that this line is primarily resonance broadened, with a Lorentzian line shape having a width proportional to the number density of sodium atoms. Normalization of the

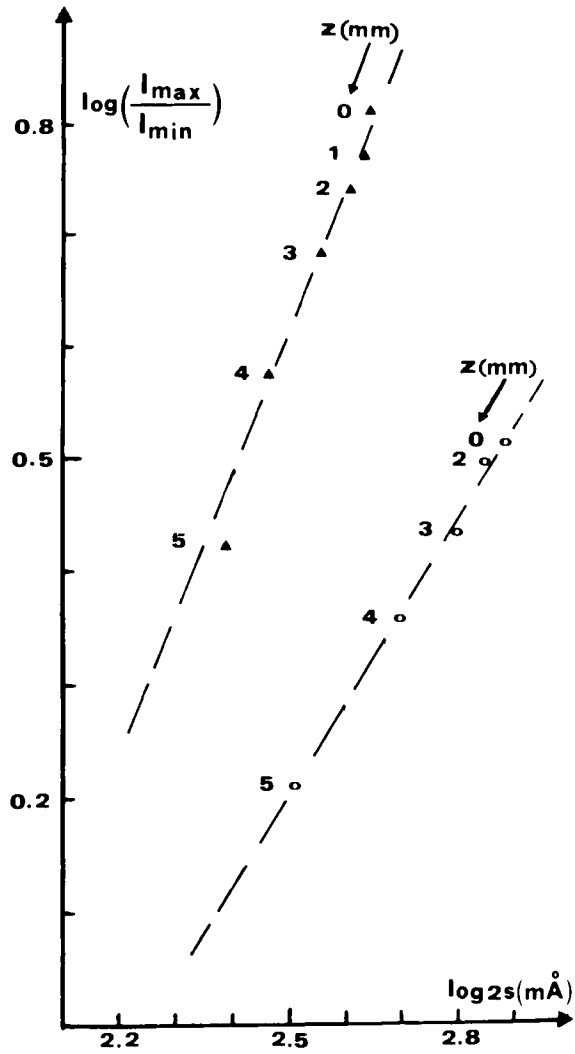


FIGURE 33. Log(R) vs. log(s) for chordal scans of Hg 546 nm line, at chordal distance z from the axis; two different discharges, with different mercury pressures; reprinted from Karabourniotis and Damelincourt (1982), with the permission of the author and of the Journal of Applied Physics.

Lorentzian line profile of the absorption coefficient gives a proportionality to the number density of the absorbers divided by the width, i.e., the normalization constant is independent of the number density of sodium atoms. Thus, in a pure resonance broadened line, the line-center absorption coefficient (and hence line center optical depth) is independent of the number density of absorbers, while the width is proportional to that number density. Consequently, the wavelength deviation from line center at which a particular absorption coefficient and optical depth (i.e., about unity) will be reached is simply proportional to the number density of absorbers, i.e., to sodium pressure. As already noted, the maxima in the wings of a self-reversed line occur at those wavelengths for which the optical depth from the arc axis is about unity.

Since the work of Ozaki, a great deal of work has gone into developing a sound fundamental basis for determination of sodium partial pressures from measurements of the spectral distribution in the sodium D-lines. A recent publication is that of Reiser and Wyner (1985), who also give adequate reference to earlier literature. The subject is also well discussed in the book by de Groot and van Vliet (1986). A central correction to Ozaki's results is that the line shape is actually not symmetrical, but has additional broadening to the red side; the red-wing width is dependent on mercury pressure as well as sodium pressure. This is the result of the admixture of both resonance broadening by sodium and van der Waals' broadening by mercury. The former is symmetrical, affecting both wings equally, while the latter affects only the red wing.

It was first shown in modelling calculations by Waymouth (1977) that sodium pressure only affected the blue-wing reversal maximum shift from line center, while red-wing shift was determined by both sodium and mercury. Consequently, measurement of the two shifts separately could be used to determine both sodium and mercury pressures. Reiser and Wyner (1985) deduce these results from first-principles calculations using published values of the resonance and van der Waals' broadening constants. (Huennekens and Gallagher, 1983 and Stormberg, 1980). Figure 34 displays the principal result of Reiser and Wyner, showing contours of constant sodium and mercury vapor densities on the plane defined by total-maxima-separation versus red-blue-maximum-shift-ratio. Similar results have been obtained by de Groot and van Vliet. See the book, *The High Pressure Sodium Lamp* by these authors (1986).

One of the problems in this diagnostic is that the line-center absorption coefficient is of the order of 10^5, so that the reversal maxima may occur between 50 and 200 line widths from line center. For this case, then, the true intermolecular potentials between absorber and perturber must be

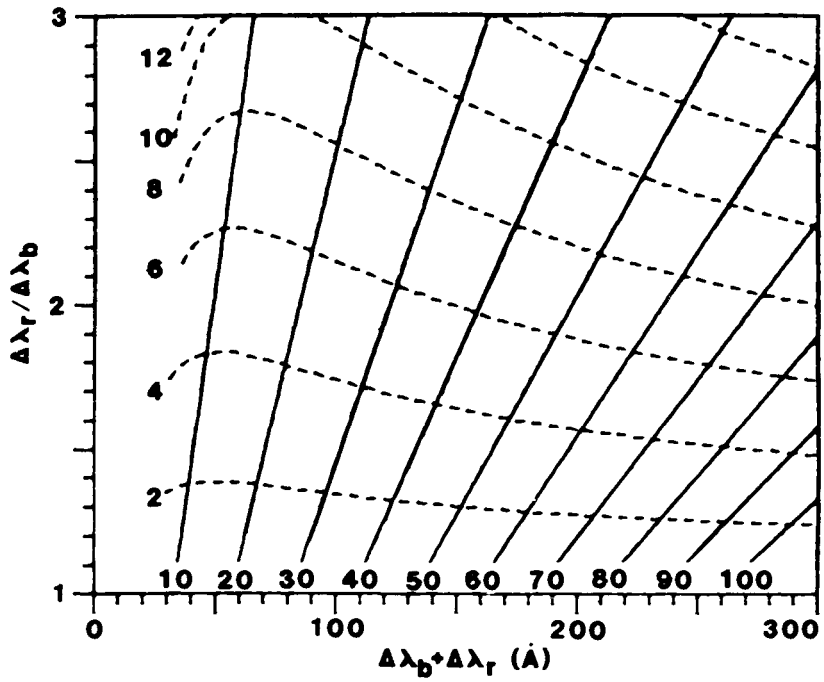


FIGURE 34. Contours of constant mercury vapor: dashed lines (mg/cm^3), and sodium vapor densities: solid lines (mg/cm^3), on $\Delta\lambda_R/\Delta\lambda_B$ vs. $\Delta\lambda_R + \Delta\lambda_B$ plane; reprinted from Reiser and Wyner (1985), with the permission of the author and of the Journal of Applied Physics.

considered, and the use of broadening constants may be no longer applicable, since the potential variation is no longer linear with any power of the internuclear separation. For the Na-Hg case, the influence of NaHg excimers is strong, and in fact shows up as an emission band in the far red wing of the broadened line, at about 680 nm; see, for example, Zollweg (1982) and Woerdman, Schlejen, et al. (1985). Woerdman and de Groot (1981 and 1982a) have also studied the Na-Na and Na-Xe contributions to the spectrum.

In addition, unlike the case of nonresonance self-reversed lines, no useful information may be extracted from the line reversal minimum. As shown by Waszink (1973), the plasma very close to the wall is not in LTE with regard to the excited state concentrations because of the strong absorption of resonance radiation emitted from the inner portions of the arc. Thus, the observed spectral radiance in the reversal minimum for an operating discharge is several orders of magnitude higher than the blackbody value at

the wall temperature. However, in the afterglow plasma after lamp shutoff, the gas temperature relaxes to wall temperature in a few milliseconds, the spectral radiance at D-line center becomes equal to the blackbody value, and may be used to determine wall temperature, as first shown by deGroot (1972). Wyner (1979) has extended the results of deGroot to measuring the rate of wall temperature decay, which can be interpreted to yield the radiant power emittance of the arc tube (for radiatively cooled arc tube walls), a very useful datum for energy balance measurements.

d. Comparison of Calculated and Observed Spectral Line Profiles. A final application of self-reversed line diagnostics is the comparison between calculated and observed line shapes. Provided radial temperature profile and axial concentrations are known, Stormberg (1980) showed that the calculated line shapes for the Tl 535 nm line and the 589 nm sodium line exhibited sufficient sensitivity to radial concentration profiles such that the agreement between theory and experiment could be used to confirm the correctness of those profiles calculated from an ambipolar diffusion model. Other authors have used agreement between calculated and measured spectral line shapes to validate model calculations as well (see the bibliography).

For a system in which partial pressure of the radiating species is accurately known and for which accurate values of all broadening constants are available, the comparison between calculated and observed spectral line profiles has been used to deduce the radial temperature profile. Stormberg and Schafer (1985) have applied this technique using the 546 nm and 436 nm lines to the determination of the radial temperature profile of a high-pressure mercury arc and find good agreement with that determined from measurements of the intensity of optically-thin 577 nm radiation. Their technique uses an expansion of radial temperature profile in a series of Bessel functions of order zero and determines the values of these coefficients by a routine minimizing the rms error between calculated and observed spectral line profile for a diametral ray. According to these authors, no more than six terms of the Bessel function series are required.

However, the spectral radiance at reversal maxima is determined primarily by axis temperature and is very insensitive to radial temperature profile; wavelength separation between reversal maxima is primarily determined by species concentration. The radial temperature profile, therefore, has little influence on the dominant features of the spectral line shape and will reveal itself only in the secondary details. Very accurate spectral measurements, a closely-spaced calculational network of radial grid points as well as spectral grid points, and many iterations of the calculations are

required for adequate accuracy of resulting calculated radial temperature profile. Consequently, if other techniques are available, this method would not be the preferred one.

E. Laser Diagnostics in High-Pressure Discharge Lamps

1. Optical Absorption

Optical absorption has been used by several authors as a diagnostic. Seehawer and Zauter (1973) have used absorption at 633 nm (from a He-Ne laser) to determine concentration profile of molecular iodine in an indium-iodide metal halide arc. More recently, Allen, Lagushenko, Maya, and Keeffe (1987) have used optical absorption in the blue wing of the sodium D-line in a scandium-sodium metal halide lamp to determine the radial concentration profile of neutral sodium. The technique is particularly advantageous for measurements near the tube wall, where emission spectroscopy yields no useful information because of the low plasma temperature. Figure 35 shows Allen's principal result. Advantage was taken in this work of the frequency stability and tunability of a dye laser; non-sodium-

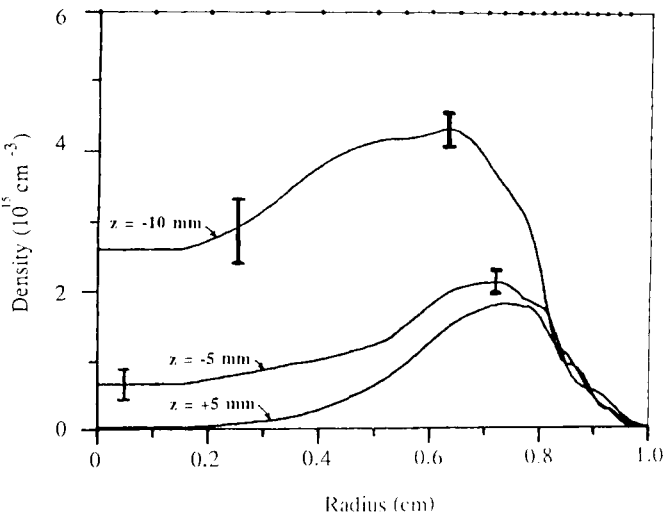


FIGURE 35. Number densities of ground-state sodium atoms vs. radial position, at three different axial positions of a vertically-operated scandium-sodium metal-halide lamp (reprinted from Allen, Lagushenko, et al. 1987, with the permission of the author and of the Journal of the Illuminating Engineering Society).

dependent absorption and scattering could be corrected for by detuning the laser far enough from resonance that absorption by sodium atoms could be neglected. Abel inversion routines developed by Lagushenko were successfully used for reducing experimental chordal absorption data to radial density profiles despite the axial minima in neutral sodium concentrations.

Weber (1986) has used absorption at 257.2 nm from a frequency-doubled argon-ion laser to measure mercury ground-state number density in a mercury-xenon LTE arc. An empirical representation of the broadening constants of the Hg 253.7 nm resonance line by Hg and Xe atoms was used, with an independent calibration experiment to determine the values of the constants.

Data was Abel-inverted to determine radial dependence of number densities, and measurements were made at 12 axial positions.

2. *Laser-Raman Scattering and Rayleigh Scattering*

Vriens (1973, 1974, 1977), Vriens and Adriaanz (1974, 1975, 1976), and Adriaanz and Vriens (1976) have extensively studied laser scattering techniques as diagnostics for high-pressure discharge lamps. Laser-Raman scattering experiments can yield information about ground-state populations of a variety of species and have the distinct advantage of providing spatially-resolved information without the necessity for Abel inversion of data from successive chordal scans. However, because of the extremely small cross sections for the Raman scattering process, the scattered light is very weak, and serious experimental problems in eliminating background radiation at the detected wavelength must be solved. The same is even more true of near-resonant Rayleigh scattering, in which the scattered wavelength is identical with the probing laser wavelength, and the additional problem of avoiding scattered laser light from other sources, such as enclosure walls or the like, is added to the already difficult signal-to-noise problems. To date, only workers at the Philips research laboratory in Eindhoven have succeeded in solving these problems and applying these techniques to high-pressure arcs in vapors of interest to light source technology. Figure 36 is an example of the type of data which may be obtained, however, showing indium ground state radial concentration profiles obtained using Laser-Raman scattering by Zaengel and Groiss (1980).

3. *Optogalvanic Effects*

The optogalvanic effect (OGE) in high-pressure discharge light sources generally does not yield information about elementary ionization and

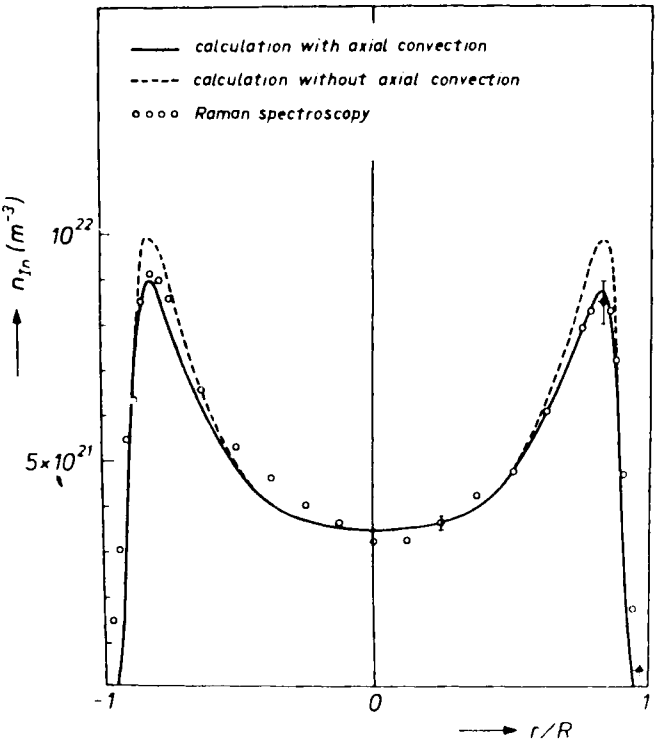


FIGURE 36. Radial concentration profile of indium ground-state atoms in a vertically-burned Na-Tl-In-I-Hg arc; points obtained by laser-Raman spectroscopy, curves are calculated from theoretical models (from Zaengel and Groiss, 1980) Reprinted from van den Hoek (1983) with the permission of the author and of Philips Journal of Research.

excitation processes because of the strong coupling between excited states and the electron gas. Absorption of radiation is manifested by an increase in electron temperature. However, OGE has been employed to study thermal relaxation effects in high-pressure mercury and sodium discharges by van den Hoek and Visser (1980). The perturation of a particular transition as a result of absorption of incident radiation in an LTE discharge results in a temporary increase in electron temperature to which the upper state is strongly coupled. The rate of collisional relaxation of electron temperature to its steady-state value can be determined from the dependence of the OGE effect on chopping frequency. Similarly, Jongerius, Ras, and Vrehen (1984) have used the optogalvanic effect to study the frequency spectrum of acoustic resonances in high-pressure sodium lamps. In the vicinity of an acoustic resonance, the OGE effect due to a tempera-

ture perturbation caused by optical absorption is magnified. Not only can the resonance frequency and width be determined, but a special acoustical mode may be selectively excited by choice of the axial position of the laser beam.

F. Continuum Measurements

In the case of other high pressure discharges, measurements of continuous-spectrum radiation yields useful information about electron densities. This technique has not been useful for this purpose in discharges used in light sources because of the presence of additional continua whose origins are not fully understood (but are probably molecular) which generally exceed electron continuum by factors of three to ten or more. This was first noted for high-pressure mercury discharges by Elenbaas (1951) and has been observed in most other high-pressure light source discharges since then.

G. Stark Shift

Van den Hoek and Visser (1981) have observed Stark shifts of the 616-nm, 569-nm, and 498-nm lines of sodium in laser-induced fluorescence of a high-pressure sodium lamp. Application of Griem's formulae (Griem, 1974) gives electron densities of order 6×10^{15} cm^{-3}, with results from the three lines agreeing within $\pm 20\%$. Observed line widths are equal to or greater than the observed Stark shifts.

III. Summary and Conclusions

Nearly all plasma diagnostic techniques available have been employed in discharges of interest to the science and technology of light sources. In many cases, several different techniques have been applied to the determination of the same quantities, with quite satisfactory agreement. Similarly, where diagnostic techniques have been used to validate model calculations, satisfying agreements have been reported. Diagnostic techniques applicable to the determination of electron concentrations in high-pressure discharge light sources are limited, and measurements of this quantity must be conceded to be in an unsatisfactory state.

IV. Acknowledgements

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References

(Includes additional citations not specifically referenced in text)

Adriaansz, M. (1973). *J. Phys. E.* (Sci Inst) **6**, 743–745.
Adriaansz, M. and Vriens, L. (1976). *Philips Res. Rpts.* **31**, 193ff.
Allen, G. R., Lagushenko, R., Maya, J., and Keeffe, W. M. (1987). *J. IES* **16**, 13ff.
Barnes, B. T. (1962). *J. Appl. Phys.* **33**, 3319–3322.
Barnes, B. T. (1966). *J. Appl. Phys.* **37**, 2679–2683.
Bartels, H. (1950a). *Zeits Phys.* **127**, 243–73.
Bartels, H. (1950b). *Zeits Phys.* **128**, 546–73.
Bigio, L. (1988). *J. Appl. Phys.* **63**, 5259–69.
Boyd, R. L. and Twiddy, N. D. (1959). *Proc. Roy. Soc.* (*Lond*) **250**, 53.
Cayless, M. A. (1960). *Brit J. Appl. Phys.* **11**, 2ff.
Chen, F. F. (1965). Chapter 4 in *Plasma Diagnostic Techniques* (R. H. Huddlestone and S. L. Leonard, eds.). Academic Press, New York.
Chung, P. M., Talbot, L., and Touryan, K. J. (1975). *Electric Probes in Stationary and Flowing Plasmas: Theory and Application*. Springer-Verlag, New York.
Cornelissen, H. J. (1986). Laser Spectroscopy in Low-Pressure Sodium-Neon Discharges. Thesis, Tech. Hogeschool Eindhoven, Netherlands.
Cornelissen, H. J. and Burgmans, A. L. (1982). *Optics Comm.* **41**, 187–190.
Cornelissen, H. J. and Merks-Eppingbroek, H. J. (1986). *J. Appl. Phys.* **59**, 2324–2331.
Cowan, R. D. and Dieke, G. H. (1948). *Rev. Mod. Phys.* **20**, 418ff.
Dakin, J. T. and Gilliard R. P. (1986). *J. Appl. Phys.* **60**, 1281–1288.
Davydov, B. and Zmanovskaya, L. (1936). *Tech. Phys. USSR* **3**, 715ff.
de Groot, J. J. (1972). *IEE Conf. Publ.* **90**, 124–126.
de Groot, J. J. (1974). Investigation of The High Pressure Sodium and of Mercury Tin-Iodide Arcs. Thesis, Univ Utrecht, Netherlands.
de Groot, J. J. and Jack, A. G. (1973a). *Journ. Quant. Spectr. and Rad. Transfer* **13**, 616–26.
de Groot, J. J. and Jack, A. G. (1973b). *J. Phys. D.* (*Appl. Phys.*) **6**, 1477–85.
de Groot, J. J. and van Vliet, J. A. (1986). *The High Pressure Sodium Lamp*. Kluwer Technical Books, Netherlands.
Easley, M. A. (1951). *J. Appl. Phys.* **22**, 590–593.
Elenbaas, W. (1951). *The High Pressure Mercury Vapour Discharge*. North Holland, Amsterdam.

Elenbaas, W. (1972). *Light Sources*. Macmillan Press, London.
Fromm, D. C., Seehawer, J., and Wagner, W. J. (1977). *Proceedings of the Symposium on High Temp. Metal Halide Chemistry* (D. Hildebrand and D. D. Cubiciotti, eds. *Proc. ECS 78-1*, 61ff.
Fromm, D. C., Seehawer, J., and Wagner, W. J. (1979). *Lighting Res. & Tech.* **11**, 1ff.
Funk, W. and Kloss, H. G. (1977). *Proc. Int'l. Conf. on Ioniz. Gases* (Berlin) 149ff.
Funk, W. and Kloss, H. G. (1973). *Beitrage Plasma Phys.* **13**, 101ff.
Godyak, V. (1985). Pulse Measurement of Electron Energy Distribution in a Noisy Plasma. Paper CB-16, 38th Annual Gaseous Electronics Conference, Monterey, Calif., Oct. 15–18.
Godyak, V. and Lagushenko, R. I. (1986). Electron Energy Distribution in Noble Gas Mercury Discharge. Paper #62, 4th International Symp. on Sci. and Technol. of Light Sources, Universitat Karlsruhe, Federal Republic of Germany.
Godyak, V., Lagushenko, R. I., Maya, J., and Pai, R., (1986). A Novel Low Pressure Discharge with Partitions Containing an Orifice. Paper No. CB-4, 39th Annual Gaseous Electronics Conference, Madison, Wisconsin, Oct 7–10, 1986.
Griem, H. R. (1974). *Spectral Line Broadening by Plasmas*, Academic Press, NY.
Gulidov, S. S., Kagan, Y. M., Kolokolov, N. B., and Milenin, V. M. (1969). *Tekh. Fiz.* **39**, 1321–1322 (translation: *Soviet Phys.-Tech. Phys.* **14**, 993–994 (1970)).
Hall, R., Preston, B., and Stojko, S. (1978). *IEE Colloquium on High Intensity Discharge Lamps*, London.
Huennekens, J. and Gallagher, A. (1983). *Phys. Rev.* **A27**, 1851.
Jen, Teh-Sen, Hoyaux, M. F., and Frost, L. S. (1969). *J. Quant. Spectr. & Rad. Transfer* **9**, 487ff.
Jongerius, M. J., Ras, A. J., and Vrehen, Q. H. (1984). *J. Appl. Phys.* **55**, 2685–2692.
Kalyazim, Y. F., Milenin, V. M., and Timofeev, N. A. (1981). *Zh. Tekh. Fiz.* **51**, 1607–1611 (translation: *Soviet Phys.-Tech. Phys.* **26**, 922–923 (1981).
Karabourniotis, D., Karras, C., et al. (1982). *J. Appl. Phys.* **53**, 7259ff.
Karabourniotis, D. and Damelincourt, J. J. (1982). *J. Appl. Phys.* **53**, 2965ff.
Karabourniotis, D., Couris, S., and Karras, C. (1985). *J. Appl. Phys.* **58**, 2786ff.
Karabourniotis, D., Couris, S., Damelincourt, J. J., and Aubes, M. (1986). *IEEE Transactions on Plasma Science, Special Issue on Arcs* (Aug, 1986).
Kazui, T., Matsuno, H., and Yamane, M. (1977). *J. Light & Vis. Env.* **1**, 10ff.
Keeffe, W. M. (1980). *IEE Proc.* **127**, 181ff.
Keeffe, W. M., Morris, J. C., and Walter, W. (1978). *Journ. IES* **7**, 249ff.
Kenty, C. (1950). *J. Appl. Phys.* **21**, 1309ff.
Kenty, C., Easley, M. A., and Barnes, B. T. (1951). *J. Appl. Phys.* **22**, 1006ff.
Koedam, M. and Kruithof, A. A. (1962). *Physica* **28**, 80–100.
Koedam, M., Kruithof, A. A., and Riemens, J. (1963). *Physica* **29**, 565ff.
Lagushenko, R., Grossman, M. W., and Maya, J. (1985). *Chem. Phys. Lett.* **120**, 21–23.
Lagushenko, R. I. and Maya, J. (1984). *Journ. IES.* **14**, 306–314.
Ligthart, F. A. and Keijser, R. A. (1980). *J. Appl. Phys.* **51**, 5295ff.
Little, R. G. and Waymouth, J. F. (1966). *Phys. Fl.* **9**, 801–808.
Lochte-Holtgreven, W. (ed.) (1968). *Plasma Diagnostics*. North-Holland, Amsterdam.
Milosevic, S., and Pichler, G. (1985). *Z. Phys. D.-At. Mol. & Clust.* **1**, 223–229.
Mitchell, A. C. and Zemansky, M. W. (1934). Cambridge University Press, London (reprinted 1961).
Morgan, W. L. and Vriens, L. (1980). *J. Appl. Phys.* **51**, 5300ff.
Moskowitz, P. (1987). *Appl. Phys. Lett.* **50**, 891ff.
Outred, M. (1978). *J. Phys. Chem. Ref. Data* **7**, 1ff.
Ozaki, N. (1971). *J. Appl. Phys.* **42**, 3171–3175.

Partlow, W. D., Johnson, C. T., McRoberts, M., and Feldman, R. B. (1981). *Optical Eng.* **20**, 267ff.
Pilz, W. and Seehawer, J. (1975). *Proc. 12th Int'l. Conf. on Phenom. in Ion. Gases*, Eindhoven, Part I, p. 146.
Pilz, W. and Popp, H.-P. (1973). *Proc. Int'l. Conf. Ion. Gases* (Prague) 409ff.
Polman, J., van der Werf, J. E., and Drop, P. C. (1972). *J. Phys. D: Appl. Phys.* **5**, 266–279.
Polman, J. and Drop, P. C. (1972). *J. Appl. Phys.* **41**, 1577–1580.
Post, H. A. (1985). Excitation and Radiative Decay of the 184.9 nm Hg Resonance Line in Low-Pressure Mercury Noble-Gas Discharges. Thesis, Tech. Hogeschool Eindhoven, Netherlands.
Post, H. A. (1986). *Phys. Rev. A.* **33**, 2003–2016.
Post, H. A., van de Weijer, P., and Cremers, R. M. (1986). *Phys. Rev.* **A33**, 2017–2024.
Reiser, P. A. and Wyner, E. F. (1985). *J. Appl. Phys.* **57**, 1623–1631.
Reiser, P. A. and Wyner, E. F. (1986). Estimation of the Broadening Constants for the 3S-3P Sodium Resonance Line in a Sodium-Mercury Arc. Paper #JB-6, 39th Annual Gaseous Electronics Conference, Madison, Wisconsin, Oct 7–10, 1986.
Rothwell, H. L. and Keeffe, W. M. (1980). *Journ. IES* **10**, 40ff.
Rozhdestvenskii, D. S. (1912). *Ann. Phys.* **39**, 307ff.
Seehawer, J. and Zauter, M. (1973). *Technische Wissenshaftliche Abhandlung der Osram-Gesellschaft* **11**, 147–162.
Stormberg, H.-P. (1980). *J. Appl. Phys.* **51**, 1963ff.
Stormberg, H.-P. (1981). *J. Appl. Phys.* **52**, 3233ff.
Stormberg, H.-P. and Schafer, R. (1985). *J. Quant. Spectrosc. Rad. Transfer* **33**, 27–33.
Swift, J. D. and Schwar, M. J. (1969). *Electrical Probes for Plasma Diagnostics*. Iliffe Books, London.
Uvarov, F. A. and Fabrikant, V. A. (1965). *Opt. Spectry.* **18**, 541–546.
van de Weijer, P. and Cremers, R. M. (1982). *J. Appl. Phys.* **53**, 1401–1408.
van de Weijer, P. and Cremers, R. M. (1985a). *Optics Comm.* **53**, 109–112.
van de Weijer, P., and Cremers, R. M. (1985b). *J. Appl. Phys.* **57**, 672–677.
van den Hoek, W. J. (1983). *Philips J. Res.* **38**, 188–213.
van den Hoek, W. J. and Visser, J. A. (1979). *Appl. Phys. Lett.* **34**, 357ff.
van den Hoek, W. J. and Visser, J. A. (1980a). *J. Appl. Phys.* **51**, 174ff.
van den Hoek, W. J. and Visser, J. A. (1980b). *J. Appl. Phys.* **51**, 5292ff.
van den Hoek, W. J. and Visser, J. A. (1981). *J. Phys. D.: Appl. Phys.* **14**, 1613ff.
van den Hoek, W. J., van den Nieuwvenhuizen, H. C., and Stormberg, H.-P. (1982). *Philips J. Res.* **37**, 129ff.
van Tongeren, H. and Heuvelmans, J. (1974). *J. Appl. Phys.* **45**, 3844–3850.
van Tongeren, H. (1975). *Philips Res. Rpts. Supp.* **3**, (entire volume).
van Vliet, J. A. and de Groot, J. J. (1981). *IEE Proc.* **128A**, 415ff.
Verweij, W. (1961). *Philips Res. Rpts. Sup.* **2**, 1–112.
Vriens, L. (1973). *Phys. Rev. Lett.* **30**, 585ff.
Vriens, L. (1973). *J. Appl. Phys.* **44**, 3980ff.
Vriens, L. (1974). *Opt. Comm.* **11**, 396.
Vriens, L. (1974). *J. Appl. Phys.* **45**, 4422ff.
Vriens, L. (1977). *J. Appl. Phys.* **48**, 653–661.
Vriens, L. and Adriaansz, M. (1974). *Opt. Comm.* **11**, 402ff.
Vriens, L. and Adriaansz, M. (1975). *J. Appl. Phys.* **46**, 3146ff.
Vriens, L. and Adriaansz, M. (1976). *Appl. Phys.* **11**, 253ff.
Waszink, J. H. (1973). *J. Phys. D.* **6**, 1000–1006.
Waymouth, J. F. (1959). *J. Appl. Phys.* **30**, 1401–1412.

Waymouth, J. F. (1964). *Phys. Fl.* **7**, 1843–1854.
Waymouth, J. F. (1971). *Electric Discharge Lamps*. MIT Press, Cambridge, Mass.
Waymouth, J. F. (1977). *Journ. IES.* **6**, 131–140.
Waymouth, J. F. and Bitter, F. (1956). *J. Appl. Phys.* **27**, 122–131.
Waymouth, J. F., Keeffe, W. M., et al. (1975). Paper No. 14, 1st International Symposium on Science and Technology of Light Sources, Univ. Loughborough, U.K., April 8–11.
Weber, B. (1986). Paper No. 61, 4th International Symposium on Science and Technology of Light Sources, Universitat Karlsruhe Germany, April 6–10, 1986.
Wesselink, G., de Mooy, D., and van Gemert, M. J. C. (1973). *J. Phys. D.* (*Appl. Phys.*) **6**, L27–L30.
Woerdman, J. P. and de Groot, J. J. (1981). *Chem. Phys. Lett.* **80**, 220ff.
Woerdman, J. P. and de Groot, J. J. (1982a). *J. Chem. Phys.* **76**, 5653ff.
Woerdman, J. P., de Groot, J. J. (1982b). ASC Symposium Series, No. 197.
Woerdman, J. P., Schlejen, J., et al, (1985). *J. Phys. B.*: *At. Mol. Phys.* **18**, 4205–4221.
Work, D. E. (1978). *Journ. IES* **7**, 118.
Wyner, E. F. (1979). Total Emittance Measurement of Arc Tubes in High Pressure Sodium Lamps. Paper #2, 2nd International Symposium on Science and Technology of Light Sources, Enschede, Netherlands, April 9–12, 1979.
Yamane, M., Suenaga, M., and Miyashita, T. (1978). *J. Light & Vis. Env.* **2**, 9ff.
Zaengel, T. and Groiss, E. (1980). *Bull. Am. Phys. Soc.* **25**, 99.
Zaengel, T. and Groiss, E. (1983). Laser Scattering from Metal Halide Arcs. Paper No. 37, Third International Symposium on Science and Technology of Light Sources, Univ Paul Sabatier, Toulouse, France, Apr. 18–21, 1983.
Zollweg, R. J. (1978). Proc. 10th Mat'l. Res. Symp. on Char. of High Temp. Vapors and Gases. NBS Special Publ. 561, US Department of Commerce, National Bureau of Standards.
Zollweg, R. J. (1978). *J. Appl. Phys.* **49**, 1077ff.
Zollweg, R. J. (1982). *Metal Bonding and Interactions in High Temperature Systems*. ACS Symposium Series No. 179, (Gole and Stwalley, eds.), pp. 407–420.
Zollweg, R. J. and Liebermann, R. W. (1978). *High Temperature Metal Halide Chemistry* (D. L. Hildebrand and D. D. Cubicciotti, eds.). *Proc. Symp. Electrochem. Soc.* **78-1**, pp. 52–60.

3 How Langmuir Probes Work

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I. Introduction 114
II. Interpretation of Semi-Infinite Planar Langmuir Probe Data 116
A. General Expressions 116
B. Nondrifting Maxwellian Electrons 117
C. Isotropic Monoenergetic Electrons 121
D. Drifting Maxwellians 122
E. Ion Current 125
F. Floating Potential 127
III. Effects of Finite Probe Size and Shape 130
A. Probe Sheaths 130
B. Orbital Theory 132
C. Ion Beam Effects on I-V Characteristics of Non-Semi-Infinite Planar Probes 134
IV. Effects of Ion-Neutral Collisions 138
A. General Considerations 138
B. Thin Sheaths 140
C. Thick Sheaths 140
D. Probe at Plasma Potential 141
E. Flowing Plasma 141
V. Probes in Magnetic Fields 142
VI. Secondary Electron Effects 144
VII. Emissive Probes 146
A. Simplified Theory 146
B. Floating Point in Strong Emission 149
C. Inflection Point in the Limit of Zero Emission Method 150
D. Emissive Probe in a Vacuum 152
E. Differential Emissive Probe 153
F. Secondary Emission Capacitive Probes 153
G. Self-Emissive Probes 156
VIII. Probe Perturbations 157
A. General Considerations 157
B. Probe Measurements in Sheaths 158

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C. Paddle Probes 160
D. Glow Discharge Data—Ionization 160
IX. Fluctuating Plasma Potentials 162
A. Simple Model 162
B. Sinusoidal RF 164
C. Techniques 168
X. Probe Contamination 171
XI. Langmuir Probe Circuits 172
A. Simple Circuit 172
B. Curve Tracer 173
C. Op Amp Circuit 174
D. Emissive Probe Circuit 176
E. Probe Construction 176
XII. Direct Electron Temperature Measurements 176
A. Double Probes 176
B. Admittance Probe 178
XIII. Directional Probe 180
XIV. Conclusion 181
XV. Acknowledgements 181
References 181

I. Introduction

Plasma diagnostics must serve a wide variety of roles. At one extreme, for applications in basic plasma experiments, they are needed to determine the details of the electron and ion distribution functions. In fusion experiments they are needed to determine moments of the distribution functions such as temperature. At the other extreme, in plasma processing control, they may be needed to just give an indication that a plasma processing device has the same plasma characteristics as on a previous occasion, but it may not be necessary to know the characteristics. In addition, interesting plasmas exist with densities ranging from a few particles per cm^3 to greater than $10^{14}\,cm^{-3}$. It is extraordinary that one type of diagnostic, Langmuir probes, have been used to serve the full range of roles over such a wide range of plasma densities.

Langmuir probes, in their simplest form, amount to no more than a bare wire inserted into a plasma. They are often used as plasma diagnostics because of their apparent simplicity. They are easy to construct. They measure electrical currents which depend on their bias voltage with respect to the plasma potential. Over a very wide range of situations, the details of the current-voltage (I-V) characteristics can be related to the plasma

parameters that are present in the absence of the probe. Although probes perturb their local surroundings, it is still possible to determine electron temperature T_e, electron density n_e, plasma potential V_p, and electron and ion beam energy over an extremely wide range of parameters. Probes have been successfully operated at plasma densities ranging from less than 10^0 to 10^{13} cm^{-3}, at temperatures ranging from 0.1 eV to hundreds of eV, at plasma potentials of 0.1 V to many kV and at neutral pressures of less than 10^{-6} torr to greater than 1 torr.

The interpretation of Langmuir probe characteristics is complicated by the probes' perturbation to their local surroundings because of currents collected by and emitted from the probes. The current "collected" by the probe consists of a combination of collected ion and electron currents from the plasma and secondary electron current produced by electron, ion, and photon bombardment of the probe.

There is now an extensive literature on the ideal characteristics of probes, and many excellent review articles give standard results (Chen, 1965; Cherrington, 1982; Schott, 1965; Swift and Schwan 1970). Probe literature has concentrated on ideal characteristics, even though there has been a general awareness that such theory does not apply in many cases. For example, when a magnetic field is present, probe characteristics depend on electron diffusion into a flux tube depleted by the probe. The details of the diffusion are strongly dependent on particular device characteristics, so in practice it is usually necessary to calibrate probes with other diagnostic techniques. It is not generally realized that measurements without B fields also have a large number of complications. This article attempts to present the most important results of probe theory and also to present a description of the many of the most serious problems.

There are some remarkable results in probe theory. 1) The ion saturation current in a plasma where $T_e \gg T_i$, depends on T_e and not on the ion temperature T_i. 2) Ion beams can show up as large increases in electron saturation current. 3) The floating potential of a probe (i.e., the potential at which the probe collects no net current) in a plasma with hot electrons can be close to the plasma potential because of secondary electron emission. 4) The probe draws unequal electron and ion currents when biased at the plasma potential but still can determine the plasma potential.

Some serious problems are often not recognized. Among these are: 1) Collecting probes don't work well in sheaths. 2) Collecting probes do not distinguish between kinetic energy and plasma potential. 3) Secondary electron emission can cause electron temperatures to be underestimated. 4) The energy spread of beams is not equal to the beam temperature. 5) Stationary high pressure probe theory does not usually apply. 6) Positively biased probes can produce significant ionization in glow discharge plasmas.

Even though probes are an old subject, techniques for their use are continually evolving. This article attempts to combine some well established theory, which is likely to have a considerable measure of truth, with many new developments that have not yet made their way into other reviews. The emphasis will be on how probes actually work and on the real life problems associated with their use. Unfortunately, because of the large number of results and workers in this area, it is likely that many new techniques will still be unmentioned.

II. Interpretation of Semi-Infinite Planar Langmuir Probe Data

A. General Expressions

Most of the current collected by a probe biased near the plasma potential is electron current because the ion mass is much greater than the electron mass. We can describe plasma electrons by the distribution function $f(\underline{x}, \underline{v}, t)$, which gives the number of electrons per unit volume with velocities between $\underline{v}$ and $\underline{v} + d\underline{v}$ at position $\underline{x}$ and time t. The electron density at $\underline{x}$ and t is given by

$$n_e(\underline{x}, \underline{t}) = \int f(\underline{x}, \underline{v}, t)\, dv_x\, dv_y\, dv_z . \tag{1}$$

Probes provide a way to get information about electron distribution functions because the electron current density to the probe can be written

$$j = e \int f(\underline{v})\, \underline{v} \cdot \hat{n}\, d^3v, \tag{2}$$

where $\hat{n}$ is the normal to the probe surface. Without detailed knowledge of $f(\underline{v})$ or additional assumptions, it is hard to proceed further. However, the system under study often allows some approximations or imposes some symmetry on the problem which aids in the interpretation of experimental data.

For many interesting systems, especially those employed in plasma processing, sheath dimensions (see later discussion) are small compared to the probe dimensions and it is a good approximation to assume that the probe is approximately planar. In terms of Cartesian coordinates, we can write the probe current density to a negatively biased semiplanar probe as

$$j = e \int_{v_{\min}}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(v_x, v_y, v_z)\, v_z\, dv_x\, dv_y\, dv_z , \tag{3}$$

where $v_{\min} \equiv [2e(V_p - V_B)/m_e]^{1/2}$, V_B is the probe (bias) voltage, and m_e is the electron mass. We can integrate over the two velocity components perpendicular to the probe to give

$$j = e\int_{v_{\min}}^{\infty} f_z(v_z)\, v_z\, dv_z, \tag{4}$$

where $f_z(v_z) \equiv \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(v_x, v_y, v_z)\, dv_x\, dv_y$. The probe current density has been reduced to a one-dimensional problem. This is both good and bad. It simplifies the algebra, but it also means that information about the distribution function in the perpendicular direction has been lost. This can be a serious omission if the distribution function is not isotropic. This point will be considered later.

If we express the plasma density in spherical polar coordinates

$$n = \int f(v, \theta, \phi)\, v^2 \sin\theta\, dv\, d\theta\, d\phi. \tag{5}$$

If the distribution function is *isotropic*

$$n = 4\pi\int v^2 f(v)\, dv = 4\pi\sqrt{\frac{2}{m^3}}\int f(\epsilon)\sqrt{\epsilon}\, d\epsilon, \tag{6}$$

where $\epsilon = mv^2/2$, and $f(\epsilon)$ is obtained by replacing v in $f(v)$ by $\sqrt{2\epsilon/m}$. From Eq. (6) we can identify the energy distribution function $f_E(\epsilon)$ as

$$f_E(\epsilon) = 4\pi\sqrt{\frac{2}{m^3}}\, f(\epsilon)\sqrt{\epsilon}. \tag{7}$$

The current density to a negatively biased planar probe can be written in terms of spherical polar coordinates as

$$j = e\int_{v_{\min}}^{\infty}\int_{\theta_{\min}}^{\pi/2}\int_0^{2\pi} v\cos\theta f(v, \theta, \phi)\, v^2 \sin\theta\, dv\, d\theta\, d\phi, \tag{8}$$

where $\theta \le \cos^{-1}\sqrt{v_{min}^2/v^2}$. For *isotropic* distribution functions $f(v, \theta, \phi) = f(\epsilon)$, and Eq. (8) reduces to

$$j = e\frac{2\pi}{m^2}\int_{e(V_p - V_B)}^{\infty} \epsilon f(\epsilon)\left(1 - \frac{e(V_p - V_B)}{\epsilon}\right) d\epsilon. \tag{9}$$

B. Nondrifting Maxwellian Electrons

If we restrict our discussion to a nondrifting Maxwellian electron distribution function (Chen 1984) with electron temperature T_e, electron mass m_e

and electron density n_e

$$f(\underline{x}, \underline{v}, t) = n_e\left(\frac{m_e}{2\pi T_e}\right)^{3/2} \exp\left[-\left(\frac{m_e v^2}{2T_e}\right)\right]. \tag{10}$$

The electron current per unit area to a planar probe biased at V_B can be calculated by taking the integral

$$\begin{aligned} j_e(V_B) &= e\int \mathrm{f}(\underline{x}, \underline{v}, t)\, v_z\, dv_x\, dv_y\, dv_z \\ &= en_e\left(\frac{m_e}{2\pi T_e}\right)^{3/2} \int_{v_{\min}}^{\infty} \exp\left[-\left(\frac{m_e v^2}{2T_e}\right)\right] v_z\, dv_z. \end{aligned} \tag{11}$$

Carrying out the integral and multiplying by the probe area S gives an expression for the probe current I_e

$$\begin{aligned} I_e(V_B) &= I_e^* \exp\left[\frac{-e(V_p - V_B)}{T_e}\right], && V_B \le V_p \\ I_e &= I_e^*, && V_B > V_p, \end{aligned} \tag{12}$$

where $I_e^* = Sn_{e\infty}e\sqrt{T_e/2\pi m_e}$, $n_{e\infty}$ is the plasma electron density, and S is the collecting area of the probe. I_e^* is commonly referred to as the electron saturation current. An identical result can be obtained by integrating Eq. (9).

Equation (12) is graphed in Figure 1a. The most prominent feature of this curve is the "knee" near the plasma potential. Figure 1 suggests a very simple technique for determining the plasma density and plasma electron temperature. The slope of a semilog graph of I_e vs. V_B on a semilog plot is e/T_e. Given the electron temperature, the electron saturation current I_e^* gives the density. The plasma potential V_p can be approximated by the intersection of two straight lines—one parallel to the sloping part and the other parallel to curve above the knee, as shown in Figure 1a.

For example, for a 1/4 inch diameter two-sided disc probe, the plasma density given by Eq. (12) is

$$n(\mathrm{cm}^{-3}) = \frac{5.9 \times 10^8\, I_e^*(\mathrm{mA})}{\sqrt{T_e(\mathrm{eV})}}, \tag{13}$$

where the electron saturation current I_e^* is measured in milliamps. This procedure is easy to use and can give quite reproducible results. It can give

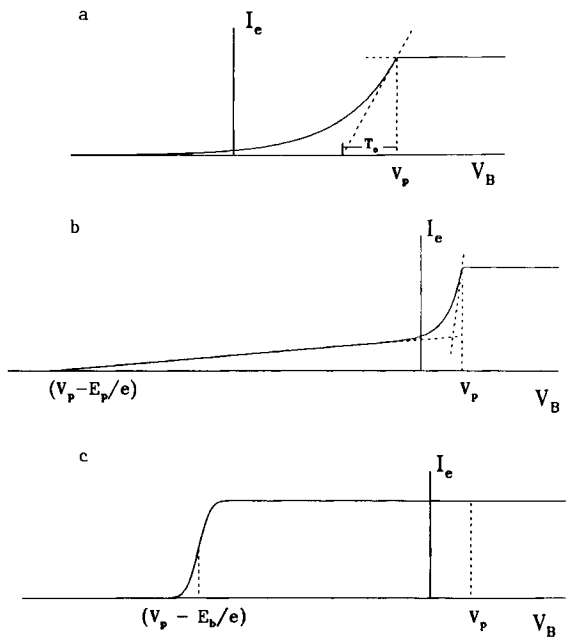


FIGURE 1. Ideal electron current (I_e) versus probe bias (V_B) for a variety of plasma conditions. In all cases the plasma potential V_p has been chosen to be equal to $+3T_e$. (a) Only bulk electrons are present. (b) Bulk electrons in addition to primary electrons with energy $E_p = 30\ T_e$ are present. (c) Only an electron beam with energy $E_b = 20\ T_e$ is present. (d) The electron beam plus bulk electrons are present. (e) Ion saturation current is added to the currents shown in Figure 1d. The amount of ion saturation current is exaggerated for clarity. (f) The electron current associated with ion beams with energy E_{bi} are added to the currents shown in 1e.

a good measurement of T_e, n_e, and V_p if a variety of conditions are satisfied. A partial list of these is: the plasma is not drifting, the electron distribution function is Maxwellian, the plasma density is sufficiently high that sheaths are small compared to the probe dimensions, significant fluctuations are not present, the probe surface is clean, magnetic fields are not present, secondary electron emission is not large, and electron neutral collision lengths are large compared to probe dimensions.

In many applications it is important to know the details of the electron distribution function. In particular $f(\underline{v})$ is very important for kinetic theory and $f(\epsilon)$ is important in consideration of particle confinement by electrostatic potentials. For the large planar probe case that we are considering, it is easy to determine $f_z(v_z)$, if $f_z(v_z)$ is a nondrifting Maxwellian. The probe

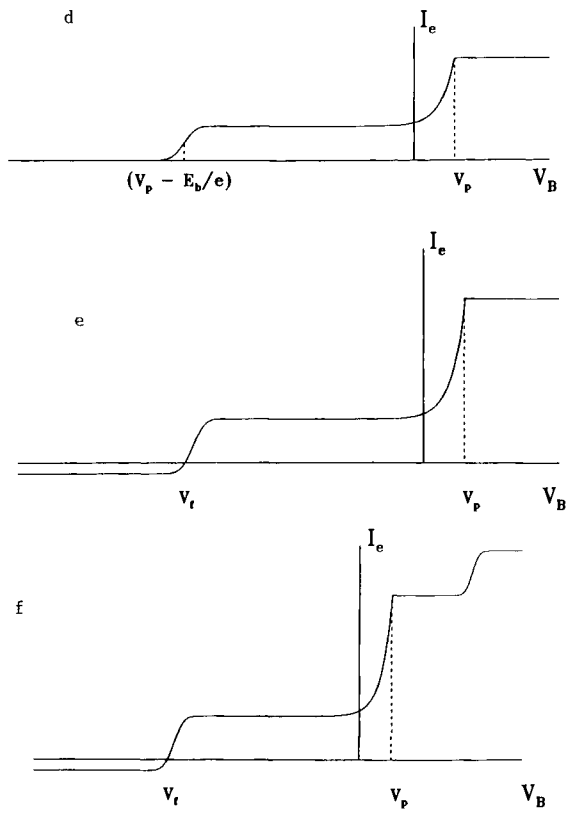


FIGURE 1. Continued.

current and each of its derivatives with respect to the probe voltage are proportional to f(ϵ), i.e.,

$$I_e = I_e^* \exp\left[\frac{-e(V_p - V_B)}{T_e}\right] \propto f(\epsilon) \tag{14}$$

$$\partial I_e/\partial V_B = \frac{eI_e^*}{T_e} \exp\left[\frac{-e(V_p - V_B)}{T_e}\right] = \frac{ef_z(v_z)}{T_e} \propto f(\epsilon) \tag{15}$$

and

$$\frac{\partial^2 I_e}{\partial V_B^2} = \frac{e^2 f_z(v_z)}{T_e^2} \propto f(\epsilon). \tag{16}$$

When non-semi-infinite planar probes are used, I_e and $\partial I_e/\partial V_B$ are no longer proportional to $f(\epsilon)$, but Druyvesteyn (1930 and 1935) was the first to show that the second derivative of the probe current with respect to the probe bias voltage is still proportional to $f(\epsilon)$ and to the energy distribution function under a variety of situations. It can be shown that for *isotropic* distribution functions with *arbitrary convex* probe geometry or for *anisotropic* distributions with spherical probes, the second derivative is still proportional to $f(\epsilon)$ and related to the energy distribution function by

$$\frac{\partial^2 I_e}{\partial V_B^2} = \frac{2\pi e^3 S}{m_e^2} f(\epsilon)\Big|_{\epsilon = e(V_p - V_B)}$$

$$= \tfrac{1}{4} S e^2 \left(\frac{2e}{m_e(V_p - V_B)}\right)^{1/2} f_E(\epsilon)\Big|_{\epsilon = e(V_p - V_B)} . \qquad (17)$$

The reader should be cautioned that $f(\epsilon)$ or $f_E(\epsilon)$ cannot be converted to $f(v)$ unless the plasma potential V_p is known. Many investigators assume that the knee in the I-V characteristic indicates the plasma potential, but this is only true when the plasma is nondrifting. To fully sort out the distribution function, a separate measurement is required to determine V_p. Fortunately, this can be done in many situations with emissive Langmuir probes (see later discussion). In general, electron distribution functions may not be Maxwellians at rest. Two important alternatives that are often found are monoenergetic, isotropic "primary" electrons and one-dimensional electron beams.

C. Isotropic Monoenergetic Electrons

The I-V characteristics often suggest Maxwellian distribution functions even when there is no particular reason to expect they would be present. In general, errors tend to result in Gaussian distributions, and it is relatively easy to fit semilog graphs of data to straight lines. When more than one electron species is present, data is often fit to the sum of two Maxwellians.

An example which exhibits one extreme is the two electron species plasma found in low pressure ($\leq 10^{-3}$ torr) devices used for basic plasma experiments and plasma processing. Plasma is produced by energetic "primary" electrons, which are energized by acceleration at a cathode sheath. The distribution function of the primary electrons then depends on the details of the particular production technique. Multidipole devices (Leung et al., 1975) operated at very low pressure ($\leq 10^{-5}$ torr) are a particularly simple case to consider, because the primary electrons are

confined by surface magnetic fields and bounce many times before they ionize the neutral gas. After several bounces the primary electrons become isotropic and are still monoenergetic with energy E_p. Thus the primary electrons form a shell in velocity space at a speed $v_p = \sqrt{2E_p/m_e}$.

The I-V characteristic for isotropic monoenergetic primary electrons in a multidipole plasma were first given by Leung et al. (1976). The three dimensional distribution function can be approximated by

$$f(v) = \frac{n_p \delta(v - v_p)}{4\pi v_p^2}, \tag{18}$$

where δ is a Dirac delta function and n_p is the primary electron density. It is easy to show that the corresponding distribution function $f(\epsilon)$ is given by

$$f(\epsilon) = \frac{n_p m_e}{4\pi v_p} \delta(\epsilon - E_p). \tag{19}$$

The probe current due to the primary electrons is then given by Eq. (9)

$$\begin{aligned} I &= \frac{n_p e S v_p}{4} \equiv I_p^*, && V_B > V_p \\ &= I_p^* \left(1 - \frac{2e(V_p - V_B)}{m_e v_p^2}\right), && V_p - \tfrac{1}{2}\frac{m_e v_p^2}{e} \le V_B \le V_p \\ &= 0, && V_B \le V_p - \tfrac{1}{2}\frac{m_e v_p^2}{e}. \end{aligned} \tag{20}$$

Figure 1b shows an I-V characteristic which combines Maxwellian electrons (with temperature T_e) to primary electrons with energy E_p. Given this I-V characteristic, T_e and n_e can be approximately obtained by first subtracting the current due to the primary electrons. This can be done by matching the straight line portion of the I-V characteristic for biases much more negative than T_e/e and continuing the line to V_p. This procedure is subject to considerable error because it is difficult to estimate the tail of the Maxwellian and because there is no guarantee that the distribution function in question is just a simple combination of only one species of primary electrons and background Maxwellian electrons.

D. Drifting Maxwellians

There has been much concern in the literature over the appropriate way to interpret probe characteristics for cylindrical and spherical probes. On the other hand, there has been very little concern over the appropriate way to

interpret probe characteristics in drifting plasmas. In that case the center of the distribution function is not at the plasma potential.

Consider a drifting Maxwellian with a drift velocity v_b that is much greater than the velocity spread of the half width of the Maxwellian, i.e., the electron drift energy E_b (defined $E_b \equiv m_e v_b^2/2$) is much greater than T_e. This situation could correspond to a plasma with a plasma source in a region of low potential that is followed by an increasing potential. The key point that must be recognized, and often is not, is that a large negative bias must be applied to repel the drifting electrons. An electron beam with mean velocity v_b, parallel temperature $T_{b\|}$ and perpendicular temperature $T_{b^\perp}$ can be described by

$$f(\underline{v}) = n_b\left(\frac{m_e}{2\pi}\right)^{3/2} \frac{1}{T_b\sqrt{T_{b\|}}} \exp\left[\frac{-m_e(v_z - v_b)^2}{2T_{b\|}}\right] \times \exp\left[\frac{-m_e(v_x^2 + v_y^2)}{2T_{b^\perp}}\right]. \tag{21}$$

The electron current collected by a planar probe oriented perpendicular to the electron beam and biased at V_B is given by

$$I = n_b\left(\frac{m_e}{2\pi T_{b\|}}\right)^{1/2} \int_{v_{min}}^{\infty} \exp\left[\frac{-m_e e(v_z - v_b)^2}{2T_{b\|}}\right] v_z\, dv_z, \tag{22}$$

where $v_{min} = \sqrt{2e/m_e(V_p - V_B)}$. This can be shown (Swift and Schwar 1970) to be equal to

$$I = en_b\sqrt{\frac{T_b}{2\pi m_e}} \exp(-x_m^2) + \frac{en_b v_b}{2}[1 + erf(x_m)], \tag{23}$$

where

$$erf(x_m) \equiv 2\sqrt{\pi}\int_0^{x_m} \exp(-x^2)\, dx \tag{24}$$

and

$$x_m = \sqrt{\frac{m_e}{2T_e}}[v_b - v_{min}]. \tag{25}$$

The electron current corresponding to an electron beam with parallel temperature $T_{b\|}$ given by Eq. (23) is graphed in Figure 1c. Note that the beam I-V characteristic resembles an electron species at a plasma potential close to $V_p - E_b/e$. The current which corresponds to an electron beam plus a background plasma is shown in Figure 1d. Note that *two* knee-like structures are present, with the one on the right corresponding to V_p. Also

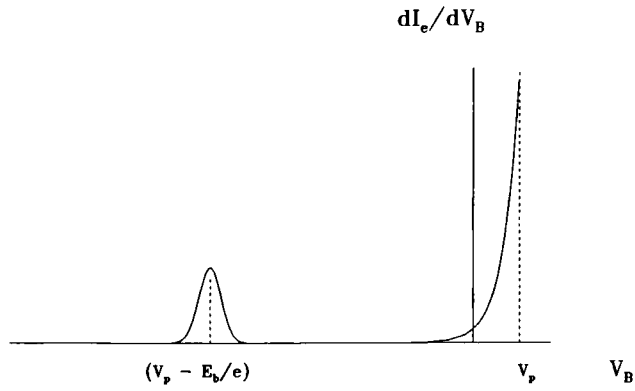


FIGURE 2. The derivative dI/dV_B versus V_B of I shown in Figure 1d.

note that the presence of an electron beam adds a constant current that must be subtracted off *before* drawing ln I vs. V to determine T_e and n_e.

The derivative dI/dV_B of the curve shown in Figure 1d is graphed in Figure 2. Taking the derivative of I (given by Eq. (23)) with respect to V_B gives

$$\frac{dI}{dV_B} \propto -\exp\left[\frac{-m_e(v_{min} - v_b)^2}{2T_b}\right]. \tag{26}$$

If bulk electrons are present as well, the derivative becomes

$$\frac{dI}{dV_B} \propto \left\{ n_b \exp\left[\frac{-(v_{min} - v_b)^2}{2T_{b\|}}\right] + n_e \exp\left[-\left(\frac{v_{min}^2}{2T_e}\right)\right]\right\}. \tag{27}$$

It is apparent that in the case of one dimensional beams dI/dV_B is proportional to the electron distribution function with v replaced by v_{min}.

The temperatures T_e and $T_{b\|}$ have been chosen so that the energy spreads of the background and beam shown in Figure (2) are comparable. The one-dimensional velocity distribution function $f_z(v_z)$ can be obtained by converting the horizontal scale to velocity using $v = v_{min} = [2e(V_p - V_B)/m_e]^{1/2}$. This is easy to do because V_p is known. The one-dimensional distribution function is $f_z(v_z)$ graphed in Figure 3.

The probe I-V characteristic when an electron beam is the only species present (see Figure 1c) has a strong resemblance to that of a plasma, with a plasma potential equal to $V_p - E_b/e$. In practice, it is often difficult to distinguish the plasma potential from the energy of drifting electrons. In such cases the best solution is to determine the plasma potential with emissive probes (see later discussion) which are insensitive to electron drifts.

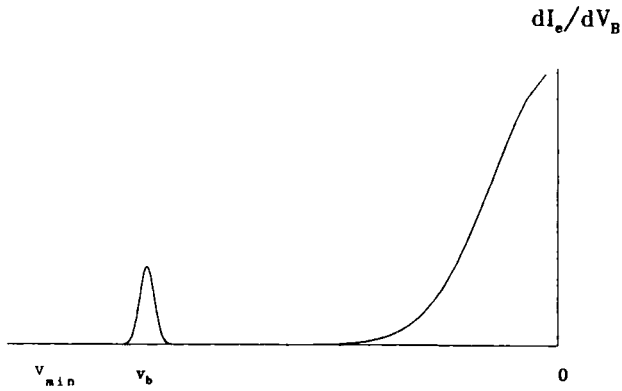


FIGURE 3. The derivative dI/dV_B shown in Figure 2 graphed versus v_{min}.

A mistake that is often made is to designate the *energy* spread of the beam as the beam "temperature." It is easy to see that this is not a valid procedure. Consider the I-V characteristic shown in Figure 1d and dI/dV_B vs. V_B shown in Figure 2. Assuming that the beam velocity v_b is much greater than the beam thermal spread in velocity, it follows that

$$T_{b\|} \simeq \left(\frac{\Delta E_b}{4E_b}\right) \Delta E_b , \tag{28}$$

where ΔE_b is the measured spread in the beam energy which corresponds to the half width of the energy distribution function at the e^{-1} point. Note that parallel beam temperatures are much lower than the measured energy spread. The parallel beam temperature in Figures 2 and 3 is only 0.03 eV, while the background electron temperature is 1 eV.

E. Ion Current

The total current to the probe is the sum of the electron and the ion current. One is tempted to write the ion saturation current to the probe by Eq. (12) with e replaced by i. Such a procedure is valid when T_i is comparable to T_e (i.e., in fusion plasmas or in thermal plasma processing plasmas) but does not work when $T_i \ll T_e$. In that case the ion saturation current is given by the Bohm current (Bohm 1949)

$$I_i^* \simeq 0.6 n_{i\infty} e S \sqrt{\frac{T_e}{m_i}} , \tag{29}$$

where m_i is the ion mass. Chen (1965) has provided a simple way to

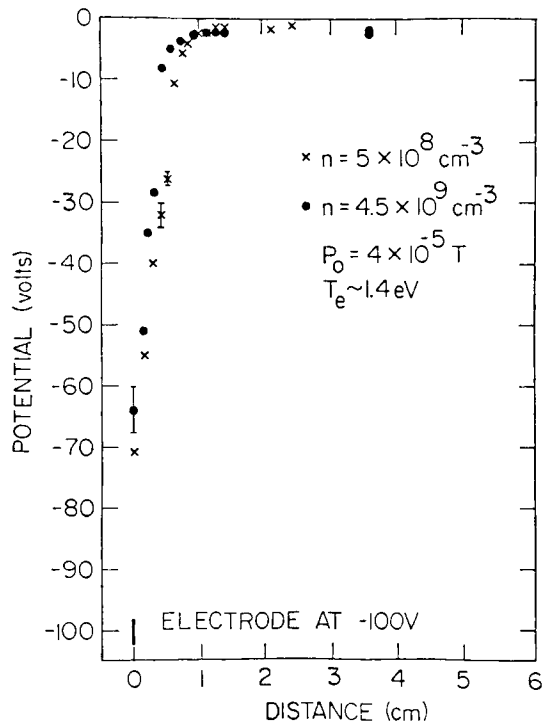


FIGURE 4. Plasma potential versus position along the axis of a 0.9 cm disc probe biased at −100 V (Coakley 1979).

understand the Bohm current. The plasma potential near a negatively biased planar probe (Coakley, 1979) is shown in Figure 4. Note that the curvature of the potential is negative. Poisson's equation then indicates that the ion density must exceed the electron density near the probe. Assuming Boltzmann electrons

$$n_e(V) = n_o \exp\left[\frac{e(V - V_o)}{T_e}\right], \tag{30}$$

where n_o and V_o are the electron density and the plasma potential at the sheath edge. Assuming that ions have a drift velocity u_o directed toward the probe at the edge of the sheath and assuming conservation of energy and flux, the ion density can be expressed

$$n_i(V) = \frac{n_o}{\sqrt{1 + \dfrac{2e(V_o - V)}{T_e}\left(\dfrac{c_s}{u_o}\right)^2}}, \tag{31}$$

where the ion acoustic velocity $c_s \equiv \sqrt{T_e/m_i}$ and m_i is the ion mass. The

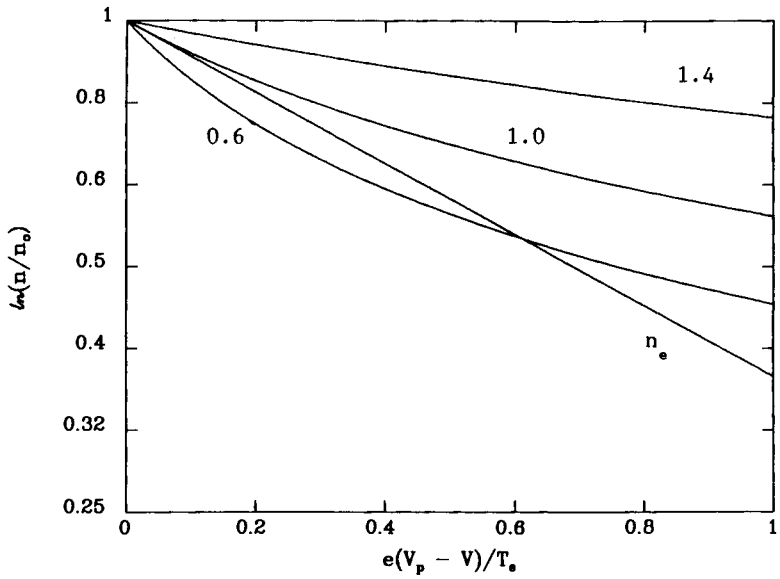


FIGURE 5. Graph of $\ln n$ versus $(V_o - V)$. Curves are drawn for electrons n_e and for ions with $u_o/c_s = 0.6$, 1.0, and 1.4.

ion density has also been chosen to be equal to n_o at the edge of the sheath. Electron density and ion density for several values of u_o are graphed in Figure 5 versus V. Note that when $u < c_s$, the electron density is initially greater than the ion density corresponding to positive curvature. The negative curvature that must be present requires that $u_o \geq c_s$. This means that ions must have a directed velocity at the sheath edge, even when ions have $T_i = 0$. Bohm argued that this directed velocity was provided by a "presheath" potential drop of $T_e/2$ which occurs over a long distance in the plasma. Emmert et al. (1980) have provided the most recent extension of this analysis.

The sum of the electron and ion currents leads to the I-V characteristic shown in Figure 1e. Clearly, all electron currents measured at $V_B < V_p$ must be measured with respect to the ion current ($\approx I_i^*$) rather than zero current.

F. Floating Potential

The I-V characteristic has several features that make it easy to estimate plasma parameters. These are the knee, the slope below the knee, the electron saturation current, and the floating potential.

The floating potential is defined as the bias voltage at which the probe draws no net current. Note that, unlike a conductor, the floating potential of an insulator can vary point-by-point. Secondary electrons complicate the situation because emitted current from one point on an insulator can return to another point. The floating potential of a conductor is determined by the balance of electron and ion current to and from the probe. If only plasma electrons and ions are present, assuming $T_e > T_i$ and taking the Bohm current for I_i^*

$$n\sqrt{\frac{T_e}{2\pi m_e}}\exp\left[\frac{-e(V_p - V_f)}{T_e}\right] = 0.6n\sqrt{\frac{T_e}{m_i}}, \tag{32}$$

where V_f is the floating potential. This gives $V_p - V_f \simeq (3.3 + 0.5 \ln \mu)$ T_e/e, where μ is the ion mass ratio normalized to hydrogen.

When energetic electrons (e.g., primary electrons or electron beams) are also present, the floating potential can be much more negative. Consider a situation in which the saturation current from the energetic electrons I_p^* is equal to α multiplied by the bulk electron saturation current I_e^*, i.e., $I_p^* = \alpha I_e^*$, where α is the order of 1.0 and the energetic electron density $n_p \ll n_e$. Using Eq. (20), the current balance can be written

$$I_e^* \exp\left[\frac{-e(V_p - V_f)}{T_e}\right] + \frac{I_p^*\left[E_p - e(V_p - V_f)\right]}{E_p} = 0.6n_i eS\sqrt{\frac{T_e}{m_i}}$$

$$\simeq 0.6I_e^*\sqrt{\frac{2\pi m_e}{m_i}}. \tag{33}$$

For $E_p \gg T_e$, Eq. (33) has a solution

$$e(V_f - V_p) \simeq -E_p\left(1 - \frac{0.035}{\alpha}\right) \simeq -E_p, \tag{34}$$

i.e., the floating potential is close to the plasma potential minus the energy of the primary electrons divided by e and does not depend on T_e.

The floating potential given by Eq. (32) or Eq. (34) is independent of the probe area S. Unfortunately, probe circuits do not measure the true floating potential. What is actually measured in the laboratory is the intersection of a circuit load line with the probe I-V characteristic. As shown in Figure 6, increasing the probe's area increases the collected current and causes the measured floating potential to move closer to the true floating potential. This effect can be minimized by choosing load resistances that are sufficiently large.

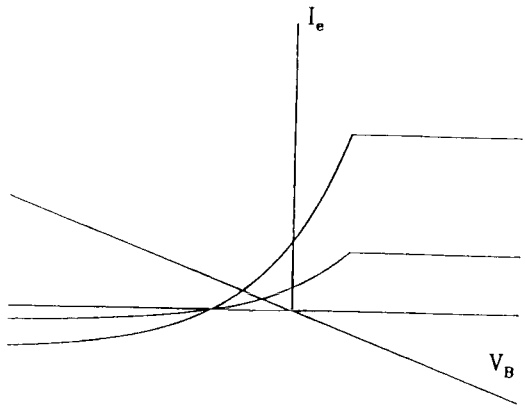


FIGURE 6. Current versus voltage for probes, with relatives areas 3 : 1, in the same plasma. Note that the load line $V = IR$ intersects the two curves at different values of V_B. Ion saturation current has been increased relative to electron saturation current.

Nam et al. (1988) have shown that when secondary electron emission from energetic primary electrons, with energy and density E_p and n_p, is present (see later discussion), the floating potential of a Langmuir probe can be double valued. This result means that the floating potential must be used with extreme care until the details of the I-V characteristic are determined. Prokopenko and Laframboise (1980) have presented similar results.

The I-V curve for a dirty planar tantalum Langmuir probe (diameter = 5 cm), mounted in a multidipole device, is shown in Figure 7. Plasma parameters were $n = 7 \times 10^8$ cm^{-3}, $T_e = 6$ eV, and $E_p = 300$ eV. This curve occurs because energetic primary electrons are present in addition to cold plasma electrons. As the bias voltage is chosen to be more negative, the plasma electrons are all reflected and only primary electrons can reach the probe. The secondary electron emission coefficient $\sigma(E)$, defined as the ratio of emitted current to incident current at energy E, increases with E and can be greater than 1.0. As the bias voltage is chosen to be more negative, the incident energy is reduced and the secondary electron current is also reduced.

Note that this curve crosses the $I = 0$ axis *three* times, so that three values of the floating potential might be expected. Nam et al. have demonstrated that only two of the crossings, those for which $dI_e/dV_B > 0$, are stable. When the probe is initially biased more positive than C in Figure 7, the probe floats at A. When the probe is initially biased more negative than C, the probe floats at B.

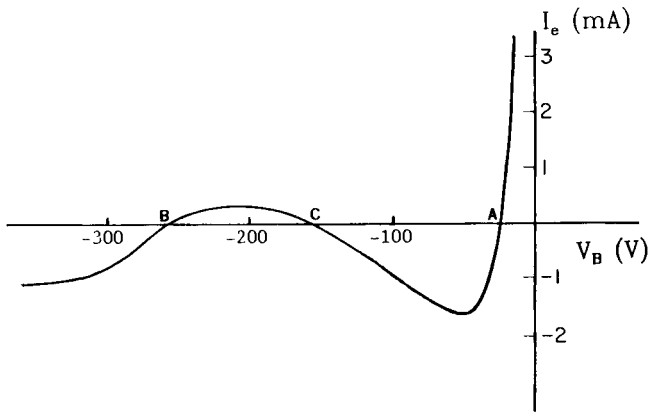


FIGURE 7. I-V characteristics of a tantalum probe with 300 eV primary electrons and 5 eV plasma electrons at an argon neutral pressure of 2×10^{-5} torr. The net current equals 0 at A, B, and C.

III. Effects of Finite Probe Size and Shape

A. Probe Sheaths

The discussion so far has avoided considering real probe geometry by mainly considering semi-infinite planar probes, which are large compared to sheath dimensions. This procedure has the advantage that it exhibits many probe characteristics and also provides results that are useful in many situations. However, before these results can be used it is necessary to consider how big is big. In this section we consider sheaths and effects of orbital particle motion. In the following section we will consider effects of ion-neutral collisions on probe characteristics. Probe geometry will be restricted to (planar) discs, cylinders, and spheres.

The relative size d of a probe sheath, the probe radius a, and the collision mean free path λ in the plasma surrounding the probe determine the probe characteristics. Both the probe radius and the collision mean free path are easy to measure or estimate, but how thick is the sheath? Probe sheaths are often assumed to be approximately equal to the electron Debye length $\lambda_D \equiv \sqrt{\epsilon_0 T_e/ne^2}$, but this approximation can underestimate the sheath thickness for large negative bias.

We can get an estimate of the sheath thickness for large negative bias as follows.

For $e(V_p - V_B)/T_e \gg 1$, the sheath region near the probe consists primarily of ions, if primary electrons can be neglected. In this region the ion current is space charge limited and can approximately be described by

the Child-Langmuir current (Chen, 1984)

$$j_{CL} = \frac{\frac{4}{9}\sqrt{\frac{2e}{m_i}}\,\epsilon_0 (V_p - V_B)^{3/2}}{d^2\beta^2}, \tag{35}$$

where d is the separation of the probe surface and the plane at which the electron density cannot be neglected. The factor β^2 equals 1.0 for planar geometry and is the order of d/a for cylindrical and spherical probes for values of $d/a < 3$ (Langmuir and Blodgett, 1923). If we equate the Child-Langmuir current (with $\beta = 1$) to the Bohm current Eq. (29) we find

$$\tilde{d} \simeq 1.1\eta^{(3/4)}, \tag{36}$$

where $\eta \equiv e(V_p - V_B)/T_e$ and $\tilde{d} \equiv d/\lambda_D$. The Debye length can be written

$$\lambda_D(\text{cm}) = 743\sqrt{\frac{T_e(eV)}{n(\text{cm}^{-3})}}, \tag{37}$$

so for electron temperatures the order of 1 eV and densities the order of 10^{10} cm^{-3}, $\lambda_D \simeq 7 \times 10^{-3}$ cm. For such plasmas a probe would normally be large in relative size.

We can expect that Eq. (36) gives a good indication of d (for large negative bias) when the collisional mean free path λ is much greater than d. Most usage normally involves bias voltages of only several T_e/e, and in that case the sheath will be only a few Debye lengths thick. Occasionally it is necessary to bias probes to large negative voltages to measure ion saturation current, e.g., when electron beams or energetic primary electrons are present. If 100 eV primary electrons are present, as they often are in multidipole plasmas at low pressure or in sheaths near cathodes, biasing the probe to -100 V, with $T_e \sim 1$ eV would result in a sheath $32\lambda_D$ thick.

In ion-neutral collisional plasmas the sheath thickness can be estimated by evaluating the mobility current. If a constant mobility is assumed, this gives (Chapman 1980)

$$\tilde{d} \propto \eta^{2/3}. \tag{38}$$

When the sheath thickness is comparable to probe dimensions, the probe geometry has a big influence on the probe I-V characteristics. We can represent the probe by an effective radius and an effective collecting area S_{eff} located parallel to the probe surface. Planar probes with one dimensional sheaths are an idealization that does not apply to real situations because such a probe determines the entire boundary to the system. In practice, the probe can be large compared to the sheath dimension but small compared to the dimensions of the plasma. In this case the effective collecting area increases as the probe is biased.

B. Orbital Theory

1. *Single Particle Theory*

The characteristics of a cylindrical or spherical probe with radius small compared to the sheath dimension can be treated using "orbital theory." This theory is important for the interpretation of emissive probes, which are normally wires with radius of from .001 to .003 cm. Orbital theory was developed by Langmuir and Mott-Smith (1924). Consider a probe biased at V_B. The simplest situation is a particle with charge q, initial velocity v_0, and impact parameter p at a point far from the probe where the potential is V_p. At closest approach its radius is r_c and its velocity is v_c. Conservation of energy and angular momentum gives

$$p = r_c\left[1 + \frac{(V_c - V_p)}{V_o}\right]^{1/2}, \tag{39}$$

where V_c is the plasma potential at the point of closest approach and $-qV_o \equiv 1/2mv_o^2$. Taking r_c as the probe radius a, any particle with $p < a[1 + (V_B - V_p)/V_o]^{1/2} \equiv p_a$ will be collected.

For monoenergetic particles at infinity (forming either a beam or a cylindrical or spherical shell in velocity space), the probe current is given by

$$\begin{aligned} I &= j_r 2\pi p_a 1 \qquad \text{for cylinders of length 1} \\ &= j_r 4\pi p_a^2 \qquad \text{for spheres,} \end{aligned} \tag{40}$$

where j_r is the random current density of particles crossing a unit area in one direction in the plasma. Substituting for probe parameters gives

$$\begin{aligned} I &= 2\pi a l j_r\left[1 + \frac{(V_B - V_p)}{V_o}\right]^{1/2} \qquad \text{for cylinders} \\ &= 4\pi a^2 j_r\left[1 + \frac{(V_B - V_p)}{V_o}\right] \qquad \text{for spheres} \end{aligned} \tag{41}$$

2. *Maxwellian Distributions*

Langmuir and Mott-Smith (1929) have worked the problem for Maxwellian distributions and shown that

$$I = Sj_r\left(\frac{d}{a}\right)\left[erf(\eta')^{1/2} + e^{\eta}\left[1 - erf(\eta + \eta')^{1/2}\right]\right] \qquad \text{for cylinders}$$

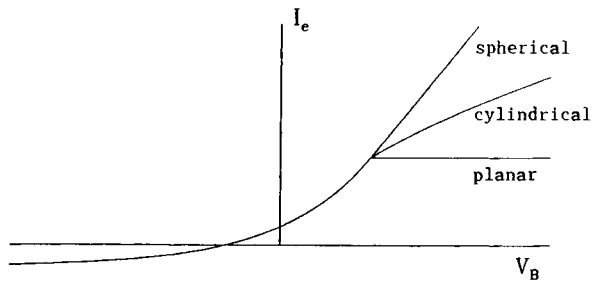


FIGURE 8. Ideal I-V characteristics for planar, cylindrical, and spherical probes.

and

$$I = Sj_r\left(\frac{d^2}{a^2}\right)\left[\left[1 - e^{-\eta'}\right] + e^{-\eta'}\right] \qquad \text{for spheres,} \tag{42}$$

where

$$\eta' = \frac{a^2}{d^2 - a^2}\eta. \tag{43}$$

For small sheaths ($d/a \simeq 1$) Eq. (42) reduces to the thin sheath value to $I \simeq Sj_r$. In the thick sheath limit Eq. (42) reduces to (Chen, 1965)

$$\begin{aligned} I &\simeq \frac{Sj_r^2(\eta + 1)^{1/2}}{\sqrt{\pi}} && \text{for cylinders} \\ &\simeq Sj_r(\eta + 1) && \text{for spheres.} \end{aligned} \tag{44}$$

Equation (41) is graphed in Figure 8. It is apparent that the knee is not well defined for spherical probes.

Equation (44) shows that the probe currents, i.e, the effective probe areas, increase approximately linearly for small increases in $\eta \equiv e(V_B - V_p)/T_e$. One major difference between probe characteristics in real plasmas as opposed to those of ideal semi-infinite planar probes shown in Figure 1 is that the electron saturation current continues to increase (approximately linearly) above the plasma potential because of contributions from the edges of finite size disc probes, as shown in Figure 9. As the sheath increases in size, planar disc probes behave more and more like spherical probes. We can easily take account of this behavior by determining the plasma potential (assuming nondrifting plasma) from the intersection of two lines—one drawn parallel to the I-V characteristic curve at the steepest

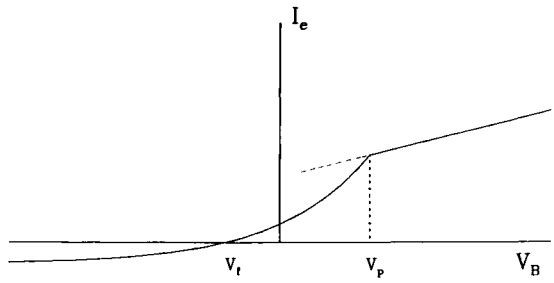


FIGURE 9. I-V characteristic showing increase in current for $V > V_p$. In laboratory plasmas, there would also be some rounding near the knee as a result of plasma noise.

point and the other drawn parallel to the flat portion below the knee. This procedure is illustrated in Figure 9.

C. Ion Beam Effects on I-V Characteristics of Non-Semi-Infinite Planar Probes

1. Additional Collected Electron Current

It has recently been found (Weber et al., 1979; Skøelv et al., 1984) that the contribution of ion beams to probe I-V characteristics can be quite large and is sensitive to sheath expansion. Just as electron beams show up as increased electron current for bias voltages that are insufficient to repel them, we might also expect ion beams to show up as reductions in ion beam current or increases in electron beam current at bias potentials more positive than the plasma potential for which the ion beams are reflected. However, the ratio of ion beam current to electron saturation current is only the order of $(n_b/n_e)(4\pi E_b m_e/m_i T_e)^{1/2}$, where n_b and E_b are the ion beam density and energy, respectively. It seems obvious that unless E_b is very large, the increase in current associated with the ion beam "knee" in a Langmuir probe I-V trace would be more than an order of magnitude smaller than the increase associated with the knee at the plasma potential. From this there are then two apparent conclusions: 1) Langmuir probes can not be used to detect ion beams in the presence of background plasma unless the beam current is comparable to the electron saturation current and 2) ion beams do not add much confusion to the interpretation of swept Langmuir probe characteristics if the ion beam current is much less than the electron saturation current.

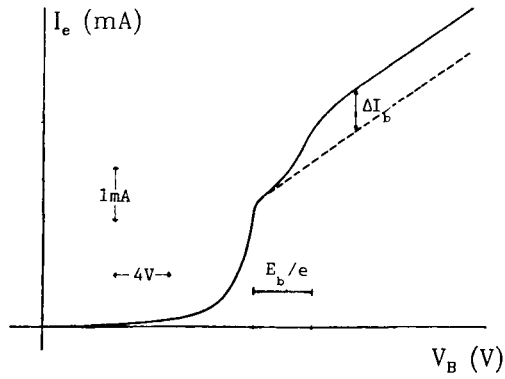


FIGURE 10. Laboratory I-V characteristic showing the presence of an ion beam.

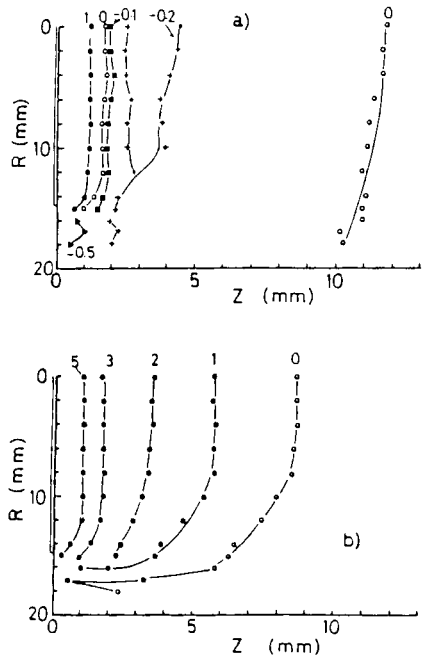


FIGURE 11. Equipotential contours of a 3 cm diameter disc probe in the presence of a 10 V ion beam. (a) The probe is biased at 8 V, below E_b/e. (b) The probe is biased at 15 V, above E_b/e. The OV contour has expanded by 6.5 mm.

Experiments by Weber et al. (1979) in an unmagnetized plasma have demonstrated that both of these conclusions are false and that the increase in current associated with the ion beam knee can be comparable to the increase at the plasma potential, even when ion beam current density is well below the electron saturation current density. This means that Langmuir probes can be used to detect the presence of ion beams in plasmas. Weber et al. have also shown that it is possible to determine the beam "temperature" from a swept Langmuir probe trace.

Results obtained by Skøelv et al. are given in Figure 10. They employed a plasma with a density of $10^9\,\mathrm{cm}^{-3}$, $T_e = 1$ eV, $T_i = 0.2$ eV, and with beam energies the order of 10 eV. As seen in Figure 10, they found that the electron saturation current was amplified by the presence of ion reflection.

Weber et al. argued that the presence of the ion beam increases the electron current to the probe by modifying the space charge sheath in front of the probe. When the probe bias is sufficiently large to stop and reflect the beam, the reflected ions provide a large space charge region which enlarges the sheath thickness (and with it the effective probe collecting area) and allows electrons to be "drawn in sideways" (see Figure 11). Nakamura et al. (1981) have argued that in an unmagnetized plasma the sheath

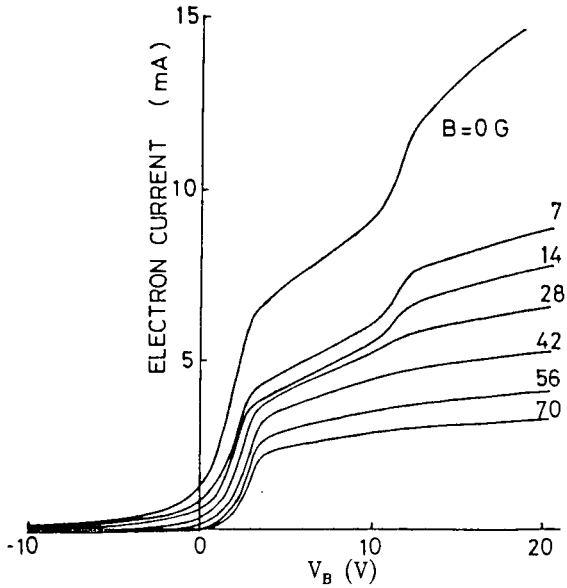


FIGURE 12. I-V characteristics of a 2 cm diameter disc probe with different axial magnetic fields. The ion beam has $E_b \approx 10$ eV.

collecting area S_{eff} can then be written

$$S_{\text{eff}} = S_0\left[1 + \frac{6L}{D} + \frac{5L^2}{D^2}\right], \tag{45}$$

where D is the probe diameter and L can be identified as the sheath thickness defined as separation between the probe and a plane in the plasma $1V$ below the probe potential. Assuming $L \propto (V_B - V_p)$, it follows that $\Delta I \propto (V_B - V_p)$ for small increases in V_B. The effect of adding ion beams to a background plasma with an electron beam is shown in Figure 1f.

When a magnetic field perpendicular to the probe is present, the sheath still expands, but the effective collecting area remains S_0 and the ion beam does not significantly enhance the collected current and is not readily detected, as shown in Figure 12 (Nakamura et al., 1981).

2. *Reflected Ion Beam Instability*

Although reflected ion beams in strong magnetic fields do not add significant collected current, the reflected ion beams can make their presence known by another effect, the excitation of electrostatic ion cyclotron (EIC) waves (Chan et al., 1980). A swept Langmuir probe trace indicates the beam energy by the point of the onset of the EIC instability (as shown in Figure 13a). Ion beams can also be more readily detected by a variety of ion "energy analyzers."

3. *EIC Instability Without Beams*

The EIC instability can also be destabilized by a relative drift between the electron and ion components in a magnetized plasma. A sufficient electron drift can be established by biasing a probe just above the plasma potential. The instability is close to the ion cyclotron frequency and is apparent on I-V characteristics if the sweep time is sufficiently slow. This effect is readily apparent in Q-machines when probes are biased to collect electron saturation current.

I-V characteristic data demonstrating this instability, which onsets above the knee, are given in Figure 13b and were obtained with a 0.6 cm diameter one-sided probe in an argon plasma in a 0.1 T magnetic field.

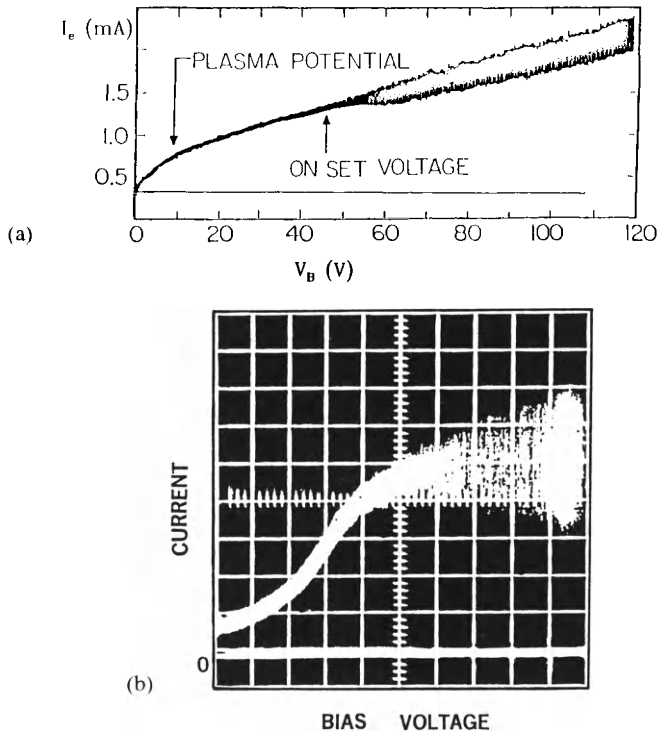


FIGURE 13. (a) I-V characteristic of probe in axial magnetic field with an ion beam. The on set voltage of EIC instability corresponds to the ion beam energy. (b) I-V characteristic showing the onset of the EIC instability at the plasma potential.

IV. Effects of Ion-Neutral Collisions

A. General Considerations

In weakly ionized plasmas at high neutral pressures, neutrals are important because they can govern the ion motion to a probe. There are two effects: 1) When the ion-neutral mean free path becomes comparable to the electron Debye length, the ion motion in the sheath differs from free fall motion. 2) Shadowing effects can be important when the probe dimensions become comparable to the collision mean free path.

The ratio of the Debye length to ion-neutral mean free path can be written

$$\frac{\lambda_D}{\lambda} = 2.4 \times 10^{19} \sigma p \sqrt{\frac{T(\text{eV})}{n(\text{cm}^{-3})}}\,, \tag{46}$$

where p is the neutral pressure in torr and σ is the ion-neutral collision cross section (in cm^2). For plasmas with densities on the order of $10^{11}\,cm^{-3}$ with $T_e = 2$ eV and cross sections on the order of $10^{-16}\,cm^2$ (Brown, 1966), this ratio equals 1.0 at a pressure on the order of 10^2 torr. In practice, short mean free paths are only of interest in plasma processing applications close to atmospheric pressure, i.e., in "thermal" plasmas which also have the characteristic that $T_e = T_i$.

Shadowing can become important for large probes at lower neutral pressures. The ratio of the mean free path to probe radius a can be written $\lambda/a = (3.3\sigma(10^{-16}\,cm^2)p(\text{torr})a(\text{cm}))^{-1}$. This means that for large probes with $a = 0.5$ cm, shadowing effects can become important for neutral pressures less than 1 torr.

We consider negatively biased probes. In this case electrons are normally in thermal equilibrium and are not greatly affected by collisions in the vicinity of probes. The extreme case of ion motion that is everywhere dominated by ion-neutral collisions is considered, i.e., the mean free path is assumed to be short compared to any other length of the problem. In this situation the ion current density j_i can be written

$$j_i = e(D\nabla n + \mu_i n \nabla V), \tag{47}$$

where D is the diffusion coefficient and μ_i is the ion mobility. A similar expression can be written for the electrons. Poisson's equation, determines the potential. Since the probe current I is conserved, Eqs. (47) and Poisson's equation can be solved simultaneously to determine n and V. Using Einstein's relation for classical diffusion,

$$\mu_i = \frac{eD}{T_i}, \tag{48}$$

and assuming that the scale lengths of the gradients of n and V are comparable, it follows for large negative bias voltages that the second term in Eq. (47) is much greater than the first term and that the ion density satisfies

$$n_i \approx \frac{j_i}{(e\mu_i \nabla V)}. \tag{49}$$

For a spherical probe the ion current I_i can be written

$$I_i = 4\pi a^2 j_i. \tag{50}$$

Su and Lam (1963) have approximately solved Poisson's equation by obtaining separate sheath and plasma (quasineutral) solutions and matching them at the sheath edge in a transition region. The region far from the probe, where $V_p - V$ is small, is considered first. Following Smy (1976) and Su and Lam (1963), we can write down an approximate solution for the

density in this quasi-neutral region

$$n_i = n_o\left(1 - \frac{r_s}{r}\right). \tag{51}$$

Note that this solution has the nonphysical characteristic that the ion density vanishes at the sheath boundary r_s rather than at the probe ($r = a$). Taking Boltzmann electrons and equating n_i and n_e at the sheath boundary where $V = V_s$ gives

$$\eta = -\frac{r_s}{r}, \tag{52}$$

where $\eta \equiv e(V - V_p)T_e$, Eqs. (47), (50), and (51) then give

$$I_i \approx 4\pi\mu_i n_o r_s[T_i + T_e]. \tag{53}$$

The electron current can be written with a similar expression but can be neglected (for negative probe biases) in comparison to ion current because $\mu_e/\mu_i \ll 1$.

B. Thin Sheaths

The result of Eq.(53) is not useful until r_s is determined. This requires solution of the sheath equation and appropriate matching. However, if the sheath thickness is small compared to the probe radius a, there is no need to solve the sheath equation. The radius r_s can be replaced by the probe radius a in Eq. (53) yielding

$$I_i \simeq 4\pi\mu_i n_o(T_i + T_e)a. \tag{54}$$

This expression is useful in thermal plasmas because $T_e = T_i$ and can be determined spectroscopically as well as from the I-V characteristic.

C. Thick sheaths

The sheath expands as the probe bias voltage is increased and eventually becomes large compared to the probe radius. Sheath expansion is accompanied by increased ion saturation current to the probe. For a very thick sheath, the solution for space charge limited flow between two spherical surfaces can be written (following Smy, 1976)

$$I_i \simeq \frac{6\pi\mu_i\epsilon_o a^2(V_p - V_B)^2}{r_s^3}. \tag{55}$$

Combining Eqs. (53) and (55) gives an expression for the asymptotic current to a thick sheath (close to that given by Su and Lam 1963)

$$I_i = (6\pi\epsilon_0)^{1/4}\left((V_p - V_B)a\right)^{1/2}(4\pi n T_e)^{3/4}\mu_i\left(1 + \frac{T_i}{T_e}\right)^{3/4}. \quad (56)$$

Manipulation of Eqs. (52), (53) and (56) shows that thick and thin sheaths i.e., $r_s \gg a$ or $r_s \simeq a$ correspond to $\eta/\tilde{a} \gg 1$ and $\eta/\tilde{a} \ll 1$, respectively, where $\tilde{a} \equiv a/\lambda_D$. In general, numerical solutions are necessary (Su and Lam 1963). Fendell (1970) and Baum and Chapkis (1970) have derived analytic solutions to the thick sheath problem. Kiel (1969) gives an expression for the case when a is the order of r_s.

In order to be considered stationary and for Eqs. (54) and (56) to be valid, the ambipolar ion diffusion velocity must be greater than the plasma flow velocity. For most thermal plasmas this condition is not satisfied.

D. Probe at Plasma Potential

When the probe is biased at the plasma potential, Eq. (47) reduces to

$$j = eD\nabla n. \quad (57)$$

Following Chen (1965) consider a surface one mean free path from the probe, parallel to the probe. The plasma can be considered to be collision dominated outside this surface and collisionless inside this surface. Taking D as a constant

$$\nabla \cdot \mathbf{j} = -D\nabla^2 n = 0 \quad (58)$$

This is just Poisson's equation for n and the problem of determining n_λ, the density at the surface, becomes analogous to determining the capacitance of a conductor in a vacuum. Matching the outside solution to the inside solution (in the limit $\lambda/a \ll 1$) gives (see Chen, 1965)

$$I_e = \left(\frac{3}{4}\frac{\lambda}{a}\right)\frac{n_0 evS}{4}. \quad (59)$$

It is apparent that when $\lambda/a \ll 1$, the electron saturation to a probe is reduced by $\frac{3}{4}\frac{\lambda}{a}$.

E. Flowing Plasma

Smy (1976) has noted that in order to be considered stationary a collisional plasma must satisfy the condition that the ambipolar ion diffusion velocity must be greater than the plasma flow velocity v_f. The ion diffusion velocity

is the order of $(T_e/ea)\mu_i$, where μ_i is the ion mobility, so a plasma is stationary only if

$$\mu_i T_e/eav_f > 1. \tag{60}$$

For typical parameters this condition implies that the plasma flow velocity be less than $0.1 \rightarrow 1$ m/s. However, most high pressure plasmas have flow velocities in excess of these values and therefore must be considered to be flowing.

Smy (1976) has derived expressions for the asymptotic current to negatively biased probes in *flowing* plasmas.

For spherical probes with thin sheaths

$$I_i = 12(\mu_+\epsilon_0)^{1/4}(nev_f)^{3/4}(V_p - V_B)^{1/2}a^{5/4}; \tag{61}$$

for spherical probes with thick sheaths

$$I_i = \pi(nev_f)^{0.6}(6\epsilon_0\mu_+)^{0.4}\big((V_p - V_B)a\big)^{0.8}; \tag{62}$$

for cylindrical probes with a thick sheath, Clements and Smy (1969) find

$$I_i = \frac{2(\pi\mu_+\epsilon_0)^{1/3}}{[\log(I_+/2nev_f a)]^{2/3}}\big(nev_f(V_p - V_B)\big)^{2/3}. \tag{63}$$

Equations (61)–(63) were found to be accurate within $\pm 50\%$.

V. Probes in Magnetic Fields

The motion of electrons and ions in magnetic fields can be described in terms of their gyroradii

$$r_{e,i} \equiv \frac{\sqrt{T_{e,i}m_{e,i}}}{eB}. \tag{64}$$

Specifying T_e in eV and B in gauss gives

$$r_e = \frac{2.4\sqrt{T_e(\text{eV})}}{B(\text{gauss})}$$

$$r_i = \frac{103\sqrt{T_i(\text{eV})\mu}}{B(\text{gauss})}, \tag{65}$$

where μ is the mass of the ion normalized to hydrogen and $r_{e,i}$ are in cm.

The ratio of electron to ion gyroradii is $0.02\sqrt{T_e/T_i\mu}$, so electrons usually have much smaller gyroradii than ions. Even for relatively weak B fields, electron gyroradii can be quite small. For example, r_e for a 1 eV electron in

a 100 gauss field is only equal to 0.02 cm. In fusion plasmas, probes cannot be made too small because of the need to provide cooling. This suggests that probe radii often will be greater than the electron gyroradius. Details of probe electron current collection will then depend strongly on how electrons populate the magnetic flux tube which connects to the probe. Ion current collection will be unaffected until much stronger fields are present. In general, the relative amount of ion current to electron current will increase.

Rubinstein and Laframboise (1983) have generalized orbital theory in collisionless plasmas to include the presence of magnetic fields. They have provided numerical solutions for cylindrical and spherical geometries in a series of papers (see Rubinstein and Laframboise (1983) for references). However, when magnetic fields are present in plasma processing and fusion plasmas, probe characteristics are dominated by diffusion into the probe's flux tube and the generalized orbital theory rarely applies.

If probes are located on magnetic islands, probe characteristics will be dominated by the processes which fill in the islands once they are depleted.

In plasma processing applications, electron diffusion is dominated by collisions with neutrals and can be described by a diffusion constant $D_{\|}$

$$D_{\perp} = D_{\|}\left(1 + \Omega^2\tau_c^2\right)^{-1}, \tag{66}$$

where $\Omega \equiv eB/m_e c$ is the electron cyclotron frequency, τ_c is the mean collision time, and $D_{\|}$ is the diffusion coefficient. Chen (1965) has considered this case in some detail. He shows that the electron saturation current at the plasma potential can be approximated by

$$I_e^* = \frac{n_0 v_e S}{4}\frac{4}{3}\frac{\lambda}{a}\sqrt{\frac{D_{\perp}}{D_{\|}}}\,. \tag{67}$$

Stangeby (1982) has treated the effects of cross-field diffusion into the probe's flux tube on the electron saturation current portion of the probe characteristic in strongly magnetized "fusion" plasmas with high fractional ionization. An important feature of his model is the inclusion of a potential hill in the presheath between the probe and the background potential. When the probe is biased to collect net electrons, the potential hill reduces the ion flux and aids in pulling electrons to the sheath edge against Coulomb collisions. Based on this work, it can be shown that the total current collected by the probe can be approximated by (LaBombard and Lipschultz 1986)

$$I(\eta_p) = I_{sat}\left[e^{-\eta_h/\tau} - \frac{1}{\alpha}\left(\frac{r}{1 + re^{(\eta_h - \eta_p)}}\right)\right], \tag{68}$$

when $\eta_p \leq \tau \ln[(1 + r)/r)]$, and

$$I(\eta_p) = I_{sat}\left[e^{-\eta_h/\tau} - \frac{1}{\alpha}\left(\frac{r}{1+r}\right)\right], \tag{69}$$

when $\eta_p > \tau \ln[(1 + r)/r]$. Here α is defined as

$$\alpha = 4f(\tau)\sqrt{\frac{Z_i(1+\tau)\pi m_e}{8m_i}}, \tag{70}$$

and η_h is given by

$$\eta_h = \tau \ln(r^{-1}e^{(\eta_p - \eta_h)} + 1). \tag{71}$$

The parameters $\tau \equiv T_i/T_eZ_i$, and $I_{sat} \equiv eSn_oC_sf(\tau)\sqrt{1+\tau}$ and $\eta_p \equiv eV_p/T_e$. The factor $f(\tau) \simeq 1/2$ accounts for the presheath acceleration. The electron saturation reduction factor, r, is related to the rates of parallel and perpendicular transport along the collection flux tube. Much of the complicated physics is put into r, the electron saturation current reduction factor, defined

$$r = 16\lambda_{ei}\left(\frac{D_\perp}{D_\parallel}\right)^{1/2}\frac{(1+\tau)}{2\pi a}, \tag{72}$$

where $D_\parallel$ is the diffusion coefficient parallel to the magnetic field and λ_{ei} is the mean free path for electron-ion collisions. In terms of r the parallel electron saturation current density can be written

$$j_{\parallel e} = \frac{\frac{1}{4}n_e v_e r}{(1+r)}. \tag{73}$$

The parameter r varies from device to device. In practice separate calibration of probes is always required.

VI. Secondary Electron Effects

Secondary electron emission plays a key role in plasma production in many plasma processing devices. It should be no surprise that it also can be important in determining the probe characteristics. In fact, it is likely that electron temperatures are often underestimated because of secondary electron emission. Secondary electrons can result from either electron or ion bombardment. Secondary electron emission coefficients $\sigma(E)$ are defined as the ratio of emitted to incident current and depend on the incident particle energy. For clean surfaces, they can be greater than 1.0 for ion energies greater than 1 keV or for incident electron energies of several hundred eV (Gibbons, 1966). Insulators and dirty surfaces tend to have higher sec-

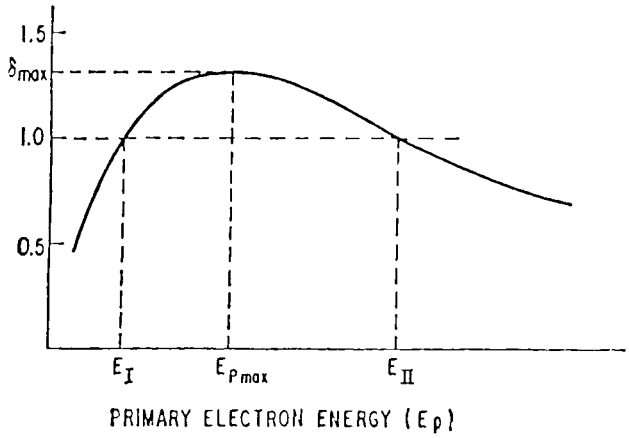


FIGURE 14. Schematic description of the variation of the electron secondary electron emission coefficient versus incident electron energy.

ondary electron emission coefficients than conductors. Photoemission can also produce significant electron currents.

In the large voltage cathode sheaths of plasma processing plasmas, ions are accelerated to high energies before hitting the cathode. These ions produce electrons by secondary emission. These electrons are accelerated away from the cathode and produce plasma by ionization. In fusion plasmas, both energetic electrons and ions are present within the bulk of the plasma and may also be present near boundaries.

Consider secondary electron emission caused by monoenergetic primary electrons. The secondary electron emission coefficient σ of most metals increases monotonically with incident energy to maximum value somewhat greater than 1.0 at an incident electron energy the order of several hundred eV. Further increases in incident electron energy results in a slow reduction in secondary electron emission (see Figure 14).

The effect on Langmuir probe I-V characteristics can be estimated by representing the curve in Figure 14 by $\sigma \simeq \delta\epsilon/E_o$, where $E_o \simeq 200$ eV for tantalum. This is a reasonable approximation for $\epsilon < 150$ eV. Using Eq. (9) the secondary electron current density can be written

$$j_s(V_B) = \frac{e2\pi}{m_e^2}\int \epsilon f(\epsilon)\left[\frac{1 - e(V_p - V_B)}{\epsilon}\right]\sigma(\epsilon_w)\, d\epsilon \qquad (74)$$

where σ is evaluated at ϵ_w, the kinetic energy that the incident primary

electrons have at the probe. Conservation of energy gives

$$\epsilon_w = E_p - e(V_p - V_B). \tag{75}$$

Using Eq. (19) for f(ϵ) gives

$$j_s(V_B) = \left(\frac{e}{m_e}\right)\left(\frac{n_p}{2v_p}\right)\frac{E_p^2}{E_o}\,\delta\left[1 - e(V_p - V_B)/E_p\right]^2. \tag{76}$$

The ratio of secondary current to primary current can be written

$$\frac{j_s}{j_p} = \frac{E_p}{E_o}\,\delta\left[1 - e(V_p - V_B)/E_p\right]. \tag{77}$$

If j_p is comparable to the bulk electron background current density, the condition for significant secondary electron emission is that E_p is the order of E_o. It can be shown that a similar result holds for Maxwellian electrons (Wang et al., 1986a).

Representative data for a tantalum probe immersed in a plasma with monoenergetic electrons (with energy of 300 eV) are shown in Figure 7. The secondary electron emission current has the appearance of additional ion collection current. As the probe bias voltage is swept, the curve dips below 0 A, indicating significant electron emission. Note that secondary electron emission gives rise to a greater slope below the knee and would normally be interpreted as a *lower* plasma electron temperature than is actually present. When energetic electrons are present, it is critical that the contribution of secondary electrons be subtracted.

VII. Emissive Probes

A. Simplified Theory

Electron emission from probes provides an opportunity for direct measurements of the plasma potential. Although several techniques have been developed for interpreting emissive probe characteristics, they all are based on one principle. When the probe is biased more positive than the local plasma potential, electrons emitted from the probe are reflected back to the probe. When the probe is biased negative with respect to the local plasma potential, electrons can escape to the plasma and appear as an effective ion current. This process is *not sensitive* to plasma flow because it depends directly on plasma potential rather than electron kinetic energy and it is less sensitive to probe surface contamination when heated surfaces provide

electron emission. Unlike collecting probes, emissive probes do not give useful data on plasma density or electron temperature.

Although emissive probes were first discussed by Langmuir (1929) and a technique has been described for obtaining the plasma potential from the floating potential of a strongly emitting probe with $I_{eo}/I_{co} \gg 1$ (Kemp and Sellen, 1966), they have not found widespread use because of problems associated with strong electron emission and because of generally negative comments about them in previous review articles (e.g., see Chen, 1965). Recently it has become clear that other options are present for using emissive probes (Wang et al., 1985, 1986a, 1986b, 1986c). These options include the inflection point method in the limit of zero electron emission (Smith et al., 1979), differential emissive probes (Wang et al., 1985), emissive-capacitive probes which depend on secondary electron emission (Wang et al., 1986c), and self emissive probes (Hershkowitz et al., 1983). Because of these new options it now appears that the only limitation on their use is whether or not the probe can survive in the plasma environment, a problem shared by all probes. Contamination of emissive probes is less serious than in general because the hot surface tends to stay clean.

The current-voltage characteristics of a wire probe as a function of heating voltage is shown in Figure 15. It is apparent that several features are present which might be used to determine the plasma potential. Chen (1965) has pointed out that the two I-V characteristics tend to separate from each other near the plasma potential and that the bias voltage corresponding to the separation can be taken as an indication of the plasma potential. Another indication is the probe floating potential, which moves toward more positive potentials (and close to the plasma potential) as the probe is heated. However, Langmuir (1929) has derived the theory for the potential near a strongly emitting probe and found that space charge effects can limit the usefulness of the latter technique. He shows that a double sheath sets up near the probe and that the electron emission is space charge limited.

For fixed probe bias voltage V_B, space charge effects become important as electron emission is increased (see Figure 15). Space charge effects are also sensitive to probe geometry because they depend on the local electric field near the probe. For example, electric fields near cylindrical or spherical probes are much stronger than those near planar probes biased at the same value of V_B, (if the radius is small), so space charge limitation of current emitted by planar probes becomes important at smaller values of V_B. Note that for sufficiently high V_B, space charge effects are also not important.

In order to simplify the description of emissive probe characteristics, we first consider weakly emitting probes for which we can assume that space

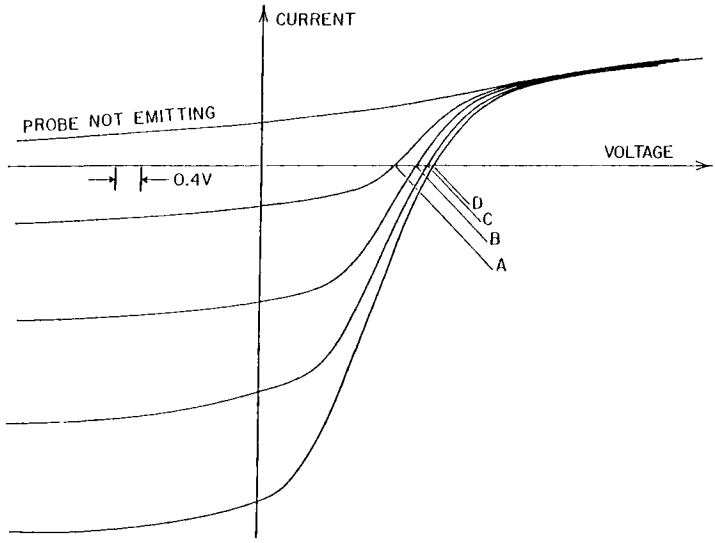


FIGURE 15. Emissive probe I-V characteristics at four probe temperatures, increasing from A to D.

charge effects are not important. This approach is useful because it provides an understanding of both the inflection point method in the limit of zero emission and the differential emissive probe technique. It also provides qualitative results for situations with strong emission when double sheaths are present. Neglecting space charge effects, the emission current from a hot wire can be written

$$I_e = I_{eo}\exp\left[\frac{-e(V_B - V_p)}{T_w}\right] g(V_B - V_p), \qquad V_B > V_p$$

$$= I_{eo}, \qquad V_B < V_p, \qquad (78)$$

where I_{eo} is the electron emission saturation current and T_w is the hot wire temperature. The quantity $g(V_B - V_p)$ accounts for the orbital angular momentum and depends on the hot wire radius and on the sheath radius (see earlier discussion). The emission current appears as an apparent ion collection current.

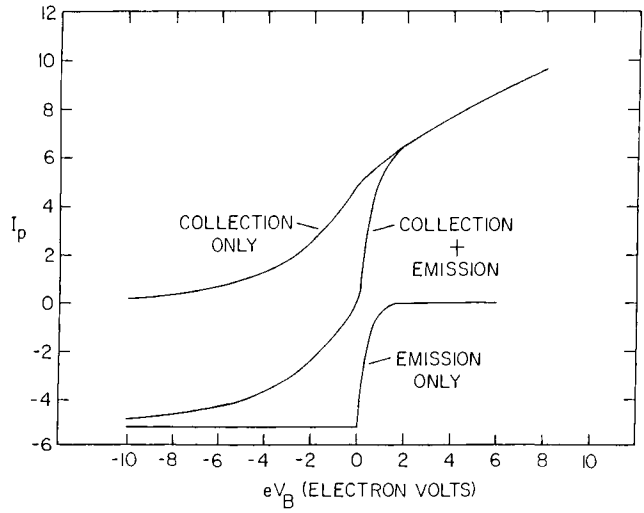


FIGURE 16. Collecting and emitting I-V characteristics for a cylindrical wire probe in a plasma when the probe radius $\ll$ sheath thickness. Space charge is neglected. The electron and wire temperatures are $T_e = 3$ eV, $T_w = 0.3$ eV.

Neglecting space charge effects, the electron current collected from a Maxwellian plasma with temperature T_e can be written (see Eq. (44))

$$I_c = \begin{cases} I_{co} \exp\left[\dfrac{-e(V_p - V_B)}{T_e}\right], & V_B \leq V_p, \\ I_{co} g'(V_B - V_p), & V_B \geq V_p. \end{cases} \tag{79}$$

Ideal collection and emission current-voltage characteristics are given in Figure 16.

B. Floating Point in Strong Emission

For strong emission, the current is space charge limited by a double sheath, a potential dip between the probe and the background plasma. A dip of only several T_w/e reduces the emitted current by an order of magnitude. Since $T_w \ll T_e$, we expect the principal effect of the space charge would be to lower the floating potential by several T_w/e below the value which corresponds to no space charge (i.e., with no potential dip). The "no

potential dip" floating potential can be estimated by balancing the electron collection current against the sum of the electron emission current and the ion saturation current I_{io}. Since $V_f > V_p$, this gives

$$I_{eo}\exp\left[\frac{-e(V_f - V_p)}{T_w}\right]g(V_f - V_p) + I_{io} = I_{co}g'(V_f - V_p). \quad (80)$$

For cylindrical wires with radius small compared to sheath dimensions, g can be approximated (see Eq. 44)

$$g = \left[\left(1 + \frac{e(V_f - V_p)}{T_w}\right)\right]^{1/2} \quad \text{and} \quad g' \simeq \left[\left(1 + \frac{e(V_f - V_p)}{T_e}\right)\right]^{1/2}. \quad (81)$$

Note that typical emissive probe radii range from 3×10^{-3} cm to 1.3×10^{-2} cm. For $e(V_f - V_p) \ll T_e$ we can approximate $g' = 1$. Neglecting I_{io}, the hot probe floating potential satisfies

$$\ln\left(\frac{I_{eo}}{I_{co}}\right) = \frac{e(V_f - V_p)}{T_w} - \tfrac{1}{2}\ln\left(1 + \frac{e(V_f - V_p)}{T_w}\right). \quad (82)$$

For example, $I_{eo}/I_{co} = 4.0$ and 10, $V_f - V_p = 1.9\ T_w/e$, and $3\ T_w/e$, respectively. In general, the floating potential is more positive than the plasma potential by several T_w/e and becomes more positive as the electrons emission increases. This behavior is apparent in the data shown in Figure 15.

C. Inflection Point in the Limit of Zero Emission Method

The basic idea of the inflection point technique (Smith et al., 1979) is to follow the inflection point of the I-V characteristic as the emission is varied to the point of zero emission. Examination of Eqs. (78) and (79) shows that when space charge effects can be neglected, the inflection point corresponds to the plasma potential. Space charge effects associated with the emitted electrons go to zero in the limit of zero electron emission. The inflection point is readily determined by taking the derivative dI/dV_B. In practice, the peak in slope is often small compared to noise and op amp turn-on transients. The peak is readily determined because the peak moves with varying emission. Data for the measured inflection point as a function of electron emission are given in Figure 17. Note the shift in the inflection

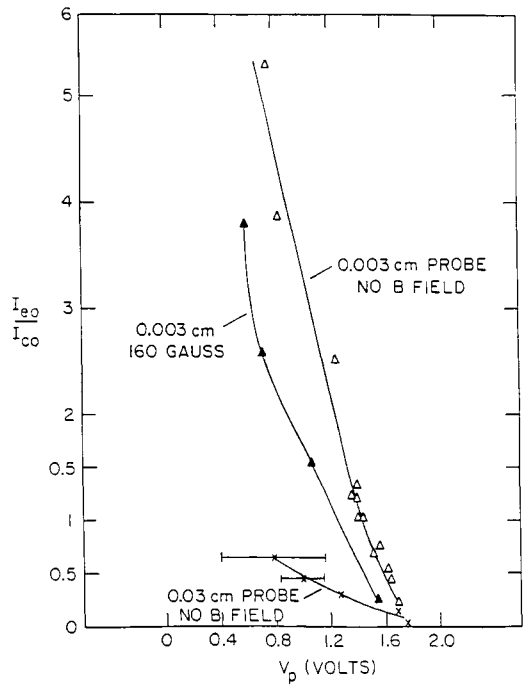


FIGURE 17. Ratio of electron emission to collection saturation current versus the inflection point V_p. Data are given for a .003 cm diameter probe with $B = 0$ and 160 G and for a .03 cm probe with $B = 0$.

point associated with increases in emission (i.e., in space charge) and that the shifts become greater as the probe radius is increased. This is because the radial electric field at the probe decreases as the probe radius is increased. Clearly, space charge effects can be reduced by choosing probes of sufficiently small radii. Note that both curves are asymptotic to the same value. The total probe current I_p is given by

$$I_p = I_{co} \exp\left[\frac{e(V_B - V_p)}{T_e}\right] - I_{eo} \qquad \text{for} \qquad V_B \leq V_p \tag{83}$$

$$I_p = I_{co}\left[\left(1 + \frac{e(V_B - V_p)}{T_e}\right)\right]^{1/2} - I_{eo} \exp\left[\left(\frac{-e(V_B - V_p)}{T_w}\right)\right] \times \left[1 + \frac{e(V_B - V_p)}{T_w}\right]^{1/2}, \qquad \text{for } V_B > V_P. \tag{84}$$

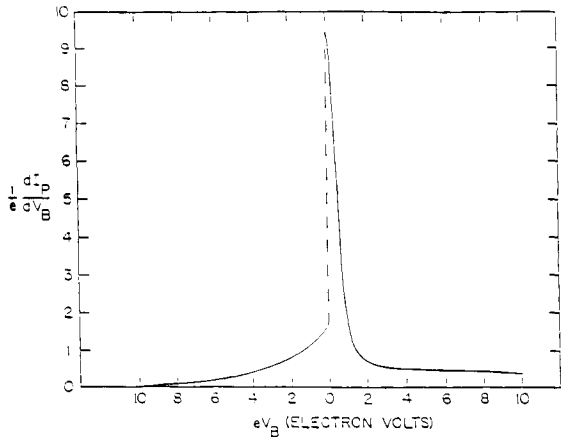


FIGURE 18. Computed dI/dV_B for an ideal cylindrical probe. $T_e = 3.0$ eV, $T_w = 0.3$ eV.

Differentiating I_p in Eq. (83) with respect to V_B gives

$$\frac{dI_p}{dV_B} = \frac{eI_{co}}{T_e} \exp\left(\frac{e(V_B - V_p)}{T_e}\right), \qquad \text{for } V_B \leq V_p \tag{85}$$

and

$$\frac{dI_p}{dV_B} = \frac{eI_{co}}{T_e}\tfrac{1}{2}\left(1 + \frac{eV_B}{T_e}\right)^{-1/2} - \frac{eI_{eo}}{T_w}\exp\left[\frac{-e(V_B - V_p)}{T_w}\right] \times\left[\left(\tfrac{1}{2} + \frac{eV_B}{T_w}\right)^{-1/2} - \left(1 + \frac{eV_B}{T_w}\right)^{1/2}\right], \qquad \text{for } V_B \geq V_p. \tag{86}$$

Equations (85 and 86) are graphed in Figure 18 for parameters representative of laboratory multi-dipole plasmas. Note the peak at $V_B = V_p = 0$.

D. Emissive Probe in Vacuum

Cho et al. (1984) have shown that the inflection point method in the limit of zero emission can be used to determine the plasma potential in a vacuum. The potential was determined between the plates of a parallel plate capacitor (plate separation equal 10 cm), with an applied potential difference of 10 V. In this case very weak probe emission is possible because small probe diameters are used, which result in relatively large electric fields at the probe.

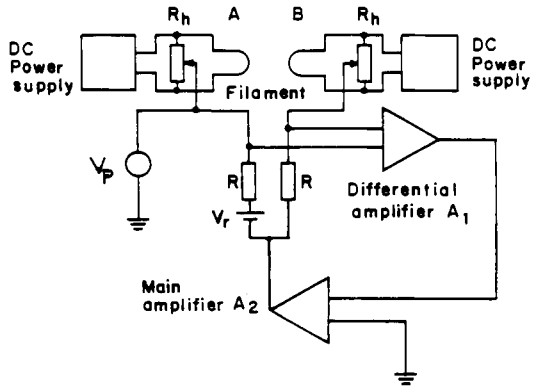


FIGURE 19. Differential probe feedback loop control circuit to give a direct indication of the plasma potential.

E. Differential Emissive Probe

Wang et al. (1985) have shown that a feedback circuit can be used to control the bias voltage of two emissive probes which are connected to a differential amplifier, as shown in Figure 19. This circuit can be used to follow fluctuations in the plasma potential. One probe operates as a warm collecting probe, heated only to reduce surface impurities. The other probe was heated to electron emission. The probe is based on an idea given by Chen (1965), who pointed out that emitting characteristics tend to diverge from collecting characteristics near the plasma potential. The voltage offset V_r of the input to the differential circuit is set to track the point of separation. This diagnostic gives a continuous readout of the plasma potential and have been operated at potentials ranging from several V to 1 kV with respect to ground.

F. Secondary Emission Capacitive Probes

The floating potential V_f gives an indication of the time variation of the plasma potential. Unfortunately, V_f depends on T_e, n_e, and n_p, in addition to V_p, so temporal variation of any of these parameters shows up as variations in V_f. Equation (32) depends on 3.3 T_e, so T_e variations are amplified. The floating potential of an insulating probe can improve this situation somewhat, because the secondary electron emission coefficient of insulators can be much greater than 1.0 for sufficiently high incident electron energy, resulting in floating potentials much closer to V_p.

Glass has been employed to construct capacitive probes to measure AC floating potentials in cold plasmas (Benjamin 1982). In plasmas with electrons having energies greater than 50 eV, the secondary electron emission coefficient $\sigma \approx 1$, so it is possible that V_f of such probes can be close to V_p. In this case fluctuations in V_f are a much better measure of fluctuations in V_p.

The case of strong secondary electron emission (where $I_{eo} \approx I_{co}$) is interesting. Sizonenko (1981) has studied this case and argued that two solutions are possible—one in which the probe floats slightly positive with respect to the plasma potential, with the potential varying monotonically between the probe and the plasma, and the other with the probe approximately T_e/e negative of the plasma potential. The second solution also has an additional potential dip between the probe and the plasma associated with a double sheath. He argues the solution that the plasma finds depends on the initial conditions.

Wang et al. (1986c) have suggested that Sizonenko's first solution is not possible because it does not simultaneously satisfy the floating condition (of zero net current) and Poisson's equation without the presence of additional ion density. Sizonenko treats the ratio of background ion density to background electron density as a free parameter which can adjust to a large density of emitted electrons, even if the emitted density is comparable to the ambient density.

Probes which employ secondary electron emission normally have much larger radii than hot wire emissive probes. In most cases the secondary emissive probe radius is greater than λ_D and the electron gyroradius and it is reasonable to treat the problem as one-dimensional.

For $V_f > V_p$ the requirements that the probe be floating and the electron and ion density balance *far* from the probe give

$$n_E v_E = I_E \exp\left[\frac{-e(V_p - V_f)}{T_E}\right] = I_c \tag{87}$$

and

$$n_E = \frac{I_c}{v_E} \sim \frac{n_e v_e}{v_E}, \tag{88}$$

where I_E is the current emitted by the probe when $V_p = V_f$, I_c is the collected electron current, and n_E is the density of emitted electrons *far* from the probe. Since the velocity of emitted electrons v_E is usually *much* smaller than v_e, the ratio n_E/n_e exceeds 1.0. Thus, no solution is possible unless the background ion density can adjust. It is possible, however, to find a solution if there is a potential dip between the probe and the background plasma or if the probe floats at a sufficiently negative potential with respect to the background plasma. In this case the emitted electron

density far from the potential minimum is reduced by $\sqrt{T_E/2e\,\Delta V}$, where the ΔV is the potential difference between the minimum and the background potential.

Sizonenko has solved this problem assuming that ions are monoenergetic with energy u_o far from the probe (i.e., there is a presheath). Wang et al. (1986c) gave Sizonenko's solution for ΔV as a function of u_o. They show that the probe tends to float more negative than the plasma potential by approximately $(0.7 \pm 0.2)T_e/e$, plus any presheath voltage drop.

Wang et al. (1986c) have made use of this property to determine the plasma potential with a capacitive-secondary emission probe in a "fusion" plasma. In this case, electrons with $T_e \geq 60$ eV caused the secondary emission coefficients of small glass tubes to be greater than 1.0, so the probe floating potential was approximately $V_p - 0.7T_e/e$. A probe was constructed (see Figure 20) which could follow fluctuations with frequencies ranging from 1 Hz to 20 MHz.

In Figure 21 experimental results obtained with a capacitive-secondary emissive probe are compared to those obtained with a self emissive probe (Hershkowitz et al. 1983), which gives a good measure of the plasma potential. Note that there is good agreement between the self-emissive and capacitive-secondary emissive probes. The straight line through the data shows that the capacitive probe floating potential is shifted below the plasma potential by approximately $0.7T_e/e$.

We can demonstrate that the secondary electron emission coefficient is sufficiently high by comparing the floating potential V_{fc} of the capacitive-

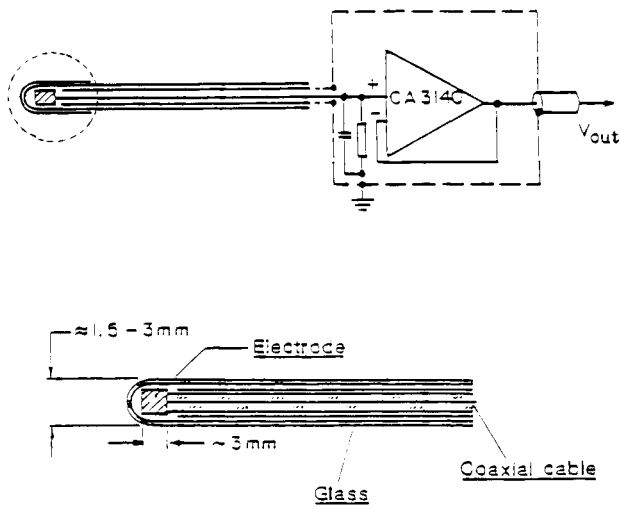


FIGURE 20. Physical construction of a capacitive probe together with probe circuit.

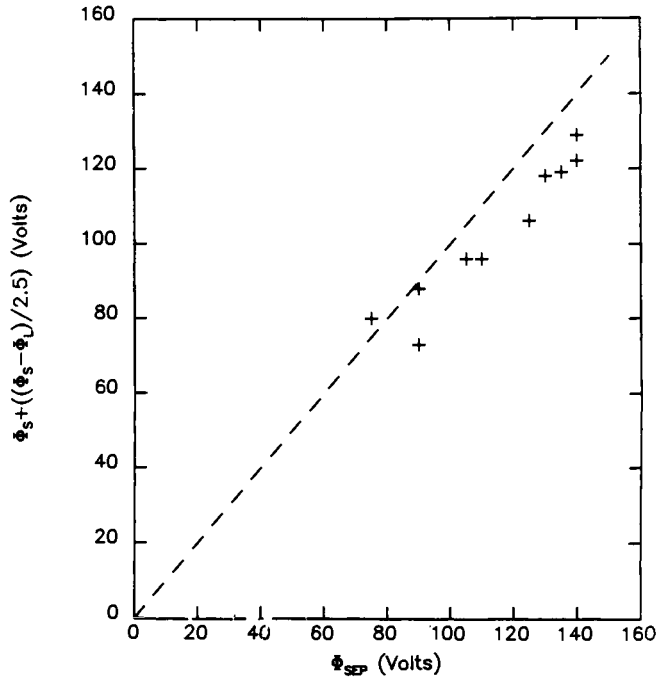


FIGURE 21. Comparison of the floating potential of capacitive-second emissive probe to the plasma potential determined by a self-emissive probe.

secondary emission probe to the floating potential V_{fm} of a metal probe (with low σ) in the same plasma. In hydrogen we expect the difference $\Delta V_f \equiv V_{fc} - V_{fm}$ to be approximately equal to $(3.3 - 0.7)T_e \simeq 2.6T_e$ when the capacitive probe has $\sigma \geq 1$. For glass, if $\Delta V_f/2.6 \geq 40$ V, then $\sigma \sim 1$. If $\sigma < 1$, V_{fc} will approach V_{fm} and $\Delta V_f/2.6$ will take on a smaller value. A value of ΔV_f of 40 V or more indicates that both the electrons are hot enough and that the secondary electron emission is also high enough for the technique to be valid.

G. Self-Emissive Probes

One problem encountered in the use of floating emissive probes in high density-high temperature plasma is that it is difficult to emit a current that is comparable to the collected electron current. A simple solution to this problem is to make use of the plasma to heat the probe. Hershkowitz et al. (1983) have shown that the floating potential of a fine wire "self-emissive"

probe is a good measurement of the plasma potential in a plasma with 4×10^{12} cm^{-3}, $T_e \sim 20$ eV, $T_i \sim 1$ keV. Self-emissive probes were found to give values of the plasma potential that agreed with gridded ion end loss analyzer energy measurements in an open magnetic mirror plasma.

Like capacitive probes, self-emissive probes have the advantage that they do not require a heating circuit and can consist of only one wire. They have the disadvantage that they require high-energy density plasmas and so they often melt and must be replaced.

The self-emissive probe used for the data shown in Figure 21 consisted of a 0.003 cm diameter tungsten wire heated to thermionic emission (and to melting) by a plasma with $n \simeq 2 \times 10^{12}$ cm^{-3}, $T_e \simeq 40 - 50$ eV. The self-emissive probe was found to be accurate by comparison with ion end loss analyzer data. Because of its small size, the self-emissive probe is not subject to space change effects, floats close to the plasma potential, and can have good frequency response. If it were not for its melting it would be preferable to capacitive probes.

VIII. Probe Perturbations

A. General Considerations

Probes always perturb their local surroundings, so they always must be used with care. The extent of the perturbation is determined by a number of characteristic lengths. These are: the probe radius a (or other characteristic dimension if the probe is not round), the electron Debye length λ_D, the mean free paths for ionization λ_i and charge exchange λ_{cx}, the ion gyroradius r_{ci}, and the electron gyroradius r_{ce}.

First consider perturbations associated with the probes physical dimensions. One simple form comes from plasma losses to the probe and probe supports. We can make a rough estimate of the extent of such perturbation by considering a probe in an unmagnetized plasma with volume V and surface loss area A_w. The particle loss times τ and τ_w associated with a particle of velocity v in a plasma, to a probe shaft plus probe of loss area A and to the walls of the system can be written

$$\frac{nV}{\tau} = nvA \qquad \text{and} \qquad \frac{nV}{\tau_w} = nvA_w. \tag{89}$$

This gives loss time $\tau = V/Av$ and $\tau_w = V/A_w v$. The total confinement time τ_T can be expressed

$$\frac{1}{\tau_T} = \frac{1}{\tau_w} + \frac{1}{\tau}. \tag{90}$$

So long as the ratio $\tau/\tau_w = A_w/A$ is large, the probe perturbation caused by losses to the probe will not be significant. However, considerable effort has been spent to reduce the effective loss area A in many fusion plasmas because improved confinement has been the goal. For example, magnetically trapped hot electrons in magnetic mirrors can have lifetimes of many seconds corresponding to effective loss areas much less than 1.0 cm^2. In such systems it is only possible to use probe diagnostics at the plasma boundaries where probe lengths can be kept to a minimum size.

Probe shadows are another form of perturbation. Shadows can become important when probes are located near boundaries and when magnetic fields are present with sources at only one end of the device. The major difficulty is that probes located near boundaries prevent plasma ions and electrons from reaching the back side of the probe, so the plasma there is quite different than it would be if no probe was present. This problem is particularly serious for probes located in sheaths, because ions acquire a directed velocity which produces more well defined shadows.

Another major perturbation by probes is caused by the desorption of gases on the probe shaft and probe by pulsed plasmas. This can result in a significant impurity concentration in the plasma near the probe. If probes are chosen to be too small, the probes can melt, resulting in the evolution of significant impurities, etc.

In magnetized systems, probes can deplete electrons from the flux tubes they are connected to and drastically change local plasma parameters.

When probe shafts are constructed out of insulators, it sometimes happens that ions trapped in potential dips in plasmas can drain out to the insulating probe shafts. This can result in potential structures in plasmas that depend on the presence of the probe for their existence (Forest and Hershkowitz, 1986).

B. Probe Measurements in Sheaths

Probe measurements of sheath characteristics are a very important application for two reasons. In fusion plasmas, sheaths are the only region where probes can survive, because plasma energy density is lowest there. In plasma processing plasmas, sheaths often play a critical role.

Sheaths are always present at plasma boundaries. It is tempting to insert probes into sheaths and to measure I-V characteristics, but this must be done with a great deal of care. The major problem encountered by probe measurements is that the probe strongly perturbs the sheath. A positively biased probe can "short out" the sheath and collect electron current far in excess of the electron current present when the probe is not present. An

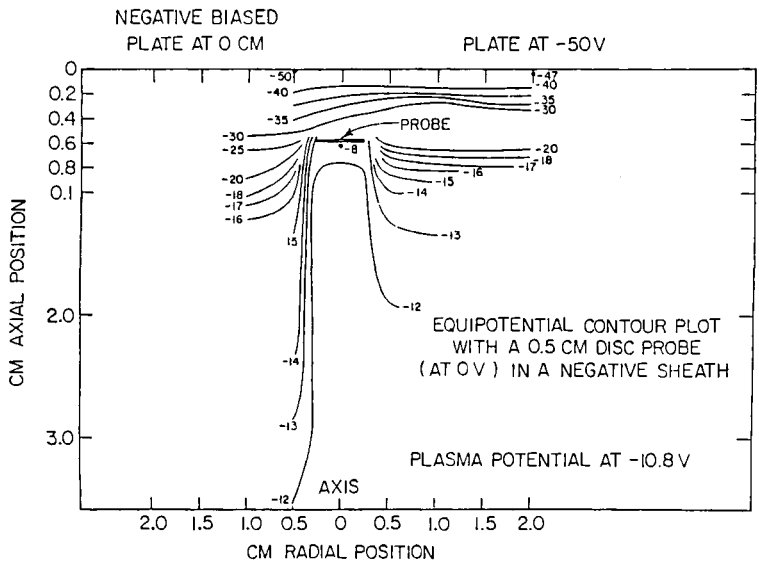


FIGURE 22. Two dimensional equipotential contours near a probe in a plasma wall sheath. The probe was biased much more positively than the potential in the absence of the probe.

additional problem encountered in sheaths is spatial variation of secondary electron emission and sputtering.

The interaction of probes with sheaths is illustrated in Figure 22 (Coakley et al., 1979). In this example a 2 cm diameter probe, biased at 0 V, was inserted into a sheath near a plasma boundary that was biased to -50 V. The electron Debye length was chosen to be 0.1 cm. The equipotential contours, measured with an emissive probe, show the probe's perturbation. The contours between the probe and the plasma are very different from those at the sides of the probe, which more closely correspond to those present without the probe. It appears as if the probe has "sucked in" or "shorted out" the contours.

Note that the floating potential of a probe in a sheath is determined by the balance of electron and ion current to the probe. For planar one-dimensional sheaths ion current is conserved, so the current balance for $V_B > V_s$ where V_s is the local sheath potential in the probe's absence (see Eq. (30)

$$0.6\, nc_s \simeq 0.4\, nv_e \exp\left[\frac{e(V_f - V_p)}{T_e}\right], \tag{91}$$

where $v_e \equiv \sqrt{T_e/m_e}$, which implies that the probe will float at approximately $3 - 4T_e/e$ more negative than the background potential, indepen-

dent of the true sheath potential, and will not indicate the true sheath potential. It appears, at least to a first approximation, that a planar probe in a sheath has the same I-V characteristic as a probe in the background plasma far from the sheath.

Even if the size of a probe is chosen to be sufficiently small to avoid perturbations, secondary electron emission associated with ion bombardment of the probe must be taken into account. This is a nontrivial problem when the probe is not clean.

C. Paddle Probes

In plasma, where a net drift velocity is present, it is sometimes useful to make use of "paddle probes." Paddle probes consist of two collecting Langmuir probes mounted parallel to each other with an insulator placed between them. The difference in electron or ion saturation current to the two probes gives a measure of the plasma drift. Note that although such probes consist of two probes, their operation is quite different from that of "double probes" (see later discussion).

The principal problem encountered with the use of paddle probes is the perturbation caused by the probe's shadow, especially when magnetic fields are present. If the drift velocity is substantially greater than the beam thermal velocity spread, it is unlikely that the probe mounted on the downstream side will detect any current. If the drift velocity is comparable to the thermal velocity, shadowing will depend on the details of the probe geometry, the presence or absence of magnetic fields, and on the characteristics of the plasma. In general, such probes give an indication of plasma drifts but require a separate calibration.

D. Glow Discharge Data—Ionization

When high neutral pressure ($\geq$ 100 mtorr), low fractional ionization plasmas are present, ionization mean free paths become comparable to probe dimensions. Then ionization near the probe can be significant. This additional ionization becomes possible when the probe is biased at $V_p + \epsilon_i$, where ϵ_i is the ionization potential. With short mean free paths the disturbance is not felt far beyond the probe. The local ionization produces increased electron collection and is also apparent as an additional glow discharge near the probe.

Figure 23a gives the I-V characteristic for a DC glow discharge operated at 100 mtorr [Cho, 1987]. The plasma potential determined by using an

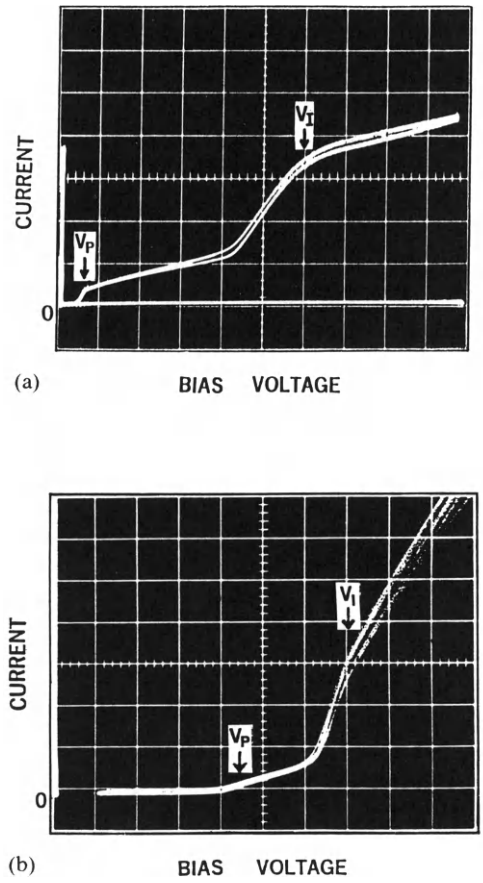


FIGURE 23. (a) I-V characteristic for argon DC glow discharge plasma. The plasma potential is indicated by V_p. The knee at V_I corresponds to V_p + the ionization potential. (b) I-V characteristic for argon AC glow discharge.

emissive probe is indicated by V_p. The knee at V_I corresponds approximately to $V_p + \epsilon_i$. It is interesting to observe that it has the appearance of an ion beam (see Section III.C.1). The relative size of the ionization knee is found to increase with neutral pressure. It is easy to misidentify the knee at the ionization potential as the knee at the plasma potential, because the latter can be quite a bit larger than the former. For more positive bias voltages, the probe current continues to increase as the sheath increases.

Similar I-V characteristics are present for V_B near the most negative values of $V_p(t)$ in capacitive rf discharges, as shown in Figure 23b (Cho, 1987).

IX. Fluctuating Plasma Potentials

A. Simple Model

Plasmas are rarely quiet at all frequencies. Abundant sources of free energy are available (e.g., energetic electron beams from cathodes in processing plasmas, unfilled loss cones in mirror plasmas, energetic ion beams, E × B drifts, etc.) The free energy drives instabilities over a wide range of frequencies. Another prominent source of noise is rf applied to heat and to create plasmas. Ignoring the effects of plasma noise can, and probably often does, lead to serious errors in the interpretation of probe data.

The effect of plasma potential fluctuations on measured I-V characteristics is sensitive to the plasma sheath impedance and the probe circuit's impedance to ground. The sheath impedance for $\omega \ll \omega_{pi}$ is

$$Z_s^{-1} \equiv \frac{\partial I}{\partial V}. \tag{92}$$

For Maxwellian plasmas $Z_s \simeq T_e/I_e^*$. For the simple circuit for determining the I-V characteristics shown in Figure 24, the circuit impedance to ground is R_L. The operating point of the circuit, with a battery supplying a voltage V, is given by the intersection of the load line

$$V_B = V - I_e R_L \tag{93}$$

with the I-V characteristic.

We can get an idea of the role of the fluctuations by considering the ideal I-V characteristics for a planar Langmuir probe shown in Figure 25. In the presence of a fluctuating plasma potential $V(t) = V_p + \tilde{V}_p(t)$, the trace can be thought of as oscillating in the horizontal direction. The extremes of this oscillation are indicated in Figure 25. If we choose $R_L \ll Z_s$, the load line given by Eq. (93) is approximately the vertical line $V = V_B$.

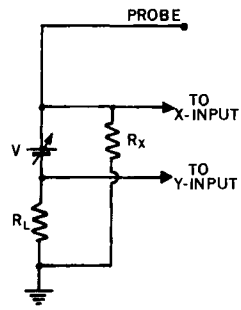


FIGURE 24. Simple Langmuir probe circuit for determining I-V characteristic using an *xy* recorder.

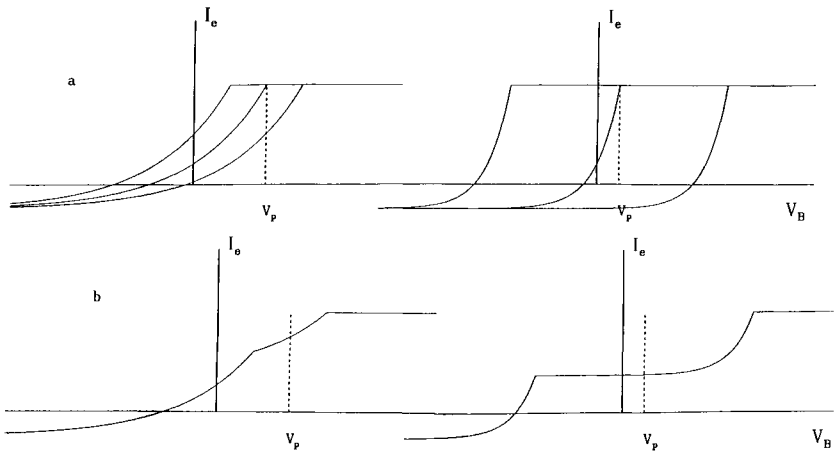


FIGURE 25. I-V characteristics in the presence of an oscillating plasma potential together with characteristic with $\tilde{V} = 0$. (a) A square wave oscillating potential is assumed with $V_p \pm T_e/2$ and $V_p \pm 4T_e$. (b) Averages of the curves shown in Figure 23a. Such averages would be observed if a very low impedance probe circuit is used.

When rf is present we can write the probe current as

$$I_e = \left(I_e^* + \tilde{I}\right) \exp\left[\frac{-e\left(V_p + \tilde{V}_p - V_B\right)}{\left(T_e + \tilde{T}_e\right)}\right], \tag{94}$$

where $\tilde{I}$, $\tilde{V}_p$, and $\tilde{T}_e$ are the fluctuating components of the saturation current, plasma potential, and electron temperature, respectively. Assuming that $\tilde{T}_e = 0$, the time average of I_e measured with R_L can be written

$$\langle I_e \rangle = I_e^* \exp\left[\frac{-e\left(V_p - V_B\right)}{T_e}\right] \left\langle \left(1 + \frac{\tilde{I}}{I_e^*}\right) \exp\left[\frac{-e\tilde{V}_p}{T_e}\right] \right\rangle. \tag{95}$$

It is easy to see the qualitative effect of a fluctuating potential by considering a square wave fluctuation in V_B and assuming that $\tilde{I} = 0$ and that T_e remain constant during the fluctuation. In that case the time average is equal to

$$\begin{aligned} \langle I_e \rangle &= I_e^* \exp\left[-e\left(V_p - V_B\right)/T_e\right] \cosh\left[e\tilde{V}_p/T_e\right], && V_B < V_p - \tilde{V}_p \\ &= I_e^* \exp\left[-e\left(V_p - V_B\right)/T_e\right], && V_p - \tilde{V}_p < V_B < V_p + \tilde{V}_p \\ &= I_e^*, && \tilde{V}_p + V_p < V_B. \end{aligned} \tag{96}$$

Equation (96) is graphed in Figure 25b. This result can be thought of as due to the rectification of the rf voltage by the nonlinear conductivity of the probe sheath.

B. Sinusoidal RF

The result of current averaging over a sinusoidal waveform (with amplitude $\tilde{V}$) is shown in Figure 26. It is evident that the most negative portion of the curves resembles an I-V characteristic without rf. It is easy to show that this portion of the curve is proportional to $\exp[-e(V_p - V_B + \tilde{V})/T_e]$ and so can be used to determine T_e. The curve continues to increase and eventually saturates for $V_B \geq V_p + \tilde{V}$. Nonideal probe characteristics differ significantly from that shown in Figure 26 because of ionization and sheath effects when V_B is much more positive than V_p.

The results of current averaging over a half-wave rectified sine wave is shown in Figure 27. Note that the knee in the time averaged curve corresponds to V_p, the flat value of the oscillating plasma potential. It is also apparent that the electron saturation current is half the value for the DC case. It is easy to show that when the waveform is only a partially rectified sine wave, the saturation current gives $1 - \alpha$ times the DC

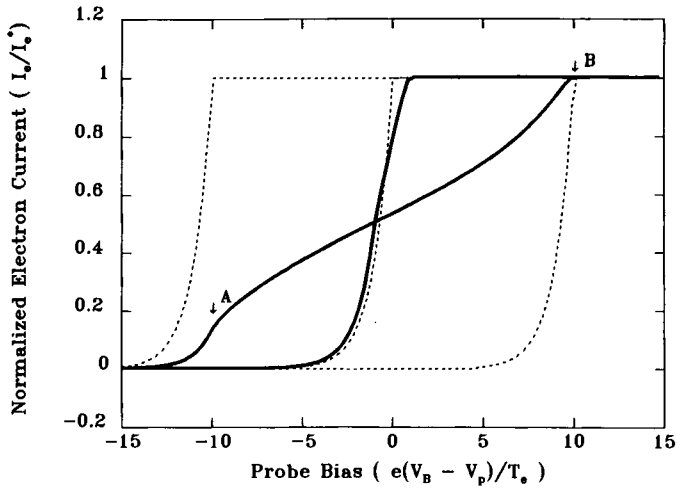


FIGURE 26. Current averaged I-V characteristic corresponding to a sine wave plasma potential variation. $\tilde{V}$ is chosen to be T_e/e and $10\,T_e/e$.

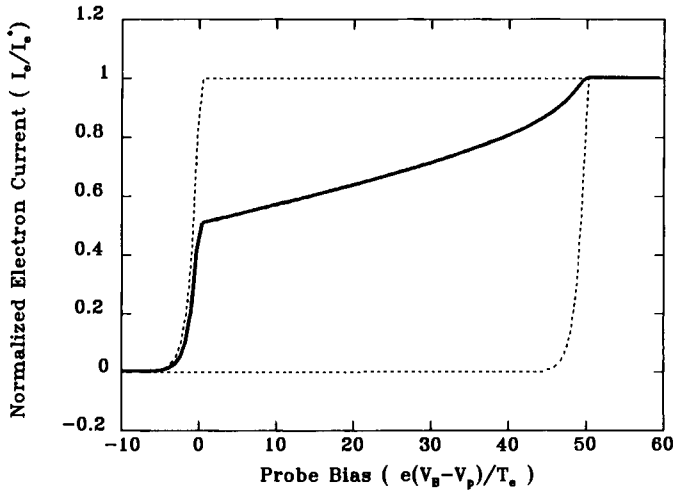


FIGURE 27. Current averaged I-V characteristic for half-wave rectified sine wave plasma potential variation.

saturation current, where $1 - \alpha$ is the fraction of the period that the waveform is constant.

For sinusoidal fluctuations, the apparent knee tends to flatten and move to a more positive potential, and the floating potential becomes more negative. If $V_B < V_p - \tilde{V}$, the time-averaged quantity does not depend on V_B, so T_e determined from the slope of ln I vs. V_B remains unchanged. For large $e\tilde{V}/T_e$, the *apparent* electron temperature, obtained from the average slope of the I-V characteristic, increases and the apparent plasma density decreases.

For sinusoidal rf (Crawford 1963) with $\tilde{I} = 0$, the electron current is shifted by

$$\langle I_e \rangle = I_e^* \exp\left[\frac{-e(V_p - V_B)}{T_e}\right] I_o\left(\frac{e\tilde{V}}{T_e}\right), \tag{97}$$

where I_o is the Bessel function of the imaginary argument. Again the electron temperature, determined from the straight portion of the (ln I) vs V characteristic, will be accurate if $e\tilde{V}/T_e \leq 1$. Balancing electron and ion current to the probe gives a change ΔV_f in the probe's floating potential

$$\Delta V_f = -\left(\frac{T_e}{e}\right) \ln\left[I_o\left(\frac{e\tilde{V}}{T_e}\right)\right]. \tag{98}$$

This phenomena is known as rf self-bias (Garscadden and Emeleus, 1962). If the rf voltage $\tilde{V} \gg T_e/e$, significant rounding occurs at the knee, and for very large $\tilde{V}$ very little remains of the original I-V characteristic. For large values of $e\tilde{V}/T_e$ the Bessel function can be approximated by

$$I_o(e\tilde{V}/T_e) \approx \left[1/\sqrt{\frac{2\pi e\tilde{V}}{T_e}}\right] \exp\left[\frac{e\tilde{V}}{T_e}\right]. \tag{99}$$

When high frequency rf is present, stray capacitance to ground normally results in an AC impedance to ground, which can be much smaller than R_L. The measured current and voltage fluctuations then depend on both the DC load line and the AC load line shown in Figure 28. The AC load line has slope $|(1/R_L + j\omega C)|$, where $j \equiv \sqrt{(-1)}$ and C is the stray capacitance. We define the *AC current* as the difference in current between the AC operating loadline and the operating point (see Figure 28). The operating point of the circuit is determined by the intersection of the DC load line with the AC load line in such a way that the time-averaged AC current at the operating point is zero. The AC load line can be thought of sliding along the DC load line (while maintaining constant slope) until this condition is satisfied. For large values of ωC, the AC load line becomes approximately vertical (i.e., $Z_L \to 0$), while at low frequencies $Z_L \to R_L$. For low frequency operation, the floating potential can be measured by

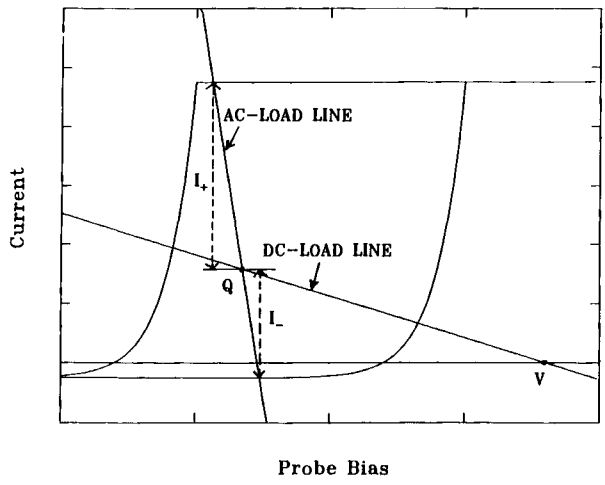


FIGURE 28. I-V characteristic together with high impedance DC load line and low impedance AC load line.

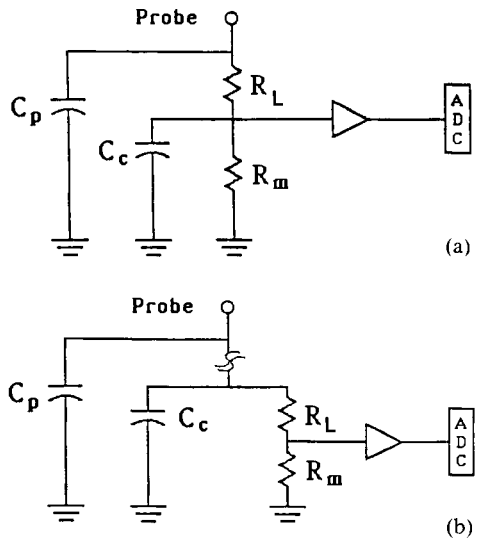


FIGURE 29. Probe bias circuit with load resistor R_L mounted close to the probe tip. (b) Probe bias circuit with R_L at the end of a coaxial cable.

choosing $R_L \gg Z_s$. Then the load line is close to the horizontal axis and intersects the I-V characteristic at a point close to where $I_e = 0$. It is important to note that because of the sheath rectification of the rf the time-average floating potential can be quite different from the floating potential of the time-average current.

If we assume sinusoidal fluctuations, the time-average floating potential is equal to V_p, the value of the floating potential if no rf were present. This can be compared to the floating potential for the time-average I-V characteristic shown in Figure 26.

When stray capacitance becomes large, the measurements of the floating potential depend on the placement of the resistor R_L. Figures 29a and 29b correspond to situations in which R_L is placed at the probe and buffer amplifier end of a coaxial cable, respectively. The cable and probe capacitances to ground are C_c and C_p, respectively, and R_m is the buffer amplifier impedance to ground. Assuming that $R_m \gg 1/(\omega C_c)$ and $R_L \gg 1/(\omega C_c)$, the corresponding AC load lines have slopes $|j\omega C_p + 1/R_L|$ and $|j\omega(C_p + C_c)|$, respectively. If $\omega C_p R_L \ll 1$, then putting the resistor close to the probe results in an AC load line slope equal to $1/R_L$. If R_L is also large compared to the sheath capacitive impedance, it gives a good measurement of the time-average floating potential. Putting the resistor at the end of the

coaxial cable or choosing $R_L \ll (1/C_c\omega)$ can lead to a near vertical AC loadline, which then results in a floating potential that is much more negative than the time-average floating potential.

C. Techniques

A variety of techniques have been developed to extract data from probes in noisy environments. They can be divided into several categories: 1) slow, time-averaged I-V scans, 2) fast I-V scans, 3) time-sampled I-V scans, 4) following techniques, and 5) filtering techniques. All of these techniques have been applied to both collecting and emitting probes.

1. Slow, Time-Averaged I-V Scans Using Emissive Probes

In principle it is possible to extract the rf fluctuation level from time-averaged collecting probe traces. However, the interpretation of traces, such as those shown in Figure 26, are complicated by the need for additional information about the electron distribution function. Emissive probes simplify the interpretation of rf fluctuation data because when probes are operated with $I_{eo}/I_{co} \gg 1$, their characteristics depend mostly on electron emission. This is even more true for differential emissive probes (Wang et al., 1985) for which the electron collection contribution is cancelled out.

Unlike collecting probes, emissive probes can be used to give a good indication of the fluctuations in the plasma potential. Figure 30a shows emissive probe traces with and without the presence of rf (Wang et al., 1986b). It has been shown that the spread between peaks in the differentiated I-V trace (shown in Figure 30b) is given by

$$V_{p2} - V_{p1} \simeq 2\tilde{V} + V_e - V_c, \tag{100}$$

where $\tilde{V}$ is the amplitude of the rf fluctuation, V_e the potential corresponding to the onset of electron saturation current, and V_c the point corresponding to the onset of ion saturation current. For large fluctuation levels ($e\tilde{V}/T_e \gg 1$), the peak separation is approximately equal to twice the rf fluctuation level.

2. Fast I-V Scans

A simple solution to the problem of fluctuations is to sweep the probe faster than the fastest fluctuation. Unfortunately, this is often difficult to do

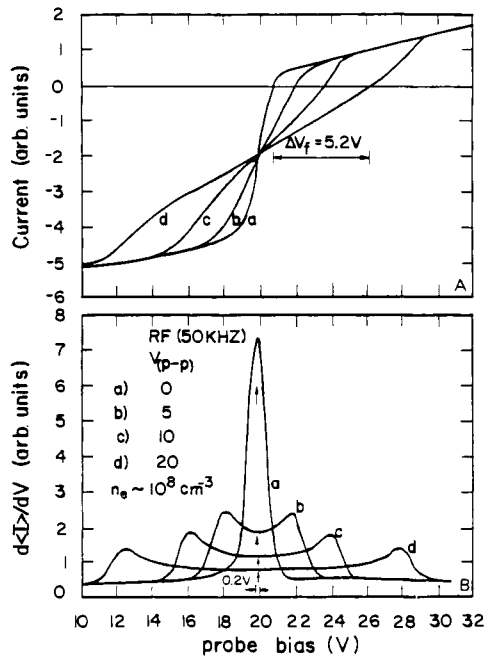


FIGURE 30. Time-averaged emissive probe characteristics in the presence of rf. (a) $\langle I \rangle$ versus V_B for several rf amplitudes. (b) $d\langle I \rangle / dV_B$ for the same rf amplitudes as in Figure 25a.

because of the inevitable stray capacitance of the probe sweep circuit. The main contribution is usually the capacitance of the cable connecting the probe to the amplifier. The time constant of the circuit is then limited to $1/RC$, where R is the sum of the values of R_L and Z_s.

Fluctuations in the floating potential can give a good measure of the fluctuations in the plasma potential. For floating potential measurements, it is easy to improve the frequency response of the circuit by using a doubly shielded probe (Chen 1985) and by placing a unity gain amplifier across the stray capacitance C (see Figure 31). This technique (known as capacitance neutralization) reduces the potential drop across C to nearly zero, so no current flows through C. This circuit increases the frequency response by a factor of $(1 - G)^{-1}$, where G is the gain of the unity gain amplifier. For G very close to 1.0, the frequency response is limited by the frequency response and/or input capacitance of the op amp. The op amp must have an output impedance sufficiently low to drive the capacitance C_1.

For low frequency operation, the floating potential can be measured by choosing $R_L \gg Z_s$. Then the load line is close to the horizontal axis and

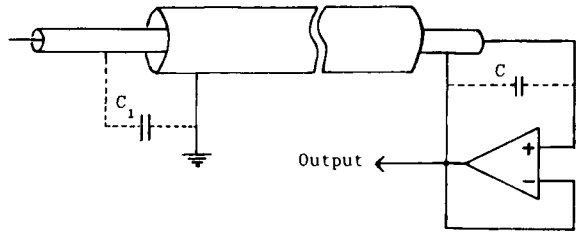


FIGURE 31. Circuit to improve frequency response by adding a unity gain amplifier across the stray capacitance.

intersects the I-V characteristic at a point close to where $I_e = 0$. At low frequencies, plasma parameters may vary throughout the cycle. It may not be valid to assume that T_e or I_e^* is constant since plasma production can change throughout the cycle.

3. *Time-Sampled I-V Scans*

Time sampling using standard boxcar averaging techniques provides a convenient way to measure I-V characteristics with very high time resolution. This technique can be expected to work well if noise is periodic and reproducible (e.g., when the noise is associated with an rf source rather than plasma instabilities).

4. *Following Techniques*

In the "following technique" (Gulidov et al., 1970) rf is added to a slowly swept probe bias voltage. By varying the amplitude and phase of the coupled rf, it is possible to vary the bias applied to the probe so that it follows the rapid changes in the plasma potential. A transformer can provide a convenient way to couple the rf. This technique can be useful if the probe noise is dominated by one frequency (e.g., the rf frequency used to produce an rf plasma).

5. *Filtering Technique*

A blocking choke coil can be inserted into the probe circuit in order to filter out the rf component $\tilde{V}$. If a high rf choke AC impedance is combined with

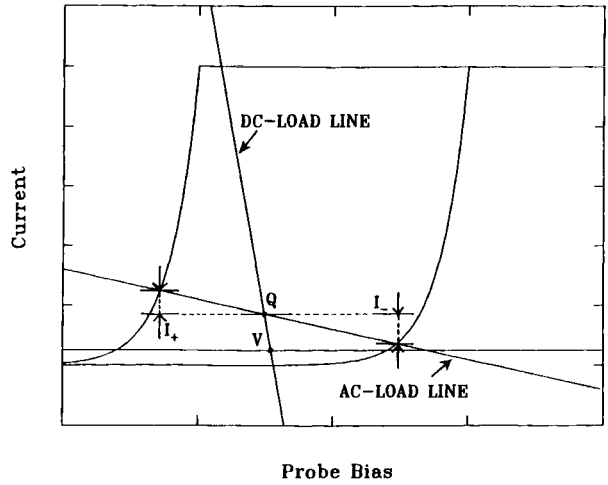


FIGURE 32. I-V characteristic together with high impedance AC and low impedance DC load lines.

a small DC impedance, the resulting load lines are given in Figure 32. In this case, the requirement that the time average of positive and negative AC currents be zero results in an I-V characteristic which corresponds to $\tilde{V} = 0$, i.e., to the I-V characteristic that would be present if *no* rf were present.

This technique requires the AC impedance of the coupling element Z_c to be much greater than the maximum probe sheath impedance, which occurs near the floating potential. Godyak and Popov (1977) have expressed this condition as

$$\frac{e\tilde{V}}{T_e} \leq \frac{Z_c}{Z_s}. \tag{101}$$

Note that in experiments where a variety of frequencies are employed, the condition may only be satisfied at high frequencies.

X. Probe Contamination

Consider a probe which has a resistive layer with resistance R on the front surface as the result of surface impurities. If a bias voltage V_B is applied to the probe, the bias that is applied to the surface exposed to the plasma will be reduced to $V_B - IR_r$, where I is the net current. If the probe is biased near the floating potential, the net current will be small and the applied bias

will be close to the actual bias. On the other hand, when the probe is biased at V_p, the actual potential will be $V_p - I_e{*}R_r$. The actual bias will not equal V_p until the applied bias is greater than $V_p + I_e{*}R_r$. It is clear that the apparent plasma potential, determined from the knee of the measured I-V characteristic, will be high, that the apparent electron temperature will also be high, and that the error will depend on the value R_r of the resistive layer.

Another source of error in determination of the plasma potential from the knee is variations in the surface work function caused by impurities. Variations in the surface work function can be as large as 1 V. Probe contamination is often apparent in hysteresis in I-V characteristics (see XI.B)

Heating probes red- to white-hot by biasing probes more positive than V_p can provide clean probe surfaces. However, if base vacuums are $\sim 10^{-6}$ torr, monolayers of contaminants can form in several minutes and cleaning must be carried out often.

XI. Langmuir Probe Circuits

A. Simple Circuit

The easiest way to determine I-V characteristics is to use the circuit shown in Figure 24. The voltage V_B at the probe is measured across R_x and the current to the probe is measured by the voltage drop across resistor R_L. The voltage is swept by varying the power supply V. The current measured for applied voltage V is given by the intersection of the load line with the probe's I-V characteristic. By applying the voltage across R_L to the vertical scale of an xy recorder and the voltage V_1 to the horizontal scale, it is easy to trace out the I-V characteristic. The only concern is to make sure that the input impedance R_x of the xy recorder is much greater than R_L. This problem is particularly noticeable when looking at the effects of energetic primary electrons, which can have energies in excess of 100 eV and so require bias voltages more negative than -100 V. In that case V_B/R_x, where R_x is the input impedance of the x input, may be comparable to the collected current.

Another simple circuit (Nelson 1986) which has been used for measuring probe characteristics in pulsed systems (with $n \simeq 10^{12}\ \mathrm{cm}^{-3}$, $T_e \simeq 20$ eV) is given in Figure 33. In this circuit the probe current and voltage are

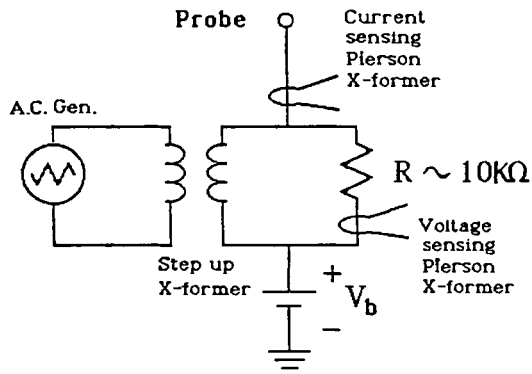


FIGURE 33. Simple circuit for determining I-V characteristic of pulsed high density plasma.

determined by measuring currents with Pierson transformers, and the probe bias is varied by using a transformer.

B. Curve Tracer

Transistor curve tracers (e.g., a Tektronix model 575) provide packaged circuitry that is convenient for determining I-V characteristics. All that is necessary is to connect the probe across the collector and base leads. The curve tracer may also provide an easy way to clean probes if they are not too large. One need only bias the probe with a sufficiently large positive bias (usually several hundred volts). The electron input power may then be sufficiently large to heat the probe white-hot. Type 575 curve tracers have the advantage that they can provide quite high currents and voltages. The advantage of using the curve tracer to bias the probe is that variations in work function are immediately apparent as shifts in the knee of the I-V trace and are also apparent as changes in electron saturation current. One disadvantage of the curve tracer is that I-V characteristics are swept at a fixed frequency, 120 Hz.

Hysteresis can be observed in I-V traces because the curve tracer sweeps the probe bias voltage in both directions. This hysteresis is an indication of dirty probes. Double valued I-V traces (see Figure 34) indicate that the probe surface characteristics are changing with applied bias voltage. Two possibilities are that the surface work function is changing, resulting in a horizontal shift in the I-V characteristic and that there are changes in

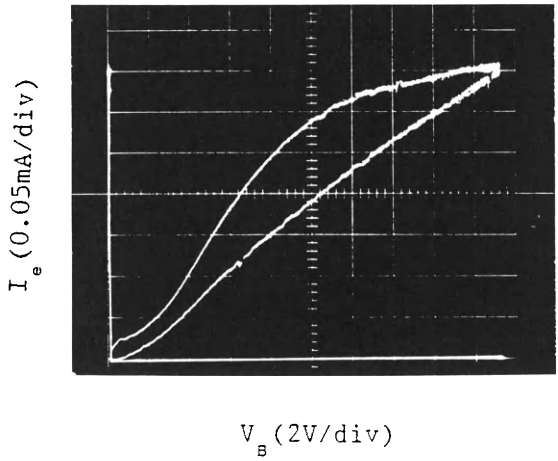


FIGURE 34. I-V characteristic of a dirty probe (drawn from curve tracer data) showing hysteresis.

surface impurity or perhaps oxide level, resulting in a change in secondary emission coefficient.

C. Op Amp Circuit

Another useful circuit (Wong, 1977) developed at UCLA is shown in Figure 35. This circuit makes use of battery powered op amps and uses the sawtooth voltage of an oscilloscope to provide the voltage ramp which biases the probe. By displaying both the ramp voltage and the voltage across a resistor between the battery common lead and ground, the I-V characteristics can be determined. It has the interesting feature that the probe current is carried by the battery common lead. Circuit current is limited by either the current capabilities of the batteries used to power the circuit or the output current capabilities of the op amp. The probe bias voltage swing is limited by the voltage capability of the op amps and the batteries. This circuit has the advantage that the only limitation to the sweep frequency besides the slew rate of the op amp is the time response determined by the probe's capacitance, the output impedance of the op amp, and the load resistor. By operating the oscilloscope time base at a high frequency, the I-V characteristic can rapidly be swept. Probe operation with short pulses at low duty cycles has a distinct advantage in situations where long applied biases more positive than the floating potential can deplete flux tubes or overheat the probe because of high power dissipation.

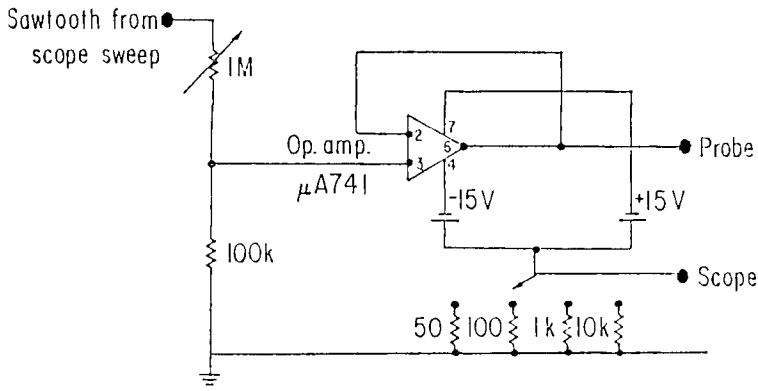


FIGURE 35. Op amp circuit for determining I-V characteristic by making use of an oscilloscope sawtooth.

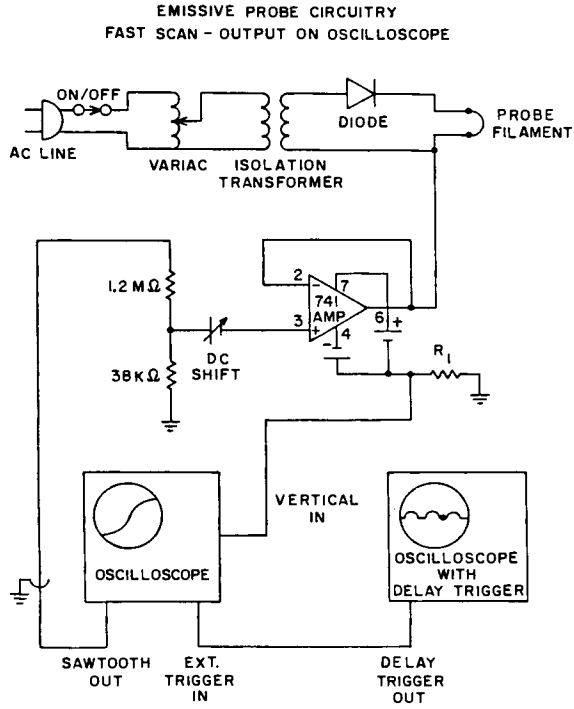


FIGURE 36. Circuit for fast I-V scans of an emissive probe. The oscilloscope sawtooth is used to generate the probe bias voltage.

D. Emissive Probe Circuit

A convenient circuit for determining emissive probe I-V characteristics is shown in Figure 36. This circuit half wave rectifies the heating voltage so that data can be taken during the half of the cycle when the heating voltage is 0. The hot wire cools down to e^{-1} of its initial temperature in approximately 5 ms, so 60 Hz is a convenient frequency to work at and a transformer–diode rectifier is readily employed. The I-V characteristic can then be measured with the modification of the circuit shown in Figure 35, as shown in Figure 36.

E. Probe Construction

Probes are normally constructed out of refractory materials (tungsten, tantalum, graphite) or materials which are chemically inert (platinum). These choices enhance their ability to survive and allow heating to high temperature for cleaning.

Tantalum and tungsten disc probes are readily constructed by spot welding metal discs to tungsten- or nickel- (or gold-) plated wire. Spherical probes are constructed by melting the end of a platinum wire with a high voltage arc. In some designs provision is made for water or air cooling the probe tips.

Probe shafts can be constructed out of conductors (stainless steel is convenient for low density temperature plasmas) or out of ceramic or glass insulators. Vacuum seals can be made either at the probe or at the end of the probe shaft farthest from the probe. The latter choice has the advantage of removing the seals from the high-energy density plasma and is more reliable in practice. The former choice allows reduction of stray capacitance by placing electronic components close to the probe.

XII. Direct Electron Temperature Measurements

A. Double Probes

In some situations (i.e., when a large electrode is not present), a Langmuir probe can become the dominant loss area of plasma from a system. In that case the probe determines the plasma potential. This situation can be avoided by making use of double probes (Johnson and Malter, 1950). Double probes consist of two identical Langmuir probes. A bias potential is

applied between them and the entire system is allowed to float, so there is no net current. The current which flows *between* the probes is measured as a function of bias voltage. Using such probes it is possible to determine electron temperature and plasma density.

The current measured by each probe is given by the expression for single Langmuir probes. The potential profile between the two probes is given schematically in Figure 37. Since the system is floating, the ion and electron currents to the probe must satisfy

$$i_{1+} + i_{2+} = i_{1-} + i_{2-}. \tag{102}$$

The current I in the loop can be written

$$I = i_{1+} - i_{1-} = i_{2-} - i_{2+}. \tag{103}$$

Using Eq. (12) with V_p taken equal to zero gives

$$i_{1-} = i_{1-}^{*} \exp\left(\frac{eV_1}{T_e}\right). \tag{104}$$

Equations (102)–(104) can then be manipulated to give

$$I = i_{+} \tanh\left(\frac{eV}{2T_e}\right), \tag{105}$$

where $i_{1+} \simeq i_{2+}$, valid if the two probes have the same area, has been written as i_+, and V is the relative bias voltage defined $V \equiv V_1 - V_2$. Equation (105) is graphed in Figure 38. Note that the slope dI/dV at zero relative bias equals $e/2T_e$ and is inversely proportional to T_e.

It is easy to show (Chen 1965) that when the two probes have different areas, the derivative dI/dV evaluated at zero relative bias voltage is

$$\left[\frac{dI}{dV}\right]_0 = \frac{e}{T_e} \frac{i_{1+} i_{2+}}{i_{1+} + i_{2+}}. \tag{106}$$

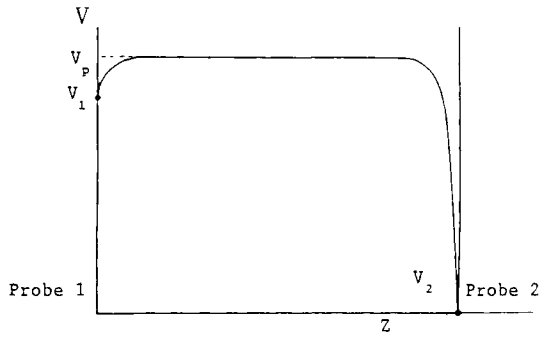


FIGURE 37. Schematic of the plasma potential between the two probes of a double probe.

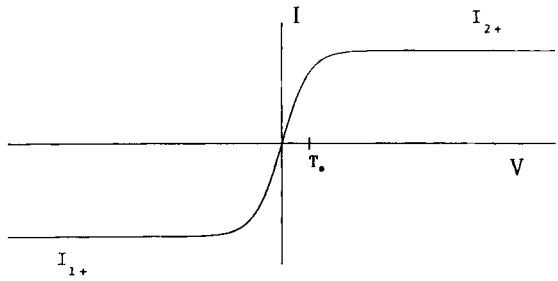


FIGURE 38. Current-voltage characteristic for an ideal double probe. Here the voltage is the relative voltage between the two probes. In real plasmas, finite probe size results in finite slope in the I-V characteristic above the knee.

Measurements of the currents i_{1+} and i_{2+} can be combined with $dI/dV|_0$ to give a continuous readout of T_e. Note that this technique provides a measurement of T_e at the floating potential. Double probes have proven to be extremely useful in determining the temperature in fluctuating systems because they float with the fluctuations. The floating potential, however, is sensitive to fast electron populations and to changes in electron density and temperature.

Godyak et al. (1967) have shown that in the presence of rf a probe floating potential shifts by

$$\Delta V_f = \frac{-T_e}{e} \frac{\ln\left[I_o\left(\frac{e\tilde{V}}{T_e}\right)\right]}{(1+\beta)}, \tag{107}$$

where β is defined

$$\beta \equiv \left(\frac{T_e}{eI_i^*}\right)\frac{\partial I_p}{\partial V}\bigg|_{e(V_f - V_p) \ll T_e}, \tag{108}$$

with I_p equal to the instantaneous current. The factor β takes into account the increase in collected ion saturation current as V_f is made more negative.

B. Admittance Probes

Admittance probes (Sprott, 1968) were invented to provide measurements of the direct electron temperature. Like double probes, admittance probes determine T_e from the slope of the I-V characteristic at the floating

potential. The plasma admittance Y_s is determined using a capacitive bridge. The plasma sheath admittance can be defined: $Y_s \equiv 1/R_s \equiv -(dI/dV_B)$ evaluated at V_f. For Maxwellian electrons (see Eq. (12)), $-dI/dV_B = eI_e/T_e = eI_i^*/T_e$ at the floating potential. Once the slope is known at the floating potential, T_e is known if I_i^* is measured separately. Using the Bohm current for I_i^* (valid when $T_e \gg T_i$), the sheath impedance can be written

$$R_s(\Omega) = 1.0 \times 10^{13} \frac{\sqrt{T_e(eV)\mu}}{n(\text{cm}^{-3})S(\text{cm}^2)}, \tag{109}$$

where μ is the ion mass normalized to hydrogen. Note that R_s is inversely proportional to the probe area and plasma density. The admittance probe is connected to one side of a capacitive bridge, as shown in Figure 39. The bridge output is nulled in the absence of plasma by adjusting the impedance Z_2. The presence of plasma unbalances the bridge because of the addition of the complex admittance Y_s in parallel with Z_3.

$$Y_s = 1/R_s + j\omega C_s = \frac{1}{Z_s}, \tag{110}$$

where ω is the oscillator frequency and C_s is the sheath capacitance. The bridge output is

$$V_{\text{out}} = \left[\frac{Z_2 \Delta Z_3}{(Z_2 + Z_3)^2}\right] V_{\text{osc}}, \tag{111}$$

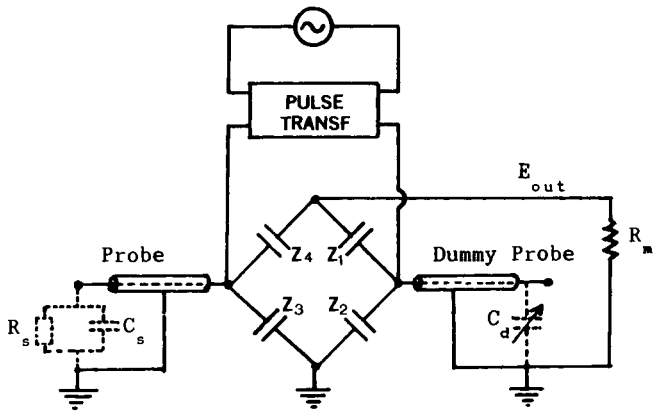


FIGURE 39. Admittance probe bridge circuit.

where ΔZ_3 is the change in the impedance introduced by the plasma

$$\Delta Z_3 = Z_3 - \frac{Z_3 Z_s}{(Z_3 + Z_s)}. \tag{112}$$

When $Z_s \gg Z_3$, $\Delta Z_3 = Z_3^2/Z_s$, so V_{out} is proportional to V_{osc}. If ω is chosen to be sufficiently small, Y_s can be approximated by $1/R_s$. Then

$$V_{out} \simeq \left(\frac{1}{R_s}\right) \frac{Z_2 Z_3^2}{(Z_2 + Z_3)^2} V_{osc} \propto T_e^{-1/2}. \tag{113}$$

This technique then gives a direct measurement of T_e as a function of time. Absolute measurements of T_e require a separate simultaneous measurement of I_i*, which can be obtained with a separate low impedance probe.

One problem with this technique and with the use of double probes is that measurements are made at the floating potential and therefore are indicative of tail rather than bulk temperature, when a tail is present (as it often is in fusion plasma).

XIII. Directional Probe

As a final topic, we consider an important new development—directional probes. Stenzel et al. (1983) have developed a directional velocity analyzer for determining electron velocity distribution functions. This device can determine anisotropic distribution functions from the first derivative of the collected current. The analyzer selects a narrow cone in velocity space by addition of a multichannel plate (containing many cylindrical holes (radius $R \sim 0.1$ mm, length $L \sim 1.5$ mm) and several retarding grids in front of a planar Langmuir probe. The current density is given by

$$j = e \int_{v_{min}}^{\infty} \int_0^{\theta_{max}} \int_0^{2\pi} v \cos\theta \, \mathrm{f}(v, \theta, \phi) v^2 \sin\theta \, dv \, d\theta \, d\phi, \tag{114}$$

where the angular integration is limited by geometry to satisfy $v \cos\theta_{max} = v_{min} \equiv \sqrt{2e(V_p - V_B)/m}$ or $\tan\theta_{max} = R/L$, whichever gives a smaller angle. For small angles the integral can be approximated

$$j \simeq e\pi\theta_{max}^2 ev^2 \frac{\mathrm{f}(v)}{2} \tag{115}$$

Note that Eq. (115) holds for any distribution function as long as the anisotropy is within the detector resolution. Using such detectors, it is often possible to determine distribution functions without the need for plasma potential measurements, so long as measurements are made over a 4π solid angle. The main difficulty with such detectors is they are very insensitive and tend to require relatively high plasma densities.

XIV. Conclusion

Langmuir probes have the widest range of plasma parameters for any plasma diagnostic. They come in several varieties—collecting, emissive, capacitive, planar, cylindrical, and spherical. They can determine, or at least give an indication of, the plasma potential density, electron temperature, and both electron and ion beam energies, and can give some details of the electron distribution functions $f(v)$ and $f_E(\epsilon)$.

In recent years considerable progress has been made in probe diagnostic techniques. Among these are new emissive probe techniques, detection of ion beams from enhanced electron current, and the invention of directional probes to determine $f(v)$. Secondary electron emission has been identified as a major culprit in confusing the interpretation of probe data as is rf (which has been well known). This article has attempted to present important new results as well as some standard results.

This article has tried to give an idea of how probes work and to identify major problems. It has concentrated on probe characteristics in unmagnetized systems. With B fields there always appears to be a need to provide a separate calibration because results are sensitive to the particular geometry and system in which they are employed.

In many situations probes have not been used because either they were not understood or it was believed that they would be perturbing to the system they were designed to measure. This has probably been a mistake, because even when they are not understood, their characteristics are often quite reproducible. This author believes that, as a general rule, if probes survive and do not destroy the plasma, they should be used.

XV. Acknowledgements

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References

Baum, E. and Chapkis, R. L. (1970). *AIAA Jl.* **8**, 1073.
Benjamin, N. (1982). *Rev. Sci. Instrum.* **53**, 1541.

Bohm, D. (1949). *The Characteristics of Electrical Discharges in Magnetic Fields* (A. Guthrie and R. K. Wakerling, eds.) McGraw-Hill, New York.
Brown, S. C. (1966). *Basic Data of Plasma Physics*, MIT Press.
Chan, C., Lonngren, K. E., and Hershkowitz, N. (1980). *Phys. Lett.* **78A**, 79.
Chapman, B. (1980). *Glow Discharge Processes*. John Wiley & Sons, New York, p. 109.
Chen, F. F. (1986). Institute of Plasma Physics, Report No. IPPJ-750, Nagoya University, Nagoya, Japan.
Chen, F. F. (1984). *Introduction to Plasma Physics and Controlled Fusion*, Vol. 1. Plenum Press, New York.
Chen, F. F. (1965). In: *Plasma Diagnostic Techniques* (R. H. Huddlestone and S. L. Leonard, eds.). Academic Press, New York, p. 113.
Cherrington, B. E. (1982). *Plasma Chemistry and Plasma Processing* **2**, 113.
Clements, R. M. and Smy, P. R. (1969). *J. Appl. Phys.* **40**, 4553.
Cho, M. H. (1987), private communication.
Cho, M. H., Chan, C., Hershkowitz, N., and Intrator, T. (1984). *Rev. Sci. Instrum.* **55**, 631.
Coakley, P. (1979). *Large Electrostatic Potential Variations in Laboratory Plasmas*, Ph.D. Thesis, University of Iowa-Iowa City.
Coakley, P., Johnson, L., and Hershkowitz, N. (1979). *Phys. Lett.* **70A**, 425.
Crawford, F. W. (1963). *J. Appl. Phys.* **34**, 1897.
Druyvésteyn, M. J. (1930). *Z. Phys.* **54**, 790.
Druyvésteyn, M. J. and Warmoltz, N. (1935). *Phil. Mag.* **17**, 1.
Emmert, G., Wieland, R., Mense, A., and Davidson, J. (1980). *Phys. Fluids* **23**, 803.
Fendell, F. (1970). *Comb. Sci. Tech.* **1**, 331.
Garscadden, A. and Emeleus, K. G. (1962). *Proc. Phys. Soc.* (London) **79**, 535.
Gibbons, D. J. (1966). In: *Handbook of Vacuum Physics*, Vol. 2 (A. H. Beck, ed.) Pergamon, New York.
Godyak, V. A., Ivanov, A. I., and Kuzovnikov, A. A. (1967). *Sov. Phys. Tech. Phys.* **12**, 766.
Godyak, V. A. and Popov, O. A. (1977). *Sov. Phys. Tech. Phys.* **22**, 461.
Gulidov, S. S., Kagan, Yu. M., Kolokolov, N. B., and Milenin, V. M. (1970). *Sov. Phys. Tech. Phys.* **14**, 993.
Hershkowitz, N., Nelson, B., Pew, J., and Gates, D. (1983). *Rev. Sci. Instrum.* **54**, 29.
Johnson, E. O. and Malter, L. (1950). *Phys. Rev.* **80**, 58.
Kiel, R. E. (1969). *J. Appl. Phys.* **40**, 3668.
Kemp, R. E. and Sellen, J. M., Jr. (1966). *Rev. Sci. Instrum.* **37**, 455.
LaBombard, B. and Lipschultz, B. (1986). *Rev. Sci. Instrum.* **57**, 2415.
Langmuir, I. (1961). *The Collected Works of Irving Langmuir* (G. Suits, ed.). Pergamon Press, New York.
Langmuir, I. (1929). *Phys. Rev.* **33**, 954.
Langmuir, I. and Blodgett, K. B. (1924). *Phys. Rev.* **24**, 49.
Langmuir, I. and Blodgett, K. B. (1923). *Phys. Rev.* **22**, 347.
Langmuir, I. and Mott-Smith, H. M. (1926). *Phys. Rev.* **28**, 727.
Langmuir, I. and Mott-Smith, H. M. (1924). *Gen. Elec. Rev.* **27**, 449, 583, 616, 726, 810.
Langmuir, I. and Mott-Smith, H. M. (1923). *Gen. Elec. Rev.* **26**, 731.
Leung, K. N., Hershkowitz, N., and MacKenzie, K. R. (1976). *Phys. Fluids* **19**, 928.
Leung, K. N., Samec, T. K., and Lamm, A. J. (1975). *Phys. Lett.* **51A**, 490.
Nakamura, Y., Nomura, Y., and Stenzel, R. L. (1981). *J. Appl. Phys.* **52**, 1197.
Nam, C. H., Hershkowitz, N., Cho, M. H., Intrator, T. (1988). *J. Appl. Phys.* **63**, 5674.
Nelson, B. A. (1986), private communication.
Prokopenko, S. M. L. and Laframboise, J. G. (1980). *J. Geophys. Res.* **85**, 4125.
Rubinstein, J. and Laframboise, J. G. (1983). *Phys. Fluids* **26**, 3624.

Schott, L. (1968). Chapter 11 in *Plasma Diagnostics* (W. Lochte-Holtgreven, ed.). John Wiley & Sons, New York.
Sizonenko, V. L. (1981). *Sov. Phys. Tech. Phys.* **26**, 1345.
Skøelv, A., Armstrong, R. J., and Trulsen, J. (1984). *Phys. Fluids* **27**, 2744–2751.
Smith, J. R., Hershkowitz, N., and Coakley, P. (1979). *Rev. Sci. Instrum.* **50**, 210.
Smy, P. R. (1976). *Adv. Phys.* **25**, 517.
Sprott, J. C. (1967). *Rev. Sci. Instrum.* **39**, 1569.
Stangeby, P. C. (1982). *J. Phys. D.* **15**, 1007.
Stenzel, R. L., Gekelman, W., Wild, N., Urrutia, J. M., and Whelan, D. (1983). *Rev. Sci. Instrum.* **54**, 1302.
Su, C. H. and Lam, S. H. (1963). *Phys. Fluids* **6**, 1479.
Swift, J. D. and Schwar, M. J. (1969). *Electrical Probes for Plasma Diagnostics*, American Elsevier, New York.
Tonks, L. and Langmuir, I. (1929). *Phys. Rev.* **34**, 876.
Wang, E. Y., Hershkowitz, N., and Intrator, T. (1986a). *Rev. Sci. Instrum.* **57**, 1085.
Wang, E. Y., Hershkowitz, N., Intrator, T., and Forest, C. (1986b). *Rev. Sci.* **57**, 2425.
Wang, E. Y., Hershkowitz, N., Intrator, T. (1986c). *J. Appl. Phys.* (in press).
Wang, E. Y., Hershkowitz, N., and Intrator, T. (1985). *Rev. Sci. Instrum.* **56**, 519.
Weber, W. J., Armstrong, R. J., and Trulsen, J. (1979). *J. Appl. Phys.* **50**, 4545.
Wong, A. Y. (1977). *Introduction to Experimental Plasma Physics*, University of California at Los Angeles, Physics Department.

4 Mass Spectrometry of Plasmas

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I. Introduction 185
II. Plasma Flux Analysis 187
III. Partial Pressure Analysis 190
IV. Flux Analysis of Plasmas: Energy Selection and Mass Filtering 191
A. Ion Flux and Sheaths 192
B. Techniques for Energy Selection 195
V. Mass Filtering 201
A. Electron-Impact Post-Ionization 202
B. Mass Filtering and the Flux Analysis of Ions 202
C. Ion Flux Analysis: General Considerations 205
D. Ion Flux Analysis: Experimental Considerations 209
VI. Flux Analysis: Neutral Species 210
VII. Experimental Considerations and Applications: Flux Analysis of Plasmas 214
VIII. Experimental Considerations and Applications: Partial Pressure Analysis 219
A. Pressure-Reduction versus Dynamic Range 220
B. System Response 223
C. Calibration Procedures 224
D. Applications to Magnetic Fusion Devices 228
IX. Acknowledgments 232
References 232

I. Introduction

Mass spectrometric analysis is a routine vacuum measurement that has become increasingly useful as a plasma diagnostic technique for both low-temperature (process) plasmas (Flamm, 1979) and high-temperature

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(fusion) plasmas (Poschenrieder, 1982a; Dylla, 1983). Mass spectrometry is the primary diagnostic for optimizing and process monitoring of the plasma discharge cleaning techniques (Dylla, 1980; Mathewson, 1974) which have been developed for conditioning the large vacuum vessels of fusion devices and particle accelerators. The relative simplicity of the mass spectra of the gaseous components in a plasma process, in comparison to the optical emission spectra (Griem, 1964; Pearse and Gaydon, 1963; Greene, 1978) makes mass spectrometry useful for quantifying plasma composition and impurity concentrations in the study of gas-phase plasma reactions and plasma-surface interactions.

One can distinguish two general schemes for mass spectrometric analysis of plasmas: flux analysis and partial pressure analysis. Flux analysis samples the plasma directly by line-of-sight emission of plasma particles through a small aperture in the plasma container. Collisionless trajectories lead directly to the ion optics of the mass spectrometer, as shown in Figure 1. This method is best suited for the identification of plasma species and for energy analysis. Partial pressure analysis samples gases via a connection between the plasma containment vessel and the mass analysis chamber in which wall collisions take place, as shown in Figure 2. Partial pressure analysis does not measure line-of-sight plasma particles and hence allows simpler vacuum connections between the mass spectrometer and plasma chamber, and generally requires less stringent differential pumping than flux analysis. Partial pressure analysis is the method of choice for monitoring glow discharge treatments of vacuum vessels or large area materials processing, because the measured signals are representative of the integrated effects of the relevant plasma-surface interactions.

The application of mass spectrometric diagnostics to plasmas covers a very broad spectrum of topics, which precludes a complete review. We have chosen, instead, to concentrate on two major areas of practical importance, viz, the "process" type plasmas, in which heterogeneous chemical reactions are made possible or enhanced by low temperature gas discharges, and the fusion plasmas which are high temperature and homogeneous but are subject to the gas phase species produced by the impingement of plasma particles on the vacuum vessel walls. Our treatment of these subjects divides quite naturally into two analysis schemes noted above: flux analysis and partial pressure analysis. Flux analysis is the most common technique employed for sampling glow-discharge type process plasmas, since in the vast majority of applications, there is no strong magnetic field for confinement and therefore no significant restriction on the distance between the plasma vessel and spectrometer. The use of strong magnetic fields for confinement of fusion plasmas has stimulated the development of the

partial pressure analysis technique, since the spectrometer is situated generally at a distance which renders the effect of the magnetic field negligible.

We consider only one type of mass spectrometer in this treatise, the quadrupole, since its use is so widespread. The general techniques used in sampling directly from a plasma are covered very briefly, since these topics have been adequately covered in a previous review (see Drawin, 1968). Attention has been called to two specific areas: 1) the combination of energy selection and the quadrupole mass spectrometer in flux analysis, and 2) the importance of calibration on an absolute basis in flux analysis and in partial pressure analysis. The treatment of energy selection and the ion optics of the quadrupole is qualitative, since a rigorous treatment would be a complete work in itself.

The discussion of applications is confined to a few examples which are current and illustrate the type of information which can be obtained and the types of problems that are encountered. We have indicated to some extent the historical perspective of the motivations for mass spectrometric analysis of plasmas, and we also indicate the most promising directions of current efforts.

II. Plasma Flux Analysis

Plasma flux analysis employs instrumentation which samples plasma particles (ions and neutrals) which stream through a sampling aperture located at or near the plasma vessel wall. After extraction from the plasma, the particle flux is energy and/or mass analyzed. Figure 1 shows a schematic diagram of the typical apparatus required for plasma flux analysis. There are four stages shown in Figure 1: 1) the particle extraction/collimation optics, 2) the energy analyzer, 3) the mass analyzer, and 4) the ion detector. The most complicated component in the system, in terms of its design and effect on signal collection and interpretation, is the extraction optics. The simplest extraction system employs an aperture which separates the plasma vessel from a separately-pumped chamber containing the energy and mass spectrometers. The resulting flux of neutral particles from the plasma (background atomic and molecular neutrals and plasma charge-exchange neutrals) will be collimated by the aperture according to the appropriate kinetic-gas laws. Once collimated by the sampling aperture, the flux of neutrals can be ionized to allow energy and mass analysis by electrostatic and/or magnetic deflection. For particles with thermal or near thermal energies (0.02 – 10 eV), the ionization is usually performed with an electron bombardment ionizer (Austin et al., 1976), whereas stripping cells

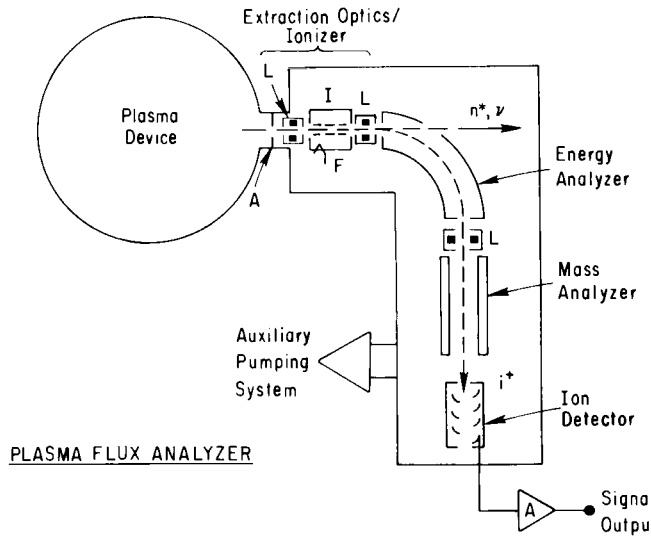


FIGURE 1. Schematic diagram of the typical apparatus required for plasma flux analysis. A sampling aperture (A) separates the plasma device from a vacuum enclosure housing energy and mass spectrometers. The aperture may include ion optical elements or be followed by additional ion lenses (L) for ion extraction or by an ionizer (I) for ionization of neutral particles effusing through the aperture. The ion detector (shown here as an electron multiplier) should be positioned off the axis of the aperture to prevent background problems caused by high-energy neutrals (n^*) and photons (ν) emitted by the plasma (from Dylla (1983) with permission).

(Fleischmann and Tuckfield, 1968) are used for ionizing the higher energy charge exchange neutrals (100 eV – 100 keV) that are present in contemporary fusion devices (Parsons and Medley, 1974; Wagner, 1982). It is important to notice that these two conventional ionization techniques leave a gap in the energy spectrum (10 – 100 eV). To bridge this shortcoming, surface ionization detectors have been developed with time-of-flight analyzers (Voss and Cohen, 1982), which have extended the detected charge-exchange energy spectrum down to 5 eV for H_2 (Voss et al., 1984).

Flux analysis of plasma ions is not as straightforward as the analysis of neutrals because of the electrostatic interaction between the sampling aperture and the plasma. The nature of the charged particle flux through the aperture depends on the plasma conditions in the vicinity of the aperture and the geometry and electrostatic potential of the aperture. This problem is not solvable in general, however, a few special cases of practical importance are well treated in the literature. Drawin (1968) discusses conditions for extracting plasma ions from an electrostatic plasma (i.e., glow discharge) that depend on the relative magnitude of the ion mean-

free-paths, λ_i, the Debye shielding length, λ_D, and the extraction aperture diameter, D. The simplest case is collisionless extraction through a plasma sheath, which occurs when $\lambda_i \gg D, \lambda_D$. In this case, if the extraction aperture is biased negatively (or is simply an aperture in the cathode if the plasma is a DC glow discharge), the sampled flux is described by the ion saturation current $I^+ \sim n_i[k/M_i(T_i + T_e)]^{1/2}$, derivable by electrostatic probe theory (Langmuir and Mott-Smith, 1923), where n_i, T_i, and M_i are the density, temperature, and mass of the positive ionic species. The case of collision dominated extraction through a plasma sheath, (where $\lambda_i < \lambda_D, D$) is more complicated. Collision processes between plasma particles can lead to the development of an extracted ion flux with a very different ion population from the unperturbed plasma far from the vicinity of the aperture.

Flux analysis of magnetized plasmas is more difficult. The magnetic field complicates the analysis of the extraction optics and, in addition, can cause practical problems with the design of the downstream instrumentation because fringing or stray magnetic fields from the plasma magnetic system have to be shielded. For flux sampling of neutral emissions from magnetized plasmas, the usual experimental approach is to move the apparatus a sufficient distance from the plasma vessel to insure that the effects of stray magnetic fields on the spectrometer are negligible or can be minimized by shielding. A concomitant loss in signal intensity and a more stringent differential-pumping requirement are necessary with this approach because of the longer neutral flight path. This approach is impractical for sampling ion fluxes from magnetized plasmas. Because of the confinement of ions along magnetic field lines, the ion flux effusing through an aperture will vary with the geometry of the aperture relative to the magnetic field. If the aperture axis is parallel to the magnetic field, the streaming flux will be sampled; whereas, if the aperture axis is perpendicular to the magnetic field, the cross-field diffusive flux will be sampled. Both fluxes can be related to the local plasma ion density, but the analysis is complicated. This problem has been considered in the design of probes for space plasma research (Krawec, 1980), and more recently by Staib (1980) in the design of a minature E × B mass spectrometer for in situ measurements of ion fluxes in the edge-plasma of fusion devices. Note that by using in situ spectrometers which make use of the ambient magnetic fields, the problem of shielding a high magnetic field from a sensitive instrument is circumvented.

With proper design of the sampling aperture and extraction optics, plasma ions and neutrals that are representative of the plasma composition in the vicinity of the aperture can be sampled such that the extracted species is unaffected by subsequent gas-phase and surface collisions. The primary disadvantages of the flux analysis technique are basically practical

concerns: 1) The technique requires careful design of the sampling and extraction optics, which cannot always be made nonperturbing to the plasma device. 2) The differential-pumping requirements to maintain collision-free conditions may be nontrivial, especially for sampling the higher pressure process plasmas. 3) For fusion device applications, the magnetic shielding requirements for the spectrometers can be severe in terms of the design tradeoff between the level of required shielding and the loss of signal intensity due to remote separation of the spectrometer from the plasma device.

III. Partial Pressure Analysis

The alternative experimental scheme for mass spectrometry of plasmas, which employs the same or somewhat simpler instrumentation as required for flux analysis, is partial pressure analysis. Figure 2 shows a schematic diagram of the typical experimental arrangement for partial pressure analysis in which a quadrupole mass spectrometer (QMS) is indicated for the partial pressure measurement. Depending on the operating pressure regime of the plasma device, there is either a high-conductance or low-conductance connection tube between the plasma device and the QMS chamber, which is usually differentially-pumped.

The instrumentation for partial pressure analysis can be made simpler than that for flux analysis because the extraction optics and energy analyzing stages are unnecessary. Usually, plasma particles are allowed to thermalize before reaching the QMS chamber by repeated collisions on the connection tube wall. Thus, for the typical electron-bombardment ionizer of a QMS, the mass analyzed ion current, which is proportional to the local gas density, can be calibrated in terms of a partial pressure equilibrated to the temperature of the QMS chamber walls.

The primary experimental attribute of partial pressure analysis is that the technique samples volatile reaction products resulting from the effects of plasma-wall interactions summed over the entire surface area of the plasma chamber. In contrast, the flux analysis technique samples only the reaction products that are representative of plasma-material interactions in the vicinity of the sampling aperture. This advantage of partial pressure analysis is particularly useful when mass spectrometry is used as a diagnostic or monitor of 1) plasma-etching (Flamm and Donnelly, 1981) of large area samples; 2) discharge cleaning (Dylla, 1980) for large vacuum vessels; 3) plasma-wall interactions (McCracken and Stott, 1979) in fusion devices, such as impurity generation from wall sputtering and hydrogen isotope replacement processes in wall materials.

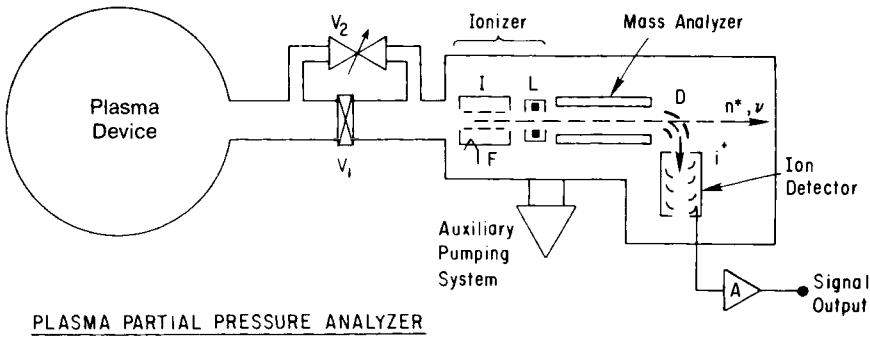


FIGURE 2. Schematic diagram of the typical apparatus required for partial pressure analysis of plasma devices. The mass spectrometer measures the neutral density in the analyzer chamber which can have a high conductance (V_1) or low conductance (V_2) connection to the plasma device. The indicated mass spectrometer components: a hot-filament (F), electron-bombardment ionizer (I), ion lens (L), quadrupole mass filter, and deflection electrode (D) for an off-axis (electron multiplier) ion detector are typical for commercially available quadrupole mass spectrometers (from Dylla (1983) with permission).

The combination of the latter two applications and the advantages of simpler instrumentation with commercial availability have resulted in the wide application of partial pressure analysis for mass spectrometric studies in process plasma systems and fusion research of the past decade. Specific examples from fusion research are reviewed in Section VIII.D.

IV. Flux Analysis of Plasmas: Energy Selection and Mass Filtering

We now consider specific aspects of the energy selection and mass filtering of the particles sampled directly from the plasma. There have been numerous accounts of flux analysis measurements over the past 20 years, encompassing a wide variety of techniques and objectives. Very few experimental arrangements use the general scheme for energy and mass selection shown in Figure 1, and thus it is appropriate to briefly review what has been attempted. The emphasis will be on the measurements performed on process type plasmas, where the neutral particles have thermal energies (0.02–0.04 eV) or near-thermal energies ($\sim$ 1–10 eV) as a result of charge exchange, and the ions have energies up to the full sheath potential (10–20 eV). The energies of ions sampled through electrode sheaths may be considerably higher, ranging up to several hundred eV, depending upon the type of discharge employed.

Energy selection in flux analysis of plasmas was originally employed to limit the spread of energy of the ions incident on magnetic mass selectors. The widespread use of quadrupole mass filters in recent times has relaxed this restriction, but measurement of the energy distribution of ions can provide important information about materials modifications. Thus, energy analysis of the ion flux becomes an important diagnostic of the plasma. The coupling of energy selection and quadrupole mass filtering is of prime concern when a comprehensive diagnostic approach is taken: The interpretation of data will depend on the technique used for energy selection and the mode of coupling.

A. Ion Flux and Sheaths

Positive ions are accelerated across the sheath between the plasma and the surface containing the sampling aperture. (The chapter on Langmuir probes in this text contains a complete treatment of sheaths, as well as Franklin, 1976.) The potential difference between the plasma and the sampling aperture will be the floating potential, V_f, if the sampling aperture is electrically isolated. The magnitude of this potential is given approximately by (Chen, 1965; Uman, 1964; Franklin, 1976)

$$V_f = \frac{kT_e}{2e} \ln\left[\frac{2\pi m}{M_i}\right], \tag{1}$$

with the assumptions that $T_e \gg T_i$ and secondary electron emission processes are ignored ($\gamma_e = 0$). V_f is typically 10 to 20 eV for glow discharge plasmas in which $T_e \sim 3$ to 5 eV. In Eq. (1) the symbol T_e is the electron temperature, M_i and m are the ion and electron masses, and k is the Boltzmann constant. The sheath thickness will vary markedly with the plasma density, but the potential variation with distance (Uman, 1964; Franklin, 1976), χ, can be approximated by the following

$$V(x) = V_f e^{-x/X}, \tag{2}$$

with the characteristic length X being given by

$$X = \left(\frac{\epsilon_0 kT_e}{e^2 n_e}\right)^{1/2}. \tag{3}$$

The term n_e in Eq. (3) refers to the plasma electron density, essentially at infinite distance from the plasma-sheath boundary, and ϵ_0 is the permittiv-

ity of free space. The sheath potential of a representative set of plasma parameters is plotted in Figure 3. If there are no collisions of any type within the sheath and the criterion that $D < \lambda_D$ is fulfilled so that the sheath is unperturbed (Drawin, 1968), then the ions would arrive at the sampling orifice with a sharp energy distribution centered around V_f, and the species observed would be representative of the bulk plasma. There is no *a priori* way of knowing whether this situation dominates the sampling characteristics, and, in general, collisions within the sheath may be expected to be the norm rather than the exception at total pressures of a few tenths of a torr and above.

Collisions within the sheath will modify the energy distribution and composition of the sampled ions. The types of collisions we may expect to dominate at relatively low kinetic energies are simple momentum transfer collisions with neutral atoms or molecules and resonant symmetric charge transfer (McDaniel, 1964). Exothermic or thermoneutral ion-molecule reactions will be important also when the accelerating ion still has relatively low kinetic energy, since the cross sections for these processes are quite large (McDaniel et al., 1970; Su and Bowers, 1979). Additional reaction channels which change the identity of the ions (asymmetric charge transfer or dissociative charge transfer) become possible, as the ion gains kinetic energy as it traverses the sheath.

Knowledge of the energy distribution of any particular ion sampled from the plasma provides an additional piece of diagnostic information, which is as valuable as the identity or the intensity of that ion. For example, the maximum kinetic energy that an ion can have with the sampling aperture at the wall potential will be simply V_f. Bimodal or structured energy distributions indicate a charge or energy transfer process with high probability under the existing sheath conditions (Coburn, 1970; Davis and Vanderslice, 1963). Ions which have energy distributions that peak at significantly lower energies than V_f (and have no tail which extends to values approaching V_f) are most likely created within the sheath by electron impact or by ion-molecule reactions which occur at distances close to the sampling aperture. Thus by measuring the kinetic energy distribution of several ions of interest, we may determine whether the ions originate within the body of the plasma or are created by collisional processes within the sheath.

Mass spectrometric sampling may require that the sampling aperture be held at a potential other than the wall potential. This is the case in process plasmas, where interactions of the ions with materials on the electrode surfaces are of prime interest. Flux analysis of ions sampled through an electrode surface of either an rf or DC discharge requires that we consider the magnitude of the potential drop across the sheath and the processes that can result. The cathode fall in a DC discharge is almost as large as the total

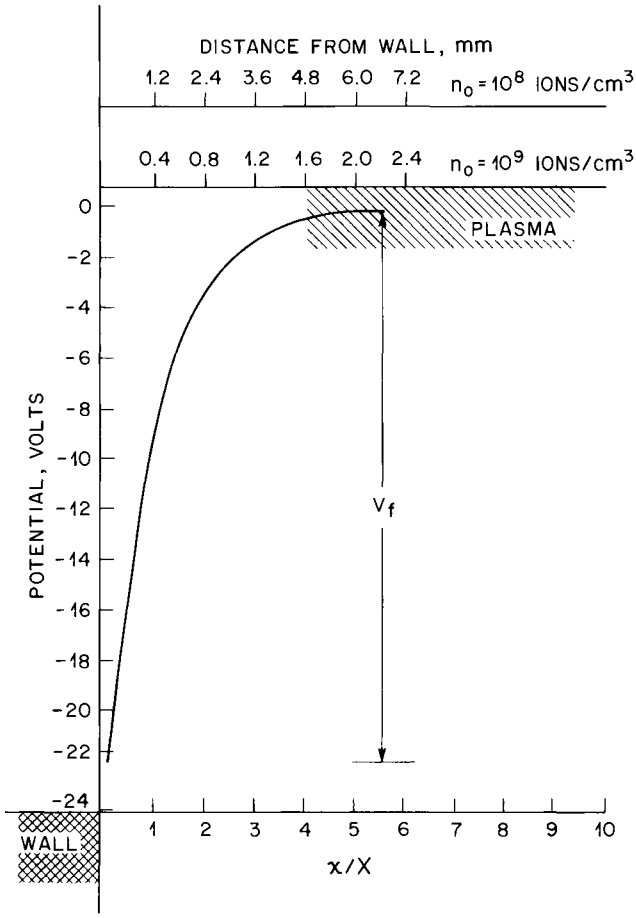


FIGURE 3. Plot of the plasma to wall potential variation in terms of the reduced parameter x/X, for $T_e = 5$ eV, $T_i = 300$ °K, $M = 40$ amu. The absolute distance scales at the top of the plot are for $n_o = 10^8$ ions/cm^3 and $n_0 = 10^9$ ions/cm^3.

potential applied between the electrodes, and the sheath thickness will be considerably greater than that of the wall sheath. The probability of transmutations within the electrode sheath will be considerably higher under these conditions than within the wall sheath. Electron impact ionization must also be considered as a major source of the observed ions. Interpretation of mass spectral data obtained when sampling through the

cathode sheath therefore requires more detailed knowledge of the energy distribution of the ions than sampling through the wall.

Discharges excited by rf potentials applied to internal or external electrodes present a somewhat different situation with regard to the magnitude of the sheath potentials. The most common rf discharge configurations use symmetrically arranged internal electrodes, with one of the electrodes (the excitation electrode) carrying the radio frequency potential and the other electrode at ground. The plasma potential will depend upon how the rf waveform is coupled to the excitation electrode: This electrode may be held at DC ground through an inductor, or may be biased at a DC potential applied through an inductor from a DC power supply, or the coupling may be purely capacitive. The magnitude of the time-averaged potential difference between the plasma and the walls, the excitation electrode, or the ground electrode will be a function of the frequency of the alternating potential and the system geometry and it will be different for each coupling scheme. Köhler et al. (1985a and 1985b) have investigated each of these coupling schemes using the energy distribution of the ion flux through the grounded electrode as one of the diagnostics of the time-averaged plasma potential, (Köhler et al., 1985a and 1985b), and show that, in general, there are two components to sheath potential: One is due to the effect of the time-averaged alternating potential and the other is due to the floating potential resulting from the normal electron energy distribution (c.f., Eq. (1)).

The increased kinetic energy of the ions sampled through an electrode effects the performance of the mass filter. The mass filtering characteristics of quadrupole mass filters is independent of the ion kinetic energy only to a first order (Austin et al., 1976). Ions with energies greater than $\sim$ 50 eV with respect to the reference potential of the quadrupole create a broadening at the base of the mass peak, leading to poor baseline resolution and to a reduction in the dynamic range. Sheath potentials can reach 40 to 50 eV for some rf discharge configurations, thus some type of energy selection or ion retardation is necessary for optimum mass filter performance as well as for data interpretation.

B. Techniques For Energy Selection

There are numerous descriptions of mass spectrometric sampling apparatus which have no energy selection capabilities except for simple retardation which reduces the incident energy of the ions (Tardy et al., 1981; Aita, 1985; Keren et al., 1976; Brown and Bell, 1974; Armour, 1974). Sampling without retardation allows the full spectrum of ion energies into the mass

filter, and the result may be a degradation of resolution, as previously discussed. Retardation of the incoming ions results in the loss of the low-energy portion of the spectrum and the information contained therein. These techniques are satisfactory if the only objectives of the measurements are to identify the species incident on the walls of the discharge vessel and to measure their relative currents. Conclusions regarding the origin of the observed ionic species (e.g., from the bulk plasma vs. the sheath) can only be drawn when the gas pressure is low ($\lesssim 10^{-2}$ torr) and it is known that the sheath thickness is small compared with the mean free path for ion-molecule reactions.

Retarding potential techniques, where a potential ramp is applied to one or several mesh electrodes, have been used as a convenient means to measure the energy distributions of both the ions and electrons fluxes from discharges (Neiswender and Kohout, 1972; Hiraoka and Kamada, 1971; Franklin et al., 1968). The derivative of the current measured on a suitable collector with respect to the potential change gives the energy distribution of particles. This use of planar retarding grids results in distortions arising from the off-axis energy components which affect the measured distribution (Simpson, 1961). The use of curved grids improves the fundamental performance with a suitable collector geometry (Neiswender and Kohout, 1972; Wei and Kupperman, 1969), but care must be taken to avoid potential penetration effects (Wei and Kupperman, 1969), scattering at the flux source (Neiswender and Kohout, 1972) and the buildup of insulating films on the grid electrodes (Wei and Kupperman, 1969). The retarding potential measurements on plasmas cited previously employ Faraday cup collectors to measure the current-voltage characteristic.

The use of retardation techniques as an energy prefilter with a quadrupole mass spectrometer involves a very different ion-optical situation. The ion beam emerging from the sampling aperture will, in general, contain a distribution of kinetic energies. A sharp distribution function will only be obtained from a collision-free sheath, which is a rare occurrence in most applications. This ion beam will also be divergent, owing to the single lens effect of the sampling aperture and the plasma sheath (Hasted, 1975). A nondistorted, ideal energy distribution measured by retarding grids would require that the divergent beam be made parallel prior to retardation and also that all the components of the energy spectrum surmounting retardation have trajectories which fall within the electrical aperture of the QMS. The focusing effect of the lens electrodes cannot, however, result in the trajectories required for the ideal retarding voltage-current characteristics, as shown in Figure 4a. The real situation is more accurately represented by the trajectories shown in Figure 4b. There will be a distribution of focal

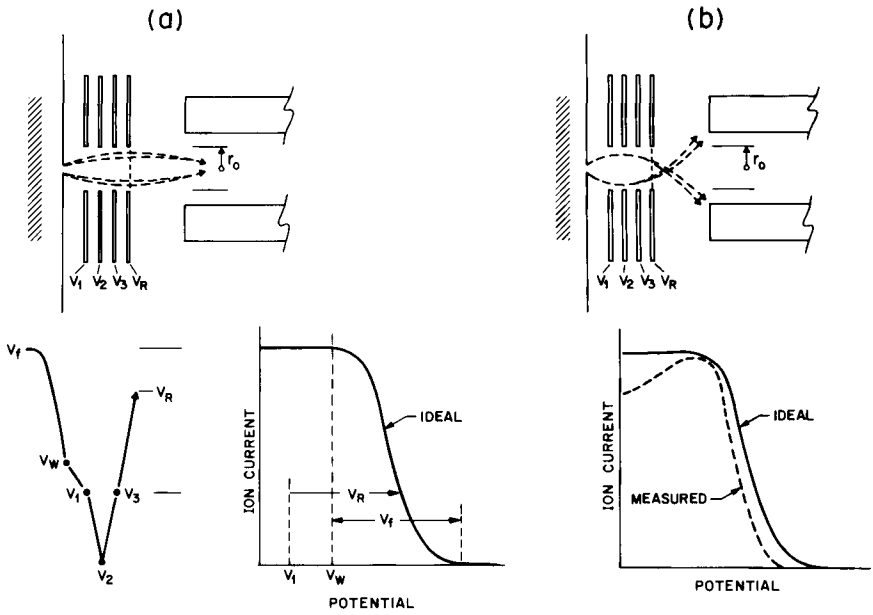


FIGURE 4. Diagrammatic representation of ion trajectories resulting from the coupling of a focusing-retarding lens system with a quadrupole mass filter. (a) All ion trajectories which surmount the retard potential fall within the electrical aperture, r_0, for transmission through the quadrupole. The result is the ideal current-potential relationship, which can be differentiated to yield the energy distribution of the ions. (b) The effect of the continuously changing focal point and of ion trajectories which fall outside the electrical aperture of the quadrupole, resulting in a distorted kinetic energy distribution.

points of the incoming ion beam as a result of the spread in energy of the ion beam, and some of the ion trajectories which have sufficient energy to pass the retardation electrode V_R, will be outside the radius r_0 (the effective electrical aperture for transmission through the quadrupole). Distortions in the measured energy spectrum indicated in Figure 4b are further aggravated by the continuously changing focal points resulting from the potential ramp applied to the electrode V_R. Some improvement on the aberrations in retarding potential measurements can be obtained with a combination of cylindrical lens assemblies (Klemperer and Barnett, 1971; Harting and Read, 1976) with retarding grids. The basic idea is to utilize the properties of cylindrical lenses to produce a nearly parallel beam of ions for the range of energies expected from the plasma. The performance of such systems with acceleration-deceleration grid electrodes placed before (Vasile, 1980) or after (Kohout and Neiswender, 1970; Vasile and Smolinsky, 1973) the

cylindrical lens assemblies does not appear to satisfy the quadrupole entrance requirements for a nondistorted energy distribution after mass analysis. The retarding potential technique is therefore of minimal value if the objective of the measurements is to obtain an accurate representation of the ion kinetic energy distribution. Retardation with focusing lens elements is useful and convenient for maximizing the ion signals while reducing the overall kinetic energy.

The optimum method for energy selection employs hemispherical sector analyzers, as indicated in Figure 1. Hemispherical analyzers have the advantage of being energy dispersive; they can be constructed with a predetermined range of energy resolution, and they eliminate the complications due to line-of-sight trajectories to the detector.

The 90° hemispherical sector selects ions according to the relationship (Riviere, 1983)

$$E = \frac{\Delta V}{(r_1/r_2 - r_2/r_1)}, \tag{4}$$

where ΔV is the potential difference between two spheres of radius r_1 and r_2. The energy resolution is given by

$$\frac{\Delta E}{E} = \frac{\omega}{R} + \alpha^2, \tag{5}$$

where R is the mean sphere radius, ω is the width of the defining aperture after energy selection, and α is the entrance angle into the analyzer. The energy resolution of this device is a function of both the geometry and the energy selected for transmission (pass energy). Two modes of energy analysis are possible: The device may be operated at a constant pass energy with an appropriate electrode arrangement to accelerate or decelerate the particles prior to energy selection (commonly called the constant transmission mode). Alternatively, the entire energy spectrum is admitted to the analyzer, and the energy dispersion is achieved by varying ΔV. The latter method has the consequence of an increasing bandwidth with increasing energy and therefore weights the high-energy portion of the distribution. This distortion in the measured energy distribution can be removed by mathematical deconvolution (Morrison, 1963), since the form of the distorting function is known (Eq. (5)). Scanning the energy spectrum by varying ΔV results in variable energy ion injection into the quadrupole, with the concomitant focal length problem already described. The potential difference across the hemispherical analyzer to provide a uniform focal length, but the ion energy through the quadrupole will be variable, unless the DC reference level of the quadrupole is also varied with the pass energy of the analyzer.

Energy selection in the constant transmission mode (fixed pass energy) requires acceleration of the ions such that the lowest energy ions in the distribution have a kinetic energy equal to the pass energy of the filter. The entire accelerated distribution is then retarded by an appropriate electrode structure to move the incoming energy distribution across the pass energy of the analyzer. Köhler et al. (1985a) have described such a system for energy analysis of ions sampled through the sheath of rf plasmas. In their system, the DC reference for the quadrupole mass filter and the hemispherical analyzer are floated together so that energy analyzed ions are injected into the quadrupole at constant energy. This arrangement eliminates the focal length problem at the mass filter entrance and maintains a constant transmission from the entrance aperture of the energy analyzer through the mass filter. The only distortions introduced into the measured energy spectrum would arise in the acceleration/deceleration region, but these should be minimal relative to those introduced by a retarding grid placed in front of the aperture of the mass filter.

Examples of the energy distributions measured by Köhler et al. (1985a and 1985b) for ions extracted through the grounded anode in an rf discharge are shown in Figure 5. The ions sampled were Ar_2^+ and ArH^+, which are not subject to resonant charge transfer within the sheath, and the pressure was sufficiently low to allow collision-free transit across the sheath (0.02 torr to 0.05 torr). Figure 5a shows the energy distributions obtained for Ar_2^+ at 0.02 torr, with an excitation frequency of 13.56 MHz. The sharp distribution at $V_p \cong 20$ V is obtained when the coupling of the rf is purely capacitive (Köhler et al., 1985a), and the broader distribution centered at $V_P \cong 170$ V occurs when the excitation electrode is set to DC ground. Figure 5b shows the effects of the excitation frequency (Köhler et al., 1985b) on the ion kinetic energy distribution. The ion energies range from zero to $\sim$ 70 eV when the excitation is 100 kHz, because this frequency is low enough for the ions to follow the sinusoidal potential excursion. These results illustrate several important points: 1) The plasma to wall potential in an rf discharge is critically dependent upon the method used to couple the rf to the electrodes. 2) Collision-free sheaths do result in very narrow energy distribution. 3) The sheath potential can be measured by such a technique. 4) One can tell immediately if sheath collisions occur for other species extracted from the plasma by comparison of the energy spectrum to those shown in Figure 5.

The resolution requirements for energy analysis of ions sampled from a discharge need not be better than a few eV. Thus, the usual transmission losses associated with narrow exit apertures need not be incurred. The application of hemispherical energy analyzers to the flux analysis from plasmas has been limited (Coburn, 1970; Köhler et al., 1985a and 1985b;

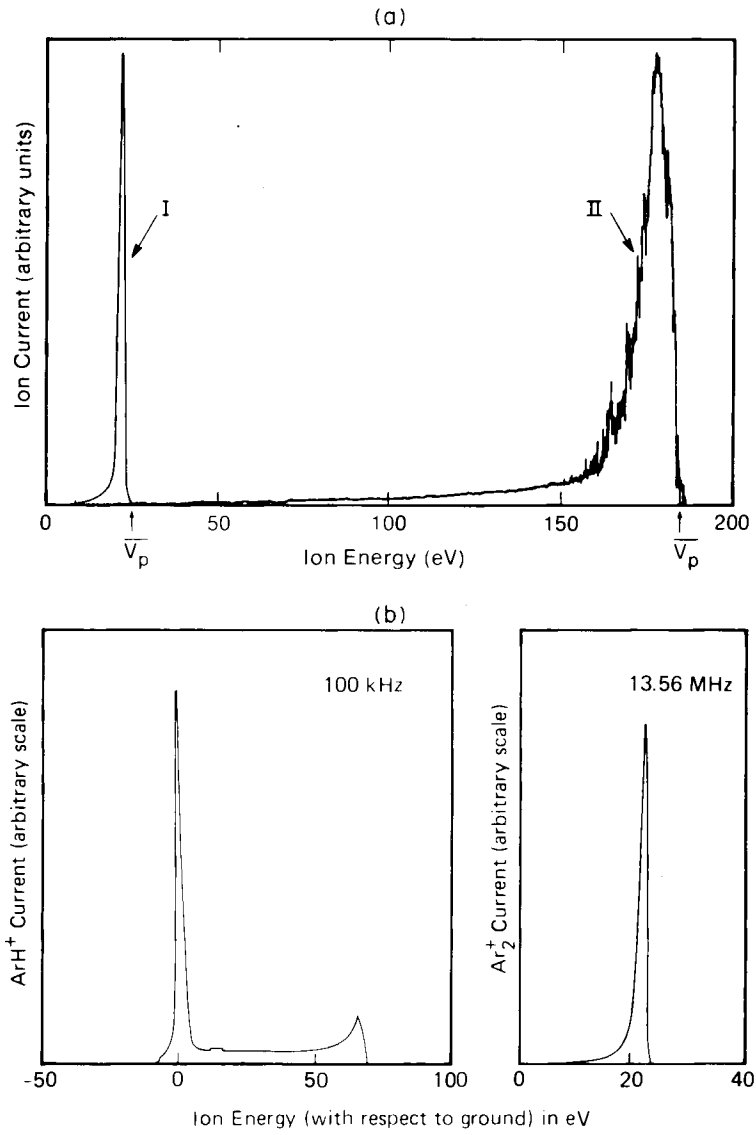


FIGURE 5. (a) The energy distributions of ions extracted through the grounded anode of a 13.56 MHz rf planar diode discharge system for coupling of the rf potential in a capacitive mode (I) and coupling of the rf potential in the symmetric mode with the rf excitation electrode at DC ground (II). (b) Energy distributions extracted through the ground electrode of a planar diode rf glow discharge system excited at 100 KHz and at 13.56 MHz. Argon pressure = 50 mtorr. ArH^+ and Ar_2^+ are selected so that resonant charge transfer is not a factor. After Köhler et al. (1985a and 1985b), with permission.

Coburn and Kay, 1972), presumably due to difficulties of design and construction. Small hemispherical segment analyzers with dimensions and a design that are highly compatible and convenient for plasma sampling apparatus are now commercially available (Warmack et al., 1978), and it is very likely that this technique for energy selection will be widely utilized in the next generation of ion sampling devices. A complete plasma sampling apparatus using most of the design features shown in Figure 1 (except for the electron impact ionizer) is now commercially available, with the principal application being the post-ionization of sputtered materials by the plasma.

V. Mass Filtering

In the following discussion of mass filtering only the quadrupole mass spectrometer will be considered, since it is the obvious choice for mass spectrometer sampling of plasmas. The key features which make the QMS so useful for this application are the first-order independence of the mass filtering on the kinetic energy of the ion and the control over the mass filtering characteristics (e.g., resolution vs. transmission) obtained by variation of electrical parameters only. Other significant advantages of the QMS are the small dimensions, ease of mounting and alignment, and low overall cost when vacuum chamber structure and pumping requirements are considered.

The theory of operation of the quadrupole has been thoroughly covered in the literature, and very neatly and concisely brought together in Austin et al. (1976). These are some considerations in the sampling of plasmas which are unique and must be discussed at this point. The performance characteristics of the QMS must be optimized for each of the three possible applications: 1) flux analysis of neutral particles, 2) flux analysis of charged particles, or 3) partial pressure analysis of equilibrated gases. The choice of a quadrupole system which allows the variation of frequency, reference potential, and resolution over the mass range of interest is therefore advantageous, particularly for flux analysis applications.

Common to all QMS is a characteristic transmission vs. mass curve. The instrument transmission is a function of the ion velocity; hence at a fixed energy, ions of high mass will be transmitted less efficiently than ions with low masses. This phenomenon is a general one regardless of the source of the ions which are filtered. It is therefore unavoidable in all the modes of plasma sampling mentioned above. Mass spectrometric diagnostics of plasmas often requires determining the quantitative relationship between

species from several different groups of masses (e.g., masses 1–5, 26–32, 39–44, etc.). The falloff in sensitivity of the quadrupole with increasing mass must therefore be a consideration when setting up the experimental parameters, particularly in the calibration.

Operating the quadrupole mass filter in the constant resolution mode over the entire mass range will produce the normal decrease in sensitivity associated with the action of the fringe fields on the more massive, lower velocity ions (Austin et al., 1976). Operation in the constant ΔM mode, where the resolution increases with the mass, compounds the effect of the instrument transmission. The result is a mass spectrum with sensitivity skewed toward the lower masses, since at $\Delta M = 1$, the resolution required to separate adjacent peaks at mass 30 is simply 30, while at mass 250 the resolution must increase to 250 to separate adjacent peaks. In practice, most quadrupole systems can be electrically tuned to operate between the constant resolution and the constant ΔM modes. Once a convenient mode has been selected in accord with the experimental object, calibration with known compounds or standard gas mixtures is necessary. This calibration should include resolution measurements as well as peak intensity measurements over the mass range of interest (see Section VIII.C for additional discussion of calibration).

A. Electron-Impact Post-Ionization

The correction or proper calibration for mass discrimination effects of the quadrupole are well defined for partial pressure analysis or for flux analysis of neutral particles. In these situations, the ionization of the neutrals by electron impact is localized, resulting in an ion beam with a narrow range of energies. Ion trajectories at the entrance to the quadrupole are reproducible and independent of the conditions existing in the plasma. The use of fringe field correction devices (Siegel and Vasile, 1981), or an ion energy programmed with the mass scan is also helpful for reducing mass discrimination by the QMS in this situation.

B. Mass Filtering and the Flux Analysis of Ions

Flux analysis by mass filtering of the ions from the plasma is a different situation from flux analysis of post-ionized neutral species. We have already discussed the effect of the variable focal lengths produced by simple

electrostatic lenses on the measurement of energy distributions of any particular ion. The relative abundances of ions observed after mass filtering will also be a function of the sheath characteristics (i.e., collisionless or collision dominated), which determines the kinetic energy distribution of each ion in the spectrum. The coupling of the energy filtering with the ion optics at the entrance aperture of the quadrupole will always affect the relative abundances of the ions observed, except in the rare case where no electrostatic focusing is used and the sheath is collisionless for each species in the spectrum.

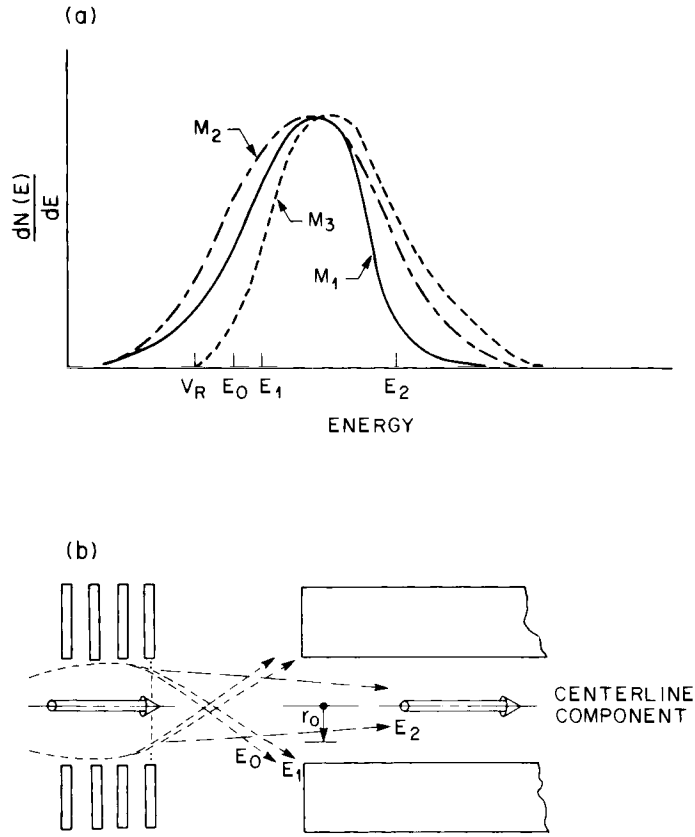


FIGURE 6. The nonuniform discrimination of ions resulting from a focusing-retard arrangement with a quadrupole mass filter. The ions of masses M_1, M_2, and M_3 having energy distributions shown in (a) will be focused as indicated in (b). Off-axis components with energies E_0 and E_1 for each ion will have trajectories which will not be transmitted by the mass filter.

The general situation for flux sampling of ions results in a collision-dominated sheath for some of the species and a collisionless sheath for others, in accord with the probabilities of resonant charge exchange and with ion-molecule reactions which have large cross sections. The use of focusing electrodes with a small retarding field on incoming ions of masses M_1, M_2, and M_3, all of which have similar kinetic energy distributions, is

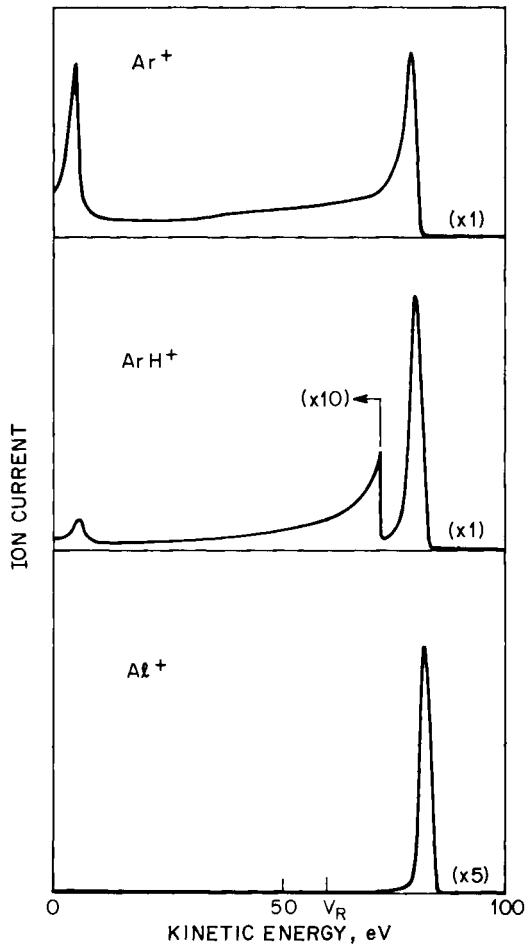


FIGURE 7. Measured kinetic energy distributions of Ar^+, ArH^+, and Al^+ from an Ar discharge at 0.1 torr. Ar^+ undergoes extensive charge transfer, evidenced by the low energy component of its kinetic energy distribution. These energy distributions were measured with a dispersive type energy filter (unpublished data, after Coburn and Köhler). The retard potential for a focussing/retarding ion sampling system referred to in the text is indicated at V_R.

shown in Figure 6. The center-line components with energies greater than V_R for masses M_1, M_2, and M_3 will be transmitted without ion-optical losses, but the off-axis components, at energies represented by E_0 and E_1 for each of the masses, will not be transmitted by the mass filter since their trajectories carry them outside the acceptance aperture of the quadrupole. The resulting mass spectrum will have signals which exaggerate the abundance of M_3 and under-represent the abundances of M_1 and M_2.

Figure 7 shows the kinetic energy distributions of Ar^+, ArH^+, and Al^+ measured with an energy dispersive device (Köhler et al., 1985) after traversing a 75 V sheath in an Ar discharge at 0.1 torr. The Ar^+ distribution is strongly bimodal, with the low energy component due to resonant charge transfer near the sampling orifice. If a retarding potential system were used to sample this discharge, the Ar^+ flux would be underestimated by a factor of two (with $V = V_R$). Hence the relative abundances of the ions measured with such a system will not represent the true flux from the plasma. Integration of the energy spectra shown in Figure 7 yields the true fluxes of each of the ionic species and thus allows for a more accurate interpretation of the data. The volume of data generated by such a procedure is a consideration, since a complete energy spectrum is needed at each mass. This was a prohibitive factor as little as 15 years ago, but with present capabilities for computer acquisition and manipulation, obtaining a complete mass and energy analysis of the ion flux is a practical goal.

C. Ion Flux Analysis: General Considerations

The mass spectra of plasma ions which are flux analyzed give direct qualitative information on the energetics of the inelastic electron-atom or electron-molecule collisions taking place at the sampling position. The species observed and their relative signal intensities will reflect the electron population with energies greater than the respective ionization potentials of components of the neutral gas. This will be the case in many situations, even if the ion sampling optics are less than ideal. Sheath collisions can alter the ionic species observed from those in the plasma by ion-molecule reactions, or by asymmetric charge transfer, but it is improbable that either of these reactions will render the precursor ion totally unobservable. For example, many process plasmas have He, Ne, or Ar as one of the major neutral components. Observation of these ions or one of their ion-molecule reaction products indicates that a significant population of electrons exist, with energy greater than the ionization potential of the rare gas. Doubly charged rare gas ions, Ar^{+2}, and Xe^{+2} have been observed in DC and in rf

discharges (Kohout and Neiswender, 1970; Vasile and Smolinsky, 1973; Gilkinson et al., 1969; Knewstubb and Tickner, 1962a), which indicate that relatively high-energy processes are operating in these discharges. The observance of these doubly charged ions, even at low levels, was contrary to general expectations.

Sequential lower energy processes leading to ionization must also be considered when qualitatively interpreting mass spectra. Ion production paths such as stepwise ionization

$$Ar + e \rightarrow Ar^* + e\,(11.6\text{ eV}) \tag{6}$$

$$Ar^* + e \rightarrow Ar^+ + 2e^- (\sim 5\text{ eV}) \tag{7}$$

or metastable-metastable interactions

$$Ar^* + Ar^* \rightarrow Ar^+ + Ar + e^- \tag{8}$$

$$He^* + He^* \rightarrow He^+ + He + e^- \tag{9}$$

may have to be considered when the rare gas is the major neutral component.

The comparison of ionic species at various sampling points with respect to the electrodes is another indicator of energy transfer mechanisms within the plasma or within the sheath. Mass spectrometric sampling at a fixed radial point with movable electrodes has been used to determine the ionic composition in DC glow discharges from the beginning of the negative glow to points well into the positive column (Knewstubb and Tickner, 1962a and 1962b). The radial variation of ions in rf generated sputtering discharges has been sampled using a fixed sampling point in the grounded electrode and a movable rf driven electrode (Coburn, 1970; Coburn and Kay, 1972). Considerably different distributions of species in methane discharges have been observed when sampling ions axially through an rf electrode or radially through a floating wall probe (Vasile and Smolinsky, 1975). The species observed indicate energetic electron collisions with simple molecules in the rf electrode sheath and ion-molecule reaction products as well as threshold ionization of molecules at the wall sheath. Figure 8 shows the results in the methane discharge. The differences observed in the ionic species in the methane discharge were observed also in rf discharges of a homologous series of hydrocarbons.

The effects of pressure or input power have frequently been diagnosed by the resulting changes in both the species and intensity of the ions extracted from the plasma. The objective of such measurements in plasmas of unreactive neutral gases has often been the elucidation of the predominant

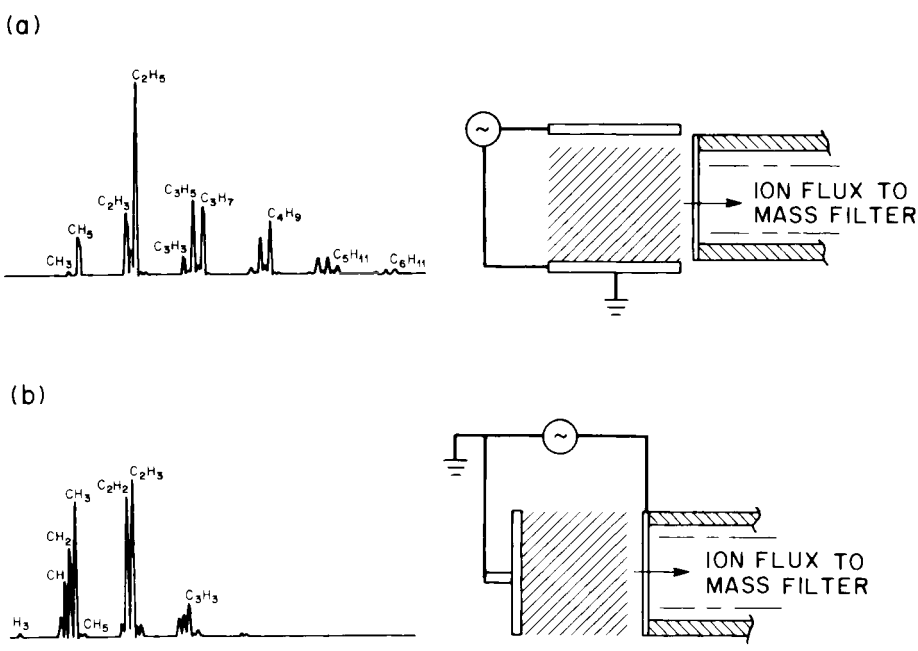


FIGURE 8. The ionic species observed in a 13.56 MHz discharge in CH_4 at 0.45 torr. 150 V p-p rf potential applied to the electrodes in the configurations shown: (a) Parallel plate excitation with ions sampled through a floating wall probe. (b) Parallel plate excitation with ions sampled through the electrode carrying the 150 V rf potential (from Vasile and Smolinsky (1975), with permission).

energy transfer mechanisms which yield ions (Gilkinson et al., 1969; Knewstubb and Tickner, 1962a and 1962b), and the determination of sampling conditions which allow collision-free ion transit across the sheath (Franklin et al., 1968; Bohme and Goodings, 1966; Seguin et al., 1972). Pressure variations in plasmas of unreactive neutral gases which readily cluster in the ionic state have been used to study the clustering process itself (Dawson and Tickner, 1964 and 1966; Studniarz and Franklin, 1968), and to study ion-molecule reactions in simple gases such as CH_4, CO_2, N_2, or NH_3 (Seguin et al., 1972; Dawson and Tickner, 1966; Studniarz and Franklin, 1968; Schildcrout et al., 1970). The attractive feature for these early studies of ion-molecule reactions in the plasma state is that the reactions take place at thermal energies in a field-free region. Such conditions cannot be achieved in high pressure mass spectrometers. Unfortunately, the efficiency of secondary and tertiary sequential reactions in the

plasma, and the sheath processes makes the extraction of rate constants intractable. The techniques of ion cyclotron resonance (Beauchamp, 1971), flowing afterglows, and tandem mass spectrometry (McDaniel et al., 1970; Franklin, 1972) have emerged to provide the ion-molecule reaction data base, which is now used to help interpret the ionic species observed when sampling plasmas.

Flux analysis of ions from plasmas containing condensing or reactive gases was stimulated by the useful and unusual properties of the films formed by the action of the plasma. Practical applications of such films include passivation coatings, solar cells, optical light guides, and semipermeable membranes. Ion flux analyses have been performed on plasmas containing simple organic molecules that include the series CH_4, C_2H_4, C_2H_6, C_2H_2, and C_6H_6 (Smolinsky and Vasile, 1975, 1976, 1977; Vasile and Smolinsky, 1976, 1977a). The more complex organic systems such as styrene, phenylacetylene, and mixtures of these with each other and with rare gases have also been studied recently (Ferriero, 1986). The ionic species in plasmas containing silane (Haller, 1980; Turban et al., 1979, 1980, 1982; Perrin et al., 1984), organosilanes (Smolinsky and Vasile, 1973; Vasile and Smolinsky, 1974), siloxanes (Schmidt et al., 1979), and fluorocarbons (Vasile and Smolinsky, 1977b) have also been studied. The general experimental approach in these studies is to observe the ionic species at low total pressure and then follow the pressure dependence of the subsequent condensation reactions. The interpretation of the data is complicated by simultaneous neutral condensation reactions, the products of which ionize within the plasma to give ionic species which often are indistinguishable from the ion-molecule reaction products. The neutral chemistry in such systems must also be studied, either by flux analysis or partial pressure analysis. Practical considerations in the flux analysis of plasmas containing condensing molecules are the gas phase formation of particulate matter and the rate of solid film formation on the discharge tube walls. The sampling aperture may be instantaneously blocked by a solid particle or reduced to infinitesimal dimensions by a rapidly growing film.

Sputtering phenomena in both DC and rf discharges have been extensively studied by ion flux analysis. The conditions of low pressure, long mean free path, and the limited possibilities for ion and neutral reactions make the data interpretation less ambiguous in comparison to condensing systems. The principal objectives of ion flux analysis in sputtering systems are the identification of species involved in reactive sputter deposition (Shiniki and Itoh, 1974; Aita et al., 1981; Coburn and Kay, 1972), the correlation of the resulting film properties with ionic species (Coburn and Kay, 1972; Coburn et al., 1975) or ion energy (Köhler et al., 1985a and 1985b; Coburn et al., 1974a; Eckstein et al., 1975) at the deposition site,

and the definition of the decomposition of compound sputtering targets (Aita, 1985; Eckstein et al., 1975). These studies of the ionic species in reactive and nonreactive sputtering systems have emphasized the importance of energy transfer mechanisms in low-pressure rare gas plasmas (Coburn et al., 1974a; Eckstein et al., 1975), and have all led to the development of a practical solids analysis technique based on these principles.

D. Ion Flux Analysis: Experimental Considerations

The formation of ions is the most energetic atomic process in glow discharges, and the steady-state ion density is small compared to neutral particle densities. The ionic species observed are highly susceptible to the presence of small amounts of impurities or residual gases via two principal mechanisms: 1) the presence of an impurity with a large ionization cross section and an ionization threshold significantly lower than the major gas constituents; 2) impurities or residual gases that participate in ion-molecule reactions or resonant charge transfer processes with the ions that are formed by electron impact within the plasma. Residual gases such as H_2O, CO, CO_2, or H_2 in plasmas containing rare gases usually satisfy one or both of the above conditions, and early studies of flux analysis of ions (Knewstubb and Tickner, 1962a) clearly show the extent to which these species can act as the heavy charge carrier in the plasma. The residual gas composition of the discharge tube *at the pumping speed* used to provide the gas flow in the discharge is the critical variable. Since the pumping speed on the discharge vessel is reduced in many experimental situations, the outgassing rate under these throttled pumping conditions with no gas flow must be low enough to yield a pressure of $\sim 10^{-4}$ times the total pressure when the gas is flowing in order to minimize the participation of residual gases in the ionic processes. Thus, the discharge tube must be leak-tight and well preconditioned prior to taking data, either through use of a bakeout or discharge cleaning procedure.

Standard mass spectra and reference values of absolute ion currents on selected species are recommended for use in the flux analysis of ions. In the past, it is evident that little use has been made of ion signals on an absolute basis. It is far simpler to use relative signals for observing the effects of variables such as pressure or applied power. The value of a reference set of absolute signal intensities on a few ions and a reference mass spectrum under "standard" conditions lies in the fact that when such conditions are established prior to a change in parameters, we may be confident that

several important criteria are being fulfilled: 1) Energy coupling to the gas is reproduced. 2) The sampling aperture and ion optics are functioning in a reproducible manner. 3) The residual gas contributions are minimal. If such reference conditions are not achieved, comparisons with previous results may be of little value. Indeed, the data interpretation process will be unnecessarily complicated by extraneous factors.

VI. Flux Analysis: Neutral Species

Plasmas of practical interest in materials processing (e.g., etching or deposition) are characterized by chemistry initiated by electron impact on molecules of some complexity. The resulting reaction products can be observed with a flux analysis system shown in Figure 1 by arranging the potentials to repel ionic species and then ionizing the neutral flux. The sampling system is essentially a molecular beam apparatus when operated in this manner, and the requirements for an accurate representation of the effusing species are those which apply to molecular beam systems (Ramsey, 1969) with mass spectrometric detection (Vasile et al., 1975).

The sampling aperture diameter must be small with respect to the mean free paths of atom–atom, atom–molecule, and molecule–molecule collisions of the gases which are present in the plasma. This is a concern in the flux sampling of neutral species because of the effects of formation of van der Waal's molecules by isentropic expansion. (Note that in the case of flux analysis of ions, the concern was not to disturb the sheath potential.) Van der Waal's molecules, bound only by 0.01–0.2 eV, often ionize on electron impact to form stable molecular ions. These ionic species (particularly dimers) may be difficult to distinguish from those resulting from the ionization of the neutral gas phase products of the plasma (Dyke et al., 1972). This effect should be negligible for sampling low pressure discharges with systems designed for flux analysis of ions as well as neutrals, since the criterion of the orifice $D < \lambda_D$ will also satisfy the mean free path requirements for effusive rather than isentropic flow.

Collimation of the incoming neutral flux is difficult to achieve in systems that are intended for flux analysis of both ionic and neutral species. Ion optical lens components (cylindrical lenses in particular) have large apertures and confining surfaces which admit and ultimately scatter a larger solid angle of the neutral flux than is optimal for ideal molecular beam formation. Ionization of the neutral flux before significant scattering occurs is therefore essential in such systems.

Modulation of the neutral flux is highly recommended to improve signal-to-background ratios. Phase shift measurements or flight time analyses have generally not been performed in the flux analysis of neutral species from plasmas, since the geometry of most sampling systems precludes obtaining significant phase lags between different species that can be obtained with other types of molecular beam systems. Modulation is necessary primarily to enhance low level signal detection, which is often required in this application, and for the discrimination of directly sampled molecules from scattered molecules. Pumping speed on the QMS chamber is also a significant factor since the time required for a complete set of measurements is usually quite long. Ideally, the pumping speed should be high enough to avoid the buildup of background gas over the duration of the experiment. This is essential for systems which do not employ modulation of the neutral flux. For modulated systems, a high pumping speed preserves the depth of modulation over the duration of the experiment and makes the detection of minor components unambiguous with respect to background gases.

Flux analysis of neutral species by electron impact ionization and mass analysis gives us the rare opportunity to directly compare the results of plasma processes obtained on different reactors or to compare experimental results with modelling calculations. (The calibration procedures previously described for the flux analysis of ions are limited to internal consistency checks only.) Absolute calibration of the neutral gas components can do much to alleviate one of the major difficulties encountered by workers in the field of plasma chemistry, namely, that of comparison of results between different reactor geometries operating on the same gas mixtures.

The calibration procedure of most general utility is the establishment of the signal level for any particular gas in the discharge vessel as a function of the partial pressure. This is accomplished by separately admitting each gas of interest to the discharge vessel and measuring the signal for a prominent ion in the cracking pattern over a range of pressures and/or flow rates without a discharge. This procedure should include any inert "carrier" gases such as Ar, Ne, etc. Establishment of the partial pressures of the gases with the discharge on may then be done from the calibration, provided there are no products which yield interferences at the calibrator mass numbers. A self consistency check is available also, since

$$\sum_i P_i = P_{\text{TOTAL}}, \tag{10}$$

and the total pressure in the discharge vessel is independently measured. The use of this procedure in the flux analysis of neutrals is not widespread, presumably because of the convenience of using relative methods. Abso-

lute calibrations have been used in discharges of simple hydrocarbons (Smolinsky and Vasile, 1976; Vasile and Smolinsky, 1977), in the studies of fluorocarbons (Truesdale and Smolinsky, 1979; Truesdale et al., 1980), and in CF_4/O_2 etching plasmas (Smolinsky and Flamm, 1979; Turban and Rapeaux, 1983).

Ionization of the neutral flux with electrons with energies (60–80 eV) usually employed in mass spectrometers produces significant fragmentation of molecules. In plasma applications, this often results in fragment ions from one species, which interfere with the ions of another, since many of the molecules produced by the plasma are either multimers or very simple fragments of the starting molecules. Electron energies of the order of 20–30 eV have been found to be helpful to reduce this effect (Vasile and Smolinsky, 1976; Truesdale and Smolinsky, 1979; Manory et al., 1983) and simplify the mass spectra in general. The requirement for using reduced electron energy is that the signal-to-background ratio must be adequate, since the absolute signal levels will decrease in accord with smaller ionization cross sections. The extent to which the electron energy should be reduced depends upon the appearance potential functions of the various fragment ions of the molecule of interest. For example, the fragmentation patterns of SiH_4 show that an electron energy in the range of 20 eV reduced the SiH^+ and Si^+ ions to a very low level with respect to their 60 eV values, while the ionization efficiency of SiH_3^+ and SiH_2^+ at 20 eV actually increases somewhat relative to the 60 eV value (Turban et al., 1980). A similar effect is apparent in $SiCl_4$ for the ionization efficiencies (Manory et al., 1983) of $SiCl^+$ and $SiCl_2^+$ relative to $SiCl_3^+$ and $SiCl_4^+$. The appearance potentials of the fragments from ethylene at 18–20 eV, on the other hand, show a substantial loss in the $C_2H_4^+$ signal is necessary if the relative intensities of the fragments $C_2H_2^+$ and $C_2H_3^+$ are to be reduced vs. their 60–80 eV values (Catherine et al., 1982).

Fragmentation patterns for each of the molecules of interest must be measured with the QMS, regardless of the electron energy used to ionize the neutral flux. This procedure ensures that variations in ion transmission characteristics with mass and/or resolution will not obscure the identification of any given species. Alternatively, the transmission function of the mass spectrometer can be determined against the reference spectrum of a calibration compound (Longway et al., 1984), and corrections subsequently applied to the data.

There are two disadvantages to adhering to the calibration procedures outlined above, which have apparently discouraged their widespread use in the past. The first is the sheer volume of calibration data needed for both qualitative (cracking patterns and appearance potential) analysis and for

quantitative analysis (sensitivities in absolute terms, i.e., signal strength vs. partial pressure). The widespread use of computers for rapid data acquisition should make this tedious and overwhelming task far less forbidding than it was as little as 10 years ago, and there are clear signs in the more recent literature that such is the case. The second deterrent to the implementation of absolute calibration procedures is the required capability of the vacuum system to maintain the low-level background signals, especially when the products of the discharge are strong chemisorbers, as is frequently the case. Pumping speeds and QMS chamber conductances sufficient to reach the 10^{-9} torr range (Smolinsky and Flamm, 1979), combined with modulation of the incoming flux, are required (Wagner and Veprek, 1982).

The identification and measurement of neutral products are iterative procedures, since the calibration for many of the species produced by the discharge will not be available *a priori*. The mass spectrum observed from the plasma products must be interpreted first in terms of the stable neutral products which are possible from the starting gas or gas mixture. Additional calibration measurements with molecules that are deduced as products are then performed, both for sensitivity factors and for the cracking pattern at the relevant electron energy. In some cases, the discharge products cannot be obtained or are not stable molecules. Indirect methods (Longway et al., 1984) for the determination of sensitivity factors must then be used. A final consistency check for the overall mass spectrum must be made to insure that no interferences occur and that any sensitivity factors obtained by indirect methods are not unreasonable in magnitude.

Discharges in some gases result in relatively high steady-state concentrations of radicals, which can be observed directly by the flux analysis of neutral species. A difference in the fragmentation pattern indicates the possibility of a radical species. Appearance potential measurements (McDowell, 1963) on the ion fragment that is suspected of having a radical precursor are necessary, with the discharge on and with the discharge off. The measurements with the discharge off are necessary to establish the appearance potential of the ion fragment to an accuracy of about 0.5 eV. Appearance potential measurements have been used to establish the presence of the CF_2 radical in the CHF_3 discharge, in the 60% H_2/CF_4 discharge (Truesdale and Smolinsky, 1979), and in the C_2F_4 discharge (Vasile and Smolinsky, 1977). The same techniques show that measurable concentrations of CF_2 could not be detected in pure CF_4 discharges (Truesdale and Smolinsky, 1979), nor could any direct evidence for CF_3 radicals be found in pure CF_4 discharges (Smolinsky and Flamm, 1979). Direct evidence for C_2H_3 radicals in the rf electrode sheath of C_2H_4 discharges (Smolinsky and Vasile, 1976) has been found by the imbalance

in the 20 eV cracking pattern of the neutral species. The detection of radicals by flux analysis requires that their steady-state concentration within the discharge be high enough so that the radical flux to the walls is roughly equivalent to that of the minor stable gas phase products. Gas phase reactions which act as sinks for the radical population can result in the nonobservance of a suspected radical intermediate. Thus, a long path between the plasma edge and the sampling point reduces the probability of observing free radicals produced in the plasma. This may be the reason why CF_2 and CF_3 were not observed in the CF_4 discharges mentioned previously (Truesdale and Smolinsky, 1979); Smolinsky and Flamm, 1979). However, wall reactions cannot be invoked as the cause for nonobservance. A well designed flux sampling apparatus should detect radicals arriving at walls, regardless of whether or not the wall surface is the major sink for the radical. Detection of radical species by the appearance potential technique is a matter of absolute sensitivity and the rejection of background signals. These vanishly small signals require the signal-to-noise rejection provided by modulation with phase sensitive detection or gated pulse counting. Line-of-sight trajectories to the particle multiplier must also be avoided since the "dark current" or null signal must be minimal for these measurements.

VII. Experimental Considerations and Applications: Flux Analysis of Plasmas

The impetus for the studies of ion chemistry in plasmas by mass analysis during the 1960s came from indications of ion-propagated chain reactions in radiation chemistry (Dawson and Tickner, 1964 and 1966). Plasmas also afforded a nearly field-free environment for the study of ion-molecule reactions, which was another attractive prospect at that time (Studniarz and Franklin, 1968; Schildcrout et al., 1970). These early studies emphasized flux analysis of the ionic species and were more inclined to define the discharge itself, rather than provide information on a plasma chemical process for materials transformations. The conclusions regarding similarities between ionic chain reactions and their relationship to radiation chemistry were supportive, although qualitative. Similarly, the measurement of ion-molecule reaction rate constants could be done, but only in cases where the ion chemistry and the discharge gas composition was unambiguously defined (Schildcrout et al., 1970).

Flux analysis of ions from plasmas did not totally fulfill the early expectations with regard to radiation chemistry or ion-molecule chemistry,

but the information gained from these studies proved to be of considerable value for subsequent studies. The flux analysis of ions from a variety of discharges (either rf or DC driven) during the era from about 1963 to 1975 clearly emphasized several features of the physical chemistry in discharges which were not previously appreciated or known: 1) The ionic species observed were, in most cases, not representative of the species expected from electron impact on the gases present in the plasma. Secondary ionization mechanisms (such as Penning ionization) had to be considered. 2) Ion-molecule condensation reactions proceeded to a considerable degree within a plasma, and the distribution of ions representing the species within the body of the plasma could only be obtained at low pressure ($P \lesssim 0.05$ torr) due to collisions within the sheath. 3) The species observed were strongly dependent upon the sampling location. 4) The choice of rare gas used to dilute a reactive gas severely affected the ionization mechanisms. 5) The interpretation of the spectra in terms of a few dominant ionization and clustering mechanisms became obscured at elevated pressures, especially when chemical reactions among the neutral components could be initiated by low-energy electrons.

The complexity of an active (i.e., non-afterglow) plasma in relatively simple gas mixtures precludes an unequivocal interpretation of the data obtained by ion flux analysis, even when the sheath is collision-free. A different view of the plasma was taken, stimulated mainly by the need to know more about how the neutral gas chemistry proceeds within the plasma. The emphasis then shifted to flux analysis of the neutral species in plasmas where the conversion of starting materials was relatively high. This approach treats the plasma as a nonequilibrium reactor (Truesdale and Smolinsky, 1979; Truesdale et al., 1980; Smolinsky and Flamm, 1979), and the reaction mechanisms must be inferred by analogy with photochemistry and radiation chemistry. This concept is not universally applicable, since in many critical plasma processes the conversion of starting material into stable gaseous products is low, and the detection of neutral products by flux analysis becomes difficult. Recourse has been made to limited analytical data on the neutral products, supplemented by a flux analysis of the ionic species (Picard and Turban, 1985; Coburn and Kay, 1977).

A comprehensive picture of the physical chemistry within the plasma is possible with flux analysis, and this is the feature that makes it so attractive. There are examples where the flux analysis of ions and neutrals has provided the most detailed description of material production or modification by plasmas available. These descriptions are by no means complete or quantitative, but they are significantly better than could be obtained by treating the plasma as a flow reactor with only inference about the mechanism. A good case in point is the chemistry of SiH_4 in rf or DC discharges.

The formation of amorphous hydrogenated silicon (a-SiH_x, with $x \sim 0.1$ to 0.3) by plasma processes has potential as a method for producing photovoltaic devices (Brodsky, 1979; Knights and Lucovsky, 1980). The deposition of such films has been studied by flux analysis of ions, or of ions and neutrals in DC discharges, DC multipole plasmas, and rf excited discharges (Haller, 1980; Turban et al., 1979, 1980, 1982; Perrin et al., 1984; Longway et al., 1984). The findings of all of these studies indicate that the ion chemistry observed is basically that of the ion-molecule system (Yu et al., 1972; Kenis et al., 1972) initiated by SiH_2^+ and SiH_3^+ on SiH_4, such as

$$SiH_2^+ + SiH_4 \rightarrow SiH_3^+ + SiH_3 \tag{11}$$

$$SiH_2^+ + SiH_4 \rightarrow Si_2H_n^+, \qquad n = 2 \rightarrow 5 \tag{12}$$

$$Si_2H_n^+ + SiH_4 \rightarrow Si_3H_m^+ \cdots \tag{13}$$

with ionic species condensed (Haller, 1980; Turban et al., 1980) as far as $Si_6^+H_x$ and $Si_7^+H_y$, depending upon conditions. In cases where the neutral gas phase products were measured (Turban et al., 1982; Longway et al., 1984), the ionic chain could not be accounted for by electron impact ionization of the neutral species, especially for species $Si_3H_n^+$ and above. The neutral products of the SiH_4 discharge are H_2, Si_2H_6, and Si_3H_8 in the gas phase and a-SiH_x as a solid product. The investigations are in agreement with the supposition that the Si_2H_6 and Si_3H_8 are the end products of free radical reactions. NO scavenging experiments (Longway et al., 1984; Schmidt and Lampe, 1969) indicate that there may also be an ionic path to these products. Addition of a radical scavenger to the SiH_4 discharge results in total inhibition of solid formation (Longway et al., 1984), and the consensus is that a-SiH_x is formed by the diffusion of SiH_2 and possibly SiH_3 to the walls. A mechanism for the film growth stimulated by photolysis of SiH_4 to yield SiH_2 has been given in the recent literature (Motooka and Greene, 1986), but it is not clear how the SiH_3 radical might participate. The specific film formation mechanism that prevails in a plasma is also apparently dependent upon the experimental conditions (Tsai et al., 1986), such as the mean free path and the partial pressure of silane. A comprehensive picture of the a-SiH_x film formation has not been established by the studies that have been done thus far, since the radical flux to the walls, which presumably forms the solid film at an appreciable rate, remains undetected (Turban et al., 1980 and 1982). Many authors who have studied the formation of a-SiH_x by mass analysis have been reluctant to rule out an ionic contribution to the film growth (Haller, 1983; Weakliem

et al., 1987), especially in view of the large ion-molecule cross sections for this system (Yu et al., 1972; Kenis et al., 1972) and considering that NO scavenging does not eliminate all gas phase products (Schmidt and Lampe, 1969). Mass spectrometric flux analysis of the products in hydrogen plasmas (Turban et al., 1980; Wagner and Veprek, 1983; Catherine et al., 1981) on a-SiH_x have shown that a definite "back-reaction" occurs; i.e., the material deposited can be vaporized back into the discharge by the action of an H_2 plasma. Further implications of the residence time in the plasma and the type of film grown have also been shown (Longway et al., 1984).

An interesting outgrowth of the studies in SiH_4 are the mass spectrometric studies of SiH_4/CH_4 and SiH_4/C_2H_4 mixtures (Catherine et al., 1981 and 1982). The SiH_4/CH_4 mixture shows a chemistry in which the SiH_4 and the CH_4 react independently of each other (Catherine et al., 1981). The SiH_4/C_2H_4 system is in direct contrast with considerable cross products observed in both the ion chemistry and the neutral gas phase products (Catherine et al., 1982). Flux analysis has also been used to study the action of plasmas in $SiCl_4$ (Manory et al., 1983 and 1984), which also results in a solid silicon film.

Flux analysis of ionic and neutral species from process plasmas generally supplies an overwhelming array of data, much of which appears contrary to expectation or intuition based on radiation chemistry or photolysis studies. These plasmas are far more complex than any comparable or analogous chemical reaction system. The gas phase chemistry is both free radical and ionic in nature and, furthermore, is not described by equilibrium processes. Heterogeneous processes often act as sources or sinks, and of course a variety of electron impact processes (excitation, dissociation, and ionization) have to be considered. The only realistic method for explaining the experimental results observed in flux analysis of these systems is through computer modelling with sensitivity analysis. Large numbers of concurrent gas phase reactions initiated by electron impact can be calculated with boundary conditions that simulate the experimental situation: The stable products can then be determined, and the importance of each of the individual reactions within the initial set can be tested. The results of the calculations are then compared to experimental measurements, and, when reasonable agreement has been achieved, the major contributors to the discharge chemistry have been defined.

The emergence of computer simulations of plasma processes has begun recently, with attempts to make a detailed description of the fluorocarbon-oxygen systems used in the plasma etching of silicon (Kushner, 1982; Edelson and Flamm, 1984; Plumb and Ryan, 1986; Ryan and Plumb, 1986). Kushner's model (1982) included an ambitious accounting of ionic

and neutral gas phase chemistry and heterogeneous chemistry. The modelling was reasonably successful at reproducing the qualitative features of the etching of Si and SiO_2 by a CF_4/H_2 plasma and quite successful in accounting for the gas phase products from a CF_4/H_2 plasma (Truesdale and Smolinsky, 1979). Edelson and Flamm (1984) constructed a model utilizing neutral chemistry only and boundary conditions which were a very close approximation to experimental measurements of the CF_4 etching of Si (Smolinsky and Flamm, 1979). The quantitative agreement between the model and the experimental results can be considered to be extremely good, but, more importantly, the sensitivity analysis indicated which of the 40 input reactions were dominant. Plumb and Ryan (Plumb and Ryan, 1986; Ryan and Plumb, 1986) have constructed successful models which reproduce the major features of the experimental results of the CF_4/O_2 plasma measured by flux analysis (Smolinsky and Flamm, 1979). The effluent gas composition was predicted within a factor of two, without adjustable parameters, as shown in Figure 9. Furthermore, their original base of 49 reactions could be reduced to 13 without a significant effect on

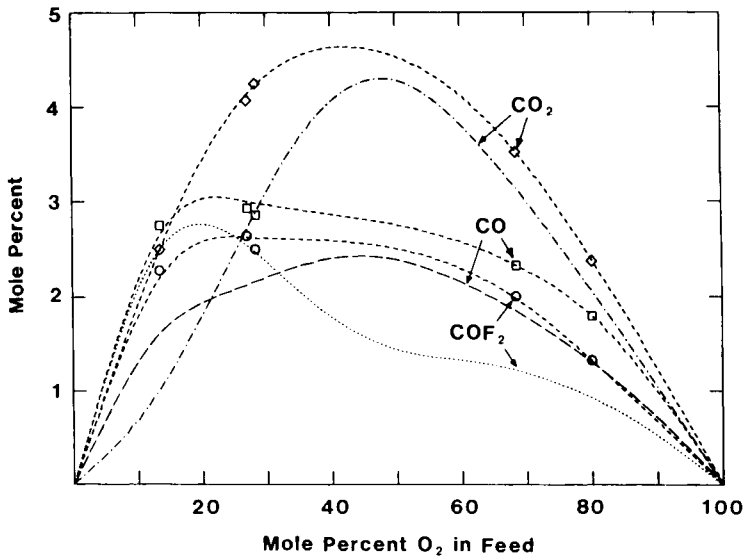


FIGURE 9. Computer simulation of the gas phase products of a CF_4/O_2 rf discharge compared with a mass spectrometric flux analysis of the same. (After Plumb and Ryan (1986), with permission.) The dashed curves with symbols are the experimentally determined values, and the curves without symbols are the results of the computer simulation.

the accuracy of the simulation. The computer simulations for the etching of Si in a CF_4 plasma (Ryan and Plumb, 1986) are also remarkably successful. The key to an accurate simulation lies in identifying those reactions which have the most impact upon the overall process and the accurate measurement of their rate constants (Ryan and Plumb, 1984a and 1984b).

The systematic interpretation of the data provided by flux analysis of plasma requires a computer simulation. The computing power in terms of the appropriate software and hardware is clearly available. The major impediment to general applicability of computer simulations of plasmas is the incomplete data base. Additional measurements of electron impact cross sections for radical production and ionization, for rate constants of radical reactions, and for reaction probabilities of heterogeneous processes are necessary. The thrust of experimental measurements in the past several years has been to measure the appropriate rate coefficient or cross sections of the individual processes which occur in plasmas. For example, the ionization cross sections of the halogen atoms and radicals such as CD_2 and CD_3 have been measured (Hayes et al., 1987; Baiocchi et al., 1984), the chemisorption of Cl_2 on Si has been studied (Margaritondo and Rowe, 1980), and numerous measurements of the reaction probabilities of plasma species with Si have been made (Tu et al., 1983; Flamm et al., 1981; Vasile, 1982; Miyake et al., 1982). Computer simulations will certainly improve as more reliable input data become available. For plasma chemical systems of considerable technological impact, such as etching mixtures or a-SiH_x deposition, we may see a comprehensive description in the near future, with the ionic and neutral chemistry clearly defined. It would then be possible to go further and use the simulations to predict the results of parameter variations, thus taking the dreadful empiricism out of our present approach to plasma processing. We hasten to point out, however, that the most successful descriptions of plasma chemistry have relied upon flux analyses which had been calibrated on an absolute basis.

VIII. Experimental Considerations and Applications: Partial Pressure Analysis

The interface of a partial pressure analyzer to a plasma device has been poorly addressed in the literature and commercial documentation, and improper interfacing can lead to confusing and erroneous data interpretation. The following sections will address these interface problems in detail.

A. Pressure-Reduction versus Dynamic Range

Let us consider the operating specifications of the typical high performance quadrupole mass spectrometer (QMS) that could be used for partial pressure analysis. The operational pressure range with a secondary electron multiplier ion detector is $10^{-11} - 10^{-4}$ torr. The lower limit is considerably above the minimum partial pressures which have been detected with a QMS ($< 10^{-13}$ torr), but does represent a practical detection limit defined as a signal-to-noise ratio of approximately unity after integration of the minimal output current ($10^{-11} - 10^{-12}$ A) of a 14 stage CuBe multiplier over a period of 1 second. The high pressure limit (10^{-4} torr) is usually defined as the pressure at which gas scattering becomes important in the ion source and the signals become nonlinear with total pressure. It is a rare experimental situation in which one has the luxury of the full factor of 10^7 in dynamic range that such a QMS affords, because of the presence of background signals at mass numbers of practical interest. Background levels are almost always a concern when sampling plasma systems because of the chemically active gases that are involved and the increased chemical reactivity of volatiles caused by dissociation, excitation, and ionization in the plasma.

To rescale the operational pressure range of the QMS so that the upper pressure limit encompasses the regime of glow discharge plasmas ($10^{-3} - 1$ torr), it is necessary to use a pressure reduction stage between the plasma chamber and QMS chamber. This pressure reduction can be accomplished with a low conductance connection tube with differential-pumping of the QMS chamber, as shown in Figure 2. By such means, the upper pressure limit can be scaled arbitrarily, and the full dynamic range of measurement is available over the rescaled range. The minimum detectable pressure in the plasma chamber would increase in proportion to the rescaled upper pressure limit, assuming there were no increases in background levels. However, in practice the pressure reduction cannot be accomplished without a commensurate increase in background levels because of the high flow of the primary plasma gas species through the throttling conductance and the QMS. Figure 10 plots the tradeoff between the pressure reduction factor and the minimum detectable partial pressure in the plasma chamber for a QMS with the specifications given above. Also shown is the decrease in system dynamic range as a function of the background partial pressure.

A practical example: The application of partial pressure analysis to hydrogen plasmas illustrates the possible severity in loss of dynamic range when sampling reactive gas systems. When molecular H_2 is introduced into the typical electron-bombardment ionizer, of the order of (0.1%–0.5%) of the H_2 will be converted to the residual gases H_2O, CO, and CH_4 because

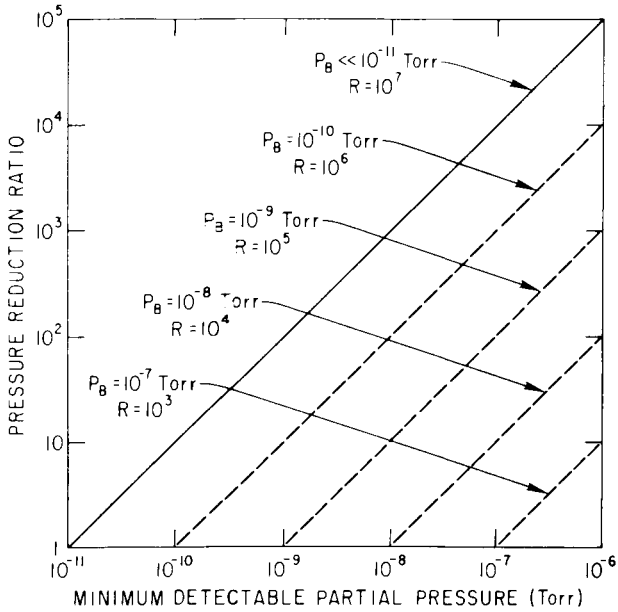


FIGURE 10. Tradeoff of the pressure reduction ratio (the ratio of static pressure in the plasma chamber to QMS chamber) versus the minimum detectable partial pressure in the plasma chamber. The QMS is assumed to have a minimum detectable partial pressure of 10^{-11} torr and a dynamic range of 10^7. Also shown (with the dotted lines) is the loss in dynamic range (R) with increasing background pressure (P_B) in the QMS (from Dylla (1983) with permission).

of the formation of atomic hydrogen on the ionizer filaments and the subsequent reaction of H_2 with the electrode surfaces.

This problem has been studied in detail (Dylla and Blanchard, 1983; Boucher et al., 1986), and Figure 11 shows the results of background measurements as a function of ambient H_2 pressure for a commonly used QMS ionizer. Without special conditioning of this ion source, the background levels of H_2O, CO, and CH_4 are such that the dynamic range is reduced to 10^3 for H_2O and 10^4 for CO/CH_4 for measurements of partial pressures at masses 18, 28, and 16 amu in H_2 plasmas. Pressure reduction would exacerbate this situation, because the necessary throttling valve would preferentially conduct the lower mass H_2 in comparison to the higher mass residual gases, thus decreasing the effective signal-to-background level for these gases. This enhanced background problem is not limited to the case where the primary plasma species is a chemically active

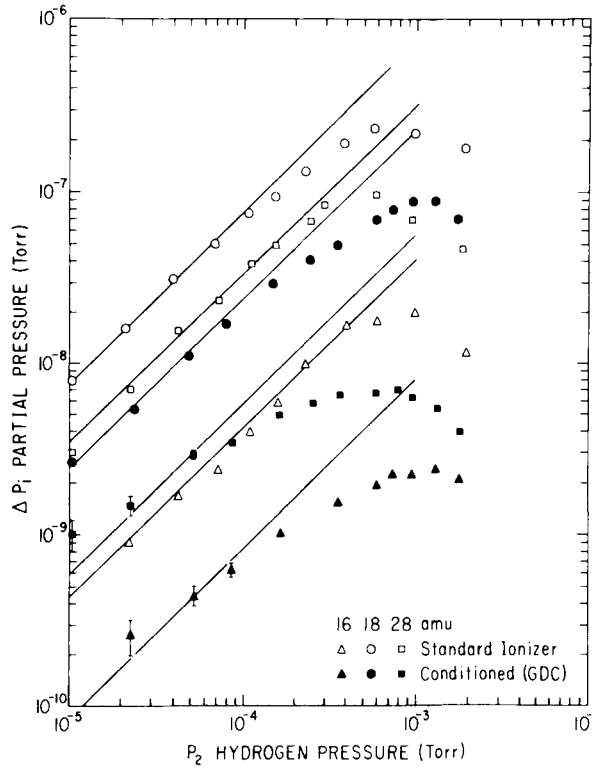


FIGURE 11. Change in partial pressure of the dominant residual gases produced in the extranuclear 041-1 ion source as a function of the H_2 pressure for the standard ion source (open points) and after Cr plating and H_2 glow discharge conditioning (filled points) (from Dylla and Blanchard (1983) with permission).

gas. The effect also has been documented in the sampling of inert gas (argon) plasmas (Rozgonyi, 1966).

Standard UHV practice in the design and handling of the QMS components can reduce the background problem. In addition, there are several alternative schemes to minimize the problem: 1) The ionizer can be treated to minimize the offending gas-surface interactions. The interaction of H_2 with standard QMS ionizers was traced to the interaction of atomic hydrogen with iron oxides on the electrode surfaces. Figure 11 shows that a factor of five improvement in background generation was obtained when the electrode surfaces were chrome-plated and then conditioned by exposure to a H_2 glow discharge. 2) For background problems that can be traced to the presence of a hot filament in the ionizer, a low temperature

emitter can be substituted, or an externally generated electron beam can be used as the ionization source, allowing the removal of the filament from the QMS chamber (Ertl and Taglauer, 1973; Hofer and Seggern, 1984). A disadvantage of the common low-temperature emitters (Th/W and LaB_6) is that their emission characteristics are unstable in active gases. The primary disadvantage of the compound ionizer with a separated electron source is the considerable complication in experimental design over the relatively simple arrangement shown schematically in Figure 2. 3) A last resort solution to an intolerable background problem in a partial pressure measurement is to revert to a flux type measurement. By introducing an effusion aperture on the line-of-sight between the plasma chamber and QMS, and modulating the effusing beam with a mechanical chopper, the signal-to-background ratio of the QMS can be improved significantly using the well known phase-sensitive detection technique (Foner and Hudson, 1953) on the QMS output.

B. System Response

Figure 12 shows the experimental scheme of Figure 2 redrawn with the important kinetic-gas factors in the system identified: the plasma chamber with volume V_D, is pumped with speed S_P, and may have one or more gas input sources Q_i; the QMS chamber with volume V_Q, is pumped with a separate pump with speed S_Q, and is connected to V_D through a tube with conductance C_T. The quantities S_P, S_Q, and C_T can be a function of the

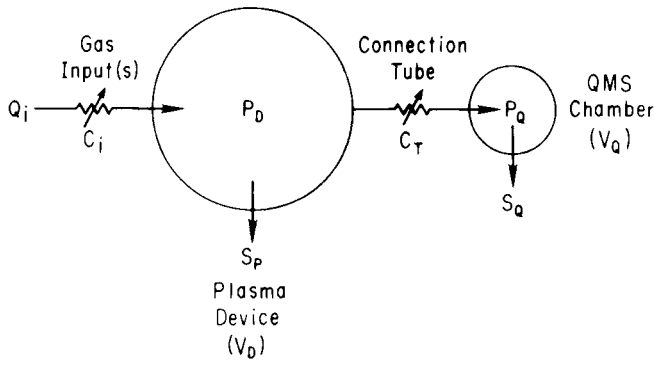


FIGURE 12. Simplified vacuum schematic for the partial pressure analyzer shown in Figure 2 (from Dylla (1983) with permission).

gas temperature T, and the pressure P, mass M, and viscosity of the gas depending on the flow regime and type of vacuum pump.

For proper calibration of a partial pressure analyzer, the relationship between the pressure change in the plasma device $\Delta P_D(t)$ and the pressure change in the QMS chamber $\Delta P_Q(t)$ needs to be evaluated. First, we consider the simplified case of nonreacting gases, i.e., gas-surface interactions within the connection tube and the QMS are ignored, so that only the kinetic-gas relationship between $\Delta P_D(t)$ and $\Delta P_Q(t)$ need be considered. For static measurements ΔP_D and ΔP_Q are related simply by a multiplicative throttling factor, ρ, which is the same number as the pressure reduction ratio used as a parameter in Figure 10:

$$\Delta P_D = \rho(\Delta P_Q). \tag{14}$$

ρ is inversely proportional to the net series conductance (C_T) of the connection tube and any intervening apertures or flow restrictions, and proportional to the differential-pumping speed, S_Q.

For dynamic measurements, a differential equation describing the time response of the system must be solved:

$$P_T(t) = \left(1 + \frac{S_Q}{C_T}\right) P_Q(t) + \frac{V_Q}{C_T} \frac{d[P_Q(t)]}{dt} - \left(1 + \frac{S_Q}{C_T}\right) P_B, \tag{15}$$

where the last term represents an explicit background pressure P_B correction in the QMS. Equation (15) usually has to be solved numerically. Solutions of interest to partial pressure analysis have been given by Poschenrieder et al. (1982) and Waelbroeck et al. (private communication).

Waelbroeck et al. (1984) have considered the additional complication to system time response that results when the gas molecule of interest has a non-ignorable sticking time on the surface of the connection tube in comparison to transit times. They found for the case of the transmission of H_2O through a stainless steel connection tube that the equilibration time between a step function increase in P_D and a static value of P_Q was greater than an hour for $T \simeq 300$ °K and declined to ~ 10 minutes for $T \simeq 430$ °K. Thus, absolute dynamic measurements for P_{H_2O} in such a system would be difficult on time scales shorter than the equilibration time because of the variable dependence of the equilibration rates on the surface conditions of the connection tube.

C. Calibration Procedures

A common occurrence in the vacuum science literature is the presentation of partial pressure data in an uncalibrated format, i.e., the signal output of a QMS is displayed as raw data (the electron multiplier or Faraday cup

output current) or in arbitrary intensity units. Without, as a minimum, a calibration that encompasses the entire system, the interpretation of QMS data can be very misleading. Consider the simplified vacuum system schematic shown in Figure 12. A system calibration determines the functional relationship between P_D and P_Q for one or more gases over a pressure range of interest. In addition, the relationship between P_Q and the measured output current of the QMS must be determined.

In certain situations the calibration procedure can be quite complicated. For example, if the pumping speed of the plasma chamber or QMS chamber pump is not a well defined function of gas species and pressure, if the calibration must span a pressure regime which causes the conductance of the connection tube to overlap more than one kinetic flow regime, or if the gas species of interest has significant physical or chemical reactions with the vessel wall materials; then the calibration procedure is not straightforward.

There are several examples in the literature of experimental arrangements for partial pressure analysis that are difficult to calibrate, unstable with respect to maintaining a calibration, or capable of being calibrated only over a limited range of parameters. The use of an ion pump for the differential-pump on the QMS chamber (Raby, 1978) is a bad choice because of the large variation of pumping speed with gas species, the tendency of the pumping speed to decrease with time and total throughput, and the potential for gas re-emission at high throughput. The closed-cycle cryopumps operating at 20 °K have the same disadvantages and also should be avoided for this application (Bolker et al., 1981).

The optimum type of pump for this application is either a turbo pump or a diffusion pump, because the pumping speed for both types of pumps can be characterized as function of the mass of the gas and the gas pressure, and, for all practical purposes, the speed does not vary with time or total throughput. This property greatly simplifies the calibration of a QMS system over a wide range in mass and pressure.

A second common pitfall which complicates calibration procedures is the use of needle-type throttling valves in the connection tube to accomplish the pressure reduction. The mechanical hysteresis that is inherent in the design of most valves of this type makes repeatability difficult. A better choice is the use of a series of well defined apertures. There exists no optimal calibration procedure for the partial pressure analysis scheme shown in Figure 12, nor have the standards organizations addressed this experimental situation. However, a number of relatively simple approaches can be taken which yield satisfactory results. If the process to be monitored is a static process or otherwise slowly varying with respect to the characteristic vacuum time constants, then a static calibration will suffice. In this case, a predetermined gas mixture can be introduced to V_D, the amount of

introduced gas is quantified either by means of a calibrated leak (Q_i) or by comparison with a calibrated total pressure gauge on the volume V_D, and, finally, the response of the QMS as a function of mass is recorded. For dynamic measurements, single component gases must be introduced individually, and the system differential equation [Eq. (16)] can be obtained numerically from the measured pumping transients (V_D/S_D, V_Q/S_Q) and the form of $dP_Q(t)/dt$.

Specific calibration procedures have been described in the recent literature involving the use of equimolar gas mixtures with absolutely calibrated total pressure gauges (Blanchard et al., 1986), and single component molecular leaks (Ellesfon et al., 1987), quantified gas bursts (Dobrozemsky, 1985), and vibrating quartz crystal flux monitors (Winters, 1985). An important distinction between various calibration methods is whether or not the calibration involves a flow vs. partial pressure measurement of the calibration gas species. This choice usually is determined by the intended use of the instrument. It is usually simpler and less susceptible to error to have the calibration method and intended use be both flow-type or both partial pressure-type measurements. The methodology of both types of calibration procedures have been described recently by Santeler (1987). For most plasma diagnostic applications of mass spectrometry, the useful form of calibrated output is a mass flow rate for a specific species (i.e., the number of molecules/second generated by the surface or gas-phase reactions in the plasma). Calibrated output in this form can be obtained by either type of calibration procedure. If the mass spectrometer vacuum pump is the only pump in the system (i.e., the plasma chamber pumping speed, $S_p = 0$, in Figure 12), then a flow calibration will yield the specific mass flow rate directly. With the more general case of $S_p \neq 0$, partial pressure calibrations are more useful, and specific mass flows, Γ_m, in the plasma chamber are equal to the product:

$$\Gamma_m = S_m \zeta P_Q, \tag{16}$$

where S_m is the plasma chamber pumping speed for species m, ζ is the pressure reduction ratio, and P_Q is the measured partial pressure in the mass spectrometer chamber.

Most calibrations rely on the use of one or more absolutely calibrated total pressure gauges. Many type of gauges are available at higher pressures (above 10^{-3} torr) that can be conveniently calibrated against widely used low pressure standards such as oil or mercury-filled manometers (Heydemann et al., 1977; Tilford, 1983; Nash and Thompson, 1983) or dead-weight balances (Mitchel, 1981). However, in the high vacuum range ($10^{-7} - 10^{-3}$ torr) there exist no universally recognized high vacuum standards, although there is much activity among the various national standards organizations concerning the suitability of the recently developed

spinning rotor gauge (Fremerey, 1985) for use as a high vacuum standard over the range $10^{-7} - 10^{-2}$ torr and the suitability of high accuracy capacitance manometers (Sullivan, 1985) for use as secondary (or transfer) standards over the range of $10^{-5} - 1$ torr. Recent studies (Messer et al.) at the PTB (Berlin) of the long-term stability of the spinning rotor gauge have demonstrated accuracies of $\pm 2\%$ for this gauge over a wide high vacuum range. Similar studies at the NBS (Hyland and Tilford, 1985) of capacitance manometers have demonstrated accuracies of better than 1% at the high end of operation. For use as transfer standards, capacitance manometers can be used to calibrate ionization gauges or mass spectrometers within the overlapping range of operation ($10^{-5} - 10^{-3}$ torr) with an accuracy of $\pm 5\%$.

There are numerous instrumental factors which can affect the overall sensitivity of a mass spectrometer once it has been calibrated. These factors include the stability of the ion optics in the ion source and extraction optics, the ion optical stability of the mass filter or mass selection element, and the stability of the ion detector (Dawson, 1986; Leck, 1985). In most situations variations in the ion source and sensitivity of the ion detector account for the observed variations in overall sensitivity of a modern mass spectrometer. Ion source variations can often be traced to interfering gas-surface interactions within the ion source from active gas species, such as H_2O, H_2, and halogens (Dylla and Blanchard, 1983; Blanchard et al., 1986). For example, water vapor and oxygen-containing molecules that are cracked on the hot cathode surfaces can transport cathode material to colder portions of the ion source and change the ion optics of the electrode array.

The most commonly used ion detector, the secondary electron multiplier, has an overall sensitivity which is dependent on the surface conditions of the dynodes. For the discrete dynode CuBe type multiplier, the sensitivity initially drops by more than an order of magnitude during the first few weeks of operation (with other conditions being constant). The sensitivity then stabilizes and drops more slowly with subsequent use. Figure 13 shows the measured long-term sensitivity of a quadrupole mass spectrometer over a 20-week period (Dylla, 1983) where the vacuum, ambient temperature, and electrical biasing were kept constant. The change in spectrometer sensitivity in this case is dominated by the decrease in sensitivity of the secondary electron multiplier. Any change in the ambient conditions, such as a bakeout cycle or a brief vent to atmospheric pressure, will change the surface conditions of the multiplier and hence lead to significant changes in sensitivity (Blanchard et al., 1986). If the high gain and fast response of the secondary electron multiplier detector is sacrificed with the use of the simple Faraday cup detector, then a significant increase in stability of calibration can be realized (Blanchard et al., 1986).

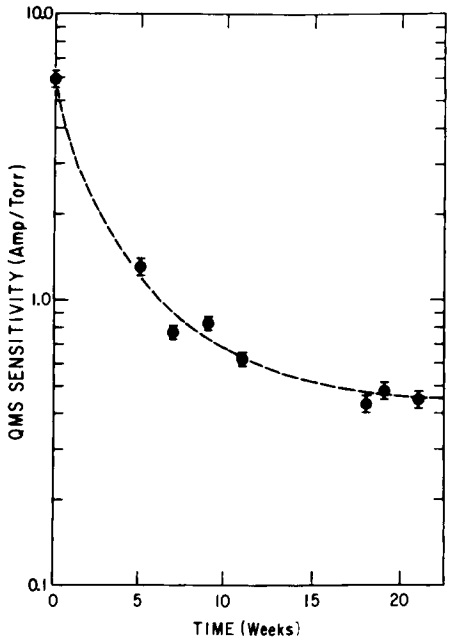


FIGURE 13. Observed change in sensitivity for N_2 as a function of operating time for a quadrupole mass spectrometer operated in the partial pressure analysis mode. These data were obtained with an extranuclear 041-1 ionizer operated at 1 mA emission, an extranuclear 324-9 quadrupole mass filter, and a 14 stage CuBe electron multiplier operated at 2.0 kV (from Dylla (1983) with permission).

D. Applications to Magnetic Fusion Devices

Mass spectrometry (usually in the form of partial pressure analysis) has had wide application in the field of magnetic fusion research for both the general task of monitoring the vacuum integrity of magnetic fusion devices and as a plasma diagnostic of important boundary plasma-vessel wall interactions. Understanding the primary mechanisms of plasma-wall interactions and developing specific plasma control procedures and wall materials to modify and minimize plasma-wall interactions have been key successes in fusion research (Post and Behrisch, 1986). The typical plasma conditions of the boundary plasma of a hydrogen isotope-fueled tokamak device include plasma densities in the range of $10^{11} - 10^{13}$ cm^{-3} and plasma electron temperatures of 10 – 100 eV (Post and Lackner). Plasma ions that are accelerated through the sheath, which develops at the plasma contact with vessel first-wall structures, can cause significant desorption and

sputtering of the wall material. If these surface processes are not controlled, then the fusion reactivity of the central plasma can be seriously degraded by dilution and excessive radiative energy loss by the sputtered or desorbed wall species. The most successful techniques for minimizing plasma impurities caused by plasma-wall interactions have involved judicious choices of first-wall materials and the development and optimization of in situ wall conditioning techniques usually involving some form of hydrogen discharge cleaning (Dylla, 1980; Cohen, 1986).

The technology of discharge cleaning, which also has important applications in accelerator vessel conditioning (Mathewson, 1974) and materials processing, has been developed primarily with the aid of mass spectrometry. For the case of conditioning magnetic fusion devices, hydrogen discharge conditioning procedures have been developed that involve simple DC or rf-assisted glow discharges, electron-cyclotron resonance (ECR) generated discharges, or rapid-rate pulse discharges produced by coil systems intrinsic to the fusion device. In any of the above plasma generation schemes, the generated discharge is at relatively low plasma temperature ($T_e = 2$ eV). The aim of the technique is to produce a relatively large flux of incident hydrogen ions (or atoms) onto the vessel wall, which promotes the formation of volatile compounds (H_2O, CO, CH_4, etc.) with surface impurities. The low plasma temperature prevents the subsequent ionization of the impurity species so that these species are removed from the vessel via pumping ducts rather than recycled to the vessel wall. Partial pressure analysis was the essential diagnostic technique for the initial development and optimization of both pulse discharge cleaning (Oren and Taylor, 1977) and glow discharge cleaning techniques (Dylla, 1980; Dylla et al., 1980) for tokamak vessels. This diagnostic is now an essential and routine monitor of the necessary conditioning procedures on all contemporary fusion devices, including the large precursor tokamak reactors (Dylla et al., 1984a and 1984b; La Marche et al., 1987; Dietz et al., 1984; Dietz, 1983; deKock et al., 1987; Kishimoto et al., 1987; Arai et al., 1987) and other magnetic fusion device configurations (Allen et al., 1984; Post et al., 1987; Jackson et al., 1984; Cayton et al., 1987).

The earliest application of mass spectroscopy to fusion devices was for the measurement of the residual gases which outgassed from the vessel following the cessation of a plasma discharge. This use was first described on the C-stellerator in 1966, (Martin and Lewin, 1966), which was the first magnetic fusion device to employ ultrahigh vacuum technology in the vessel design. This application continued with the fast pace of development of tokamak-type magnetic fusion devices during the 1970s. Early partial pressure analysis studies of the gases formed by plasma-wall interactions were performed in the ALCATOR (Parker et al., 1976), DITE (Burt et al.,

1976), TFR, and PLT (Dylla, 1978) tokamaks. These studies confirmed the utility of the technique as a diagnostic of the vessel surface conditions and were followed by studies of the discharge cleaning techniques described above, which were developed to remove surface impurities (primarily carbon and oxygen) from the vessel walls. Discharge conditioning techniques applied to magnetic fusion devices have been the subject of several recent review articles (Dylla, 1980; Cohen, 1986). The technology was developed largely on the basis of combined mass spectrometric and surface analysis studies on various tokamak devices including the MACROTOR (Oren and Taylor, 1977), PLT (Dylla et al., 1978), PDX (Dylla et al., 1980), ORMAK (Clausing et al., 1976), ISX (Gomay et al., 1979), JFT-2 (Gomay et al., 1978), PULSATOR (Staib and Staudenmaier, 1976), ASDEX (Poschenrieder, 1980), TFR (TFR Group, TFR, 1981), and TEXTOR (Waelbroeck et al., 1983; Winter et al., 1984) tokamaks.

With the basic mechanism of hydrogen discharge cleaning being understood (Dylla, 1980; Cohen, 1986; Staib et al., 1980; Dietz et al., 1979), subsequent discharge cleaning efforts in the magnetic fusion devices of the 1980s have relied on partial pressure analysis as the control and monitoring diagnostic for the cleaning process. Examples of these recent studies include the discharge conditioning experience of the TFTR (Dylla et al., 1984a and 1984b; LaMarche, 1987), DIII (Brooks and Peterson, 1987), JET (Dietz et al., 1984; Dietz, 1983; deKock, 1987), JT-60 (Kishimoto, 1987; Arai, 1987), JFT-2M (Sakamoto, 1980; Matsuzaki, 1984), JIPPT-II (Noda et al., 1982), HT-6B (Fu et al., 1984), and HL-1 (Yang et al., 1987) tokamaks, and the TMX-U mirror device (Allen et al., 1984).

Figure 14 shows an example partial pressure analysis taken from the initial hydrogen glow discharge conditioning of the large (86 m^3) stainless steel vessel of the TFTR tokamak (Dylla et al., 1984a). The plots in Figure 14 show the time dependence of the primary residual gases (H_2O, CH_4, CO, and C_2H_4) which are produced by the interaction of the hydrogen glow discharge with the stainless steel surfaces. The approximate $1/t$ time dependence of the residual gas production is generally observed and is thought to represent kinetics which are rate-limited by a multicomponent diffusion of oxygen and carbon species in the passivation oxide of the steel (Dylla, 1980; Dylla et al., 1980). Since the mass spectrometer measurements are absolutely calibrated, the progress of the discharge cleaning can be monitored quantitatively, and integration of the data in Figure 14 yields the total amount of surface carbon and oxygen removed from the vessel (Dylla et al., 1984a).

Additional applications of mass spectrometry to magnetic fusion devices include the study of the important interactions of hydrogen isotopes with first-wall materials. The process of hydrogen recycling (Marmar, 1978;

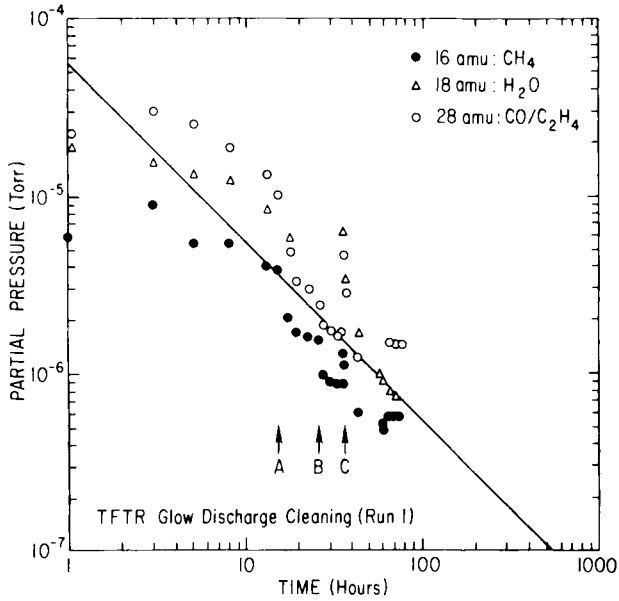


FIGURE 14. Observed time dependence of the partial pressures of the predominant impurity gases produced during the initial glow discharge conditioning of the large (86 m^3) stainless steel vessel of the TFTR fusion device (from Dylla et al. (1984a) with permission).

Howe, 1980) refers to the reflection, absorption, retention, and reemission of hydrogen isotopes from first-wall structures during a discharge in a fusion device. Recycling phenomena are important to understand because they affect the fueling and isotopic composition of the discharge and the isotopic inventory in the wall materials. Mass spectrometry has been used to investigate hydrogen isotope exchange in numerous devices—DITE (McCracken et al., 1978), PLT (Cohen et al., 1978; McCracken, 1979), TFR (1980 and 1982), TFTR (LaMarche et al., 1986), and JET (Ehrenberg et al., 1987) and has also been used to monitor the isotopic composition during measurements of hydrogen isotope recombination rates in the ISX (Langley and Howe, 1984), TFTR (Dylla et al., 1984c), and TEXTOR (Winter et al., 1984) devices.

Most applications of mass spectrometry to magnetic fusion devices have been partial pressure analysis type measurements and have used an experimental configuration similar to the setup shown in Figure 2. There have been no attempts to use the direct flux measurement scheme shown in Figure 1 for general mass spectrometric observations on fusion devices, primarily because of the difficulty of shielding the ion optics and ionizer for

operation in the high ambient magnetic field environment. However, the scheme shown in Figure 2 is routinely used for measurement of the high-energy hydrogen charge exchange neutral emission from a magnetic fusion device. With the use of stripping cells as the ionizer, the charge exchange flux from tokamaks has been examined from neutral energies greater than 50 keV to detection thresholds of approximately 100 eV (Wagner, 1982). This lower limit has been pushed to 10–20 eV using a combination of surface ionizers and time-of-flight momentum analyzers (Voss and Cohen, 1982; Verbeek, 1986).

There have been several attempts in the literature (Staib, 1982; Kojima et al., 1984) to place mass spectrometers within the edge plasma of a device to take advantage of the high ambient magnetic fields and the relatively low operating pressure ($< 10^{-4}$ torr). The EXB analyzer described by Staib (1980, 1982) and a similar device described by Kojima et al. (1984) demonstrated direct flux measurements of hydrogenic and impurity ion species within fusion device boundary plasmas. An interesting hybrid experimental configuration has been described by Poschenrieder et al. (Poschenrieder et al., 1982b; Poschenrieder, 1987), who use a moveable "sniffer" probe in the boundary plasma of the ASDEX tokamak to sample the plasma ions. The sniffer probe is interfaced to standard quadrupole mass spectrometer and acts as a flux to partial pressure converter.

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References

Aita, C. R. (1985) *J. Vac. Sci. and Technol.* **A3**, 625.
Aita, C. R., Myers, T. A., and LaRocca, W. J. (1981). *J. Vac. Sci. and Technol.* **18**, 324.
Allen, S. L., et al. (1984). *J. Nucl. Mater.* **128 / 129**, 26.
Armour, D. G. (1974). *J. Phys. B., Atom. Molec. Phys.* **7**, 1213.
Arai, T. et al. (1987). *J. Nucl. Mater.* **145–147**, 686.
Austin, W. E., Holme, A. E., and Leck, J. H. (1976). In: *Quadrupole Mass Spectrometry and its Applications* (P. H. Dawson, ed.). Elsevier Scientific, Amsterdam, NY, pp. 121–152.
Baiocchi, F. A., Wetzel, R. C., and Freund, R. S. (1984). *Phys. Rev. Lett.* **53**, 771.

Beauchamp, J. L. (1971). *Ann. Rev. Phys. Chem.* **22**, 527.
Blanchard, W. R., McCarthy, P. J., Dylla, H. F., LaMarche, P. H., Simpkins, J. E. (1986). *J. Vac. Sci. Technol.* **A4**, 1715.
Bohme, D. K. and Goodings, J. M. (1966). *J. Appl. Phys.* **37**, 4261.
Bolker, B. F. et al. (1981). *J. Vac. Sci. Technol.* **18**, 328.
Boucher, C. et al. (1986). *J. Vac. Sci. Technol.* **A4**, 1766.
Brooks, N. H. and Petersen, P. (1987). *J. Nucl. Mater.* **145–147**, 770.
Brodsky, M. H. ed. (1979). *Amorphous Semiconductors.* Springer Verlag, Berlin.
Brown, L. C. and Bell, A. T. (1974). *J. Chem. Phys.* **61**, 666.
Burt, J., McCracken, G. M., Stott, P. E. (1976). *Proceedings of the International Symposium on Plasma-Wall Interactions Julich*, 1976. Pergamon Press, p. 457.
Catherine, Y., Turban, G. and Grolleau, B. (1981). *Thin Solid Films* **76**, 23.
Catherine, Y., Turban, G. and Grolleau, B. (1982). *Plasma Chem. Plasma Proc.* **2**, 81.
Cayton, T. E. et al. (1987). *J. Nucl. Mater.* **145–147**, 71.
Chen, F. F. (1965). In: *Plasma Diagnostic Techniques.* (R. H. Huddlestone and S. L. Leonard, eds.), Academic Press, New York, pp. 177–183.
Clausing, R. E., Emerson, L. C., Heatherly, R. J. and Colchin, J. (1976). *J. Nucl. Mater.* **63**, 495.
Coburn, J. W. (1970). *Rev. Sci. Instrum.* **41**, 1219.
Coburn, J. W., Eckstein, E. W., and Kay, E. (1975). *J. Vac. Sci. Technol.* **12**, 151.
Coburn, J. W. and Kay, E. (1972). *J. Appl. Phys.* **43**, 4965.
Coburn, J. W. and Kay, E. (1977). *Proc. 7th Intern. Vac. Congr. and 3rd Int. Conf. Solid Surfaces* (Vienna 1977) p. 1257.
Coburn, J. W., Taglauer, E., and Kay, E. (1974a). *J. Appl. Phys.* **45**, 1779.
Coburn, J. W., Taglauer, E., and Kay, E. (1974b). *Japanese J. Appl. Phys.*, Suppl. 2, Pt 1, 501.
Cohen, S. A. (1986). In: *Physics of Plasma-Wall Interactions.* (D. E. Post and R. Behrish, eds.). Plenum Press, New York, p. 773.
Cohen, S. A., Dylla, H. F., Rossnagel, S. M., Picraux, S. T., Bordens, J. A., and Magee, C. W. (1978). *J. Nucl. Mater.* **76 / 77**, 459.
Davis, W. D. and Vanderslice, T. A. (1963). *Phys. Rev.* **131**, 219.
Dawson, P. H. (1986). *J. Vac. Sci. Technol.* **A4**, 1709.
Dawson, P. H. and Tickner, A. W. (1964). *J. Chem. Phys.* **40**, 3745.
Dawson, P. H. and Tickner, A. W. (1966). *J. Chem. Phys.* **45**, 4330.
deKock, K. et al. (1987). *J. Nucl. Mater.* **145–147**, 26.
Dietz, K. J. (1983). *Proc. IX Intern. Vacuum Congress, Madrid*, 1983. A.S.E.V.A., Madrid, p. 706.
deKock, K. et al. (1987). *J. Nucl. Mater.* **145–147**, 26.
Dietz, K. J., Alikhan, I., Waelbroeck, F., Wienhold, P. (1981). *Proc. 4th Intern. Symp. on Plasma Chemistry Zurich* 1979. IUPAC Communications, 1981.
Dietz, K. J. et al. (1984). *J. Nucl. Mater.* **128 / 129**, 10.
Dobrozemsky, R. (1985). *Vacuum* **35**, 630.
Drawin, H. W. (1968). In: *Plasma Diagnostics.* (W. Lochte-Holgreven, ed.). North-Holland, Amsterdam, pp. 777–841.
Dyke, T. R., Howard, B. J., and Klemperer, W. (1972). *J. Chem. Phys.* **56**, 2442.
Dylla, H. F. (1978). *J. Vac. Sci. Technol.* **15**, 734.
Dylla, H. F. (1980). *J. Nucl. Mater.* **93 & 94**, 61.
Dylla, H. F. (1983). *Proc. IX Intern. Vacuum Congress, Madrid*, 1983. A.S.E.V.A., Madrid, p. 578.
Dylla, H. F. and Blanchard, W. R. (1983). *J. Vac. Sci. Technol.* **A1**, 1297.

Dylla, H. F., Blanchard, W. R., Krawchuk, R. B., Hawryluk, R. J., Owens, D. K. (1984a). *J. Vac. Sci. Technol.* **A2**, 1188.
Dylla, H. F., Blanchard, W. R., Hawryluk, R. J., Hill, K. W., Krawchuk, R. B. et al. (1984b). *J. Nucl. Mater.* **128 / 129**, 861.
Dylla, H. F., Cecchi, J. L., and Knize, R. J. (1984c). *J. Nucl. Mater.* **121**, 243.
Dylla, H. F., Bol, K., Cohen, S. A., Hawryluk, R. J., Meservey, E. B., Rossnagel, S. M. (1978). *J. Vac. Sci. Technol.* **16**, 752.
Dylla, H. F., Cohen, S. A., Rossnagel, S. M., McCracken, G. M., Staib, P. (1980). *J. Vac. Sci. Technol.* **17**, 291.
Eckstein, E. W., Coburn, J. W., and Kay, E. (1975). *Int. J. Mass Spectrom. Ion Phys.* **17**, 129.
Edelson, D. and Flamm, D. L. (1984). *J. Appl. Phys.* **56**, 1522.
Ehrenberg, J. et al. (1987). *J. Nucl. Mater.* **145–147**, 551.
Ellesfon, R. E., Cain, D., and Lindsay, C. N. (1987). *J. Vac. Sci. Technol.* **A5**, 134.
Ertl, K. and Taglauer, E. (1973). *J. Vac. Sci. Technol.* **10**, 204.
Ferriero, L. (1986). Doctoral Dissertation, University of Minnesota, March 1986.
Flamm, D. L. (1979). *Sol. State Technol.* **4**, 109.
Flamm, D. L. and Donnelly, V. M. (1981). *Plasma Chem. and Plasma Proc.* **1**, 317.
Flamm, D. L., Donnelly, V. M., and Mucha, J. A. (1981). *J. Appl. Phys.* **52**, 3633.
Fleischmann, H. H. and Tuckfield, R. G., Jr. (1968). *Nucl. Fusion* **8**, 81.
Foner, S. N. and Hudson, R. L. (1953). *J. Chem. Phys.* **21**, 1374.
Franklin, J. L., ed. (1972). *Ion-Molecule Reactions*, Vol. 2, Plenum, New York, Chaps. 8, 11.
Franklin, J. L., Studniarz, S. A., and Ghosh, P. K. (1968). *J. Appl. Phys.* **39**, 2052.
Franklin, R. N. (1976). *Plasma Phenomena in Gas Discharges*. Clarendon Press, Oxford.
Fremerey, J. K. (1985). *J. Vac. Sci. Technol.* **A3**, 1738.
Fu, J. K. et al. (1984). *J. Nucl. Mater.* **128 / 129**, 851.
Gilkinson, J. L., Held, H., and Chanin, L. M. (1969). *J. Appl. Phys.* **40**, 2350.
Gomay, Y., Clausing, R. E., Colchin, R. J., Emerson, L. C., Heatherly, L., Namkung, W., Simpkins, J. E. (1979). *J. Vac. Sci. Technol.* **16**, 918.
Gomay, Y., Tazima, T., and Fujisawa, N. (1978). *J. Vac. Sci. Technol.* **15**, 103.
Greene, J. E. (1978). *J. Vac. Sci. Technol.* **15**, 1718.
Griem, H. R. (1964). *Plasma Spectroscopy*. McGraw Hill, New York.
Haller, I. (1980). *Appl. Phys. Lett.* **37**, 282.
Haller, I. (1983). *J. Vac. Sci. Technol.* **A1**, 376.
Harting, E. and Read, F. H. (1976). *Electrostatic Lenses*. Elsevier, New York.
Hasted, J. B. (1975). *Int. J. Mass Spectrom. Ion Phys.* **16**, 3.
Hayes, T. R., Wetzel, R. C., and Freund, R. S. (1987). *Phys. Rev.* **A35**, 578.
Heydemann, P. M., Tilford, C. R., and Hyland, R. W. (1977). *J. Vac. Sci. Technol.* **14**, 597.
Hiraoka, K. and Kamada, H. (1971). *Japanese J. Appl. Phys.* **10**, 339.
Hofer, W. O. and Seggern, J. V. (1984) *J. Nucl. Mater.* **128 / 129**, 943.
Howe, H. C. (1980). *J. Nucl. Mater.* **93 / 94**, 17.
Hyland, R. W. and Tilford, C. R. (1985). *J. Vac. Sci. Technol.* **A3**, 1738.
Jackson, G. et al. (1984). *J. Nucl. Mater.* **128 / 129**, 531.
Kenis, J. M., Stewart, G. W., Tripodi, M. K., and Gaspar, P. P. (1972). *J. Chem. Phys.* **57**, 389.
Keren, H., Avivi, P., and Dothan, F. (1976). *Phys. Lett.* **56A**, 85.
Kishimoto, H. et al. (1987). *J. Nucl. Mater.* **145–147**, 41
Klemperer, O. and Barnett, M. E. *Electron Optics*, 3rd ed. (1971). Cambridge Univ. Press.
Knewstubb, P. F. and Tickner, A. W. (1962a). *J. Chem. Phys.* **36**, 684.
Knewstubb, P. F. and Tickner, A. W. (1962b). *J. Chem. Phys.* **36**, 674.
Knights, J. C. and Lucovsky, G. (1980). *CRC Crit. Revs. Solid State Mater. Sci.* **9**, 211.
Köhler, K., Coburn, J. W., Horne, D. E., and Kay, E. (1985a). *J. Appl. Phys.* **57**, 59.

Köhler, K., Horne, D. E., and Coburn, J. W. (1985b). *J. Appl. Phys.* **58**, 3350.
Kohout, F. C. and Neiswender, D. (1970). *Int. J. Mass Spectrom. Ion Phys.* **4**, 21.
Kojima, H., Sugai, H., Mori, T., Toyoda, H., and Okuda, T. (1984). *J. Nucl. Mater.* **128 / 129**, 965.
Krawec, R. (1980). *NASA Technical Note* TN D-5747.
Kushner, M. J. (1982). *J. Appl. Phys.* **53**, 2923, 2939.
LaMarche, P. H. et al. (1986). *J. Vac. Sci. Technol.* **A4**, 1198.
LaMarche, P. H. et al. (1987). *J. Nucl. Mater.* **145–147**, 781.
Langley, R. A. and Howe, H. (1984). *J. Nucl. Mater.* **121**, 237.
Langmuir, I. and Mott-Smith, H. M. (1923, 1924). *Gen. Electr. Rev.* **26**, 731 (1923); **27**, 449 (1924).
Leck, J. H. (1985). *Vacuum* **35**, 629.
Longway, P. A., Estes, R. D., and Weakliem, H. A. (1984). *J. Phys. Chem.* **88**, 73.
Margaritondo, G. and Rowe, J. E. (1980). *J. Vac. Sci. Technol.* **17**, 561.
Manory, R., Grill, A., Carmi, U., and Anvi, R. (1983). *Plasma Chem. Plasma Proc.* **3**, 235.
Manory, R., Grill, A., and Anvi, R. (1984). *Plasma Chem. Plasma Proc.* **4**, 261.
Marmar, E. S. (1978). *J. Nucl. Mater.* **76 / 77**, 59.
Martin, G. and Lewin, G. (1966). *J. Vac. Sci. Technol.* **3**, 6.
Mathewson, A. (1974). *Vacuum* **24**, 505.
Matsuzaki, Y. et al. (1984). *J. Nucl. Mater.* **145 / 146**, 851.
McCracken, G. M. (1979). *J. Nucl. Mater.* **85 / 86**, 943.
McCracken, G. M., Fielding, S. J., and Stott, P. E. (1978). *Nucl. Fusion* **18**, 35.
McCracken, G. M. and Stott, P. E. (1979). *Nucl. Fusion* **19**, 889.
McDaniel, E. W. (1964). *Collisional Phenomena in Ionized Gases*. John Wiley & Sons, New York, pp. 240–260.
McDaniel, E. W., Cermák, V., Dalgarno, A., Ferguson, E. E., and Friedman, L. (1970). *Ion Molecule Reactions*. Wiley Interscience, pp. 315–360.
McDowell, C. A. (1963). *Mass Spectrometry*. McGraw-Hill, New York, p. 513 et seq.
Messer, G., Rohl, P., Gross, G., and Jitschin, J. (1987). *J. Vac. Sci. Technol.* **A5**, 2440.
Mitchel, A. R. (1981). *Transducer Technol.* **4**, 9.
Miyake, K. et al. (1982). *J. Appl. Phys.* **53**, 3214.
Motooka, T. and Greene, J. E. (1986). *J. Appl. Phys.* **59**, 2015.
Morrison, J. D. (1963). *J. Chem. Phys.* **39**, 200.
Nash, P. J. and Thompson, T. J. (1983). *J. Vac. Sci. Technol.* **A1**, 172.
Neiswender, D. D. and Kohout, F. C. (1972). *Rev. Sci. Instrum.* **43**, 1475.
Noda, N., Tanahashi, S., Kawahata, K. et al. (1982). *J. Nucl. Mater.* **111 / 112**, 498.
Oren, L. and Taylor, R. J. (1977). *Nuclear Fusion* **17**, 1143.
Parker, R. R., Pappas, D. S., Apgar, E. (1976). *Bull. Am. Phys. Soc.* **21**, 1141.
Parsons, C. R. and Medley, S. S. (1974). *Plasma Phys.* **16**, 267.
Pearse, R. W. and Gaydon, A. G. (1963). *The Identification of Molecular Spectra*. John Wiley & Sons, New York.
Perrin, J., Floret, A., de Rosney, G., and Schmitt, J. P. (1984). *Int. J. Mass Spectrom. Ion Proc.* **57**, 249.
Picard, A. and Turban, G. (1985). *Plasma Chem. Plasma Proc.* **5**, 333.
Plumb, I. C. and Ryan, K. R. (1986). *Plasma Chem. Plasma Proc.* **6**, 205.
Poschenrieder, W. (1987). *J. Vac. Sci. Technol.* **A5**, 2265.
Poschenrieder, W., Staudenmaier, G., and Staib, P. (1980). *J. Nucl. Mater.* **93 / 94**, 322.
Poschenrieder, W. et al. (1982a). *J. Nucl. Mater.* **111 & 112**, 29.
Poschenrieder, W., Venus, G. et al. (1982). *J. Nucl. Mater.* **111/112**, 29.
Post, R. S. et al. (1987). *J. Nucl. Mater.* **145–147**, 81.

Post, D. E. and Behrisch, R., eds. (1986). *Physics of Plasma-Wall Interactions*. Plenum Press, New York.
Post, D. E. and Lackner, K. (1986). In: *Physics of Plasma-Wall Interactions* (D. E. Post and R. Behrisch eds.). Plenum Press, New York, p. 627.
Raby, B. A. (1978). *J. Vac. Sci. Technol.* **15**, 205.
Ramsey, N. F. (1969). *Molecular Beams*. Oxford University Press, London, p. 361 et seq.
Riviere, J. C. (1983). In *Practical Surface Analysis* (D. Briggs and M. P. Seah, eds.). John Wiley and Sons, New York, pp. 80–82.
Rozgonyi, G. A. (1966). *J. Vac. Sci. Technol.* **3**, 187.
Ryan, K. R. and Plumb, I. C. (1984a). *Plasma Chem. Plasma Process* **4**, 271.
Ryan, K. R. and Plumb, I. C. (1984b). *Plasma Chem. Plasma Process* **4**, 141.
Ryan, K. R. and Plumb, I. C. (1986). *Plasma Chem. Plasma Proc.* **6**, 231.
Sakamoto, Y. Ishibe, Y., Yano, K. et al. (1980). *J. Nucl. Mater.* **93/94**, 333.
Santeler, D. J. (1987). *J. Vac. Sci. Technol.* **A5**, 129.
Schildcrout, S. M., Collins, J. G., and Franklin, J. L. (1970). *J. Chem. Phys.* **52**, 5767.
Schmidt, J. F. and Lampe, F. W. (1969). *J. Phys. Chem.* **73**, 2706.
Schmidt, M., Seefeldt, R., and Pohle, G. (1979). *J. de Phys.* **40**, C7-461.
Seguin, J. G., Dugan, C. H., and Goodings, J. M. (1972). *Int. J. Mass Spectrom. Ion Phys.* **9**, 203.
Shiniki, F. and Itoh, A. (1974). *Japanese J. Appl. Phys.*, Suppl. 2, pt 1, 505.
Siegel, M. W. and Vasile, M. J. (1981). *Rev. Sci. Instrum.* **52**, 1603.
Simpson, J. A. (1961). *Rev. Sci. Instrum.* **32**, 1283.
Smolinsky, G. and Flamm, D. L. (1979). *J. Appl. Phys.* **50**, 4982.
Smolinsky, G. and Vasile, M. J. (1973). *Int. J. Mass Spectrom. Ion Phys.* **12**, 147.
Smolinsky, G. and Vasile, M. J. (1975). *Int. J. Mass Spectrom. Ion Phys.* **16**, 137.
Smolinsky, G. and Vasile, M. J. (1976). *Int. J. Mass Spectrom. Ion Phys.* **21**, 171.
Smolinsky, G. and Vasile, M. J. (1977). *Int. J. Mass Spectrom. Ion Phys.* **24**, 311.
Staib, P. (1980). *J. Nucl. Mater.* **93 & 94**, 351.
Staib, P. (1982). *J. Nucl. Mater.* **111/112**, 109.
Staib, P., Dylla, H. F., Rossnagel, S. M. (1980). *J. Vac. Sci. Technol.* **17**, 291.
Staib, P. and Staudenmaier, G. (1976). *J. Nucl. Mater.* **63**, 37.
Studniarz, S. A. and Franklin, J. L. (1968). *J. Chem. Phys.* **49**, 2652.
Su, T. and Bowers, M. T. (1979). *Gas Phase Ion Chemistry*, Vol. 1. (M. T. Bowers, ed.). Academic Press, New York, pp. 83–118.
Sullivan, J. J. (1985). *J. Vac. Sci. Technol.* **A3**, 1721.
Tardy, J., Poitevin, J. M., and Lemperiere, G. (1981). *J. Phys. D., Appl. Phys.* **14**, 336.
TFR Group (1980). *J. Nucl. Mater.* **93/94**, 272.
Epe. TFR (1981). *J. Nucl. Mater.* **99**, 334.
TFR Group (1982). *J. Nucl. Mater.* **111/112**, 199.
TFR Group (1976). In: *Proceedings of the International Symposium on Plasma-Wall Interactions*. Pergamon Press, New York, p. 465.
Tilford, C. R. (1983). *J. Vac. Sci. Technol.* **A1**, 152.
Truesdale, E. A. and Smolinsky, G. (1979). *J. Appl. Phys.* **50**, 6594.
Truesdale, E. A., Smolinsky, G., and Mayer, T. M. (1980). *J. Appl. Phys.* **51**, 2909.
Tsai, C. C., Knights, J. C., Chang, G., and Wacker, B. (1986). *J. Appl. Phys.* **59**, 2998.
Tu, Y. Y., Chuang, T. J., and Winters, H. F. (1983). *Phys. Rev.* **B23**, 823.
Turban, G., Catherine, Y., and Grolleau, B. (1979). *Proc. 4th Int. Symp. on Plasma Chem* (S. Veprek and J. Hertz, eds.) Univ. of Zurich, 1979, p. 164.
Turban, G., Catherine, Y., and Grolleau, B. (1980). *Thin Solid Films* **67**, 309.
Turban, G., Catherine, Y., and Grolleau, B. (1982). *Plasma Chem. Plasma Proc.* **2**, 61.

Turban, G. and Rapeaux, M. (1983). *J. Electrochem. Soc.* **130**, 2231.
Uman, M. A. (1964). *Introduction to Plasma Physics*. McGraw-Hill, New York.
Vasile, M. J. (1980) *J. Appl. Phys.* **51**, 2503.
Vasile, M. J. (1982, 1983). *J. Appl. Phys.* **54**, 6697 (1983); **53**, 3799 (1982).
Vasile, M. J. and Smolinsky, G. (1973). *Int. J. Mass Spectrom. Ion Phys.* **12**, 133.
Vasile, M. J. and Smolinsky, G. (1974). *Int. J. Mass Spectrom. Ion Phys.* **13**, 381.
Vasile, M. J. and Smolinsky, G. (1975). *Int. J. Mass Spectrom. Ion Phys.* **18**, 179.
Vasile, M. J. and Smolinsky, G. (1976). *Int. J. Mass Spectrom. Ion Phys.* **21**, 263.
Vasile, M. J. and Smolinsky, G. (1977a). *Int. J. Mass Spectrom. Ion Phys.* **24**, 11.
Vasile, M. J. and Smolinsky, G. (1977b). *J. Phys. Chem.* **81**, 2605.
Vasile, M. J., Stevie, F. A., and Falconer, W. E. (1975). *Int. J. Mass Spectrom. Ion Phys.* **17**, 195.
Verbeek, H. (1986). *J. Phys.* **E.19**, 964.
Voss, D. E, and Cohen, S. A. (1982). *Rev. Sci. Instr.* **53**, 1696.
Voss, D. E. et al. (1984). *Nucl. Fusion* **24**, 1490.
Waelbroeck, F. et al. (1983). *Proc. IXth Intern. Vacuum Congress Madrid* 1983. A.S.E.V.A., Madrid.
Waelbroeck, F., Winter, J., and Wienhold, P. (1984). *J. Vac. Sci. Technol.* **A2**, 152.
Waelbroeck, F. et al. (KFA/Julich, FRG), private communication.
Wagner, F. (1982). *J. Vac. Sci. Technol.* **20**, 1211.
Wagner, J. J. and Veprek, S. (1983). *Plasma Chem. Plasma Proc.* **3**, 219.
Wagner, J. J. and Veprek, S. (1982). *Plasma Chem. Plasma Process.* **2**, 95.
Warmack, R. J., Stockdale, J. A., and Compton, R. N. (1978). *Int. J. Mass Spectrom. Ion Phys.* **27**, 239.
Weakliem, H. A., Estes, R. D., and Longeway, P. A. (1987). *J. Vac. Sci. Technol.* **A5**, 29.
Wei, P. S. and Kupperman, A. (1969). *Rev. Sci. Instrum.* **40**, 783.
Winter, J., Waelbroeck, F. G., Wienhold, P., Rota, F., and Banno, T. (1984) *J. Vac. Sci. Technol.* **A2**, 679.
Winter, J. et al. (1984). *J. Nucl. Mater.* **128/129**, 841.
Winters, H. (1985). *Vacuum* **35**, 634.
Yang, H.-R. et al. (1987). *J. Nucl. Mater.* **145–147**, 676.
Yu, T. Y., Cheng, T. M., Kampter, V., and Lampe, F. W. (1972). *J. Phys. Chem.* **76**, 3321.

5 Microwave Diagnostics

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I. Introduction 239
II. How to Get Started: A Simple Microwave Diagnostics Apparatus 242
III. Cavity and Waveguide Diagnostic Techniques 249
A. Cavity and Waveguide Loading by a Plasma 250
B. Resonant Cavity Techniques 253
IV. Interferometric Techniques 264
A. Basic Relations 264
B. Types of Interferometers 265
C. Abel Inversion and Multichord Interferometers 271
D. Interferometer Design 272
E. Phase Detection Techniques 279
V. Microwave Scattering and Radiometry 280
A. Microwave Scattering 281
B. Radiometry 283
VI. Plasma Waves and Their Applications in Microwave Diagnostics 284
A. The Dielectric Tensor of a Cold Plasma 285
B. Maxwell's Equations 287
C. Wave Equations and Wave Solutions 289
D. Boundary Conditions 294
E. Examples of Diagnostic Interest 297
F. Optical Approximations: Diagnostic Applications 304
References 307

I. Introduction

Similar to optical diagnostics microwave plasma diagnostic techniques employ the measurement of (1) the wave reflection from, (2) the wave transmission through, (3) the wave absorption in a plasma, or any combina-

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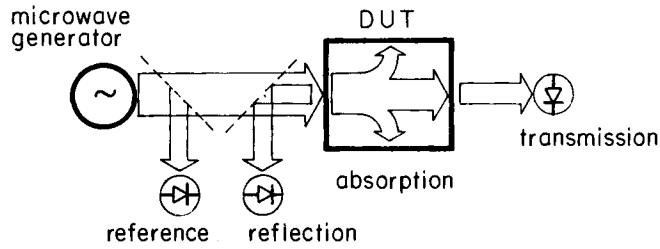


FIGURE 1. Microwave measurements that characterize a device under test (DUT).

tion thereof (see Figure 1). Exceptions to this are the case of interferometry, where only the effect of the plasma electron density upon the *phase* of the wave is of importance, and also the case of radiometry, whereby the plasma luminosity is recorded and possibly spectrally resolved. However, unlike in optics, for reflection, transmission, or absorption measurements in the microwave regime, the plasma is typically placed, or generated, in a cavity or a waveguide. Consequently, the goal of the measurement is to extract from the resulting loading effect some information about certain plasma properties, notably the *plasma density* and *collision frequency*, and any properties associated with them.

At lower frequencies, up to about 1 GHz, commonly referred to as *rf frequencies*, the loading characteristics of a circuit element are mainly determined with the aid of a bridge circuit. This procedure presupposes that the element under investigation can be idealized as a single lumped circuit element. Resistors, capacitors, and inductors are common examples that are measured with an LRC bridge. However, for the frequency range of *microwaves*, customarily defined as 1 GHz to 30 GHz (in the more restrictive sense of the word), free-space wavelengths are on the order of centimeters. The lumped circuit model has to be replaced more and more by a distributed circuit model, consisting of an array or network of individual components. Naturally, this applies even more so for *millimeter waves* for frequencies above 30 GHz.[1] The measurement circuits are basically of a bridge-type configuration and, for many applications, are referred to as network analyzers (Masi and Phillips, 1986). There is a great number of such microwave bridge schemes, although they are variants of only a few basic designs. Hardware cost and technical support requirements can vary tremendously between a stand-alone cavity setup or a one-channel interferometer for a small table-top plasma device and a fully computer-inter-

[1]For the purpose here, we will frequently use the more loosely defined term *microwaves* for the entire frequency range of diagnostic interest.

faced high-power scattering apparatus or the multichannel interferometer employed on present-day large scale machines for fusion research.

While waveguide and resonant-cavity experiments are typically performed at a few gigahertz, the range for interferometry extends to above 100 GHz, and microwave scattering and radiometry frequencies up to about 200 GHz are used in today's large fusion machines. Which frequency range to use depends to some extent on the parameters of the plasmas under investigation, especially for the latter two techniques.

Even after more than 40 years since the introduction of microwave techniques, their technology and engineering are still rapidly developing, nowadays particularly in the sector of active solid-state devices (Morgan and Howes, 1980). Nevertheless, most of the components, like antennae, mixers, oscillators, amplifiers, detectors, and so on, are still serving essentially the same principal functions, while some new diagnostic designs may have become more feasible. This fact will permit us to emphasize the various diagnostic schemes that are commonly used, rather than their technological specifics. Moreover, there is ample and also introductory literature available that has kept pace with new technological developments in the field (Gandhi, 1981; Bhartia and Bahl, 1984; Roddy, 1986). Analog and digital techniques for data acquisition, evaluation, and display have rapidly emerged and are now widely used, and some details will prove useful to the diagnostician.

Quite a number of books and survey articles on the use of microwaves for plasma diagnostics exist that cover some aspects of the subject presented here (Wharton, 1957, and 1965; Hermansdorfer, 1968; Podgornyi, 1971; Lisitano, 1975; Heald and Wharton, 1978; Golant, 1984; Luhmann and Peebles, 1984). This chapter should be considered as a complement to these works, which however, will assume no background in plasma physics and which is intended for a larger audience of scientists and engineers performing research with any type of plasma. Plasmas are encountered whenever the average energy or gas temperature is on the order of characteristic ionization energies or in the presence of ionizing radiation. While in the past decades, plasma research was mainly directed towards the achievement of controlled fusion, other applications have been as diverse as flames in combustion devices and plasmas in gas lasers, in high-current ion sources, or in microwave power sources. Very recently, plasmas have been harnessed for material processing, exploiting the capability of energetic plasma charges for removal (etching or sputtering), deposition, or implantation when applied to material surfaces (Chapman, 1980).

These plasmas span a large parameter range in plasma and neutral background densities, temperature, plasma constituents of different ion mass and charge, and so on. The plasma dynamics are strongly influenced

by all of these parameters. Collisions among plasma or gas particles, where energy and momentum can be exchanged, are an example (Spitzer, 1967; Chapman, 1980).

A magnetic field is often used in plasmas for magnetic containment that limits the cross-field motion of the plasma charges. Magnetic fields are also used to induce forced cyclotron motion by applying high frequency electromagnetic radiation for plasma formation and heating (Herlin and Brown, 1948). This technique is not only employed in many fusion devices, but also more and more for process type plasmas (Asmussen et al., 1974; Mejia et al., 1985; Paraaszczak et al., 1985).

In this chapter we restrict ourselves to the most commonly used microwave diagnostic techniques. The various sections are written such that they can be used independently. We first discuss the more experimental aspects of the various diagnostic methods in Sections III through V, where we cover the cavity techniques, then reflectometry. A large part is dedicated to the various interferometric schemes, and, finally, scattering methods are also presented. The material cannot be complete without a brief and basic, yet comprehensive introduction to waves in cold plasmas, with or without a static (i.e., time-constant) embedded magnetic field, as they apply to microwave diagnostics in a plasma. This is covered in Section VI. As in any diagnostic technique, the applied power levels are assumed to be small enough so as not to alter the plasma properties. However, similar techniques can be used for *plasma formation and heating* at higher microwave power, and Section VI should provide access to these topics as well. However, they are not part of the present chapter.

Before presenting all of these subjects in more detail, we would like to give a few hints to the researcher who is uninitiated in microwave and millimeter techniques as to how to get started and spare the specifics and diversity for later sections.

II. How to Get Started: A Simple Microwave Diagnostic Apparatus

Laboratory plasmas show resonant behavior when subjected to electromagnetic waves in the frequency range of microwaves. Among these resonances, the one at the plasma frequency ω_p plays an especially important role since it allows the determination of the plasma (electron) density n_e:

$$\omega_p^2 = \frac{n_e e^2}{\epsilon_0 m}, \tag{1}$$

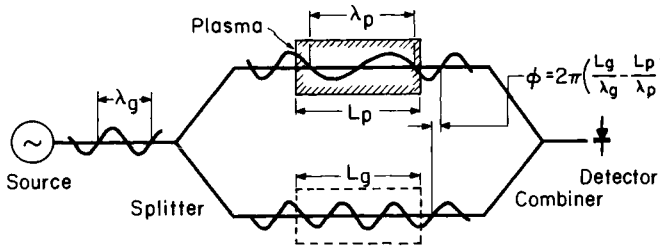


FIGURE 2. Interferometer scheme.

where e and m are the electron charge and mass, and ϵ_0 is the vacuum permittivity. As for any resonance, the plasma resonance markedly alters the microwave propagation through the plasma. Microwave techniques can determine any such changes in reflection, absorption, or transmission. The plasma behaves like an unknown load in the microwave circuit. The coherent quality of microwave radiation can be used to extract the phase shift that the wave will experience when propagating through a plasma under conditions close to a resonance. The plasma load is usually determined with a resonant cavity setup. The plasma alters the cavity's resonance frequency. Another possibility is the standing wave meter, since the plasma is capable of reflecting microwaves. Phase shifts, on the other hand, are determined by interferometry: one compares the absolute phases of 1) a microwave signal that has travelled through the plasma and of 2) an otherwise identical signal that meanwhile has travelled through vacuum along an equivalent path length (Figure 2). Regardless of the specific technique used, there are a number of microwave elements that fulfill a similar function in each method. As an example for this introductory outline, we will focus on a conceptually very simple interferometer.

First we have to recall that microwaves cannot travel in free space, not even vacuum, without appreciable dispersion. Microwaves are therefore guided in wavepipes. These waveguides come in all sorts of sizes and shapes. We will consider only the commonly used guides with rectangular cross section. Each size is suitable only for a certain frequency band. In Table 1 the standard frequency band designations are given and also the waveguide inner dimensions, together with the frequencies and wavelengths.[2]

[2]Much useful information on technical component details can be found in the various catalogs and handbooks from the manufacturers and suppliers. The pertinent trade directories and indexes can help locate these manufacturers and suppliers. Several trade journals specializing in microwaves and radio frequencies provide an introduction to existing and new technologies.

Table 1 Frequency bands.

Class	Microwaves						Millimeter Waves					
Band Letter Designation	L	S	C	X	Ku	K	Ka	A	M	V	/	D
Band Generic Designation				"3 cm"			"8 mm"		"4 mm"		"2 mm"	
Frequency Region (GHz)	1.1–1.7	2.6–3.9	5.8–8.2	8.2–12.4	12.4–18	18–26.5	26.5–40		50–75		110–170	
Vacuum Wavelength (mm)	150–300	75–150	36.6–75	24.2–36.6	16.6–24.2	11.3–16.6	7.5–11.3		4.0–6.0		1.8–2.7	
Waveguide inner dim. (inches)	6.50 × 3.25	2.80 × 1.34	1.37 × 0.62	0.90 × 0.40	0.62 × 0.31	0.42 × 0.17	0.28 × 0.14		0.15 × 0.07		0.07 × 0.03	

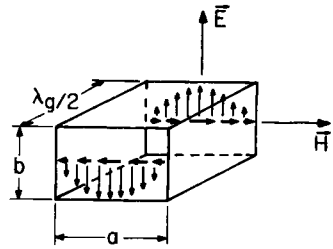


FIGURE 3. The TE_{10} mode in a rectangular waveguide.

The wavelength for microwave propagation inside a guide, λ_g, differs from the one in free space, given by $\lambda_0 = cf_0$, for a source of fixed frequency $f_0 = \omega/2\pi$:

$$\lambda_g = \lambda_0\left[1 - \left(\frac{\lambda_0}{2a}\right)^2\right]^{-1/2} = \left[\left(\frac{f_0}{c}\right)^2 - \left(\frac{1}{2a}\right)^2\right]^{-1/2}. \qquad (2)$$

The inside wide dimension of the waveguide is denoted by a, c.f., Table 1 and Figure 3, and c is the vacuum speed of light. The relationship (2) holds for the dominant transverse electric waveguide mode, TE_{10} (Baden Fuller, 1979). For this TE_{10} mode the *transverse* mode structure of the electric and magnetic fields are shown in Figure 3. The *longitudinal* wavelike structure has the periodicity λ_g. This periodicity becomes infinite when a is reduced to $\lambda_0/2$. For $a < \lambda_0/2$ no propagation in the guide is possible for the given frequency (cutoff).

When building a microwave system it is important to choose a suitable waveguide size (or frequency band). Hardware is commonly available from microwave manufacturers and surplus houses for the S-band, the X-band, and also the Ka-band. The final choice of which band to use will depend somewhat on the available components, on the specific diagnostic technique, and, of course, on the envisaged plasma density range. For cavity techniques, the cavity has to be resonant at the employed frequency. This favors the S-band, since a larger cavity, which is easier to build, implies a lower frequency. For interferometers, the spatial resolution is of the order of the waveguide width, suggesting therefore the use of the X and Ka-bands. The plasma imposes a *lower* frequency limit: Microwaves at frequencies below the plasma frequency cannot propagate in or through the plasma. For many types of glow discharges with typical densities of the order of 10^{15} to 10^{16} m^{-3}, this limit is usually of no concern because of the low plasma frequency. For many other plasma applications, the frequency constraint posed by the plasma frequency may have to be considered. The

upper useful frequency limit is due to the technical limitations in resolving small frequency shifts in cavities or small phase shifts in interferometers.

This interferometric phase shift ϕ is indicated in Figure 2. We note that the wavelength in the plasma is always longer than in vacuum

$$\lambda_p = \lambda_0 \left(1 - \frac{\omega_p^2}{\omega^2}\right)^{-1/2} . \tag{3}$$

For diminishing plasma densities λ_p approaches λ_0. The phase difference between when there is a plasma and when not is then (c.f., Figure 2)

$$\Delta\phi = 2\pi L_p \left(\frac{1}{\lambda_0} - \frac{1}{\lambda_p}\right), \tag{4}$$

where L_p is the effective path length the microwaves travel through the plasma. This provides the basis of the simplest technique to measure the plasma density: 1) measure the microwave phase without plasma; 2) measure the phase shift with the plasma turned on; 3) subtract the two phases; 4) finally, relate the phase difference via the plasma frequency, Eq. (1), and the effective path length L_p of the microwave in the plasma to the plasma density n_e.

But how is the phase of microwave radiation measured in the first place? In Section IV we present a variety of sophisticated methods that permit phase extraction to high accuracy. These methods can extend the range of the measurable plasma densities at the low and at the high ends. The simple technique given here is to serve as an illustration to provide the novice with some first experience with microwaves. It can also easily be put together from surplus parts. Overzet and Verdeyen (1986) have used a similar system for the plasma density determination in a parallel plate rf discharge device.

The essential components for a microwave phase measurement are a summing device, and a square-law diode detector. Figure 4 shows two standard versions of such summing devices: the hybrid (or magic) T-junction (see, e.g., Baden Fuller, 1979) and the hybrid ring junction (see, e.g., Collin, 1966). If we feed into either device at port 1 the signal $S_1 = S \sin \omega t$, and at port 3 the signal $S_3 = S \sin(\omega t + \phi)$, then the sum of both signals will appear at port 2: $S_2 = S_1 + S_3 = 2S \sin(\omega t + \phi/2)\cos(\phi/2)$. This signal S_2 is measured with a square-law detector. Such a detector time averages out the fast microwave oscillatory part, $\sin^2(\omega t + \phi/2)$, and produces as output the squared input: $S_D = \overline{S_2^2} = CS^2(1 + \cos\phi)$. The detector sensitivity is characterized by the constant C. The detector output does not provide directly the phase ϕ, but rather its cosine. As soon as the phase exceeds the value π, a unique phase determination is no longer possible. (The phase extraction techniques of Section IV allow phase

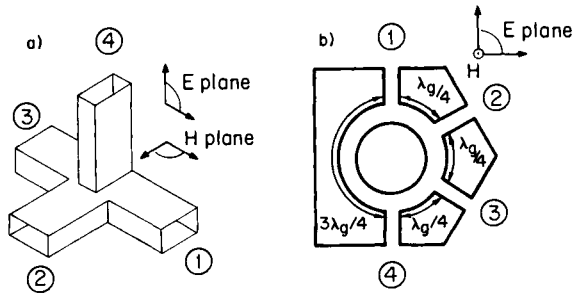


FIGURE 4. Two summing devices: (a) magic "T", (b) hybrid ring.

measurements that exceed π.) However, if one deals with low plasma densities, where $\omega_p \ll \omega$, the phase change caused by the plasma is small, and $S_D \approx CS^2(2 - \phi^2)$. To get an output that is linear in the phase, one can insert into either of the interferometer arms an adjustable phase shifter device (see, e.g., Roddy, 1986), and set its phase to 90°. This makes the detector output to $S_D = CS^2[1 + \cos(\phi \pm 90°)] = CS^2(1 \mp \sin\phi) \approx CS^2(1 \mp \phi)$. The sign depends on which arm carries the shifter and on the shifter itself. The difference ΔS_D between the detector output with plasma and when the plasma is turned off removes the constant offset CS^2 and is thus directly proportional to the phase difference $\Delta\phi$ and, therefore, the plasma density n_e:

$$\Delta S_D = S_D(\text{plasma}) - S_D(\text{no plasma})$$

$$\approx CS^2\Delta\phi \approx CS^2\pi\left(\frac{e^2}{\epsilon_0 m}\right)\left(\frac{L_p}{\lambda_0}\right)\left(\frac{1}{\omega^2}\right)n_e \tag{5}$$

or

$$n_e(\mathrm{m}^{-3}) = 1.184 \times 10^6 f_0(\mathrm{Hz})\left[L_p(\mathrm{m})\right]^{-1}\left[CS^2(\mathrm{V})\right]^{-1}\Delta S_D(\mathrm{V}). \tag{6}$$

Again, Eqs. (5) and (6) are only valid for low plasma densities, where $\omega_p \ll \omega$.

Figure 5 shows a layout for an interferometer that allows measuring the plasma density on the basis of Eq. (6). The microwave source output (of at least 50 mW) is divided into two equally powered arms by means of a 3 dB directional coupler (splitter), see, e.g., Baden Fuller (1979). All links and interconnections are of course made from waveguides with appropriate flanges. The plasma arm is opened up to give way to the plasma vessel, which is preferably made from glass or quartz to reduce interference and reflection of the microwaves. The open waveguide ends are flared out to

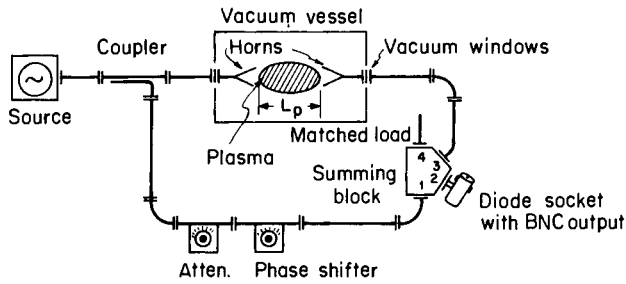


FIGURE 5. Simple interferometer lay-out.

horns, which make a more efficient waveguide to free space transition. By use of suitable, commercially available waveguide vacuum feedthrough windows, the horns should reach into the vacuum vessel yet sufficiently clear the plasma region. This is to avoid any microwave absorption in and reflections off glass windows or the vessel itself. An adjustable phase shifter with minimum adjustability of 180° is inserted into the reference arm. The plasma arm, the reference arm, and the crystal detector are attached to the above specified ports of a summing block (Figure 4), and the remaining port in the block has to be closed with a matched load (Baden Fuller, 1979).

The first test is to check the interferometer's phase characteristic without plasma by recording the detector output versus the phase: The phase of the phase shifter ϕ_{ps} is manually adjusted from 0° to at least 180°, resulting in a detector DC output, displayed with an oscilloscope or a voltmeter, of the form $S_D = CS^2(1 + \cos\phi_{ps})$ in volts. Note that any slight length difference between the two arms also leads to a phase shift. This phase shift is a constant and can be disregarded by shifting the zero of the detector output plot on the phase abscissa to get the $(1 + \cos\phi_{ps})$ type variation with ϕ_{ps}. By changing ϕ_{ps} one can determine the proportionality constant CS^2 (in volts) needed for Eq. (6). The phase shifter position of 90° corresponds to the turning point signal level $S_D(\phi_{ps} = 90°) = CS^2$. A next useful test is to insert between the horns, instead of a plasma, a slab of styrofoam ($\epsilon \sim 1.03$) or balsa wood ($\epsilon \sim 1.3$), and watch the detector reading change. This method can be useful for a calibration check of the apparatus (Archer et al., 1984). The phase change from plasma is, however, opposite in sign from the phase change for these two materials, since the effective dielectric constant ϵ of a plasma is usually less than one.

Last, we have to point out a few subtleties inherent to this simple design. We used the fact that the two microwave signals that arrive through both arms at the mixer block are balanced. That is to say, they must have the

same amplitude S. Horns, some cheaper versions of phase shifters, and the waveguides themselves, especially the waveguide connections (flanges), contribute to microwave power losses. This then suggests the use of as few flange connections as possible, possibly matched (choke) flanges (see Gandhi, 1981), and only smooth and gradual waveguide bends where $r_{\text{bend}} \gtrsim 5a$. All waveguide lengths should be minimized. Inadequately aligned flanges lead to standing or reflected waves that can interfere with the measurement. To ensure balance of both arms, signal levels should be measured at the point of the summing block by the use of a square-law detector directly attached to the waveguide flanges. Insufficient balance can be corrected by exchanging the 3 dB coupler for another of unequal branching ratio in order to beam more power through the plasma arm, which usually has more losses because of the horns and the unguided propagation between them. Insertion of an adjustable attenuator in the reference arm can further help to correct imbalance. All not specifically stabilized microwave sources have power drifts in time that affect the value of S^2 and, therefore, the constant CS^2 needed in Eq. (6). The crystal diode detector sensitivity usually does not stay constant during its lifetime. Frequent recalibrations of the constant CS^2 are therefore advisable. Moreover, the detectors are not exactly square-law but may also produce higher order products, depending on the signal level. All of these issues affect the final accuracy of the phase determination. In any case the detectable phase shift limit is a fraction of a degree ($\sim 10^{-3}$ radians). A plasma source that is pulsed at high frequency can lower this limit by about a factor of 5 (Chen, 1960; Overzet and Verdeyen, 1986). In this case, care should be taken that the detector has enough frequency bandwidth. An oscilloscope, triggered in synchronization with the plasma modulation, can be used for display of the signal. For DC discharges, phase-locked loop techniques or heterodyne interferometry (see Section IV) improve the resolution even further and are also not sensitive to any drift in the power source.

III. Cavity and Waveguide Diagnostic Techniques

The loading effect of a plasma that is present in a microwave circuit provides the means to deduce the plasma density and, sometimes, the collision frequency and electron temperature. Microwave circuits are made up from waveguides, waveguide components, and resonators (cavities). We therefore have to consider briefly the propagation or resonance properties of microwave radiation impinging upon a plasma situated in a waveguide or a cavity, respectively. While these phenomena are discussed in more detail in Section VI, we restrict the discussion in Section III.A to the basic ideas.

The extraction of the plasma parameters from the results of a circuit analysis becomes particularly simple by means of a perturbation theory (Section III.B.1), and Section III.B.2 introduces the technical tools to characterize a resonant cavity response. The discussion of the less common technique of microwave reflectometry concludes Section III.

A. Cavity and Waveguide Loading by a Plasma

A cavity resonance can directly be observed by insertion of a small probe (Section III.B). The presence of a plasma in a waveguide or a cavity can be measured by the microwave transmission, absorption, or reflection that the plasma causes. In any case, the plasma can be interpreted as a load. We include both a *dissipative* load, which converts electromagnetic energy into thermal energy, and a *reactive* load, which, like an inductor or a capacitor, only temporarily stores energy, usually in an oscillatory fashion, cycling energy into and out of the rest of the microwave circuit. Once the notion of a load (or impedance) is in hand, the reduction to a lumped circuit scheme is simpler (Adler et al., 1960; Fano et al., 1960). Therefore, we first determine the flow of energy in and out of a plasma-cavity or waveguide configuration.

1. *Poynting's Theorem*

The energy associated with the wave fields follows from Maxwell's equations (c.f., Section VI.B):

$$\frac{1}{2}\frac{\partial\left(\epsilon_0 E^2 + \mu_0 H^2\right)}{\partial t} + \nabla \cdot \left(\vec{\mathbf{E}} \times \vec{\mathbf{H}}\right) = -\vec{\mathbf{j}} \cdot \vec{\mathbf{E}}. \tag{7}$$

The vectors $\vec{\mathbf{E}}$ and $\vec{\mathbf{H}}$ are the electric and magnetic fields, respectively, and $\vec{\mathbf{j}}$ is the current density. The Poynting's theorem (7) is nothing else than the continuity equation for the energy density of the electromagnetic fields, $u = \epsilon_0 E^2/2 + \mu_0 H^2/2$; its changes are due to energy convection, given by the Poynting vector $\vec{\mathbf{E}} \times \vec{\mathbf{H}}$, or due to the work done by the electric field on all the charged particles and vice versa. The latter can be seen when equating $\vec{\mathbf{j}} \cdot \vec{\mathbf{E}} = n\vec{\mathbf{F}} \cdot d\vec{\mathbf{x}}/dt = ne\vec{\mathbf{E}} \cdot d\vec{\mathbf{x}}/dt$. For spatial and temporal field oscillations where the Fourier amplitudes are still complex, the equation corresponding to Eq. (7) has a somewhat different form:

$$i\omega\left(\epsilon_0 |E|^2 - \mu_0 |H|^2\right) = \nabla \cdot \left(\vec{\mathbf{E}} \times \vec{\mathbf{H}}^*\right) + \vec{\mathbf{j}}^* \cdot \vec{\mathbf{E}}. \tag{8}$$

Equation (7) is equivalent to (8), once it is integrated over one temporal or spatial period. Equation (8) provides a direct interpretation: 1) Magnetic and electric energy are 180° out of phase. 2) The *real part* of its RHS accounts for, as expected, a damping or a dissipative load, while the *imaginary part* gives rise to a reactance. In light of the complex plasma conductivity that enters via Ohm's law into the term $\vec{\mathbf{j}}^* \cdot \vec{\mathbf{E}}$, the latter is of importance for plasmas with low collision frequency. The complex conductivity is a consequence of the fact that the plasma electrons can pick up and return field energy. They generate displacement currents, while their collisions lead to dissipation of the microwave power.

2. *The Quality Factor or Q Value*

A plasma situated in a cavity will alter the cavity resonance frequency and mode spectrum. Before discussing the diagnostic applications of this fact, we will first consider the resonance properties of a cavity. These may be characterized by the quality factor Q, defined as the ratio of the (energy stored in the cavity) $\times$ (resonance frequency, when no damping is assumed) to the average dissipated power *loss*. In order to compute these quantities, Poynting's theorem (7) is integrated over the cavity volume to get

$$\frac{1}{Q} = \frac{1}{\omega_0} \frac{\int dV \vec{\mathbf{j}} \cdot \vec{\mathbf{E}} + \int da\, \hat{\mathbf{n}} \cdot (\vec{\mathbf{E}} \times \vec{\mathbf{H}})}{\mu_0 \int dV H^2}, \tag{9}$$

where, in the denominator, the fact was used that at resonance $\omega = \omega_0$, and after period averaging, electric-field energy = magnetic-field energy = $\frac{1}{2}$ total electromagnetic energy. Note that had Eq. (8) been used instead of Eq. (7), the Q value would then be defined as the *real part* of the resulting complex products, analogous to those of Eq. (9), relating Q to the dissipation.[3] Clearly, for an *empty* cavity whose walls have infinite conductivity, $Q \to \infty$, invoking the idealized boundary conditions at a metallic wall. For the realistic case of a very large, but finite wall conductivity σ_c (Jackson, 1975; Gandhi, 1981), the average power loss in the numerator of the RHS of Eq. (9) becomes, to leading order of $1/\sigma_c$,$1/4\mu_0\omega\, \delta H_{\|}^2$, where $H_{\|}$ is the magnetic field just outside the conductor, tangent to the cavity wall, and δ is the skin depth, $\delta^2 = 2/(\mu_0\omega\sigma_c)$. The field $H_{\|}$ follows from a mode calculation similar to the one presented in Section VI, where the boundary conditions (c.f., Section VI.D) remain again to leading order in $1/\sigma_c$, unaltered. This implies that Q depends on the chosen mode. There are

[3]Sometimes, the term Q value is also used for a cavity with a dielectric plasma present, which makes Q complex (see Section III.B.1.).

various simplifications possible for specific modes, e.g., the TM modes, and the Q factor can commonly be reduced to the form (Jackson, 1975)

$$Q = \left(\frac{V_c}{S_c \delta}\right) \times (\text{Geometrical factor}), \tag{10}$$

with V_c/S_c being the cavity's volume-to-surface ratio. Typical Q values are large ($> 10^3$). The importance of the quality factor to the resonance width will be illustrated in the following section. The Q value *in the presence of plasma* also follows from Eq. (9) by inserting the fields and the plasma conductivity. Specifically, plasma effects will contribute to the real part of the Q value if there are ohmic losses in the plasma or if the wave fields are evanescent inside the plasma, which happens for $\omega < \omega_p$.

3. *The Lumped Circuit Analog*

a) Resonance Frequency for Free Oscillations. The discussion of the preceding section can be illustrated for an LRC-circuit. There, the magnetic energy (in the inductor L) is $W_m = \frac{1}{2}LI^2$, I being the current, while the electric energy (in the capacitor C) is $W_e = \frac{1}{2}q^2/C$, q being the charge, with $dq/dt = I$ according to Kirchhoff's law. The power dissipated by the resistor R is given by Joule's heat, $P_t = RI^2$. Thus, conservation of energy (or Poynting's theorem) requires $d(W_m + W_e)/dt = -P_t$. Consequently, the current I obeys the equation for free oscillations

$$L\frac{d^2I(t)}{dt^2} + R\frac{dI(t)}{dt} + \frac{I(t)}{C} = 0, \tag{11}$$

which, with $I(t) \propto e^{i\Omega t}$, gives

$$\frac{\Omega}{\omega_0} = \frac{i}{2Q} \pm \sqrt{1 - \left(\frac{1}{2Q}\right)^2}. \tag{12}$$

The lossless resonance frequency is denoted by $\omega_0 = 1/\sqrt{LC}$, whence the quality factor $Q = 2\omega_0 W_m/P_t = \omega_0 L/R$ is derived, using $W_m = W_e$ for $\omega = \omega_0$. For a large Q value, which is equivalent to only small losses in the cavity, Eq. (12) yields the solution for the familiar damped oscillations of an LRC circuit, i.e. $\Omega = \omega_0 + i\omega_0/2Q = \omega_0 + iR/2L$. Equation (12) can be solved for $1/Q$ to yield

$$i\left(\frac{\Omega}{\omega_0} - \frac{\omega_0}{\Omega}\right) + \frac{1}{Q} = 0. \tag{13}$$

b) Resonance Line Width for Forced Oscillations. Meanwhile, for oscillations *forced* by an in-series drive, the RHS of Eq. (11) becomes $dU(t)/dt$.

With $U = U_0 e^{i\omega t}$, the (stationary) solution for the current is then

$$I(t) = \frac{U_0}{\omega_0 L}\left[\frac{1}{Q} + i\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)\right]^{-1} e^{i\omega t}. \tag{14}$$

Writing I in terms of amplitude and phase $|I|\, e^{i\Phi}$, the amplitude attains its largest (resonant) value for $\omega = \pm\omega_0$. While current and voltage are in phase for $\omega = \omega_0$, the current *lags* the voltage and vice versa, respectively, for $\omega > \omega_0$ (due mainly to the inductor) and for $\omega < \omega_0$ (due mainly to the capacitor). A different situation arises if the circuit is driven in *parallel.*

Equation (14) can be used to get the resonance line shape versus the frequency ω. Since the power of forced cavity oscillations is proportional to $|I|^2$, we have for the line shape

$$|I|^2 \propto \left[\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)^2 + \frac{1}{Q^2}\right]^{-1}. \tag{15}$$

Again, in the case of large Q, the resonance width will be very narrow, i.e., $\omega/\omega_0 - \omega_0/\omega$ may be replaced by $2(\omega - \omega_0)/\omega_0$, and the line shape will then be a Lorentzian (Jackson, 1975), with a full width $\Delta\omega$ at half-maximum $\Delta\omega = \omega_0/Q$.

These examples show that, while the quality factor Q is directly related to a circuit's dissipation, the resonant properties, like resonance frequency and half width, also depend, aside from Q, on *how* the currents are permitted to *flow*. In a cavity, this is dependent upon the choice of mode, as already mentioned. Hence, a lumped circuit has to be chosen in accordance with a certain mode or mode family.

Lumped circuits provide not only model descriptions for resonant cavities, but also for the mentioned transmission, reflection, and absorption measurements usually far away from any resonance. Terms as impedance, with its real part, the resistance, and its imaginary part, the reactance (or, inversely, the admittance, with conductance and susceptance, respectively), have all their counterpart for waveguides and can be calculated from the wave fields via Poynting's theorem (7) or (8) (Adler et al., 1960; Fano et al., 1960; Jackson, 1975).

B. Resonant Cavity Techniques

One of the earliest plasma diagnostic techniques, the resonant cavity method, was developed by Slater (1946) and first applied to the characterization of ambipolar diffusion in a helium plasma (Biondi and Brown, 1949). As already mentioned, the presence of the plasma in the cavity will result in both a shift of the cavity's resonance frequency and a change in the Q

value. These two effects are, respectively, related to the plasma frequency and, therewith, the electron density, and to the frequency of collisions of the electrons with other plasma or neutral particles. The early success of this technique was based on the fact that the experimental results had a simple theoretical interpretation when employing a first order perturbation theory, which obviated a machine computation, something rather involved in those days. However, in order for the perturbation approach to be valid, a series of conditions on configuration and parameters of the plasma have to hold, which cannot always be met under laboratory conditions (Chen et. al., 1968). The rigorous (in the limit of a homogeneous cold plasma model) methods of Section VI must then be employed to compute the resonance frequencies and the mode structures in a cavity, which are dependent on the plasma density n_e, collision frequency ν, plasma radius r_0, and the physical dimensions of the cavity.

1. Perturbation Theory Formulae

a) Cavity Resonance Shift. The assumption is made that the plasma, or the displacement currents in the plasma, modify only very little the resonant wave fields of an empty cavity. Following Slater (1946), we describe the damped cavity oscillations, just as in Section III.A.3, by means of the real and imaginary part of the angular frequency ω, as given by Eq. (8). There, the energy flow in and out of the interior of the *cavity* is determined by $\nabla \cdot (\vec{\mathbf{E}} \times \vec{\mathbf{H}}^*)$, the imaginary part of which accounts for the wave reflections off the walls and thus gives rise to the cavity resonance frequency, while its real part accounts for dissipation in the wall, leading to damping, characterized by Q. Assuming at first that the currents $\vec{\mathbf{j}} = \vec{\vec{\sigma}}\vec{\mathbf{E}}$ in the *plasma* are mainly nondissipative displacement currents, with negligible collision frequency $\nu \ll \omega_p$, we can obtain from Eq. (8) the small relative frequency shift $\delta\omega$ due to the plasma, not to be confused with the cavity line width $\Delta\omega$ of Section III.A.3 (Biondi and Brown, 1949; Biondi, 1951; Buchsbaum, et al., 1960)

$$\frac{\delta\omega}{\omega_0} = -\frac{1}{2\omega_0\epsilon_0} \frac{\mathrm{Im} \int dV\, \vec{\mathbf{E}}^* \cdot \vec{\vec{\sigma}}\vec{\mathbf{E}}}{\int dV |E|^2}, \tag{16}$$

where Eqs. (9) and (13), and $(\omega - \omega_0)/\omega_0 \ll 1$ were used. Im stands for the imaginary part. The explicit form for the complex conductivity tensor $\vec{\vec{\sigma}}$ can be found in Section VI.A, and the integration is to be taken over the cavity volume; the resonant frequency ω_0 and the wave fields $\vec{\mathbf{E}}$ are those for the empty cavity. When dealing with a nonmagnetized plasma, where σ is a

scalar, Eq. (16) reduces further to

$$\frac{\delta\omega}{\omega_0} = -\frac{1}{2\omega_0\epsilon_0}\frac{\int dV\,\mathrm{Im}\,\sigma|E|^2}{\int dV|E|^2} = \frac{\bar{n}e^2}{m\omega_0^2\epsilon_0}G. \tag{17}$$

This result also applies to magnetized plasmas, when a TM_{0m0} cavity mode is used. Then the conductivity component along the magnetic field is to be used in Ohm's law and Eq. (7). Thus, aside from a yet to be determined geometrical factor G, the average plasma density $\bar{n}$ can directly be inferred from measuring the relative resonance frequency shift $\delta\omega/\omega_0$. Various authors have derived or used approximate formulae for G (Biondi 1951; Buchsbaum et al., 1960; Golant, 1960; Anissimov et al., 1965; Bisschops and de Hoog, 1985), defined in terms of the known, unperturbed electric field distribution $\vec{\mathbf{E}}(\vec{\mathbf{r}})$ and the *assumed* (or determined by other diagnostic measurements) spatially dependent density distribution $n(\vec{\mathbf{r}})/\bar{n}$

$$G = \frac{\int dV\left|\vec{\mathbf{E}}(\vec{\mathbf{r}})\right|^2 n(\vec{\mathbf{r}})/\bar{n}}{\int dV\left|\vec{\mathbf{E}}(\vec{\mathbf{r}})\right|^2}, \tag{18}$$

and a similar, although somewhat more complicated definition holds if $\overleftrightarrow{\sigma}$ is still a tensor. Plasmas with a slight *axial density variation* may be described by this perturbation approximation (Persson, 1957).

b) Limitation of Parameters. A perturbation approximation is usually applicable when the plasma modifies but little the empty cavity fields. This is the case whenever 1) $\omega_p \ll \omega$ (i.e., low plasma density), $\nu \ll \omega$ (i.e., few collisions of plasma particles or neutrals), and 2) if a magnetic field is used for the cyclotron frequency $\omega_c \ll \omega$ (i.e., low magnetic field), or 3) the radius of the plasma column, r_0, is much smaller than the cavity radius, r_c, *irrespective of how large the plasma density*, as long as the penetration depth ε for the evanescent cavity fields $\varepsilon \gg r_0$. The depth ε is, in the absence of collisions, not related to the skin depth used in Section III.A.2, but is of the order of (Buchsbaum et al., 1960)

$$\varepsilon \sim \frac{c}{\omega}\frac{1}{\sqrt{\frac{\omega_p^2}{\omega^2} - 1}}. \tag{19}$$

c) Example: TM_{010} Mode. As an example, we evaluate Eq. (16) or (17) for the TM_{010} mode. The electric field is then $E_z \propto J_0(r\omega/c)$ (Gandhi, 1981; see also Sections VI.C.2 and VI.E.3), where $r_c\omega/c = 2.405$ in order to meet the radial boundary condition $E_z(r_c) = 0$. Furthermore, since $k = m = 0$,

$\vec{\sigma} \to \sigma = -i\epsilon_0\omega_p^2/\omega$ in terms of the plasma frequency ω_p, which is non-zero only within the plasma column, $r \leq r_0$. For $r_0 \ll r_c$, the integral in the numerator may be expanded and evaluated by the mean-value theorem to yield $(\bar{\omega}_p^2/\omega^2)(r_0^2/2)$, while the integral over the cavity radius in the denominator is evaluated by standard procedures for Bessel functions. As the final result, we get the average electron density in terms of the frequency up shift $\delta\omega$, the resonant frequency ω_0, and the plasma and cavity radii, r_0 and r_c

$$\bar{n}_e = 0.54 \frac{\epsilon_0 m}{e^2} \omega_0 \delta\omega \frac{r_0^2}{r_c^2}, \tag{20}$$

or, with $\omega = 2\pi f$ (Hz),

$$\bar{n}_e = 6.7 \times 10^{-3} f \delta f \frac{r_0^2}{r_c^2} \quad (\mathrm{m}^{-3}). \tag{21}$$

d) Collision Frequency Effect. Collisions may be due to any gas or charge species present in the plasma vessel. High neutral background can lead to collision frequencies ν on the order of or exceeding the microwave frequency ω. The various collisions, collision frequencies, and their dependence on the plasma and gas parameters are discussed by Spitzer (1967) and Chapman (1980). We restrict ourselves again to TM_{0m0} modes (or, even simpler, to nonmagnetized plasmas) where the conductivity entering Eq. (9) via the Ohmic term $\vec{j} \cdot \vec{E}^* = \sigma|E|^2$ has the real and imaginary parts (see Section VI for more details)

$$\mathrm{Re}\,\sigma = \frac{\nu\epsilon_0}{\omega^2 + \nu^2} \frac{n_e e^2}{m} \qquad \mathrm{Im}\,\sigma = -\frac{\omega\epsilon_0}{\omega^2 + \nu^2} \frac{n_e e^2}{m} \tag{22}$$

in terms of the plasma density n_e, the electron charge e, and mass m. In order to use the resonance line width (Section III.A.3(b)) to determine the collision frequency, it is convenient to write the combined cavity and plasma Q value, i.e., the real part of Eq. (9), in the form $1/Q = 1/Q_c + 1/Q_p$. In the spirit of the just discussed perturbation approach, the cavity contribution $1/Q_c$ is approximately unchanged compared to when no plasma is present. The plasma part is given by (Buchsbaum and Brown, 1957)

$$\frac{1}{Q_p} = \frac{1}{\omega_0} \frac{\int dV\, \mathrm{Re}\,\sigma|E|^2}{\epsilon_0 \int dV |E|^2} = \frac{\bar{n} e^2 \bar{\nu}}{m(\omega_0^2 + \bar{\nu}^2)\epsilon_0} G' \tag{23}$$

where

$$G' = \frac{\omega_0^2 + \bar{\nu}^2}{\bar{\nu}} \frac{\int dV \left|\vec{\mathbf{E}}(\vec{\mathbf{r}})\right|^2 \nu(\vec{\mathbf{r}}) n(\vec{\mathbf{r}}) / \left(\omega_0^2 + \nu(\vec{\mathbf{r}})^2\right) \bar{n}}{\int dV \left|\vec{\mathbf{E}}(\vec{\mathbf{r}})\right|^2}. \tag{24}$$

In principle, the collision frequency may depend on the location, but only a spatially averaged value $\bar{\nu}$ can reasonably be expected from Eq. (23). If the spatial dependence of ν is ignored, G' reduces to G from Eq. (18). Correspondingly, Eq. (17) has to be modified to read $\bar{n}e^2\omega_0 G''/[m(\omega_0^2 + \bar{\nu}^2)\epsilon_0]$, with a G'' suitably defined similar to Eq. (24). For lower collision frequencies, $\nu < \omega$, and spatially independent ν, Eq. (17) remains the same. Thus by determining the cavity's resonance width and shift with plasma, an averaged plasma density and collision frequency can be extracted (see Figure 6).

In conclusion, whenever the abovementioned conditions for a perturbation approximation apply, then the *quantitative* range of validity of Eqs. (16) through (18) can be determined by appropriate Taylor expansions of a general resonance condition derived in Section VI.E.5 (for specific examples, see Buchsbaum et al., 1960; Agdur and Enander, 1962; Lieberman and Lichtenberg, 1969). This procedure is only correct in the absence of axial density gradients (Persson, 1957). For plasmas with such gradients, only qualitative arguments can be given, usually invoking the electrostatic ap-

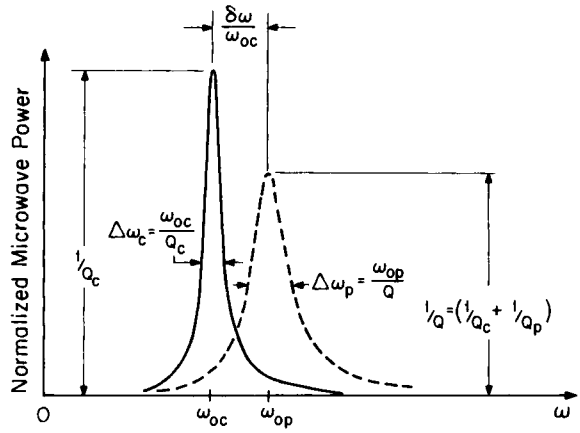


FIGURE 6. Cavity resonance lines: (a) without (solid line) and (b) with plasma (dashed line). The frequency widths $\Delta\omega$ and shift $\delta\omega$ are shown exaggerated for illustration.

proximation, which is generally not valid. In such a case, the validity limits cannot be quantified, potentially giving rise to an appreciable systematic error when interpreting resonant cavity measurements by means of Eqs. (16) or (17).

2. *Cavity Configurations and Microwave Circuitry*

A number of cavity configurations are commonly in use, generally in the shape of a right circular cylinder (Biondi, 1951), although rectangular cavities can of course also be used (Schulz and Brown, 1955). Depending on the configuration and technique, the density range roughly suggests what microwave frequency to use. Frequencies between 1.5 and 10 GHz, and, rarely, up to 35 GHz have been reported in the literature. Since the cavity has to be resonant at the chosen or available frequency and mode, this will determine the overall dimensions.

a) Empty Cavity. The resonance frequency of an empty cavity for an (m, n, p) mode[4] is given by

$$f_r = \frac{c}{2}\sqrt{\frac{\alpha_{mn}^2}{b^2} + \frac{p^2}{l}} \,. \tag{25}$$

The axial length of the cavity is denoted by l, while the parameter α_{mn} depends on the transverse geometry. For a rectangular resonator with major and minor dimensions a and b, $\alpha_{mn}^2 = m^2 + n^2b^2/a^2$ for both TE and TM modes. For a circular cylindrical resonator of radius $r_c = b$, $\pi\alpha_{mn}$ is the nth zero of $J_m'(\pi\alpha_{mn}) = 0$ for TE modes; and the nth zero of $J_m(\pi\alpha_{mn}) = 0$ for TM modes. If the transverse dimensions a, b, or r_c are much smaller than the axial dimension l (or vice versa), the first (second) term under the square root in Eq. (25) becomes dominant, assuming a low order mode, where $\{m, n, p\} = \{0, 1, 2\}$. Thus, for $r_c \ll l$, the resonance frequencies become virtually independent of the cavity length. This is always the case for the TM_{mn0} mode family, since $p = 0$.

b) Open Cavity. Open cavities (Figure 7(a)), i.e., cavities with no axial end plates, are based on this fact. However, at high wave numbers p (shorter axial wavelengths), they behave more like waveguides. This type of cavity

[4]The notation (m, n, p) characterizes, respectively, the two transverse (for cylindrical geometry: azimuthal and radial) and the axial mode modulations.

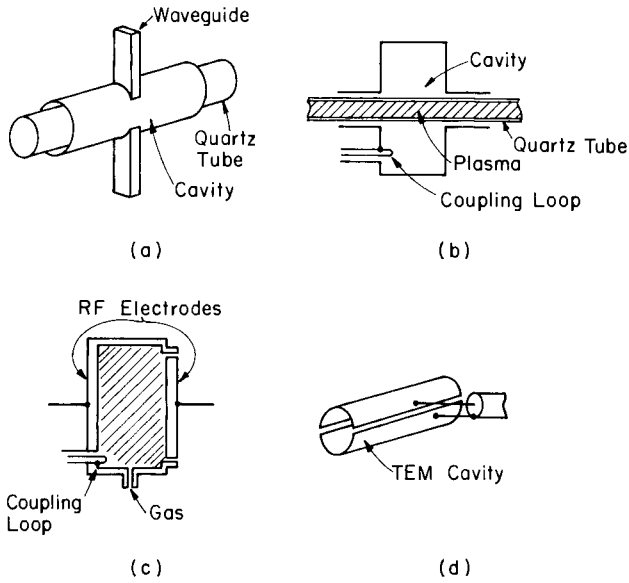


FIGURE 7. Common cavity configurations.

and its diagnostic applications are discussed by Vainstein (1963) and Anissimov et al. (1965).

c) Closed Cavity. The most common type is the closed cavity. Two typical configurations are shown in Figures 7b and 7c. To interpret the measurement for high densities ($\omega_p > \omega$) by means of the perturbation theory of the preceding section, the radius of the nonconductive plasma vessel has to be much smaller than the cavity radius (Figure 7b). This is also advisable to minimize the effect upon the Q value of the two holes in both end plates that serve as feed-throughs for the quartz tube. Circular waveguides, which are well below cutoff for the cavity resonance frequency, usually extend from these holes to reduce radiation losses. Choosing a thin-walled quartz tube minimizes its dielectric effect in the cavity. All these deviations from an ideal cavity may distort the fields near the holes (fringe fields). A change in Q and the resonance frequency, compared with their calculated ideal values, is the result; and a recalibration, possibly with a dielectric rod instead of the plasma column, may be necessary (Chen et al., 1968).

These problems can largely be avoided if the plasma vessel and the cavity are the same (Figure 7c). Such a configuration requires that both the

microwave power that generates the plasma and the probing microwave radiation are fed into the same cavity, although each may form a different cavity mode (Schulz and Brown, 1955; Asmussen et al., 1974). Frequently, plasmas are also generated by power application in the radio-frequency (rf) range, for instance in a plasma reactor with plane parallel electrodes. Both electrodes can be configured such that they close to a microwave cavity (Bisschops and de Hoog, 1985), see Figure 7c, possibly using appropriate choke connections that are matched for the microwave frequency, while constituting a high impedance for the rf frequencies in the MHz range. When using such a plasma-vessel-cavity configuration where the plasma fills most of the cavity, it is permissible to apply the perturbation formulae of Section III.B.1 only for low plasma densities, whenever $\omega_p \ll \omega$.

Most measurements in closed circular cylindrical cavities of either type of Figure 7b or 7c are based on the TM_{0n0} mode family, usually for low order modes, $n = 1$ or 2. The reason for this is the simple interpretation of the results that is possible in this case. Such pure TM_{0n0} modes can exist in a cavity, be it empty, partially, or entirely filled, with a plasma of arbitrary density or collision frequency and with or without an embedded axial magnetic field that is present in many experimental configurations (c.f., Sections VI.E.3 to E.5). The empty cavity Q value for these modes is easily calculated (Section III.A.2), and the perturbation formulae become particularly simple (Section III.B.1). By choosing the dimensions such that $r_c/l > 0.493$, c.f., Eq. (25), the TM_{010} mode will have the lowest cavity resonance frequency, which can be helpful for unambiguous mode determination. The most commonly selected mode is the TM_{010} mode, although there are also benefits in choosing the TM_{020} mode instead (Harris and Balfour, 1965). Bishops and de Hoog (1985) employed the TM_{020} mode to optimize the cavity Q in the presence of the isolated rf electrodes (Figure 7c). Since the resonance frequency for any of the TM_{mn0} modes is independent of the cavity length l (since $k = p = 0$), the cavity lacks simple tunability for such modes. Modes with $p \neq 0$ can be tuned by means of sliding endplates (Asmussen et al., 1974).

d) Large Plasma Vessel. Sometimes, the experimental or technical situation necessitates a rather large plasma vessel. If it is simultaneously to be used as a cavity, high mode numbers m, n, and p will be the consequence whenever the dimensions a, b, and l are much larger than the vacuum wavelength for the used microwave frequency. However, as long as the quality factor of the vessel is sufficiently large, the density can still be determined by the plasma-induced frequency shift (Akulina, 1965; Fessenden and Smullin, 1965). A high cavity Q is required in order to well separate any of these high order modes that are narrowly spaced in

frequency and can be accomplished by low-resistance chamber-wall seams and joints and a minimum of feed-through orifices.

e) Capacitor Resonator. Frequencies in the low microwave or the high rf range are used to excite dipole resonances in a plasma column situated in a capacitor-type resonator (Figure 7d). Such a resonator is more akin to a TEM structure than a cavity. The dipole resonances are closely related to the plasma frequency (Tonks, 1931; Dattner, 1961) and can, therefore, also be used to measure the plasma density. For this low frequency range the electrostatic limit is invoked that ignores all the rf magnetic fields in comparison with the electric fields. The employed diagnostic frequency must still be much higher than any characteristic ion frequency. The electrostatic potential obeys Laplace's equation. For a cylindrical plasma column of radius r_0 placed into a plane-parallel capacitor, with plate spacing $d \gg r_0$, the electric field induced in the plasma by the capacitor field E_0 is then given by (Crawford et al., 1963)

$$E = \frac{2E_0}{\left(2 - \frac{\omega_p^2}{\omega^2}\right)}, \tag{26}$$

relating the resonance frequency to the plasma frequency, $\omega_r = \omega_p/\sqrt{2}$. It is thus *less* than the plasma frequency. Crawford et al. (1963) addressed in particular the modifications of Eq. (26) due to the presence of a static magnetic field, and Aničin (1965) considered the plasma-capacitor coupled oscillations in an LC circuit. Experimentally, higher order resonances also emerge, and Parker et al. (1964) have achieved good agreement between their measurements and a nonuniform hot-plasma model description. They also treated the quadrupole oscillations of a plasma column.

f) Power Feed. The geometry of the microwave power feed and its location in the cavity or waveguide wall is decisive for the excited mode structure. The various feed schemes are essentially the same as with 1) waveguide-to-waveguide junctions or 2) coaxial line-to-waveguide junctions (Ishii, 1966; Gandhi, 1981). A loop coupling (c.f., Figure 7) induces locally a magnetic field perpendicular to the loop. Local electric field excitation can be achieved by a coaxial line whose outer conductor is connected to the cavity wall, while its inner conductor protrudes slightly into the cavity (Baden Fuller, 1979). For waveguide-waveguide or waveguide-cavity junctions, iris couplings and slot couplings are used. The latter select a particular electric field polarization. The *location* of the power feed should match the location of the field maximum for the desired mode, for waveguides usually $\lambda/4$

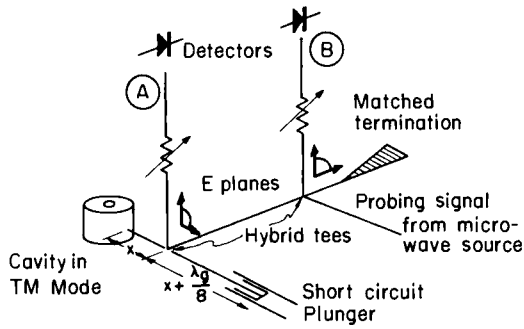


FIGURE 8. Discriminator circuit.

(electric coupling) or $\lambda/2$ (magnetic coupling) away from the end wall (Mumford, 1953). To drive a specific higher order mode, possibly several feeds at appropriate positions may be necessary in order to select against other higher order modes of similar resonance frequency but different mode structure.

g) Circuitry. The aim of the microwave circuitry is to detect the change in resonance frequency due to the susceptance change, $\mathrm{Im}(\sigma)$, and/or the change in Q value due to the conductance change, $\mathrm{Re}(\sigma)$. Both changes are caused by the presence of the plasma within the cavity (Harris and Balfour, 1965). A microwave discriminator (Pollard, 1980; Gandhi, 1981; Somlo and Hunter, 1985) provides the standard scheme for such standing-wave-ratio measurements that determine, respectively, the plasma density and the collision frequency. The discriminator consists of an oscillator (microwave generator), two hybrid tees (Figure 4a), to which detectors and/or matched loads are attached, and a short-circuit plunger to balance the reflections from the cavity under off-resonance condition (Figure 8). Such a microwave discriminator circuit is analyzed in detail by Gandhi (1981). The discriminator voltage, composed of detector outputs A and B (Figure 8), is simply $V_D = V_B - 1/2V_A \propto \delta\omega$ when the discriminator is operated close to resonance. A simplified, frequently used scheme retains one directional coupler (instead of the hybrid tee) and isolates the source by in-line attenuators (Biondi, 1951). Another method to directionally couple out the measured signal is the use of inductive or magnetic pick-up loops in the cavity, similar to the driving couplers discussed earlier (Schulz and Brown, 1955). The latter are a natural consequence when coaxial lines or TEM waveguides are used.

h) Source. A microwave source with some tunability, which depends on the envisioned density range, is required, be it manual or by sawtooth type frequency sweep. In the case that the plasma source is modulated or pulsed, the manual operation is simple but time consuming; the frequency swept operation requires a trigger in synch with the plasma source, possibly with an additional adjustable time delay provision (Biondi, 1951; Lieberman and Lichtenberg, 1969; Bisschops and de Hoog, 1985). The tuning range is limited by the frequency dependency of the discriminator components, particularly of the short-circuit plunger.

3. *Microwave Reflectometry*

Transmission and reflection measurements, c.f., Figure 1, can be performed on plasmas that partially fill a waveguide or by irradiating the plasma surface with microwave power (Vadnjal and Buffa, 1965). Transmission measurements are rare, since the density range where this would be feasible is narrow. In one case, the plasma is tenuous or optically thin at the wavelength employed, and virtually all the microwave power is transmitted. Then the *dispersive* effects and, to a lesser extent, the *refractive* effects are dominant. The dispersive effects are the basis for most cavity measurements[5] and for interferometry. Exploiting the refractive effects for diagnostics would have to rely on complicated inverse ray tracing calculations and has, therefore, not been employed so far. In the other case, the plasma density is close to or has reached the cutoff level, *and/or* it is much thicker than a penetration length. Then essentially all of the microwave power gets reflected, which is used for reflectometry (Anissimov et al., 1960). This technique was used to characterize moving or stationary plasmas filling a waveguide segment (Bethke and Ruess, 1964; Minami and Takeda, 1969). Roughly speaking, the plasma acts as a waveguide termination, which can be determined by a bridge circuit or discriminator. If the plasma is not optically thick, but terminated by a shorting (wave reflecting) endplate that closes off the waveguide end, the same bridge technique can determine the *phase shift* after a double pass and the *attenuation constant* the microwave has suffered by the plasma. This technique is a hybrid between a transmission measurement and interferometry, and requires that no appreciable power is reflected *off the plasma*. Bhattacharya et al. (1967) used this technique to monitor the ionizing radiation in a nuclear reactor. They also used the same bridge scheme for a radiometric determination of the electron temperature (see Section V).

[5]Even for $\omega_p > \omega$, the plasma still has to be optically thin to apply perturbation theory.

IV. Interferometric Techniques

Microwave interferometry is one of the methods used to obtain the line integrated density of a plasma. This measurement and the knowledge of the plasma dimensions give the average plasma density, one of the basic parameters needed to characterize a discharge. In contrast with probes (Chen, 1965), which can perturb the plasma to be measured, interferometry is nonintrusive.

A. Basic Relations

As a consequence of the dispersion properties of a plasma at microwave frequencies discussed in detail in Section VI, a wave propagating through a plasma undergoes a phase shift relative to a wave propagating in vacuum. For a wave with an electric field parallel to the magnetic field in the plasma, the propagation properties are independent of the magnitude of the magnetic field. For this case, known as the ordinary mode, the wavelength in the plasma is given by Eq. (3). In practice, for a waveguide carrying the fundamental TE_{10} mode, the electric field is parallel to the minor waveguide dimension.

In interferometry, the plasma whose density we would like to measure is inserted in one of the arms of a Mach-Zehnder interferometer, as shown for example in Figure 9. The resulting phase shift between the waves traveling along paths of equal length, one through the plasma and the other in the waveguide, is measured to determine the plasma line density. The phase shift $\Delta\phi$ is related to the plasma parameters by

$$\Delta\phi = \int_{-x_0}^{x_0} (k_0 - k_x)\, dx = k_0 \int_{-x_0}^{x_0} \left[1 - \left(1 - \frac{\omega_p^2}{\omega^2}\right)^{1/2}\right] dx, \qquad (27)$$

where the integral is along the plasma path. The plasma frequency ω_p is given by Eq. (1). We have assumed that the collision frequency is much smaller than the microwave frequency and, therefore, the effects of collisional damping can be neglected. For $\omega_p^2 \ll \omega^2$ or, equivalently, for a plasma well below cutoff $n_e \ll n_c$, with n_c the critical density given by $n_c = \epsilon_0 \omega^2 m/e^2$, the phase shift in units of 2π, i.e., the number of fringes, is

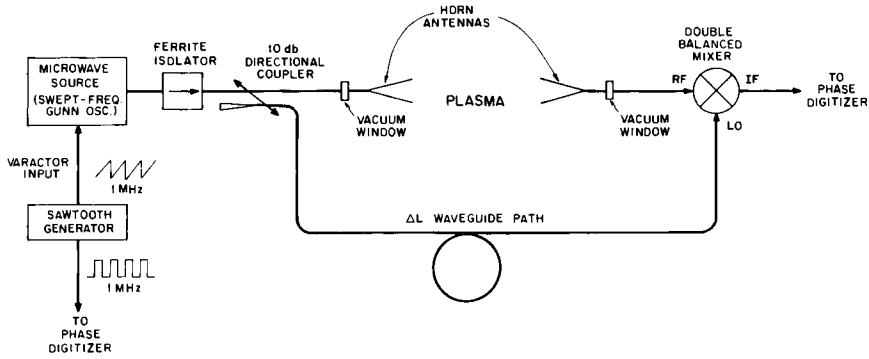


FIGURE 9. Swept-frequency interferometer schematic.

given by

$$N = \frac{\Delta\phi(t)}{2\pi} = 4.48 \times 10^{-18}\lambda_0\ (\text{cm}) \int n_e(x, t)\, dx\ (\text{m}^{-2})$$

$$= 1.34 \times 10^{-16} \frac{1}{f_0\ (\text{GHz})} \int n_e(x, t)\, dx\ (\text{m}^{-2}), \tag{28}$$

where λ_0 and f_0 are the microwave vacuum wavelength and frequency, respectively.

B. Types of Interferometers

Although the basic interferometer configuration is common to all systems there is a great variety of methods to determine the phase shift. Next, we discuss some of the techniques typically used.

1. Bridge Interferometers

When the expected fringe shift $N \ll 1$, that is, for low density plasmas, a bridge interferometer (Hotston and Seidl, 1965) is a simple system to set up. In this interferometer, the microwave power from a source goes through a ferrite isolator and is then divided into two branches. One is used as a power monitor, the other goes through another directional coupler to obtain the two arms of the interferometer. Each leg contains a variable attenuator used to adjust the signal amplitudes in each leg. The reference leg contains,

in addition, a phase shifter also used for the initial adjustments. A hybrid tee with two diode detectors after ferrite isolators is used as the detector assembly. A detailed discussion of the adjustments required is given by Hotston and Seidl. With a proper setup, direct phase measurements are possible, since for small phase angles $\sin \Delta\phi \approx \Delta\phi$. Their system was limited to phase shifts $\Delta\phi \leq \pi/9$. One advantage is that the time response of the system is limited only by the detector response, which can be several MHz (see also Overzet and Verdeyen, 1986).

Another system (Lindberg and Eriksson, 1982) extended the phase evaluation to arbitrary phase angles by using exact analytical expressions of the detector response. A detailed discussion of the phase evaluation is discussed in their paper. In their system there is still an ambiguity at some values of the phase angle where it is not possible to determine whether the phase increases or decreases. They propose that an alternate density measurement such as probes could be used to eliminate this ambiguity. In cases where probes are impossible to use, phase quadrature techniques can be employed, as discussed in Section IV.B.5.

2. *Frequency Swept Interferometers*

A schematic diagram for a serrodyne, frequency swept or "zebra" interferometer is shown in Figure 9. The operation of the interferometer is as follows: An additional length of waveguide ΔL is introduced in one of the arms of the interferometer, which produces a phase shift $\Delta\Phi = k_g \Delta L$ between the arms, where $k_g = 2\pi/\lambda_g$. The guide wavelength λ_g is frequency dependent and is given by

$$\lambda_g = \frac{1}{\left[\left(\frac{f_0}{c}\right)^2 - \left(\frac{1}{2a}\right)^2\right]^{1/2}}, \tag{29}$$

where c is the speed of light and a the major dimension of rectangular waveguide.

The microwave source, either a klystron or Gunn oscillator, is swept in frequency using a sawtooth of frequency f_{swp}. After an isolator, which prevents reflections back to the source, a side-arm coupler is used to split the power into the two interferometer arms. The detector is a double balanced mixer (Roddy, 1986), a device whose intermediate frequency (IF) output has two different frequency components given by the sum and difference of the input frequencies of the local oscillator (LO) and signal (RF) inputs. The sum frequency is naturally rejected by the frequency

response of the circuitry used after the IF. In this interferometer, the plasma leg is connected to the RF input, and the reference leg with the additional phase shift is connected to the LO, but the arrangement can be reversed. The microwave power in the LO leg should be high enough to drive the mixer. Its value depends on the unit and is typically 3 to 7 dBm.

The phase information from the plasma is contained in the difference frequency signal of the IF output, in which the microwave source frequency cancels. This term is proportional to $\cos(k_g(t)\,\Delta L - \Delta\phi(t))$. The time dependence of the argument is a consequence of the frequency sweep, i.e., k_g, and $\Delta\phi$, the phase shift produced by the plasma. For times less than a sawtooth period, the guide wave number is given by:

$$k_g(t) = k_g(0) + \frac{\Delta k_g}{2\pi}\Omega t = k_g(0) + \frac{1}{2\pi}\frac{dk_g}{df_0}\Delta f_0\,\Omega t, \qquad (30)$$

where Δf_0 is the amount the source frequency is swept and $\Omega = 2\pi f_{\text{swp}}$.

Substituting for this value of k_g, the IF output has the form $\cos(n\Omega t - \Delta\phi + \text{constant})$, where n is given by

$$n = \frac{1}{2\pi}\frac{dk_g}{df_0}\Delta f_0\,\Delta L = \frac{\Delta L}{\lambda_g}\left(\frac{\lambda_g}{\lambda_0}\right)^2\frac{\Delta f_0}{f_0}. \qquad (31)$$

The interferometer is adjusted to obtain integer values of n, usually $n = 1$ or one complete cycle. In this case the IF output is modulated at the sweep frequency.

Since the oscillator has a limited frequency sweep range of typically several hundred MHz for Gunn oscillators, Eq. (31) is used to determine the required ΔL. In practice, ΔL is chosen taking into account the practical limitations in Δf_0, and the amplitude of the sawtooth voltage is adjusted until a clean sine wave is obtained at the IF port. Usually, the output frequency of the source is not linear with the applied voltage over the entire range of adjustment. When the required frequency sweep exceeds the linear range, there will be some distortion of the sine wave at the IF port. If possible, ΔL should be chosen such that the amount of frequency sweep occurs over the linear range, however, this may require an excessively long waveguide and attenuation may become a problem. In addition, at the sawtooth resets there will be "fly-back" glitches in the signal. A tunable amplifier or a bandpass filter can eliminate the problems associated with the distortion and fly-back.

Finally, the IF output from the double balanced mixer is pre-amplified and sent to a phase detection system. The phase between a reference square wave from the sawtooth generator is compared to the square wave of the IF output that results from a bandpass filter and limiting amplifier. The phase

sensitivity is $2\pi/16$, limited by the sweep frequency of 1 MHz and the response of the components used in the phase detection circuit. More details about phase detection techniques are given in Section IV.E.

3. *Heterodyne Interferometers*

As the plasma density and size increase, as in present day fusion experiments, the microwave frequency required to avoid cutoff and refraction problems, discussed in detail in Section IV.D.1, increases. It becomes necessary to use millimeter and submillimeter sources and techniques. At these higher frequencies, the available source power decreases rapidly, the losses in the transmission elements become significant, and waveguide alignment needs to be accurate. It is therefore important to improve the sensitivity of the detection system. Heterodyne techniques, which involve the use of two different frequency sources, are quite powerful in this respect.

One example of such an interferometer (Cummins, 1970) operates as follows: A microwave source of frequency f_0 is used as a local oscillator, and a different source operating at frequency $f_0 + f_m$ provides the plasma signal. Care has to be taken to keep the LO and main source locked to each other so that the difference frequency remains constant. This is accomplished using automatic frequency control (AFC) loops. The signals are combined in a mixer, which provides an output at the intermediate frequency f_m, which, in turn, contains the phase shift produced by the plasma. A local oscillator at frequency f_m is used to detect this phase shift using phase quadrature techniques.

Some super-heterodyne receivers use several frequencies to eliminate frequency drifts in the transmitter and local oscillator (Efthimion et al., 1985), using up and down conversion with appropriate filter stages. Other receivers (Coffield et al., 1981, 1983) use two different intermediate frequencies (IF). A high radio frequency (several hundred MHz), is used to separate the microwave sources. The proper combination of mixers and local oscillators provides down conversion to another IF output that contains the plasma phase information. This second IF output, at a frequency of tens of MHz, is sent to phase comparator circuits to obtain the phase shift, which is then stored in transient digitizers.

The complexity and cost of these systems increase tremendously in comparison with those previously described. The reader is referred to the references for more details on these techniques. Super-heterodyne receivers are widely used in scattering and radiometry, which will be discussed in more detail in Section V.

4. Frequency Modulated Interferometers

A system developed by Ernst (1964), which operates on a similar principle as the heterodyne technique, uses a microwave source of frequency f_0 that is split using a side arm coupler with one arm going to the LO input of a mixer. The other arm goes through an isolator into a single side band modulator that generates an output with frequency $f_0 + f_m$, where f_m is an intermediate frequency used in the modulation source. This wave traverses the plasma, undergoes a phase shift $\Delta\phi$, and goes to the RF input to the mixer. The mixer output of frequency f_m and which contains the plasma phase information is compared with a reference signal from the modulation source. Again, basic phase detection techniques can be used. The modulation frequency can be tens of MHz, which allows excellent time resolution and the capability of using FM demodulation techniques (Jacobson, 1982) to obtain the phase. Phase digitizers are commercially available in this frequency range. One problem with this system is that single side band modulation at millimeter wavelengths has a conversion loss on the order of 30 dB. In addition, care should be taken in the suppression of the carrier or unwanted side bands generated in the modulator, as discussed in detail by Ernst.

A variant of the technique is given by Wallington and Beynon (1969). The standard microwave bridge, with a side arm coupler to split the power into the main and reference legs, is used. However, two modulators are added in the plasma arm, one single side band modulator and a balanced modulator. Each is driven by frequency synthesized signals differing in frequency by 1 kHz, in their case, the single side band unit driven at 111.101 MHz and the balanced at 100.100 MHz. The reader is referred to their article for a detailed explanation of the different frequencies generated in these modulators. The net result is that the signal traversing the plasma differs by 1 kHz from the microwave source, and when both are combined in a mixer, a 1 kHz output results. The reference 1 kHz is obtained from mixing the two frequencies that drive the modulators. The phase difference is then determined with a counting circuit. High phase accuracy $2\pi/1000$ is obtained with a 1 MHz clock that drives the counting circuit.

5. Quadrature Interferometers

In these systems, signal outputs that are proportional to the sine and the cosine of the phase angle are generated simultaneously. Each signal can be digitized and a computer program used to calculate the phase as a function of time. The digitization rate should be high enough to prevent a phase

change due to the plasma in excess of π between samples. If this is the case, the program can evaluate the phase uniquely at each point and assume the smallest change in phase between adjacent time points in order to determine the direction in which the phase is changing.

In practice, such a system has been implemented by Kaiser et al. (1962). In their system the reference leg of the interferometer contains two mixer networks, one of which is phase shifted by 90° with respect to the other. Brenning (1984) uses a similar method and describes in detail the initial setup required. A detailed analysis of the calculations needed to obtain the phase is presented in this reference.

An alternate method is used by Lister et al. (1982), in which a fast phase shifter (FPS) is used in the reference leg to provide a phase shift of 90° in a time much faster than the time rate of change of the density. A single detector is used whose output signal is proportional to the sine of the phase during one time period, for example, when the FPS is off, and then to the cosine during the next period when it is on. In practice, a four port circulator is used, since at the frequency used (140 GHz) fast phase shifters are not available. The whole system is controlled by a microprocessor which also calculates the phase as a function of time. The time response of the system is limited by the number of computer cycles needed to evaluate the phase and the clock rate of the processor itself to 24 μsec. As in any quadrature method changes in phase should be less than π in each cycle which gives the upper limit in the line density changes measurable with this system.

6. *Other Interferometers*

When the time rate of change of the plasma is slow and the density is low, so that fringe shifts do not exceed 2π, commercially available lock-in amplifiers can be employed to obtain the phase (Levine et al., 1965; Brown et al., 1970).

A variant of the frequency swept systems (Section IV.B.1) can be used in which the waveguide path difference between the two legs can be substituted by a fast phase shifter (FPS) in one of the arms (Ernst, 1967). The microwave source is not swept in this case. After a directional coupler, the plasma or reference leg contains a y-junction circulator that routes the microwaves through the FPS, and a sawtooth is used to modulate the phase in the FPS. A double balanced mixer can be used as the detector. The IF output is identical to that previously described in Section IV.B.1. A similar system was used by Uckan (1984) to obtain a direct readout phase system. In his setup the voltage to the FPS is continuously adjusted using a

feedback loop to keep the arms nulled. The output from the phase detector serves as the input to a current driver to the FPS. The voltage produced by this current across a resistor is therefore proportional to the phase change due to the plasma.

Another variant of the serrodyne system is obtained using a phase locked loop that changes the frequency of the microwave source (Strawitch et al., 1986) such that the change due to the plasma is cancelled. In this case a measurement of the voltage at the source is required to infer the phase. Some systems have used a Gunn oscillator simultaneously as source and detector (Bartlett and Brand, 1973).

C. Abel Inversion and Multi-chord Interferometers

Interferometry provides only a line-integrated density measurement so that several radial chords need to be measured in order to obtain the density profile. Consider a cylindrical plasma of radius R that is azimuthally symmetric and a phase measurement along a chord located at a distance h from the axis. The phase shift at each chord location is given by

$$\Delta\phi(h) = 2\int_h^R \frac{n_e(r)\,r\,dr}{\sqrt{r^2 - h^2}}. \tag{32}$$

The inversion of this integral equation for $n_e(r)$, known as the Abel inversion formula, is

$$n_e(r) = -\frac{1}{\pi}\int_r^R \frac{d\Delta\phi}{dh}\frac{dh}{\sqrt{h^2 - r^2}}. \tag{33}$$

Even in the case of azimuthal symmetry considered here, it becomes immediately apparent that since the value of the derivative of the phase shift is required, a large number of chord measurements is needed to infer the density. In practice, it is useless to increase the number of channels too much since the accuracy of the measurement is limited by the accuracy in $\Delta\phi$, which is typically 5 to 10%. In addition, the complexity of the system and its cost increase usually limit the number of channels to the range of 5 to 10. When the number of available channels is small (< 5), there is often not enough information to infer the profile using Abel inversion. For these cases the line density is calculated using assumed density profiles (Demas et al., 1985) with several fitting parameters. Alternatively, constant density, trapezoidal, parabolic, or Gaussian density profiles can be used to calculate the line density. These profiles involve only a few parameters, which can be determined using least squares fitting techniques. In this manner, the profile

and plasma parameters that best fit the experimental data can be obtained. Another useful technique involves the use of a derivative-free inversion formula (Deutsch and Beniaminy, 1982), which can decrease the errors incurred in the differentiation of experimental data. When the assumptions of azimuthal symmetry cannot be made, it becomes necessary to use more than one view of the plasma (Sauthoff and von Goeler, 1979; Wetzer, 1983). This is the same technique used in tomographic reconstruction, for which inversion algorithms are readily available (Huesman et al., 1977).

Experimentally, in cases where the plasma is in steady state or it is highly reproducible, a movable single channel system can be sufficient (Uckan, 1983). In this system, a hornlens (Uckan, 1981) arangement with 45° mirrors is used to eliminate attenuation problems caused by material wall flaking. In transient discharges, however, multichord systems are often required to obtain density profiles for each discharge. Such information is needed to understand the plasma properties and transport in each experiment. In a frequency swept multichord system using a single source (Lisitano, 1975), there can be crosstalk between adjacent channels. In this case, it is possible to use different values of ΔL for each chord. With this method, several harmonics of the sawtooth frequency are obtained, and a tunable amplifier for each channel is used to eliminate crosstalk. Waveguide attenuation as ΔL increases may become important, especially at higher microwave frequencies. In these cases it becomes necessary to use overmoded waveguide to reduce the power losses. When several channels are fed from a single microwave source, there should be enough power available for each channel. Alternatively, each channel can be fed using individual sources (Sevillano et al., 1984). The power requirements depend on the detection method and the components chosen, which we discuss in more detail in Section IV.D.2.

A horn-lens arrangement is used by Casper et al. (1986) to generate a microwave beam that scans the plasma. In this system, two perpendicular views are used to obtain tomographic reconstruction information. An array of closely spaced horns is used as receivers in this system.

D. Interferometer Design

A practical and simple approach to interferometer design is given that should be useful to the novice in microwave interferometry and techniques. A basic knowledge or expectation of the plasma density and dimensions is required at the outset. There is also a need to specify the requirements we would like to be met by our system, which vary in each application. These may include the time response needed, accuracy of the measurements,

stability of the system with respect to long time drifts, etc. The issues that need to be considered when designing an interferometric system can be divided into

a) Physics issues related to the properties of the plasma under investigation, such as wave propagation properties, plasma size and its relation to probing wavelength, refraction effects, cutoff, and plasma lifetime.
b) Practical issues such as phase sensitivity, unambiguous density tracking, microwave source power, electrical noise immunity, maintenance, ease of operation, cost, convenient data acquisition, and display.
c) Specific issues that may be unique to the application, such as vacuum needs, elimination of reflections, access to the plasma chamber, etc. The large variety of such specific requirements exceeds the scope of this chapter and must be addressed on a case-by-case basis. Therefore, meeting them is left to the ingenuity of the experimentalist.

1. Choice of Microwave Frequency

This is the most basic yet most important step in the design. The microwave frequency required is determined by the physical properties of the plasma under investigation. In addition, its value has important implications on a variety of other issues such as phase sensitivity, system cost, availability of power sources, etc. We first consider the physics of the microwave propagation and then some of the practical issues.

From the physics standpoint, we need to consider 1) plasma size, 2) cutoff, and 3) refraction. From a practical point of view, especially in the case of low density plasmas, 4) phase sensitivity may be an important limitation.

a) Plasma Size. The plasma size needs to be large in comparison with the wavelength of the probing beam. In this case, the assumptions of a plane wave are still valid. From a practical point of view, the spatial resolution of the system is limited by the fact that it is not possible to focus a microwave beam to a size smaller than a few wavelengths (Section VI.F). In addition, the wavelength in the plasma increases due to its dispersion properties. Typically, it is required that the characteristic transverse plasma size d must satisfy $d > 5\lambda_0$ (Ernst, 1972).

b) Cutoff. If the plasma density increases such that $\omega_p^2 > \omega^2$ or $n_e > n_c$, the microwave beam does not propagate through the plasma; it is cut off. Although a measurement of the transmitted power can be used to de-

termine when the microwave beam is cut off, and thus the density at that time, this technique only provides a lower bound on the density. In interferometry, it is useful to determine the density unambiguously at all times during the discharge. Therefore, the frequency chosen must be high enough to avoid cutoff. It is usually enough to choose $n_c > 3n_e$. Note that near cutoff Eq. (28) needs to be replaced by the exact form Eq. (27).

c) Refraction. Refractive effects usually pose a more stringent condition on the microwave frequency than cutoff. It is of particular importance for the design of multichord systems where refraction near the edge may deflect the beam out of the receiver. In the presence of density gradients a microwave beam is refracted according to Snell's law (Born and Wolf, 1980). The total angle of refraction can be calculated by integration over the path length. It is given in terms of the plasma refractive index n_r (c.f. Eq. 119) by (Shmoys, 1961)

$$\alpha = \int_{-x_0}^{x_0} \frac{\nabla n_r}{n_r} \, dx \,. \tag{34}$$

To estimate the refraction angle some assumptions about the plasma density profile must be made. A parabolic density profile given by $n_e = n_0(1 - r^2/r_p^2)$, where n_0 is the density on axis and r_p the plasma radius, can be solved analytically. In this case, the maximum refraction angle calculated from Eq. (34) is given by $\alpha_{\max} = \sin^{-1} n_0/n_c$. This maximum refraction occurs near the edge. Although, because of symmetry, α vanishes for the chord on axis, refraction effects should be considered, even when designing a single channel on axis. Under some conditions, due for example to the presence of instabilities, density gradients can move across the beam, causing refraction.

When designing a multichord system the maximum angle is limited by deviations from a chord measurement and the possibility of crosstalk between adjacent channels. The maximum refraction angle may therefore be limited by the total number of channels in the system M. The maximum refraction angle is in this case inversely proportional to M. In systems with high gain horns, commonly used to provide beam directivity and because of power transmission requirements, the refraction angle α should be smaller than the acceptance angle of the horn, otherwise no power will be coupled. The acceptance angle of a horn depends on its gain (Gandhi, 1981; Bhartia and Bahl, 1984).

It is useful to consider a practical example of a plasma with peak density 4×10^{18} m^{-3}, using 25 dB gain optimum horns. Usually the plasma is uniform along one direction (otherwise we would also need to measure the density profile in that direction). Therefore, either the H-plane refraction angle or the E-plane needs to be considered. For a cylindrical plasma with a

magnetic field along the axis of the cylinder, H-plane refraction is the important one. For an optimal horn of gain G, the half-power beam width is given by $\alpha_h = 80°/\sqrt{G/(2\pi 0.81)}$ (Gandhi, 1981), which translates to a condition on the critical density of $n_c > 5.9n_0$. This is a factor of 2 larger than the required cutoff condition. The microwave frequency is then $f_0 > 44$ GHz.

d) Phase Sensitivity. In the case of low density plasmas, the phase shift introduced by the plasma can be small. In this situation, cutoff and refraction are not a concern. However, an important consideration in the design is the required accuracy with which we desire to measure the density and, given the plasma dimensions, the minimum phase shift to be measured. In practice, one can measure fractions of a degree ($\sim 10^{-3}$ radians) with some of the digital techniques described in Section IV.E. As in every design, there are competing factors; the phase sensitivity is better at lower microwave frequencies, however, the plasma size relative to wavelength discussed in a) above pose a lower limit to the frequency. Other factors, such as spatial resolution, may also come into play that may pose a still higher frequency requirement.

Next we consider some of the practical issues. As in most designs, there are conflicting requirements. Whereas cutoff and refraction indicate the use of higher microwave frequencies, phase sensitivity decreases with frequency as in Eq. (28). Thus selecting a frequency that is too high may limit the dynamic range of the interferometer. System cost is almost always an important consideration, and, as the microwave frequency increases, microwave components cost increases rapidly. In addition, power sources and levels are more scarce. If possible, it may be better to select frequencies commonly used commercially, such as, for example, 10 GHz or 24 GHz, for which many inexpensive components are readily available.

2. *Choice of Interferometer Type*

Aside from the practical issues such as cost, maintenance, ease of operation, etc., there are also important physical limitations to consider, such as phase sensitivity and time resolution. Of the latter, time resolution may be the most important consideration when choosing a particular interferometer system.

Bridge interferometers are comparatively inexpensive but require careful initial setup and adjustments (Hotston and Seidl, 1965). Unless used for low densities, where the fringe shifts are small, $N \ll 1$, the evaluation of

the phase angle can be ambiguous (Lindberg and Eriksson, 1982). Their main advantage is a good frequency response, several MHz, limited by the detectors used.

The time response of serrodyne systems is limited to the sweep rate of the source frequency and the availability of digital techniques to evaluate the phase shift. As previously mentioned, a 1-MHz system is easily accomplished using commercially available phase digitizers with a fringe resolution of $2\pi/16$. Better fringe resolution can be obtained at lower sweep frequencies.

Frequency modulated systems can have good frequency response, on the order of the modulation frequency. One advantage of these systems is that quadrature phase detection techniques are easy to implement using off-the-shelf components. Commercial phase digitizers are also available. An important disadvantage is the low conversion efficiency of the side band modulators in the millimeter range. Heterodyne systems have similar time response and advantages, albeit they can be complicated. Their sensitivity makes them the natural choice when power levels are not high, usually at high microwave frequencies.

3. *Power Requirements*

The minimum power that is acceptable at the detector depends, among other things, on the experimental noise environment, the type of interferometer chosen, and the detector sensitivity. It is not possible to give a number that applies to all situations. The value of the signal-to-noise ratio at the detector should be decided upon and, working back through the microwave network, the losses caused by the components, waveguide transmission, plasma and vacuum propagation losses, etc. are calculated to obtain the desired output power at the source. We discuss some examples.

a) Component Losses. It is important to consider insertion losses, which give a measure of the power loss in transmission through a device or network. Single side band modulators mentioned earlier have a large insertion loss, on the order of 30 dB. Reflection losses caused by impedance mismatch between components can be calculated using the voltage standing wave ratio (VSWR) of the device. In this case the voltage reflection coefficient ρ_v is given by

$$\rho_v = \frac{\text{VSWR} - 1}{\text{VSWR} + 1}. \tag{35}$$

Since power goes as the square of the voltage, the reflected power is given by ρ_v^2.

b) Transmission Losses. We consider waveguide and vacuum losses. As the microwave frequency increases, dissipation caused by the finite resistivity of the waveguide material becomes important. The theoretical attenuation can be calculated in terms of this resistivity, waveguide dimensions, and microwave wavelength. Attenuation in rectangular waveguides is discussed in detail by Benson (1969). When using the theoretical formula, it is important to know that the discrepancy between the calculated and measured value increases with frequency, particularly in the case of copper guides. Measured values of the attenuation for copper guides at 35 GHz and 70 GHz are, respectively, 0.65 dB m^{-1} and in the range 3.5–5.3 dB m^{-1}. Etching of the waveguide followed by hydrogen annealing gives closer agreement with the theoretical values. This shows the importance of waveguide etching and cleaning techniques required when waveguide sections are built rather than purchased. The number of flanges, elbows, twists, etc. should be minimized to avoid losses at each connection. Waveguide benders can be fabricated to allow more complicated waveguide paths with a minimum of breaks in the guide.

The attenuation values given above indicate that waveguide losses in the fundamental mode become unacceptable at high microwave frequencies. A simple solution for low-loss transmission over moderate distances up to about 30 m is the use of an overmoded guide (Robson, 1969). Commercially available tapered transitions are used to connect the fundamental to the larger, overmoded waveguide. Important considerations when using overmoded guides are losses, mode conversion in bends, tapers and other discontinuities, and trapped-mode resonances. Experimental data on losses in overmoded guides are not as extensive as for fundamental guides (Robson, 1969). To avoid mode conversion in bends, gradual bending is required, with radius of curvature greater than 20 times the wavelength. Trapped-mode resonances (Robson, 1969; Bhartia and Bahl, 1984) in configurations when there are transitions to a fundamental guide at each end of the transmission line, can reduce substantially the transmitted power due the excitation of standing cavity modes in the larger guide.

Power attenuation caused by the plasma, which can become serious near cutoff, needs to be estimated. In addition, there will be losses in the transmission through the vacuum. To obtain plane waves incident upon the plasma, the distance d from the horn to the plasma should satisfy $d \gtrsim 2D^2/\lambda_0$, where D is the largest horn dimension. In this case, the received power P_r at the antenna is related to the transmitted power P_t by

$$P_r = \frac{\lambda_0^2 G_t G_r}{16\pi^2 R^2} P_t, \tag{36}$$

where G_t and G_r the gains of the horns, and R their separation. The gains

are usually given in reference to an isotropic radiator. Open ended waveguides are sometimes used instead of horns (Efthimion et al., 1985). As for horns, the open ended gain can also be computed (Gandhi, 1981).

4. *Choice of Components*

It is outside the scope of this chapter to discuss in detail the great variety of microwave components available. Microwave technology books that have kept up-to-date (Heald and Wharton, 1978; Gandhi, 1981; Bhartia and Bahl, 1984; Roddy, 1986) are instead valuable sources that can help in the understanding of the operation of microwave devices. Examples of possible applications are also given in such references. We will restrict our discussion to a few examples.

Ferrite isolators are frequently used in interferometer systems to eliminate reflections back to the source or in detector assemblies. When working in regions of high magnetic field, often found in fusion experiments, it is necessary to provide magnetic shielding to such devices. A high permeability material—soft iron, for example—surrounding the device is commonly used. The material should be thick enough to be below magnetic saturation (Jackson, 1975).

Power dividers frequently used include side arm couplers and hybrid tees. Tees are also employed in detector assemblies.

Attenuators and phase shifters are needed in some of the bridge and quadrature techniques discussed above in order to balance each of the interferometer arms. Usually, they are of limited use in serrodyne systems. Phase shifters can simulate the presence of the plasma and therefore can be used to check and calibrate the system.

When selecting horns, size may be an important consideration because of restricted access to the plasma chamber. In those cases where a particular design is needed, the properties of horn antennas can be found in the literature (Love, 1976; Heald and Wharton, 1978; Gandhi, 1981; Bhartia and Bahl, 1984; Roddy, 1986). Microwave component manufacturers can usually fabricate them to order, although this may be expensive. Generally, standard gain horns are adequate. They have the advantage of being commercially available, and their gain and emission patterns are readily available from the manufacturer. If length is a problem, the commercial horns can be easily cut, which would be cheaper than ordering a special unit. The gain of the shortened horn should be calculated or measured.

Mixers are an important component in most systems. There are several kinds (Roddy, 1986) depending on the number of diodes used in the mixing process. Single diode mixers can be used when the signal levels are rela-

tively high and noise is not a problem. Balanced mixers use two diodes and provide no isolation between the RF and IF ports. Double balanced mixers provide isolation between all the ports and are the preferred choice when signal levels are small. Isolation between the ports, especially between LO and RF, and LO and IF, is an important parameter when specifying a mixer. We recall that the LO power is typically much higher than the RF power (~ 7 dBm), the higher this ratio the better is the isolation. The noise figure, the required signal-to-noise ratio, and the isolation are used to determine the lowest acceptable input signal.

E. Phase Detection Techniques

Digital, quadrature, and FM demodulation techniques can be used to obtain the values of the phase. In digital techniques, the phase is obtained using comparator circuits or counters. For frequency swept interferometers, with $f_{swp} < 1$ MHz, a system developed at the Princeton Plasma Physics Laboratory (Greenberger, 1978) can be used. Detailed schematics and an explanation of the operation are given in that reference. Briefly, with a master clock at 16 MHz, limited by the frequency response of the components used, the phase sensitivity is $2\pi/16$ at 1 MHz. Improved fringe resolution is obtained at lower sweep frequencies. Using this technique a phase digitizer for CAMAC is commercially available.

In frequency modulated or heterodyne interferometers, the phase information is contained in a higher frequency carrier, typically tens of MHz. A phase comparator circuit using fast ECL logic has been developed (Coffield et al., 1981). The phase comparison is made by a pair of flip-flops, one for the LO reference and the other for the signal, clocked to give an output pulse the duration of which is given by the time difference between the input signals. The outputs from the flip-flops are low-pass filtered and go to a differential amplifier that gives an output voltage proportional to the phase shift. The phase range, given by the number of total fringes, is adjustable.

At tens of MHz, quadrature phase detectors can also be used. In these, the sine and cosine of the phase angle are generated and stored in digitizers. The phase is then computed from the arctangent. The digitization rate should be large enough to avoid phase changes greater than π between time samples. If this is the case, the phase can be determined assuming the smallest phase change between samples. Phase detectors are commercially available, but they can also be built using discrete components. A schematic diagram for a quadrature phase detector is given in Figure 10. In this circuit, the phase difference at the mixer inputs should be carefully adjusted

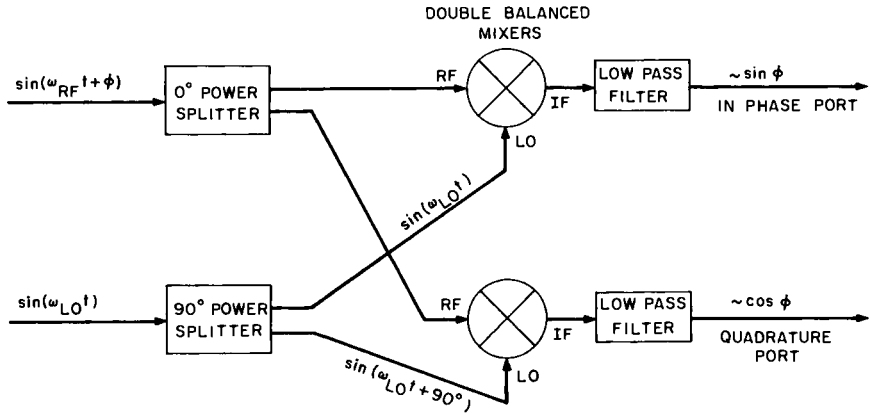


FIGURE 10. Quadrature phase detector schematic.

to 90°. At these frequencies, proper grounding techniques are required when building these circuits.

An output proportional to the phase can also be obtained using FM demodulation techniques (Jacobson, 1982).

V. Microwave Scattering and Radiometry

The measurement of the scattering of electromagnetic radiation by the electrons in a plasma is a powerful diagnostic used in the determination of the electron temperature and density, ion temperature, and electron density fluctuation spectra. In addition, the emission of radiation at the electron cyclotron frequency is used in fusion devices in the determination of the electron temperature. This is particularly useful in tokamaks where the magnetic field depends on the major radius of the device, thus providing a radial profile measurement of the electron temperature.

To cover in detail the theory of scattering of electromagnetic waves or the emission of radiation in plasmas is outside the scope of this chapter. For details on scattering the reader is referred to Evans and Katzenstein (1969) or Sheffield (1975). A simple treatment is also given in Tsukishima (1979) and a review by Cano (1979). Radiation from plasmas is covered in Bekefi (1966). A recent review of cyclotron radiation measurements is given in Bornatici et al. (1983). A review of the instrumentation used in both techniques is given by Luhmann and Peebles (1984).

A. Microwave Scattering

Electrons accelerated by the electric field of the incident wave reradiate such that the resulting scattered signal satisfies energy and momentum conservation given by

$$\omega_s = \omega_0 \pm \omega \qquad \text{and} \qquad \vec{\mathbf{k}}_s = \vec{\mathbf{k}}_0 \pm \vec{\mathbf{k}}, \tag{37}$$

where $(\omega_s, \vec{\mathbf{k}}_s)$ are the angular frequency and wave vector of the scattered wave, $(\omega_0, \vec{\mathbf{k}}_0)$ those of the incident monochromatic wave, and $(\omega, \vec{\mathbf{k}})$ characterize the fluctuations being studied. The amplitude of the wave number for the fluctuations is obtained from trigonometric relations for the triangle formed by the three wave vectors $\vec{\mathbf{k}}$, $\vec{\mathbf{k}}_s$, and $\vec{\mathbf{k}}_0$ that satisfy (37). It is assumed that the amplitude of the scattered wave vector is the same as the incident, which implies the momentum transfer to the electron is negligible. The resulting relation is known as the Bragg condition

$$k = 2k_0 \sin\left(\frac{\theta}{2}\right), \tag{38}$$

where θ is the scattering angle chosen.

The scattering parameter α determines the scattering regime under investigation. It is defined as

$$\alpha = \frac{1}{k\lambda_D} = \frac{\lambda_0}{4\pi\lambda_D \sin\left(\frac{\theta}{2}\right)}, \tag{39}$$

where $\lambda_D = (\epsilon_0 kT/n_e e^2)^{1/2}$ is the Debye length. There are two regimes of α to consider: 1) $\alpha \ll 1$, which applies for the density range in present day fusion experiments for incident wavelengths in the visible or near infrared unless $\theta \sim 0$. For this case, the scattering comes from individual unscreened electrons, and the spectrum determines the electron temperature. 2) $\alpha \geq 1$; in this regime scattering can be used to determine the ion temperature or the level of fluctuations in the plasma. For wavelengths in the microwave region, this condition is satisfied for nearly all of the scattering angles θ chosen.

The scattering cross section in a plasma can be calculated from classical electrodynamics. The resulting cross section is proportional to the quantity $S(\vec{\mathbf{k}}, \omega)$, the dynamic form factor, defined as

$$S\left(\vec{\mathbf{k}}, \omega\right) = \frac{1}{2\pi n_e V_s} \int_{-\infty}^{\infty} d\tau\, e^{i(\omega_s - \omega_0)\tau} \overline{n\left(\vec{\mathbf{k}}, t\right) n^*\left(\vec{\mathbf{k}}, t + \tau\right)}, \tag{40}$$

where n_e is the electron density, V_s the scattering volume, $n(\vec{\mathbf{k}}, t)$ is the

Fourier transform of the electron density, the asterisk denotes the complex conjugate, and the bar implies a time average.

The differential scattered power per unit solid angle $d\Omega$ and angular frequency can be written in terms of the form factor as follows

$$\frac{d^2P_s}{d\omega\, d\Omega} = n_e V_s p_0 \sigma_T S(\vec{\mathbf{k}}, \omega), \tag{41}$$

where $p_0 = (1/2)c\epsilon_0 E_0^2$ is the power density of the incident radiation, $\sigma_T = (e^2/mc^2)^2(1 - \sin^2\theta\cos^2\phi_0)$ is the Thomson cross section for scattering of electromagnetic radiation by electrons, and ϕ_0 is the angle between the electric field vector of the radiation and the plane of observation (usually one chooses $\cos\phi_0 = 1$) (Sheffield, 1975).

In terms of the density fluctuations the total scattered power is given by

$$P_s = p_0 \sigma_T V_s^2 \overline{|n_{\vec{\mathbf{k}}}(t)|^2}\, \Delta\Omega_r, \tag{42}$$

where $\Delta\Omega_r$ is the angle subtended by the receiver. An absolute calibration of the scattered power is usually performed using sources at known temperatures. The value of the fluctuations can thus be obtained. Because the Thomson cross section is so small and the microwave power available is not large (less than 1 kW), it is necessary that the fluctuations are not of thermal origin but instead caused by instabilities or collective plasma motion to be observable.

1. Measurement Techniques

Some of the techniques already mentioned in interferometry are directly applicable to scattering. The frequency used for the scattering must be high enough so that refraction is not significant. In addition, the scattering parameter $\alpha \gg 1$ to measure collective fluctuations and the frequency response of the detector must be in the range of the expected instability. A spectrum analyzer is commonly used to obtain all the frequency information. Basically there are two methods of detection: homodyne and heterodyne. Homodyne systems are simpler and involve only one microwave source, examples are given in the literature (Okabayashi and Arunasalam, 1977; Paris and Hollenstein, 1983; Rohatgi et al., 1985). In heterodyne systems (Doane, 1980; Machuzak et al., 1986), more than one microwave source is used, leading to more complexity. The sensitivity of these systems

is greatly improved over homodyne and, when measuring fluctuations that also involve plasma rotation, they allow the determination of the direction of rotation.

At each scattering angle only a particular value of $\vec{k}$ is obtained. In many cases it is necessary to measure the scattering at several scattering angles to properly identify the mode present in the plasma (Goldston et al., 1976; Paris and Hollenstein, 1983).

B. Radiometry

The measurement of the radiation from a plasma can be used to obtain information about the electron temperature. For a Maxwellian plasma, the radiation intensity is given by

$$I(\omega) = \frac{\omega^2 k T_e}{8\pi^3 c^2} \frac{(1 - e^{-\tau})}{(1 - \zeta e^{-\tau})}, \tag{43}$$

where τ is the optical depth and ζ the chamber reflectivity that depends on the optical system. Frequently, the frequency is chosen so that the plasma behaves as an optically thick ($\tau \to \infty$) object to the radiation. The temperature of the emitter can then be determined. The temperature can also be determined using the ratio of the emission intensity from successive harmonics (Celata and Boyd, 1977). The density can also be obtained from optically thin harmonics if the temperature is known (e.g., from an optically thick harmonic). When the electron energy distribution is non-Maxwellian, the radiation intensity needs to be calculated from a knowledge of the distribution function. Examples of such calculations are given in the literature (Bekefi et al., 1961a and 1961b). In these cases, care should be taken that the temperature inferred from the distribution chosen is not too sensitive to the details of the model.

1. Measurement Techniques

Conventional millimeter heterodyne receivers are the most commonly used (Taylor et al., 1985 and 1986). When broadband detectors like InSb are employed, a frequency analyzing element is also needed. Examples are Fourier-transform spectrometers (Costley et al., 1974; Bartlett et al., 1978),

Fabry-Perot interferometers (Walker et al., 1981), or grating spectrometers (Rutgers and Boyd, 1977).

VI. Plasma Waves and Their Applications in Microwave Diagnostics

Most treatises dealing with plasma waves usually have to discuss a variety of specific cases of waves. This is a consequence of the many parameters involved, which we enumerate here for electron plasma waves: the plasma angular frequency ω_p, which depends on the electron density n_e; the cyclotron frequency ω_c, depending on the magnetic field; and the collision frequency, ν, for electrons with other plasma particles or neutrals. Then there is a set of wave numbers, or inversely, wavelengths, one parallel and generally two perpendicular to the static magnetic field vector, as well as the microwave frequency, $f = \omega/2\pi$. The electromagnetic wave fields are vectors characterized by their projections in and normal to the direction of propagation, or of the static magnetic field. All these quantities are interrelated according to propagation laws, or dispersion relations, and this allows the plasma parameters, n_e and ν, to be inferred by measuring the propagation characteristics. If the plasma is spatially bounded, boundary conditions may significantly modify the wave dispersion. Such bounded plasmas are usually confined by metallic waveguides or cavities, commonly used for microwaves. A typical waveguide situation imposes the frequency, e.g., by an oscillator, and lets the wave numbers assume their permitted values. In a cavity, the wavelengths are set by the cavity dimensions, which therefore determine the resonant frequencies.

In this section, the propagation properties, i.e., the critical and resonance conditions, are developed in a form suitable for most diagnostic applications by invoking the cold plasma approximation, with or without an embedded homogeneous magnetic field. Both bounded and unbounded, homogeneous plasmas are considered, but, as an exception, the inhomogeneous plasma column is also treated in Section VI.E.4. The justifications for a homogeneous approximation are: 1) microwave measurements yield only a spatially integrated density or collision frequency and 2) with the above simplifications, the entire family of wave modes can be determined analytically.

While the earlier literature frequently also gave results in graphical form, this was omitted here. With today's widespread computerization, manipulation routines for, e.g., a 6×6 matrix and Bessel functions of all kinds are readily available on libraries for computers of any performance level.

A. The Dielectric Tensor of a Cold Plasma

The dielectric tensor is a measure of response of a dielectric medium to external electric fields. To derive an explicit form for the dielectric tensor, we consider the case where the plasma response is solely due to the plasma electrons, which move with the same speed. The first idealization implies that we are primarily interested in oscillatory electric fields of such high frequencies that the plasma ions with their much larger mass cannot follow these field oscillations. Microwaves are in this frequency range. The latter assumption precludes electrons having a sizable thermal motion of their own. Hence, the description addresses the cold plasma. While quite simplistic, these idealizations capture, to a large extent, the characteristics of the plasma response to external microwave fields.

We consider an electron of charge $-e$ and mass m that moves under the influence of the Lorentz force in an electric field $\vec{\mathbf{E}}$ and a magnetic field $\vec{\mathbf{B}}$, with velocity $\vec{\mathbf{v}}$, and of a damping force proportional to $\vec{\mathbf{v}}$:

$$\frac{d\vec{\mathbf{v}}}{dt} = -\frac{e}{m}(\vec{\mathbf{E}} + \vec{\mathbf{v}} \times \vec{\mathbf{B}}) - \nu\vec{\mathbf{v}}. \tag{44}$$

The damping is usually caused by collisions. The quantity ν can be identified in the simple picture presented here as a collision frequency and is assumed to be velocity independent.[6] We are specifically interested in electromagnetic wave fields, i.e., we write $\vec{\mathbf{E}}$ and $\vec{\mathbf{B}}$ in terms of their Fourier amplitudes:

$$\vec{\mathbf{E}}(t) = \vec{\mathbf{E}}\, e^{i\omega t}(a), \qquad \vec{\mathbf{B}}(t) = \vec{\mathbf{B}}_0 + \vec{\mathbf{B}}e^{i\omega t}(b). \tag{45}$$

We also assume $\vec{\mathbf{v}}$, $\vec{\mathbf{E}}$, and $\vec{\mathbf{B}}$ to be "small" quantities in the sense of a perturbation theory. Then the second order term $\vec{\mathbf{v}} \times \vec{\mathbf{B}}$ may be ignored[7] and we can set $\vec{\mathbf{v}} \propto e^{i\omega t}$ to reduce Eq. (44) to an algebraic equation. This $\vec{\mathbf{v}} \times \vec{\mathbf{B}}$ term and a second neglected nonlinear term, which also emerges from Eqs. (44) and (45), give rise to the ponderomotive force. It is of importance for the understanding of microwave scattering in a plasma, which was discussed in Section V (Chen, 1984). Any DC electric fields are not considered in Eq. (45), because the high mobility of the charges inside the plasma tends to short out such fields, except in the edge regions (or sheaths) not considered here.

[6] In reality, the collision frequency is energy or velocity dependent via the scattering cross section (Spitzer, 1967; Chapman, 1980). It also depends on the nature of the collisions. Usually it is adequate to consider these dependencies in the final formulae derived below.

[7] More succinctly, we will consider the self-consistent electric and magnetic fields in the plasma, as they result from the plasma response to just these fields.

The alternating electronic motion of Eq. (44) results in an alternating current density $\vec{\mathbf{j}}$

$$\vec{\mathbf{j}} = -en_e\vec{\mathbf{v}} = \overleftrightarrow{\sigma}\vec{\mathbf{E}}, \tag{46}$$

where n_e is the number density (m^{-3}) of the electrons and $\overleftrightarrow{\sigma}$ is the (AC) conductivity tensor. Its tensor character arises from the cross product $\vec{\mathbf{v}} \times \vec{\mathbf{B}}_0 = v_y B_0 \hat{\mathbf{e}}_x - v_x B_0 \hat{\mathbf{e}}_y$, where the z-axis is chosen along the direction of the (homogeneous) magnetic field $\vec{\mathbf{B}}_0$. An appropriate superposition of v_x and v_y, or j_x and j_y, for that matter, will make $\overleftrightarrow{\sigma}$ diagonal

$$X_{\left\{\begin{smallmatrix} L \\ R \end{smallmatrix}\right\}} = \frac{1}{\sqrt{2}}(X_x \mp iX_y) \leftrightarrow X_{\left\{\begin{smallmatrix} x \\ y \end{smallmatrix}\right\}} = \left\{\begin{smallmatrix} 1 \\ i \end{smallmatrix}\right\} \frac{1}{\sqrt{2}}(X_L \pm X_R). \tag{47}$$

In analogy to optics (Born and Wolf, 1980), the indices L and R stand for left and right circular polarization. Varying notations can be found in the literature, including unit vectors, which are self-orthogonal. With the transformation (47), the diagonalized conductivity tensor is made up of the elements

$$\sigma_{\left\{\begin{smallmatrix} L \\ R \end{smallmatrix}\right\}} = -i\frac{\epsilon_0\omega_p^2}{\omega - i\nu \pm \omega_c}; \qquad \sigma_P = -i\frac{\epsilon_0\omega_p^2}{\omega - i\nu}, \tag{48}$$

introducing the plasma frequency, ω_p, and the Larmor frequency, ω_c

$$\omega_p^2 = \frac{n_e e^2}{\epsilon_0 m} \tag{49}$$

$$\omega_c = \frac{eB_0}{m}. \tag{50}$$

The (relative) effective dielectric tensor, as it follows from Maxwell's equations (c.f., Section VI.B) can then be expressed in the three diagonal elements

$$\epsilon_{\left\{\begin{smallmatrix} L \\ R \\ P \end{smallmatrix}\right\}} = 1 - \frac{i}{\epsilon_0\omega}\sigma_{\left\{\begin{smallmatrix} L \\ R \\ P \end{smallmatrix}\right\}}. \tag{51}$$

In Cartesian components, the dielectric tensor takes the form

$$\overleftrightarrow{\epsilon} = \begin{pmatrix} S & -iD & 0 \\ iD & S & 0 \\ 0 & 0 & P \end{pmatrix}, \tag{52}$$

where

$$S = 1 - \frac{\omega_p^2(\omega - i\nu)}{\omega\left[(\omega - i\nu)^2 - \omega_c^2\right]}; \qquad D = \frac{\omega_p^2\omega_c}{\omega\left[(\omega - i\nu) - \omega_c^2\right]};$$

$$P = 1 - \frac{\omega_p^2}{\omega(\omega - i\nu)}. \tag{53}$$

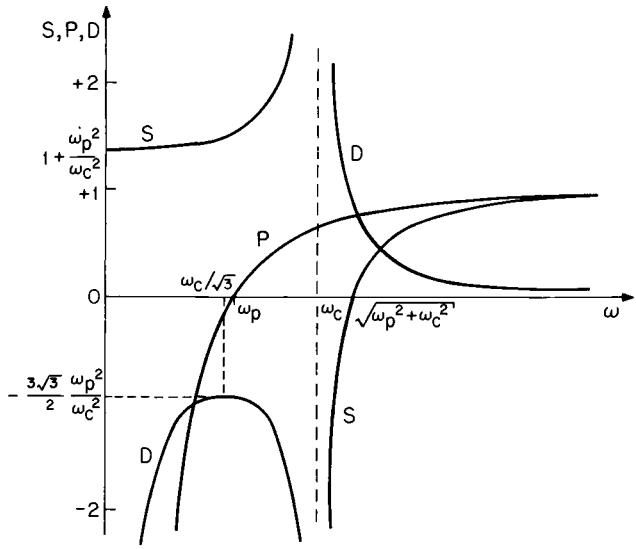


FIGURE 11. The three independent components of the dielectric tensor for a cold plasma vs. frequency.

Figure 11 shows qualitatively these quantities as a function of ω, for $\nu = 0$. The zeros of S and P and the singularities of S, D, and P signify, nonpropagation and bulk resonances in the plasma, respectively. In particular, the (longitudinal) dielectric constant P vanishes, given a certain frequency ω, and neglecting the collision frequency ν, for a critical density n_c, for which

$$n_c = \frac{\epsilon_0 \omega^2 m}{e^2}. \tag{54}$$

In terms of S, D, and P, the diagonalized tensor elements ϵ_L, ϵ_R, and ϵ_P can be written as

$$\epsilon_{\left\{\begin{smallmatrix} L \\ R \end{smallmatrix}\right\}} = S \pm D; \qquad \epsilon_P = P. \tag{55}$$

B. Maxwell's Equations

The two homogeneous Maxwell's equations do not contain explicitly any currents or charges

$$\nabla \times \vec{\mathbf{E}} = -\frac{\partial \vec{\mathbf{B}}}{\partial t} \tag{56}$$

$$\nabla \cdot \vec{\mathbf{B}} = 0. \tag{57}$$

The remaining two equations, in terms of current and charge densities, $\vec{\mathbf{j}}$ and ρ, and the relative dielectric tensor $\overleftrightarrow{\epsilon}$, according to Eqs. (46) and (51), read

$$\nabla \times \vec{\mathbf{B}} = \mu_0 \left(\vec{\mathbf{j}} + \epsilon_0 \frac{\partial \vec{\mathbf{E}}}{\partial t} \right) = \mu_0 \epsilon_0 \frac{\partial \overleftrightarrow{\epsilon} \vec{\mathbf{E}}}{\partial t} \tag{58}$$

$$\nabla \cdot \epsilon_0 \vec{\mathbf{E}} = \rho. \tag{59}$$

By vector identity, we conclude from Eq. (58) that

$$\nabla \cdot \overleftrightarrow{\epsilon} \vec{\mathbf{E}} = 0, \tag{60}$$

which is also a direct consequence of the continuity equation for the current, $\nabla \cdot \vec{\mathbf{j}} + \partial \rho / \partial t = 0$, and of Eq. (59). With $\vec{\mathbf{B}} = \mu_0 \vec{\mathbf{H}}$, Eqs. (56) and (59) can be cast into the form,

$$-\nabla \times (\nabla \times \vec{\mathbf{E}}) = \mu_0 \epsilon_0 \frac{\partial^2 \overleftrightarrow{\epsilon} \vec{\mathbf{E}}}{\partial t^2} \tag{61}$$

$$-\nabla \times (\nabla \times \vec{\mathbf{H}}) = -\epsilon_0 \frac{\partial \nabla \times \overleftrightarrow{\epsilon} \vec{\mathbf{E}}}{\partial t}. \tag{62}$$

For waves in infinite plasmas, the usual procedure is to expand $\vec{\mathbf{E}}$ and $\vec{\mathbf{H}}$ in plane waves with wave vector $\vec{\mathbf{k}}$, i.e., $\vec{\mathbf{E}}$, $\vec{\mathbf{H}} \propto e^{i(\omega t - \vec{\mathbf{k}} \cdot \vec{\mathbf{r}})}$. The operator $\nabla \times (\nabla \times)$ can then be replaced by $-\vec{\mathbf{k}} \times (\vec{\mathbf{k}} \times)$. The ensuing equation of the form $\overleftrightarrow{M} \vec{\mathbf{E}} = 0$ has only nontrivial solutions for $\vec{\mathbf{E}}$ when $\|M\| = 0$, where M depends on $|\vec{\mathbf{k}}|$, the angle $(\vec{\mathbf{k}}, \vec{\mathbf{B}}_0)$, and the various elements of the dielectric tensor (Chen, 1984). The assumption of a plasma of infinite spatial dimension and, therefore, of plane waves, is justified whenever the characteristic wavelength is much smaller than the extent of the plasma under investigation. However, for diagnostic purposes we are interested not only in plane wave propagation, but also in waves in bounded plasmas, as they occur in waveguides and cavities. There, e.g., the boundary conditions at curved surfaces will not admit plane wave solutions, as we will see in the subsequent sections. Still, essentially all configurations of diagnostic interest permit a plane wave expansion along the magnetic field direction. We pursue, therefore, a more general approach that is akin to the one used in standard waveguide theory (see, e.g., Gandhi, 1981). Since the goal is to provide formulae to interpret microwave measurements in plasmas, the plasma parameters, and notably the dielectric tensor, are assumed to be homogenous but by no means isotropic. As an exception to this, in Section VI.E.4 we also will treat a particular case where ϵ depends on the radius alone.

C. Wave Equations and Wave Solutions

1. Waves in a Magnetized Homogeneous Plasma

All vector quantities are separated into generally two perpendicular components and a parallel (or z-) component. For Cartesian geometry one sets

$$\vec{\mathbf{X}}(x, y, z, t) = \left(\vec{\mathbf{X}}_\perp(x, y) + X_z(x, y)\hat{\mathbf{e}}_z\right)e^{i(\omega t - kz)}, \tag{63}$$

while for cylindrical geometry one uses, instead,

$$\vec{\mathbf{X}}(r, \varphi, z, t) = \left(\vec{\mathbf{X}}_\perp(r) + X_z(r)\hat{\mathbf{e}}_z\right)e^{i(\omega t + m\varphi - kz)}. \tag{64}$$

The Fourier amplitudes $\vec{\mathbf{X}}_\perp$ and X_z carry still a tag for ω, k, and, if applicable, for m.[8] In the following, the arguments (x, y) or (r), and the Fourier modulation factors will be omitted unless they are needed for clarity.

The z-components of Eqs. (61) and (62), respectively, turn, with the aid of Eqs. (52) and (56) through (60), into

$$\left(\nabla_\perp^2 - k^2\frac{P}{S} + \omega^2\mu_0\epsilon_0 P\right)E_z = \left(\nabla_\perp^2 + \kappa_E^2\right)E_z = -i\omega k\frac{D}{S}\mu_0 H_z \tag{65}$$

and

$$\left(\nabla_\perp^2 - k^2 + \omega^2\mu_0\epsilon_0\frac{S^2 - D^2}{S}\right)H_z = \left(\nabla_\perp^2 + \kappa_H^2\right)H_z = i\omega k\frac{DP}{S}\epsilon_0 E_z, \tag{66}$$

with the transverse gradient defined, in analogy to Eqs. (63) and (64), as $\nabla_\perp = \nabla + ik\hat{\mathbf{e}}_z$. Equations (65) and (66) are coupled for E_z and H_z. As a consequence, in a magnetized plasma, generally no pure *TM* or *TE* modes familiar from (empty) waveguides and cavities can exist. Only for 1) electrostatic modes, where $\omega \to 0$, 2) modes at cutoff, where $k \to 0$, or 3) modes in unmagnetized plasmas where $D \to 0$, this classification may possibly still be valid. However, the mixture of TE and TM modes that is here due to the bulk properties can also arise from a coupling in the boundary condition (c.f., Section VI.D).

The longitudinal Eqs. (65) and (66) do not depend on any of the transverse components of the fields and may thus be solved independently. Therefore, before proceeding to the transverse equations, we first observe some general properties of Eqs. (65) and (66). As long as $\omega kD/S \neq 0$, we

[8] For comparison with the literature, attention has to be paid to whether the Fourier modulation factors in (63), (64), or their conjugate complex are used.

can combine them in the two forms:

$$\left(\nabla_\perp^2 + \kappa_H^2\right)\left(\nabla_\perp^2 + \kappa_E^2\right)\begin{Bmatrix} E_z \\ H_z \end{Bmatrix} = k_0^2 k^2 \frac{D^2 P}{S^2}\begin{Bmatrix} E_z \\ H_z \end{Bmatrix}. \tag{67}$$

The wave number $k_0 = \omega/c$ characterizes the free-space propagation with the speed of light $c = 1/\sqrt{\mu_0 \epsilon_0}$. Anticipating the solutions of Eq. (67) in terms of Bessel functions for cylindrical geometry, or plane waves for Cartesian geometry, we make the substitution $\nabla_\perp^2 \to -\kappa^2$, where κ is the transverse propagation constant. The two characteristic values for κ^2 follow from the now biquadratic Eq. (67) as

$$\kappa^2_{\begin{Bmatrix} 1 \\ 2 \end{Bmatrix}} = \frac{1}{2}\left(\kappa_E^2 + \kappa_H^2\right) \pm \sqrt{\frac{1}{4}\left(\kappa_E^2 - \kappa_H^2\right)^2 + \left(k_0 k \frac{D}{S}\right)^2 P}. \tag{68}$$

In all generality, there will be four characteristic solutions to Eq. (67), the linear superposition of which will still have to obey the boundary conditions. The indices $\{1,2\}$ in Eq. (68) were chosen such that for $k\omega/c = kk_0 \to 0$, we have $\kappa_1^2 \to \kappa_E^2$, $\kappa_2^2 \to \kappa_H^2$. We mentioned already in Section VI.A that for a critical density n_c, or for $P = 0$, the wave can no longer propagate or, in other words, is cut off. This is, in view of Eqs. (65) and (66), only rigorously correct in the unmagnetized plasma. Another critical case follows from Eqs. (67) and (68), when two of the modes are transversely cut off, for which $\kappa^2 \to 0$. The condition for this to happen is

$$|k| = |k_0|\sqrt{\epsilon_{\begin{Bmatrix} L \\ R \end{Bmatrix}}}, \tag{69}$$

which still describes propagation of plane waves along the magnetic field, with a modified speed characterized by the dielectric constants ϵ_L or ϵ_R given by Eqs. (55). Again, such modes may still be disallowed by the boundary conditions. Aside from plane waves, where $\kappa = 0$, Eqs. (68) and (69) lead also to waves with nontrivial transverse wavenumbers

$$\kappa^2_{\begin{Bmatrix} L \\ R \end{Bmatrix}} = \mp k^2 \frac{\epsilon_L - \epsilon_R}{\epsilon_L + \epsilon_R}\left(1 + \frac{\epsilon_P}{\epsilon_{\begin{Bmatrix} L \\ R \end{Bmatrix}}}\right). \tag{70}$$

We now specifically consider cylindrical geometry. There, the general nonsingular solutions to Eq. (67) read, aside from a common Fourier modulation factor

$$E_z = \sum_j J_m(\kappa_j r)\mathscr{E}_j \tag{71}$$

and

$$H_z = \sum_j J_m(\kappa_j r)\mathscr{H}_j. \tag{72}$$

The coefficients $\mathscr{E}$ and $\mathscr{H}$ are, by virtue of Eqs. (65) and (66), still subject to the relation

$$\mathscr{E}_j = e(\kappa_j)\mathscr{H}_j \tag{73}$$

or

$$\mathscr{H}_j = h(\kappa_j)\mathscr{E}_j, \tag{74}$$

where

$$e(\kappa_j) = -i\frac{S}{\epsilon_0 \omega k D P}\left(\kappa_H^2 - \kappa_j^2\right) \tag{75}$$

and

$$h(\kappa_j) = i\frac{S}{\mu_0 \omega k D}\left(\kappa_E^2 - \kappa_j^2\right). \tag{76}$$

Clearly, from Eqs. (67) and (68), one gets $e(\kappa_j)h(\kappa_j) = 1$. In the limiting case $\omega k \to 0$, one has $h(\kappa_1) \to 0$; $|h(\kappa_2)| \to \infty$; $|e(\kappa_1)| \to \infty$, and $e(\kappa_2) \to 0$. Regarding the solutions (71) and (72), any basis of coefficients $(\mathscr{H}_1, \mathscr{H}_2)$, or $(\mathscr{E}_1, \mathscr{E}_2)$, or $(\mathscr{E}_1, \mathscr{H}_2)$, or $(\mathscr{E}_2, \mathscr{H}_1)$ may be chosen, while all others follow via Eqs. (73) to (76).

The equations for the transverse components can be cast into the following form, making use of Eqs. (56) through (60) and the symmetry properties of $\vec{\vec{\epsilon}}$

$$\left(k_0^2 \vec{\vec{\epsilon}}_\perp - k^2 \vec{\vec{I}}_\perp\right)\vec{\mathbf{E}}_\perp = -ik \nabla_\perp E_z + i\mu_0 \omega (\hat{\mathbf{e}}_z \times \nabla_\perp) H_z \tag{77}$$

and

$$\left(k_0^2 \vec{\vec{\epsilon}}_\perp - k^2 \vec{\vec{I}}_\perp\right)\vec{\mathbf{H}}_\perp = -ik \nabla_\perp H_z - i\epsilon_0 \omega \vec{\vec{\epsilon}}_\perp (\hat{\mathbf{e}}_z \times \nabla_\perp) E_z. \tag{78}$$

The subscript $\perp$ refers again to the (x, y) or (r, φ) components only, and $\vec{\vec{I}}$ is the unit diagonal tensor. Therefore, once the solutions for E_z and H_z have been determined from Eqs. (65), (66), (67), and (71) through (76), the inhomogeneous particular solutions for all the transverse components can be constructed. In Eqs. (77) and (78), we encounter again the component mixing property of the cross product. It can be disentangled by a complex principal axis transformation of the sort of Eq. (47), which can also be written in cylindrical coordinates

$$X_{\left\{\begin{smallmatrix} L \\ R \end{smallmatrix}\right\}} = \frac{1}{\sqrt{2}} e^{\mp i\varphi}\left(X_r \mp i X_\varphi\right) \leftrightarrow X_{\left\{\begin{smallmatrix} r \\ \varphi \end{smallmatrix}\right\}} = \left\{\begin{smallmatrix} 1 \\ i \end{smallmatrix}\right\} \frac{1}{\sqrt{2}}\left(X_L e^{i\varphi} \pm X_R e^{-i\varphi}\right). \tag{79}$$

In any case, independently of the geometry chosen, Eqs. (77) and (78) readily reduce, also by virtue of Eqs. (63) or (64), to the four algebraic equations for E_L, E_R, H_L, and H_R

$$\begin{Bmatrix} E_\ell \\ H_\ell \end{Bmatrix} = \frac{1}{k_0^2\epsilon_\ell - k^2}\partial_\ell\left(-ik\begin{Bmatrix} E_z \\ H_z \end{Bmatrix} \pm \omega s_\ell \begin{Bmatrix} \mu_0 H_z \\ \epsilon_0\epsilon_\ell E_z \end{Bmatrix}\right), \tag{80}$$

where $\ell = \{L, R\}$, and $s_L = +1$, $s_R = -1$. The differential operator ∂_ℓ, acting on the known functions E_z and H_z, is defined in analogy to Eqs. (47) and (79). In particular, for cylindrical geometry one has

$$\partial_\ell \mathscr{C}_m(\kappa r)e^{im\varphi} = \langle s_\ell \rangle \frac{\kappa}{\sqrt{2}} \mathscr{C}_{m-s_\ell}(\kappa r)e^{i(m-s_\ell)\varphi}. \tag{81}$$

The $\mathscr{C}_m$'s stand for any cylinder function J_m, Y_m, I_m, $e^{im\pi}K_m$ (for the latter two modified cylinder functions the factor $\langle s_\ell \rangle$ on the RHS is to be omitted). For Cartesian geometry, the action of ∂_ℓ on plane waves is trivial via the substitution $\partial_\ell \to (-1/\sqrt{2})(ik_x + s_\ell k_y)$. In microwave plasma diagnostics, the most common configuration is cylindrical, to which we will restrict ourselves. The Cartesian or slab configuration can be dealt with analogously with the simpler property of ∂_ℓ or by the plane wave method discussed in Section VI.B.

On the basis of Eqs. (71), (72), (79), and (81), we find the solutions to Eq. (80) in terms of the radial and azimuthal components.

$$E_{\begin{Bmatrix} r \\ \varphi \end{Bmatrix}} = \sum_j \left(k\mathscr{A}_{\begin{Bmatrix} 1 \\ 2 \end{Bmatrix}}(\kappa_j, r) \pm \omega\mu_0 \mathscr{A}_{\begin{Bmatrix} 2 \\ 1 \end{Bmatrix}}(\kappa_j, r)h(\kappa_j)\right)\mathscr{E}_j \tag{82}$$

and

$$H_{\begin{Bmatrix} r \\ \varphi \end{Bmatrix}} = \sum_j \left(k\mathscr{A}_{\begin{Bmatrix} 1 \\ 2 \end{Bmatrix}}(\kappa_j, r) + \omega\epsilon_0 \mathscr{B}_{\begin{Bmatrix} 1 \\ 2 \end{Bmatrix}}(\kappa_j, r)e(\kappa_j)\right)\mathscr{H}_j, \tag{83}$$

which still have to be subjected to Eqs. (73) to (76). The radial dependency is in the functional coefficients $\mathscr{A}(\kappa_j, r)$, $\mathscr{B}(\kappa_j, r)$

$$\mathscr{A}_{\begin{Bmatrix} 1 \\ 2 \end{Bmatrix}}(\kappa, r) = -\frac{i\kappa}{2}\sum_{\ell=L,R}\begin{Bmatrix} s_\ell \\ i \end{Bmatrix}\frac{J_{m-s_\ell}(\kappa r)}{k_0^2\epsilon_\ell - k^2} \tag{84}$$

$$\mathscr{B}_{\begin{Bmatrix} 1 \\ 2 \end{Bmatrix}}(\kappa, r) = -\frac{\kappa}{2}\sum_{\ell=L,R}\begin{Bmatrix} 1 \\ is_\ell \end{Bmatrix}\frac{\epsilon_\ell J_{m-s_\ell}(\kappa r)}{k_0^2\epsilon_\ell - k^2}. \tag{85}$$

These sums can be expanded in terms of J_m and $J'_m = dJ_m(x)/dx$. For this we recall that $s_L = +1$, $s_R = -1$, and make use of standard recursion relations for cylinder functions, and of Eqs. (55), (65), and (66)

$$\begin{Bmatrix} \mathscr{A}_1(\kappa, r) \\ \mathscr{A}_2(\kappa, r) \end{Bmatrix} = -\frac{i\kappa \dfrac{S}{P}}{k_0^2(\kappa_H^2 - k^2)S + k^4} \begin{pmatrix} \kappa_E^2 & ik_0^2 \dfrac{PD}{S} \\ -ik_0^2 \dfrac{PD}{S} & \kappa_E^2 \end{pmatrix} \times \begin{Bmatrix} J'_m(\kappa r) \\ \dfrac{im}{\kappa r} J_m(\kappa r) \end{Bmatrix} \tag{86}$$

$$\begin{Bmatrix} \mathscr{B}_1(\kappa, r) \\ \mathscr{B}_2(\kappa, r) \end{Bmatrix} = \frac{\kappa S}{k_0^2(\kappa_H^2 - k^2)S + k^4} \begin{pmatrix} k^2 \dfrac{D}{S} & i\kappa_H^2 \\ -i\kappa_H^2 & k^2 \dfrac{D}{S} \end{pmatrix} \times \begin{Bmatrix} J'_m(\kappa r) \\ \dfrac{im}{\kappa r} J_m(\kappa r) \end{Bmatrix}. \tag{87}$$

The $\times$-product is to be understood between the matrix ( ) and the column { }.

Equations (71) and (72), in conjunction with Eqs. (73) to (76), and Eqs. (82) and (83), in conjunction with Eqs. (84) to (87), constitute the entire family of solutions of waves in a cold plasma suitable for cylindrical geometry. To construct a specific solution, their linear superposition has to obey the boundary conditions, which will be addressed in Section VI.D.

2. *Guided Electromagnetic Waves in Vacuum*

For completeness, we shall briefly quote the wave solutions in a vacuum layer between the plasma edge and the wall. They are of importance for the propagation and resonance properties in waveguides or cavities that are only partially filled with plasma. Along the very same lines as in the preceding section, however, with the simplification that now $D = 0$, $P = S = 1$, we can expand the solutions in the form

$$\begin{Bmatrix} E_z^v \\ H_z^v \end{Bmatrix} = \sum_i \begin{Bmatrix} \mathscr{F}_i \\ \mathscr{G}_i \end{Bmatrix} \mathscr{C}_m^{(i)}(\kappa_v r), \tag{88}$$

where $\mathscr{C}_m^{(1,2)} = (J_m, Y_m)$ are the regular and singular Bessel functions, and the vacuum transverse propagation constant is given by $\kappa_v^2 = \omega^2/c^2 - k^2 = k_0^2 - k^2$. The common Fourier modulation factors were omitted in Eq.

(88). The resulting transverse components are then

$$E^{v}_{\begin{Bmatrix} r \\ \varphi \end{Bmatrix}} = \sum_i \left(k\mathcal{K}^i_{\begin{Bmatrix} 1 \\ 2 \end{Bmatrix}} \mathcal{F}_i \pm \omega\mu_0 \mathcal{K}^i_{\begin{Bmatrix} 2 \\ 1 \end{Bmatrix}} \mathcal{G}_i \right) \tag{89}$$

and

$$H^{v}_{\begin{Bmatrix} r \\ \varphi \end{Bmatrix}} = \sum_i \left(\mp \omega\epsilon_0 \mathcal{K}^i_{\begin{Bmatrix} 2 \\ 1 \end{Bmatrix}} \mathcal{F}_i + k\mathcal{K}^i_{\begin{Bmatrix} 1 \\ 2 \end{Bmatrix}} \mathcal{G}_i \right), \tag{90}$$

with the radial functions $\mathcal{K}$ given in terms of the Bessel functions and their derivatives

$$\mathcal{K}^i_1(\kappa_v r) = -\frac{i}{\kappa_v} \mathcal{C}^{(i)\prime}_m(\kappa_v r) \qquad \text{and} \qquad \mathcal{K}^i_2(\kappa_v r) = \frac{m}{\kappa_v^2 r} \mathcal{C}^{(i)}_m(\kappa_v r). \tag{91}$$

D. Boundary Conditions

The boundary conditions follow directly from integrating Maxwell's equations through any arbitrary interface and are therefore consistent with the wave equations of Section VI.C. However, since not only Maxwell's equations, but also the equation of motion for the electron plasma, Eq. (44), were used, caution has to be exercised. Any sufficient and linearly independent set of conditions that is consistent with both Maxwell's equations and the equation of motion may be chosen to uniquely determine the wave solutions.

1. General Conditions

We consider first the conditions resulting from Maxwell's equations (56) and (57). They require that, on any physical or fictitious surface, characterized by its local normal vector $\hat{\mathbf{n}}$, the normal component of the magnetic field and the tangential component of the electric field be continuous

$$\hat{\mathbf{n}} \cdot (\vec{\mathbf{B}}^{<} - \vec{\mathbf{B}}^{>}) = 0; \qquad \hat{\mathbf{n}} \times (\vec{\mathbf{E}}^{<} - \vec{\mathbf{E}}^{>}) = 0. \tag{92}$$

Equation (58) leads to

$$\hat{\mathbf{n}} \times (\vec{\mathbf{B}}^{<} - \vec{\mathbf{B}}^{>}) = -\mu_0 \vec{\mathbf{I}}_s. \tag{93}$$

Unlike in hot plasmas, there is no singular surface current in a cold plasma, i.e., $\vec{\mathbf{I}}_s = 0$. Equation (60) yields the requirement for the quantity $\hat{\mathbf{n}} \cdot (\overleftrightarrow{\epsilon}\vec{\mathbf{E}})$ to be continuous. For a cylindrical boundary of radius r_0 where

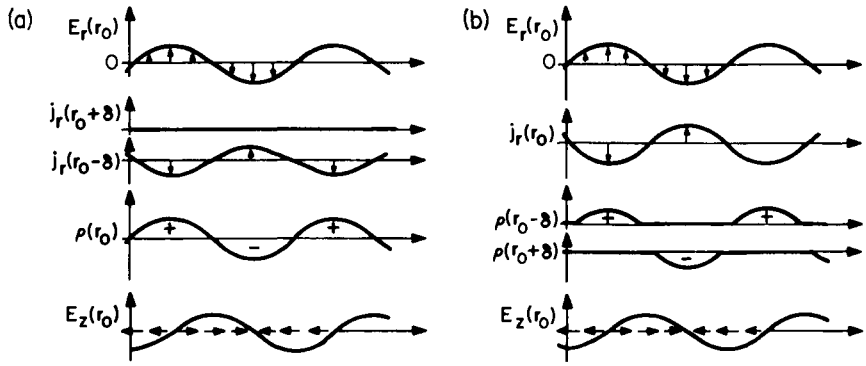


FIGURE 12. Wave coupling in the boundary: surface waves (a) plasma motion stops at the boundary; (b) plasma swings through the boundary.

$\hat{\mathbf{n}} = \hat{\mathbf{e}}_r$, this implies that

$$SE_r^< - iDE_\varphi^< = E_r^> \qquad \text{at} \qquad r = r_0, \tag{94}$$

which, in the isotropic unmagnetized case, where $D \equiv 0$, $S = P$, reduces to the condition familiar from electrostatics: $PE_r^< = E_r^>$ at $r = r_0$. Equation (94) neglects the presence of any surface charge Σ, for if we consider Ohm's law through the surface, $j_r = [\vec{\sigma}\vec{\mathbf{E}}]_r$, the current has to drop instantly to zero at $r = r_0 + \delta$. This will lead to charge buildup according to the continuity equation $\nabla \cdot \vec{\mathbf{j}} + (\partial\rho/\partial t) = 0$. Consequently, the electrostatic condition (94) may not be used to determine the solution.

We briefly point out the importance of the surface charge for the coupling of waves in the boundary: The resulting charge distribution in the surface is such that whenever there exists a wave mode with finite axial wavelength (i.e., $k \neq 0$), radial electric fields will always be accompanied by axial electric fields (and vice versa), as they are induced by the plasma vacuum boundary. Additionally, it is so regardless of whether the static magnetic field is zero or not. This situation is schematically depicted in Figure 12. A similar statement applies for any non-zero azimuthal mode number (i.e., $m \neq 0$). It should be recalled that the bulk properties of the plasma lead to such coupling only for a non-zero magnetic field (c.f., Section VI.C.1). Even if the physically more reasonable assumption is made that the electrons may move across the boundary, the situation still remains the same, as can be argued from Figure 12b. In reality, there is never such a sharp plasma edge. In a more diffuse plasma-vacuum transition, a polarization charge is induced and the above applies at least qualitatively. If the plasma, however, is in contact with a conducting wall, then the axial E-fields are largely shorted out, as will be discussed now.

2. Conditions on a Conducting Wall

At an infinitely conducting wall, Eqs. (92) reduce to

$$\hat{\mathbf{n}} \cdot \vec{\mathbf{B}}^{<} \to 0; \qquad \hat{\mathbf{n}} \times \vec{\mathbf{E}}^{<} \to 0, \tag{95}$$

since no electric field, nor any oscillatory magnetic field may exist inside the conductor. For waveguides or cavities, where the plasma is not in contact with the conducting wall, the boundary condition $\partial B_z/\partial r(r = r_c) = 0$ is also used (Jackson, 1975). More generally, from the φ-component of Eq. (58) we have, invoking the conditions (95), for a plasma in contact with the wall

$$\frac{\partial B_z}{\partial r} = \frac{1}{2}\omega\mu_0\epsilon_0(\epsilon_L - \epsilon_R)E_r = \frac{1}{2}\omega\mu_0\epsilon_0 DE_r, \qquad \text{at} \qquad r = r_c. \tag{96}$$

The RHS vanishes for a nonmagnetized plasma, just as in the case for the empty waveguide.

The discussion so far has been concerned explicitly with the boundary conditions in transverse directions. The boundary conditions (92) and (95), applied in axial direction, where $\hat{\mathbf{n}}$ is parallel to the z-axis, are easily met by an appropriate superposition of the axial plane waves. Clearly, considering the symmetry properties of Eqs. (71), (72), (82), (83), (88), (89), and (90), with respect to the change of sign of k, solutions can always be constructed such that, say, H_z, E_r, $E_\varphi \propto \sin kz$ and E_z, H_r, $H_\varphi \propto \cos kz$ to ensure that the field projections (95) vanish. It is, therefore, not necessary to differentiate between running waves, as they occur in waveguides that are not subject to axial conditions, and standing waves, that occur in an axially bounded plasma in a cavity, with, however, one exception: For $k = 0$, the situation is different. As a result, TE_{mn0} modes[9] cannot exist in cavities, with or without plasma.

The homogeneous conditions of the type (95) cannot hold everywhere on the conductive enclosure of the plasma for the necessity to feed in the microwave power. The effect is that 1) there will be a certain field configuration imposed at the feed point in the wall, which usually leads to selective mode excitation. Locally, 2) the induced field will possibly be distorted, because the nodal plane formed by the wall is interrupted there. The latter effect can be minimized by an appropriate feed geometry, while the first underscores the necessity to know all field components from Section VI.C

[9] The notation m, n, p (here $p = 0$) characterizes, respectively, the two transverse (here azimuthal and radial), and the axial mode modulations.

in order to know which mode is being excited, in particular, for nearly degenerate modes with only slightly different resonance frequencies (Section VI.E.3–5).

E. Examples of Diagnostic Interest

1. *Free Propagation in a Magnetized Plasma: Microwave Interferometry*

We shall consider now a typical configuration used in interferometry in a magnetized plasma column of large diameter, i.e., several wavelengths wide. From Figure 11 it is clear that a magnetic field even for $\omega_c > \omega_p$ has essentially no effect on the dielectric tensor and therefore on the microwave propagation for $\omega \gg \omega_{\text{hybrid}} = \sqrt{\omega_c^2 + \omega_p^2}$, for which we have $D \to 0$, $S \to P \to 1$. However, this also implies that the plasma density must go to zero, a case that is not desirable for a density measurement. To extend the useful range, even in the presence of a magnetic field, the geometry schematically shown in Figure 13 is employed: A waveguide carrying a probing microwave power in TE_{10} mode is expanded into a horn. As a result, the emerging wave vector is largely parallel to the waveguide (Figure 13a). The waveguide with horn antenna is pointed at the plasma, oriented such that the wave's electric field vector is parallel to the static magnetic field $\vec{B}_0$ that is embedded in the plasma (Figure 13b). The wave vector component along

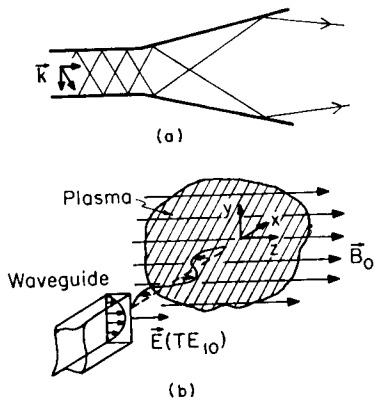


FIGURE 13. Geometry for microwave interferometry.

$\vec{\mathbf{B}}_0$ vanishes approximately (i.e., $k \to 0$), and the wave will propagate inside the plasma as a pure TE mode with a wave vector, say, k_x. From Eqs. (65) and (66) we see that then

$$\left(k_x^2 - k_0^2 P\right) E_z = 0 \tag{97}$$

$$B_z \to 0. \tag{98}$$

Inserting this into Eqs. (77), (78), or (80) we conclude that $B_L = -B_R = ik_x E_z/\sqrt{2}$, i.e., $B_x \to 0$, $B_y = -k_x E_z$. The characteristic wave propagation, or dispersion follows from condition (97) for nontrivial E_z

$$k_x^2 = k_0^2 P = \mu_0 \epsilon_0 \left(\omega^2 - \frac{\omega_p^2}{1 - \dfrac{i\nu}{\omega}} \right). \tag{99}$$

As already discussed in Section VI.A, whenever $n_e > n_c$ (disregarding ν at this point), the wave cannot propagate. Then, namely, $k_x^2 < 0$, resulting in solutions that spatially decay exponentially in the x-direction. Otherwise, measuring the wavelength in the plasma, $\lambda_p = 2\pi/k_x$, the electron density n_e can be determined via the plasma frequency (49).

As simple as the above derivation is, it forms the basis of any microwave interferometry used to determine the plasma density. It is valid under the assumption that 1) the plasma diameter be large, i.e., many plasma wavelength across and 2) there be no sizable diffractive plasma effects that could deflect the wave from its direction of propagation. Both assumptions can become violated when $n \to n_c$, since then, from Eq. (99), $k_x \to 0$, or $\lambda_p \to \infty$, a reason why we have always stressed the wavelength in plasma. We shall discuss these questions briefly in Section VI.F, while all aspects of interferometry were covered in Section IV.

2. *Free Propagation in a Magnetized Plasma: Faraday Rotation*

We now consider a plane wave travelling along the magnetic field and assume no boundaries nearby. From Eq. (69) we see that with a left-polarized and a right-polarized wave, each travels with different speed. Consequently, the polarization vector of a linearly polarized microwave will rotate along the magnetic field direction. This Faraday-rotation effect is used to measure the line integrated density along the field lines, supposing that the magnetic field strength is known. For simplicity we disregard any collisionality effects. From Eq. (69), the angle of rotation γ that results for

propagation through an axial distance z_0 is given by

$$\gamma = 2\pi \frac{z_0}{\lambda_0}\left[\sqrt{1 - \frac{\omega_p^2}{\omega(\omega + \omega_c)}} - \sqrt{1 - \frac{\omega_p^2}{\omega(\omega - \omega_c)}}\right]. \quad (100)$$

The vacuum wavelength of the microwave radiation was denoted by λ_0. For low densities where $\omega_p \ll \omega$, the term in the brackets can be approximated by $[\ ] \approx \omega_p^2\omega_c/\omega(\omega^2 - \omega_c^2)$, making γ proportional to the electron density n_e [c.f., Eq. (49)]. A more detailed account of the Faraday effect in plasmas and of the suitable microwave diagnostic techniques are given by Künstler (1965).

3. *Cavity Resonances: The Unmagnetized Homogeneous Plasma*

To gain some insight in resonances in plasma filled cavities and the accompanying mode structures, in this section we treat the simplest possible case, where the unmagnetized plasma fills the cavity entirely. Then, and only then, by the arguments given in Sections VI.C.1 and VI.D, pure TE and TM modes exist, as they are familiar from empty cavities (Gandhi, 1981). All the equations for the plasma-filled cavity reduce here to the equations governing the modes in an empty cavity, only that the following substitution is to be made

$$\epsilon_0 \to \epsilon_0 P. \quad (101)$$

Mode structures and resonance frequencies of empty cavities of various cross sections and configurations can be found in the microwave literature. As a specific example, we choose the frequently used TM_{0n0} modes in a cylindrical cavity of radius r_c, for which $k = m = 0$. There the resonance frequency is given by the condition [c.f. Eq. (65)]:

$$J_0(\kappa_E r_c) = J_0\left(\omega\sqrt{\mu_0\epsilon_0 P(\omega)}\, r_c\right) = 0. \quad (102)$$

Specifically, the mode with no radial nodes, labeled by $n = 1$, which arises from the first zero of Eq. (102), displays the resonance frequency, disregarding collisional effects, by virtue of Eq. (53)

$$f_{010} = \frac{\omega_{010}}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{2.405^2 + n_e e^2 \mu_0 r_c^2/m}{\epsilon_0 \mu_0 r_c^2}}, \quad (103)$$

from which the electron density n_e can be extracted.

4. *Cavity Resonances: The Unmagnetized Inhomogeneous Plasma*

Everywhere up to this point in Section VI, we have dealt with a homogeneous plasma, whose properties, as they enter the dielectric tensor, are spatially constant. In order to get an impression of how much any spatial variation of $\vec{\epsilon}$ may influence the resonance frequency, we also give now the equations for an isotropic, nonmagnetized but inhomogeneous plasma, as they can be derived from Maxwell's equations (56) to (59)

$$\nabla^2\vec{\mathbf{E}} + k_0^2\epsilon(\vec{\mathbf{r}})\vec{\mathbf{E}} + \nabla(\vec{\mathbf{E}}\cdot\nabla)\ln\epsilon(\vec{\mathbf{r}}) = 0 \tag{104}$$

and

$$\nabla^2\vec{\mathbf{B}} + k_0^2\epsilon(\vec{\mathbf{r}})\vec{\mathbf{B}} - (\nabla\times\vec{\mathbf{B}})\times\nabla\ln\epsilon(\vec{\mathbf{r}}) = 0. \tag{105}$$

The analysis of these equations becomes rather simple for the case where (1) the fields have no axial or azimuthal dependence (i.e., $k = m = 0$),[10] and 2) the dielectric constant depends solely on r, $\epsilon = \epsilon(r)$. For comparison with the TM_{0n0} modes discussed in Section VI.E.3, Eqs. (104) and (105) reduce, with $\vec{\mathbf{E}} = E\hat{\mathbf{e}}_z$, $\vec{\mathbf{B}} = B\hat{\mathbf{e}}_\varphi$, to

$$\left(\nabla_\perp^2 + k_0^2\epsilon(r)\right)E = 0 \tag{106}$$

$$\left(\nabla_\perp^2 + k_0^2\epsilon(r)\right)B - i\omega E\frac{\partial\epsilon(r)}{\partial r} = 0. \tag{107}$$

If we assume that the plasma is in contact with the wall, we can immediately find the resonance condition by imposing the radial boundary condition (95) upon E at $r = r_c$, i.e., $E(r_c) = 0$. For this, it is sufficient to seek solutions only to Eq. (106), as it does not depend on B, and B does not enter the boundary conditions (95).

For a parabolic plasma density profile, we have

$$n_e(r) = \hat{n}\left(1 - \delta\frac{r^2}{r_c^2}\right), \qquad \text{or} \qquad \epsilon(r) = \epsilon_1 + \epsilon_2\frac{r^2}{r_c^2}, \tag{108}$$

where

$$\epsilon_1 = 1 - \frac{\hat{\omega}_p^2}{\omega(\omega - i\nu)}; \qquad \epsilon_2 = \delta\frac{\hat{\omega}_p^2}{\omega(\omega - i\nu)}; \qquad \hat{\omega}_p = \omega_p(n_e \to \hat{n}). \tag{109}$$

[10] Another reason to consider only modes with $k = m = 0$ is that otherwise no pure TM or TE modes can exist for a radially varying plasma, as discussed in Section VI.D. This, of course, would also follow directly from Eqs. (104) and (105).

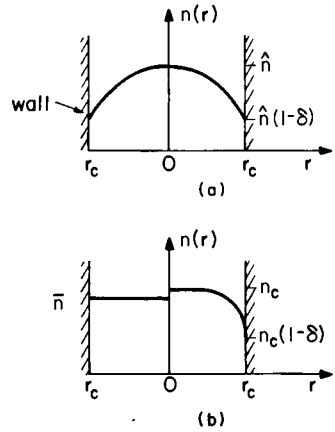


FIGURE 14. Radial plasma profiles in a cavity or waveguide: (a) parabolic, (b) power-law (right half) and equivalent rectangular (left half) profile.

The quantities $\hat{n}$ and δ follow from Figure 14a, and are entirely arbitrary, although a physically reasonable choice would assume $\hat{n} < n_c, 0 < \delta < 1$. Using the abbreviation $\hat{k}_0^2 = \hat{\omega}_p^2 \mu_0 \epsilon_0 \delta / r_c^2$, the solution

$$\vec{\mathbf{E}}(r) = \hat{\mathbf{e}}_z E_0 \, Re \, e^{\pm i r^2 \hat{k}_0} \, {}_1F_1\left[\frac{1}{2}\left(1 \mp i\epsilon_1 k_0^2/\hat{k}_0\right), 1, \mp i r^2 \hat{k}_0\right] \qquad (110)$$

can be written in terms of confluent hypergeometric functions, whose zeros determine the resonance condition. Since this is somewhat awkward to discuss, we choose a slightly different power-law profile. This will permit to make a direct comparison with the results of the preceding section to estimate the effect of the density profile on the TM_{0n0} resonance frequency. We assume the worst possible case, where the density on axis $\hat{n}$ takes the critical value n_c from Eq. (54), for which $\epsilon_1 = 0$.[11] Clearly, for a rectangular profile, as assumed elsewhere in Section VI, no oscillatory fields could form in the cavity for $\epsilon = 0$. Therefore, to determine the influence of the profiles we will instead require that the radially integrated density $\int r\,dr\, n_e(r)$ be the same for the rectangular as for the power-law profile that is defined by

$$n_e(r) = n_c\left[1 - \delta\left(\frac{r}{r_c}\right)^{2\lambda}\right], \qquad \text{or} \qquad \epsilon(r) = \delta\left(\frac{r}{r_c}\right)^{2\lambda}. \qquad (111)$$

[11] Because of the nonlinear (square-root type) dependency of the resonance frequency on P, and therefore on n_e—c.f., Eqs. (102) and (103)—the profile properties will play their greatest role for $\hat{n} \to n_c$.

Again δ and $\lambda \geq 0$ are arbitrary (c.f., Figure 14b). The equivalent mean density is then $\bar{n} = n_c[1 - \delta/(\lambda + 1)]$. The solution for a profile of the form (111), takes the form

$$\vec{\mathbf{E}}(r) = \hat{\mathbf{e}}_z E_0 J_0\left(\omega\sqrt{\mu_0\epsilon_0\delta}\,\frac{r_c}{(\lambda+1)}\left(\frac{r}{r_c}\right)^{\lambda+1}\right). \tag{112}$$

The ratio of the resonance frequencies for the rectangular profile of Section VI.E.3 and of the power-law profile (111) with the same mean density,

$$\frac{\omega^r}{\omega^p} = \sqrt{1 - \frac{\lambda\delta}{(\lambda+1)^2}}\,, \tag{113}$$

is rather insensitive to the choice of δ and λ, where $0 \leq \delta \leq 1$ and $\lambda > 1/2$. Consequently, the error in the density determination, incurred by assuming a rectangular rather than a diffuse profile is, at least for the TM_{0n0} modes, small. Agdur and Enander (1962) discuss this topic in more detail.

5. *The General Cavity Resonance Condition: The Homogeneous Plasma Column*

We have discussed so far the resonance conditions for an unmagnetized plasma, both homogeneous and inhomogeneous, which fills entirely a cylindrical cavity. Now we will consider the general situation: A circular magnetized, homogeneous plasma column of radius r_0 is coaxially situated in a cylindrical waveguide or cavity. Again, we will not differentiate axially standing (case of the cavity) or running waves (case of the waveguide). In addition to the critical condition given below, the cavity case will require that the axial wavelength $\lambda = 2\pi/k$, be determined by the length of the cavity, l_c, via

$$k = \frac{p\pi}{l_c}, \tag{114}$$

in order to match the axial boundary conditions. Only, for the TM_{0n0} family, this is not the case. There all the field components that are subject to these axial conditions vanish identically.

Recalling the host of wave solutions and boundary conditions and, in particular, the fact that, for $k \neq 0$ no pure TE or TM modes can exist, it may appear at first glance a hopeless venture to treat the general case.

However, all that is needed in order to find the specific solutions and the corresponding set of critical conditions is that the 6×6 coefficient determinant vanish. The six homogeneous equations, or the matrix of coefficients, for the six independent unknowns of Eqs. (71), (72), (82), (83), (88), (89), and (90), here chosen to be $\mathscr{E}_1$, $\mathscr{H}_2$, $\mathscr{F}_1$, $\mathscr{F}_2$, $\mathscr{G}_1$, and $\mathscr{G}_2$, can be composed in the compact scheme of Table 2. The omitted arguments of the functions $\mathscr{A}_i$ and $\mathscr{B}_i$ [c.f., Eqs. (84) and (85)], e and h [c.f., Eqs. (75) and (76)], and $\mathscr{K}_j^i$ [c.f., Eq. (91)] follow from their respective rows and columns. The empty squares are meant to be zero. As an example, the equation for $E_z(r_0)$ (first row) would thus read:

$$J_m(\kappa_1 r_0)\mathscr{E}_1 - J_m(\kappa_v r_0)\mathscr{F}_1 - Y_m(\kappa_v r_0)\mathscr{F}_2 + e(\kappa_2)J_m(\kappa_2 r_0)\mathscr{H}_2 = 0. \quad (115)$$

As in the two preceding sections, an infinite set of characteristic solutions will ensue because of the multivaluedness of the principal solutions, i.e., here the Bessel functions. For $k \to 0$ and $m \to 0$, a case that was discussed in detail by Buchsbaum et al. (1960), we see, in light of the limit values of $\mathscr{A}$, $\mathscr{B}$, e, and h that the 6×6 matrix of Table 2 becomes reducible into

Table 2 Matrix of Coefficients

	$\mathscr{E}_1$	$\mathscr{F}_1$	$\mathscr{F}_2$	$\mathscr{H}_2$	$\mathscr{G}_1$	$\mathscr{G}_2$	
E_z	J_m	$-J_m$	$-Y_m$	eJ_m			r_0
H_φ	$k\mathscr{A}_2 h + \omega\epsilon_0\mathscr{B}_2$	$-\omega\epsilon_0\mathscr{K}_1^1$	$-\omega\epsilon_0\mathscr{K}_1^2$	$k\mathscr{A}_2 + \omega\epsilon_0\mathscr{B}_2 e$	$-k\mathscr{K}_2^1$	$-k\mathscr{K}_2^2$	r_0
E_z		J_m	Y_m				r_c
H_z	hJ_m			J_m	$-J_m$	$-Y_m$	r_0
E_φ	$k\mathscr{A}_2 - \omega\mu_0\mathscr{A}_1 h$	$-k\mathscr{K}_2^1$	$-k\mathscr{K}_2^2$	$k\mathscr{A}_2 e - \omega\mu_0\mathscr{A}_1$	$\omega\mu_0\mathscr{K}_1^1$	$\omega\mu_0\mathscr{K}_1^2$	r_0
$\frac{\partial H_z}{\partial r}$					J_m'	Y_m'	r_c
	κ_1	κ_v	κ_v	κ_2	κ_v	κ_v	

two 3×3 matrices. Hence, pure TM modes (where $\mathscr{H}_2 = \mathscr{G}_1 = \mathscr{G}_2 \equiv 0$) or pure TE modes (where $\mathscr{E}_1 = \mathscr{F}_1 = \mathscr{F}_2 \equiv 0$) are then permitted, regardless of whether the plasma is magnetized or not. But while TM_{0n0} modes can exist in an axially bounded cavity, the TE_{0n0} modes exist only as cutoff modes TE_{0n} in a waveguide. The electrostatic limit for which $\omega \to 0$ yields similarly a reducible blockform, which is most easily seen when the rows for $E_\varphi(r_0)$ and $H_\varphi(r_0)$ are interchanged.

Finally, for comparison with the TM_{0n0} mode family in an unmagnetized plasma discussed in the two preceding sections, we consider the simpler case resulting from Table 2 when the plasma fills the cavity completely ($r_0 = r_c$). The subset of conditions from Table 2 is then

$$J_m(\kappa_1 r_c)\left[k\mathscr{A}_2(\kappa_2, r_c)e(\kappa_2) - \omega\mu_0\mathscr{A}_1(\kappa_2, r_c)\right]$$
$$= e(\kappa_2)J_m(\kappa_2 r_c)\left[k\mathscr{A}_2(\kappa_1, r_c) - \omega\mu_0\mathscr{A}_1(\kappa_1, r_c)h(\kappa_1)\right]. \quad (116)$$

Particularly, for $k = 0$, $m = 0$, and by virtue of Eqs. (68), (75), (76), (86), and (87), this reduces to

$$J_0(\kappa_E r_c)J_1(\kappa_H r_c) = 0, \quad (117)$$

permitting either $\mathscr{E}_1 \equiv 0$ for $J_1(\kappa_H r_c) = 0$, or $\mathscr{H}_2 \equiv 0$ for $J_0(\kappa_E r_c) = 0$. The latter is, as above, the condition for the TM_{0n0} cavity modes where, as in the unmagnetized case Eq. (101), the substitution

$$\epsilon_0 \to \epsilon_0 P \quad (118)$$

makes the connection to the empty-waveguide or cavity solution. The second condition can only be fulfilled for a TE_{0n} waveguide mode at cutoff.

In Section III.B.1, we presented approximate perturbation techniques to simplify the results of the present section. They may be used whenever the plasma introduces a small perturbation to the fields of a cavity, either by a low plasma density or by a thin plasma column, $r_0 \ll r_c$.

F. Optical Approximations: Diagnostic Applications

Starting from Maxwell's equations as they apply to dielectric media, light refraction and diffraction can be suitably described by geometric optics or by wave optics. Both deal with the modification of the light propagation by the dielectric properties of matter and its boundary surfaces. Wave optics

are appropriate whenever the wavelength is of the same order of the magnitude as the characteristic dimensions of the dielectric body and plane waves are a reasonable assumption. Geometric optics are applicable when the wavelength is much smaller than any characteristic gradient length (Cornbleet, 1976 and 1984). Given the appropriate range of validity, the same formulations may be applied to microwave propagation. They yield efficient tools for a qualitative and, often times, excellent quantitative description of the propagation phenomena. Microwave lenses and mirrors, and beam refraction or deflection by plasma density gradients are common examples for the use of geometric optics. On the other hand, diffraction patterns emerging from horn antennae, the Fabry-Perot transmission and reflection properties of a plasma layer, or the coherent Bragg scattering of microwaves off a plasma density wave are well described by wave optics.

Rather than justifying the theoretical basis of these optical approximations, which can be found in an abundant literature (Cornbleet, 1976 and 1984), we will only briefly cite a few applications useful to the diagnostician.

1. Microwave Horn Antennae and Microwave Imaging

While microwave power is best transported in waveguides, it is frequently necessary that it be transmitted into a plasma sample for investigation. For this, horn antennae are used (Love, 1976; Milligan, 1985). Lenses and mirrors are employed when better imaging is necessary, for instance for improved spatial resolution (Primich et al., 1965). Microwave lenses are usually made from a polymeric compound of suitable refractive index (Brown, 1953; Gandhi, 1981; Milligan and McDonnell, 1986). Multiple reflections off the lens surfaces that can substantially degrade focusing properties, especially in arrays of several lenses, can be reduced by reactive coating or quarter wavelength matching (Jones et al., 1956; see also Section VI.F.3 below). Another possibility for a lens is to use a plano-concave metallic channel plate, which, via Huygens' principle, transforms, say, a spherical wavefront into a plane wave front, and vice versa (Uckan 1981). Mirrors serve not only for imaging purposes, but also for double path transmission of interferometric measurements in low-reflection plasmas (Efthimion et al., 1985; Overzet and Verdeyen, 1986); and mitre box type mirrors are, for example, used for microwave beam scanning applications (Uckan 1983). Partially reflecting mirrors and grid polarizers permit experimental setups essentially equivalent to their counterparts in light optics (Fellers, 1962; Davis and Patsakos, 1983; Suvorov et al., 1984).

2. *Optical Refraction in Plasma Columns*

Associated with the dielectric constant of a plasma there is also a refractive index defined by

$$n_r = \frac{c}{v_{\text{phase}}} = \frac{kc}{\omega} = \frac{k}{k_0}. \tag{119}$$

In the geometry of Section VI.E.1, this becomes, using Eqs. (97), (98), and (53),

$$n_r = \sqrt{P} = \sqrt{1 - \frac{\omega_p^2}{\omega(\omega - i\nu)}}. \tag{120}$$

As long as $\nu \ll \omega$, the effective refractive index of a plasma is $n_r < 1$, unlike in most other optical materials. With the identification (120), we can characterize the refractive imaging properties of a homogeneous plasma column in analogy to the optics of a cylinder lens (Meuth, 1986). This simple calculation is usually sufficient to estimate the deflection that the beam may suffer in multichannel or scanning microwave interferometers or the divergence in transmission measurements. For inhomogeneous plasmas of circular or elliptic cross section, the more realistic, but also more involved, ray-tracing method is available, based on Eqs. (97) or (106) (Shmoys, 1961; Jones and Wooding, 1965; Faugeras, 1965; Tallents, 1984, see also Section IV.D.1).

3. *Fabry-Perot Methods*

Similar to the resonances in cavities and the propagation characteristics in waveguides, whereby the electromagnetic waves bounce back and forth, reinforcing themselves constructively, there are such resonance and transmission phenomena in any bounded dielectric, be it enclosed by a metal wall, another dielectric, or by vacuum. Dielectric waveguides are such an example. When these are immersed into a plasma, the altered wave propagation in the waveguide can be used to determine the plasma density interferometrically with high spatial resolution (Stewart and Robson, 1965; Lisitskaya and Shustin, 1981; Armand et al., 1982).

As in the case of an optical interference filter, a plasma slab or column displays transmission and reflection characteristics for microwave radiation near the cutoff that are oscillatory with frequency or plasma density. This effect is due to multiple internal reflections in the plasma (Heald and Wharton, 1978). With the plasma density approaching its critical value

given by Eq. (54), the wavelength in the plasma becomes large according to Eq. (99). Hence, the situation where the wavelength is of the order of the plasma slab thickness or column radius may easily arise. Moreover, if the plasma is situated in a conductive chamber, microwave reflections off nearby walls can make a similar additional effect, making the interpretation of measurements difficult. This is of particular importance when the plasma density, and therefore the plasma wavelength, vary with time. The Fabry-Perot effect can be exploited to reduce such unwanted reflections from metallic walls by methods similar to those used for antireflection coatings in optics (Severin, 1956; Kato and Hutchinson, 1984). The most desirable situation, especially for microwave interferometry, is that the plasma density be low, while its thickness accommodate many wavelengths. However, this cannot always be achieved. For higher plasma densities, $n_e \to n_c$, and smaller column width, $\lambda \sim r_0$, a Fabry-Perot cell, in conjunction with focusing lenses, can be employed (Bize et al., 1965; Primich et al., 1965; Davis and Patsakos, 1983).

References

Adler, R. B., Chu, L. J., and Fano, R. (1960). *Electromagnetic Energy, Transmission and Radiation*. Wiley, New York.

Agdur, B. and Enander, B. (1962). *J. Appl. Phys.* **33**, 575.

Akulina, D. K. (1965). *Proceedings VIIth International Conference on Phenomena in Ionized Gases*, Beograd, Vol. 3, p. 74.

Aničin, B. A. (1965). *Proceedings VIIth International Conference on Phenomena in Ionized Gases*, Beograd, Vol. 2, p. 143.

Anissimov, A. I., Vinogradov, N. I., Golant, V. E., and Konstantinov, B. P. (1960). *Zhurn. Tekhn. Fiz.* **30**, 1009.

Anissimov, A. I., Budnikov, V. N., Vinogradov, N. I., and Golant, V. E. (1965). *Proceedings VIIth International Conference on Phenomena in Ionized Gases*, Beograd, Vol. 3, p. 86.

Archer, B. T., Meuth, H., and Lieberman, M. A. (1984). University of California, Berkeley, ERL Report M84-104.

Armand, N. A., Lisitskaya, A. A., Rogashkov, S. A., Rogashkova, A. I., Chmil', A. I., and Shustin, E. G. (1982). *Sov. J. Plasma Phys.* **8**, 212.

Asmussen, J., Jr., Mallavarpu, Raghuveer, Hamman, J. R., and Park, Hee Chung. (1974). In: *Proceedings of the I.E.E.E.* **62**, 109.

Baden Fuller, A. J. (1979). *Microwaves*, 2nd ed., Pergamon, Oxford.

Bartlett, D. V. and Brand, G. F. (1973). *J. Phys.* **E**: **6**, 1213.

Bartlett, D. V. Costley, A. E., and Robinson, L. C. (1978). *Infrared Phys.* **18**, 749.

Bekefi, G. (1966). *Radiation Processes in Plasmas*. John Wiley and Sons, New York.

Bekefi, G., Hirshfield, J. L., and Brown, S. C. (1961a). *Phys. Fluids* **4**, 173.

Bekefi, G., Hirshfield, J. L., and Brown, S. C. (1961b). *Phys. Rev.* **122**, 1037.

Benson, F. P. (1969). Attenuation of Rectangular Waveguides. In: *Millimeter and Submillimeter Waves* (F. P. Benson, ed.). Iliffe Books Ltd., London.

Bethke, G. W. and Ruess, A. D. (1964). *Phys. Fluids* **7**, 1446.

Bhartia, P. and Bahl, I. J. (1984). *Millimeter Wave Engineering and Applications*. Wiley, New York.

Bhattacharya, A. K., Verdeyen, J. T., Adler, F. T., and Goldstein, L. (1967). *J. Appl. Phys.* **38**, 527.
Biondi, M. A. (1951). *Rev. Sci. Instr.* **22**, 500.
Biondi, M. A. and Brown, S. C. (1949). *Phys. Rev.* **75**, 1700.
Bisschops, T. J. and de Hoog, F. J. (1985). *Pure Appl. Chem.* **57**, 1311.
Bize, D., Cadart, M., and Consoli, T. (1965). In: *Proceedings VIIth International Conference on Phenomena in Ionized Gases*. Beograd, Vol. 3, p. 74.
Born, M. and Wolf, E. (1980). *Principles of Optics*, 6th ed., Pergamon, Oxford.
Bornatici, M., Cano, R., De Barbieri, O., and Engelmann, F. (1983). *Nucl. Fusion* **23**, 1153.
Brenning, N. (1984). *J. Phys.* **E**: **17**, 1018.
Brown, I. G., Compher, A. B., and Hopkins, D. B. (1970). *Rev. Sci. Instr.* **41**, 600.
Brown, J. (1953). *Microwave Lenses*. Wiley, New York.
Buchsbaum, S. J. and Brown, S. C. (1957). *Phys. Rev.* **106**, 196.
Buchsbaum, S. J., Mower, L. and Brown, S. C. (1960). *Phys. Fluids* **3**, 806.
Cano, R. (1979). In: *Diagnostics for Fusion Experiments* (E. Sindoni and C. Wharton, eds.). Pergamon Press, Oxford, New York, pp. 117, 191.
Casper, T., Stever, R., Rice, B., and Zammit, R. (1986). *Bull. Am. Phys. Soc.* **31**, 1619.
Celata, C. M. and Boyd, D. A. (1977). *Nucl. Fusion* **17**, 735.
Chapman, B. (1980). *Glow Discharge Processes*. Wiley, New York.
Chen, F. F. (1965). Electric Probes. In: *Plasma Diagnostic Techniques* (R. H. Huddlestone and S. L. Leonard, eds.). Academic Press, New York, p. 113.
Chen, F. F. (1984). *Plasma Physics and Controlled Fusion*, Vol. 1, 2nd ed. Plenum, New York.
Chen, F. F., Etievant, C., and Mosher, D. (1968). *Phys. Fluids* **11**, 811.
Coffield, F. E., Stever, R. D., and Lund, N. P. (1981). In: *Proceedings of the 9th Symposium on Engineering Problems of Fusion Research*. Chicago, Vol. 1, p. 961.
Coffield, F. E., Thomas, S. R., Lang, D. D., and Stever, R. D. (1983). In: *Proceedings of the 10th Symposium on Fusion Engineering*. Philadelphia, Vol. 2, p. 1920.
Collin, R. E. (1966). *Foundations for Microwave Engineering*. McGraw-Hill, New York.
Cornbleet, S. (1976). Microwave optics. In: *Pure and Applied Physics*, Vol. 41 (H. S. Massey and K. A. Brueckner, eds.). Academic Press, London, New York.
Cornbleet, S. (1984). *Microwave and Optical Ray Geometry*. Wiley, New York.
Costley, A. E., Hastie, R. J., Paul, J. W., and Chamberlain, J. (1974). *Phys. Rev. Lett.* **33**, 748.
Crawford, F. W., Kino, G. S., and Cannara, B. A. (1963). *J. Appl. Phys.* **34**, 3168.
Cummins, W. F. (1970). *Rev. Sci. Instr.* **41**, 234.
Dattner, A. (1961). In: *Proceedings Vth International Conference on Ionization Phenomena in Gases*. Munich, Vol. 2, p. 1477.
Davis, L. W. and Patsakos, G. (1983). In: *Proceedings of SPIE- the International Society for Optical Engineering*. Vol. 423: Millimeter Wave Technology II (J. C. Wiltse, ed.), Bellingham, p. 40.
Demas, N. G., Ma, C. H., Hutchinson, D. P., and Staats, P. A. (1985). *I.E.E.E. Trans. Plasma Sci.* **PS-13**, 41.
Deutsch, M. and Beniaminy, I. (1982). *Appl. Phys. Lett.* **41**, 27.
Doane, J. L. (1980). *Rev. Sci. Instr.* **51**, 317.
Efthimion, P. C., Taylor, G., Ernst, W., Goldman, M., McCarthy, M., Anderson, H., and Luhmann, N. C. (1985). *Rev. Sci. Instr.* **56**, 908.
Ernst, W. P. (1964). *I.S.A. Trans.* **3**, 9.
Ernst, W. P. (1967). Princeton Plasma Physics Laboratory Report MATT-535.
Ernst, W. P. (1972). *I.E.E.E. Trans. Nucl. Sci.* NS-19, 740.
Evans, D. E. and Katzenstein, J. (1969). *Rep. Prog. Phys.* **32**, 207.
Fano, R. M., Chu, L. J., and Adler, R. B. (1960). In: *Electromagnetic Fields, Energy and Forces*. Wiley, New York.

Faugeras, P. E. (1965). In: *Proceedings VIIth International Conference on Phenomena in Ionized Gases*. Beograd, Vol. 3, p. 119.
Fellers, R. G. (1962). *The Microwave J.* **5**, 80.
Fessenden, T. J. and Smullin, L. D. (1965). In: *Proceedings VIIth International Conference on Phenomena in Ionized Gases*. Beograd, Vol. 3, p. 83.
Gandhi, O. P. (1981). *Microwave Engineering and Applications*. Pergamon, New York.
Golant, V. E. (1960). *Zhurn. Tekhn. Fiz.* **30**, 1265.
Golant, V. E. (1984). Methods of diagnostics based on the interaction of electromagnetic radiation with a plasma. In: *Handbook of Plasma Physics*, Vol. 2 (M. N. Rosenbluth and R. Z. Sagdeev, eds.). Elsevier, New York.
Goldston, R. J., Mazzucato, E., Slusher, R. E., and Surko, C. M. (1976). In: *Proceedings of the VIth International Conference on Plasma Physics and Controlled Nuclear Fusion Research*. Berchtesgaden, IAEA Vienna, Vol. I, p. 371.
Greenberger, A. (1978). Princeton Plasma Physics Laboratory Report PPPL-1466.
Harris, J. H. and Balfour, D. (1965). In: *Proceedings VIIth International Conference on Phenomena in Ionized Gases*. Beograd, Vol. 3, p. 79.
Heald, M. A. and Wharton, B. C. (1978). *Plasma Diagnostics with Microwaves*. Robert E. Krieger Publishing Company, New York.
Herlin, M. A. and Brown, S. C. (1948). *Phys. Rev.* **74**, 291.
Hermansdorfer, H. (1968). Microwave diagnostic techniques. In: *Plasma Diagnostics* (W. Lochte-Holtgreven, ed.). North-Holland, Amsterdam, p. 478.
Hotston, E. and Seidl, M. (1965). *Rev. Sci. Instr.* **42**, 225.
Huesman, R. H., Gullberg, G. T., Greenberg, W. L., and Budinger, T. F. (1977). *Donner Algorithms for Reconstruction Tomography*. Lawrence Berkeley Laboratory, publication 214.
Ishii, T. K. (1966). *Microwave Engineering*. Ronald Press, New York.
Jackson, J. D. (1975). *Classical Electrodynamics*, 2nd ed. Wiley, New York.
Jacobson, A. R. (1982). *Rev. Sci. Instr.* **53**, 918.
Jones, A. R. and Wooding, E. R. (1965). In: *Proceedings VIIth International Conference on Phenomena in Ionized Gases*. Beograd, Vol. 3, p. 122.
Jones, E. M., Morita, T., and Cohn, S. B. (1956). *I.R.E. Trans.* **AP1**, 31.
Kaiser, J. A., Smith, H., Pepper, W., and Little, J. (1962). *I.R.E. Trans. MTT* **10**, 548.
Kato, K. and Hutchinson, I. H. (1984). M.I.T. Plasma Fusion Center Report PFC/RR-84-11.
Künstler, F. (1965). In: *Proceedings VIIth International Conference on Phenomena in Ionized Gases*. Beograd, Vol. 3, p. 93.
Levine, A., Kuckes, A., and Skislak, A. (1965). Princeton Plasma Physics Laboratory Report MATT-369.
Lieberman, M. A. and Lichtenberg, A. J. (1969). *Phys. Fluids* **12**, 2109.
Lindberg, L. and Eriksson, A. (1982). *J. Phys.* **E**: **15**, 548.
Lisitano, G. (1975). Density distribution measurements in large CTR devices. In: *Course on Plasma Diagnostics and Data Acquisition Systems* (A. Eubank and E. Sindoni, eds.). CNR Euratom, Varenna, p. 17.
Lisitskaya, A. A. and Shustin, E. G. (1981). *Instrum. Exp. Tech. (USSR)* **24**, 158.
Lister, J. B., Means, R. W., and Oberson, P. (1982). *Rev. Sci. Instr.* **53**, 600.
Love, A. W. (ed.) (1976). *Electromagnetic Horn Antennas*. I.E.E.E. Press Selected Reprint Series, New York.
Luhmann, N. C. and Peebles, W. A. (1984). *Rev. Sci. Instr.* **55**, 279.
Machuzak, J. S., Woskoboinikow, P., Mulligan, W. J., Cohn, D. R., Gerver, M., Guss, W., Mauel, M., Post, R. S., and Temkin, R. J. (1986). *Rev. Sci. Instr.* **57**, 1983.
Masi, C. G. and Phillips, T. G. (1986). *Test and Measurement World*, Vol. 60.
Mejia, S. R., McLeod, R. D., Pries, W., Shuftlebotham, P., Thomson, D. J., White, J., Shellenberg, J., Kao, K. C., and Card, H. C. (1985). *J. Non-Cryst. Solids* **77** and **78**, 765.

Meuth, H. (1986). In: *Proceedings of SPIE- The International Society for Optical Engineering*, Vol. 701, *Optics, Optical Systems and Applications*, S. Sottini, S. Trigari, eds.), Bellingham, p. 22
Milligan, T. A. (1985). *Modern Antenna Design*. McGraw-Hill, New York.
Milligan, T. A. and McDonnell, J. (1986). *Microwaves and RF* **25**, 103.
Minami, Kazuo and Takeda, S. (1969). *Phys. Fluids* **12**, 1089.
Morgan, D. V. and Howes, M. J., eds. (1980). *Microwave Solid State Devices and Applications*. Peter Peregrinus, New York.
Okabayashi, M. and Arunasalam, V. (1977). *Nucl. Fusion* **17**, 497.
Overzet, L. J. and Verdeyen, J. T. (1986). *Appl. Phys. Lett.* **48**, 695.
Paraszczak, J., Heidenreich, J., Hatzakis, M., and Moisan, M. (1985). *Microelectronic Eng.* **3**, 397.
Paris, P. J. and Hollenstein, C. (1983). *Rev. Sci. Instr.* **54**, 825.
Parker, J. V., Nickel, J. C., and Gould, R. W. (1964). *Phys. Fluids* **7**, 1489.
Persson, K. B. (1957). *Phys. Rev.* **106**, 191.
Podgornyi, I. M. (1971). *Topics in Plasma Diagnostics*. Plenum, New York.
Pollard, R. D. (1980). In: *Microwave Solid State Devices and Applications* (D. V. Morgan and M. J. Howes, eds.). Peregrinus, New York, p. 59.
Primich, R. I., Auston, D. H., Hayami, R. A., McLeod, J. D., and Zivanovic, S. (1965). In: *Proceedings VIIth International Conference on Phenomena in Ionized Gases*. Beograd, Vol. 3, p. 126.
Robson, P. N. (1969). Overmoded Rectangular Waveguides. In: *Millimeter and Submillimeter Waves* (F. P. Benson, ed.). Iliffe Books Ltd., London.
Roddy, D. (1986). *Microwave Technology*. Prentice Hall, Englewood Cliffs.
Rohatgi, R., Chen, K. I., Bekefi, G., Kaplan, R. D., Luckhardt, S. C., Mayberry, M., Porkolab, M., and Villasenor, J. N. (1985). In: *Radiofrequency Plasma Heating* (D.G. Swanson, ed.) American Institute of Physics, New York, p. 174.
Rutgers, W. R. and Boyd, D. A. (1977). *Phys. Lett.* **62A**, 498.
Sauthoff, N. R. and von Goeler, S. (1979). *I.E.E.E. Trans. Plasma Sci.* **PS-7**, 141.
Schulz, G. J. and Brown, S. C. (1955). *Phys. Rev.* **98**, 1642.
Severin, H. (1956). *I.R.E. Trans.* **AP4**, 385.
Sevillano, E., Brau, K., Goodrich, P., Irby, J., Mauel, M., Post, R. S., Smith, D. K., and Sullivan, J. (1984). *Rev. Sci. Instr.* **56**, 960.
Sheffield, J. (1975). *Plasma Scattering of Electromagnetic Radiation*. Academic Press, New York.
Shmoys, J. (1961). *J. Appl. Phys.* **32**, 689.
Slater, J. C. (1946). *Rev. Mod. Phys.* **18**, 441.
Somlo, P. I. and Hunter, J. D. (1985). *Microwave Impedance Measurements*. Peregrinus, London, I.E.E.E. Electrical Measurement Series.
Spitzer, L. (1967). *Physics of Fully Ionized Gases*, 2nd ed. Interscience, New York.
Stewart, R. D. and Robson, P. N. (1965). In: *Proceedings VIIth International Conference on Phenomena in Ionized Gases*. Beograd, Vol. 3, p. 109.
Strawitch, C., Yujiri, L., Richardson, R., and Lang, A. (1986). *Bull. Am. Phys. Soc.* **31**, 1543.
Suvorov, E. V., Fedyanin, O. I., Fraiman, A. A., and Kholńov, Yu.V. (1984). *Sov. J. Plasma Phys.* **10**, 73.
Tallents, G. J. (1984). *J. Phys.* **D**: **17**, 721.
Taylor, G., Efthimion, P. C., McCarthy, M., Fredd, E., Goldman, M. A., and Kaufman, D. (1985). *Rev. Sci. Instr.* **56**, 928.
Taylor, G., Efthimion, P. C., McCarthy, M. P., Fredd, E., and Cutler, R. C. (1986). *Rev. Sci. Instr.* **57**, 1974.

Tonks, L. (1931). *Phys. Rev.* **37**, 1458; *Phys. Rev.* **38**, 1219.
Tsukishima, T. (1979). In: *Diagnostics for Fusion Experiments* (E. Sindoni and C. Wharton, eds.). Pergamon Press, Oxford, New York, p. 255.
Uckan, T. (1981). *Rev. Sci. Instr.* **52**, 21.
Uckan, T. (1983). *Rev. Sci. Instr.* **54**, 1420.
Uckan, T. (1984). *Rev. Sci. Instr.* **55**, 1874.
Vadnjal, M. and Buffa, A. (1965). In: *Proceedings VIIth International Conference on Phenomena in Ionized Gases*. Beograd, Vol. 3, p. 105.
Vainstein, L. V. (1963). *Zhurn. Exp. Teor. Fiz.* **44**, 1050.
Walker, B., Baker, E. A., and Costley, A. E. (1981). *J. Phys.* **E: 14**, 832.
Wallington, J. R. and Beynon, J. D. (1969). *J. Plasma Phys.* **3**, 371.
Wetzer, J. M. (1983). *I.E.E.E. Trans. Plasma Sci.* **PS-11**, 72.
Wharton, C. B. (1957). UCRL Report 4836.
Wharton, C. B. (1965). Microwave techniques. In: *Plasma Diagnostic Techniques* (R. H. Huddlestone and S. L. Leonard, eds.). Academic Press, New York, p. 477.

6 Paramagnetic Resonance as a Plasma Discharge Diagnostic

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I. Introduction 313
II. Basic Theory 314
A. Atoms 315
B. Molecules 321
C. Selection Rules 327
D. Polyatomic Molecules 330
III. Application 332
A. Gas Phase Paramagnetic Resonance Apparatus 332
B. Qualitative and Quantitative Results 339
IV. Discussion 344
A. Species 344
B. Examples of Paramagnetic Resonance Diagnostics 345
C. Summation 346
V. Acknowledgments 346
References 347

I. Introduction

Plasma processing is now a widely used technology in the microelectronics industry. Fundamental to this technology are processes employing radio or microwave frequency discharges in a low-pressure gas, often containing several components. The discharge fragments and ionizes the gas, and the highly reactive species produced do the actual processing. Probably the most widespread application of plasma processing is in the etching of integrated circuits; however, the basic technology has been used for general

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cleaning of components, stripping of photoresists, and the deposition of various materials such as silicon.

What all these diverse processes have in common is that the chemical reactions that accomplish the desired objective are driven by free atoms, radicals, and ions created in the discharge. Because of the extreme reactivity of many of these chemical intermediates, their concentration is low, and they are difficult to detect. Add to this the fact that many of the reactions are complicated and delicately balanced, often leading to nonreproducible or unexpected yields, and one has a process where understanding is far from complete. Indeed, there is often a temptation to treat plasma processing as a "black box" to be "twiddled and tinkered with" if something goes wrong.

To gain a better understanding of the processes occurring in plasma reactors, it is necessary to be able to monitor the reactive chemical intermediates produced in the plasma. Towards that end, this chapter will review two closely-related, magnetic resonance spectroscopic techniques, electron paramagnetic resonance (EPR) and laser magnetic resonance (LMR).

The rationale for using EPR and LMR for plasma diagnostics is that they are well suited for detecting at low concentrations the atoms, ions, and radicals of interest. These species must be derived from the feedstock gases in the discharge, e.g., H_2, N_2, O_2, F_2, Cl_2, Freons^{TM}, etc., the semiconductor materials such as Si, Ge, As, Ga, etc. and the various exposed metal surfaces, e.g., Al, Cu, etc. As will be seen in the tables and text of this review, these are precisely the elements of which many of the radicals, ions, and atoms detected by magnetic resonance techniques are composed.

There have been, in the past few years, several reviews and summaries dwelling on various aspects of magnetic resonance. In particular, we mention works on EPR by Carrington, Levy, and Miller (1970), Brown (1972), and Westenberg (1973). Similar works on LMR have been written by Evenson and coworkers (1980 and 1981), Davies and Evenson (1975), McKellar (1981), Davies (1981), Thrush (1981), and Hills (1984). This chapter is devoted to the application of magnetic resonance to plasma processing problems. We recommend the above works and references therein for more detailed discussions of the spectroscopic techniques and underlying theory.

II. Basic Theory

Both EPR and LMR depend on the fact that most reactive species of interest to plasma processing possess a significant magnetic moment. This

moment arises from unpaired electron spins, unquenched electronic orbital angular momentum, or a combination thereof. The interaction of this atomic or molecular magnetic moment with an externally applied magnetic field fundamentally alters the quantum states of the atom or molecules.

In most forms of spectroscopy, the species of interest have well defined quantum states between which discrete quanta of radiation are absorbed or emitted. In these traditional forms of spectroscopy, the radiation frequency (and hence the quantum size) is experimentally varied, and the frequencies at which radiation is absorbed or emitted are determined.

As is explained in more detail in Section III, it is experimentally much more convenient and sensitive to use radiation of fixed frequency in EPR and LMR. Thus, the one external variable at the experimenter's control is the magnitude of the external magnetic field. By utilizing the interaction of the atom's or molecule's magnetic moment with the variable external magnetic field, it is possible to tune the atomic or molecular energy levels through resonance with the fixed microwave or laser frequency for EPR and LMR, respectively. In this case, the magnetic fields at which absorption (or in some cases, emission) of the fixed frequency radiation occur define the spectrum of the particular atom or molecule. This spectrum, in turn, enables the investigator to determine the presence or absence of a particular reactive species and to follow the concentration either as a function of spatial position or time.

To understand the appearance of the spectra of various species, e.g., F, Cl, O, or other atoms, or HCl^+, OH, ClO, or similar molecules requires some basic quantum mechanics, with emphasis on the interaction of atoms and molecules with magnetic fields. For simplicity, we shall first deal with atoms and then treat molecules.

A. Atoms

1. General Energy Structure and Zeeman Effect

To calculate the energy levels of an atom in a magnetic field, it is necessary to write the Hamiltonian, $\hat{H}^a$, for the atom plus field. It can be concisely written as

$$\hat{H}^a = \hat{H}^a_{zf} + \hat{H}^a_z, \tag{1}$$

where $\hat{H}^a_{zf}$ refers to that part of the atomic Hamiltonian appropriate for an isolated atom in zero external field while $\hat{H}^a_z$ describes the interaction of that atom with an external magnetic field, the so-called Zeeman effect.

To be more detailed concerning the form of $\hat{H}_{zf}^{a}$ and $\hat{H}_{z}^{a}$, it is useful to recall some of the properties of atomic angular momentum for a given electronic state of an atom. (Since all EPR and LMR transitions occur within a given electronic state, usually the ground state, it is only necessary to consider a single electronic state with well defined angular momenta.) There are basically two kinds of angular momentum that contribute significantly to the atom's magnetic moment, the electron spin angular momentum, **S**, and the electron spin orbital angular momentum, **L**. For light atoms, such as are of interest in plasma processing, Russell-Saunders coupling is an appropriate description of the interaction of these angular momenta. In Russell-Saunders coupling, a resultant angular momentum, **J**, is defined as the vector sum of **L** and **S**, i.e., $\mathbf{J} = \mathbf{L} + \mathbf{S}$.

The term symbols characterizing atomic electronic states are determined by the values of S, L, and J. Specifically, the term symbol is written ${}^{(2S+1)}L_J$, where for $L = 0, 1, 2$, etc. the letters S, P, D etc. are used. For example, the ground states of some of the common reactive atoms in plasmas are given in Table 1.

In examining Table 1, clearly some electronic states consist of only one energy level, e.g., H with ${}^{2}S_{1/2}$ and P with ${}^{4}S_{3/2}$, while others consist of more than one fine structure level, e.g., O with ${}^{3}P_0$, ${}^{3}P_1$, ${}^{3}P_2$ and Cl with ${}^{2}P_{1/2}$, ${}^{2}P_{3/2}$. These fine-structure levels correspond to the possible values of **J** allowed by the vector addition of **L** and **S**, i.e., $J = |L - S|$, $|L - S + 1|, \ldots |L + S|$. Referring back to our Hamiltonian operator in Eq. (1) that defines the overall energy of atoms, we see that the only part of $\hat{H}_{zf}^{a}$ that need concern us in an EPR or LMR experiment is that portion which determines the zero-field fine-structure splittings. Thus, in quantum mechanical language, $\hat{H}_{zf}^{a}$ can be treated as an entirely diagonal operator in our basis set of the zero-field electronic eigenfunctions for the atom. The numerical value of the diagonal elements or energies can be found in several tabulations of atomic energy levels (Moore, 1949; Baskin and Stoner, 1978).

The key term of Eq. (1) for EPR and LMR experiments is $\hat{H}_{z}^{a}$, describing the interaction of the external field with the atomic magnetic moment. From simple magnetic theory, the potential energy of a dipole in a field **H**, is

$$\hat{H}_{z}^{a} = -\mu \cdot \mathbf{H}, \tag{2}$$

where μ is the electronic dipole moment of the atom. It can be shown that the expectation value of the dipole moment is related to the total angular momentum by

$$\langle \mu \rangle = -g_J \beta \mathbf{J}. \tag{3}$$

Hence letting **H** define the z axis of the coordinate system we have

$$\langle \hat{H}_{z}^{a} \rangle = g_J \beta H M_J, \tag{4}$$

Table 1 Ground-State Term and Characteristics of Common Reactive Atoms

Atom	L	S	Fine Structure Term	g_J	Observed
H	0	1/2	$^2S_{1/2}$	2	EPR
C	1	1	$^3P_2, ^3P_1, ^3P_0$	3/2, 3/2, 0	LMR
N	0	3/2	$^4S_{3/2}$	2	EPR
O	1	1	$^3P_0, ^3P_1, ^3P_2$	0, 3/2, 3/2	EPR, LMR
F	1	1/2	$^2P_{1/2}, ^2P_{3/2}$	2/3, 4/3	EPR
Al	1	1/2	$^2P_{3/2}, ^2P_{1/2}$	4/3, 2/3	—
Si	1	1	$^3P_2, ^3P_1, ^3P_0$	0, 3/2, 3/2	LMR
P	0	3/2	$^4S_{3/2}$	2	EPR
S	1	1	$^3P_0, ^3P_1, ^3P_2$	0, 3/2, 3/2	EPR
Cl	1	1/2	$^2P_{1/2}, ^2P_{3/2}$	2/3, 4/3	EPR, LMR
Ga	1	1/2	$^2P_{3/2}, ^2P_{1/2}$	4/3, 1/2	—
Ge	1	1	$^3P_2, ^3P_1, ^3P_0$	3/2, 3/2, 0	—
As	0	3/2	$^4S_{3/2}$	2	EPR
Br	1	1/2	$^2P_{1/2}, ^2P_{3/2}$	2/3, 4/3	EPR
Sb	0	3/2	$^4S_{3/2}$	2	EPR
Se	1	1	$^3P_2, ^3P_1, ^3P_0$	3/2, 3/2, 0	EPR
I	1	1/2	$^2P_{3/2}, ^2P_{1/2}$	4/3, 2/3	EPR
Na	0	1/2	$^2S_{1/2}$	2	EPR
K	0	1/2	$^2S_{1/2}$	2	EPR
Rb	0	1/2	$^2S_{1/2}$	2	EPR
Cs	0	1/2	$^2S_{1/2}$	2	EPR

where β is the Bohr magneton and M_J denotes the allowed values of the projection of **J** along $z(M_J = |J|, |J-1| \ldots -|J|)$.

Simple atomic theory gives an expression for g_J,

$$g_J = \frac{g_L}{2J(J+1)}[L(L+1) - S(S+1) + J(J+1)] + \frac{g_S}{2J(J+1)}[S(S+1) - L(L+1) + J(J+1)]. \quad (5)$$

g_L is the electronic orbital g factor, and g_S is the electronic spin g factor.

For precise numerical work,

$$g_L = 1 - \frac{m}{M} \tag{6a}$$

$$g_S = 2.00232, \tag{6b}$$

where m/M is the ratio of the electron's mass to that of the nucleus. For qualitative work, it is a very good approximation to take $g_L = 1$ and $g_S = 2$. Using these latter values of g_L and g_S and Eq. (5), we have calculated and listed the g_J factors for all the atoms in Table 1.

The dominant effect of $\hat{H}_Z^a$, then, is to raise the $(2J + 1)$-degeneracy of the fine structure states. Thus, in this approximation the Zeeman energy is linear in H and M_J so that as the field strength increases, the levels become more separated.

The fundamental difference between the EPR and LMR experiment is the frequency of the applied radiation field that drives transitions within the atom's ground electronic state. For EPR the microwave frequency is typically ≈ 10 GHz. If we refer to Eq. (4), it is clear that the energy separation between adjacent M_J states is $g_J\beta H$. From Table 1, g_J is of the order of unity. The Bohr magneton, β, has a numerical value in these units of 13.996 GHz/Tesla. Hence it is clear that the resonance condition for microwave energy absorption is satisfied for applied magnetic fields of the order of 1 Tesla, a typical laboratory value. Thus the EPR transitions take place within a fine-structure J level, between the adjacent M_J levels whose degeneracy has been raised by the applied field.

LMR transitions are of a rather different nature. The IR and far-IR lasers typically used in LMR experiments have frequencies in the range 1–100 THz (1 THz = 10^3 GHz). Thus to separate the adjacent M_J levels of a given J state sufficiently to be in resonance with one of these lasers would require $H \geq 100$ Tesla, an impossibly large value. Fortunately the fine-structure splittings between adjacent J states for the atoms listed in Table 1 typically are of the order of 1–100 THz. Since both the laser and fine-structure frequencies are very precisely defined, an accidental resonance between them is very unlikely. However, if the laser frequency for a particular LMR experiment is sagely chosen to match approximately the fine-structure interval, the Zeeman tuning provided by $\hat{H}_Z^a$ (of order 10 GHz) of the M_J states of each fine-structure level can achieve the precise resonance conditions.

2. *A Specific Example—The Oxygen Atom*

To see more clearly the theoretical details of an EPR or LMR experiment, it is useful to choose from Table 1 one atomic species and examine its spectrum. Our choice is the O atom which, as can be seen from the Table 1,

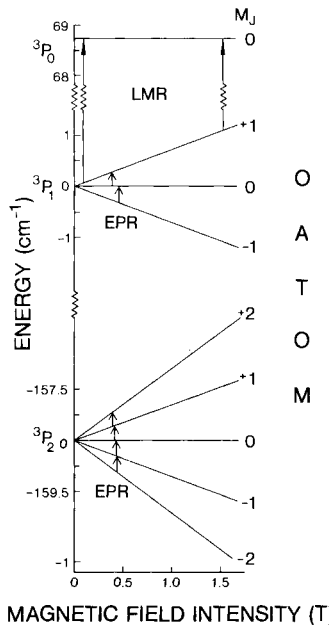


FIGURE 1. Energy level diagram for the O atom showing the fine structure states, $^3P_2$, $^3P_1$, $^3P_0$. EPR transitions are shown within the $^3P_2$ and $^3P_1$ states and LMR transitions for two different laser frequencies (see text) are shown between the $^3P_1$ and $^3P_0$ levels.

has three fine-structure states $^3P_0$, $^3P_1$, and $^3P_2$, the latter being lowest in energy. Figure 1 shows an energy level diagram, determined by the Hamiltonian, Eq. (1), as a function of magnetic field EPR transitions are indicated taking place within both the $^3P_2$ and $^3P_1$ levels. To the approximation to which Figure 1 is drawn, one would expect the four indicated transitions within the $^3P_2$ state to be superimposed, as well as the two transitions within the $^3P_1$ level. Indeed, since from Table 1, the g_J factors for $^3P_2$ and $^3P_1$ oxygen happen to be the same, one would expect all six transitions to be superimposed. ($^3P_0$ has only one M_J level whose energy is nearly independent of field; therefore, it gives rise to no EPR spectrum.)

Figure 2 shows the actual EPR spectrum. One sees that all transitions shown in Figure 1 are actually at least partially resolved. The four closely spaced transitions in the center of Figure 2 occur within the $^3P_2$ fine-structure state, while the pair of outermost transitions occurs within the $^3P_1$ level. The reason the transitions are distinct resides in a facet of $\hat{H}_z^a$ that we have not yet described. Until now we approximated the effect of $\hat{H}_z^a$ on the energy levels by its diagonal matrix elements, that is, by only its expectation value. $\hat{H}_z^a$ also has non-zero matrix elements between the different fine-

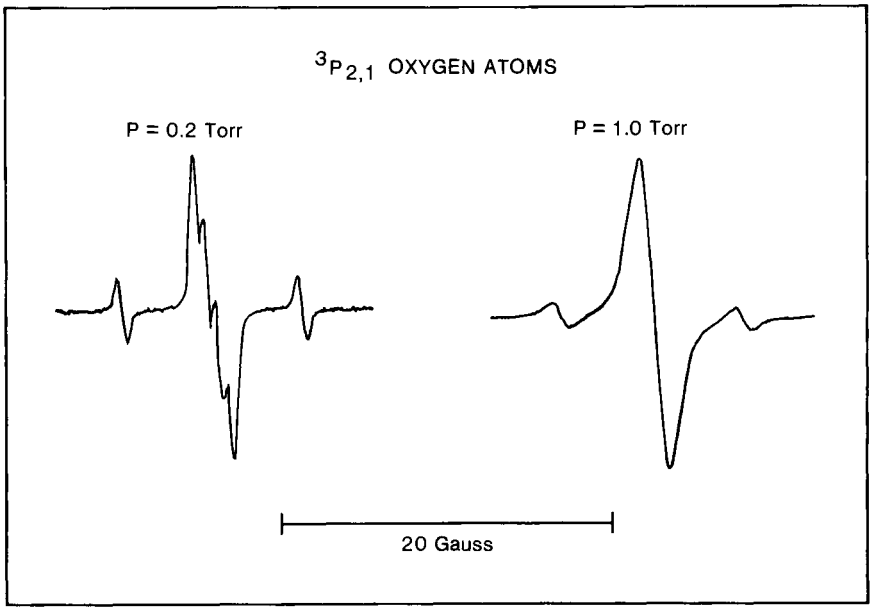


FIGURE 2. EPR spectral trace of $^3P_2$ and $^3P_1$ O atoms. The spectrum on the right is taken at higher pressure, and only the two outer $^3P_1$ lines are resolved from the quartet of $^3P_2$ transitions. In the lower pressure spectrum at the left, at least partial resolution of each of the four distinct $^3P_2$ EPR transitions is achieved. (20 Gauss = 2 mTesla).

structure J states. By using second-order perturbation theory (or other techniques) it can be easily found that the effects of these terms on the O-atom eigenenergies (Radford and Hughes, 1959), while not large, are also not negligible on the scale of the EPR experiment. This nonlinear Zeeman effect shifts slightly the M_J levels. These shifts cause the precise magnetic fields required for resonance to be slightly different for each of the six EPR transitions indicated in Figure 1.

LMR transitions are also indicated in Figure 1 between the $^3P_1$ ($M_J = 1$) level and the $^3P_0$ level. For the LMR experiments (Davies et al., 1978b), the CH_3OH and CH_3NH_2 far IR lasers have been employed. The frequencies of these lasers are 2.0581418 and 2.0277526 THz, respectively. These frequencies are very slightly smaller than the $^3P_0 - {}^3P_1$ zero-field separation in the ^{16}O atom of 2.0600688 THz. Thus one can tune this transition into resonance with the applied field. The actual transitions in the LMR spectrum are shown in Figure 3.

Before concluding this section, it is worth noting that up to this point we have not mentioned the effect of nuclear spin on either the LMR or EPR

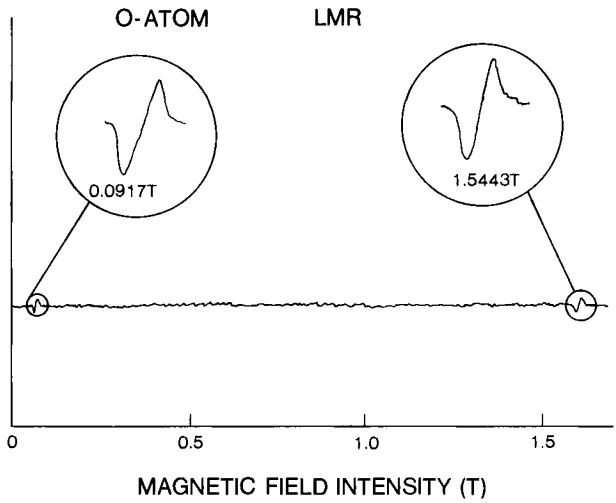


FIGURE 3. LMR of the O atom between the $^3P_1$ and $^3P_0$ fine structure states. The low and high field lines are obtained with the CH_3NH_2 and CH_3OH lasers, respectively.

spectra of atoms. Since the O atom has a nucleus spin of zero, there is no effect upon its spectrum; for non-zero-spin nuclei, however, the effect can be significant, although fairly simply accounted for.

The coupling of the nuclear spin to the angular momentum **J** is usually very weak compared to its coupling to the external field **H**. Hence the primary effect of a non-zero nuclei spin is to split each line of the LMR or EPR spectrum into a multiplet, each member corresponding to one of the allowed orientations, M_I, of the nuclear spin angular momentum, **I**, along the field direction. There will be $2I + 1$ nearly equally spaced lines, the separation being referred to as the hyperfine interaction. The magnitude of the hyperfine splitting will usually be a few milliteslas or less for atoms of interest in plasma processing.

B. Molecules

1. Linear Molecules

The fundamental principles governing LMR and EPR spectroscopy change little in going from atoms to diatomic molecules. One can still write the basic Hamiltonian for the system as in Eq. (1),

$$\hat{H}^{\mathrm{m}} = \hat{H}^{\mathrm{m}}_{\mathrm{zf}} + \hat{H}^{\mathrm{m}}_{\mathrm{z}}, \tag{7}$$

where the m's indicate that we will now apply this operator to molecules; specifically, in this section, we shall consider linear molecules.

However, even linear molecules differ in a number of ways from atoms. For example, the electronic states are differently defined. As in Table 1, we can still deduce the electronic spin S, but we can no longer define L because of the reduction from spherical to cylindrical symmetry. Instead, we must be content with defining the projection, denoted Λ of $\mathbf{L}$ along the internuclear axis.

If $\Lambda = 0$ then the state is referred to as a Σ state and its behavior with respect to magnetic resonance is very similar to that of the polyatomic molecules considered in the next section, so we postpone detailed discussion of Σ states until then. If $\Lambda \neq 0$ then linear molecules of interest to plasma processing generally obey a coupling scheme (Herzberg, 1950) known as Hund's case (a). The electronic states of case (a) molecules can be denoted as ${}^{(2S+1)}\Lambda_{\Omega}$, where, in analogy with atoms, the numerical values of Λ are replaced by letters. However, for molecules, the letters are Greek; for $\Lambda = 1, 2, \ldots$ the states are denoted, respectively, by $\Pi, \Delta, \ldots$. The Ω denotes the fine-structure states with the definition that $\Omega = \Lambda + \Sigma$, with Σ being the projection of $\mathbf{S}$ along the internuclear axis. In Table 2 we list the Λ and $\mathbf{S}$ values and the term symbols for the ground states of a number of linear molecules.

The zero-field part, $\hat{H}_{zf}^{m}$ of the Hamiltonian is much more complex for linear molecules than it is for atoms. In the general case

$$\hat{H}_{zf}^{m} = \hat{H}_{v} + \hat{H}_{R} + \hat{H}_{fs} + \hat{H}_{LD} + \hat{H}_{CD} + \hat{H}_{hfs}. \tag{8}$$

The term $\hat{H}_{v}$ determines the vibration levels of the molecule; $\hat{H}_{R}$ represents the effect of molecular rotation; $\hat{H}_{fs}$ denotes the interactions, principally spin-orbit coupling, causing the molecular fine-structure splitting, and $\hat{H}_{CD}$ represents centrifugal distortion effects upon any of the above interactions. $\hat{H}_{LD}$ splits apart levels with the same value of Λ and Ω, but the opposite sign; this is the so-called Λ-doubling interaction. $\hat{H}_{hfs}$ denotes hyperfine interactions; although these interactions are more complex than in atoms, it is still true that the nuclear spin is usually decoupled by the magnetic field. The net effect of $\hat{H}_{hfs}$ is then simply to split each magnetic resonance transition into a multiplet with the various components labeled by the possible M_I values.

The details of the various contributions to $\hat{H}_{zf}^{m}$ are well beyond the scope of this review. For most plasma processing applications, these details and precision data will be available in the literature. It should be noted that generally these details for zero-field energy levels will vary considerably from molecule to molecule, but some qualitative observations can be made

Table 2 Linear Molecules Observed by EPR and LMR Techniques

Molecule	S	Λ	Fine Structure Terms	Observations
CH	1/2	1	$^2\Pi_{1/2}, ^2\Pi_{3/2}$	LMR
SiH	1/2	1	$^2\Pi_{1/2}, ^2\Pi_{3/2}$	LMR
GeH	1/2	1	$^2\Pi_{1/2}, ^2\Pi_{3/2}$	LMR
OH	1/2	1	$^2\Pi_{1/2}, ^2\Pi_{3/2}$	EPR, LMR
SH	1/2	1	$^2\Pi_{1/2}, ^2\Pi_{3/2}$	EPR, LMR
NH	1	0	$^3\Sigma$	LMR
AsH	1	0	$^3\Sigma$	LMR
SeH	1/2	1	$^2\Pi_{1/2}, ^2\Pi_{3/2}$	EPR, LMR
TeH	1/2	1	$^2\Pi_{1/2}, ^2\Pi_{3/2}$	EPR
HCl^+	1/2	1	$^2\Pi_{1/2}, ^2\Pi_{3/2}$	LMR
HBr^+	1/2	1	$^2\Pi_{1/2}, ^2\Pi_{3/2}$	LMR
CF	1/2	1	$^2\Pi_{1/2}, ^2\Pi_{3/2}$	EPR, LMR
NF	1	0	$^3\Sigma$	LMR
SF	1/2	1	$^2\Pi_{1/2}, ^2\Pi_{3/2}$	EPR
SeF	1/2	1	$^2\Pi_{1/2}, ^2\Pi_{3/2}$	EPR
SO	1	0	$^3\Sigma$	EPR, LMR
SeO	1	0	$^3\Sigma$	EPR
FO	1/2	1	$^3\Sigma$	LMR
ClO	1/2	1	$^2\Pi_{1/2}, ^2\Pi_{3/2}$	EPR, LMR
BrO	1/2	1	$^2\Pi_{1/2}, ^2\Pi_{3/2}$	EPR, LMR
IO	1/2	1	$^2\Pi_{1/2}, ^2\Pi_{3/2}$	EPR
NO	1/2	1	$^2\Pi_{1/2}, ^2\Pi_{3/2}$	EPR, LMR
NS	1/2	1	$^2\Pi_{1/2}, ^2\Pi_{3/2}$	EPR
NSe	1/2	1	$^2\Pi_{1/2}, ^2\Pi_{3/2}$	LMR
CN	1/2	0	$^2\Sigma$	LMR
AsO	1/2	1	$^2\Pi_{1/2}, ^2\Pi_{3/2}$	LMR
PO	1/2	1	$^2\Pi_{1/2}, ^2\Pi_{3/2}$	LMR
NCO	1/2	1	$^2\Pi_{1/2}, ^2\Pi_{3/2}$	EPR, LMR
O_2	1	0	$^3\Sigma$	EPR, LMR

that will allow for a physical understanding of the EPR and LMR spectra of most linear molecules.

The magnitude of the splittings engendered by the various terms of $\hat{H}_{zf}^{m}$ is roughly $\hat{H}_{v} \geq \hat{H}_{fs} \geq \hat{H}_{R}$, with $\hat{H}_{LD}$, $\hat{H}_{CD}$, and $\hat{H}_{hfs}$ usually representing relatively small corrections to the three leading terms. It may be remembered that, for atoms, LMR transitions usually take place between different fine-structure levels tuned slightly by the Zeeman interaction, $\hat{H}_{z}^{a}$. For molecules, once again, LMR can take place between different fine-structure states, but now because of the comparability of $\hat{H}_{v}$, $\hat{H}_{fs}$, and $\hat{H}_{R}$, it is just as likely that they will take place between different vibrational or rotational levels. Transitions which involve a change in two or more degrees of freedom, e.g., fine-structure combined with rotation, are also quite likely.

As we shall see below, the Zeeman interaction is comparable in atoms and molecules. Thus, EPR transitions normally occur between levels arising from Zeeman-split, degenerate states. However, it is possible in molecules to have EPR transitions in which transitions between nondegenerate zero-field states, e.g., Λ-doublets and/or rotational levels, are tuned to resonance by the Zeeman interaction.

In analogy with Eqs. (2–4), we write

$$\hat{H}_{z}^{m} = g_{J} B \mathbf{J} \cdot \mathbf{H} \tag{9}$$

$$\langle \hat{H}_{z}^{m} \rangle = g_{J} \beta H M_{J}, \tag{10}$$

where **J** is the molecule's total angular momentum exclusive of nuclear spin. The definition (Herzberg, 1950) of g_J in Hund's case (a) is

$$g_{J} = \frac{(g_{S}\Lambda + g_{L}\Sigma)(\Lambda + \Sigma)}{J(J+1)}. \tag{11}$$

The electronic g-factors, g_L and g_S, are again defined by Eqs. (6a and 6b) and can usually be approximated adequately by 1 and 2, respectively. For each of the linear molecules listed in Table 2, Λ is given and Σ equals $|S|, |S-1| \ldots |-S|$. The precise value of Σ can be determined by the relationship $\Sigma = \Omega - \Lambda$, with Ω being defined in Table 2 by the fine-structure state.

Eqs. (10) and (11) define the expectation value of $\hat{H}_{z}^{m}$, but, as with atoms, there are also off-diagonal terms of $\hat{H}_{z}^{m}$ between different fine-structure and rotational states, the latter being particularly important owing to the relative nearness of adjacent rotational levels. As in atoms, these nonliner Zeeman interactions resolve what would otherwise be a superposition of transitions within a J state between adjacent M_J sublevels.

2. *A Specific Example—OH*

In analogy to the atomic case, it is probably easiest to apply the general theory to explain EPR and LMR spectra by considering a specific molecule. We choose OH for several reasons: 1) both EPR (Radford, 1962) and LMR (Brown et al., 1981 and 1982; Davies et al., 1978a) spectra are well known and understood, 2) its spectra are complex enough to illustrate most of the principles encountered in EPR and LMR experiments, and 3) it is ubiquitously present in plasma processes.

Figure 4 shows an energy level diagram of OH as a function of magnetic field. It is important to note a critical difference between this diagram and

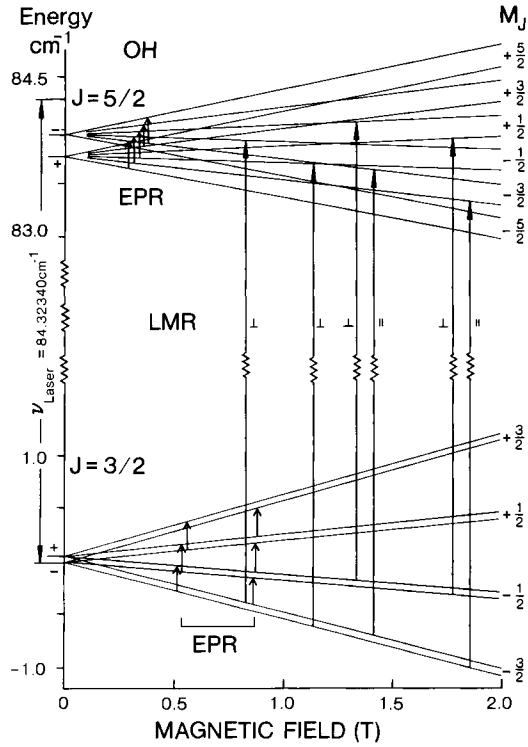


FIGURE 4. Energy level diagram of $^2\Pi_{3/2}$ OH, $v = 0$, showing the $J = 3/2$ and $5/2$ rotational levels only. EPR transitions are indicated within each of the $J = 3/2$ and $J = 5/2$ levels, and LMR transitions are indicated between the $J = 3/2$ and $J = 5/2$ levels, with the $\perp$, $\parallel$ symbols indicating relative orientation of the laser electric field and the magnetic field.

Figure 1 for the O atom. Figure 1 shows the complete energy level structure for the O ground electronic state; Figure 4 illustrates only a fraction of the energy level structure for ground state OH. Pictured are two rotational levels ($J = 3/2,\ 5/2$) of the $^2\Pi_{3/2}$ OH fine-structure component in its lowest vibrational level. Not pictured are the higher rotational levels of the $^2\Pi_{3/2}$ state, nor any of the $^2\Pi_{1/2}$ state, nor the higher vibrational levels. In principle, both EPR and LMR transitions can take place (and some have been observed) between the non-illustrated energy levels. However, for the purpose of the present discussion, the portion of the total picture illustrated by Figure 4 should suffice.

Let us first discuss the zero field structure, caused by $\hat{H}_{zf}^{m}$, depicted for OH in Figure 4. The dominant splitting is between the $J = 3/2$ and $J = 5/2$ rotational energy levels and is roughly 2.4 THz. At a smaller scale, each of the rotational levels can be seen to be split into two components labeled + and −. This is the Λ-doubling induced by the term $\hat{H}_{LD}$ in Eq. (8).

If a magnetic field is applied, the $(2J + 1)$ degeneracy of each ± component of each J state is raised. Using Eq. (11), it can be seen that the g_J factors are approximately 4/5 and 12/35 for $J = 3/2$ and $J = 5/2$, respectively. Equating the difference in Zeeman energies, $g_J\beta H$, between adjacent M_J levels to a typically EPR microwave frequency of $\nu_\mu = 9$ GHz yields a resonance field of $H \approx 0.8$ Tesla for $J = 3/2$. Taking into account nonlinear Zeeman interactions shifts the field slightly but, most importantly, resolves the three possible transitions between adjacent M_J levels. In actual fact, Figure 4 shows a total of six EPR transitions, three below and three above the expected field. Careful examination of Figure 4 shows the predicted transitions actually occur between the + and − components of the J state. (This selection rule will be more fully explained in Section II.4.) Thus, a more precise value of the resonance field would have been obtained by satisfying the equality

$$\nu_\mu \pm \nu_{LD} = g_J\beta H. \tag{12}$$

The ± signs yield two solutions for each pair of adjacent M_J levels, while ν_{LD} is the Λ-doublet splitting between paired ± components, i.e., $\nu_{LD} \approx \langle \hat{H}_{LD} \rangle$. Similar arguments yield the five EPR transitions indicated in Figure 4 for $J = 5/2$ when one employs the minus sign in Eq. (12). The value of H required for resonance with the plus sign is well beyond the diagram.

Figure 5 shows a portion of the EPR spectrum of the OH radical. Each of the transitions is actually a doublet, owing to the nuclear hyperfine interaction, $\hat{H}_{hfs}$. Since H has a spin of 1/2, the two components can be ascribed to $M_J = \pm 1/2$.

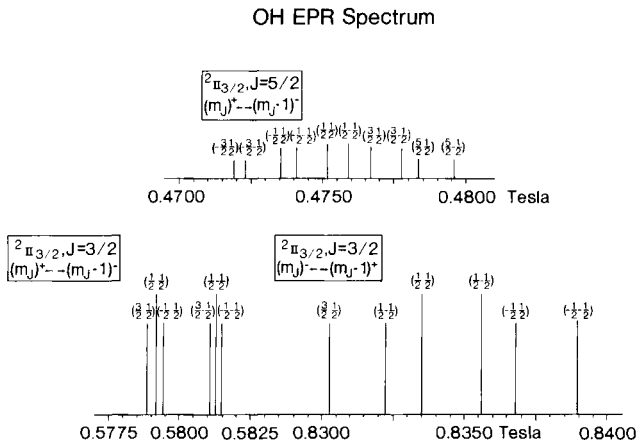


FIGURE 5. Stick diagram of observed EPR transitions below 1.0 Tesla for $^2\Pi_{3/2}$ OH.

Six LMR transitions are indicated in Figure 4 and shown in the spectrum in Figure 6. The laser employed in the experiment is the H_2O-vapor laser operating at a frequency of 2.512204 THz, which is slightly more than the zero field separation of the $J = 3/2$ and $J = 5/2$ levels. However, exact resonance can be achieved at the indicated magnetic fields because of the different Zeeman tuning rates of the various M_J levels in the two rotational states.

C. Selection Rules

The fundamentals of the magnetic resonance of atoms and linear molecules have now been discussed. The next logical step in the progression is to discuss the magnetic resonance of nonlinear polyatomic molecules. However, we shall see that this discussion is greatly simplified if we first consider a topic that we have touched on only lightly heretofore.

Most forms of spectroscopy are governed by selection rules that dictate between which pairs of levels transitions may occur. Magnetic resonance experiments, EPR and LMR, are no exception to this rule. Elementary quantum mechanics tells (Herzberg, 1950) us that transitions are induced by interactions of atomic or molecular multipole moments with a time-varying external electric or magnetic field. In the EPR case this field is provided by the microwaves; for LMR it is the lasers.

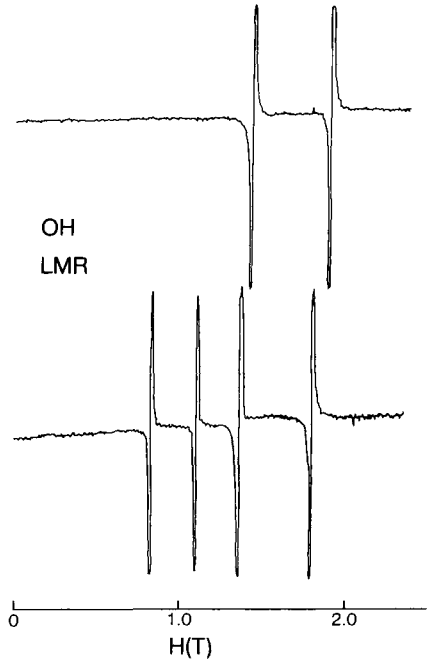


FIGURE 6. LMR transitions for OH using the H_2O-vapor laser. For the top trace, the laser electric field is directed parallel to the magnetic field, while for the bottom trace, it is perpendicular.

For magnetic resonance we need not go beyond the dipole approximation in the multipole treatment. Thus we need only consider an oscillating electric field interacting with an electric dipole and an oscillating magnetic field interacting with a magnetic dipole moment. It is well known (Herzberg, 1950) that in first-order time-dependent perturbation theory, the electric dipole interaction provides a probability factor for a transition that is $\approx 10^4$ times greater than that of the magnetic interaction.

In atoms each electronic state (configuration of electrons) has associated a wavefunction by which it is described. If the value of the wavefunction changes sign upon inversion (replacement of the coordinates x, y, and z by $-x$, $-y$, and $-z$), the wavefunction, and hence the given electronic state, is said to have odd parity. If it remains unchanged, the parity is even. Likewise, the interactions that transform one state into another ("transitions") have a parity as well, with the electric and magnetic dipole interactions being odd and even, respectively. Essentially, for a transition to occur, the product of the parities of the two states with the parity of the

interaction must then be even. Since all magnetic resonance transitions take place within a single electronic state, the probability for electric dipole transitions vanishes. On the other hand, magnetic dipole transitions do not vanish within an atomic electronic state. These considerations do not tell the entire story, however, and there are further selection rules that govern what levels can be connected by magnetic dipole transitions.

A general result (Radford and Hughes, 1959), called the Wigner-Eckart theorem, rests upon the conservation of angular momentum. The Wigner-Eckart theorem allows the calculation of transition moments (strengths) to be separated into two portions: a constant that depends only on the physical properties of the state; and a second that depends on the symmetry of the wavefunction. This symmetry or "spatial" dependence for electric or magnetic dipole transition moments is proportional to $C(J, 1, J'; M_J, q, M_J')$, the so-called Clebsch-Gordan coefficient, well known in angular momentum theory (Rose, 1957). In this expression, J and J' are the total angular momentum quantum numbers for the two states, and M_J and M_J' are the projections of the total angular momentum vectors on a space-fixed axis. An explicit formula (Rose, 1957) exists for C, and for common values of the J's numerical tabulations have been published (Condon and Shortley, 1963; Rose, 1957). However, here we need only concern ourselves with the conditions that force C to vanish, because the only transitions observable are those for which C does not vanish.

The two basic rules for the nonvanishing of C are that $\Delta(J, 1, J')$ and $M_J + q = M_J'$. The first result, the triangle rule, requires the existence of a triangle with sides of length, J, J', and 1 for C to be non-zero. For the present case this amounts to

$$J - J' = \Delta J = 0, \pm 1. \tag{13}$$

The values of q are determined by the orientation of the laser or microwave field (in this case magnetic) with respect to the H field defining the z axis. The component parallel to the axis has $q = 0$, while perpendicular components have $q = \pm 1$. Thus

$$\Delta M_J = 0 \quad \| \quad \text{transitions} \tag{14a}$$

$$\Delta M_J = \pm 1 \quad \perp \quad \text{transitions.} \tag{14b}$$

Eqs. (13) and (14) constitute the useful selection rules for magnetic dipole transitions in atoms for both EPR and LMR.

For molecules, things are different; all the levels of a given electronic state do not have the same parity, so electric dipole transitions are allowed. Indeed, in the vast majority of magnetic resonance experiments on molecules, only electric dipole transitions in molecules are not too weak to be observed. The reason for this is not that molecular magnetic moments are,

in general, smaller than for atoms. Rather, the chief reason is the large number of discrete quantum levels in a given electronic state of a molecule. Through the partition factor this abundance of states dilutes the population in any distinct level by a factor of $10 - 10^3$ compared to atoms, making magnetic dipole transitions, in all but a few cases, too weak to observe in molecules.

While all levels of a molecular electronic state do not have the same parity, the individual levels do still have parity. In the case of non-Σ linear molecules, this parity can be uniquely correlated with Λ-doublet levels and is the basis for marking these levels + and − in Figure 4, the energy level diagram for OH. The electronic dipole transitions are allowed only between levels of opposite parity ($+ \leftrightarrow -$) and Figure 4 shows that the indicated EPR and LMR transitions obey this rule.

The remaining general selection rule is based upon the Wigner-Eckart theorem and is the same for both electric and magnetic dipole transitions in both atoms and molecules. These selection rules on J and M_J are embodied in Eqs. (13–14).

D. Polyatomic Molecules

For polyatomic molecules, and linear molecules in Σ states, the general form of the Hamiltonian, Eq. (7), is still valid. For polyatomics the details of several of the terms of $\hat{H}_{zf}$ are more involved. However, of particular concern is the fact that now $\Lambda = 0$ and hence $\Omega = \Sigma$. This has considerable consequence for the term $\hat{H}_{fs}$ in the Hamiltonian; it reduces its magnitude so that usually $\hat{H}_{fs} < \hat{H}_R$. The fact is that $\Lambda = 0$ has eliminated spin-orbit coupling within the state and leaves in $\hat{H}_{fs}$ only the much weaker spin-rotation coupling. Physically this means that the electronic spin is only weakly coupled to the molecule and can, in many instances, like the nuclear spin, be uncoupled by the magnetic field. This phenomenon is called spin-uncoupling. Since it is not necessarily complete at all the fields in the magnetic resonance experiment, the resulting energy level patterns are very complicated and unique to each molecule.

A further complication arises from the fact that the electron spin carries the molecular magnetic moment and hence its tunability. However, the electric dipole moment is carried by the molecular framework. The physical result is that levels that can be magnetically tuned with respect to one another generally lack electric dipole transition probability between them.

An example is the energy level diagram shown in Figure 7. The depicted rotational level is split in zero-field into two components, F_1 and F_2, by the fine-structure or spin-rotation interaction. It is easy to see that at high

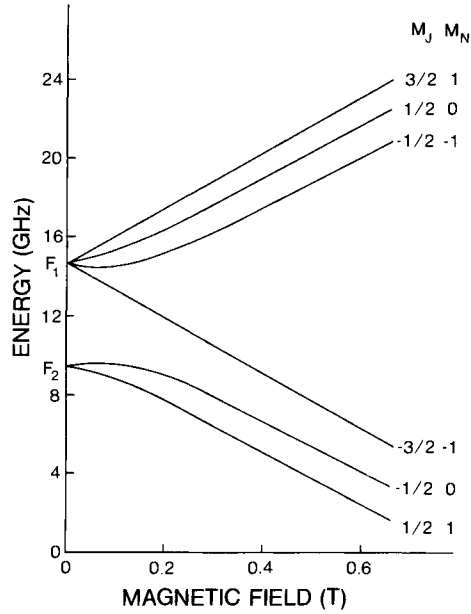


FIGURE 7. Energy level diagram depicting a pair of fine structure levels, F_1 and F_2, of a typical polyatomic radical in a doublet state. The magnetic field is seen to decouple the spin so that at moderate field, the upper levels can be described as $M_S = 1/2$ ($M_S = M_J - M_N$), while the lower levels have $M_S = -1/2$.

magnetic fields each rotational level splits into two sets of levels tuning essentially identically. This pair can be labeled by the quantum number $M_s = \pm 1/2$, and the subcomponents of each set can be labeled by the projection, M_N, of the rotational angular momentum, **N**. In the limit each M_N component with a given M_S value will tune essentially identically as a function of field. In that limit also, electric dipole transitions are rigorously forbidden between states differing in M_S. In this limit magnetic resonance experiments become impossible.

Depending upon the molecule, there is a larger or smaller region of applied magnetic field values wherein decoupling is not complete. For these fields, electric dipole transitions are tunable and magnetic resonance spectra can be observed.

LMR has a potential advantage over EPR on molecules of this type. If a laser can be found whose frequency is very close to a molecule's zero-field separation, then the small magnetic field required to tune to precise resonance will probably not induce complete uncoupling.

EPR transitions typically occur between originally degenerate levels, so the fields required to split this degeneracy by the magnitude of even a

microwave quantum may be too great to avoid spin uncoupling. Indeed, most successful EPR experiments on Σ or polyatomic molecules have been performed on triplet states where the spin-rotation coupling is usually much larger than for doublet states. For triplet Σ states, it is often possible that the fine structure level separations are comparable to a microwave quantum so that only a small magnetic field is required to tune to exact resonance.

III. Application

In this section we will consider the apparatus employed in the observation of paramagnetic resonance spectra. This will include not only the spectrometer system, but also the "plumbing" necessary for transporting the reactive species from the plasma discharge to the detection region in the spectrometer. We will also deal with both the qualitative and quantitative aspects of the observed signals, making use of the theory discussed above wherever necessary. Accurate interpretation of the signals observed in these instruments is likely the most difficult aspect of using the technique for plasma diagnostics. Part of this difficulty arises because the method is of a "downstream" nature, i.e., the species being detected are temporally and spatially removed from the plasma discharge where they were generated. As a result, the interpretation can be somewhat involved. The first step is to determine reliably the number and type of species present in the spectrometer. Secondly, these quantities have to be correlated with the number and type of species in the discharge itself so that the information obtained "downstream" reveals details of the "upstream" processes. Whether or not this second step is necessary will depend in large part on the particular nature of the experiment, i.e., on whether it is a downstream or in situ plasma process.

The emphasis throughout the remainder of this chapter will be on practical application of the technique to plasma diagnostics. As such, the purely spectroscopic aspects will be minimized except as they are necessary for the interpretations. While the spectroscopic aspects are never completely separable from the diagnostic aspects, our focus here will be primarily on what species are present and in what quantities. Section II provides the details relating the species' quantum mechanical structure and the observed magnetic resonance spectra.

A. Gas Phase Paramagnetic Resonance Apparatus

Figure 8 is a simple block diagram of the components found in typical, gas phase EPR and LMR spectrometers. As outlined above, the systems func-

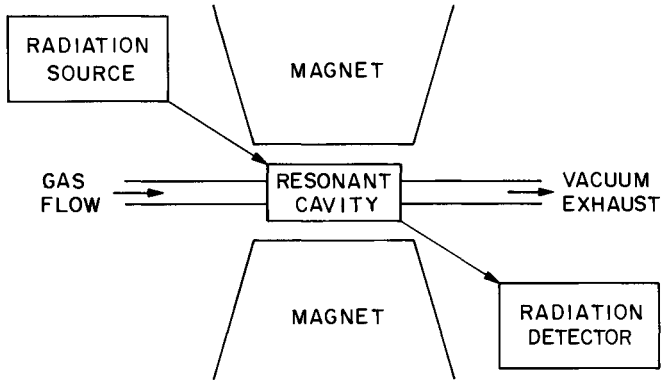


FIGURE 8. Block diagram of gas phase paramagnetic resonance experiment. The radiation source can be microwave (EPR) or (F)IR laser (LMR).

tion by subjecting the species of interest to the radiation produced by the microwave (EPR) or (F)IR laser (LMR) sources. Most often the radicals are transported, using a vacuum flow system, into a resonant cavity that is placed between the poles of a large magnet. As the magnetic field is scanned, the separations of the Zeeman sublevels of the species become coincident with the fixed energy of the radiation source, and its energy is absorbed. The amount of the absorption and the magnetic field at which it occurs can, respectively, be related to the number and type of species present. This, of course, is exactly the purpose of such a diagnostic tool.

In order to consider the details of the processes, we will examine a representative system of each type. There are numerous configurations and variations possible for each method so that we will necessarily restrict our consideration to one of each. Interested readers can consult the voluminous literature for other examples (Carrington et al., 1970; Brown, 1972; Westenberg, 1973; Evenson, 1981; Evenson et al., 1980; Davies and Evenson, 1975; McKellar, 1981; Davies, 1981; Thrush, 1981; Hills, 1984; Poole, 1983).

1. EPR Systems

a. Description of the Spectrometer. Figure 9 is a more detailed block diagram of one type of gas phase EPR system. It operates in the so-called *X* band (8 to 12 GHz) and has been used to observe atomic and molecular species. Its frequency range can extend to 35 GHz or higher, but the *X*

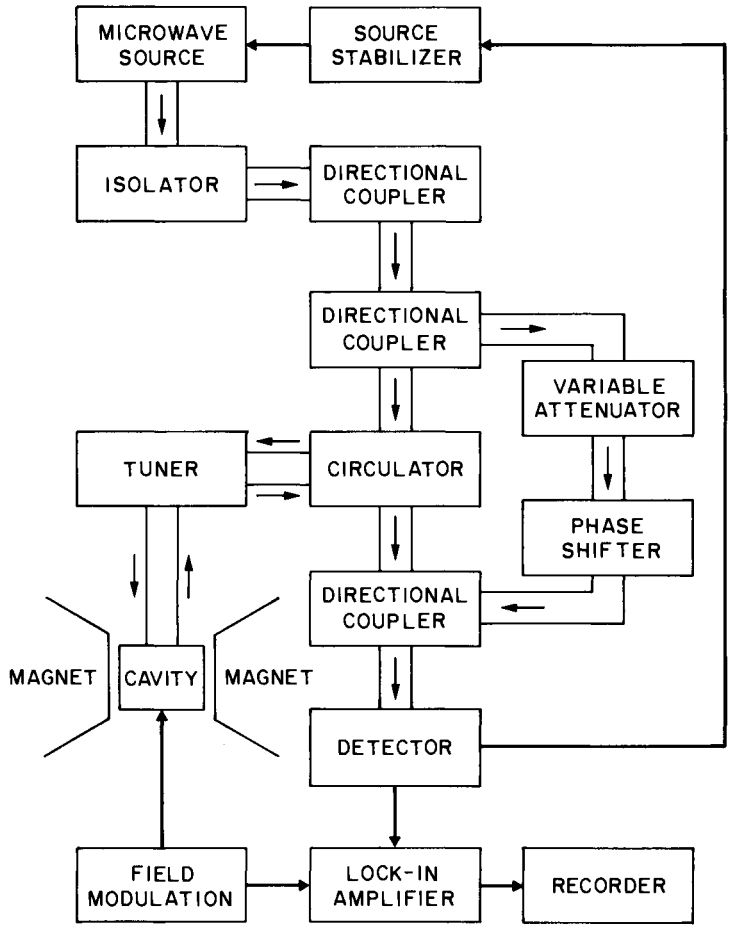


FIGURE 9. Block diagram of gas phase EPR system used for detecting atoms and molecules. Microwave radiation is transmitted in the directions of the arrows. The flow system is omitted.

band is the most common. The typical microwave source is a klystron, producing CW powers of up to several hundred milliwatts. Other sources can be and have been used, among them are backward wave oscillators (BWO's), magnetrons, and various solid state devices (Poole, 1983).

The microwave radiation, which is frequency stabilized by "locking" to the detection cavity resonance, is isolated from the source and routed through several passive circuit elements on its way to the resonant cavity.

The first directional coupler samples a small fraction of the radiation in order to measure its frequency. The second and third ones form part of the bypass around the cavity. As shown by the arrows, the circulator directs the source microwaves into the cavity and those reflected into the detector.

The cylindrical, resonant cavity, having a fairly high "Q" or quality factor, is employed here owing to its enhancement of the microwave energy density and thereby the signal intensity (Townes and Schawlow, 1975). Cavities can usually support several modes, but only the TE_{011} is employed here, since this mode allows the Q to remain high while a dielectric flow tube is present along the principal cavity axis. This mode also has the main component of its oscillating magnetic field present along this same axis (Poole, 1983), an important point when it is magnetic-dipole-allowed transitions that are being examined. The axis itself is oriented at right angles to the magnetic field of the large magnet.

Signal modulation is employed to increase the sensitivity of the system, and magnetic field modulation is found to be the more convenient method. To effect this, Helmholtz coils are placed on the sides of the cavity, parallel to the large magnet pole faces. A high frequency ($\sim$ 100 kHz) modulation is thereby superimposed on the large, slowly varying magnetic field and the signal processed by a lock-in amplifier referenced to this modulation frequency.

Microwave power not absorbed in the cavity is reflected back through the circulator and into the detector, which is generally a cartridge diode, but with bolometers finding some application. These diodes have their best signal-to-noise ratios at specific current levels that are dependent on the amount of incident microwave power. To bring the detector diode to this point is the purpose of the bypass mentioned previously. The output of the diode is then processed by the lock-in amplifier and the resulting signal displayed.

The slowly varying magnetic field used to produce the Zeeman splittings is produced by a large electromagnet. It must be able to generate fields homogeneous to a few ppm over the area of the sample, perhaps 25 cm^2, and having strengths ranging from a few tens of mTesla up to 2.5 Tesla. The gap between the pole faces is on the order of 5 to 10 cm, with the pole faces themselves being between 5 and 35 cm in diameter. The field strength is measured with a Hall effect probe for scanning purposes and with an NMR probe for very accurate determination of the resonance line positions. In instances where the magnetic field modulation is in the audio frequency range, the coils used can be placed directly on the pole caps. Higher modulation frequencies give, in general, better signal-to-noise ratios so that the coils are usually found on the cavity, as described above.

b. Flow Systems. Except in rare instances, it is not desirable to have the plasma discharge interacting with the EPR cavity. To do so makes detection difficult and can result in spurious lines arising from free electron cyclotron resonance. Conversely, when performing diagnostics on plasma processing systems, it is neither feasible nor desirable to place the discharge zone between the poles of a large electromagnet. As a result, the species of interest must be transported in some fashion from the plasma discharge that generates them into the EPR cavity. Most often this is accomplished by the use of a vacuum flow system.

The flow systems are simply tubes connecting the discharge region to a vacuum system with the detection cavity interposed. They may have provision for the addition of other gases, or other arrangements for measuring pressure, temperature, or photon emission. Oftentimes they are constructed with several 90° bends, or other geometry, in order that photons from the plasma discharge do not interact with species or substrates downstream. To minimize heterogeneous processes, the largest volume-to-surface ratio possible is employed, with the further requirement that the surface have low probability for recombining the reactive species. The volume-to-surface ratio is generally constrained by the spectrometer size so that the tubes' sizes are usually 25 mm or less in diameter. There is a bit more freedom in choosing the materials from which the flow system is constructed. Within the cavity, of course, the tube must be a dielectric material so that the microwave radiation may penetrate. Most often the tube will be of quartz or Al_2O_3 in the cavity, with these or similar materials used elsewhere as well. Metals are not used elsewhere unless their inner surfaces are coated with materials having low recombination efficiency, e.g., fluorocarbon res-

Table 3 Recombination Efficiencies for O Atoms at 20 °C

Material	Efficiency*
Ag	1.000
Cu	0.708
Fe	0.150
Ni	0.117
Au	0.022
Pyrex	0.001
Al_2O_3	0.009
Quartz	0.001

*Defined as probability of recombination per wall collision.

ins. The dielectric tubes are themselves sometimes coated, either with fluorocarbon resins or with viscous solutions of boiled phosphoric or boric acids, all of which help reduce wall recombination.

As an example of how recombination efficiency can vary, we consider the ground state oxygen atom whose recombination on various surfaces has been studied (Kaufman, 1961). Table 3 lists the relative efficiencies of several materials for recombining these ground state atoms. As can be seen, metal surfaces are quite efficient at recombining the oxygen atoms, while dielectric materials are much less so. Our own observations of perfluorocarbons and polyvinylidene fluoride indicate that they are at least as inefficient as quartz or Pyrex™. Species other than atomic oxygen may behave differently toward these materials, although the fluorocarbon resins should be acceptable in most cases. Fluorine atoms will passivate some metals, rendering them quite useful. Whenever the "typical" materials are not acceptable for some other reason, care must be taken to choose alternates that possess similar properties.

2. *Laser Magnetic Resonance*

a. Spectrometer Description. As has been previously discussed, LMR spectroscopy is quite closely related phenomenologically to EPR spectroscopy. However, the use of lasers as the fixed frequency sources obviously mandates somewhat different experimental arrangements, the magnet being the principal exception. As it is essentially the same in both cases, perhaps being a bit smaller in LMR, it will be omitted from this discussion of LMR.

Figure 10 depicts the principal components of an intracavity, LMR spectrometer used in several studies of molecular free radicals (Curl et al., 1972; Cook et al., 1976). The laser, operating in the far IR, oscillates between mirrors C and D and has its discharge isolated from the sample region by a polyethylene beamsplitter. This not only isolates the sample region from the lasing portion, but also can be used to polarize the laser radiation either parallel or perpendicular to the magnetic field as desired (see Section II for an explanation of the effect of polarization upon transition selection rules). The output of detector A is used to keep the laser centered on the lasing line. Analogously to the diode output of the EPR system, the output of detector B is processed by a lock-in amplifier referenced to the frequency of the modulation coils present on the magnet poles. (In this instance the modulation frequency is < 100 Hz so that this arrangement may be used.) Various detectors have been used, among them

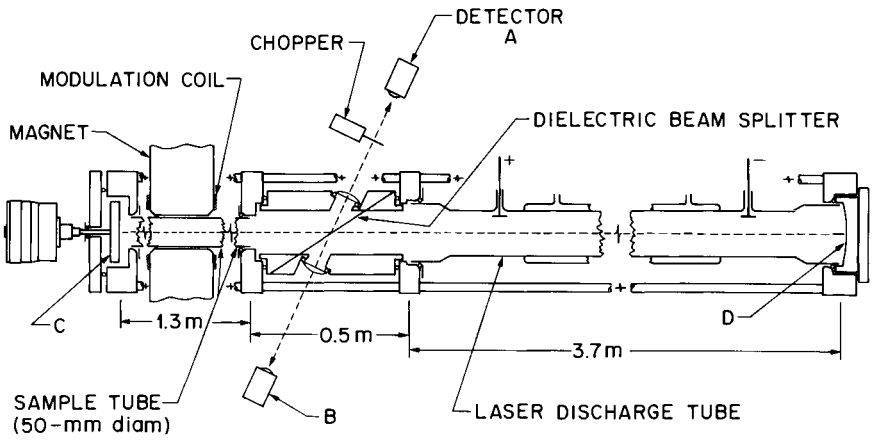


FIGURE 10. Schematic cross section of LMR spectrometer employing a discharge laser. The flow system is omitted. The various components are discussed in the text.

Golay cells, cooled bolometers, and Au:Ge types. Liquid He cooled bolometers are used here, owing to their high sensitivity and low noise properties.

The laser gain medium itself may be formed from a discharge in CO, CO_2, HCN, H_2O, D_2O, or other gas, or it may be optically pumped (Hills, 1984). Even though the lasers operate at discrete frequencies, the number of available lines is quite large and allows good coverage of the region lying between 5 cm^{-1} (150 GHz) and 2000 cm^{-1} (6000 GHz). The use of this intracavity configuration results in high sensitivity ($\sim 10^7$ cm^{-3} for OH), owing to the nonlinear modulation of the laser gain by the absorbing species as the magnetic field is tuned. The laser frequency can generally be shifted by a few MHz by changing the length of the laser cavity. This allows a determination of the tuning rate and direction of the Zeeman transitions and so can be a help in assuring the accuracy of the line assignments.

b. LMR Flow System. The flow systems used for LMR work generally differ very little from those employed in EPR investigations. The restrictions on their size are not so severe as is the case for EPR since there is no resonant microwave cavity through which they must pass. For IR lasers, the tubes may range from one or two centimeters up to seven or eight centimeters in diameter. For FIR systems, they may be even larger. Here the principal constraint is the ability to fit the flow tube between the poles of the magnet so that the volume being sampled is in a homogeneous magnetic field. Low recombination efficiency is generally still desirable, but the necessity of using a dielectric tube is removed as the radiation is either

coaxial for its length or else admitted into it through IR transparent windows.

B. Qualitative and Quantitative Results

Having discussed both the theory and apparatus of the techniques, it should be useful to consider some of the details of their application. While the methods are themselves well understood and reasonably straightforward to apply, care must still be exercised in order to obtain spectra or signals whose lineshapes and intensities are not badly distorted or otherwise perturbed. These results can then be analyzed to obtain the concentrations of the various species. A number of problems are possible in such an undertaking, and this section will be comprised of a brief discussion of a few of them. The numerous works dealing with these details for both EPR (Ulteer, 1966; Westenberg, 1969; Breckenridge and Miller, 1971; Goldberg and Laeger, 1980) and LMR (Howard and Evenson, 1974; Takars and Howard, 1984) should be consulted for in-depth descriptions of techniques and pitfalls.

1. The EPR Technique

a. Qualitative Interpretation. In all probability the species being monitored in diagnostic experiments already will have been studied and their spectroscopic details will be known. As a consequence, identification of the various signals should be straightforward. The one slight complication that can arise is a consequence of the continuous frequency coverage available in the microwave region. As described previously, the resonant magnetic field values are a function of the microwave frequency, as are the line intensities and widths. Hence, if the microwave frequency being used is not the same as that employed by previous workers, their reported patterns will occur at different field values and may vary in their appearance. This poses no real problem so long as the frequency difference is not too great, since $\partial \nu / \partial H$ (the frequency variation as a function of field) is often known and can be used to predict the resonance field. Moreover, by using the spectroscopic constants, it is not all that difficult to predict the positions and intensities from the theory. It may be prudent, at least initially, to reproduce the conditions used in the spectroscopic studies whenever possible, in order to ensure the identity of the species being observed. This approach can often alleviate problems with assignment of lines.

b. Quantitative Interpretation. The use of the observed signal intensities to obtain both relative and absolute number densities is one of the chief advantages of the EPR technique. We will consider a few of the details of obtaining these concentrations, along with some circumstances under which problems can arise.

Owing to the magnetic field modulation and phase sensitive detection employed in EPR experiments, the lineshape obtained experimentally is essentially the first derivative, with respect to magnetic field, of the "true" absorption lineshape (Westenberg, 1969; Breckenridge and Miller, 1971). Thus, in the most general case, a quantity proportional to number density would be obtained by a double integration to yield the area beneath the absorption curve. In practice, this process can often be simplified if certain conditions exist so far as pressure, field modulation, and the nature of the number density determination are concerned.

If only relative concentrations of a given species are desired, and if the pressure and modulation amplitude at which the observations are made are constant, the relative concentrations can be obtained simply by measuring the peak-to-peak height. This is shown in Figure 11 as the line from A to B. Linear instrumental effects, such as amplifier gain and recorder gain, must also be included, of course. If the line being used is actually a multiplet, this technique still yields valid results so long as the multiplet remains unresolved. This is the situation that exists for the 1 Torr data of Figure 2. If it is resolved, the individual transitions must be measured separately. For cases where the various parameters are not constant, or absolute concentrations are desired, the situation is somewhat more complicated.

If the pressure at which a given absorption line is measured should vary, the peak height alone is no longer a valid indication of even the relative concentrations of the species since the linewidth is no longer constant. This effect can be accounted for by the aforementioned double integration or by calculating the first moment of the observed line (Poole, 1983; Westenberg, 1969; Breckenridge and Miller, 1971). Each of these techniques requires some effort, however, and it has been found that multiplying the peak-to-peak height by the linewidth squared will yield equally good results so long as the transition is not saturated nor overly modulated (Breckenridge and Miller, 1971). In Figure 11 the height is the distance A to B and the width is the distance from C to D. In practice, the most significant source of error in this method often arises from difficulty in accurately measuring the linewidth, owing to noise in the signal and difficulty in locating the maxima and minima. Should modulation amplitude become a significant fraction of the linewidth, more distortion can occur but can be accounted for by methods described by Westenberg (1969) and Wahlquist (1961). In general, when two measurements are made under differing conditions of modulation

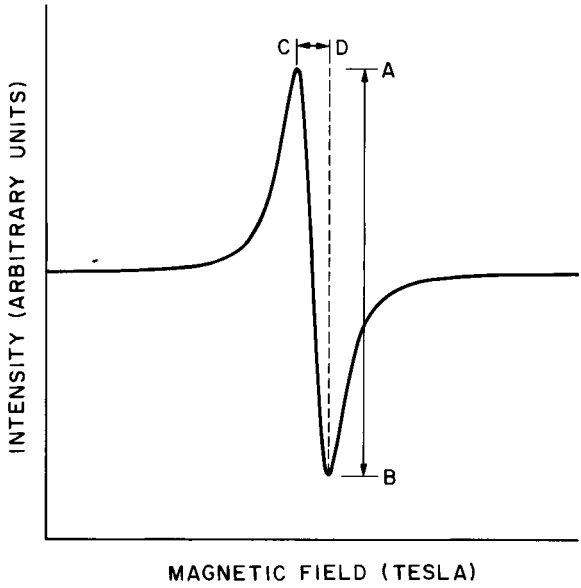


FIGURE 11. Lineshape observed with conventional EPR and LMR spectrometers employing magnetic field modulation. The distance A to B is used for relative concentration measurements at fixed pressure. The distance C to D squared times the distance A to B can often be used in place of the double integral.

amplitude, incident microwave power, and instrumental gain, their measured absorptions can be related by Westenberg's expression (1969).

$$I_{\mathrm{REF}} = I_{\mathrm{MEAS}} \times \left(\frac{M_{\mathrm{REF}}}{M_{\mathrm{MEAS}}} \right) \times \left(\frac{W_{\mathrm{REF}}}{W_{\mathrm{MEAS}}} \right)^{1/2} \times \left(\frac{G_{\mathrm{REF}}}{G_{\mathrm{MEAS}}} \right). \qquad (15)$$

In this equation the subscripts *REF* and *MEAS* refer to the first and second measurements respectively. The variables M, W, and G represent modulation amplitude, microwave power and instrumental gain, respectively. Correcting I_{MEAS} by the given factors allows the two measurements to be compared directly.

Absolute number densities may be obtained by essentially the same methods, but with the additional need for measurement of the absorption of a known concentration of a stable calibration gas. The stable calibrants are either O_2 for magnetic dipole or NO for electric dipole transitions, respectively. That gaseous calibrants be used is essential so that their recorded spectra represent species that "see" the same cavity environment as does the reactive species. Otherwise, the strong dependence of the

recorded signal on all the various parameters described above would invalidate the results. Additionally, the use of a stable gas as the calibrant allows its number density to be inferred from its pressure. Having measured the spectra, the absolute number density of the reactive species can be determined by taking the ratio of the measured absorption intensities and applying factors related to the theoretical absorption intensities (Westenberg, 1969). These theoretical quantities have been calculated for a number of species, including O, Cl, Br, N, and H atoms, as well as the molecular fragments SO and OH (Westenberg, 1969; Breckenridge and Miller, 1971). These, along with the actual procedures used, should be obtained from the original works (Westenberg, 1969; Breckenridge and Miller, 1971).

2. *Laser Magnetic Resonance*

a. Qualitative Analysis. Because of its great sensitivity and the availability of numerous IR and FIR laser lines, LMR is also an excellent tool for detecting short-lived, transient species such as occur in plasma discharges. However, while detection of the species is not too difficult, the identification and assignment of the numerous lines can be. The theory is well developed (see above) but somewhat involved, particularly for polyatomic molecules. Consequently, the technique is probably best suited to be used as a diagnostic tool for obtaining quantitative results on species whose lines have already been assigned. However, qualitative measurements can certainly be performed, and several of the reviews and references therein have discussed the applicable methods (Evenson et al., 1980; Hills, 1984).

Once the lines have been assigned for a particular species, the fact that the lasers provide discrete lines rather than broadly tunable radiation, as in the case of microwave sources, makes using the results of others straightforward. Even for polyatomics, where there may be many lines, identification from known assignments and/or fitted constants can be done without undue difficulty.

b. Quantitative Analysis. The use of LMR as a quantitative tool for investigations of reactive species is not nearly so well developed as is the EPR method. Part of this stems from the relative newness of the technique and the fact that most LMR experiments have been of a spectroscopic, rather than kinetic, nature. Moreover, calculation of the theoretical intensity relationships, in analogy with EPR, is more complicated, owing in part to the multiplicity of states involved. As a result, the method has been used for mostly relative measurements of reactive species' concentrations, but with

some absolute measurements being made by using reactions that produce known amounts of product. This latter point will be discussed in more detail shortly.

The pioneering work for using LMR as a quantitative detection tool for free radicals was performed by Howard and Evenson (1974), who studied the reactions of OH with CO, NO, and NO_2. They used NO_2 as a calibration gas and showed that the height of the absorption line, essentially a first derivative, was proportional to the molecular number density. In their work the species being measured were always present in a large excess of carrier gas (He) that allowed the total pressure to be held constant, even when the reactive species' concentrations varied markedly. Sensitivities on the order of 2×10^8 OH molecules/cm^3 were determined. These results can be applied to a number of other species so long as one is careful to maintain constant conditions between successive measurements, as was described above for EPR. In the case of LMR, it is necessary that the species being observed be in rotational equilibrium so that the specific (rotational) line measured accurately represents the concentration of species in the particular state being probed.

In the last few years several other kinetic studies have been performed where LMR has been used quantitatively (Takacs and Howard, 1984; Thrush, 1982). In these studies, too, it was found that the LMR signal was linearly proportional to the reactive species' concentration over nearly three orders of magnitude, so long as the pressure remained greater than about 100 mTorr. Determinations of the absolute concentrations of certain species, among them HO_2 and OH, were also made. In order to measure these, use was made of stoichiometric reactions known to be quite rapid. In the case of OH, such a reaction is

$$H + NO_2 \rightarrow OH + NO. \tag{16}$$

To determine the absolute concentration, a measured amount of NO_2 was mixed upstream with H atoms produced by a microwave discharge and the resulting OH signal measured. Knowing both the total pressure and the flow rate of NO_2 and that Eq. (16) was stoichiometric, allowed the concentration of OH to be determined. The resulting concentration then corresponded to the signal measured under the particular conditions, yielding a result that could be used for concentration measurements in subsequent kinetic experiments. The effects of wall loss on such measurements can be accounted for by varying the reaction-to-detector distance and radical-radical reactions minimized by the use of very low OH concentrations (Takacs and Howard, 1984). Determinations of HO_2 have been made in a similar fashion by using the reaction:

$$HO_2 + NO \rightarrow OH + NO_2 \tag{17}$$

and then comparing either the NO_2 signal to one measured using a known concentration of NO_2 or the OH signal to the one measured as described above. Application of these types of rapid, stoichiometric reactions for absolute determinations is straightforward, as long as competing reactions and other loss mechanisms are accounted for. In general, it does allow various species to be measured quite accurately over quite large concentration ranges, but does mandate some understanding of other possible reactions.

IV. Discussion

In the previous sections the theory underlying gas phase paramagnetic resonance has been discussed, and two examples of the manner in which the theory may be experimentally realized have been described. Species that have been detected and representative examples in which paramagnetic resonance has been used as a diagnostic aid will form the final section.

A. Species

Tables 1, 2, and 4 list a large number of the atoms and molecules observed by paramagnetic resonance. Several of them are known to be, or at least thought to be, important in plasma processing, either in etching or deposition reactions. F atoms are certainly known to be the primary agents in most Si etching (Flamm et al., 1984), and Cl is likewise responsible for removal of Al (Flamm et al., 1984). Organic materials are etched mainly by O atoms with $^1\Delta O_2$ possibly playing a minor role in specific cases (Cook and Benson, 1983; Cook and Hannon, 1984). F atoms can participate in these organic reactions too, increasing the removal rate by several orders of magnitude.

Reaction products in various instances undoubtedly include a number of the molecular species listed. In CF_4/O_2 plasma discharges used to etch Si, it is quite likely that species such as FO, FO_2, FCO, and CF will be present to

Table 4 Nonlinear Polyatomic Molecules Studied by LMR

CH_2, NH_2, PH_2, AsH_2, C_2H, HCO, HO_2*, HSO, NO_2*, PO_2, ClO_2*
FO_2, NF_2, FCO, ClSO, FSO, CH_3O, CH_2OH, CH_2F, CH_3, SiH_3, GeH_3

*Observed by EPR also.

varying degrees. Oxidation reactions of organic materials will certainly produce OH, CH, CH_2, C_2H, HCO, and many others. If fluorocarbons are added to accelerate the removal rate, even more species will be found, including CF, OF, and FCO.

B. Examples of Paramagnetic Resonance Diagnostics

1. Flame Diagnostics

One early example of paramagnetic resonance used as a diagnostic tool was appropriately that of Westenberg and Fristrom (1965). While this work is somewhat outside the rigorous definition of plasma processing, it nonetheless contains several aspects relevant to this discussion.

In this series of experiments, a flame formed by the combustion of C_2H_6, C_2H_4, and C_2H_2 in excess O_2 was probed by EPR. A quartz tube, having a 75 μm pinhole at one end and a vacuum pump at the other, was employed. The EPR spectrometer was interposed between the two (see above) and allowed the probe tip to cavity distance to be varied. Using this arrangement, the absolute O and H atom concentration profiles in the flames were measured using the excess, and nearly constant, O_2 concentration as a calibrant. The variable distance between the probe tip and the spectrometer allowed the quenching effects to be determined. These were found to be principally wall recombination losses, underscoring the need for careful consideration of flow tube surfaces and geometry. It was also found that the flow tube surface was not always reproducible in its effects until it had been "broken in" for some period of time. This result is in concert with the authors' own experiences and is quite often an important issue in these types of experiments.

2. Organic Oxidation

Another instance in which paramagnetic resonance has been used as a diagnostic tool is the work of Cook et al., who studied the oxidation of photoresist and graphite by-products of an oxygen discharge (Cook and Benson, 1983; Cook and Hannon, 1984). Their work employed a microwave discharge to produce O atoms, $^1\Delta O_2$, F atoms, and other species that then reacted with organic substrates placed downstream. Further downstream were located the EPR spectrometer and vacuum system, respectively. They were able to follow the concentrations of the various reactive species using

EPR while the conditions in the discharge and at the substrate were varied. In this fashion, they determined which species were responsible for removal and what parameters most affected their production and transport. Activation energies of removal for differing conditions were also determined by measuring the EPR signal attenuation, and the results were found to agree very well with those obtained by more traditional methods.

C. Summation

We have attempted here to describe two paramagnetic resonance techniques that could have application for diagnostics in various plasma processes. In most instances they would find greatest utility in downstream processing, i.e., where the plasma discharge and substrate are separated. In that sort of system one is interested only in identifying what species are present at the substrate, and this can be easily accomplished by placing the spectrometer at the substrate location. By this type of technique, the discharge feedstock, pressure, power, flow rates, and reactor materials can be varied and the resulting effects on species type and number recorded.

Less amenable to the techniques are the conventional, parallel plate types of systems, i.e., the in situ reactors. While diagnostics on these are still possible, it becomes more difficult to determine directly the constituents of the discharge, since downstream effluent is actually what is monitored. On the other hand, phenomenological relationships between downstream measurements and observed behavior in the discharge are often of great value. In these cases, the method would offer much. Knowledge of the species and reactions occurring downstream would further enhance the utility.

Finally, while the techniques are not commonplace in the world of plasma discharge processing, they have been used for years in kinetic and spectroscopic studies of both the species listed in the tables and others. Most of these have been prepared by discharges of one form or another. Consequently, there is a large body of knowledge that exists on use of the techniques and methods, which, when combined with well designed processing experiments, could do much to further the knowledge of plasmas and plasma processing.

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References

Bashkin, S. and Stoner, J. A., Jr. (1978). *Atomic Energy Level & Grotrian Diagrams*, Vols. I and II. North-Holland, New York.
Breckenridge, W. H. and Miller, T. A. (1971). *J. Chem. Phys.* **56**, 475.
Brown, J. (1972). *MTP Int. Rev. Sci.* **4**, 235.
Brown, J. M., Kerr, C. M., Wayne, R. O., Evenson, K. M., and Radford, H. E. (1981). *J. Mol. Spectros.* **86**, 544.
Brown, J. M., Schubert, J. E., Evenson, K. M., and Radford, H. E. (1982). *Ap. J.* **258**, 899.
Carrington, A., Levy, D. H., and Miller, T. A. (1970). *Adv. Chem. Phys.* **18**, 149.
Condon, E. U. and Shortley, G. H. (1963). *The Theory of Atomic Spectra*. Cambridge University Press, Cambridge, U.K.
Cook, J. M. and Benson, B. W. (1983) *J. Electrochem. Soc.* **130**, 2459.
Cook, J. M. and Hannon, J. J. (1984). *IEEE Trans. Components, Hybrids, Manufact. Technol.* **CHMT-7**, 328.
Cook, J. M., Evenson, K. M., Howard, C. J., and Curl, R. F. Jr., (1976). *J. Chem. Phys.* **64**, 1381.
Curl, R. F., Jr., Evenson, K. M., and Wells, J. S. (1972). *J. Chem. Phys.* **56**, 5143.
Davies, P. B. (1981). *J. Phys. Chem.* **85**, 2599.
Davies, P. B. and Evenson, K. M. (1975). *Proc. Int. Conf. Laser Spect.* **1**, 132.
Davies, P. B., Handy, B. J., Lloyd, E. K., and Russell, D. K. (1978a). *Mol. Phys.* **31**, 1005.
Davies, P. B., Handy, B. J., Lloyd, E. K., and Smith, D. R. (1978b). *J. Chem. Phys.* **68**, 1135.
Evenson, K. M. (1981). *Faraday Disc. Chem. Soc.* **71**, 7.
Evenson, K. M., Saykally, R. J., Jennings, D. A., Curl, R. F., and Brown, J. M. (1980). *Chem. Biochem. Appl. Lasers* (C. B. Moore, ed.) **5**, 95.
Flamm, D. L., Donnelly, V. M., and Ibbotson, D. E. (1984). *VLSI Electronics: Plasma Processing for VLSI* (N. G. Einspruch, ed.), Vol. 8. Academic Press, Orlando, FL, p. 189.
Goldberg, I. B. and Laeger, H. O. (1980). *J. Phys. Chem.* **84**, 3040.
Herzberg, G. (1950). *Spectra of Diatomic Molecules*. D. Van Nostrand Co., New York.
Hills, G. W. (1984). *Mag. Res. Rev.* **9**, 15–64.
Howard, C. J. and Evenson, K. M. (1974). *J. Chem. Phys.* **61**, 1943.
Kaufman, F. (1961). *Progress in Reaction Kinetics* (G. Porter, ed.), Vol. 1. Pergamon Press, New York.
McKellar, A. R. (1981). *Faraday Disc. Chem. Soc.* **71**, 63.
Moore, C. (1949). *Tabulations of Atomic Energy Levels*, Vols I, II, and III, U.S. GPO, Washington, D.C.
Poole, C. P., Jr. (1983). *Electron Spin Resonance*. John Wiley and Sons, New York.
Radford, H. E. (1960, 1962). *Phys. Rev.* **122**, 114 (1960); *Phys. Rev.* **126**, 1035 (1962).
Radford, H. E. and Hughes, V. W. (1959). *Phys. Rev.* **114**, 1274.
Rose, M. E. (1957). *Elementary Theory of Angular Momentum*. John Wiley & Sons, New York.
Takacs, G. A. and Howard, C. J. (1984). *J. Phys. Chem.* **88**, 2110.
Thrush, B. A. (1981). *Acc. Chem. Res.* **14**, 116.
Thrush, B. A. *Current Topics in Mass Spectrometry and Chemical Kinetics* (J. H. Beynon and M. L. McGlashan, eds.) Heyden, London, U.K., p. 89.
Townes, C. H. and Schawlow, A. L. (1975). *Microwave Spectroscopy*. Dover, New York.
Ultee, C. J. (1966). *J. Appl. Phys.* **37**, 1744.
Wahlquist, H. (1961). *J. Chem. Phys.* **35**, 1708.
Westenberg, A. A. (1969). *Science* **164**, 381 and references contained therein.
Westenberg, A. A. (1973). *Proc. React. Kinet.* **7**, 23.
Westenberg, A. A. and Fristrom, R. M. (1965). *Tenth Symposium on Combustion*. The Combustion Institute, Pittsburgh, p. 473.

7 Diagnostics in Thermal Plasma Processing

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I. Introduction 349
II. Methods for Production and Characterization of Plasmas 352
A. Electric Arcs 352
B. R.F. Inductively Coupled Discharges 355
III. Specific Problems Related to Measurements under Thermal Plasma Conditions . 356
A. Plasma Brightness 357
B. Steep Gradients 360
C. High Thermal Fluxes 361
D. Natural Fluctuations 361
IV. Plasma Flow Diagnostics 362
A. Enthalpy Probes 362
B. Emission Spectroscopy 364
C. Absorption Spectroscopy 398
D. Laser Spectroscopy 403
E. Plasma Velocity Measurements 406
V. Measurement of Particle Parameters in Thermal Plasmas 410
A. Particle Trajectory Distribution 410
B. Particle Velocity Measurements 415
C. Particle Size Measurements 430
D. In-flight Particle Surface Temperature Measurements 432
VI. Conclusions 439
VII. Acknowledgements 442
References 442

I. Introduction

Typical steady state thermal plasmas may be produced by high intensity arcs, plasma torches or radio frequency (R.F.) discharges at high pressures

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($p \geq 10$ kPa) (Finkelnburg et al., 1956; Pfender, 1978; Kassabji and Fauchais, 1981; Eckert, 1974; Boulos, 1985).

In the plasma column or in the discharge region the mean free path of the charged particles in general, and of the electrons in particular, is by far too short for these particles to accumulate sufficient energy between two collisions to produce ionizing collisions. Therefore charge carrier production is accomplished by thermal ionization. The plasma behaviour is mainly controlled by the balance between the energy losses at the boundaries of the discharge or arc column (by conduction and convection), as well as by radiation from the whole volume and the electric power dissipation ($\sigma \mathbf{E}^2$, where σ is the electrical conductivity of the plasma and $\mathbf{E}$ the electric field). The plasma reacts to an enhanced cooling of its fringes by a reduction of its surface (to reduce conduction and convection losses) and an increase of the energy dissipation. For a given arc current and cooling parameters, the conditions in the arc column (temperature distribution and associated thermodynamic and transport properties) adjust themselves in such a way that the field strength $\mathbf{E}$ required for sustaining this current reaches a minimum (Finkelnburg et al., 1956). Thermal plasmas with their low electric fields (below 10^4 V/m) and their high density or pressure are thus characterized by a ratio $\mathbf{E}/p$ lower than 3 V/m · Pa compared with values higher than 10^3 V/m · Pa for low pressure plasmas.

In thermal plasmas the energy is supplied to the electrons and ions by the electric field, these particles transmitting in turn their energy to the neutral ones by collisions. However, electrons, due to their lower mass, acquire much higher velocities than ions and they play a special role in the plasma. It can be assumed (Finkelnburg et al., 1956) that energy distributions of the electrons as well as of the heavy species (neutral atoms or molecules, ions) are Maxwellian. However, the mean kinetic energy of the electrons is not necessarily equal to that of the heavy species, especially if the collisions between electrons and heavy species are not numerous enough (an electron of mass m_e losses only a fraction $2m_e/m_h$ of its energy in an elastic collision with a heavy species of mass m_h). Thus the electron temperature T_e, defined by

$$m_e \overline{v_e^2}/2 = 3kT_e/2, \tag{1}$$

will be equal to the heavy species temperature T_h (defined by an expression similar to (1) with index h instead of e) only if the electron density n_e is high enough i.e., for $n_e > 10^{22}$ e/m^3 (such conditions are met only in the core of arcs or R.F. discharges and if the pressure is higher than 10 kPa). Figure 1 due to Pfender 1978 shows the evolution of T_e and T_h versus pressure for a mercury arc at a given arc current. One must be aware, however, that even if equilibrium is reached, ($T_e \sim T_h$), in the plasma core,

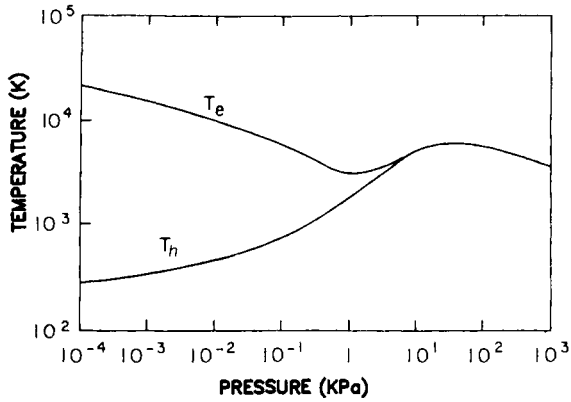


FIGURE 1. Behaviour of electron temperature (T_e) and heavy particle temperature (T_h) in an arc plasma. (Pfender, E., 1978).

that is not the case in the plasma boundaries (due to the fast diffusion of the electrons), close to walls or electrodes (not enough collisions in the boundary layers where steep gradients are encountered), or in the plume of plasma jets (where $n_e < 10^{22}\ e/m^3$). However, the non-equilibrium is never found to be as severe as that obtained in low pressure plasmas, and usually a ratio $T_e/T_h = 2$ corresponds to extreme conditions (except possibly in arcs very close to the anode or cathode).

The behaviour of thermal plasmas is mainly controlled by macroscopic parameters such as flow velocity, local composition and the two temperatures T_h and T_e which allow one to calculate composition, thermodynamic, and transport properties (Fauchais et al., 1985). The well-established methods existing already for the diagnostics of these parameters in laboratory type plasmas have to be adapted or modified to meet the specific requirements of thermal plasmas:

- complex mixtures of commercial gases: argon, hydrogen, nitrogen, air, etc., in which solid particles, chlorides, hydrocarbons, etc., are injected;
- high temperatures, ranging from 2,000 to 20,000 K, resulting in very bright media;
- high velocities ranging from 20 to 1,000 m/s;
- high gradients: up to 4,000 K/mm for temperatures, 300 m/s · mm for velocity, 104 parts/m^3 · mm for species concentration, (especially in arcs in the radial direction);
- natural fluctuations of arc roots or R.F. discharge coupling, which requires the measurement of time-average values.

Moreover, to provide results in a reasonable time, the automation of the measurements is mandatory.

Topics to be discussed in the following sections are:

a) Methods for producing thermal plasmas including the resulting range of different parameters.
b) Measurement problems related to the flow and particle parameters, as obtained by different measurement techniques.
c) The different measurement techniques for the flow and moving particles in flight and their limitations (this section will include a detailed description of the experimental techniques and actual equipment most widely used and only a short presentation of alternative possible methods).
d) Examples of typical measurements will be presented.

II. Methods for Production and Characterization of Plasmas

Temperature and velocity ranges as well as typical gradients are of primary importance to determine which measurement techniques can be used and, for a given technique, which species or which phenomenon are to be used as a probe. Thus we will first describe briefly the different production methods and the main characteristics of the resulting plasmas.

A. Electric Arcs

Arcs, known as "high intensity arcs," typically require d.c. or a.c. currents around 100 A or more and pressures in excess of 10 kPa. Most commercial devices involve d.c. currents because the electrode erosion is drastically reduced.

The principal devices used to sustain electric arcs are:

Transferred Arcs. Transferred arcs such as the one due to Young et al. (1983) represented in Figure 2, where the arc is struck between a hot thoriated tungsten cathode (thermo-electronic emission) and a ring anode. This arc is stabilized by the plasma and shroud gases injected near the cathode tip (oxidizing and carburizing gases being of course strictly forbidden near cathode tip). Such transferred arcs have been run at up to a few Megawatts for spheroidization (Thorpe and Wilks, 1971) or extractive metallurgy (Monk, 1981; Gauvin et al., 1985; Lugsheider, 1983). The gas flow rate is rather low (less than 200 l/min for 1 MW), and the energy densities may reach 2,800 MJ/m^3.

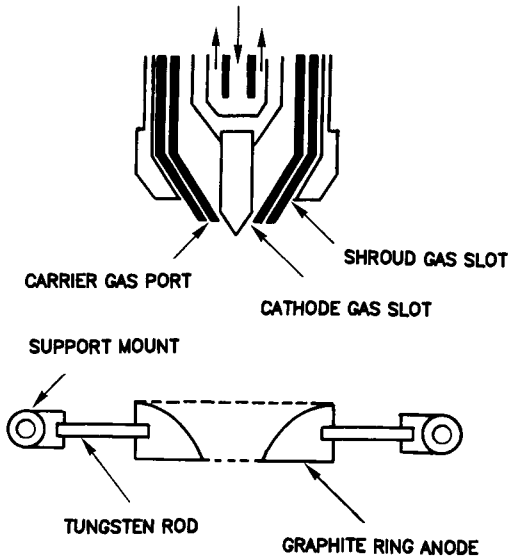


FIGURE 2. Schematic of a transferred arc. (Young, R. M., et al., 1983).

For transferred arcs (laboratory measurements with argon as plasma gas and power levels of a few tens of kW), temperatures up to 20,000 K have been measured near the cathode tip in the wall constricted plasma column (Young et al., 1983). High temperature gradients in the arc fringes have been also observed, but no velocities have been measured and only predicted values exist: up to 500 m/s near the cathode tip (McKelliget and Szekely, 1986).

Wall Stabilized Arcs with Hot (W) Cathodes. The hot cathodes are either "stick" or "button" type. The first ones are mainly used in spraying generators with power levels lower than 100 kW. A typical plasma torch of this type is shown in Figure 3 (from Vardelle, 1980). In such a generator the nozzle diameter is always less than 10 mm, and the power level ranges from 10 to 90 kW. The gas flow rate is generally below 100 l/min, and the energy densities in the hot gas may reach 145 MJ/m^3.

The temperatures are between 6,000 and 15,000 K and velocities can reach easily 900–1,000 m/s with nitrogen–hydrogen mixtures. However, radial temperature and velocity gradients are very steep in the jet fringes: up to 4,000 K/mm and up to 300 m/s · mm (Vardelle et al., 1982; Vardelle et al., 1980).

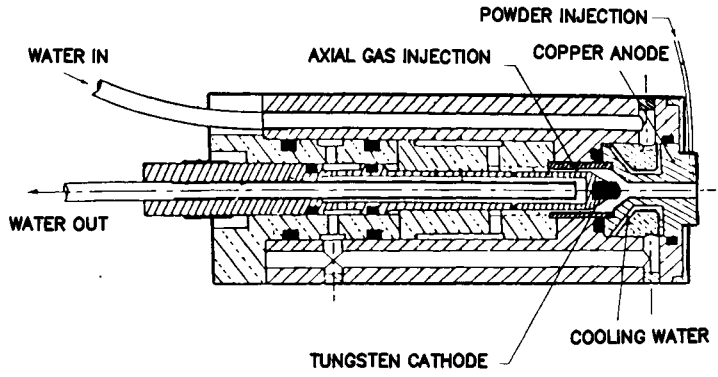


FIGURE 3. A spraying D.C. plasma torch. (Vardelle, M., 1980).

A button type cathode generator is shown in Figure 4. The plasma is stabilized by the walls and vortex injection of the gas. This vortex injection implies a high gas flow rate (for example, 300 l/min for 100 kW) and a more uniform temperature distribution in the plasma jet. Such torches have been constructed in U.S.S.R. (Joukov, 1979) with a power level up to 1 MW and specific enthalpy of the plasma gas less than 15 MJ/m^3. The lower enthalpies (compared to spraying plasma torches) obtained are due to the higher gas flow rates and the bigger nozzle diameters (at least three times for a similar power level). Recent temperature measurements (Coudert et al., 1987) performed with such torches at a power level of 50 kW have shown that temperatures are usually below 9,000 K and gradients are smaller than 1,000 K/mm. Calculated velocities are below 500 m/s.

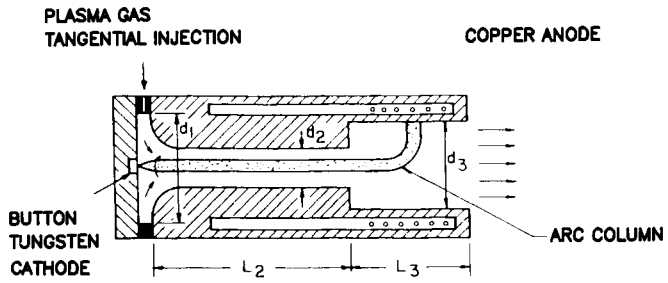


FIGURE 4. Plasma torch with a step variation in its anode diameter, a button type thermoionic cathode and a vortex injection. (Joukov, M. F., 1979).

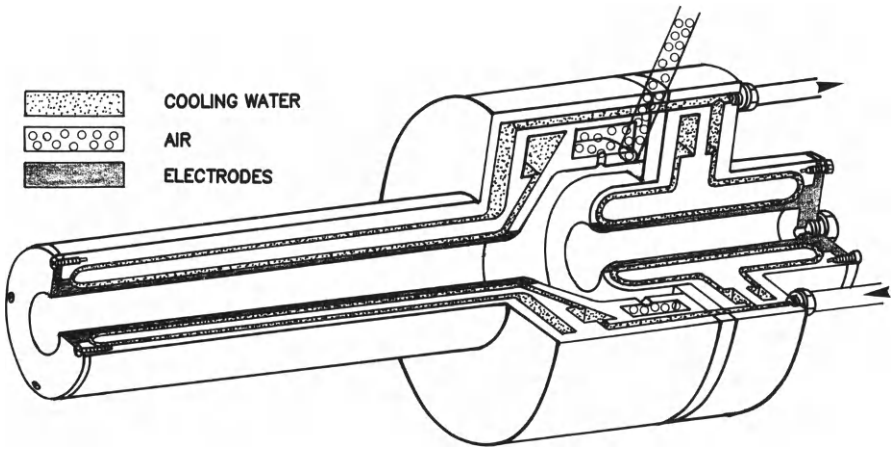


FIGURE 5. Plasma torch developed by Aerospatiale with a copper well type electrode and vortex stabilization. (Labrot, M. E., and De Logiviere, M., 1984).

Wall-Stabilized Arcs with Cold Cathodes (Cu). Cold cathode type generators use copper electrodes and vortex stabilization. A typical generator, designed by Labrot et al., (1984), is shown in Figure 5. 2.2 MW input power may be achieved with a maximum specific enthalpy of 10 MJ/m^3. These torches are mainly used for gas heating in blast furnaces, ultrafine powders production or extractive metallurgy. See, for example, Fey, 1980; Foster and Lin, 1983; George and Hare, 1984; Santen and Thornblon, 1984; Camacho, 1984.

B. R.F. Inductively Coupled Discharges

R.F. discharges are characterized by the absence of electrodes, which avoid the contamination of the plasma with metallic vapours. In most cases energy is supplied to the plasma gas by inductive coupling. Three kinds of torches using inductive coupling are shown in Figure 6 (from Akashi, 1985). The first torch, (type A), is the most conventional one. Type B is experimental, designed to improve the recovery of products and avoid plasma perturbations by the injection of the reactants. Finally, type C is a combination of a D.C. plasma jet, acting as a pilot, with the R.F. discharge, generating a very stable plasma flame (hybrid generator). An important problem for this type of generator is to find an optimum combination of the frequency, size and power which sustains a stable discharge. Nevertheless, torches of type A have been developed up to 1 MW, and their main industrial applications are in the field of spectrochemical analysis, synthesis

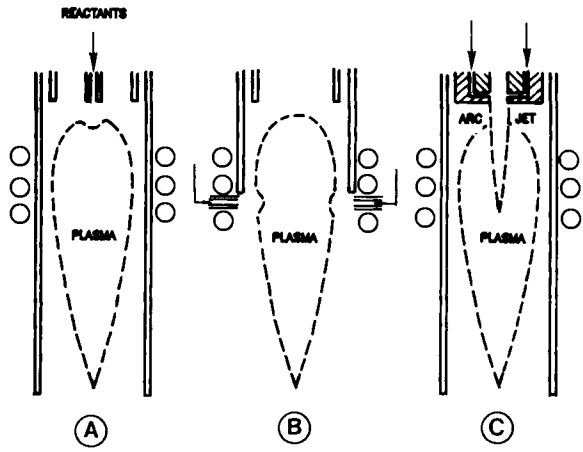


FIGURE 6. Schematic of three types of inductively coupled R.F. generators. (Akashi, K. 1985).

of high purity silicon or titanium dioxide pigments, and recently ultrafine and ultrapure powders synthesis (oxides, carbides, nitrides) (Eckert, 1974; Boulos, 1985; Akashi, 1985). Such torches have flow rates similar to those of D.C. spraying plasma torches but their diameter is almost one order of magnitude higher resulting in energy densities below 30 MJ/m^3. Measured temperatures are below 10,000 K and maximum velocities very seldom exceed 50–60 m/s. Of course with the larger torch diameter the gradients are lower than in arcs (less than 1,000 K/mm and 10 m/s · mm) (Boulos, 1985; Eckert, 1974).

III. Specific Problems Related to Measurements Under Thermal Plasma Conditions

The main measuring techniques used in thermal plasmas can be roughly divided into:

- intrusive methods which employ material probes (water cooled and/or used in a transient mode);
- passive radiation techniques which utilize radiation emitted by the plasma or plasma carried particles;
- active radiation techniques which use an external source of radiation (incoherent or coherent) to probe the system by transmission, absorption, scattering, or reflection.

However, before describing any of them, we will emphasize the main problems related to the specific characteristics of thermal plasmas, i.e.:

plasma brightness, high gradients, high thermal fluxes, natural fluctuations of the arc root, and nature of the flow (from laminar to turbulent in the same plasma jet).

A. Plasma Brightness

This point is very important because most of the techniques utilize either radiation emitted by the plasma or plasma carried particles, or an external source of radiation to probe the system. If equilibrium is assumed, plasma radiation versus temperature can be calculated for a given pressure (Fauchais et al., 1984). The radiation is due to atoms (atomic lines) or molecules (band spectra) and to continua. In the case of lines or bands, the volumetric emission coefficient ϵ_λ depends on the population of the upper emitting level. As the density of the various chemical species varies strongly with temperature, [see Figure 7 for a nitrogen plasma for example (Pateyron et al., 1986)], ϵ_λ will also strongly vary with T as illustrated in Figure 8 for the NI line 493.5 nm (Baronnet, 1978). Two points are to be emphasized from this curve:

- Temperature dependence is very steep: ϵ_λ varies by almost 3 orders of magnitude for a ΔT of 2,000 K;
- When the nitrogen atom concentration starts to decrease (see Figure 7) ϵ_λ reaches a plateau before decreasing again. For most of the plasma gases: Ar, H_2, N_2, O_2, this plateau is reached at temperatures of approximately 15,000 K. Similar results are obtained with molecular bands, the plateaus

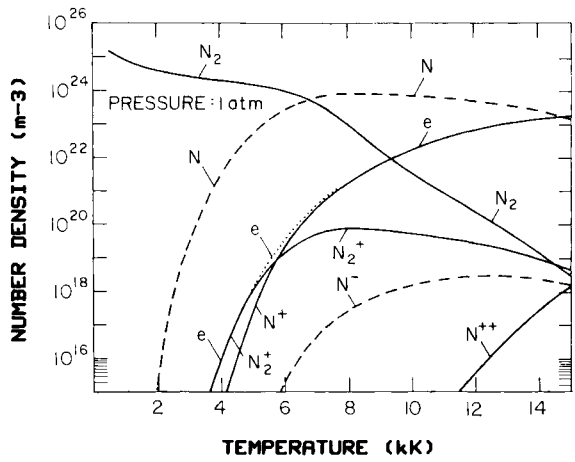


FIGURE 7. Plasma number density versus temperature evolution of the species in a nitrogen plasma at atmospheric pressure. (Pateyron, Aubreton, 1986).

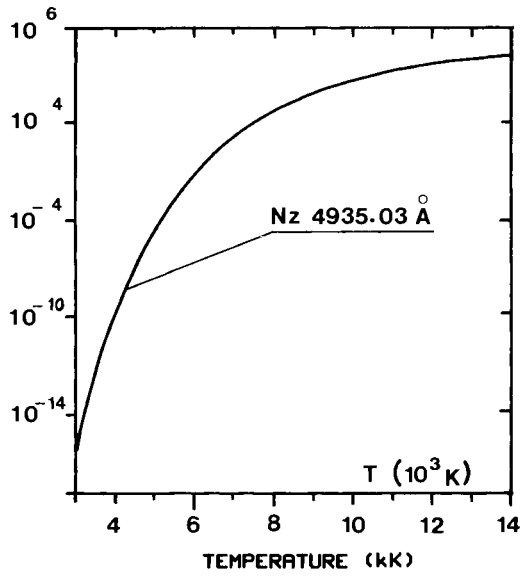


FIGURE 8. Volumetric emission coefficient (Baronnet, J. M. 1978) versus temperature evolution of the nitrogen atomic line at 493.5 nm.

for OH, CN, N_2 (which are the bands used in thermal plasmas) starting at lower temperatures: about 7,000 to 9,000 K. For the first ions of Ar, N, O, etc., the plateau is reached at approximately 22,000 K.

Similar steep variations with temperature are obtained for the continuum which depends strongly on the electron density. For example, Figure 9 shows the evolution of the total continuum radiation of argon at atmo-

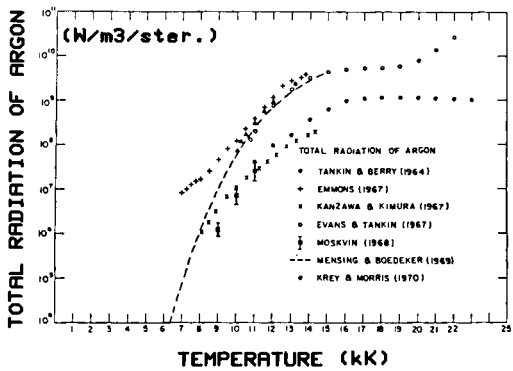


FIGURE 9. Total continuum emission of argon. (Evans, D. L., and Tankin, R. S., 1967).

spheric pressure. It can be seen that it varies from 10^3 to almost 10^{10} W/m^3 when the temperature increases from 6,000 to 14,000 K. It is to be noticed that such a curve explains why transferred arcs stabilized by a gas flow, with temperatures higher than 13,000 K, radiate more than 30% of the dissipated energy.

The consequences of this intense radiation and of its distribution among species are the following:

Emission Measurements

- Considering the different type of plasmas presented in Section II, the following species emission from the plasma itself can be used to probe temperatures in the plasma:
  - molecules in vortex gas stabilized plasma torches;
  - molecules and atoms in plasma spray torches or in R.F. discharges;
  - atoms and first ions in transferred arcs.
- When considering solid particles being carried within a plasma (assuming for simplicity they are black bodies) their flux must be at least close to or higher than that of the plasma enclosed into the measurement volume. This can be achieved only if there is no plasma line emission at the wavelength at which the measurement is performed and if plasma temperature is lower than 6,000 K. The possibility of taking the measurements depends on the particle surface temperature and apparent surface emissivity (Mishin, 1986 and Sakuta and Boulos, 1987).

For Measurements Using an External Source of Radiation

- When using an external source of light: (lamp or laser) for absorption measurements, the main problem is to use a source with an intensity high enough compared to that of the plasma. Radiation from calibrated tungsten-halogen lamps can be used up to about 3,000 K (Self et al., 1982), i.e., in the fringes of plasma jets or far downstream in the plume, hollow cathode lamps can be used up to 6,000–8,000 K (M. Vardelle, 1987) and for higher plasma temperatures only another thermal plasma source (Baronnet, 1978) or a laser can be used.
- When using laser scattering (Thomson or Rayleigh) the difficulty comes from the very low scattering cross sections: 10^{-25} cm^2 for Thomson scattering and 3 to 4 orders of magnitude less for Rayleigh scattering (Kunze, 1968). This means, for example, that the scattered light for a Rayleigh experiment is of the order of 1 photon in 10^{14} incident ones. This limits the measurements, even when using chopped lasers, polarizers and a Fabry Perot (to limit the band width), to plasma regions where temperatures are below 9,000 K.

The laser scattered signal from solid particles also has to overcome the light emitted from the plasma. Since this signal depends on the particle size, on the incident photon density, and on the direction of observation, while the intensity of the plasma emission depends on temperature and wavelength range, the parameters range has to be matched. For example, according to the lasers actually available as well as to the detection techniques, measurements can be performed only in plasmas at temperatures below 12,000 K (transferred arcs are excluded) and larger than 1 μm diameter.

B. Steep Gradients

Thermal plasmas are frequently produced by means of arcs with intensely cooled fringes (by cold walls and/or by cold gas flows). The hot core of the plasma, where a sufficient number of charged particles exist to sustain the discharge, is always at temperatures higher than about 6,000 K, which corresponds, at least for Ar, N_2, H_2, O_2, to the value above which the electrical conductivity is sufficiently high to sustain the discharge. This hot plasma core is surrounded by cold gases (at temperatures less than 3,000 K). Nonlinear variations of transport and thermodynamic plasma properties (Fauchais et al., 1987) result in strong temperature gradients, and in flows which may be laminar along the axis (for temperatures higher than 10,000 K) and highly turbulent in the fringes. The high temperature gradients result in high enthalpy gradients; almost all the enthalpy is concentrated in the hot core and, as only part of the plasma gas flow rate passes through the hot core, the rest flows in the cold regions surrounding it. The mean enthalpy of the plasma gas (energy delivered to the gas divided by the total plasma gas flow rate) is thus meaningless.

The steep gradients observed have important consequences for the measurement techniques. The dimensions of the probing volume have to be small enough to account for them. This is not necessarily true in the case of material probes which may perturb the plasma flow even under transient conditions. It may be true of laser probing systems, when the measuring volume is the intersection point of the laser beam and the optical path of the detector for emission measurements when the collected radiation is the integrated intensity of the light originating along the line of sight. Even if techniques such as Abel's inversion can overcome this problem, another point has to be accounted for: the steep variations of the volumetric emission coefficient. For example, in a plasma spray jet at the nozzle exit, temperature varies from 13,000 K at the axis to less than 6,000 K in the fringes, i.e., along a radius of 2 mm. This corresponds to almost 6 orders of

magnitude for the volumetric emission coefficient, and almost no detector can bridge this range. Moreover, in the fringes of the jet, the emitted light will be of the same order of magnitude as that scattered by windows or lenses and coming from the hot core close to the axis.

The steep gradients may also lead to demixing problems, due either to mass differences resulting in different diffusion coefficients or to differences in ionization energies (metal ions compared to plasma gas ions). It is then rather improbable that the chemical composition of the plasma at a given location will be the same as that of the initial plasma gas mixture. Moreover, as it will be shown (see Section IV.A), plasma jets flowing in a different surrounding atmosphere (for example Ar-H_2 plasma in air) pump it very fast, which modifies their chemical composition.

C. High Thermal Fluxes

For plasma jets with temperatures higher than 12,000 K, (especially those containing hydrogen as plasma gas), the heat fluxes can be higher than 5.10^8 W/m^2 and no stationary water-cooled probe can sustain such heat fluxes without evaporating. For transient small probes, for which cooling is difficult because of the dimensions, 12,000 K also seems to be the practical limit.

D. Natural Fluctuations

Whatever may be the quality of design of a plasma torch, the arc root (especially at the anode) fluctuates under the action of the cold gas flowing at its periphery (because of these fluctuations the electrodes can sustain heat fluxes of 10^9 to 10^{11} W/m^2 at the arc roots). As a consequence, the arc length varies continuously at a frequency between a few kHz and 10 kHz, resulting in a similar fluctuation of the dissipated energy. Thus concentration, velocity, and temperature measurements have only a meaning as time integrated measurements for periods at least of a few msec.

The evolution of velocity and surface temperature of solid particles carried by plasma jets, and having residence times of the order of a few msec, will not be too much affected by the energy fluctuations of the plasma jet. However, particles are injected into thermal plasmas with size and injection velocity distributions resulting in very broad trajectory distributions (A. Vardelle, 1987), and turbulences will affect the smallest ones. It is then generally impossible to follow the trajectory of a given particle and in most cases measurements are made for n particles passing through a given

measurement volume resulting in mean values

$$\bar{X} = 1/n \sum_{i=1}^{n} X_i . \tag{2}$$

IV. Plasma Flow Diagnostics

A. Enthalpy Probes

Originally, enthalpy probes were developed for space-related applications, i.e., for measuring the local enthalpies in plasma wind tunnels and for reentry material test facilities. In later years, enthalpy probes found increasing use for basic studies in plasma laboratories. They appear to be a useful technique in plasma jets (Incropera and Leppert, 1967; Grey, 1963; Hare, 1977; Dresvin and Klubnikin, 1975; Brossa and Pfender, 1987) especially when emission spectroscopy fails (below about 7,000 K for atmospheric argon plasmas, for example).

The probe which probably found the most widespread application and which is also commercially available today is the "Grey" probe, described in detail by T. Grey, 1976. The probe may consist of Cu with stainless steel supports, or it may be entirely constructed from stainless steel. The overall diameter of the probe may be as small as 1.6 mm (Pfender and Boulos, 1983). Since local heat fluxes at the front face of the probe may be in the order of 10^7 W/m^2, an effective high pressure (up to 55 atm) water cooling system is a necessity. The flow of the cooling water is indicated in Figure 10. The cooling water receives not only heat from the plasma sample passing through the central tube, but also from the plasma surrounding the probe. For eliminating the external heat transfer to the probe, a "tare" measurement is required during which the valve controlling the sample gas flow remains closed. A pair of thermocouples located at the cooling water inlet and outlet determines the temperature rise of the cooling water. A third thermocouple located opposite the cooling water outlet measures the temperature of the gas sample at that location. From these thermocouple readings, in combination with readings of the mean flow rates of the cooling water and the plasma flow through the control tube, the enthalpy of the gas sample may be obtained (Pfender and Boulos, 1983). A zirconia cell oxygen analyser, downstream of the probe, measures the oxygen partial pressure in the sample (see Figure 11), using atmospheric air as reference gas (Brossa and Pfender, 1987). The total pressure of the sample is measured at the oxygen analyser's outlet by a pressure transducer (calibrated to compensate for the pressure drop along the line). A sonic orifice allows one to determine

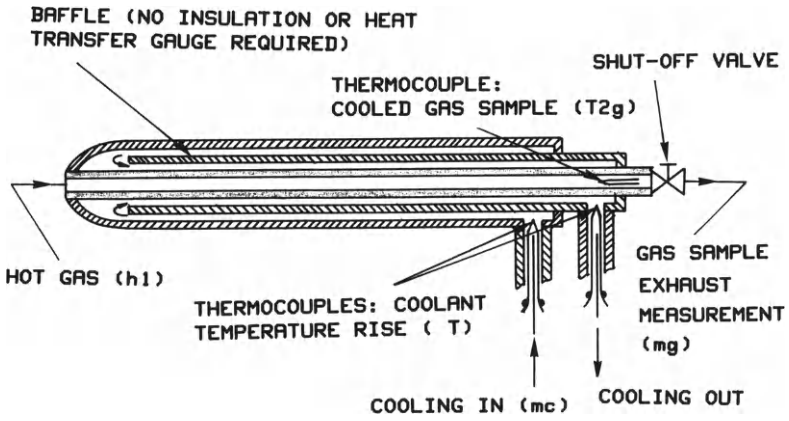


FIGURE 10. Schematic of an enthalpy probe. (Pfender, E. and Boulos, M., 1983).

the sample flow rate, which can be calculated once the pressure and temperature of the sample upstream the orifice are known. The vacuum pump, at the end of the line, ensures that a critical pressure ratio is maintained across the sonic orifice (see Figure 11). Such probes have been used to measure the diffusion of cold gases in R.F. plasmas (Dundas, 1970) or D.C. arc jets (Brossa and Pfender, 1987). In such measurements of composition, the plasma is considered as a binary mixture of plasma gas and air, which is assumed to maintain at all times the standard composition of the cold flows (no demixing of plasma gases such as Ar-H_2 mixtures, and standard composition of air at high temperature). These measurements are, for example, of particular interest for applications such as ultrafine powders synthesis from gas phase or plasma spraying where the diffusion of air rapidly cools down the plasma jet with the dissociation of oxygen. Figure 12 represents the axial decay of the argon concentration along the center-line for an argon D.C. plasma jet flowing in air at two different flow rates

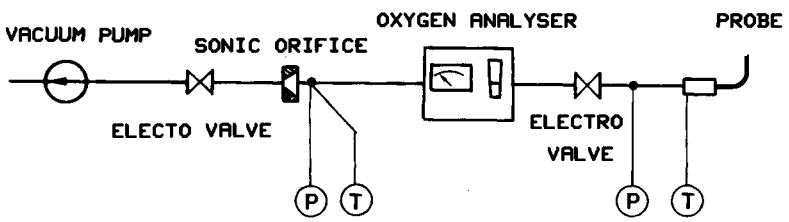


FIGURE 11. Gas sampling line of an enthalpy probe. (Brossa, M. and Pfender, E., 1987).

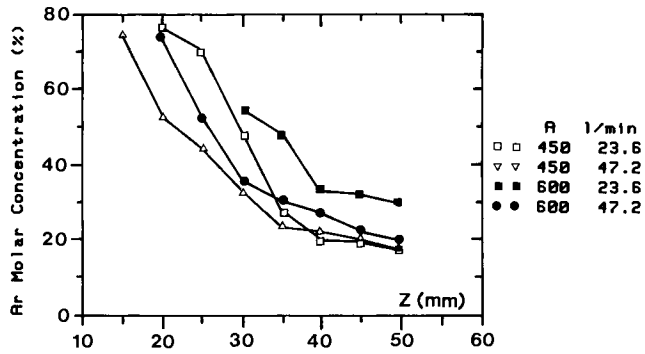


FIGURE 12. Axial decay of the argon concentration along the center line of an argon plasma jet for two arc currents (450 and 600 A) and two gas flow rates (23.6 and 47.2 1/mn). (Brossa, M. and Pfender, E., 1987).

and current intensities. This figure illustrates the importance of air entrainment and explains why spraying in controlled atmosphere chambers filled with argon at atmospheric pressure gives much better coatings due to longer plasma jets and better heat transfer to the particles in flight (see Roumilhac et al., 1987).

B. Emission Spectroscopy

1. *Principles: Emission Coefficients*

The emitted intensity of an atomic, ionic, or molecular spectral line is proportional to the population of the upper level, to the transition probability, and to the energy of the quantum (Griem, 1964)

$$\epsilon_{ul} = \frac{1}{4\pi} A_{ul} \cdot n_{i,u} \cdot h \cdot \nu_{ul}, \tag{3}$$

where

h is the Planck's constant;

$n_{i,u}$ the population density of the chemical species i, in the excited state u (m^{-3});

ν_{ul} the frequency of the photons emitted due to transition from the upper excited level u to the lower level l;

A_{ul} the corresponding transition probability per unit time (s^{-1}) given in tables such as those of Weise (1965);

ϵ_{ul} the volumetric emission coefficient of the line (W/sr · m^3).

Due to various broadening processes that will be discussed below, the line shows a spectral profile, ϵ_{ul} rather than a sharp wavelength, which is emitted over a spectral range $\Delta\lambda$, ϵ_{ul} and ϵ_λ are related by the following expressions:

$$\epsilon_{ul} = \int_{\Delta\lambda} \epsilon_\lambda \cdot d\lambda, \tag{4}$$

where ϵ_λ is expressed in $W/sr \cdot m^3$.

The measurement of ϵ_{ul} gives only the population of the upper level of the transition under interest. To determine a temperature, assumptions are necessary. For the considered chemical species i (atom, ion, molecule), Boltzmann's distribution holds for the excited states with a given temperature (T_{ex} for atoms, T_{rot} for rotational states, T_{vib} for vibrational states)

$$\epsilon_{ul} = 1/(4\pi) A_{ul} \cdot h \cdot \nu_{ul} \cdot \frac{n_i}{Z_i} \cdot g_{i,u} \exp\left(-\frac{E_{iu}}{kT_{ex}}\right) \tag{5}$$

where

E_{iu} is the energy of the upper excited state of chemical species i;

n_i the concentration of the chemical species i;

Z_i the partition function of the chemical species i calculated at T_{ex};

$g_{i,u}$ the statistical weight of the upper excited state of the chemical species i.

The temperature may be deduced from Eq. (5) using various approximations about equilibrium. It is clear, for example, that in the simple case of Complete Local Thermal Equilibrium CLTE where Z_i, n_i can be calculated versus temperature for a given pressure, (see, for example, Drellishak, 1963), the absolute measurement of ϵ_{ul} will give the unique temperature T of the plasma.

2. *Experimental Techniques and Measurement Problems*

As already emphasized (see Section III.B) one of the main problems when performing measurements using emission spectroscopy is the very steep gradients of thermal plasmas. This necessitates the deconvolution of the integrated measurements and automating them to keep data aquisition time reasonable. Care also has to be taken for the high radiation from the central part of the plasma and scattered by walls, optics.

Spectroscopic Instruments. For optical diagnostics, the plasma is imaged on the entrance slit of a monochromator, and the portion of the spectrum

falling on the exit plane is then detected using photographic plates (an old technique), photomultipliers, or a vidicon tube.

The optical arrangement generally consists of two parts: a focusing system and a spectroscopic instrument. The focusing system contains lenses, mirrors, spectral and neutral filters, the role of which is to image the plasma at known magnification on the entrance slit of the spectroscopic instrument. Diaphragms are also necessary to limit the diffusion of the very intense light from the plasma. There are three types of spectroscopic instruments: prism instruments, interferometers, and, the most common, grating monochromators. Their main characteristic lies in the resolving power which indicates the ability of the instrument to separate spectral images of two lines lying close together. If $\Delta\lambda$ is the difference between the wavelength of each line occurring at mean value λ, the resolving power R is given by the formula:

$$R = \frac{\lambda}{\Delta\lambda}. \tag{6}$$

For grating monochromators, R is a function of the slits widths, the focal length of the instrument and of the total number of grooves in the grating. (Born and Wolf, 1965; Fauchais et al., 1980). To separate two rotational lines in a molecular spectrum, a resolving power as high as 150,000 is typically required.

System calibration is a vital part of temperature measurements. The objective is to determine the response of the detector (electric current for a photomultiplier P.M., or the count number for an optical multichannel analyser) to the radiant power coming from the source through the collection optics and the dispersive instrument. The method used for the calibration is described in detail by (Fauchais et al., 1980) and (Coudert, 1985) and requires an intensity standard, commonly a tungsten ribbon lamp. The power supply driving the electric current through the lamp has to be very stable and the lamp must be run under the conditions specified by the calibrating laboratory (lamp orientation, part of the filament to be used, and ambient temperature). These lamps are calibrated in terms of radiance temperature versus current, and data on the emissivity of tungsten at various wavelengths is incorporated to derive the radiance of the lamp in each spectral region.

Other available standards include the anode crater of a carbon arc burning in argon, corresponding to a radiance temperature of about 4,000 K (Jayroe and Fowler, 1966) and Xenon lamps which have a radiance temperature of close to 5,000 K but are troublesome to stabilize (Budde, 1962).

The intensity of the thermal plasma is always higher than the specific intensity of standards. Therefore calibrated neutral density filters are used.

Frequently corrections are also made for deviations from linearity in the detector response, but fortunately photomultipliers or optical multichannel analysers are fairly linear over at least three orders of magnitude of the light intensity. This range limits the temperature range that can be measured without modifying the amplification (see Section III.A) or the filters.

Characteristics of Thermal Plasmas. In emission measurements, one difficulty is that the intensity emitted from the regions close to the axis of the plasma is more than three orders of magnitude higher than the light emitted from the edge, which is from a few millimeters to a centimeter away from the axis. This means that light coming from the core of the plasma and scattered by the optics is of the same order of magnitude as the light emitted by the edges. Hence, in the worst cases, the edge profile is swamped by scattered radiation from the center.

As a consequence of these high gradients in light intensity, the experimental data have to be corrected for the effect of light scattering by the optical system (see Figure 13). This is done by conveniently placed diaphragms blocking light which would otherwise be scattered, or by using in the signal analysis a "scattering function" deduced from the signal given by a source with a well-known intensity distribution (Coudert, 1985). Care also has to be taken for light scattered by the background such as that scattered from the walls of controlled atmosphere chambers.

A problem with emission measurements on thermal plasmas is the number of different chemical species present and the tremendous number of lines and/or molecular bands which overlap in general. For example, in plasma spray jets in ambient atmosphere a few millimeters downstream from the nozzle exit, the chemical species are Ar, N_2, O_2, H_2, N, H, O. Thus the choice of wavelength windows is critical because the continuum

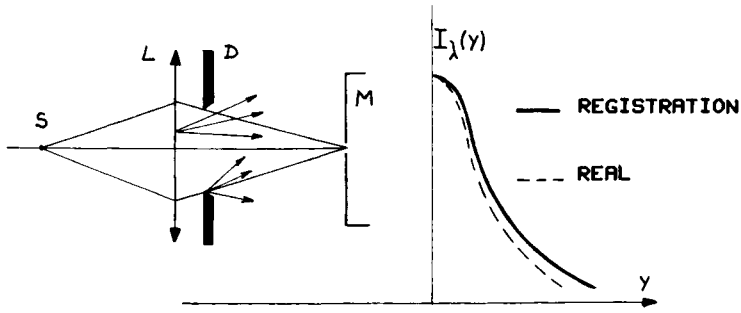


FIGURE 13. Schematic of light scattering effects.

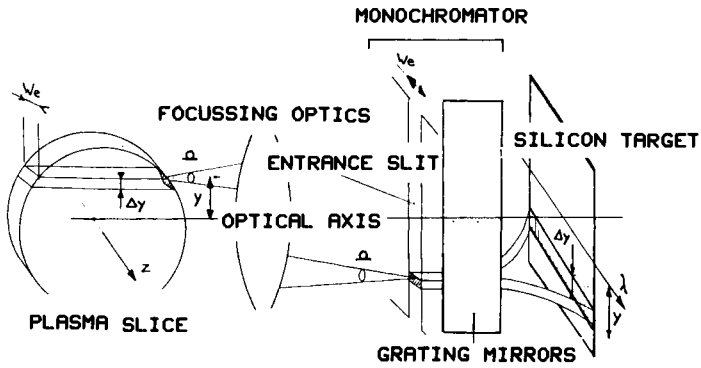


FIGURE 14. Observation of a plasma jet slice. (Coudert, J. F., 1985).

emission must be subtracted from all line intensities and overlapping lines have to be eliminated.

Abel's Inversion

Cylindrical Symmetry. What is observed spectrometrically is the integrated intensity of the light originating all along the line-of-sight (see Figure 14). Since the plasma is a very inhomogeneous (and often absorbing) medium, contributions from each local volumetric emission coefficient must be determined by deconvoluting the integrated measurements.

Light coming from the considered line of sight is in reality defined by:

- the magnification of the optical system which projects the image of the plasma onto the entrance slit of the monochromator;
- the aperture of the monochromator (mirror diameter/focal length);
- the width of the entrance slit, which is usually perpendicular to the plasma jet axis and varies from 5 to 60 μm depending on the type of analysis performed;
- the height of the entrance slit (or the height of the array in a vidicon tube), which defines the vertical dimension of the line of sight through the lens (see Figure 14).

A filter positioned in front of the entrance slit is necessary in order to reject the higher orders (diffraction from the grating) of the spectrum (in the case of grating monochromators).

If l is the thickness of the plasma along the line of sight at position y (see Figure 15a), for a given wavelength λ, the intensity seen by the optical

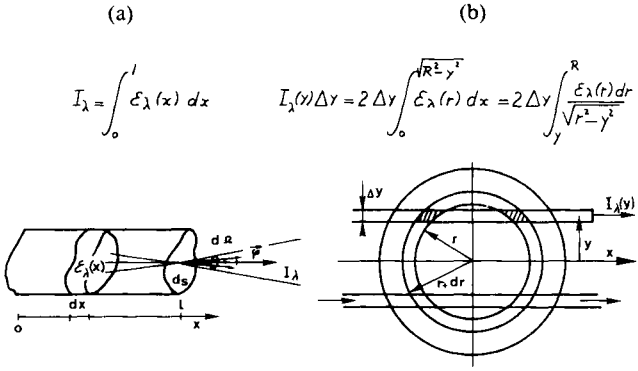


FIGURE 15. a) Relationship between ϵ_λ and I_λ. b) Abel's inversion in the case of cylindrical symmetry. (Coudert, J. F., 1985).

device is:

$$I_\lambda(y) = \int_0^l \epsilon_\lambda(x) \cdot dx, \tag{7}$$

and the Abel's inversion problem consists of determining ϵ_λ from the measured I_λ. Notice that I_λ is expressed in $W/m^3 \cdot$ ster while ϵ_λ is in $W/m^4 \cdot$ ster. From the geometry of a cylindrically symmetric plasma jet (see Figure 15b),

$$I(y) = 2\int_y^R \frac{\epsilon_\lambda(r)\,r\,dr}{\sqrt{r^2 - x^2}}, \tag{8}$$

which by inversion (Maecker, 1953) gives:

$$\epsilon_\lambda(r) = -1/\pi \int_{y=r}^R \frac{\dfrac{dI_\lambda(y)}{dy}}{\left(y^2 - r^2\right)^{1/2}}\,dy, \tag{9}$$

allowing the calculation of $\epsilon_\lambda(r)$ from the derivative of the curve $I_\lambda(y)$. Computing this derivative (via graphical, numerical, polynomial fitting) is one of the pitfalls in carrying out Abel's inversion, especially when the maximum emission of a line occurs at a temperature below that on the central axis of symmetry (Fauchais et al., 1980).

Since differentiation amplifies data noise, other methods have been proposed, such as the one by Barr (1962) which performs the differentiation after integration:

$$\epsilon_\lambda(r) = -\frac{1}{\pi}\frac{d}{dr}\int_r^R I(y) \cdot \left(y^2 - r^2\right)^{-1/2} \cdot y\,dy. \tag{10}$$

Deutsh and Beniaminy (1983) propose an expression which results from a partial integration of (8) followed by algebraic manipulation resulting in

$$\epsilon_\lambda(r) = -\frac{1}{\pi}[I_\lambda(R) - I_\lambda(r)](R^2 - r^2)^{-1/2} + \int_r^R [I_\lambda(y) - I_\lambda(r)] \cdot (y^2 - r^2)^{-3/2} y\, dy. \tag{11}$$

The two expressions (10) or (11) give fairly good results.

The precision of the Abel's inversion will also depend upon the noise with which the data are recorded as well as on the elimination of the outliers (Fauchais et al., 1984), for example, by numerical filtering using the Fourier's transform of $I_\lambda(y)$ (Kosarev and Pantos, 1983). Another important point that must be considered is that the result of the inversion will depend also on the wings of the profile $I_\lambda(y)$, which has to be accurately determined.

Quasi-Cylindrical Symmetry. In practice, emission profiles of plasmas measured perpendicularly to the plasma axis are often only approximately cylindrical. Another case where cylindrical symmetry will be destroyed is when particles are injected on one side of the jet, as under spraying conditions. Under these conditions, methods are necessary for asymmetrical plasmas.

One of the first asymmetrical methods (corresponding in fact to tomography) was developed by Maldonado and Olsen (1966). They used the basic integral equation:

$$I(\theta, x) = \int_{-\infty}^{+\infty} \epsilon(x', y') \cdot dy' \tag{12}$$

where θ denotes the angle of rotation of the source coordinates (x', y') relative to the fixed laboratory coordinates (x, y). The volumetric emission coefficient is represented by its series expansion, and the observed side-on-

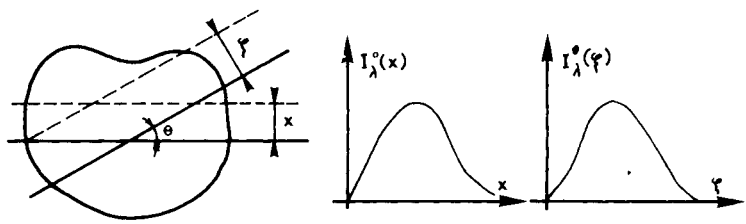


FIGURE 16. Abel's inversion in the case of quasi cylindrical symmetry. (Wetzer, J. M., 1983).

intensities $I(\theta, x)$ for various values of θ are used to derive the unknown expansion coefficients. Depending on the complexity of the shape of the plasma jet, this requires measurements along 2 to 8 directions (Gravelle et al., 1983), and then a complex mathematical treatment of the data. For this reason one prefers to use a less complex method, using two directions of the line of sight (Figure 16) (Wetzer, 1983) for simple shapes close to the cylindrical symmetry such as those obtained when injecting the particles with a single injector in spraying plasma jets.

Automation of the Measurements

General Remarks. To obtain line intensities or profiles, using Abel's inversion, intensity data are taken:

- At successive wavelengths (about 20 to 30 for a line profile, up to 100 for a rotational spectra or 2 for an atomic line intensity and its adjacent continuum). These intensities $I_\lambda(y)$ are recorded across the emitting plasma slice (see Figure 17a). Due to the high gradients, the diameter of the plasma jet has to be divided into at least 20 to 50 segments Δr (of about a few tenths of millimeters in arcs whose diameters are often smaller than 10 mm and up to millimeters in R.F. discharges or industrial plasma torches) for an accurate record;
- Next, the Abel's inversion is performed at each wavelength (see Figure 17b). If the plasma is unsymmetrical, this procedure has to be repeated at 2 to 8 angles of observation, which requires that an enormous quantity of data be treated. This is totally intractable without the help of a computer,

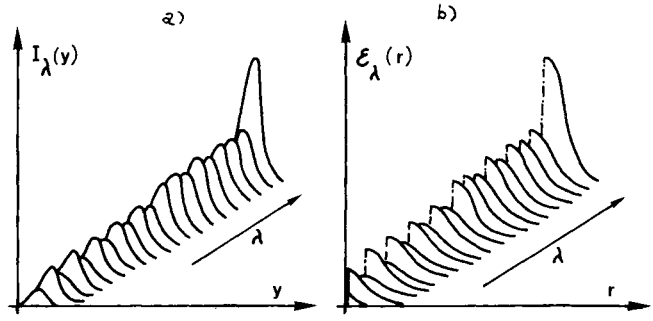


FIGURE 17. a) Registered spatial profiles for the intensities of global (P + R) lines and the head of (0-0) (1-) system of N_2^+. b) Radial profiles of the corresponding emission coefficients after inversion. (Coudert, J. F., et al., 1983).

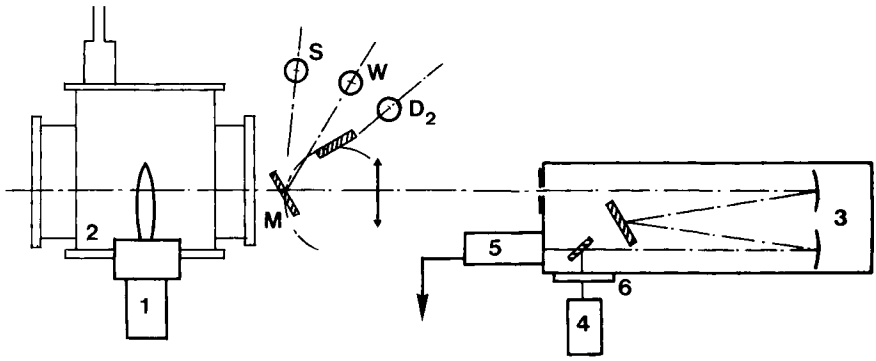


FIGURE 18. Spectroscopic set up of Baronnet, J. M., 1978.

and a number of automated methods have been developed, which differ mainly in the way the scanning is done. We will briefly describe the most important ones through examples.

Scanning by Source Displacement. In the method of Baronnet, 1978; Vardelle et al, 1980, the image of the plasma jet is focused with a magnification of 1 on the entrance slit of the monochromator with the help of a lens L 260 mm in focal length (see Figure 18). The vertical entrance slit of the monochromator is masked with a horizontal slit which defines the line-of-sight with a width of 0.2 mm. The plasma torch is moved perpendicular to the line-of-sight. The displacement is achieved using stepping motors, and the position of the analysed point is deduced from the signal given by a magnetic detector (see Figure 19). Successive recordings are obtained for the intensity profiles (at different wavelengths) of the line along the diameter of the plasma jet slice, and for the continuum plus an energy calibration (with the help of a mirror M to measure the intensity of a calibrated light source through the same lens and with the same optical path). All these operations are performed for a new position along the axis of the plasma jet while the preceding profiles are processed in the computer (smoothing, Abel's inversion) and plotted. Due to the necessary stepping times, it takes hundreds of seconds to acquire a record at a given axial position and, provided the plasma jet is stable enough, the recorded values are mean values (natural fluctuation of an arc root is in the range 1 to 10 kHz). The monochromator has a focal length of 1.5 m and uses holographic gratings (120 mm wide, with 2,400 grooves/mm for wavelengths between 2,000 and 8,000 Å). The inverse spectral dispersion is 1.3 Å/mm at 2,000 Å and 0.45 Å/mm at 7,500 Å. A rotating mirror M can intercept the optical path and

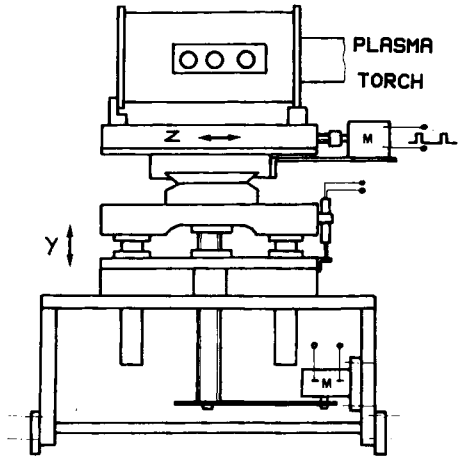


FIGURE 19. Experimental set up for scanning the jet by torch displacement. (Baronnet, 1978).

gives the image of a calibration source (spectral lamp *S*, tungsten ribbon lamp, deuterium lamp) on the entrance slit of the monochromator.

Scanning with Mirrors. Scanning of the jet (radial and longitudinal) can also be achieved with rotating mirrors (Etemadi and Pfender, 1982). Figure 20 shows a schematic view of the entire arrangement which uses an arc generated in a spherical test chamber as the plasma source. Using conventional means, the arc is focused on the entrance slit of a monochromator. An iris diaphragm reduces spherical aberrations and matches the solid angle of the incoming radiation with the aperture of the monochromator. A filter eliminates radiation of the second order.

Radiation from the plasma source passes through a mirror system which allows horizontal, as well as vertical, scanning of the plasma source. The rotating mirror, driven by a DC motor of variable speed from 3 to 100 rpm, allows horizontal scanning of the source. Since the distance between the arc and the rotating mirror is approximately 100 cm, whereas the arc diameter is the order of 1 cm, the error introduced by the changing viewing angle is negligible. Vertical scanning is accomplished by a pair of coupled stepping mirrors which move relative to the rotating mirror, keeping the length of the optical path constant; i.e., the source image on the entrance slit of the monochromator remains in focus.

For data acquisition, the rotating mirror sweeps the image of the plasma source across the entrance slit of the monochromator. This scanning may be

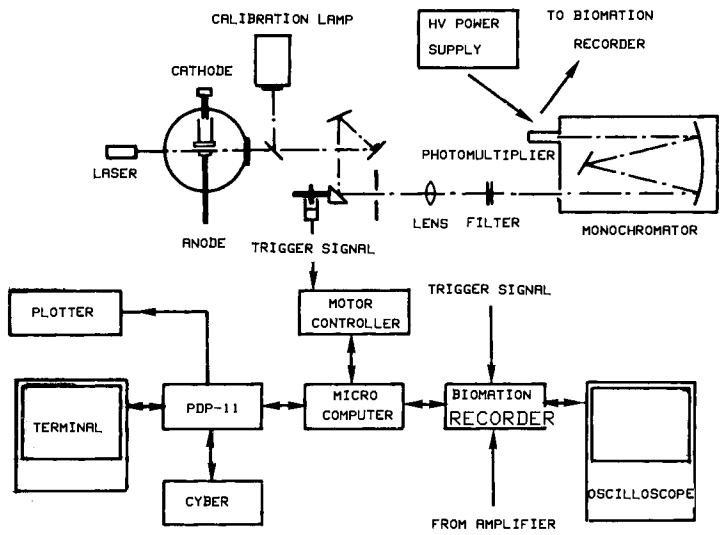


FIGURE 20. Spectroscopic set-up involving mirror-assisted scanning of the plasma. (Etemadi, K. and Pfender, E., 1982).

much faster than stepping motors automation, and the data, corresponding to a sweep of the image at a given wavelength, can be stored in a transient recorder armed by a trigger switch attached to the shaft of the motor controlling the mirror displacement. These data are then transferred to a minicomputer before the next sweep. Care must be taken not to sweep too rapidly because values then correspond to erratic random positions of the arc and intensities are not properly time averaged (see Section III.D).

Fast Temperatures Measurements Through Two Line Intensities. Using this technique, (Mehmetoglu et al., 1983), the image of an element of a plasma slice (as in the preceeding cases) is focused, through filters on two photodiodes working at two different wavelengths.

The experimental set-up is shown in Figure 21a. It consists of a two-wavelength pyrometer, a metal blocking plate (coupled to a linear motor) which varies the amount of plasma radiation seen by the pyrometer, a digital oscilloscope which stores radiation intensity signals, and an interface-minicomputer assembly which processes the pyrometer signals (as stored in the oscilloscope) to yield plasma temperature profiles. Then a metal strip (2×0.6 mm, blackened to prevent reflection of radiation) is moved linearly at a constant velocity away from the optical system through

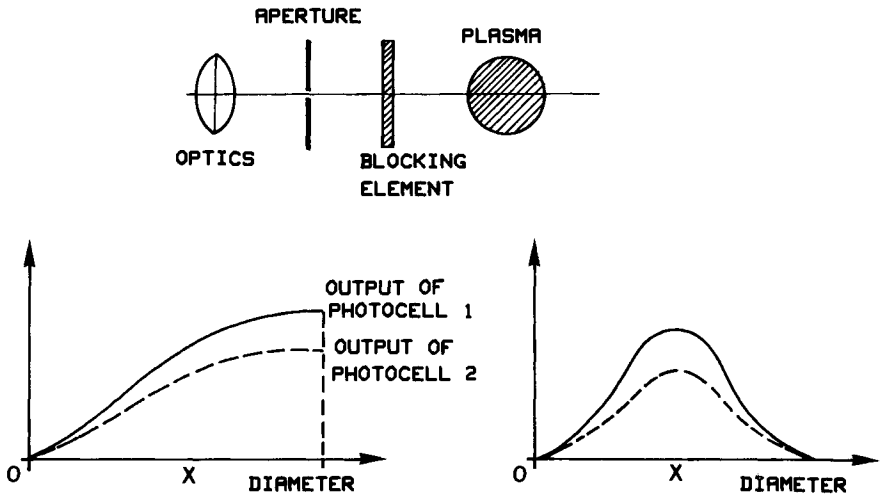


FIGURE 21a). Principle of scanning by moving a metal strip. (Mehmetoglu, M. T. et al., 1983).

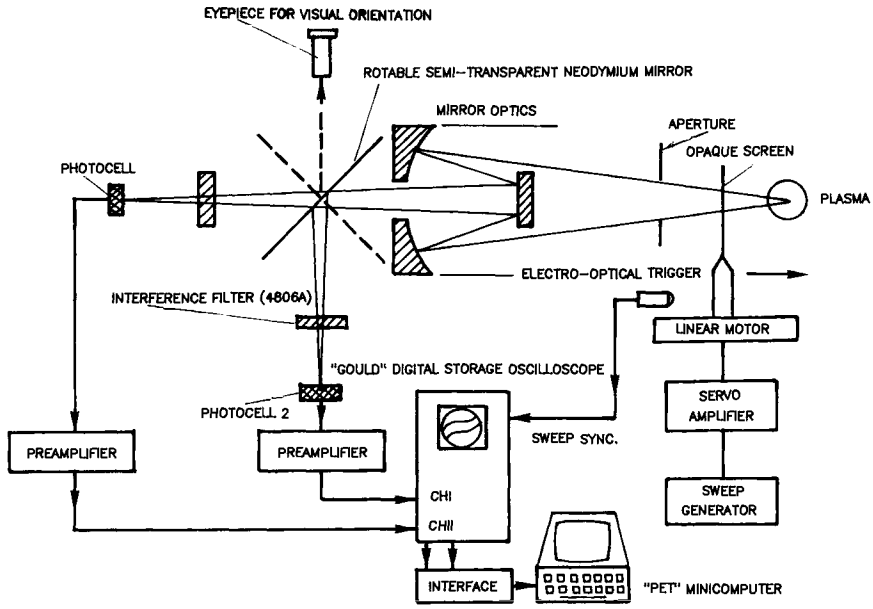


FIGURE 21b). Block diagram of complete temperature diagnostic system of Mehmetoglu (1983).

the plasma (see Figure 21b), and the two line intensity variations as a function of the metal strip position are recorded. Thus the radiating volume of the line-of-sight seen by the photodiodes increases from zero (at the beginning of the traverse) to a maximum when the strip completes its traversal movement. Then the Abel's inversion is performed and the $\epsilon_\lambda(r)$ obtained. Usually, there is a negligible perturbation of the jet by the fast transient strip which has to be accounted for.

Use of an Optical Multichannel Analyser (OMA). The use of an OMA detection system makes the analysis considerably easier (Courdert et al., 1982; Coudert et al., 1983; Coudert, 1985). The image of the plasma is focused with a magnification of unity on the entrance slit of the monochromator with the help of a lens with 260 mm focal length (see Figure 14). The detector of the OMA is located at the position of the photographic plate of the monochromator (the same as the one previously described). It consists of a 512×512 intensified silicon diode array target (SIT) scanned by a reading electron beam. The signal intensity is a measure of the integrated number of photons impinging on an element of the target matrix. The target, which is the equivalent of a photographic plate is software divided into a chosen number of tracks which can be chosen by the operator, each of them involving up to 500 channels. Each track is associated with a given spatial position, but with a different wavelength associated with each channel number within a track. Thus it is possible to simultaneously register spectra emitted in a given wavelength range across various lines-of-sight of the plasma as shown in Figure 14. A scan section of the image is thus obtained, which can be used to deduce the illumination profile, taking into account the OMA apparatus function, deduced from previous calibrations (Coudert, 1985). One-line signal conditioning includes deconvoluting the scattering function and producing numerical filtering of noise as was done for the rotational spectra represented in Figure 17. Figure 17a shows the filtered signal obtained on the OMA for a rotational spectra. Each vertical track of the detector, corresponding to a given wavelength, allows it to record the line profiles along the different lines of sight (see Figure 14). An OMA provides the only means to simultaneously, (i.e., in a few msec), sample all the data relative to rotational spectra at different positions of the plasma jet cross section. Once these intensities are stored, the numerical processing is essentially the same as in other cases, i.e., Abel's inversion is performed and the parameters such as temperature and population are deduced from the measured volumetric emission coefficients represented in Figure 17b.

3. *Temperature Measurements*

Temperature or Temperatures. Important parameters that need to be measured (see Section I) are the heavy species temperature (T_h) and the electron temperature (T_e) resulting from elastic collisions. However, what is measured are fluxes resulting from bound-bound, free-bound or free-free electronic transitions (resulting from inelastic collisions).

Even if most photons from high energy levels transitions escape plasmas (i.e., the plasma is optically thin) and Planck's law does not hold anymore, Maxwell-Boltzmann and Boltzmann distributions as well as the law of mass action may still hold for a unique temperature (often assumed to be the electrons' temperature). This type of equilibrium is called Complete Local Thermal Equilibrium (CLTE), and it may be assumed to hold at atmospheric pressure in regions where $n_e > 10^{22}$ e/m^3. It depends strongly on the electron concentration because collision between all species and electrons are the principal ways the energies are distributed between quantum states. When the electron concentration is not high enough (below 10^{22} e/m^3), as in the plume of a plasma spray jet or when a cold carrier gas is injected into the plasma core, nonequilibrium populations can be observed (Coudert, 1985) with ratios $T_e/T_h \sim 1.1$ to 1.3. Then it may be shown that quantum levels of the atoms or ions close to the ground state are no longer in equilibrium with the other states (the case of Partial Local Thermal Equilibrium, or PLTE). Only the levels close to the ionization limit show an equilibrium population related to the collisions with the electrons, i.e., to the mean kinetic energy of the electrons [see Eq. (1)]. From these excited levels close to the ionization limit, a temperature, called the excitation temperature, can be deduced.

Eddy et al. (1973) and Eddy (1976, 1985), for example, gives a critical review of spectroscopic diagnostic techniques in the case of a multi-temperature plasma where different temperatures result from collisions between the different species and from the corresponding relaxation times (Chapelle, 1971). In that case various temperatures may be distinguished:

- the electron temperature T_e (mean kinetic energy);
- the excitation temperatures (the one deduced from the excited levels close to ionization T_{ex}^{i} being different from the one deduced from the levels close to the ground state T_{ex}^{g});
- the vibrational temperature T_{vib}, related to the energy distribution among the vibrational levels;
- the rotational temperature T_{rot}, related to the energy distribution among the rotational levels;

- the kinetic temperature T_h of the heavy particles (usually one can assume that $T_{rot} = T_h$, the relaxation times between these two degrees of freedom being very short, smaller than 10^{-9} sec).

In the general case when the collisions are not frequent enough to bring about the equalisation of energies, it is expected that

$$T_e > T_{ex}^i > T_{ex}^g > T_{vib} = T_{rot} = T_h .$$

However, situations where $T_e < T_{ex}^i$ have been observed (Eddy and Sedghinasab, 1987). In conclusion, one must be aware that measurements from plasma radiation (except of T_{rot}) will provide temperatures which are not equal to the temperatures T_e and T_h.

Temperature from Absolute Atomic Line Intensity. After calibration, absolute intensity measurement is the fastest method to obtain the plasma temperature because only the intensity of one line has to be measured. For this purpose it is not necessary to register the line profile; it is sufficient to measure the signal with the monochromator turned to the wavelength of the line and using a narrow entrance slit w_1 and a wide exit slit w_2 (Born and Wolf, 1965; Fauchais et al., 1980). Thus, only one Abel's inversion is necessary at one wavelength against 20 to 30 inversions when the line profile has to be scanned (with a narrow exit slit). However, to obtain the true line intensity, the continuum intensity has to be subtracted from the measured signal (thus a second measurement has to be performed, with Abel's inversion, at another wavelength close to that of the line). It is important, however, to remember that the signal from the line is proportional to w_1 while that from the continuum is proportional to $w_1 \cdot w_2$ when working with a wide exit slit configuration. For complex mixtures such as those used in thermal plasmas, the continuum may be enhanced by the presence of a specific chemical species, (for example, Argon in a Nitrogen plasma), rending the measurement of certain lines difficult.

According to Eq. (3), the atomic line intensities are related directly to the population of the upper level and to temperature by Eq. (5) only if CLTE prevails. If PLTE prevails Eq. (5) may be considered to define a "population" temperature which is not the excitation temperature T_{ex} previously defined. Moreover, one has to keep in mind that Eq. (5) can be used only if the plasma composition at the location of the measurement is the same as that of the cold plasma gas injected in the plasma torch. This composition can be strongly modified by the surrounding gas entrainment (see Figure 12) and if it is not taken into account for the calculation of the composition, errors up to 20% can be made in temperatures deduced from absolute intensity measurements. The relative error in the temperature (with no

modification of the plasma composition) can be written

$$\frac{\Delta T}{T} = \left(\frac{\Delta \epsilon}{\epsilon} + \frac{\Delta A}{A}\right) \cdot \frac{kT}{E_{i,u}} \tag{13}$$

where A and ϵ are, respectively, the transition probability and the volumetric emission coefficient. In Eq. (13) the relative error in the partition function is not taken into account because limitation theories would be considered, and it is generally assumed that all the calculations of composition and partition function have been performed with the same partition function limitation theory (Fauchais et al., 1969; Fauchais et al., 1975). However, limitation theories and calculation of the partition functions with numerous excited levels have to be considered only at temperatures for which the intensity of the considered line reaches its plateau (see Section III.A) i.e., when the precision of the method is very poor. The term $kT/E_{i,u} \ll 1$ in (13) allows one to obtain a rather good precision for T even with significant errors in ϵ and A. Following (Chapelle, 1971) by taking $\Delta\epsilon/\epsilon = 10\%$,

$$\frac{\Delta A}{A} = 30\%, \quad E_{i,u} = 10 \text{ eV}, \quad T = 1 \text{ eV}$$
$$\frac{\Delta T}{T} = 4\%. \tag{14}$$

Temperature from Relative Line Intensities, Boltzmann's Plot. Taking the ratio R of the emission coefficients ϵ_1 and ϵ_2 of two lines emitted by the same particle (atom or ion), it can be seen, from Eq. (5), that the partition function Z_i and the particle density n_i is eliminated from R:

$$\epsilon_1/\epsilon_2 = A_1\nu_1 g_1/(A_2\nu_2 g_2) \exp\left[-(E_1 - E_2)/(k \cdot T_{ex})\right]. \tag{15}$$

By obtaining transition probabilities A_1 and A_2, the line frequencies ν_1 and ν_2, the degeneracies g_1 and g_2 as well as the energies E_1 and E_2 of the upper excited levels from tables, and ϵ_1 and ϵ_2 from measurements, T_{ex} can be calculated from Eq. (15). However, according to the equilibrium conditions, it is better to use energy levels close to the ionization limit, which can be assumed to be in equilibrium between themselves, since levels close to the ground state are often underpopulated. Then, the relative error $\Delta T_{ex}/T_{ex}$ given by:

$$\frac{\Delta T_{ex}}{T_{ex}} = \left(\frac{\Delta(\epsilon_1/\epsilon_2)}{\epsilon_1/\epsilon_2} + \frac{\Delta(A_1/A_2)}{A_1/A_2}\right) \cdot \frac{kT_{ex}}{E_1 - E_2} \tag{16}$$

may be important.

Although the ratios ϵ_1/ϵ_2 and A_1/A_2 can be obtained with a better precision than the absolute values, the error in temperature $\Delta T/T$ is always

large because when choosing levels close to ionization limit, $E_1 - E_2$ is small (less than 1 eV which is of the order of kT for thermal plasmas). Improved accuracy is supposed to be obtained by measuring a number of lines and using a Boltzmann's plot. To do this, Eq. (5) can be written:

$$\ln\left(\frac{\epsilon(\lambda)}{\nu \cdot A \cdot g}\right) = -\frac{E_u}{kT_{ex}} + K \tag{17}$$

where $K = \ln(hn_i/4\pi Z_i)$.

All the parameters in K are independent of the observed lines and, provided that A and g are known for several lines, a plot of the LHS of (17) versus E_u gives a straight line whose slope is proportional to T_{ex}^{-1}. However, one must be aware of the underpopulation of the levels close to the ground state (PLTE). This can be readily seen because the Boltzmann plot is no longer a straight line and often (Eddy, 1976) two lines with different slopes can be plotted, each one giving a different excitation temperature.

According to the quantity of data to be treated, the relative line intensity method requires four measurements (2 lines, 2 continua) and four Abel inversions, whereas the Boltzmann plot requires at least 5 to 10 lines plus 5 to 10 continua. Even though the Boltzmann plot method may be somewhat more accurate, the relative intensity method is often preferred because of its simplicity. However, in both cases accurate values of the transition probabilities have to be used.

Temperature from Rotational Spectra. Molecular spectra are far more complex than the atomic ones since the first include transitions between electronic levels (for which the mean energy difference is between 1 and 10 eV), vibrational levels (for which $\Delta E_v \sim 0.1$ eV), and rotational levels (for which $\Delta E_r \sim 10^{-4}$ eV). The spectra, for a transition between two electronic levels, will have a coarse vibrational structure with a large number of bands because there is no selection rule for the vibrational quantum numbers in that case (see, for example, Figure 22a representing a few bands of N_2 spectra). Each band by itself has a very complex rotational structure that is classified into branches according to the value of $\Delta J = J' - J'' = 0 \pm 1$ (Herzberg, 1959; Baronnet, 1978; Fauchais et al., 1980), (index $'$ is for the upper level and $''$ for the lower one).

The study of rotational spectra which gives T_h requires a high resolution monochromator because the rotational lines are very close to each other (it is not unusual to find up to 10 lines in 1 Å). In most cases the lines overlap, thus defining "global" lines from which it is possible to define a rotational temperature (Baronnet, 1978), provided overlap is only from superposition of branches within the same band. See for example Figure 22b, which

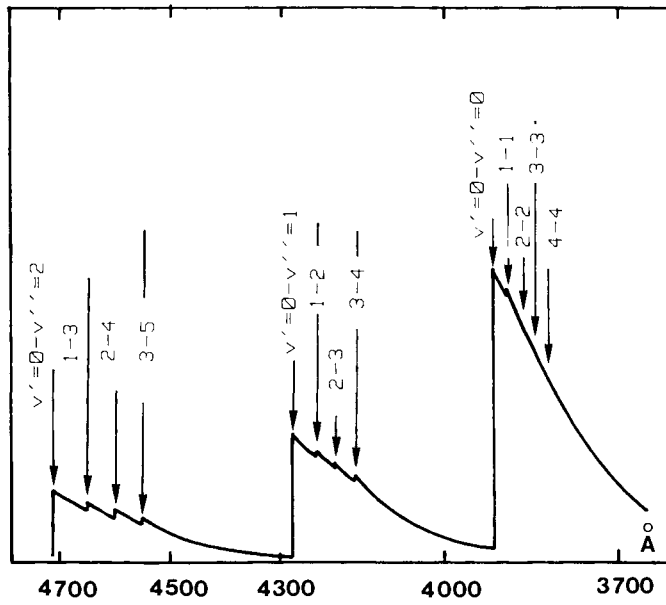


FIGURE 22a). N_2^+ vibrational bands.

shows spectra of N_2^+ at a rather high resolution (R = 300,000, see Eq. (6)) and the overlapping of the P and R branches (the band head contains only P lines and for the rotational number $K_p > 26$, a line is a global line resulting from the overlapping of a P line $K_p = K_R + 27$ and an R line). To our knowledge, only spectra of diatomic nitrogen have been used to measure the kinetic temperature for thermal plasmas. The nitrogen is introduced either as main plasma gas (N_2-H_2 mixtures), by adding less than 1% to an Ar-H_2 mixture, or by entrainment from the ambient air when a jet is flowing in free atmosphere. The measurements are performed either from N_2^+ spectra (1^-) in the spectral range 3,300–4,700 Å, or from N_2 spectra (2^+) in the spectral range 2,200–4,000 Å. The advantage of N_2^+ (1^-) is its simplicity: there are only two branches versus six branches for N_2 (2^+). These spectra allow temperature measurements between 3,000 and 10,000 K by three methods:

a) Measuring the absolute intensity of a few lines and deducing a temperature from these measurements (due to the difficulties of obtaining isolated lines, it is better to use a few lines and obtain a mean value of

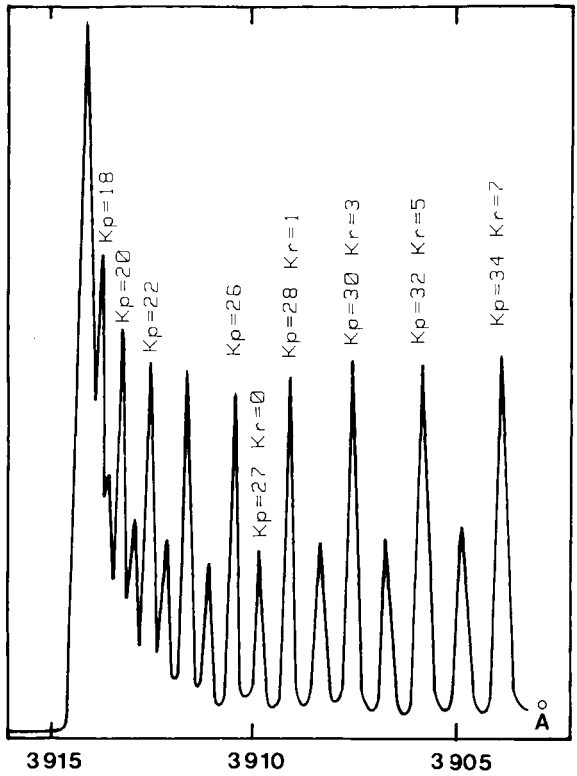


FIGURE 22b). Rotational structure of N_2^+ first negative system (0-0) band. (Baronnet, J. M., 1972).

the temperature from these measurements) (Fauchais, 1968; Baronnet, 1972; Baronnet, 1978);

b) Measuring the ratio of two adjacent groups of lines. This ratio depends only on T_h. The use of a group of lines greatly improves the accuracy of measurements;

c) Comparing simulations of integrated rotational spectra computed for a given monochromator, a given grating, and given entrance and exit slits, to the measured spectra (taking into account the Abel's inversion) under the same conditions. Such a comparison has been performed in the two following ways:

- Simulation of N_2^+ (1^-) $(v' = 0,\ v'' = 0)$ transition, whose spectral decay between 3730 and 3840 Å, plotted in logarithmic units, gives a

quite perfect straight line for each temperature. The slope of such a straight line appears to be a linear function of $(1/T_h)$ (Coudert et al., 1982);

- Simulating the N_2^+ (1^-) $(v = 0,\ v = 0)$ transition and studying for each global line J, the ratio:

$$I_j^{th}(T) = \frac{\epsilon(\lambda_j, T)}{\epsilon_{head}} \tag{18}$$

where $\epsilon_{head}(T) = \sum_{K=1}^{26} \epsilon_P(K, T)$ and $\epsilon(\lambda_i, T) = \epsilon_P(\lambda_i) + \epsilon_R(\lambda_i)$.

After recording the experimental spectra, involving N global lines (P and R overlapped lines) the temperature is calculated as the parameter which minimizes the quantity

$$E(T) = 1/N\left[\sum_{i=1}^{N}\left(I_i^{exp} - I_i^{th}\right)^2\right]^{1/2} \tag{19}$$

N being the number of global lines under analysis. See for example Figure 22b of the detailed structure of the N_2^+ O-O band for which Coudert et al., (1983) has studied the global lines corresponding to $K_P = 27$ and to $K_R = 38$.

Temperature from Vibrational Spectra. In a system of electronic bands, the intensities I of the vibrational bands $v' - v''$ are related to the population densities $n(v)$ of the vibrational levels and therefore to the vibrational temperature defined by:

$$n(v) \sim \exp\left(-\frac{G(v)hc}{kT_{vib}}\right)$$

with

$$G(v) = \omega_e(v + 1/2) - \omega_e x_e(v + 1/2)^2 + \cdots \tag{20}$$

ω_e and $\omega_e x_e$ being the vibrational constants (Herzberg, 1959).

Thus, there are three ways to measure the vibrational temperature without taking into account the rotational temperature:

a) Use the sum of the volumetric emission coefficients ϵ_v (see Figure 22a) of the bands of a sequence v' (or v''), i.e., bands having the same initial level v' (or final level v''), which is proportional to the population $n(v')$ or $n(v'')$. This sum has to be divided by ν^4 to take into account the expression for the transition probability versus frequency and band strength $S_{v''v'}^{v'v'}$ Herzberg, (1959).

For a sequence v' one finds:

$$\frac{1}{\nu^4}\sum_{v'}\epsilon_v \sim n(v''), \tag{21}$$

and according to (20)

$$\ln\left(\sum_{v'}\epsilon/\nu^4\right) = -\frac{hcG(v')}{kT_{\text{vib}}} + \text{constant}. \tag{22}$$

Thus the slope of the straight line represented by the left hand side of Eq. (22) plotted as a function of $G(v')$ provides T_{vib}.

However, although only relative volumetric emission coefficients are required, the volumetric emissions of all the bands of a system have to be measured in order to find $\epsilon_{v'}$. The main problems are associated with the overlapping of the bands (see Figure 22a), the incomplete development of the bands, and the fact that the bands extend over a wide spectral range which necessitates numerous data at different wavelengths to obtain the volumetric emission coefficients from the band profiles after Abel's inversion.

b) The absolute volumetric emission coefficient of a vibrational band is given by:

$$\epsilon_v(\lambda) = (16\pi^3/3)\cdot c\cdot \nu^4 \cdot S_{\nu''v''}^{\nu'v'}\frac{n_v(T_{\text{vib}})}{Z_v(T_{\text{vib}})}\exp\left(-\frac{hcG'(v')}{kT_{\text{vib}}}\right), \tag{23}$$

which may be calculated for a mean wavelength or frequency ν_{00} corresponding to the center of the band as:

$$\nu_{00} = \omega_e(v' - v'')\left[1 - \omega_e x_e(v' + v'' + 1)\right], \tag{24}$$

where the constants ω_e, x_e are given by Herzberg, (1959).
Again two problems arise (Chapelle, 1971) which influence the accuracy:

- the accuracy of the theoretical values $S_{\nu''v''}^{\nu'v'}$ is poor;
- the definition of the band is difficult due to overlap.

c) The relative volumetric emission coefficients of two bands vary monotonically with temperature and if their values are measured at a certain temperature it is no longer necessary to know $S_{\nu''v''}^{\nu'v'}$ to determine T_{vib}, thus:

$$\frac{\epsilon_2}{\epsilon_1} = C\cdot\exp\left(-\frac{E_{v_1} - E_{v_2}}{k\cdot T_{\text{vib}}}\right). \tag{25}$$

However the same problems of overlapping arise again.

Electron Temperature. Electron temperatures cannot easily be measured and are therefore usually calculated.

Once the electron density has been measured, it is possible to calculate the electron temperature (Eddy, 1976 and 1985) by combining the Saha equation for ionization equilibrium with Boltzmann's law for the excited levels [Eq. (5)] resulting in:

$$\frac{N_e^2}{N_j} = 2\left(\frac{2\pi m_e kT_e}{h^2}\right)^{3/2} \frac{Q^+(T_e)}{g_j} \exp\left(-\frac{\Delta E_i - \delta E_i - E_j}{kT_e}\right) \qquad (26)$$

where N_j, g_j, and E_j are respectively, the density, degeneracy, and energy of the excited level j of an atom, ΔE_i and δE_i the ionization energy of the atom and its lowering, and $Q^+(T_e)$ the partition function of the ground state ion.

The ratio N_e^2/N_j is first measured (using line profiles or absolute intensity of excited states j) and thus T_e can be calculated from Eq. (26). However, as shown by Coudert (1985), levels close to the ionization limit have to be chosen. Moreover, T_e values deduced from this equation are not very realistic in the plasma jet fringes where only a radiative collisional model (such as the one of Medani (1981) for a nitrogen thermal plasma) gives realistic values of T_e once N_e is determined. These unrealistic values (see Figures 23 and 24) are probably due to the influence of the exponential term. This influence can be reduced by extrapolation to the ionization limit as done by Eddy et al. (1973) and Eddy (1985).

4. Electron Density Measurements

From Stark Line Profiles. As each atom interacts with its neighbours, there is a shift in the energy level from which an optical transition may occur. Compared to line emission from isolated atoms, the real line shows a statistical broadening caused by collisions. When the collision partners are neutral, the interactions are short range and give rise to pressure broadening. Longer range interactions with charged particles cause Stark broadening (in that case, the main contribution is due to electrons). Moreover, owing to the velocity of these radiating atoms, the wavelength is also shifted by the Doppler effect. Since the velocity (Griem, 1974) distribution is Maxwellian, Doppler broadening leads to a Gaussian profile, with a halfwidth directly related to the kinetic temperature.

The Stark effect is important as soon as the ionized species molar fraction becomes higher than 10%. From the calculations of Griem (1964), it can be shown, for thermal plasmas, that the total width of the profile at the

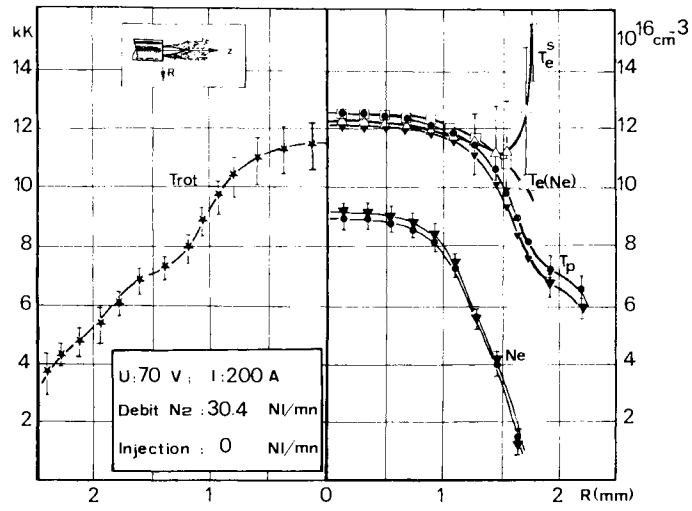


FIGURE 23. Radial profiles of the temperatures in a nitrogen plasma jet with no auxiliary gas injection. (Coudert, J. F., 1985).

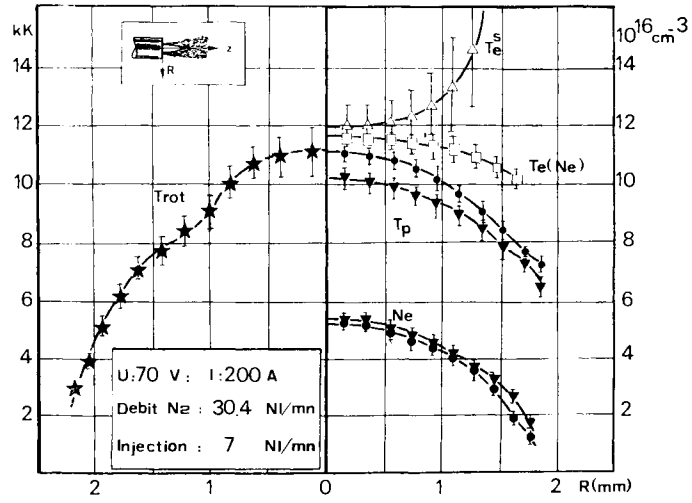


FIGURE 24. Radial profiles of the temperatures in a nitrogen plasma in which 7 l/min STP of nitrogen is injected symmetrically in the nozzle. (Coudert, J. F., 1985).

half-height $\Delta\lambda_s$ mainly depends on electron density.

$$\begin{aligned} &\Delta\lambda_s \propto n_e^{2/3} \quad \text{for linear Stark effect, eg } H_\beta; \\ &\Delta\lambda_s \propto n_e \quad \text{for quadratic Stark effect.} \end{aligned} \tag{27}$$

Thus a measurement of the profile of lines for which the Stark effect is predominant can give a direct measurement of the electron density n_e independent of whether the plasma satisfies LTE and whatever its temperature or chemical composition may be.

Measurements based on line profiles are more complex than those related to line intensities because one must determine the profile with precision, i.e., between 20 and 40 points across the profile (with the corresponding Abel's inversions). This requires a highly dispersive monochromator because the line widths are typically a few Angströms.

In thermal plasmas where $n_e > 10^{22}\ m^{-3}$ collisional broadening by neutral as well as resonance broadening can be neglected in general. The resultant line profile is a Voigt one, resulting from the convolution of the Stark and Doppler profiles (Griem, 1977).

For the Doppler effect, the full width of the line at its half maximum intensity $\Delta\lambda_D$ is proportional to the temperature of the heavy species (Griem, 1974). For a Maxwellian temperature distribution the profile is Gaussian:

$$\Delta\lambda_D = 7.16 \cdot 10^{-7}\lambda_0\sqrt{T/M}\,, \tag{28}$$

where M is the mass of the emitter (kg), T the temperature (K), and λ_0 the wavelength of the line (Å).

Usually the Doppler effect results in a small contribution to the line broadening. For example, at $T = 10{,}000$ K, $\Delta\lambda_D = 0.35$ Å for the H_β line at 4,861 Å, $\Delta\lambda_D = 0.09$ for NI nitrogen atomic line at 4,935 Å, $\Delta\lambda_D = 0.02$ for Argon atomic line at 6,965 Å.

The Stark broadening $\Delta\lambda_s$ which depends strongly on the electron density and can be calculated from:

$$\Delta\lambda_s = 2.10^{-6}\omega \cdot n_e\left[1 + 1.75\ 10^{-4}n_e^{1/4}\alpha\left(1 - 0.068\frac{n_e^{1/6}}{T_e}\right)\right] \tag{29}$$

where ω and α are tabulated by Griem, (1964), for various atomic lines. The Stark effect gives much broader lines than Doppler broadening and it is the main contribution to the Voigt profile. For example, at 20,000 K for $n_e = 10^{23}\ m^{-3}$: $\Delta\lambda = 48$ Å for H_β line, 1.14 Å for NI 7468 Å line.

The Voigt profiles of the lines are the convolution product of a Lorentzian profile (Stark effect) with a Gaussian one (Doppler effect):

$$P_v(\lambda) = P_D(\lambda) \times P_s(\lambda). \tag{30}$$

Hence a deconvolution of the line profile, taking into account the apparatus function of the monochromator must be performed to obtain $P_s(\lambda)$. However, since the true Doppler profile depends strongly on temperature, it will vary along the plasma radius and thus the deconvolution must be preceeded by an Abel's inversion.

Davis and Vaugham (1963), proposed a faster method for deconvolution of the experimental curves, which applies to thermal plasmas at atmospheric pressure where the Doppler width is about a tenth smaller than the Stark width. If $\Delta\lambda_s$, $\Delta\lambda_D$, and $\Delta\lambda_v$ are the width of the Lorentz (i.e., Stark), Doppler, and Voigt profiles respectively:

$$\Delta\lambda_s = \Delta\lambda_v\left(1 - 1.1\left(\frac{\Delta\lambda_D}{\Delta\lambda_v}\right)^2\right). \tag{31}$$

Once the temperature of the heavy particles is known, $\Delta\lambda_v$ can be calculated from (28) and $\Delta\lambda_s$, corresponding to Stark broadening, is obtained from the Voigt profile.

From Continuum Emission. The continuum radiation is mainly due to: a) photo recombination when a free electron is captured by a positive ion into a certain bound energy state, where the energy excess is converted into radiation; b) conversion of kinetic energy of a free electron into radiation when it slows down in the Coulomb field of positive ions (Bremsstrahlung). The calculation of the continuum for complex mixtures is long and tedious (Morris and Yos, 1971) because it involves many excited states and numerous photoionization cross-sections and Gaunt factors which are not necessarily known, as well as the plasma composition (again the remark about the surrounding gas entrainment is of primary importance for this calculation). For a single gas (Ar, H_2, N_2, ...) the total radiation continuum turns out to be independent of the wavelength λ for $\lambda \gg \lambda_g$ (λ_g corresponds to the threshold value for trapping an electron with near zero velocity). In that case (Kolesnikov, 1966), the volumetric emission coefficient ϵ_λ is given by:

$$\epsilon_\lambda \sim \frac{n_e n_1}{\sqrt{kT}}, \tag{32}$$

where n_e is the electron density and n_1 the density of first ion (which in most cases is equal to n_e).

Thus, for a pure plasma gas with excitations, in the visible range of spectrum, Eq. (32) can be used to determine n_e. However, for mixtures the determination of n_e from continuum measurements is far more complex.

5. *Example of Results*

Case of a D.C. Nitrogen Plasma Spray Torch. The data presented in this section are obtained with a nitrogen plasma produced by a D.C. torch. In order to see the effects of cold gas injection on the plasma behaviour, the nozzle of the torch has several holes through which an auxiliary flow of nitrogen or oxygen can be injected downstream of the arc root at the anode. The experimental arrangement consists of a high resolution monochromator (THR Jobin-Yvon) with a dispersion factor less than 2 Å/mm. The detector is a two-dimensional vidicon tube (O.M.A. 2D, P.A.R.C.) which allows both spectral resolution along the horizontal axis of the target and spatial resolution along the vertical direction. It has been calibrated over four decades by using a tungsten ribbon lamp and neutral density filters.

From the measurements, temperatures of the plasma jet are deduced from the population of atomic nitrogen levels emitting at 746.8 nm and 493.5 nm (population temperature). Electron concentrations are deduced from Stark profiles of these lines and, finally, rotational temperatures are deduced from the first negative system of the molecular N_2^+ ion. Electron temperatures are calculated either from electron densities assuming equilibrium [$T_e(N_e)$ in the figures], or from the Saha equation using the excited level population of nitrogen atoms together with the electron density [see Eq. (26) and noted T_e^s in the figures].

Comparison between Temperature Profiles. Figures 23 and 24 show the temperature profiles of a plasma produced with a nitrogen flow rate of 30 l(STP)/min and an arc current of 200 A (cylindrical nozzle 5 mm in diameter and 28 mm in length). In Figure 23, there is no auxiliary injection, but a 7 l/min flow rate of N_2 is injected in the case of Figure 24. The left part of the figure shows rotational temperature while the right shows temperatures deduced from atomic lines.

Figure 23 shows that temperatures deduced from the 746.8 nm line (full triangles), 493.5 nm lines (full circles), and from electron concentration (open triangles) are the same in the jet core ($r < 1.5$ mm). In the jet, there is some departure between the temperatures deduced from the two lines as well as an increase of the electron temperature deduced from the Saha equation, showing that Saha and Boltzmann equilibrium fail at the edges of the plasma jet.

All these effects are enhanced when cold nitrogen is injected. Here, the nonequilibrium effects illustrate the difficulty of selecting the best method for temperature measurements. Nevertheless, since the rotational temperatures are still well defined and remain a good estimate of the kinetic

temperature of heavy particles, it is our preferred choice of thermometer for $T < 10{,}000$ K.

Influence of Cold Gas Injection on the Rotational Temperature Isocontours. Although spectral analysis of molecular bands is complex and requires up to forty lines lying in a rather narrow wavelength range (less than 1 nm), the method has the great advantage that it does not require knowledge of transition probabilities or chemical composition.

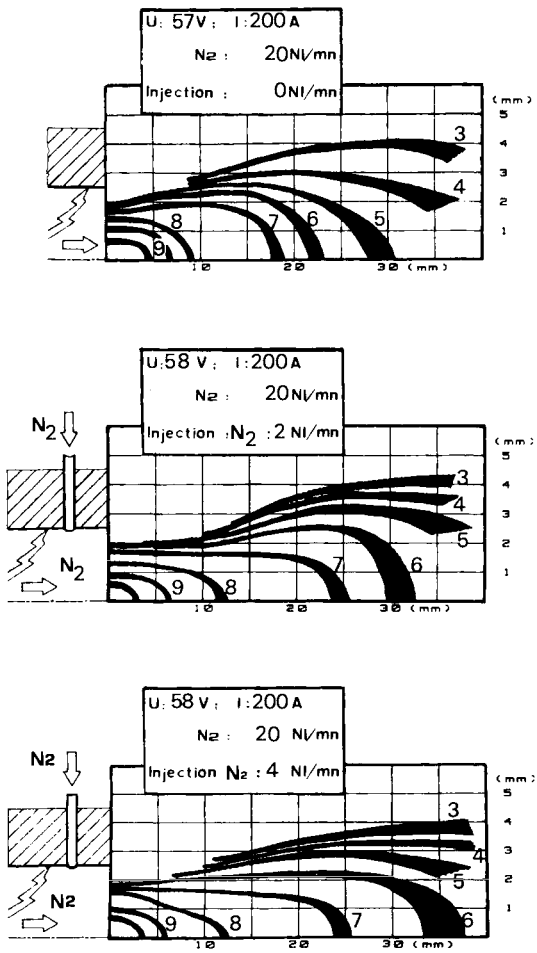


FIGURE 25. Comparison of the rotational temperature isocontours in the case of cold nitrogen injection. (Coudert, J. F., 1985).

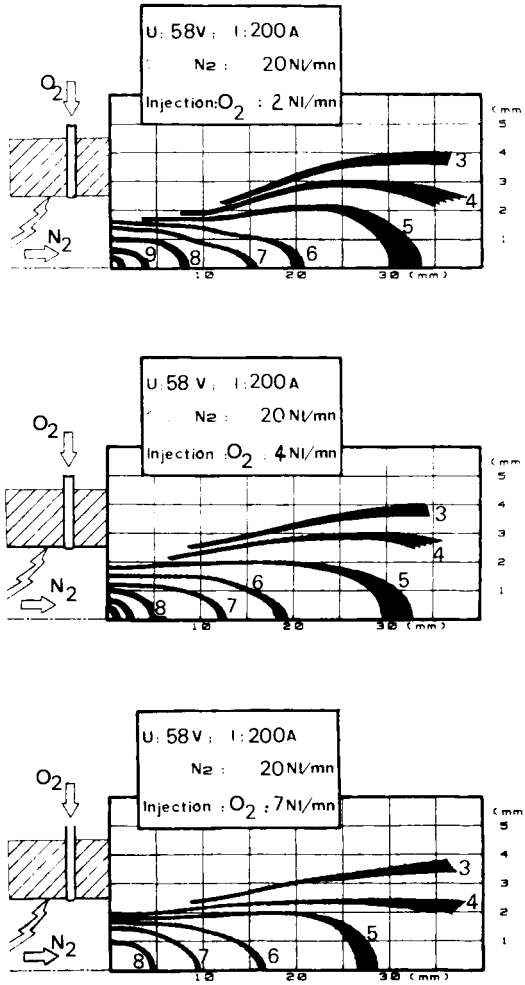


FIGURE 26. Comparison of the rotational temperature isocontours in the case of cold oxygen injection. (Coudert, J. F., 1985).

Figures 25 and 26 show the rotational temperature isocontours for almost the same plasma generation conditions as Figures 23 and 24, that is, nitrogen flow rate 20 l/min, 200 A. Figure 25a refers to a jet without auxiliary flow. Figures 25b or 25c, to plasma jets with, respectively, 2 and 4 l/min of nitrogen injected, and Figures 26a, b, c to 2, 4 and 7 l/min of oxygen. The isotherms which are drawn on these figures are ranging from 3,000 to 10,000 K. The spread of the isotherms reflects the precision of the

measurements. The region between 7,000 and 8,000 K shows the isocontours for the recombination zone where the exothermic reaction $N + N \rightleftharpoons N_2$ takes place, leading to a lowering of the temperature gradient. It may be seen clearly that cold nitrogen injection (compare Figures 25a, b, c) pinches and lengthens the isotherms while hardly modifying the temperature close to the nozzle exit.

The case of oxygen injection is very different (see Figures 26a, b and c), because of the rather low dissociation temperature of oxygen (about 3,000 K). Even in the fringes of the jet, the temperature is sufficient to dissociate O_2 and the net result is that the heat consumption during the dissociation cools down the jet (see for example, the evolution of the 8,000 K isotherm).

It should be pointed out that all these experiments were done in a controlled chamber, filled with nitrogen at atmospheric pressure. This shows the strong influence of the surrounding atmosphere on the plasma jet behaviour, and important differences may be expected when, for example, plasma spraying is performed in open air or in a chamber with a controlled atmosphere filled with argon at $p = 1$ atm (see Roumilhac et al., 1987 and M. Vardelle, 1987).

Temperature measurements by molecular spectroscopy are a very useful and sensitive method for the characterization of the jets produced by differently designed generators. Figure 27 shows the isotherms of two plasma jets produced with comparable specific enthalpies but with two different nozzle designs. The upper part of this figure relates to a plasma produced with a nozzle having a 3° divergent angle, the effect of which is to reduce recirculation of the surrounding gas and thus the cooling of the jet as demonstrated by the calculation of Pateyron et al. (1987).

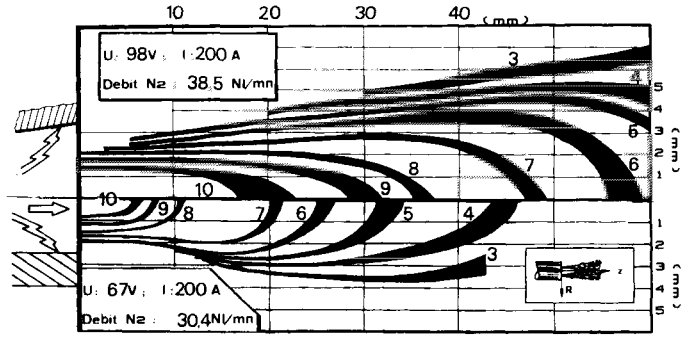


FIGURE 27. Comparison of the rotational temperature isocontours in plasma jets produced with two different generators. (Coudert, J. F., 1985).

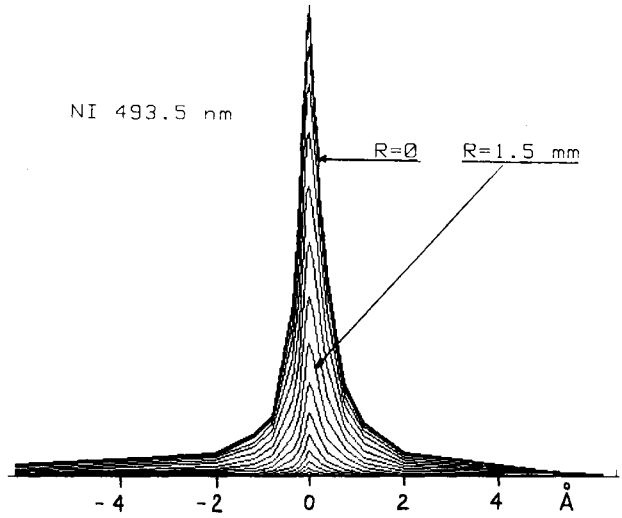


FIGURE 28. Reconstructed spectral profile of 4,935 Å NI line after Abel's inversion for each wavelength sample. (Coudert, J. F., 1985).

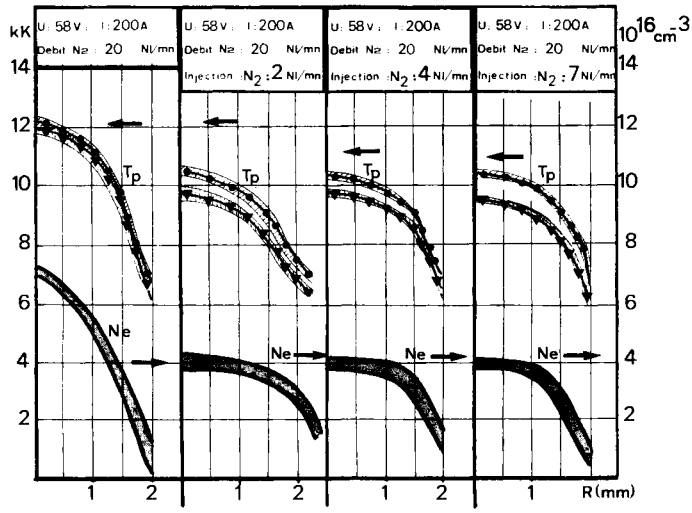


FIGURE 29. Comparison between radial profiles of electron concentration and of excitation temperatures deduced from two NI lines: 7,468 and 4,935 Å, in a nitrogen plasma jet with different auxiliary injection flow rates of cold nitrogen. (Coudert, J. F., 1985).

Electron Concentrations. Figure 28 shows the spectral profiles in emission for the 4,935 Å line of the nitrogen atom after reconstruction from the spectral volumetric emission coefficients obtained by Abel's inversion following the method mentioned above. The electron concentration N_e may be obtained for different distances from the jet axis, from the half width of these lines using Eqs. (29) and (30). Such results are presented in Figure 29 for the same plasma generation parameters given above (N_2: 20 l/mn STP, 200 A) but with different injection flow rates for the cold nitrogen : 0, 2, 4 or 7 l/mn STP. The temperature obtained from the absolute intensity of the 4,935 Å N line (full circles) and of the 7,468 Å N line (full triangles) are plotted in the figure and range from 12,000 to about 6,000 K. These figures show the fast decrease of electron densities by diffusion when the fringes of

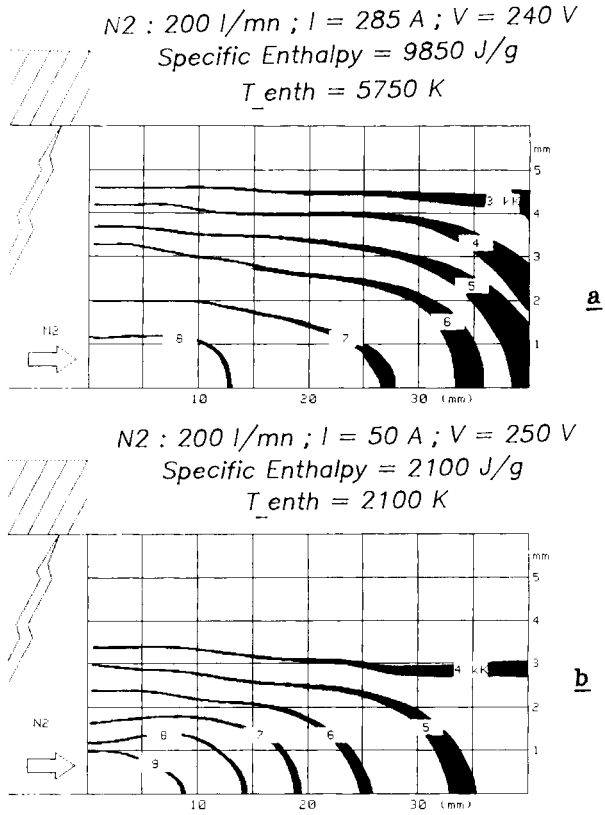


FIGURE 30. Rotational temperatures isocontours in plasma jets produced by a D.C. generator with a vortex injection. (Coudert, J. F., et al., 1987).

the jet are cooled down. It emphasizes also the close connection of the temperature deduced from atomic lines, with the electron density controlling the collisions between atoms and electrons.

Study of a Generator with Button Type Cathode and Vortex Injection. The torch used in this study was similar to the one in Figure 4 with an exit diameter of 12 mm (d_3) and a channel diameter of 6 mm (d_2). The Figures 30a and 30b show the rotational temperature isocontours deduced from N_2^+ spectral analysis by the method described before. These results are for a nitrogen flow rate of 200 l/min STP, with an arc current, of 285 A in Figure 30a and of 50 A in Figure 30b, respectively. It may be seen that the main effect of lowering the current is to reduce the radial extent of the isotherms. In the case of Figure 30b, the main part of the gas is flowing outside the jet core in which the temperatures remain similar to those in Figure 25a.

Case of a Transferred Arc in Argon. The left hand side of Figure 31 shows the experimental values of the temperatures in a transferred arc that was

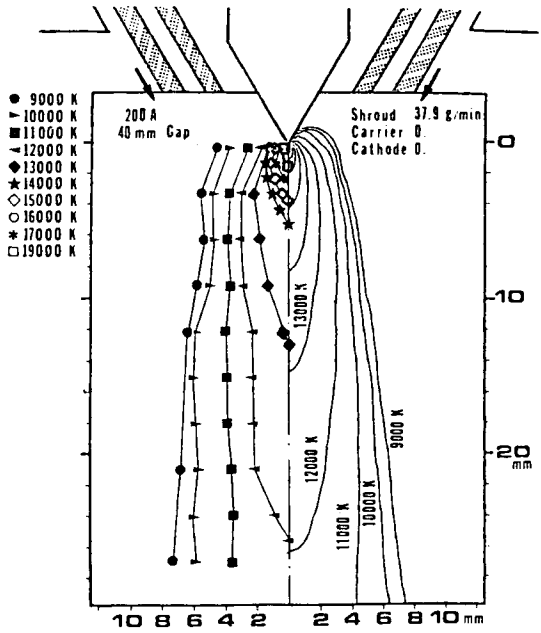


FIGURE 31. Comparison between measured and calculated isotherms in an argon plasma produced in a transferred arc reactor. (Young, R. M. et al., 1983).

obtained by using the experimental arrangement in Figure 2 (Young et al., 1983). These temperatures are deduced from the absolute intensity of the ionic line of Argon, at 4,806 Å, and from the continuum radiation at 4,312 Å for the lower temperature region. The right part of the figure is related to temperatures calculated from a model proposed by Hsu and Pfender, (1983).

Case of R.F. Plasmas. Since its development in the early sixties, the inductively coupled R.F. Plasma (ICP) has been developed for spheroidization, chemical synthesis, and mainly as an emission source in spectrochemical analysis in which it is being increasingly used on a commercial basis (see for example the reviews of Eckert, 1974; Barnes, 1978; and Trassy and Mermet, 1984). The success of these applications, however, depends on an understanding of the torch characteristics and the operating parameters which control them. Among the main parameters, the plasma temperature has been intensively studied using atomic line ratios either of argon lines or of injected elements such as Ca, Fe, and Mg, for which the ionization potentials are lower, allowing the measurement of lower temperatures than that obtained with argon (6,000 K is the lower limit of Ar). However, much

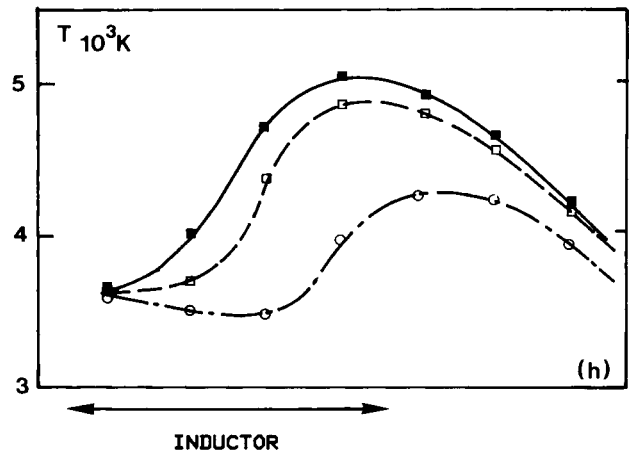


FIGURE 32. Evolution along the plasma axis of the excitation temperature deduced from Calcium atomic lines. The liquid containing Calcium is injected in the argon spectrochemical analysis plasma (DURR 56 MHz source) with different argon carrier gas flow rates D and different injector internal diameters ∅: ■∅2, D = 0.29; □∅2, D = 0.42; 0∅ = 1, D = 0.42 (∅ mm, D Nl/min). (Trassy, C. and Mermet, J. M., 1984).

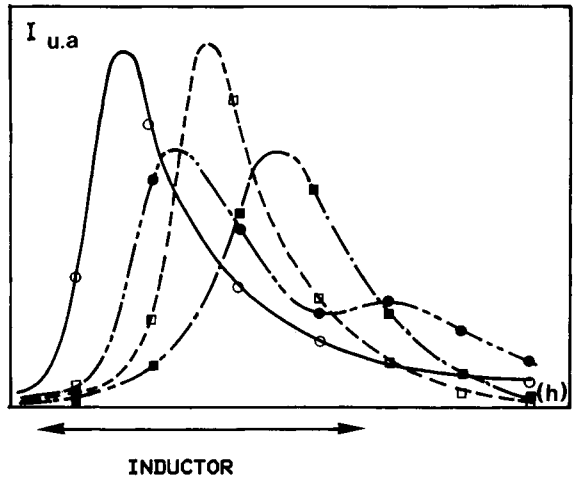


FIGURE 33. Evolution along the R.F. plasma axis of the line intensities (arbitrary units) of atomic and ionic iron lines. The liquid containing iron being injected with two different argon carrier gas flow rates D, D = 0.29, ○ FeI, □ FeII and D = 0.62, ● FeI, ■ FeII. (Trassy, C. and Mermet, J. M., 1984).

care has to be taken (Trassy and Mermet, 1984) to interpret the results obtained with lines of injected elements, because they first have to be heated, and the temperature that they reach directly controls the intensity of the lines emitted from the excited states and the population distribution among atomic and ionic levels. For a constant element flow rate, any population of the ionic states (for example of Calcium) will induce a depopulation of the atomic states and thus modify the atomic line intensities. This heating problem is illustrated in Figure 32, (due to Trassy and Mermet, 1984), showing the influence of the carrier gas for a constant flow rate of the liquid containing Calcium on the excitation of atomic Calcium along the axis of a ICP torch. The higher the carrier gas flow rate, the lower the measured temperature, and the maximum is shifted further downstream. This is probably due to the cooling of the central region of the plasma by the cold carrier gas injection. The importance of the population ratios is illustrated in Figure 33, which shows, for two carrier gas flow rates, the evolution of the relative line intensities of atomic and ionic lines of iron along the axis of the torch. In addition, the ICP torch is usually operated at relatively low currents (< 20 A) in which non-LTE has been shown to exist (Kolesnikov, 1966; Nick et al., 1984).

C. Absorption Spectroscopy

Absorption spectroscopy is the only practical way to determine ground and metastable states, populations of atoms, ions or molecules in thermal plasmas. As in emission measurements, absorption spectroscopy yields line-of-sight averages, therefore Abel's inversion techniques must be used. However, it is often more complex than for emission spectroscopy because the whole line profile has to be studied and the line shift taken into account. The main problem is to find a source which has a high enough intensity compared to that of the plasma. For thermal plasmas at $T > 6{,}000$ K, only another thermal plasma source or a laser can be used.

The traditional line-reversal method at a given wavelength, where the plasma is penetrated by radiation from a calibrated lamp, can be used for temperatures below 3,000 K. For example, an automated system based on that principle has been developed for measurements in K-seeded combustion plasmas (Self et al., 1982).

1. Absorption Technique Principle

The total absorption coefficient χ_ν (Griem, 1964) is related to the population of the lower level n and to the excitation temperature by

$$\chi_\nu = \frac{c^2}{8\pi\nu_{mn}^2} \cdot \frac{g_m}{g_n} \cdot A_{mn} \cdot n_n \left[1 - \exp\left(-\frac{h\nu_{mn}}{kT_{ex}}\right)\right] \tag{33}$$

where the exponential term represents the induced emission which diminishes the absorption.

The principle of absorption measurements is as follows (Chapelle, 1971). The spectral transmittance τ_ν of a plasma of thickness d is related to the absorption coefficient k_ν per unit length at frequency ν by

$$\tau_\nu = \int_0^d k_\nu \cdot dx. \tag{34}$$

For most spectral lines, k_ν varies significantly with ν. If the profile of an absorbing line of frequency ν is known over the total line width $\Delta\nu$, an equivalent width w_ν can be defined by integrating over the spectral absorbance a_ν of the plasma thickness considered. If the considered thickness is small enough, then the plasma is homogeneous and a_ν can be written

(Griem, 1964)

$$a_\nu = 1 - \exp(-\tau_\nu), \tag{35}$$

hence

$$w_\nu = \int_{\Delta\nu} a_\nu \cdot d\nu = \int_{\Delta\nu} [1 - \exp(-k_\nu \cdot d)]\, d\nu. \tag{36}$$

w_ν, expressed in $\sec^{-1}$, can easily be related to the total absorption coefficient via Eq. (33), when the optical thickness τ_ν remains small at the center of the line. In this case Eq. (36) reduces to

$$w_\nu = d \cdot \int k_\nu \cdot d\nu = \chi_\nu \cdot d\nu \tag{37}$$

Thus the population density of the lower level n (for example, metastable or fundamental state) is related to the absorption coefficient k_ν of the plasma at the frequency ν (or at the wavelength λ) by the transition from the lower state n to the upper state m by absorption of a photon, i.e.,

$$n_n = \frac{8\pi}{\lambda^2} \frac{g_n}{g_m} \frac{1}{A_{mn}} \int k_\nu \, d\nu \left[1 - \exp\left(\frac{h\nu}{kT_{ex}}\right)\right]^{-1} \tag{38}$$

where A_{mn} is the spontaneous transition probability from m to n and g_m and g_n are the statistical weights of the states m and n, respectively. For the line corresponding to the transition $m \rightarrow n$, one obtains by absorption measurements a curve $k_{nm}(\lambda)$ characteristic of the line and of the plasma. In fact, due to the steep temperature gradients of the plasma, $k(\lambda)$ really cannot be measured. Instead what is measured is the transmission coefficient τ_ν along the line-of-sight under consideration. If $I_s(\lambda)$ is the intensity of the variable calibration source placed behind the plasma, one finds:

$$\tau(\lambda, y) = I_{tr}(\lambda, y)/I_s(\lambda), \tag{39}$$

where $I_{tr}(\lambda, y)$ is the transmitted flux through the line-of-sight at the height y, and $\tau(\lambda, y)$ an integrated value along the line-of-sight (see Figure 15b). However, it is to be noticed that the real signal received is the transmitted one $I_{tr}(\lambda, y)$ plus the signal emitted by the plasma $I(\lambda, y)$, and it is necessary to subtract the plasma emission. This can be done for example by using a chopper (c in Figure 34) in the optical path of the source for absorption measurements (P_2 in Figure 34) and detecting the transmitted signal through the plasma (produced by P_1 in Figure 34) with a lock-in amplifier (6 in Figure 34).

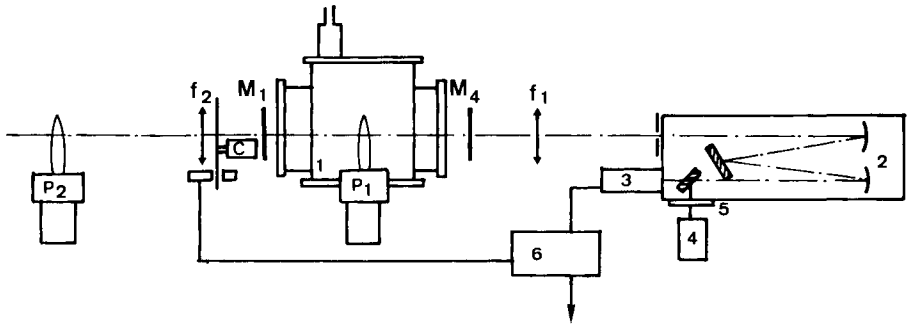


FIGURE 34. Experimental set up for absorption spectroscopy measurements. (Baronnet, J. M., 1978).

The transmitted intensity I_{tr} can easily be related to the incident intensity of the source I_s and the local absorption coefficient for an axi-symmetric plasma (see Figure 15b) by:

$$I_{tr}(\lambda, y) = I_s(\lambda)\exp\left[-2\int k(\lambda, x)\, dx\right] \tag{40}$$

with

$$x = \sqrt{r^2 - y^2} \quad \text{and} \quad dx = \frac{r\, dr}{\sqrt{r^2 - y^2}}.$$

Thus, taking the logarithm of I_{tr}/I_s and applying the Abel's inversion [see Eq. (9)] one finds

$$k_\lambda(r) = -\frac{1}{\pi}\int_r^R \frac{d}{dy}\left(\ln\frac{1}{\tau(y, \lambda)}\right)\frac{dx}{\sqrt{r^2 - y^2}}. \tag{41}$$

The value of $k_\lambda(r)$ replacing Eq. (41) into Eq. (38) will give the density of the lower level n corresponding to the line of frequency ν_{mn} in the plasma volume between r and $r + dr$ (see Figure 15b).

2. *Case of Real Plasmas*

Absorption measurements are, however, difficult to perform in thermal plasmas because:

a) Sources with intensities higher, or at least equal, to those of the plasma are not common.

b) The presence of high temperature gradients complicates the procedure required for evaluating temperature distributions [see the complexity of Eq. (41)].

c) The lines are frequently shifted along the line of sight due to the Stark effect, since the electron concentration from the external part of the plasma increases toward its center. Therefore, it makes the measurement of k_λ in the central part of the line impossible and requires the determination of the real profile of the absorbed line to calculate the mean integrated value $\int k_\lambda \cdot d\lambda$ of the absorption coefficient k_λ. Thus a high resolution monochromator must be used which evaluates $\Delta\lambda$ down to 0.25 Å for lines with widths usually less than 1 Å.

Such measurements have been performed at Limoges (Baronnet, 1978) by using a second plasma spray jet as reference source (Figure 34) for absorption measurements in a plasma spray jet. A fixed arc plasma generator P_2 is used and the intensity $I_s(\lambda)$ of the line studied and emitted by P_2 is recorded. This is done for a given line of sight, determined by a horizontal slit in front of the entrance slit of the monochromator where the image of P_2 is made, with the help of the lenses L_1 and L_2. A chopper allows the use of a lock-in amplifier to record the signal from P_2 through P_1 without the signal of P_1. A system of mirrors M_1 to M_4 (two insertable mirrors M_1 and M_4 intercepting the optical path shown in Figure 34 and two fixed ones above the plane of Figure 34 and not shown), together with a lens L_4 (not shown in Figure 34 because it is situated above P_1), permits recording simultaneously both the transmitted intensity $I_s(\lambda)$ and a reference intensity $I_s'(\lambda)$ corresponding to the same spectrum but attenuated by the reflections in the mirrors by the transmission factor of L_4 and by the longer optical path. By rotating the mirrors M_1 and M_4 at a given frequency to be recorded simultaneously, this allows the direct signal from the source I_s' (equal to I_s but with a known attenuation) and the transmitted signal I_{tr}. First P_2 is started with P_1 unenergized, and I_s and I_s' are recorded to determine the attenuation factor. Then the plasma generator P_1 is started and the transmitted intensity $I_{tr}(\lambda, y)$ is recorded for different lines of sight at a series of heights y simultaneously with the reference intensity $I_s'(\lambda)$ (the direct signal of P_1 is not registered by the lock-in amplifier). The reference signal is necessary to control the stability of the source P_2 and to obtain the wavelength of the relative minimum intensity of the absorbed line. After recording both profiles (I'_s and I_{tr}) for 20 to 30 wavelengths, a profile $\tau_\lambda(y)$ is determined for each wavelength and inverted to obtain $k_\lambda(r)$. The

inversion program approximates the recorded data with a least square parabola constructed with 5 points except for $r = 0$ and R. The calculation of $d(\ln[1/\tau(y)]/dy$ [Eq. (41)] is performed numerically along with the integration after a change of the variable $y = r\cosh(t)$:

$$k(\lambda, r) = -\frac{1}{r}\int_r^R \frac{d}{dy}\left[\ln\frac{1}{\tau_\lambda(y)}\right] d(\arg\cosh(y/r)). \tag{42}$$

For each value of r, the profile of $k_r(\lambda)$ is reconstructed to calculate $\int k_\lambda \cdot d\lambda$ and to determine n_n.

The concentrations of the argon metastable states $3P_2$ and $3P_0$ have been measured (Baronnet, 1978) under a controlled atmosphere with the follow-

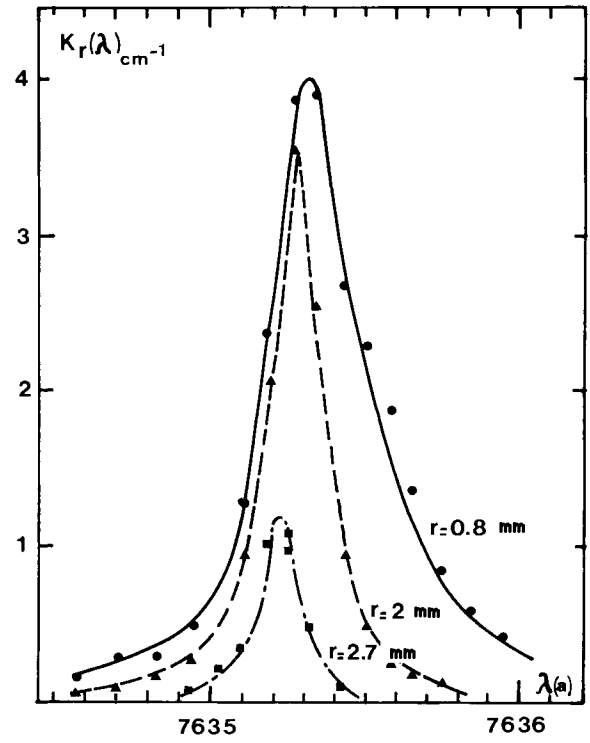


FIGURE 35. Absorption profiles of the line 7 365 Å of argon. (Baronnet, J. M., 1978).

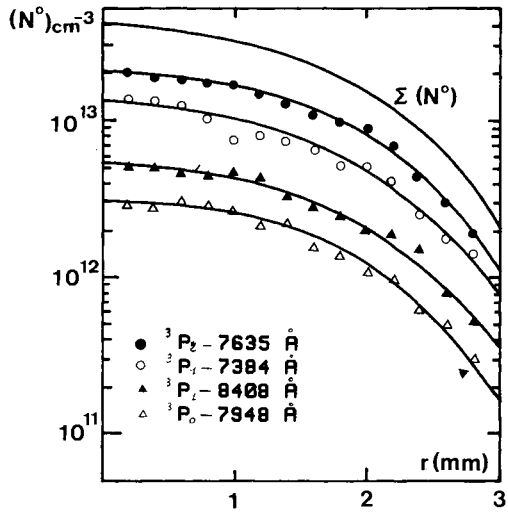


FIGURE 36. Radial profiles of the metastable $^3P_2, ^3P_0$ and resonance $^3P_1, ^1P_1$ states concentration in an argon plasma. (Baronnet, J. M., 1978).

ing conditions: Argon flow rate = 15 l/min STP, Arc current 150 A, D.C. voltage 20.5 V with the preceding absorption technique and the experimental arrangement in Figure 34. Figure 35 shows the spectral profiles of the absorbed line of Ar at 7,635 Å at different distances from the axis of the plasma jet which were obtained (notice the line shift for different distances from the jet axis). The radial profiles of metastable concentrations deduced from these measurements are shown in Figure 36.

D. Laser Spectroscopy

Regions of the plasma jet where temperatures are below 6,000 K are of primary importance for chemical reactions. Most of the products are formed in the gas phase reactions and quenched between 6,000 and 1,500 K (Akashi, 1985; Coudert, 1985). In this temperature range only emission spectroscopy with molecular spectra can be used, which provides information on molecular states with no access to the ground state population of the molecules though below 3,000 K the precision is very poor. That is why laser spectroscopy has been used in thermal plasmas, which involves techniques such as Coherent Anti-Stokes Raman Spectroscopy (CARS) and Laser Induced Fluorescence (LIF) (Garscadden, 1986).

The main advantages in using laser techniques are:

- good spatial and temporal resolution, i.e., the measuring volume is the intersection point of the excitation beam and of the optical path of the detector, and the temporal resolution results from the use of pulsed laser sources;
- that they do not require the presence of excited states and they allow one to obtain, in principle, the population of atoms or molecules in their ground or metastable states;
- that no assumption about equilibrium is required.

However, these techniques require expensive equipment, expert operators, and rely on computer fitting of the spectra to complex models.

1. *Coherent Anti-Stokes Raman Spectroscopy (CARS) Technique*

With very high power pulsed laser sources, optical effects originating from the non-linear response of molecules to incident electric fields become important. CARS diagnostics based on this effect has had a great success for combustion diagnostics, and could certainly be used for diagnostics in thermal plasmas. The main advantage of CARS with respect to emission spectroscopy is the possibility of performing the measurements in the presence of solid particles, as in plasma spraying devices or for plasma assisted coal combustion.

As explained in detail by Hall and Eckbreth (1984), the physical origin of CARS lies in the non-linear response of molecules to incident electric fields giving rise to a macroscopic polarization, proportional to the third power of the electric field amplitude. The third-order non-linear electric susceptibility $\chi(3)$ relates the macroscopic polarization to the cubic power of the electric field, and because it is dependent on molecular properties and conditions of the medium, it plays a central role for extracting diagnostic information from CARS signals.

CARS signals are observed when three light beams, two at the "pump" frequency ω_1 and one at the Stokes frequency ω_2, are mixed to produce a coherent beam at the anti-Stokes frequency $\omega_3 = 2\omega_1 - \omega_2$. The CARS process can be thought of as involving the destruction of two photons at ω_1 and the creation of photons at ω_2 and ω_3, thus leaving the molecular state unchanged. When the frequency difference $(\omega_1 - \omega_2)$ is made to coincide with the frequency of a Raman-active mode in the medium, normally rotational-vibrational transitions, the CARS signal is enhanced and the spectral signature is characteristic of the species under study.

CARS spectra have been intensively studied for diatomic molecules, namely N_2, CO, H_2, O_2, and can be used with no problems up to 6,000 K. The Electric Company in France (Magre et al., 1987) has used CARS to measure the mixing temperature (1,000 to 3,000 K) of an air plasma flow, produced by a 2 MW vortex plasma torch in which coal is injected and mixed with a high flow rate of cold air (the flow rate was 10 times that of the plasma torch). This is of course a 3D mixing (Lana and Viollet, 1987).

2. *LIF Technique*

Fluorescence is defined as spontaneous radiation emitted after stimulation of an atomic or molecular system to energies above the equilibrium state. This stimulation is obtained when pumping a lower energy level (l) of a molecule or an atom to a higher energy level (u) by absorption of a photon of energy $h\nu_{lu}$. According to the lasers available (visible and UV range), the highest energy difference between the two levels pumped with photons of energy $h\nu_{lu}$ will be around 4 eV, thus excluding pumping of the ground state of atoms for most gases used in thermal plasmas. However, Na and K can be used as tracers.

The fluorescence signal emitted by the excited state (u) is directly proportional (the rate of spontaneous decay is constant) to the number density N_u of excited particles due to the laser pumping. Thus, the fluorescence signal provides a measure of the population of the excited state u that has to be related to the parameters one wants to determine.

The technique has been developed for combustion flames by Daily, (1977), and Crosley and Smith, (1983). In the general case of the interaction of an electromagnetic radiation field with quantified particles A, it can be shown (Omenetto et al., 1979) that the density operator of the whole system (excited states, radiation, and coupling) is determined by a set of differential equations where the diagonal term represents the populations of the eigenstates of A, and where the off-diagonal terms characterize the coherence effects. When the off-diagonal terms relax much faster than the diagonal ones, the approximation of the rate equations is valid and it allows relating of the populations of the different states to each other stimulated by the laser pulse. Daily, (1977), has shown that the steady state equations are valid only when the laser pulse rises quickly compared to the collisional de-excitation rate for single-mode excitation and for certain special cases under multimode excitation. However, these conditions are not generally fulfilled in thermal plasmas and one must be very careful using results deduced from rate equations. LIF is, however, a powerful technique for determining the relative evolution of the population of the ground state of a

new molecule formed by chemical reactions in a D.C. plasma jet (see, for example, the work of Coudert (1985), related to the formation of NO when injecting cold oxygen in a nitrogen D.C. plasma jet).

3. *Thomson and Rayleigh Scattering*

To our knowledge, only a very few experiments using these techniques in thermal plasmas have been described. This is mainly due to the fact that the scattering cross sections are very low (about 10–25 cm^2 for Thomson scattering and three to four orders of magnitude lower for Rayleigh scattering (Kunze, 1968). This means, for example, that the scattered light for a Rayleigh experiment in a thermal plasma is of the order of 1 photon for 10^{14} incident ones. Haddad (1986), has measured temperatures in arc fringes by Rayleigh scattering. He has suppressed the arc light by using a chopped laser (for example, 2 W C.W. 514.5 nm AII laser), polarizers and a Fabry-Perot interferometer to limit the band width to 10^{-1} Å. However, the ratio S/N limits measurements to temperatures below 9,000 K at atmospheric pressure.

Thomson scattering has also been used to measure electron temperatures of a D.C. plasma jet (argon and nitrogen gas and low current $I < 50$ Å) (Pyatnitskii et al., 1978). Here again, the laser beam was produced by a C.W. argon ion laser at 514.5 nm with a power level of 1.6 W. The laser was chopped and the scattered emission was measured with a double-pass monochromator and using a lock-in amplifier. The wavelength was scanned at a rate of 10^{-2} Å/s between 5,096 and 5,145 Å.

E. Plasma Velocity Measurements

The velocities of plasma flows, ranging between 10 and 100 m/s for R.F. plasmas and between 60 and 1000 m/s for D.C. arc plasmas, are rather difficult to measure. Regarding temperature measurements, the measurement methods fall into two classes: intrusive and nonintrusive.

1. *Intrusive Methods*

The main intrusive method is to measure the dynamic pressure of the gas with a Pitot tube passed rapidly through the plasma, (Fizdon et al., 1976). Since the Pitot tube has to be water cooled to survive the heat transferred from the plasma, even in transient observation, this requires a diameter of

at least 2 to 3 mm, (i.e., about that of a plasma spray jet), and this perturbs the flow (see Section IV.A. for a detailed description of such probes). It is also clear that the Pitot tube method can be used only for free jets. Moreover, considering that the pressure head is very small, the measurement accuracy becomes very low for velocities higher than a few tens m/s. The calculation of the velocity from pressure measurements implies having a knowledge of the plasma temperature, and thus its density, at the point of measurement. This can induce important errors, especially in the fringes of the plasma jet, due to the steep variations of density with temperature (Hare, 1977).

2. *Optical Methods*

- A modulation technique is used for arc plasmas where a low frequency modulation of the D.C. power source results in luminosity modulation which is detected using two photodiodes with a known distance separation. The detector outputs allow one to determine the average time delay and thus the mean velocity (Cabannes et al., 1965). However, the precision of the measurement is rather poor due to both distortion and scattering of the signal. Moreover, it gives only a mean value since the modulation is averaged along the line of sight.
- Microscopic tracer particles can be added to the plasma to act as scattering centers (Self, 1987). Usually the flow is illuminated by a sheet beam of light derived from a spark or pulsed laser, and the light scattered at 90° is imaged into a conventional still or movie camera, or an electronic camera coupled to a computer. The velocity vector can be obtained from the length and direction of the image traces when the pulse is suitably long, or from the separation of adjacent point images by using short double pulses. Again, the main problem is to overcome the plasma radiation intensity.
- Doppler effect: Presently this is the most used method to measure the plasma velocity. It is based on the Doppler effect, i.e., the shift of the frequency of a signal emitted by a source moving relative to a detector. The use of plasma emission (natural or laser stimulated) is limited to high velocity (supersonic) and low pressure (below 100 Torr) plasmas. This is because the signal line width must not be too large compared to the Doppler shift (less than a tenth of Angström), and at a pressure of over 100 Torr the electron density is high so the Stark width is generally larger than the Doppler one. For this reason, lasers with their very narrow line widths are a valuable source of light scattering in plasmas. The velocity of

the plasma can be probed with lasers in two ways depending on the scattering medium.

a) When laser light is scattered by solid (or liquid) particles seeded into the plasma, the technique is known as laser Doppler anemometry (LDA) (see, for example, Durst, et al., 1976). To ensure that the particles follow the plasma flow with sufficient accuracy, they should be fine (a few microns or less) for dense gases (below 4,000 K at 1 atm). In the plasma core these particles evaporate rapidly and perturb the flow and the electrical properties of the plasma (the most marked effect is in the arc region). Even very fine particles are not able to follow the turbulent motion and the strong gradients usually present in plasmas. For particles smaller than 20 μm (the ones used are about 3 to 5 μm) the Knudsen effect must be considered (Pfender, 1985; A. Vardelle, 1987) and the particles may never reach the plasma flow velocity. At the same time, the minimum size is limited by the available laser power (the scattered light intensity decreases with the particle size). These limitations are most restrictive in the plasma core (where the light emitted by the plasma is very intense relatively to the scattered light signal). Unfortunately this is the most important region for modelling and this is where the highest precision is required. In spite of these difficulties, this is actually the most commonly used method for studying plasma spray jets (e.g., see A. Vardelle et al., 1980; A. Vardelle et al., 1982; Kruszewska and Lesinski, 1977; and Lesinski et al., 1981), or for R.F. plasmas (Lesinski et al., 1979, 1981; and Gouesbet and Trinité, 1977).

It is important to bear in mind that these measurements are only a crude approximation to the flow velocity which is always underestimated in the plasma core region. Hence they have to be carefully compared with calculations such as with the velocity distribution profile determined from temperature profile (which can be obtained with a higher precision). This has been done in D.C. plasma jets (Lee and Pfender, 1987; A. Vardelle, 1987) by assuming the following distributions at the nozzle exit:

$$\frac{T}{T_0} = \left(1 - \frac{r}{r_0}\right)^{x_1}; \quad \frac{v}{v_0} = \left(1 - \frac{r}{r_0}\right)^{x_2} \tag{43}$$

where T_0, v_0 are the maximum temperature and velocity at the axis respectively, and r_0 the radius of the plasma jet. The coefficient x_1 is determined from the measured temperature distribution and x_2 and v_0 are calculated from the measured total enthalpy of the gas and mass flowrate.

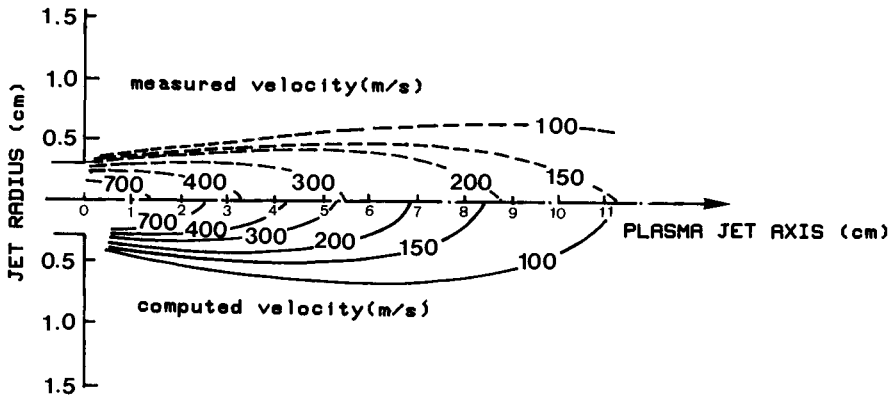


FIGURE 37. Velocity distribution measured (dotted lines) by Vardelle, A., Baronnet et al., 1980, for a N_2-H_2 D.C. plasma jet (nozzle diameter: 6 mm, D_{N_2} = 37 l(STP)/min, D_{H_2} = 11 l(STP)/min, P = 29 kW), compared with the calculated one (solid lines) by Correa, S. M. (1983).

Figure 37 shows an example of the velocity distribution measured by LDA of small particles seeded in the arc chamber for a N_2-H_2 D.C. plasma (dotted line) and the calculated one using the preceding method (Correa, 1983). It can be seen, especially in the hot core of the plasma jet, that the measured velocities do not correspond very well with the calculated ones and that the measured velocities are smaller than the calculated ones.

Agreement can be much better in the plasma plume where temperatures are below 6,000 K. Here the plasma mass density is higher, the gas mean free path smaller (minimal Kundsen effect) and light emitted from the plasma is lower.

b) When laser light is scattered by the gas, different methods can be used. With alkali atoms seeded into the plasma, resonant Doppler velocimetry provides a technique (Cheng et al., 1983). The intense resonant radiation of alkali atoms permits seeding rate as low as 10^{-12} of the total number density, limiting the perturbing effects of the seeding. This technique, if applicable, can yield the local velocity, temperature, and pressure simultaneously. Resonant Doppler velocimetry seems to be a promising technique for low pressure thermal plasmas. Stimulated Raman spectroscopy or coherent anti-Stokes Raman spectroscopy (Fuji et al., 1983) can be employed when molecules are present in the plasma. The observed line shift can be related to the velocity and the line profile to the temperature of the heavy particles for a non-equilibrium plasma.

This technique is rather complex (CARS technique) and the available signals are very weak. For supersonic flows it is possible (Self, 1987) to use the Doppler shift of laser-induced fluorescence signals and hence dispense with the need of tracer addition. Provided that pressure and Stark broadening are small, the maximum line shift equals the Doppler width of the line at Mach 1, so a sensitive means of determining small shifts of line center is required.

V. Measurements of Particle Parameters in Thermal Plasmas

A knowledge of particle velocities, trajectories, sizes, and surface temperature distributions under plasma conditions is fundamental for understanding heat, momentum, and mass transfer between the high temperature flow and entrained particles. These basic processes must be modelled, for instance, with the objective of design and control of the operation of plasma spraying equipment. The problems are complex due to the high temperatures (10 to 15,000 K in D.C. plasma jets) resulting in very intense plasma emission which makes it difficult to detect scattered light or light emitted by hot particles. The high velocities (up to 600 m/s) and very short residence times of particles in the small measurement volumes (a few thousands of cubic microns) make it difficult to track the steep gradients.

A. Particle Trajectory Distributions

The particles are injected into thermal plasmas, (downstream the arc root for D.C. plasma jets or in the active field region for R.F. discharges), through an injector with the help of a carrier gas with adjustable flow rate to give the particles sufficient momentum to penetrate into the viscous plasma. To study momentum, heat and mass transfer between plasma and particles, particles of unisize and a unique injection velocity would be ideal. Then a unique trajectory could be measured, along which the velocity and temperature evolution could be followed. However, in reality particles have size and injection velocity distributions which result in the need for statistical measurements (mean trajectories, for example) and, to correlate measurements and models, these distributions have to be taken into account.

Particles, normally processed under thermal plasma conditions, have sizes ranging between 10 and 120 μm (it is common to have a distribution of sizes in ceramic powder spraying between 10 and 45 μm). This size distribution results from the manufacturing process (for example, fused and

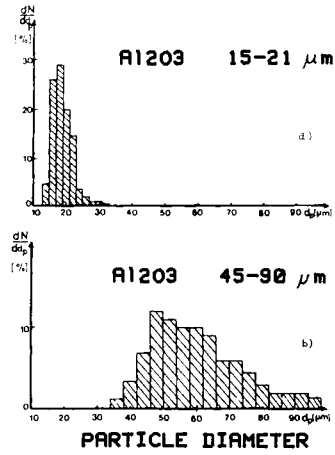


FIGURE 38. Alumina particle size distributions (Vardelle, A. et al., 1985). a) narrow distribution : $-21 + 15$ μm : Powder 1; b) large distribution : $100 + 35$ μm : Powder 2.

crushed particles) and a steep price premium is to be paid for the narrowest distributions. Figure 38 represents alumina particles with a very narrow distribution, Powder 1 : $-21 + 15$ μm*. (Figure 38a) and with a classical one, Powder 2 : $-100 + 35$ μm (Figure 38b). The price of the first is 15 times that of the second. Inside a particle injector, the carrier gas has a velocity distribution (zero velocity at the wall and maximum velocity at the axis) which, for a given injector, is the broadest for the highest carrier gas flow rate. Thus, the momentum distribution of the injected particles reflects the product of these two distributions. To have the particles penetrating to the hottest regions of the plasma jet, their momentum has to allow for the plasma viscosity and momentum. This can be done by adjusting the particles' mean size and size distribution and/or the carrier gas flow rate. For a given plasma jet and a given particle size distribution, the injection velocity (and thus carrier gas flow rate) must be the highest for the smallest particles. The trajectory distribution (along which velocity and surface temperature distributions have to be measured) will result from the injection momentum distribution, and it may be very broad with small particles even when their size distribution is very narrow. This can be seen in Figure

*The classical notation for spraying powders has been used: the first number with a minus sign is the largest diameter of the particles and the second one with a plus sign is the smallest diameter of the particles.

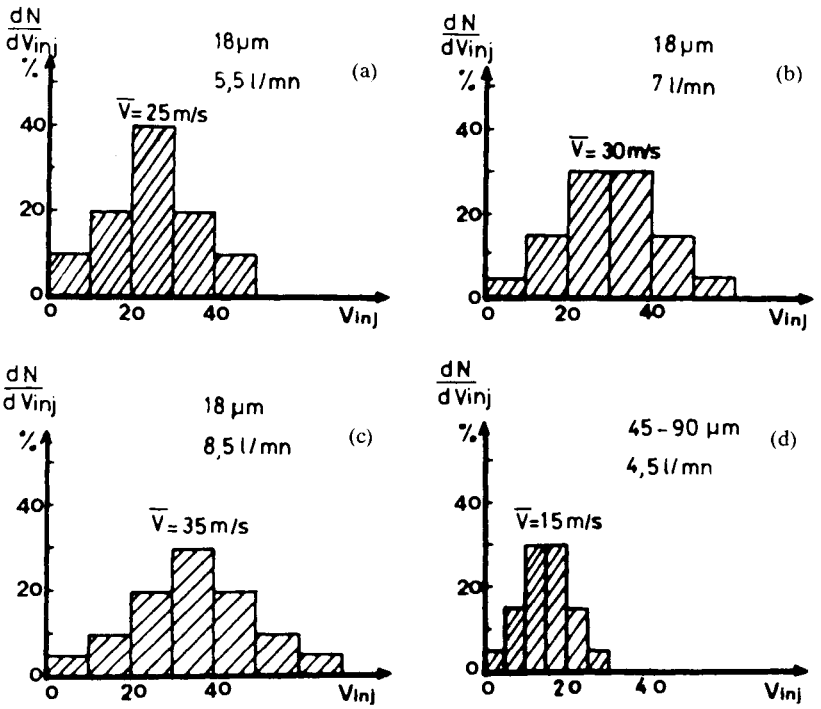


FIGURE 39. Injection velocity distribution for different carrier gas flow rates (injector internal diameter 1.6 mm). a) Powder 1 : 5.5 l(STP)/min; b) Powder 1 : 7 l/min; c) Powder 1 : 8.5 l/min; d) Powder 2 : 4, 5 l/min. (Vardelle, A. et al., 1985).

39 (A. Vardelle et al., 1985) representing the calculated particle injection velocities for the two alumina powders presented in Figure 38 for an injector inside diameter of 1.6 mm and different carrier gas flow rates. For Powder 1 injection velocities are rather high with a broad distribution, while Powder 2 injection velocities are half as great with a rather narrow distribution. Thus, the trajectory distribution inside the plasma jet will be rather broad and about the same for Powders 1 and 2. Moreover plasma flows have associated continuous fluctuations of the instantaneous plasma properties (temperature and velocity), due, for example in D.C. plasma jets, to the fluctuations of the arc root at the anode. This means that measurements on particles performed under plasma conditions will be mainly stochastic measurements and the complete statistical analysis of the results is an essential part of the experiment.

1. Particle Collection on a Fixed Substrate

One of the simplest techniques commonly used to measure particle trajectory distributions has been the collection of the molten particles on an appropriate substrate, momentarily placed across the flow. The location and the relative distribution of the material collected is a direct indication of the predominant particle trajectories and the relative flux of particles along any such trajectories. This technique has been used, for example, by Kaczmarek et al. (1984) or Champagne and Dallaire, (1987) to optimize the injection conditions, so that powders of different materials and particle sizes may be co-deposited on the same spot, in the required proportion.

However, there are two important limitations to this technique: first, since only the molten, or partially-molten particles are collected, the technique does not detect solid particles which bounce off instead of being collected on the substrate. In spraying experiments it has been shown that only about 40–60% by weight of the material injected into the plasma is collected on the substrate. Second, since the principal mechanism of particle collection is impaction, the collection efficiency as well as the deposition profile will depend on the particle inertia. This inertia effect can be especially important when dealing with fine powders (particle diameters less than 10 μm). And lastly, one never can be sure that the collection system does not significantly perturb the plasma and particle flows.

2. Laser Measurements

An alternative to the technique discussed in the previous section has been tested at the University of Limoges (A. Vardelle et al., 1980; A. Vardelle et al., 1982, Fauchais et al., 1984b; M. Vardelle, 1987). The method is based on the counting, during a given time, of the total number of bursts of scattered light obtained when particles are passing through a focused laser beam, which also can define the control volume for laser Doppler anemometry. Usually the laser beam is orthogonal to the plasma jet and the scattered light is detected orthogonally to the laser beam. The results in a rather small measurement volume but also, according to MIE theory, in a very low intensity scattered signal. The experimental set up is shown in Figure 40. A lens L_1, 100 mm in focal length, focuses the laser beam at the measurement point (diameter of the focused beam 100 μm). The scattered light is collected orthogonally to the laser beam by a lens L_2 200 mm in focal length which produces an image of the focused beam at the entrance slit of a monochromator. The total height of the measurement volume is

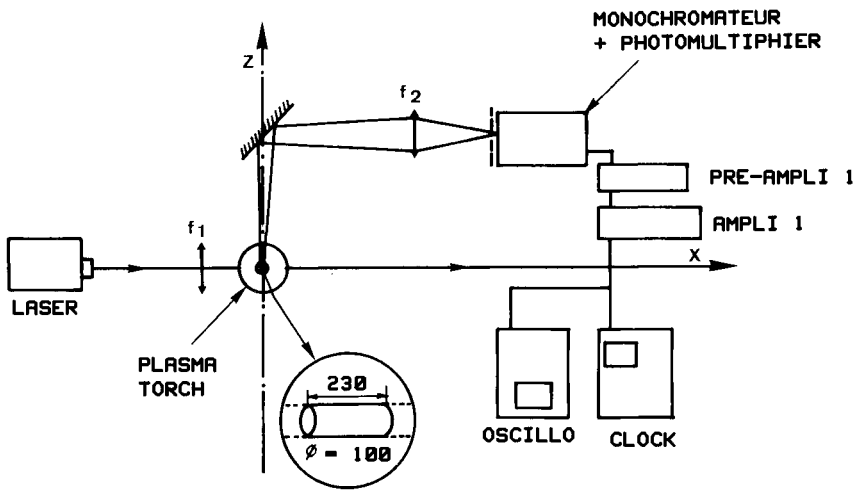


FIGURE 40. Experimental set up for particle flux measurement (Vardelle, A. et al., 1980).

limited to 400 μm by a diaphragm located in front of the slit. The monochromator has a band pass of 1 Å at 5,145 Å, the wavelength of the laser. This allows one to eliminate most of the signal emitted by the plasma. Using a C.W. argon ion laser with a 1.5 W beam, it is possible to follow particles as small as 2 μm in diameter in the core of an Ar-H_2 plasma spray jet ($T < 13{,}500$ K).

A variable-gate pulse counter connected to the photomultiplier is used to determine the number of particles passing through the measurement volume (diameter of 100 μm, length of 40 μm). For example, with 18 ± 3 μm alumina particles (see Figure 38a) injected in a 29 kW Ar-H_2 plasma, the measurements obtained (Figure 41), at 20 mm from the injection tube, show that the majority of the particles (more than 90%) pass through a region of the plasma where the temperature is higher than 5,000 K; however, the considerable dispersion of the particles in the jet, even with such a narrow size distribution, illustrates the preceding remarks about the injection velocity distribution (see Figure 39). The mean particle trajectory is obtained by moving the measurement volume (or more precisely by moving the plasma jet relative to the measurement volume) along two orthogonal directions and determining the corresponding radial distributions (see Figure 41). The position of the two maxima corresponds to the mean trajectory in the plasma "slice" considered. It should be also mentioned that the plasma temperatures presented in Figure 41 have been measured without injection of particles and carrier gas, and that it might not be the real one,

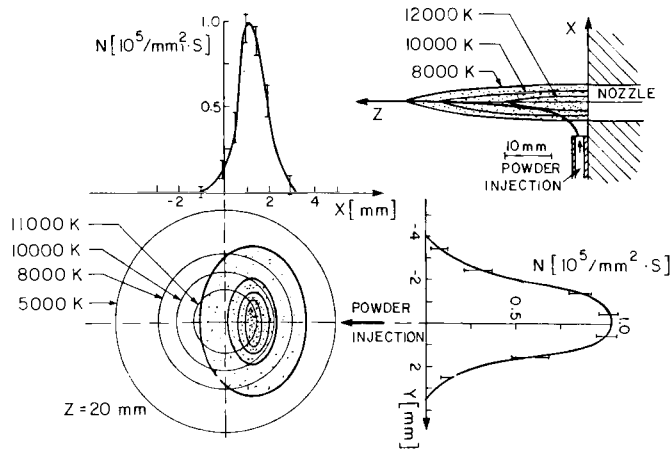


FIGURE 41. Flux of alumina particles (diameter, 18 ± 3 μm) in a "slice" of an Ar-H_2 plasma jet 20 mm downstream of the injection tube. The experimental trajectory is determined from the positions of maximum fluxes of the particles along two orthogonal directions. (Vardelle, A. et al., 1982).

the determination of which would necessitate the use of a quasi-symmetrical Abel's inversion technique.

B. Particle Velocity Measurements

In-situ optical methods are limited to conditions where the particle extinction is small and the plasma is optically thin for the wavelengths used (generally case in thermal plasmas), and various optical methods have been developed among which laser anemometry is by far the most used.

1. Photographic Methods

Some investigators have used photographic method to analyse thermal plasmas:

- Desai et al. (1968), have used a single flash technique to determine the velocity of 40–80 μm boron nitride particles in an R.F. plasma (the lengths of the particle paths during illumination allow one to calculate their velocities);

- Chase (1971), has used cine-photography to determine the pattern of motion of 10 μm alumina particles injected in an inductively coupled R.F. plasma;
- Lewis and Gauvin (1973), studied particles in D.C. plasma jets using a cine-streak photography technique. The film was transported at constant speed in a direction perpendicular to that of the predominant particle motion (each particle being photographed will create a streak across the film and the angle relative to the direction of the film gives the velocity);
- Finally Frind et al. (1983), used the same technique for the measurement of velocities of particles in a low-pressure (40 torr) D.C. plasma jet under supersonic flow conditions.

2. *Laser Anemometry*

This technique is used (Lesinski, Gagne and Boulos, 1981; Gouesbet, 1985; Steffens et al., 1985; Weissman et al., 1987; M. Vardelle, 1987) to measure velocity and size distributions of solid particles injected into a plasma jet (spraying, extractive metallurgy, vaporization, spheroidization, ceramic synthesis, etc.). Two different methods are used: laser Doppler anemometry and laser dual focus.

Laser Doppler Anemometry

General. The basic idea in LDA is to measure velocities from the Doppler frequency shift of laser light that is scattered from particles carried by the flow. The Doppler effect induces a scattered frequency different from the incident frequency of the laser. This "beating" frequency or Doppler frequency is proportional to the velocity component of the scattering center. According to Gouesbet (1985), the optical systems used for these measurements can be classified into three main types:

i) a reference system, where the incident light frequency beats with the scattered light frequency;
ii) the interference system where two laser beams are focused on the flow under study and beating is achieved between the light scattered in a given direction from each incident beam;
iii) the dual scatter system, where again a single laser beam is used with beating between scattered light received from two directions.

Principles of Interference System. The interference system ii) is the most widely used. This interference set up is very easy to adjust and align and

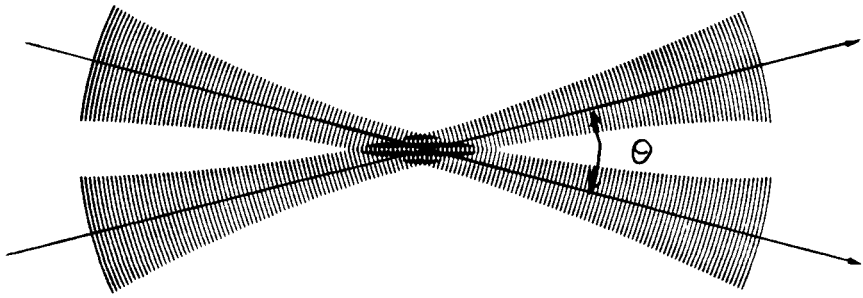


FIGURE 42. Fringe pattern of two laser beams. (Pfender, E. and Boulos, M., 1983).

rather insensitive to vibrations. Furthermore, the solid angle of collection can be made large without producing any broadening of the scattered light spectrum, thus allowing the detection of small particles. The two incident beams produce fringes in three-dimensions in the common cross-over region (see Figure 42). As a first approximation, all the fringes are parallel to the bisector plane of the beams. The fringe spacing d_f is equal to:

$$d_f = \lambda/[2\sin(\theta/2)] \tag{44}$$

where θ is the angle between the two beams (see Figure 42).

The control volume is the intersection of the focusing domain and of a viewing volume defined by the collecting optics. Typical angles of the two laser beams are in the range of 5 to 10°; the length of the control volume is much longer than its diameter (typical values are 1 mm in diameter corresponding to 10 fringes and 4 to 6 mm in length); the collection of the scattered light perpendicularly to the axis of the two laser beams reduces this length considerably. For particles which are large compared to the wavelength, the maximum intensity of the scattered light is close to the axis of the two laser beams (MIE theory) while intensity for side-scatter is several orders lower. If forward scatter is used, then there may be a problem with spatial precision of the measurements. With a typical fringe system, the spatial precision of the measurement in a D.C. plasma jet, where velocity gradients are up to 300 m/s · mm, is very poor if light is collected at 6 to 10° of the mean laser beam direction (backwards or forwards scattering). Although the precision is much better when light is collected at 90° of the mean laser beam direction, (measuring volume as short as 400 μm), the scattered light is very low, resulting in very noisy signals.

A particle crossing the control volume produces a light burst composed of a low frequency component, or pedestal, which depends on the light intensity distribution in the measuring volume, and a high frequency

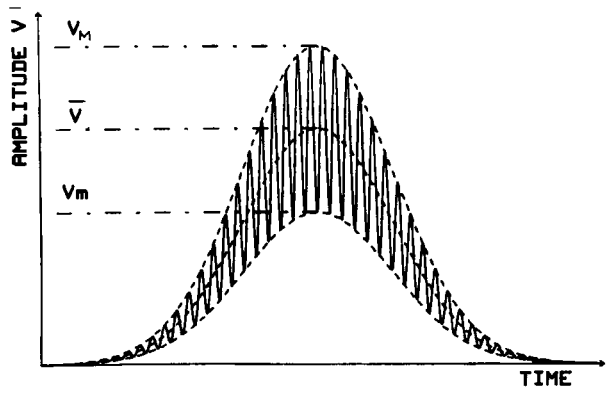


FIGURE 43. Typical Doppler Signal (modulated light burst scattered by a single particle crossing the measuring volume). (Gouesbet, G., 1985).

modulation superimposed on it (see Figure 43). The maximum amplitude of the low frequency component is denoted as V_M and the minimum amplitude is V_m. These values allow us to define the visibility V_i of the signal (see Section V.C.1.).

In the fringe system (see Figure 42), the modulation frequency f depends on the fringe spacing d_f on the particle velocity component v perpendicular to the fringes

$$f = v/d_f. \tag{45}$$

When d_f is known, (the precision of its measurement is a prerequisite), the measurement of f allows one to calculate the velocity component v. It is worthwhile noting that f is independent of the signal amplitude which depends on a number of parameters, including the light intensity distribution of the incident laser beams as illustrated by the signal deformation when particles are crossing the edge of the measurement volume (see Figure 44). Signal processors must incorporate validation circuits which reject such deformed signals.

The function of the different available signal processing techniques is to measure, as accurately as possible, the modulation frequency and to carry out appropriate statistical analysis to determine the mean velocity of the particle and the standard deviation of the velocity fluctuations, as well as velocity correlations and spectral densities.

The signal pedestal is removed by using high-pass electronic filtering. On the high frequency end, it is necessary to use low-pass electronic filtering to remove electronic noise superimposed on the modulated signal

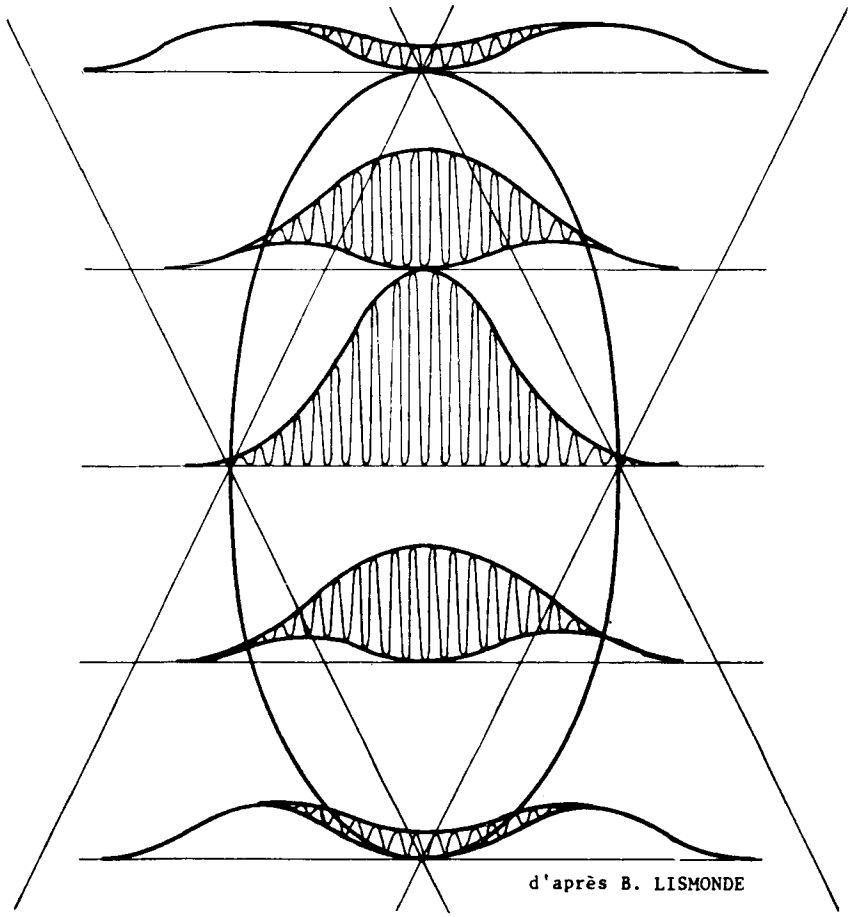


FIGURE 44. Relative intensity distribution near the crossover of two equal power, similarly polarized Gaussian laser beams. (Gouesbet, G., 1985).

(Lading, 1976). The processing system of choice and its complexity depends on the information needed, the required accuracy, and the specific situation under study. The most commonly used processor is the counter which has the ability to deal with low burst rate signals obtained in lightly seeded flows. The filtered Doppler signal is transformed in a train of square pulses with the help of a Schmitt trigger and the counter determines the duration τ for a preset number of pulses n. The frequency f is then n/τ. Counters can work up to 200 MHz. The counter is also the only technique which can give a complete statistical analysis of the results, for example by using ap-

propriate microprocessor based data acquisition and treatment systems such as the one described by Fanton et al., (1982).

As the collected power decreases, the signal becomes noiser and one enters the photon-counting regime where, instead of a modulated wave, individual random peaks are visible. Such signals cannot be processed by counters and correlation methods are necessary (these methods can be also used for processing noisy classical signals). For correlation, the photomultiplier is used as a photon counter system. Its output is a sequence of pulses corresponding to the stochastic Poisson process of photon detection. The correlation function given by the photon correlator exhibits the same periodicity as the Doppler signal, and its decay versus the lag time is closely linked to velocity fluctuations. However, such a procedure gives only the mean particle velocity, and not the full probability distribution function. Moreover, the frequency range of photon correlators is usually limited to 50 MHz.

An analysis of the precision and bias errors in LDA systems can be found in the papers of Durst et al. (1976) and Gouesbet (1985). In both cases (counter and photon correlator), the signal count rate is limited to about 100 kHz to avoid signals overlapping. For plasma jets at atmospheric pressure with small diameters (such as plasma spray jets), this requires a limitation of the particle flow rate to about 300 g/h–1 kg/h, depending on the measurement volume size.

Example of Arrangement. Figure 45 shows a typical arrangement from A. Vardelle et al., (1980). The beam from an argon ion laser (λ = 514.5 nm, P = 1.2 W) equipped with a rotating polarizer is split in two beams of equal intensity by a commercial double prism beam splitter which also focuses the two beams. The axis of the interference fringes formed at the intersection of the two beams may be rotated in two perpendicular directions by a device composed of three flat mirrors, allowing measurements of the axial and radial components of the velocity. The region of interest within the plasma jet is viewed at 90° to the incident laser beam, for optimum spatial resolution, and is focused (with the help of a camera lens 105 mm focal length) onto a diaphragm (100 μm) placed before the photocathode of a photon counting photomultiplier.

The measuring volume is a cylinder of 600 μm diameter and 1100 μm long, which is perpendicular to the plasma jet axis. The fringe spacing (100 μm) may be accurately measured as follows: the image of the fringes is focused with a magnification of 150 on a screen by a lens and retractable flat mirrors (see Figure 45). On the screen, the fringe spacing is measured with a micrometer scale with a precision of about 1%. The precision in velocity measurement is estimated to be about 5%.

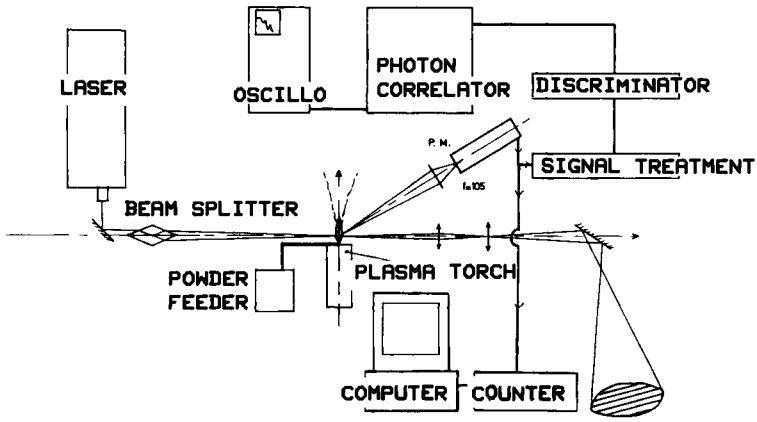


FIGURE 45. LDA experimental set-up for interference system: fringe spacing 100 μm. (Vardelle, A., et al., 1980 and Vardelle, M., 1987).

Signal processing is also very important for accurate measurements. In the arrangement of Vardelle (see Figure 45), the fringe volume is focused onto the photomultiplier by a collimator lens through an interference filter, centered at 514 nm with a half bandwidth of 0.25 nm to reduce as much as possible the light radiation from the plasma. The output of the photomultiplier after amplification and discrimination, (to shape the electric bursts), is treated by a digital photon correlator.

Two Focus Method. In this configuration, the velocity is calculated from the time between two successive pulses scattered by a particle as it crosses two narrow strips of light separated by a short distance. The intensity of the light scattered by the transient particles is higher (two orders of magnitude) than that obtained in the fringe system because of the concentration of the laser beam, so this configuration may allow measurements in situations where the signal-to-noise ratio would be too low for Doppler anemometry.

Figure 46 (A. Vardelle et al., 1980) shows, for example, the principle of this measurement technique. Two incident laser beams (λ = 514.5 nm) are focused by a lens 16 mm in focal length at two spots separated by 400 μm. At the focus, the diameter of the laser beam is about 100 μm. The two beams' foci are projected into the plasma jet by a lens (50 mm in focal length) with a magnification of unity. Each beam focus is imaged at an angle between 10 and 90° from the laser beams by a further lens (105 mm in focal length) onto a photomultiplier equipped with an interference filter (λ = 514.5 nm and $\Delta\lambda$ = 0.25 nm) to reduce the plasma light. The outputs

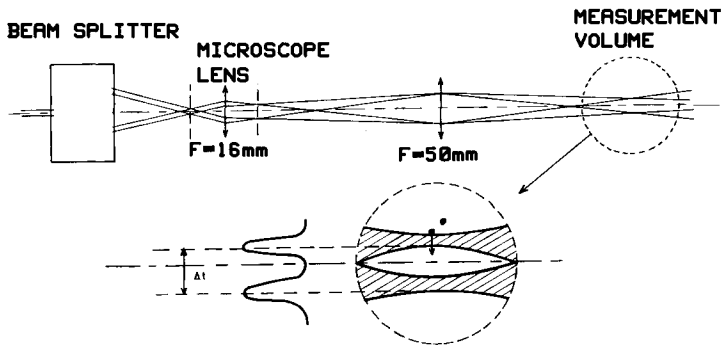


FIGURE 46. Two laser beam experimental set-up. The distance between the two spots is 400 μm. (Vardelle, A., et al., 1980 and Vardelle, M., 1987).

of the two photodetectors are cross-correlated by a photon correlator. The transmission and detection optics are in the same horizontal plane, and here scattered light is collected at a 30° angle to the incident beam. The resulting sampling volume is a cylinder with a 100 μm diameter and 200 μm length. Measurement precision is estimated to be about 5% (the relative error of the measurement is less than 1% for a distance of 0.4 mm between the two-beam spots). Moreover, this technique allows one to measure much faster particles (over 500 m/s) just by increasing the distance between the two focus points, thus increasing the flight time between them. For a distance of 1 mm, the velocity is more a mean value than the local one obtained with fringes with spacings of 0.1 mm. However, as most of the signal processing systems have a frequency limit of 200 MHz, a fringe spacing of 0.1 mm limits the velocity measurements to 500 m/s.

Light Pulse Method

Principle. The diagnostic technique developed by Sakuta and Boulos, (1987) is based on the analysis, in the time and amplitude domain, of the wave form of the light pulse generated as an emitting particle passes across a well-defined observation window. The basic principle involved is illustrated schematically in Figure 47. In Figure 47a, the relative position of a particle of diameter, d_p, is shown as it passes across a rectangular slit of height, H, and width, W. The associated light pulse obtained is given in Figure 47b. It is characterized by its trapezoidal shape with a base line shifted by a constant value, I_c, corresponding to the background plasma radiation, and a pulse height, I, corresponding to the particle emission.

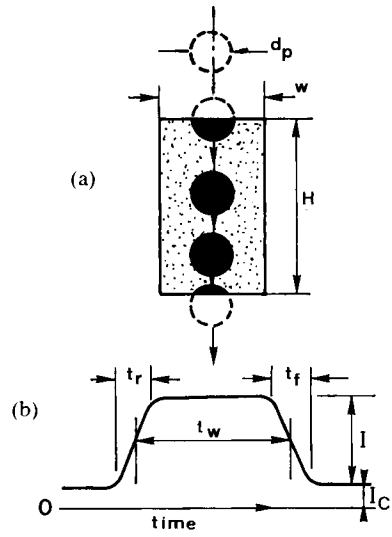


FIGURE 47. a) Measurement principle of the light pulse. b) Corresponding waveform of the generated light pulse generated. (Sakuta, T. and Boulos, M. I., 1987).

Through simple geometric considerations, the half-width, t_w, of the pulse corresponds to the time-of-flight of the particle through the slit height. The particle velocity, V_p, can accordingly be determined as

$$V_p = \frac{H}{(t_w \cdot R)}, \tag{46}$$

where R is the image ratio determined by the optical setup.

The rise time, t_r, and fall time, t_f, of the pulse corresponds to the time over which the particle is entering or leaving the observation window respectively, which depends on the particle diameter and velocity as follows:

$$t_r = t_f = d_p/V_p; \tag{47}$$

knowing V_p, Eq. (46), the diameter of the particle can be calculated as:

$$d_p = (H/R)(t_r/t_w)$$

or

$$d_p = (H/R)(t_f/t_w). \tag{48}$$

The pulse height, I, on the other hand, gives information about the surface temperature of the particle, T_p. Assuming that the particle emissivity is that of grey body, its surface temperature can be determined from the ratio of

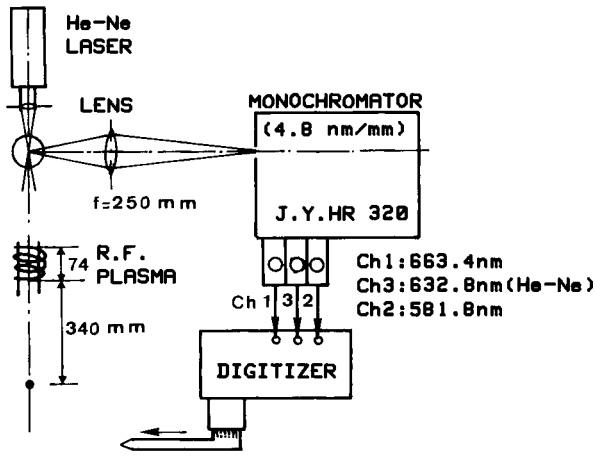


FIGURE 48. Schematic of the experimental set-up for light pulse method. (Sakuta, T. and Boulos, M. I., 1987).

the pulse height at two different wavelengths λ_1 and λ_2 as follows:

$$T_p = \frac{C_2(\lambda_1 - \lambda_2)}{\lambda_1\lambda_2\left[\ln(I_{\lambda_1}/I_{\lambda_2}) + 5\ln(\lambda_1/\lambda_2)\right]}, \tag{49}$$

where C_2 is Planck's second constant for thermal radiation.

This method, unlike the methods based on scattering, is only applicable to large particles for which one can create a resolved image with optics of reasonable aperture.

Experimental Set-up. A schematic of the experimental set-up is given in Figure 48. The principal optical axis is defined by a 75-mm diameter, objective lense, with a focal length of 250 mm, which is used to form an image of the observation volume on the entrance slit of a J.Y. HR 320 monochromator. The image ratio is 1.75, and the monochromator entrance slit is set to a 2 mm height and 1-mm width.

In order to define the location of the observation point along the principal optical axis, a He-Ne laser beam with a diameter of 0.3 mm is focused at a right angle to the optical axis, at the focal point of the objective lens. A particle passing through this point will give rise, in addition to its own continuum emission, to a pulse of scattered light at the He-Ne wavelength which can serve to discriminate it from all other parti-

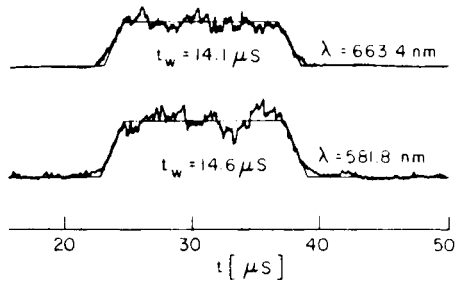


FIGURE 49. Typical wave forms of the light pulses received at wavelengths of 582.8 and 663.4 nm (Sakuta, T. and Boulos, M. I., 1987).

cles crossing the principal optical axis within the depth of field of the objective lens. The observation volume can accordingly be reduced to a parallelepiped of dimensions 1.14 × 0.57 × 0.30 mm.

The light pulses received by the monochromator are simultaneously detected by three photomultipliers (Hamamatsu R1104 with a Hamamatsu C1053 preamplifier) at the wavelengths of 581.8, 632.8 and 663.4 nm. The signal corresponding to the He-Ne wavelength (632.8 nm), Ch. 3 in Figure 48, is used to trigger a 2-channel, 10-bit fast digitizer (Tektronics 390 AD), which is used to obtain a 2,000 word digital record of each of the signals at the wavelengths, 581.8 and 663.4 nm, respectively. Typical records of such signals are given in Figure 49.

Following the acquisition of the two wave forms, the data is transferred to a PDP-11/73 microcomputer using a HSDO interface which requires about 100 ms for the complete transfer of the 4 K word data block. Due to the relatively long time required for the subsequent numerical processing of the data, particle pulse acquisition is continued uninterrupted at a rate of up to 10 particles per second, until a record of 40 particles is completed. The data block is then processed in batch form before further data acquisition is resumed.

The numerical processing of the data is carried out using a specifically developed computer algorithm to obtain, for each signal, the pulse width, t_w, its rise and fall time, t_r and t_f, and its height, I. These are used in conjunction with Eqs. 46, 47, and 49 for the computation of the velocity, diameter, and surface temperature of the particle, respectively.

For meaningful statistical analysis, a total of 25 such data blocks is treated for each measurement providing a statistical population of 1,000 particles. Under typical operating conditions, the full procedure requires between 20 and 30 minutes.

3. *Typical Examples of Measurements*

D.C. Plasma Spray Jets. To be melted upon impact, particles must travel in the hot zones of the plasma jet as close as possible to its axis. This can be achieved (see Section V.A) by injecting them with the proper mean injection velocity or carrier gas flow rate, which has to be determined for each given plasma jet and injector.

The optimum mean injection velocity varies from a few meters per second for 80 μm alumina particles to 50 ms^{-1} for 10 μm particles. For alumina, melting is controlled easily by the amount of α phase (incompletely melted particles) remaining in the deposit, while the fast solidification of completely melted particles yields the γ phase. Figure 50 shows the radial distributions along the plasma jet axis of the mean velocity of alumina particles ($-21 + 15$ μm) injected at optimum velocity for melting into a 29 kW N_2/H_2 plasma. It should be noted that, even for a powder of narrow size distribution, the radial velocity profiles are broad, indicating a wide dispersion of the particles in the flow (see Figure 41) which results from high plasma gradients and indicates a broad injection velocity owing

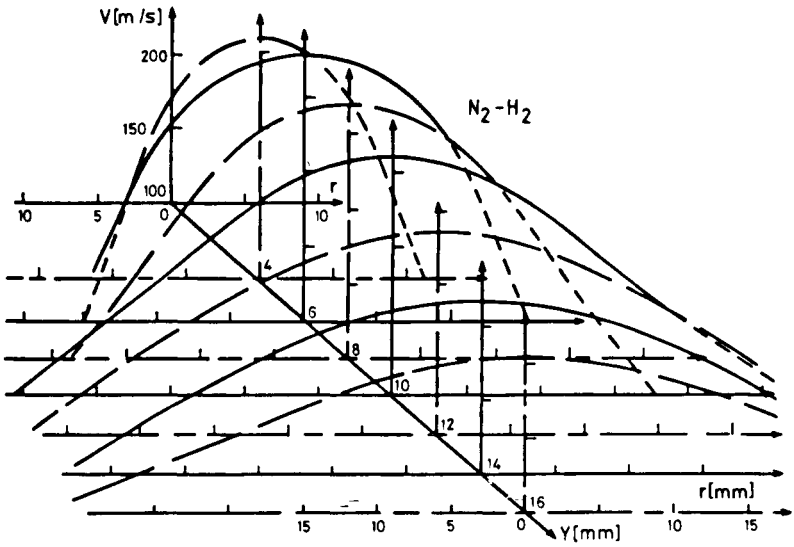


FIGURE 50. Radial distribution of the velocity of alumina particles (diameter, 18 ± 3 μm) injected with the optimum velocity into a N_2-H_2 plasma jet (37 l/min N_2, 11 l/min H_2, 290 A, 29 kW). (Vardelle, A. et al., 1982).

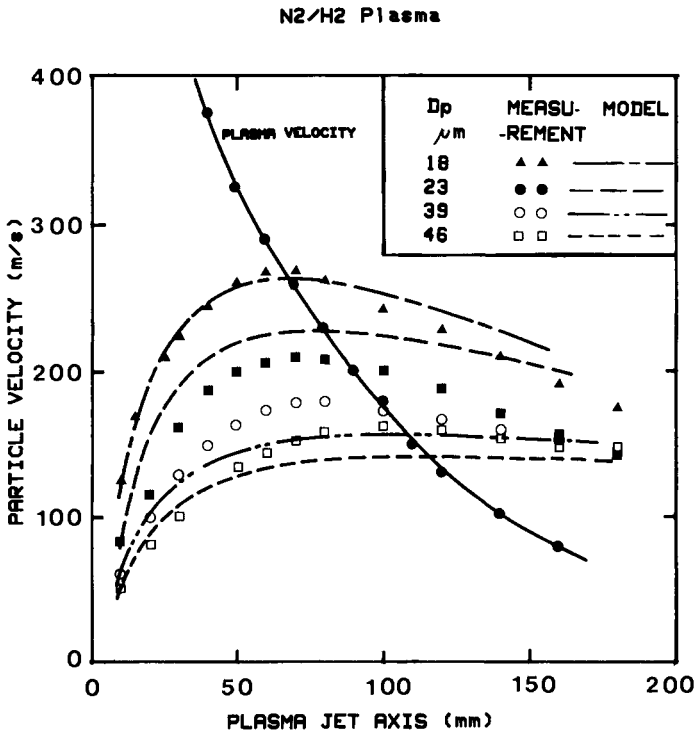


FIGURE 51. Particle axial velocities along the centerline of the N_2-H_2 plasma jet; 37 l/min N_2, 11 l/min H_2, 290 A, P = 29 kW. (Vardelle, M., 1987).

to the use of a high carrier gas flow rate to achieve a high velocity of the small particles in the injector (see Section V.A).

Measurements were made of the axial velocity profiles of different narrow cuts of pure alumina particles injected into N_2/H_2 and Ar/H_2 plasmas, under essentially the same operating conditions except for the nozzle diameter which was 6 mm for N_2/H_2 against 8 mm for Ar/H_2. The different powders were injected with the optimum mean injection velocities. The variation of the axial velocity profiles along the centerline of the torch for different powders in a N_2/H_2 plasma is given in Figure 51. As expected, the smaller particles, because of their lower inertia, are accelerated much faster than the heavier particles. However, once the heavy particles attain their maximum velocity, about 60 to 80 mm from the nozzle exist, they retain momentum which results in a higher velocity than the gas velocity 140 to 160 mm downstream from the nozzle.

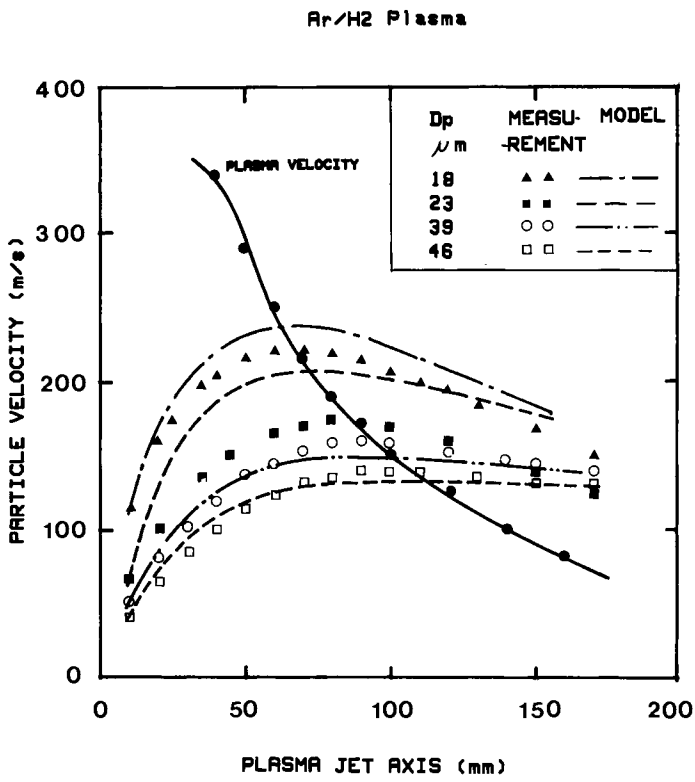


FIGURE 52. Particle axial velocities along the centerline of the Ar-H_2 plasma jet, P = 29 kW, 75 l/min Ar, 15 l/min H_2, 290 A. (Vardelle, M., 1987).

Similar results were also obtained for the Ar/H_2 plasma as shown in Figure 52, although the maximum velocity attained by the particles in this case was lower than in the N_2/H_2 plasma (275 m/s for 18 μm particles in a N_2/H_2 plasma compared to 220 m/s for the same particles in an Ar/H_2 plasma). This difference, however, is hardly noticeable for heavier particles (cf. Figures 51 and 52). A 46 μm particle attains essentially the same maximum velocity of 140 m/s at about 100 mm from the nozzle independent of the plasma gas used (N_2/H_2 or Ar/H_2).

R.F. Plasmas. R.F. plasmas have much lower velocities (10–60 m/s) than the 400–900 m/s in D.C. plasma spray jets. Lesinski et al., (1981) reported

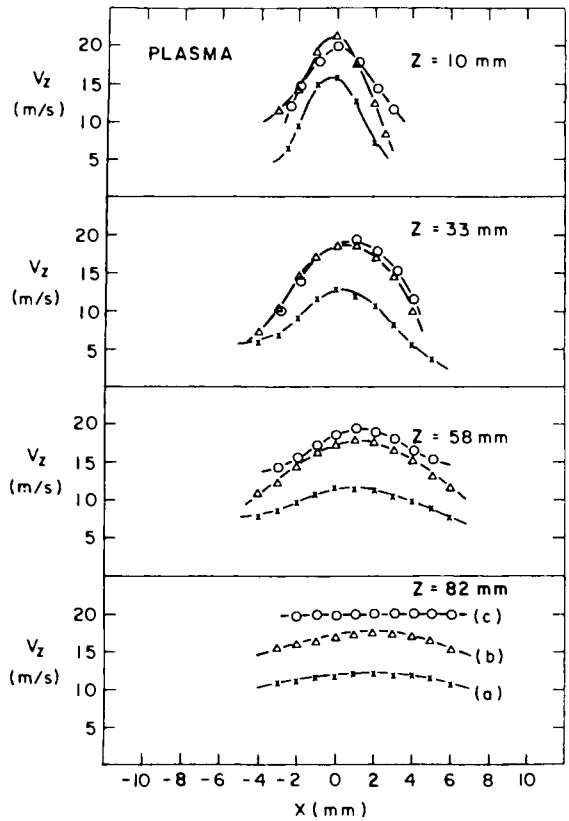


FIGURE 53. Plasma velocity profiles over the central region of a R.F. plasma torch working with argon. (Lesinski, J., et al., 1981b).

plasma and particle velocity measurements in the coil region of an inductively coupled R.F. discharge. Measurements were performed for an Ar plasma with gas flow rates of 11.7 l(STP)/mn along the generator wall and of 63 l/mn in the central part. Powders were injected along the axis of the generator with the following carrier gas flow rates and power levels in the plasma: a) Q_1 = 4.8 l/mn, P = 4.64 kW; b) Q_1 = 7.4 l/mn, P = 4.64 kW; c) Q_1 = 7.4 l/mn, P = 10.28 kW. Typical results of the velocity profiles along the torch axis are shown in Figure 53. The axial velocity profiles of the particles (×) and plasma (△), as well as the axial velocity of the gas (○), in the absence of the discharge, show distinct behaviour.

C. Particle Size Measurements

Particle size measurements have been discussed by Gouesbet, (1985), with special emphasize on techniques which can be used for simultaneous measurements of the particle size and velocity under plasma conditions. The methodology is based on the Lorenz-Mie theory, which describes the light scattered by spherical, homogeneous, isotropic, nonmagnetic particles illuminated by a plane light wave with rectilinear polarization. However, the theory is complex and the diameter to be measured is not monotonically related to the characteristics of the scattered light.

1. Visibility Technique

From the shape of a Doppler signal (see, for example, Figure 43), the visibility is defined by:

$$V_i = \frac{V_M - V_m}{V_M + V_m}. \tag{50}$$

V_i can be related to the diameter of the scattering particle by

$$V_i = 2\left|\frac{J_1(\pi d_p/d_f)}{d_p/d_f}\right|, \tag{51}$$

where J_1 is the first order Bessel function, d_p is the particle diameter and d_f the interfringe spacing. However, to obtain a simplified formula as Eq. (51), the following assumptions must be satisfied:

- A1. The wave front of the incident light must be planar.
- A2. Only para-axial forward or back scattering is considered.
- A3. The scattered intensity is calculated by assuming that the scattering center receives a mean illuminating intensity equal to a simple average of the illuminating intensity over the cross-sectional area of the scattering center.
- A4. The fringe spacing must be smaller than the radius b_0 of the focused laser beams in the control volume.
- A5. The diameter of the particle must be smaller than b_0.
- A6. Only the location of the particle at the geometric center of the probe volume is considered.

Assumption A1 is not correct for particles bigger than 30 μm; assumption A2 eliminates off axis collection systems (which are mostly used); assumption A6 makes corrections difficult (with the Gaussian intensity profile and

the uncontrolled particle trajectories similar signals are obtained with small particles passing through the center, or large particles passing through the edge of the measurement volume). Assumptions A4 and A5 are also very restrictive (see Gouesbet, 1985) and thus the precision of such measurements is usually questionable, especially when broad particle size distributions are to be considered.

2. *Pedestal Calibration Methods*

These methods are based on correlating particle diameters with the pedestal amplitude V_m (see Figure 43), through the Mie theory (Grehan and Gouesbet, 1979; 1980; 1981), taking into account the Gaussian intensity distribution of the incident beams. Here also the trajectory ambiguity (particles passing through the center or the edges of the measurement volume) is not a simple problem to solve. Because of it, simultaneous measurements of size and velocity distributions are achieved, but not simultaneous measurements of the size and velocity of single particles.

3. *Top-Hat Laser Beam Technique*

This is a new technique developed by Gouesbet, (1985), using a specially designed anti-Gaussian, metal-coated filter placed in the path of the incident laser beam in order to create a "top-hat" type of intensity distribution of the laser beam (flat intensity distribution which suppresses the trajectory ambiguity due to the Gaussian shape). This was combined in a two color system for the simultaneous measurement of particle size and velocity.

4. *Light Pulse Method*

In principle, the method developed by Sakuta and Boulos, (1987), is very simple and there is no trajectory ambiguity, the particle diameter being given by Eq. (48). However, here again the particle has to be seen in the plasma flow which limits measurements to plasma regions where $T < 6{,}000$ K. Also, as noted earlier, it is only applicable to large particles which give resolved images with available optics, since the method is subject to the Rayleigh criterion for the diffraction-limited resolution of a telescope.

D. In-Flight Particle Surface Temperature Measurements

To obtain proper melting or evaporation of particles, the surface temperature time history of injected metallic or other solid particles has to be controlled. Surface temperature data are also necessary to gain a better understanding of the momentum and heat transfer between the plasma and the particles.

Temperatures are generally measured either by calorimetric methods or by flux measurements and, to our knowledge, until now measurements have been restricted to plasma spray jets or plume of R.F. discharges.

1. Calorimetric Methods

Calorimetric methods (Houben et al., 1983; Marijinisen, 1986) sense the total heat contained in a given mass of particles that are sprayed onto a calorimetric target by a D.C. plasma jet. The heat transferred to the target by the particles and the plasma jet is measured with the calorimeter (in most cases water-cooled targets where the water temperature increase is measured). After subtracting heat transferred by the plasma jet (measured separately), the heat content of the particles is related to the temperature of the particles, their size distribution, the specific heat of their liquid and solid phases, and the latent heat of melting. However, these measurements only give averaged values and no information on the temperature distribution. Moreover, they assume that all the particles are collected in the calorimeter (that none bounce off, for example).

2. Flux Measurements

Flux measurements can either be absolute (Kruszewska and Lesinski, 1977; A. Vardelle et al., 1980) or relative (Mishin, 1985; Mishin et al., 1987; M. Vardelle, 1987; Sakuta and Boulos, 1987). The absolute radiative flux emitted by the particles is measured at a given wavelength range (using a monochromator or a filter followed by a photomultiplier), and the surface temperature can be determined, provided the diameter of the particle and its emissivity are known. However, the diameter of a particle changes as it evaporates. The size distributions used for plasma spraying, for example, are rather large (typical examples of fed powders: 5–22 μm; 20–45 μm; 20–65 μm), and thus the particle diameter is unknown. Moreover, the emission coefficient of most of the materials (especially of ceramics) is

barely known at high temperatures. Hence, methods using the ratio of two fluxes measured at two adjacent wavelengths have been developed to avoid problems associated with measuring the diameter, and to make the measurements less sensitive to the emission coefficient (assuming a "grey body" at both wavelengths is a better approximation than using uncertain and inconsistent emissivity coefficients).

Problems Related to the Measurements. Plasma emission originates both in the continuum and transitions between excited states (atoms or molecules), giving discrete lines or band spectra. Spectra regions for the measurements of the flux emitted by the particles must be chosen carefully to avoid the plasma discrete emission (lines and bands). For the argon-hydrogen plasma, entrained nitrogen and oxygen molecules (from the air) are excited by collisions with hot species and emit their corresponding spectra. The wavelength windows (with no emitted lines) are very narrow (about 160 Å) and thus monochromators have to be used within the appropriate wavelength windows to perform the particle flux measurements, where the particle emission is not obscured by plasma lines and/or bands.

The continuum emission becomes very intense for $T > 8{,}000$ K for $Ar\text{-}H_2$ or $N_2\text{-}H_2$ plasmas. This implies that the particles must emit a higher flux than that of the continuum to be seen. This flux depends on the particle size and surface temperature as well as on the wavelength range in which the measurement is performed. For example, in argon plasmas the total continuum emission between 8,000 and 11,000 K ranges from 10^6 to 10^8 W/m^3 (Evans and Tankin, 1967), which corresponds to $1.8\ 10^{-4}$ W for a measurement volume of $1.7\ 10^{-2}$ mm^3, assumed at a mean temperature of 8,000 K. By contrast, an alumina particle, 50 μm in diameter, at $T = 2{,}300$ K emits 9.91×10^{-22} W (when considered as a black body). Thus, measurements of the particles surface temperatures are possible only in the plasma plume (further than about 50 mm from the nozzle exit for a D.C. plasma jet), where the plasma temperature is below 6,000 K, and even then the particle surface temperature must be above 2,000 K (M. Vardelle, 1987).

Another difficulty stems from the high-temperature and velocity gradients in thermal plasmas. Small differences in injection velocity and size distributions can result in widely different trajectories. Even for very small measuring volumes ($\sim$ 1 mm^3), the gradients acting on particles can produce broad distributions of both velocities and surface temperatures, which means that only statistical measurements are possible.

With particle velocities up to 350 m/s (in D.C. arc spraying), the residence time of a particle in the measuring volume (usually a cylinder of less than 0.2 mm in diameter and perpendicular to the plasma jet axis) is between 0.4 to 2 μs. The consequence of these short residence times on flux

measurement is first, the very low quantity of energy received from a single particle and thus the necessity of accumulating the data from many particles to obtain a measurable signal and, second, the requirement of fast detectors (about 10 ns response time).

When the particles are subjected to thermal plasma fluxes (up to 10^8 W/m^2), with high heat transfer coefficients for plasmas containing hydrogen (Bourdin et al., 1983), the delay in heat propagation into the particle has to be taken into account. The surface of a particle (especially ceramics ones) begins to evaporate long before the core even starts to be heated, i.e., after a few millimeters of trajectory. Thus, the particle diameter changes along its trajectory (between 10 and 25% volume evaporated). Measurement of absolute fluxes is therefore difficult if this change in diameter cannot be followed (to our knowledge, it has still not been done in thermal plasmas). Moreover, the vapour from the particles can modify the plasma properties and the emitted spectra.

Experimental Set-ups. The optical system (Figure 54) developed by Mishin et al., 1987), consists of two photographic lenses (f = 50 mm) and a beam splitter arranged so that two images of the plasma jet are projected on the entrance slits of two monochromators (H10 IR—Jobin Yvon with holo-

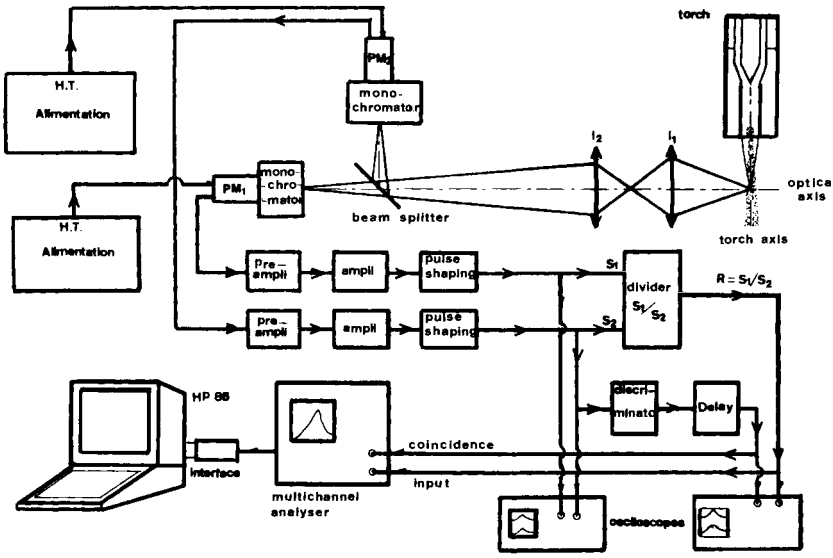


FIGURE 54. Experimental set-up for the two-color pyrometer. (Mishin, 1987).

graphic gratings, 600 grooves/mm, with a band pass width of 160 Å). The entrance slit size (8 mm × 0.5 mm) and optics determine the dimensions of the measuring plasma jet volume corresponding to a plasma slice 1 = 1.5 cm with side cross-section of 0.125 mm^2. The wavelength windows are carefully chosen to avoid the spectral distribution of the plasma radiation. For example, to measure the surface temperature of alumina particles in an argon-hydrogen plasma flowing in air, the working wavelengths are 6,850 Å (between emission lines ArI 6,714 Å and ArI 6,965 Å) and 8,310 Å (between lines ArI 8,264-ArI 8,408 Å) with windows of 100 Å.

Two photomultipliers (Hamamatsu—R 928) placed behind the exit slits of the two monochromators give a current pulse (S_1 or S_2) when a particle passes through the measuring volume. The amplitude of each pulse depends on the particle size (cross-section), surface emissivity, surface temperature, wavelength and the operating conditions (opto-electronic) (Mishin, 1985; and Mishin et al., 1987).

Signals S_1 and S_2 are amplified and shaped by a preamplifier (ORTEC 113) and a linear amplifier (ORTEC 450). They are then sent into a sensitive detector analyser (ORTEC 464), which generates an output signal, R, proportional to the ratio of amplitudes of the input signals. Calculations and measurements show R is Gaussian for the distribution of the particle temperatures. However, the shape of the two signals S_1 and S_2 vary as the root mean square of the temperature distribution. The ratio signal is sent to the input of a multichannel analyser (Northrop 5300) which provides the histogram of signal amplitude distributions.

The experimental set-up of Sakuta and Boulos, (1987) shown in Figures 47 and 48 has already been described in Section V.B.3. One of its advantages for surface temperature measurements is the small size of the measurement volume defined by means of the He-Ne laser beam.

Calibration. The calibration consists of correlating the particle temperature with the amplitude of the signal ratio.

For simulating the particle diameters in the Mishin et al., (1987) set-up, two pin-holes, of the same size, are placed in front of the entrance slit of the monochromators where a high speed chopper simulates the particule velocities. The light source, a calibration tungsten ribbon lamp or an arc lamp, provides illumination to simulate emission of a hot particle (Figure 55). The temperatures of the tungsten ribbon lamp or arc lamp are determined by a monochromatic pyrometer (IRCON), and are corrected for the emissivities at the two chosen wavelengths. For a given temperature of the tungsten ribbon lamp, the fractional signal ratio is determined by the multichannel analyser, and a calibration correspondence between the temperature and the analyser channel is mapped out.

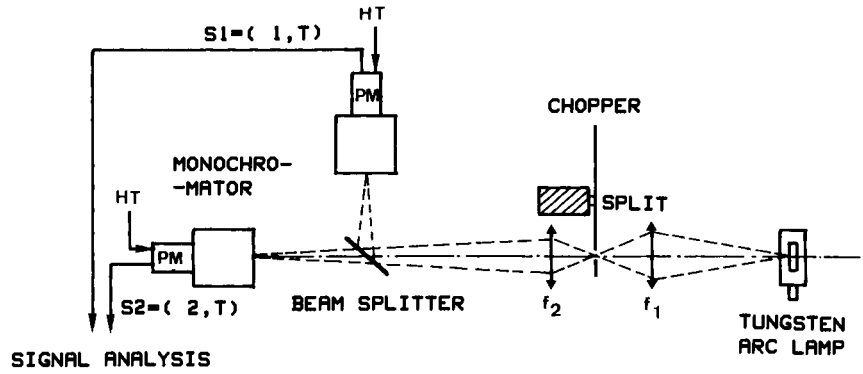


FIGURE 55. Experimental set-up for calibration. (Mishin, J. et al., 1987).

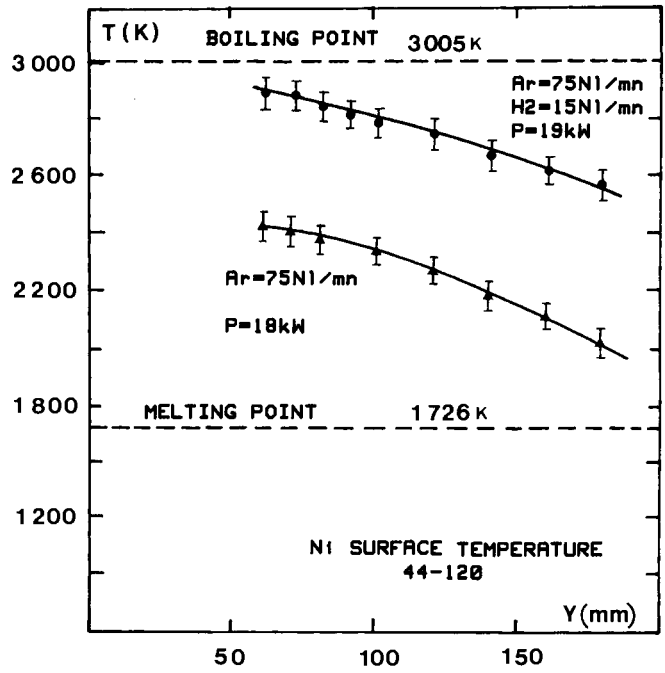


FIGURE 56. Nickel particle surface temperatures along the jet axis for two different plasma gas compositions (with 75 l/min Ar, 15 l/min H_2 or without hydrogen, 75 l/min Ar). (Vardelle, M., 1987).

The apparatus function of this system is given by the Gaussian temperature distribution recorded during the calibration with a fixed temperature of the lamp. This distribution, which ideally should be a delta function, is due to the noise in the opto-electronic system working with very high sensitivity. Using a Gaussian distribution function, it is possible to obtain the real surface temperature distribution from the measured distribution and the apparatus function distribution (Mishin, 1985).

In the Sakuta and Boulos, (1987) set-up the calibration is similar to that of Mishin et al., (1987).

Limitations and Accuracy. There are three main limitations to this technique. The first is that one cannot measure the particle surface temperatures inside the plasma core (when temperatures are higher than 7,000 K) because of the very strong radiation from the plasma. Paradoxically, it would be most interesting to determine the evolution of the particle temperature in this region because there is where heating takes place. The second limitation is that the maximum temperature range that can be measured is limited to

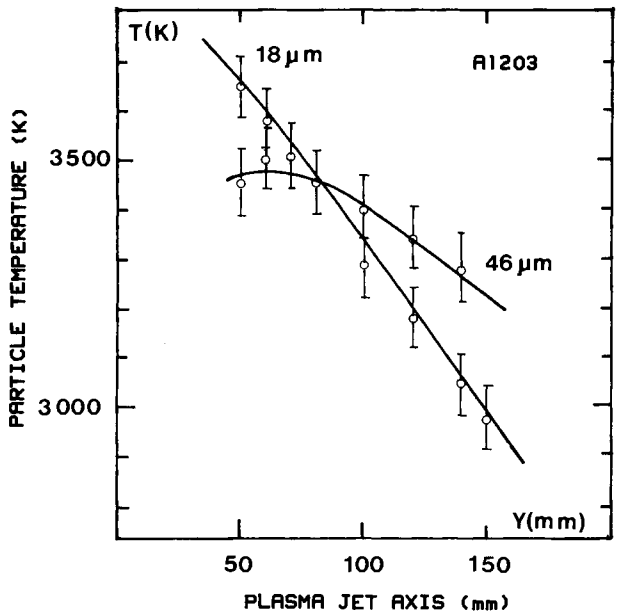


FIGURE 57. Alumina particle surface temperatures along the jet axis for two different particle mean diameters (21 + 15 μm and − 52 + 40 μm) for P = 29 kW, 75 l/min Ar, 15 l/min H_2. (Vardelle, M., 1987).

about 1,200 K by the dynamic range of input signal amplitudes accepted by an analog divider (i.e., one decade). However, by digitizing the signals and calculating their ratio directly with the computer, the temperature range has been increased to 2,000 K (M. Vardelle, 1987) and measurements can be performed between 1,800 and 3,000 K in the Sakuta and Boulos, (1987) set up. The last limitation is the number of particles in the measuring volume necessary to avoid signal overlapping. For a given particle flow rate, when the size of the measurement volume increases, the probability of having two particles in it simultaneously increases too, and the particle flow rate then has to be reduced to avoid signal overlapping. For example, in the experiment of Mishin et al., (1986), with a measuring volume as long as 15 mm, the time between two successive signals has to be higher than 30 μs and this corresponds about to a maximum powder flow rate of 250 g/h to avoid signal overlapping. In the experiment of Sakuta and Boulos, (1987), the dimensions of the measuring volume has been drastically reduced with the

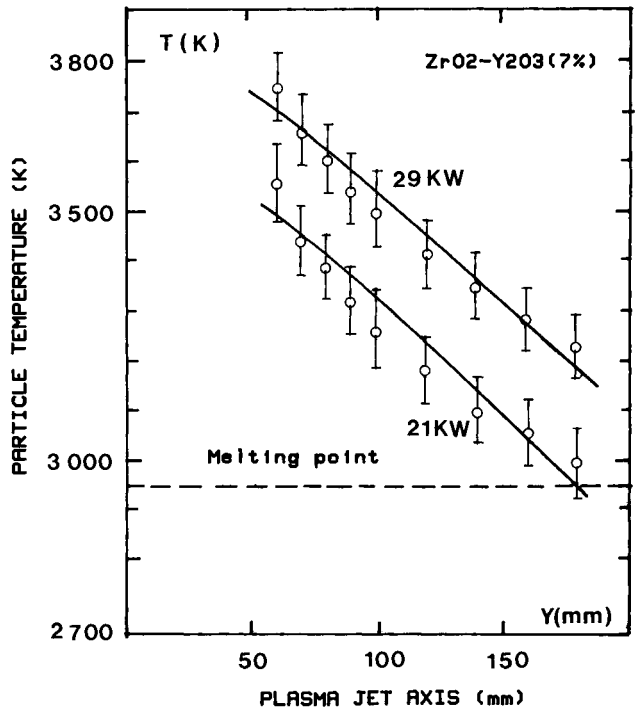


FIGURE 58. Zirconia particles surface temperatures for two different power levels (21 and 29 kW 75 l/min Ar, 15 l/min H_2). (Mishin, J. et al., 1987).

auxiliary beam from the He-Ne laser and with the larger volume of the R.F. plasma, no maximum flow rate has been determined. The overall accuracy of the method depends mainly on a knowledge of the material emissivity (about ± 150 K for the temperature), the accuracy of calibration (± 50 K) and of the measurement (± 50 K). Altogether this amounts to about ± 250 K; the stochastic error shown in the Figures 56 to 58 is about 50 K. Less important parameters such as reflected radiation from the plasma, that part of the off-normal radiation received, or nonsphericity of particles, were not taken into account in the accuracy evaluation.

3. *Examples of Results*

Measurements made with nickel particles and two different plasma gas compositions (pure argon and hydrogen-argon mixture) are shown in Figure 56 (curves 1 and 2). The mean temperature of the particles approaches their boiling point at a distance of about 60 mm from the nozzle exit and decreases slowly with distance. With a pure argon plasma, their mean temperature is always above the melting point and about 500 K less than in the case of the Ar-H_2 plasma. This is mainly caused by the increase of the thermal conductivity with H_2 addition (Bourdin et al., 1983).

Figure 57 shows the dependence of Al_2O_3 particle surface temperatures on particle diameters for a narrow size distribution range. It should be mentioned that the lowest cooling rate occurs for the largest Al_2O_3 particles.

The influence of the electric power input to the torch (working with an argon-hydrogen mixture) on the temperature of ZrO_2-8 wt% Y_2O_3 particles is shown in Figure 58. The mean temperature increases 200 K as the electrical power is increased from 21 to 29 kW. The quality of plasma sprayed deposits depends on the state of the particles at the moment of their impact on the substrate. If this powder should be sprayed at 21 kW rather than at 29 kW, the mean temperature is too close to the melting temperature, and the deposit is often more porous, due to the unmelted core of the particles (insufficient time for heat propagation).

VI. Conclusions

Most of the measurement techniques used for thermal plasma flows are quite classical; however, they have been adapted to meet the specific

Table 1a. Plasma Flow Parameters. Measured parameters and techniques available. T_h heavy species temperature, T_{ex} excitation temperature, T_e electron temperature, $n(X_2)$ molecular ground state, n_g atomic ground state, n_m atomic metastable state, n^* atomic excited state, n_{rot} rotational level population, n_{vib} vibrational level population.

Parameter	Temperature (kK) 1 3 6 9 12 15 18 21	Technique
T_h		Rayleigh Scattering
T_h		Line reversal method
T_h	?	CARS
n_g (X_2)		LIF
n_g, n_m		Absorption spectroscopy
n_e, T_e		Thomson scattering
n_e		Emission spectroscopy: continuum, Stark broadening of atomic and ionic lines
n_{rot}, n_{vib} T_h, T_{vib}		Emission spectroscopy Molecular spectra
n^*, T_{ex}		Emission spectroscopy atomic lines
n_+^*, T_{ex}		Ionic lines
enthalpy, T_h chemical composition		Enthalpy probes (non transferred arcs)

conditions encountered:

- very high radiation fluxes;
- very steep gradients (temperature, velocity, concentrations, and volumetric emission coefficients);
- flows ranging from laminar to turbulent over a few millimeters;
- natural fluctuations of arc roots or R.F. coupling with the plasma which makes only mean temporal measurements meaningful;
- very complex mixtures which, in general, contain from 2 to 60 different chemical species as molecules, atoms or ions which make the emitted spectra very complex and crowded with lines.

When particles are injected into such media, they have size and injection velocity distributions resulting in very broad trajectory distributions. Finally, according to the complexity and to the high gradients, automatized measurements must be performed.

It is worthwhile to emphasize that since the 1980s a big effort has been devoted to measurement techniques under plasma conditions, and a lot of progress has been made, allowing a better understanding of the phenomena involved and of correlations with models. However, there is no technique

Table 1b. Particle Parameters.

Parameter	Size (μm)	Plasma Temperature (kK) 1 — 6 — 12	Technique
Spatial particle distribution	> 5		Laser burst counting
Particle velocity	> 10		Laser anemometry dual beam v < 500 m/s
Particle velocity	> 3		Laser anemometry 2 focus
Particle surface temperature	> 10		Two color pyrometry (particle flux) T_{part} > 1,700 K
Particle size	> 1		Laser sizing

which allows for measurements from room temperature to 20,000 K, and each technique is only valid for a rather narrow range of parameters. A summary is presented in Tables 1a and b listing actually possible measurements and their limitations.

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References

Akashi, K. (1985). *Pure and Applied Chemistry* **57** (9), 1197, 1206.
Baronnet, J. M. (1972). Thèse 3ème cycle, Univ. of Limoges, France.
Baronnet, J. M. (1978). Thèse Doctorat Etat, Univ. of Limoges, France.
Barnes, R. M. (1978). *CRC Critical Review, Analy. Chem.* **7**, 203.
Barr, W. L. (1962). *J. Opt. Soc. Am.* **52**, 885.
Blades, M. W., Caughlin, B. L., Walker, Z. H., and Burton, L. L. (1987). *Prog. Analyst. Spectros.* **10**, 57.
Born, M. and Wolf, E. (1965). *Principles of Optics*. Pergamon Press, N.Y.
Boulos, M. (1985). *Pure and Applied Chemistry* **57** (9), 1321, 1352.
Brossa, M. and Pfender, E. (1988). *Plasma Chemistry, Plasma Processing* **8** (1), 75.
Bourdin, E., Fauchais, P., and Boulos, M. (1983). *Int. J. Heat Mass Transfer* **26**, 567.
Budde, W. (1962). *J.Q.S.R.T.* **52** (3), 343, 345.
Cabannes, F., et al. (1965). *Rev. Int. Hautes Temper. et Refract.* **2**, 297.
Camacho, S. L. (1984). *Long column plasma arc torches for pyrometallurgical applications, MINTEK* 50. MINTEK, Johannesburg, S.A.
Chapelle, J. (1971). "Spectroscopic plasma diagnostics" In: *Plasma Chemistry* (M. Venugopalan, ed.). Wiley Interscience, N.Y.
Champagne, B. and Dallaire, S. (1988). *Particle injection in plasma spraying*. National Thermal Spray Conference ASM's 1987, Orlando, FL (D. Houck, ed.) ASM Int., Metals Park, OH.
Chase, J. D. (1971). *J. Appl. Phys.* **42**, 4870.
Cheng, S., Zimmerman, M., and Miles, R. B. (1983). *Appl. Phys. Lett.* **43** (2), 143.
Correa, S. M. (1983). *ISPC6* **1**, 77 Montreal July (Prof. Boulos, ed.). Univ. of Sherbrooke, CN.
Coudert, J. F., Baronnet, J. M., and Catherinot, A. (1982). *7th Int. Conf. on Gas Discharges and their Applications, IEEE*. London.
Coudert, J. F., Baronnet, J. M., and Fauchais, P. (1983). *ISPC6* **1**, 102. (Prof. M. Boulos, ed.). Univ. of Sherbrooke, CN.
Coudert, J. F. (1985). Thèse Doctorat d'Etat. Univ. of Limoges, France.
Coudert, J. F., Roumilhac, Ph., Grimaud, A., and Fauchais, P. (1987). *ISPC8* **1**, 412. (Prof. Akashi, ed.). Univ. of Tokyo, Japan.
Crosley, D. R. and Smith, G. P. (1983). *Optical Engineering* **22**, 545.
Dailey, J. W. (1977). *Appl. Opt.* **15**, 568.

Davis, J. I., Vaugham, J. M. (1963). *Astrophys. J.* **137**.
Desai, V., Daniel, E. S., and Corcoran, W. H. (1968). *Rev. Sci. Instr.* **39**, 612.
Deutsh, M. and Beniaminy, I. (1983). *J. Appl. Phys.* **54** (1).
Drellishak, K. S. (1963). Ph.D. Thesis. Northwestern Univ., Evanston, IL.
Dresvin, S. V. and Klubnikin, V. S. (1975). *High Temperature* **13** (2), 400.
Dundas, P. H. (1970). *Induction Plasma Heating: Measurement of Gas Concentrations, Temperatures and Stagnation Heads in a Binary Plasma System.* NASA CR-1527.
Durst, F., Melling, A., and Whitelaw, J. H. (1976). *Principles and practice of laser Doppler anemometry.* Academic Press, N.Y.
Eckert, H. U. (1974). *High Temp. Sci.* **6** (2), 99.
Eddy, T. L. (1976). *IEEE Trans. Plasma Sci.* **PS4**, 103.
Eddy, T. L. (1985). *J. Quant. Spectrosc. Radiat. Transfer* **33** (3), 197.
Eddy, T. L., Pfender, E., and Eckert E. R. G. (1973). *IEEE Trans. Plasma Sci.* **PS1**, 31.
Eddy, T. L. and Sedghinasab, A. (1987). *IEEE Conf. on Plasma Science.* Arlington, Va, USA, Proc.
Etemadi, K. and Pfender, E. (1982). *Rev. Sci. Instr.* **53** (2), 12.
Evans, D. L. and Tankin, R. S. (1967). *Phys. Fluid* **10**, 1137.
Fanton, J. C., Lise, B., and Boulos, M. I. (1982). *Rev. Sci. Inst.* **53**, 656.
Fauchais, P. (1968). Thèse de Doctorat es Sciences Physiques. Univ. of Poitiers.
Fauchais, P., Vasseur, A., and Manson, N. (1969). *Rev. Int. Hautes Temp. et Réfract.* **6**, 5.
Fauchais, P., Baronnet, J. M., and Bayard, S. (1975). *Rev. Int. Hautes Temp. et Réfract.* **12**, 221.
Fauchais, P., Lapworth, K., and Baronnet, J. M. (1980). *First report on measurement of temperature and concentration of excited species in optically thin plasma.* (IUPAC Sub Committee on Plasma Chemistry) (P. Fauchais, ed.). Univ. of Limoges, France.
Fauchais, P., Coudert, J. F., Vardelle, A., Vardelle, M., and Lesinski, J. (1984a). *Mat. Res. Soc. Symp. Proc.* **30**, 37–51, Elsevier Science Publishing Co.
Fauchais, P., Vardelle, A., Vardelle, M., Coudert, J. F., and Lesinski, J. (1984b). *Thin Solid Films* **121**, 303, 316.
Fauchais, P., Vardelle, A., Vardelle, M., Coudert, J. F., and Pateyron, B. (1985). *Pure and Appl. Chem.* **57** (9), 1171–1178.
Fauchais, P., Boulos, M., and Pfender, E. (1987) "Physical and thermodynamic properties of thermal plasmas" In: *Plasma Technology in Metallurgical Processing.* (J. Feinman, ed.). Iron and Steel Society, PA, USA.
Fey, M. G. (1980). *Development of a process for high capacity arc heater production of silicon for solar array.* Contract 954 589, DOE/JPL, USA.
Finkelnburg, W. and Maecker, H. (1956). In: *Encyclopedia of Physics* (S. Flügge, ed.). Vol. XXII, p. 254. Springer Verlag, Germany.
Fizdon, J. et al. (1976). *Rev. Int. Hautes Tempér. et Réfract.* **13**, 11.
Foster, T. and Lin, G. (1983). *Plasma arc heater design.* Accurex Corp., Mountain View, CA, USA.
Frind, G., Goody, C. P., and Prescott, L. E. (1983). *6th ISPC Montreal July, Proc.* **120**. (Prof. Boulos, ed.). Univ. of Sherbrooke, CN.
Fuji, S., Gomi, M., and Eguchi, K. (1983). *J. Fluids Eng.* **105**, 128.
Garscadden, A. (1986). *Gas Discharge Laser Diagnostic Update in Radiative Processes in Discharge Plasmas.* (J. Proud and L. Niessen, eds.). Plenum Pub. Corp., N.Y.
Gauvin, W. H., Drouet, M. G., and Munz, R. J. (1985). *Recent Developments in Extractive Metallurgy Plasma Processes AIME.* TMS Technical papers Nb 186-24.
George, A. P. and Hare, A. L. (1984). *The Development of High Power Arc Material, a Progress Report* Paper C/5/3, Mintek 50, Johannesburg, S.A.

Gouesbet, G. (1985). *Plasma Chem. Plasma Proc.* **5** (2), 91.
Gouesbet, G. and Trinité, M. (1977). *Letters in Heat and Mass Transfer* **4**, 141.
Gravelle, D., Beaulieu, M., Carbone, C., and Boulos, M. (1983). *ISPC6* **1**, 108. (M. Boulos, ed.). Univ. of Sherbrooke, CN.
Greham, G. and Gouesbet, G. (1981). *Appl. Opt.* **20**, 796. See also **18** (1979), 3489, **19** (1980), 2485.
Grey, J. (1963). *Rev. of Sci. Instrum.* **34** (8), 857.
Grey, T. (1976). Probe Measurements in High Temperature Gases and Dense Plasmas. In: *Measurements in Heat Transfer, 2nd ed.* Hemisphere Publ. Corp., Washington.
Griem, H. R. (1964). *Plasma Spectroscopy*. McGraw Hill Book Company, N.Y.
Griem, H. R. (1977). *Spectral Line Broadening by Plasma*. Academic Press, N.Y.
Haddad, G. H. (1986). *Temperature Measurements in Arcs*. Presented at the Gordon Conference on Plasma Chemistry, Aug. 1986.
Hall, R. J. and Eckbreh, A. C. (1984). CARS: Application to Combustion Diagnostics. In: *Laser Applications*, Vol. 5 (J. F. Ready and R. K. Erf, eds.). Academic Press, N.Y.
Hare, A. L. (1977). *Enthalpy probes—A contribution to the theory of operation*. 3d Int. Symposium on Plasma Chem., Limoges, G.3.1 (P. Fauchais, ed.). Univ. of Limoges, France.
Herzberg, G. (1959). *Spectra of Diatomic Molecules*. Van Nostrand Company, N.Y.
Holve, D. and Self, S. A. (1979). *Applied Optics* **18**, 1632, 1646.
Houben, J. M., et al. (1983). *I.T.S.C. 10, Den Haag*. (Prof. Zaat, ed.). Eindhoven Univ. of Technology, N.L.
Hsu, K. C. and Pfender, E. (1983). *J. Appl. Phys.* **54** (8), 4359.
Incropera, F. P. and Leppert, G. (1967). *ISA Transactions*. **6** (1), 35.
Jayroe, R. R and Fowler, R. G. (1966). *J. Opt. Soc. Am.* **57** (4), 513, 516.
Joukov, M. F. (1979). *Basical Calculations of Plasmatrons*. (In Russian). Nauka, Novossibirsk, USSR.
Kaczmarck, R., Robert, W., Jurewicz, J., Boulos, M. I., and Dallaire, S. (1984). *Mat. Res. Soc. Symp. Proc.* **30**, 207. MRS, Pittsburgh.
Kassabji, F. and Fauchais, P. (1981). *Rev. Phys. Appl.* **16**, 549.
Kolesnikov, V. N. (1966). *Proc. P.H. Lebedev Phys. Inst.* **30**, 53.
Kosarev, E. L. and Pantos, E. (1983). *J. Phys. E. Sci. Instrum.* **16**, 547.
Kruszewska, B. and Lesinski, J. (1977). *Rev. Phys. Appl.* **12**, 1203.
Kunze, J. J. (1968). *Plasma Diagnostics*. (Lochte-Holtgreven, ed.), North Holland, Amsterdam.
Labrot, M. E. and De Logiviere, M. (1984). *10th Congress of UIE*. Stockholm, June, paper 6.2.
Lading, L. (1976). Processing of Laser Doppler Anemometry Signals. *DISA informations* **19**, 12.
Lana, F. and Viollet, P. L. (1987). *ISPC8* **1**, 31. (Prof. Akashi, ed.). Univ. of Tokyo, Japan.
Lee, Y. C. and Pfender, E. (1987). *Plasma Chem. Plasma Proc.* **7** (1).
Lesinski, J., Mizera-Lesinska, B., Fanton, J. C., and Boulos, M. I. (1979). *4th Int. Symp. on Plasma Chem.*, Zurich (Ch), August. (S. Veprek, ed.).
Lesinski, J., Mizera-Lesinska, B., Fanton, J. C., and Boulos, M. I. (1981a). *AIChEJ* **27**, 358.
Lesinski, J., Gagne, R., and Boulos, M. I. (1981b). *5th I.S.P.C.* Edinburgh, U.K., Aug. (B. Waldie, ed.). Univ. of Edinburgh.
Lewis, J. A. and Gauvin, W. H. (1973). *AIChEJ* **19**, 982.
Lugsheider, W. (1983). *J. Four. Electrique* **10**, 29.
Maecker, H. (1953). *Z. Phys.* **136**, 119.
Marijinisen, G. (1986). In: *Advances in Thermal Spraying*, p. 251. Pergamon Press, Oxford, G.B.
Maldonado, C. D. and Olsen, H. N. (1966). *J. Opt. Soc. Am.* **56**, 1305.
McKelliget, J. and Szekely, J. (1986). "Heat transfer and fluid flow in the welding arc." To be published in Metallurgical Transactions.

Magre, P., Moreau, P., Collin, G., and Pealat, M. (1987). Application de la méthode DRASC à la mesure de température dans une flamme turbulente, Com. 39. In: *Proc. Congrès SFT* 87 (*ed.*), SFT Paris.
Medani, A. (1981). Thèse 3ème cycle. Univ. of Toulouse, France.
Mehmetoglu, M. T., Fitzinger, F., and Gauvin, W. H. (1983). *Rev. Sci. Instrum.* **53** (2a), 285.
Mishin, J. (1986). Contribution à la mise au point d'un dispositif de mesure des températures de surface des particules "en vol" dans un jet de plasma d'arc. Thèse Doc. Ing. Univ. of Limoges, France.
Mishin, J., Vardelle, M., Lesinski, J., and Fauchais, P. (1987). *J. of Phys. E. Sci. Instr.* **20**, 620.
Monk, J. R. (1981). *ISPC5* **1**, 162 (Prof. Waldie, ed.). Univ. of Edinburgh.
Morris, J. C. and Yos, J. M. (1971). Radiation studies of arc heated plasmas. *ARL 71-0317, AFSC, 0390, 41, CR.*
Nick, K. P., Richter, J., and Helbis, V. (1984). *J. Quant. Spectrosc. Radiat. Transfer* **32** (1), 1.
Omenetto, N., et al. (1979). *Analytical Laser Spectroscopy.* J. Wiley, N.Y.
Pateyron, B., Aubreton, J., Elchinger, M. F., Delluc, G., and Fauchais, P. (1986). Internal report: thermodynamic and transport properties of Ar, N_2, He, H_2, O_2 at 1 atm. *Internal Report*, Thermodynamic and Plasma Lab., Univ. of Limoges, France.
Pateyron, B., Elchinger, M. F., and Fauchais, P. (1987). *ISPC8* **1**, 67. (Prof. Akashi, ed.). Univ. of Tokyo, Japan.
Pfender, E. (1978). In: *Gaseous Electronics* (N. N. Hirsh and H. J. Oskam, eds.). **1**, 291. Academic Press, New York.
Pfender, E. (1985). *Pure and Applied Chemistry* **57** (9), 1179.
Pfender, E. (1988). *Thermal Plasma Processing in the Nineties, Pure and Applied Chemistry* **60** (5), 591.
Pfender, E. and Boulos, M. (1983). Diagnostic Techniques. In: *Advanced Course on Thermal Plasmas Technology and Applications.* (Prof. Boulos, ed.). Univ. of Sherbrooke, CN.
Pyatnitskii, L. N., et al. (1978). *Teplofizika Yvsokikh Temperature* **16**, 202.
Roumilhac, Ph., Coudert, J. F., Leger, J. M., Grimaud, A., and Fauchais, P. (1987). *ISPC8* **1**, 419. (Prof. Akashi, ed.). Univ. of Tokyo, Japan.
Roumilhac, Ph., Coudert, J. F., Leger, J. M., Grimaud, A., Fauchais, P. (1988). *1st Plasma Technik Symp.* **1**, 105, Plasma-Technik AG, Wohlen, Switzerland.
Sakuta, T. and Boulos, M. I. (1987). *ISPC8* **1**, 371. (Prof. Akashi, ed.). Univ. of Tokyo, Japan.
Santen, S. and Thornblon, J. (1984). Application of plasma technology in metallurgical process. *10th Congress of UIE.* Stockholm, Paper 6.4.
Self, S. A., Paul, P. H., and Young, P. (1982). A Packaged, Fiber-optic Spectroradiometer for High Temperature Gases with Automatic Read-Out. *6th Int. Symp. on Temperature.* Washington, D.C., **5**, 465, March.
Self, S. A. (1987). Diagnostic Techniques for Thermal Plasmas, *Mat. Res. Soc. Symp. Proc.* **98**, 183, MRS, Pittsburgh.
Steffens, H. D., Busse, K. H., and Selbach, H. (1985). *ISPC7.* (Prof. Timmermans, ed.). Univ. of Eindhoven, NL.
Thompson, H. D., Stevenson, W. H. (eds.). (1979). *Laser Velocimetry and Particle Sizing.* Hemisphere, N.Y.
Thorpe, M. L., Jr. and Wilks, P. H. (1971). *Chem. Eng.* **15**, 117.
Trassy, C. and Mermet, J. M. (1984). Les Applications Analytiques des Plasmas HF. *Technique et Documentation.* Lavoisier, Paris, France.
Vardelle, A., Vardelle, M., and Fauchais, P. (1982). *Plasma Chemistry, Plasma Processing* **2**, 255.
Vardelle, A., Baronnet, J. M., Vardelle, M., and Fauchais, P. (1980). *IEEE Trans. Plasma Sci.* **PS8**, 417.

Vardelle, A., Vardelle, M., Pateyron, B., and Fauchais, P. (1985). *7th ISPC Eindhoven, July* 1985. **2**, 898. (Prof. Timmermans, ed.). Univ. of Eindhoven.
Vardelle, A. (1987). Etude théorique des transferts de chaleur, de quantité de mouvement et de masse entre un plasma d'arc à la pression atmosphérique et des particules solides. Thèse de Doctorat es Sciences Physiques, University of Limoges, France.
Vardelle, M. (1980). Thèse 3e cycle. University of Limoges, France.
Vardelle, M. (1987). Etude expérimentale des transferts de chaleur, de quantité de mouvement et de masse entre un plasma d'arc à la pression atmosphérique et des particules solides. Thèse de Doctorat es Sciences Physiques, University of Limoges, France.
Vardelle, M., Vardelle, A., Fauchais, P., and Trassy, C. (1987). *ISPC8* **1**, 404. (Prof. Akashi, ed.). Univ. of Tokyo, Japan.
Vardelle, M., Vardelle, A., and Fauchais, P. (1987). *ISPC8* **4**, 1970. (Prof. Akashi, ed.), Univ. of Tokyo, Japan.
Weise, W. L. (1965). *Plasma Diagnostic Techniques*. (Muddleston, R. H., Leonard, S. L., eds.). Academic Press, N.Y.
Weissman, S. H., Smith, M. F., and Chambers, W. B. (1987). Plasma diagnostics for a low-pressure plasma spray deposition system. *Spring meeting MRS*, Annaheim, CA, USA, *Mat. Res. Soc. Symp. Proc* **98**, 197, MRS. Pittsburgh.
Wetzer, J. M. (1983). *IEEE Trans. in Plasma Sci.* **PS 11**,
Young, R. M., Chyou, Y. P., Pfender, E. (1983). *ISPC6* **1**, 211, 218. (Prof. M. Boulos, ed.). Univ. of Sherbrooke, CN.

Index

A

Abel inversion, 105
  cylindrical symmetry, 368–371
  of multichord measurements, 271
  particle flux and, 415
Abinitio calculation, 87
Absolute spectral power, 88
Absorption spectroscopy
  diagnostics, 74–81, 104–105, 315, 318, 333, 340–341, 343, 398–402
  infrared, 17–19, 31–34, 41–42
  laser light and, 21–24, 31, 42–43, 76–79
  ultraviolet-visible, 15–17
Actinometry technique, 9
  infrared absorption and, 32
Adsorbates, 41–42
Allen's principle, 104
Amorphous hydrogenated silicon, 216
Anemometry, laser, 416–425
Angular momentum, 315–316, 324, 329, 338
Anharmonicities, 22
Anisotropism
  distribution functions, 121, 180
  etching, 22–24
Appearance potentials, 213
Atoms, Zeeman effect and, 315–321
Auger electron microscopy, 35
Axial profile, 91–92

B

Balanced mixers, 266, 270
  see also Microwave, summing device
Ballast resistor, 85
Bartels' method(s), 94–97
Beam
  electron, 121, 123
  ion, 115, 134, 137
  laser, 431
  molecular, 210
  parallel, 125
  thermal spread, 125
  velocity, 125
Beer's Law, 15–16
  fluorescence intensity and, 20–21
  infrared absorption and, 32
Bessel function, 103, 165
Bias
  negative, 116, 117, 126, 130
  radiofrequency self-, 166
  voltage, 128
Blackbody background emission, 4, 93–94
  spectral radiance and, 102–103
Blue-shifted emission, 8–9, 101
Bohm current, 125, 126
Bohr magneton, 317
Boltzmann equilibrium, 50, 379–380
Bombardment, ion, 26
  ellipsometry and, 38
Bragg condition, 281
  see also Microwave, scattering
Brightness, plasma, 357–360

C

Calibration, 224, 226, 227, 435–437
  pedestal, 431
Calorimetry, 432

Capacitance neutralization, 169
Capacitive bridge, 179
Capacitive probes, 181
  secondary emission, 153
  see also Langmuir probe(s)
Capacitor resonator, 261
Cathode generator, 354
Cathode sheath, 4, 121, 145
  dimensions, 116, 130, 140
  electric field, 40
  expansion, 134
  impedance, 162
  optogalvanic effects in, 34–35
  probes, 130–138, 140–141, 192
  time-resolution from, 7, 15
Cavity, *see* Microwave, cavity
Centrifugal distortion, 22
Charged species, 2, 19
Chemiluminescent recombination reaction, 5
Chopping frequency, 6
Chordal scans, 100, 105
Circuits, 172–176
  bridge, microwave, 240
  emissive probe, 128, 176
  microwave, 258, 262
  noise suppression, 61–62
  op amp, 174
  simple, 164, 172
Coherent Anti-Stokes Raman Spectroscopy (CARS), 29–30
  thermal plasma processing and, 403–405
Cold plasmas, *see* Plasma(s), cold
Collecting probes, 181
Collisional mean free path, 131, 138, 350
Collision frequency, 240, 242, 256, 286
Collision process, *see* Quenching process
Computer simulations, 2, 217
Conductivity, 223, 296–297
  nonlinear, 164
  plasma, 251, 252, 256, 286
Contamination
  probe, 146, 171–172
Continuous-spectrum radiation, 107
Continuum emission, 5
Coupling, *see* Microwave, coupling
Critical density, *see* Plasma(s), critical density
Cross-field diffusion, 143
Cross-section
  absorption, 15–16
  collision, 5, 9, 88
  Raman scattering, 28–29, 40–41
Current
  density, 116, 127
  electron, 123
  ion, 130, 148
  probe, 118
  time-averaged, 167, 168
Curve tracer, 173
Cutoff
  in plasma, 273, 289, 306
  in waveguide, 245
Cyclotron frequency, 143, 284, 286
Cylindrical probe(s), 132, 133, 181
  see also Langmuir probe(s)
Cylindrical symmetry, 368–370

D

Debye length, 59, 131, 189, 281
Decay time, 81–83
  optogalvanic resistance and, 85
Degeneracy, 318, 331
Density
  current, 116
  electron, 116, 127
  excited-state, 74–79
  ion, 127
  plasma, 239, 242, 247, 256, 264, 271, 287, 306
Deposition process, 2
  ellipsometry and, 38
  optical emission discharge and, 4
Detector
  microwave, 243, 246, 248, 249, 262, 279, 335, 337
  plasma emission(s), 3
Deuterium lamp, 16
Device isolation, 2
Diagnostics
  absorption spectroscopic, 74–81
  emission spectroscopic, 80–81, 89–103
  Langmuir probe, 58–62, 64–69
  laser, 76–79, 81–87, 105–107
  magnetic resonance, 345–346
  microwave, 72–73, 239–240, 259–263
  Rozhdestvenskii "hook", 76
  stark effects, 107
Dielectric tensor, 286
Diffusion
  cross-field, 143

electron, 115
masks, 2
Dipole, 316, 328–331, 335, 341
Directional probe, 180
Dirty probe, 129, 173
Discharge flow tube, 2, 87
Discharge(s)
absorption spectroscopy and, 15–19, 31
coupled, 355
in gas(es), 3–15, 47
glow, 115, 160
lineshape measurements, 6–12
microwave, 3, 23–24, 72–73
paramagnetic resonance, 313, 332, 336–338, 342–343, 345–346
time-resolved, 14–15, 21
see also Absorption spectroscopy; Electric discharge lamps
Discriminator, 262
Dispersion, of plasma emission(s), 3
D-lines, *see* Resonance lines, sodium
Doppler effect
broadening, 75, 84, 86, 385–387
laser anemometry, 416–425
lineshape measurements of, 6–12, 83, 84
plasma velocity and, 407–408
transition frequency, 25
width, 7, 13
see also Two-photon Doppler-free spectroscopy
Double probe, 160, 176
Double sheath, 147
Drift velocity, 123, 126

E

Electric arcs, 352
Electric discharge lamps, 47–48
fluorescent, 48–49, 57–72
mercury, high pressure, 52, 89–91
metal halide, 56, 91–93
sodium, high pressure, 54, 93–103
sodium, low pressure, 48–49, 70
spectroscopic diagnostics of, 88–107
Electric field(s)
cathode sheath, 40
coupling, 262
laser-induced fluorescence and, 26–27
Electron beam, 123
one-dimensional, 121
Electron energy loss spectroscopy, 35, 80–81, 92–93
Electron impact dissociation
plasma-induced emission and, 4
Electron impact excitation mechanisms, 2, 202, 212
absorption spectroscopy and, 74–81
and actinometry technique, 9–10, 13
emission spectroscopy and, 80–81, 92–93
energetic primary electrons and, 128, 129
laser-induced fluorescence and, 21–24
plasma-induced emission and, 4
post-ionization, 202
Electron(s)
Boltzmann, 126
changing frequencies and, 6–12
drifting, 123, 124
emission, 115, 129
energy distribution, 9, 80–81, 92–93, 116, 117
excited-state density, 74–79
fluorescent lamps and, 49, 92–93
ground-state density, 79–81
hybrid frequency, 297
Maxwellian, 59–61, 71, 122, 350
neutral collisions and, 119
non-Maxwellian, 64–69, 117
primary, 121
in radiofrequency discharges, 13
secondary, 144–146
Electrostatic ion cyclotron (EIC) instability, 137
Electrostatic potentials, 119
Ellipsometry, 37–39
Emission(s)
anti-stokes, 30
continuum, 358–359, 388, 433–434
floating point in, 149
laser-induced, 30–34, 81–84, 403, 405–406
lineshape measurements of, 6–12
spectroscopic diagnostics, 80–81, 89–103, 364–397
strong, 149
Encapsulation, 2
Endpoints, etching, 15
Energetic primary electrons, 128
Energy analyzers, 137
Enhancement, emission, 5–6
Enthalpy probes, 362–363
Equilibrium
actinometry and, 32

Boltzmann, 50
local thermodynamic, 51–52, 54, 56
Saha, 92
Etching process, 2, 217, 218, 313
anisotropic, 22–23
infrared absorption spectroscopy and, 18–19
interferometry and, 36–37
laser-induced fluorescence and, 24–25
optical emission discharge and, 4
spectrally-resolved emission and, 15
Excimer laser, 19
Excited-state density, 74–79

F

Fabre-Perot interferometer, 3, 306
high resolution spectroscopy, 86–88
light scattering and, 359
Faraday rotation, 298
Filtering techniques, 170
mass, 201–210
Fine-structure states, 317, 319–321, 323
Flame diagnostics, 345
Flashing, probe, 59
Floating potential, 58, 115, 127, 128, 147, 192
time-averaged, 167
Fluorescence
dispersed, measuring, 3
laser-induced, 14, 19–28, 42–43, 81–83, 403, 405–406
Fluorescent lamps, *see* Electric discharge lamps
Flow systems
magnetic resonance, 336–337
plasma–, diagnostics, 362–406, 440
Fluctuations, 119
natural, thermal plasma, 361
square wave, 163
Flux analysis
ion, 192–201, 203–210
neutral species, 210–214
plasma, 187–190, 191–201, 203–219, 361, 432–433
Following technique, 170
see also Filtering techniques
Fourier transform infrared, 2, 18–19
Fragmentation, 212
Free propagation, 297–298
Frequency bands, *see* Microwave, frequency band designations
Fusion
devices, 228
plasmas, 143

G

Gas composition, 2
and electric discharge lamps, 49, 52, 54, 56, 57–72, 70, 91–103
and electron collisions, 35–36
residual, 220, 221
Gas-phase
paramagnetic resonance spectrometers, 332–339
species, 2
Gaussian profile, 75, 387–388, 431
Geometry, arbitrary convex probe, 121
Glow discharge(s)
flux analysis and, 189–190
impact processes and, 4
ionization data, 160
low pressure, 1–2
plasmas, 115
Ground-state density, 79–81

H

Hamiltonian, atomic, 315–316, 319, 321, 330
H-Balmer-alpha emission, 5
lineshape measurements, 8, 10
Heating, cyclotron, 242
High intensity discharge lamps, 52
Horns, *see* Microwave, horns
Hybrid frequency, 297
Hybrid ring, 247
Hydrogen discharge cleaning, 229
Hysteresis, 173

I

Imprisonment time, 81–84
Impurities, 172
Incident light, 15–16

Inflection point method, 147, 150
Infrared
  absorption, 17–19
  spectrum, 3–4
Instability, reflected ion beam, 137
Insulator, 2, 128
Integration, 340
Intensity, fluorescence, 19–20, 342
Interferometer, 243, 297
  bridge, 265
  frequency modulation, 269
  frequency swept, 266
  heterodyne, 268
  multichord, 271
  quadrature, 269
  serrodyne, 266
  zebra-stripe, 266
  see also Fabre-Perot interferometer; Microwave, interferometer
Interferometry, laser, 36–37
Interisotopic transfer, 88
Inversion symmetry, 39
Ion beam, 115, 134
Ion impact process, 4
  ellipsometry and, 38
  flux analysis, 192–201, 203–210
  kinetic energy and, 193
  laser-induced, 22–24, 34–35
  monoenergetic, 155
  neutral collisions, 138
  probe diagnostics and, 57–61
Ionization, 2
  glow discharge data, 160
  Penning, 6, 215
  potential, 5, 13
  thermal plasma, 377
Isotropic distribution functions, 117
I–V characteristic(s), 124
  knee, 118, 134
  scans, time-averaged, 168

K

Karabourniotis methods, 97–100
Kinetics, 193, 350
  etching, 5
Kirchoff-type equation, 93
Knudsen effect, 408

L

Lambertian dependence, 88
Langmuir probe(s)
  characteristics and dimensions, 118, 130–138, 141, 142, 158, 160, 176, 180–181
  circuits, 61–62, 128, 172–176
  collision effects on, 138–142
  contamination, 171–172
  data, interpretation of, 116–130
  diagnostics, 57–61
  electron energy distribution, 59, 64–69, 81–83
  and electron temperature measurements, 176–180
  emissive, 121, 146–157, 159, 176, 181
  and fluctuating plasma potentials, 162–171
  perturbations, 59–61, 157–162
Larmor frequency, 284, 286
Laser techniques, 19
  and diagnostics, 76–79, 81–87, 105–107, 403–405
  Doppler anemometry, 416–425
  interferometry, 36–37
  laser-induced fluorescence, 14, 19–28, 81–84, 403, 405–406
  magnetic resonance, 337–339, 342–344
  optogalvanic effects, 34, 85, 105–107
  particle flux measurements, 413–415
  Raman scattering, 28–29, 40–41, 105–106
  top-hat, 431
  tunable infrared laser absorption, 2, 25, 30–34
Lifetime
  fluorescence, 5, 21–22
  of neutral species, 5
Light
  monochromatic, and plasma-induced optical emission, 3–15
  scattering, *see* Scattering, light
Limit of zero emission method, 150
Line reversal, 16, 99–104
Lineshape measurements, 6–12, 339–340
  broadening constants and, 99–103, 107
  comparisons, 103–104
  magnetic resonance, 341
  two-photon Doppler-free spectroscopy, 83, 84, 87
  velocity distribution and, 25
Load line, 129

Load resistances, 128
Lorentz force, 285
Lorentzian function (profile), 25, 75, 79
  resonance line diagnostics and, 99–102
Loss processes, 2
Lumped circuit analog, 252–253

M

Mach-Zehnder interferometer, *see* Interferometer
Magic T, 247
Magnetic field, 115, 119, 122
  coupling, 262
  fusion devices, 228–232
  paramagnetic resonance, 315, 318–319, 325–327, 330–333, 335, 337–340
  probes in, 142
  see also Plasma(s), magnetized
Magnetic moment, 314–316, 327–328, 330
Mass filtering, 201–210
Maxwell-Boltzmann maximum, 22, 377
Maxwellian distribution, 58, 59, 64–72, 117, 132
  drifting, 117, 122
  and temperature, 387–388
Maxwell's equations, 287, 304
Mercury lamp, high pressure, 52, 89–91
Metal halide lamp, 56, 91–93
Metastable species, 5
  optogalvanic effects and, 34–35
Microwave
  attenuator, 248, 265
  bridge circuit, 240
  bridge interferometer, 265
  cavity, 241, 249, 258–260, 333–336, 338, 341, 345
  components and waveguides, power losses in, 249, 276, 277
  coupling, 262, 316, 321–322, 330–332
  detector, 243, 246, 248, 249, 262
  diagnostic techniques, 239–263
  discriminator, 262
  frequency band designations, 243, 244
  horns (antennae), 248, 274, 275, 278, 297, 305
  imaging, 305
  interferometer, 243, 248, 264
  lenses, 305
  measurements, 240
  oscillator, 262, 266
  phase detection, 243, 246, 247, 248, 264, 267, 270, 271, 279
  power feed-through, 261, 296
  reflections, 249, 263, 276, 307
  scattering, 281
  source, 247, 273, 276
  summing device, 247
Microwave discharges, 3
  diagnostics, 72–73
  laser-induced fluorescence and, 23–25
Millimeter waves, 240, 245
  see also Microwaves
Modulation frequency, 5, 211, 334, 335, 337–338, 340–341
  linear absorption laser diagnostics, 76–79
  metastable species and, 6–7
  and velocity, 31–32
Molecular beam, 2
Molecules, paramagnetic resonance of, 321–332, 344–345
Monochromator, 3–15
  absorption spectroscopy and, 16
  grating, 366–367
  laser techniques and, 19–20, 28–29, 434
Monoenergetic electrons, isotropic, 121
Mott-Cabrera mechanism, 39–40
Multi-channel detectors, 2
Multidipole device, 129

N

Narrow bandwidth laser(s), *see* Tunable infrared lasers
Narrow bandwidth, 2
Neutral collisions
  electron, 119
  ion, 130, 131, 138
Neutral species, 2
Noise suppression circuit, 61–62, 162
Non-semi-infinite planar probes, 121
  see also Langmuir probe(s)
Nuclear spin, 22

O

Opacity, *see* Cutoff, in plasma
Op amp circuit, 174

Open magnetic mirror plasma, 157
Optical multichannel analyzer (OMA), 3
  thermal plasmas and, 376
Optical path length, *see* Radiation transport
Optical probes, 2, 34, 88
Optical techniques, 2, 304–307
  absorption spectroscopy and, 74–81
  emission spectroscopy and, 80–81, 89–103
  plasma-induced optical emission, 3–15
Optimization, empirical, 2
Optogalvanic effects, 26–27, 34–35
  laser diagnostics and, 85, 105–107
  spectral profile of, 85
Orbital motion
  angular momentum, 148
  particle, 130
  theory, 132
Organic oxidation, 345–346
Oscillations
  forced, 252–253
  free, 252

P

Paddle probe(s), 160
  see also Langmuir probe(s)
Paramagnetic resonance, 313–314, 344–346
  applications, 332–344
  basic theory, 314–332
Parity, 328–330
Partial pressure analysis, 190–191, 219–232
Particle(s)
  beam temperature, 125
  confinement, 119
  monoenergetic, 132
  parameters, in thermal plasmas, 410–432, 441
Penning ionization, 6, 215
Perturbation theory, 254–258
  see also Langmuir probe(s), perturbations
Phase shift, 6, 243, 246, 264, 267, 271
Photoacoustic spectroscopy, 41–42
Photodetector, 3
Photodissociation, 5
Photomultiplier tube, 3
  fluorescence and, 19–20
Photoresist, 2
Pierson transformers, 173
Piezobirefringent elements, 37–38
Pitot tube, 406–407
Planar probes, 118, 123, 131, 181
  negatively-biased, 117, 126
  semi-infinite, 116, 130
  see also Langmuir probe(s)
Plasma-induced optical emission, 3–15
  laser techniques and, 19–35
  lineshape measurements of, 6–12
  surface interactions, 35–42
Plasma(s)
  cold, 154, 284, 360
  collisional, 141
  conductivity, 251, 252, 256, 286
  critical density, 264, 287, 306
  density, 239, 242, 247, 256, 264, 271, 287, 306
  drift, 123
  flow, 142, 362–406, 440
  flux analysis, 187–190, 191–201, 203–219, 361
  frequency, 242, 261, 286
  homogenous, 299, 302
  inhomogenous, 300
  Langmuir probe diagnostics and, 59–61
  magnetized, 189–190, 242, 289, 298
  multidipole, 122
  open magnetic mirror, 157
  parameters, 127, 410–432
  potential, 114, 162
  processing, 2, 114, 313, 352–355
  reactor, 260
  real, 133, 400–403
  thermal, 356–361, 410–432
  unmagnetized, 136
Plasma-surface interactions, optical techniques for, 35
  ellipsometry, 37–38
  laser interferometry, 36–37
  photoacoustic spectroscopy and, 41
  Raman scattering from surfaces and, 40
  second harmonic generation, 39–40
Plume, plasma, 409
Polarization, 337, 404
Potential
  dip, 150
  drop, presheath, 127
  floating, 167
  plasma-, probe at, 141
Power, 2
  high, dissipation, 174
Poynting's theorem, 250

Probe(s)
admittance, 178
construction, 176
contamination, 146, 171–172
cylindrical, 132, 133, 181
diagnostics, 58–61, 59, 61–62, 64–69
directional, 180
dirty, 129, 173
double, 160, 176
emissive, 121, 146–157, 159, 176, 181
enthalpy, 362–363
of etching kinetics, 5, 22
geometry, 121, 130
laser, 28–29, 30–34, 76–79
measurements, in sheaths, 158
optical, 2, 19, 20, 88
paddle, 160
planar, 116, 117, 121, 126, 181
at plasma potential, 141
sheaths, 130–138, 140–141, 192
see also Langmuir probe(s)
Process control, 1
Product, 314, 343–344
Pulsed systems, 172
Pump beam, 76–79
Pyrometer, 374, 434

Q

Quasi-cylindrical symmetry, 370–371
Quasi-neutral region, 140
Quenching process, 21–22
broadening, 86
electric discharge lamps and, 51, 75
infrared absorption and, 32
plasma-surface interactions and, 35–36
Q-value, 251, 256, 260

R

Radial profile, 91–92, 426
Radiation transport, 57, 81–83
microwave, 245–249, 315, 318, 333–338
non-optically-thin, 93–103
optically-thin, 80–81, 89–93
Radical formation, 2
neutral, 23
Radical species, 213, 217
Radio-frequency, 2
chopping, 6
inductively coupled discharges, 355
laser techniques and, 19–35
lineshape measurements of, 6–12
modulation, 5
plasmas, 4–5, 396–397, 415–416, 428–429
self-bias, 166
sinusoidal, 164
Radiometry, 283
RAM, *see* Random access memory
Raman scattering, 28–29, 42, 409–410
laser diagnostics, 105–106
from surfaces, 40–41
Random access memory (RAM) circuitry, 1
Rayleigh scattering, 105, 359–360, 406, 431–432
Recombination reaction(s)
chemiluminescent, 5
discharge falloff and, 23–24
paramagnetic resonance, 336, 338, 345
thermal plasma processing and, 388–389
Reflectometry, 263
Refraction, 273, 274, 306
Relaxation, electron energy, 24
Resistance, 85
load, 128
Resonance lamp, *see* Electric discharge lamps
Resonance line(s), sodium, 50, 54, 56–57
diagnostics, 99–104, 107
radial temperature profile and, 96
width, 252, 257, 299

S

Scanning, 375
fast I–V, 168
with mirrors, 373–374
range, 3
by source displacement, 372–373
Scattering, light, 280–283, 406, 408, 409–410
laser Doppler anemometry, 416–425
Second Harmonic Generation (SHG), 39–40
Selection rule, 327, 329–330, 337
Semiconductor-processing plasmas, 21
Semiplanar probe, 116
see also Langmuir probe(s)
Sensitivity, 335, 338, 343
Shadowing, 139, 158

Sheath, *see* Cathode sheath; Probe(s), sheaths
Signal averaging, 16
Signal-to-noise-ratio, 19, 214
Silicon, 2
  amorphous hydrogenated, 216
  see also Etching process
Silicon integrated circuits, 2
Sinusoidal radio-frequency, 164
Skin depth
  in microwave cavity, 251
  in plasma, 255
Sodium lamps
  high pressure, 54, 93–103
  low pressure, 48–49, 70
Source function, 93
Space-resolved data, 2
  laser-induced fluorescence and, 26–27
Species concentration, 91–92
Spectrally-resolved data
  laser-induced fluorescence and, 27
  thermal plasma processing, 393
Spectrometry
  mass, 185–187
  paramagnetic resonance, 332–333, 336–338, 345–346
  see also Flux analysis
Spectrum, emission, 3, 318–322, 326–328
  and chopping frequency, 6
  Raman scattering and, 30
Spherical probes, 122, 176, 181
Spin, 315, 321–324, 326, 330–332
Splitting, 321, 326, 330–331
Spray torch, *see* Torch, plasma
Sputtering, 159, 208
Square wave fluctuations, 163
Square-law detector, 246
Stark shift, 26, 27, 107
  thermal plasma processing and, 385–388
  two-photon Doppler-free spectroscopy of, 84–85
Steep gradients, 360
Surface interactions
  and contamination, 146
  plasma–, optical techniques for, 35–42

T

TE modes
  in cavities containing plasma, 296
  coupling with TM modes, 289, 295, 302
  electric coupling of, 262, 296
  in empty cavity, 258, 296
  in rectangular waveguide, 245, 297
Temperature
  axis, determination of, 94–98
  broadening, 16, 56
  emission spectroscopic diagnostics and, 89–91
  measurements, direct electron, 115, 176
  parallel beam, 125
  particle–surface, in-flight, 432–438
  radial, 90
  thermal plasmas and, 357–359, 374–376, 377–395
  translational, 31–32
Term symbol, 316
The High Pressure Sodium Lamp (de Groot/van Vliet), 101
Thermal plasma(s), 3
  characteristics of, 367–368
  measurements, 356–361
  particle parameters in, 410–432
Thompson scattering, 406
Time-resolved data, 2, 4
  and discharge dynamics, 14–15, 59
  laser-induced fluorescence and, 21, 26–27, 81–84
  metastable species and, 6
TM modes
  in cavities containing plasma, 296
  in empty cavity, 258
  magnetic coupling, 262, 296
Torch, plasma, 349, 354, 355, 389–395
Trajectory distribution, particle, 410–414
Transferred arcs, 352–353, 395–396
Transition, magnetic resonance, 316, 318–320, 322–331, 337, 340
Translational energy, 26
Tunable infrared lasers, 2, 19–21, 30–34
Two-photon Doppler-free spectroscopy, 83, 84–87

U

Ultraviolet region
  absorption spectroscopy, 15–17
  imaging, 4
  photodissociation of, 5
  Raman scattering, 29, 40–41
  resonance lines, 52
Uncharged species, 19–20

V

Vacuum
  electromagnetic waves in, 293–294
  emissive probe in, 152
  partial pressure analyzer, 223
van der Waal's broadening, 101, 107
Velocity distribution, 25–26, 406–409, 415
Voigt profile, 75, 387–388
Vortex injection, 395

W

Wall loading, 51
Wall-stabilized arcs, 353–355
Waveguide(s), 241, 243
  attenuation, 249, 276, 277
  dielectric, 306
  directional coupler, 262
  E-plane, 247
  H-plane, 247
Wavelength
  in free space, 245
  in plasma, 243, 246, 273
  in waveguide, 243, 245, 266

X

X-ray photoelectron spectroscopy, 35

Z

Zeeman effect, 315, 324–327, 333, 335, 338

Plasma–Materials Interactions

Dennis M. Manos and Daniel L. Flamm, *Plasma Etching: An Introduction*

Orlando Auciello and Daniel L. Flamm, *Plasma Diagnostics: Volume 1, Discharge Parameters and Chemistry; Volume 2, Surface Analysis and Interactions*